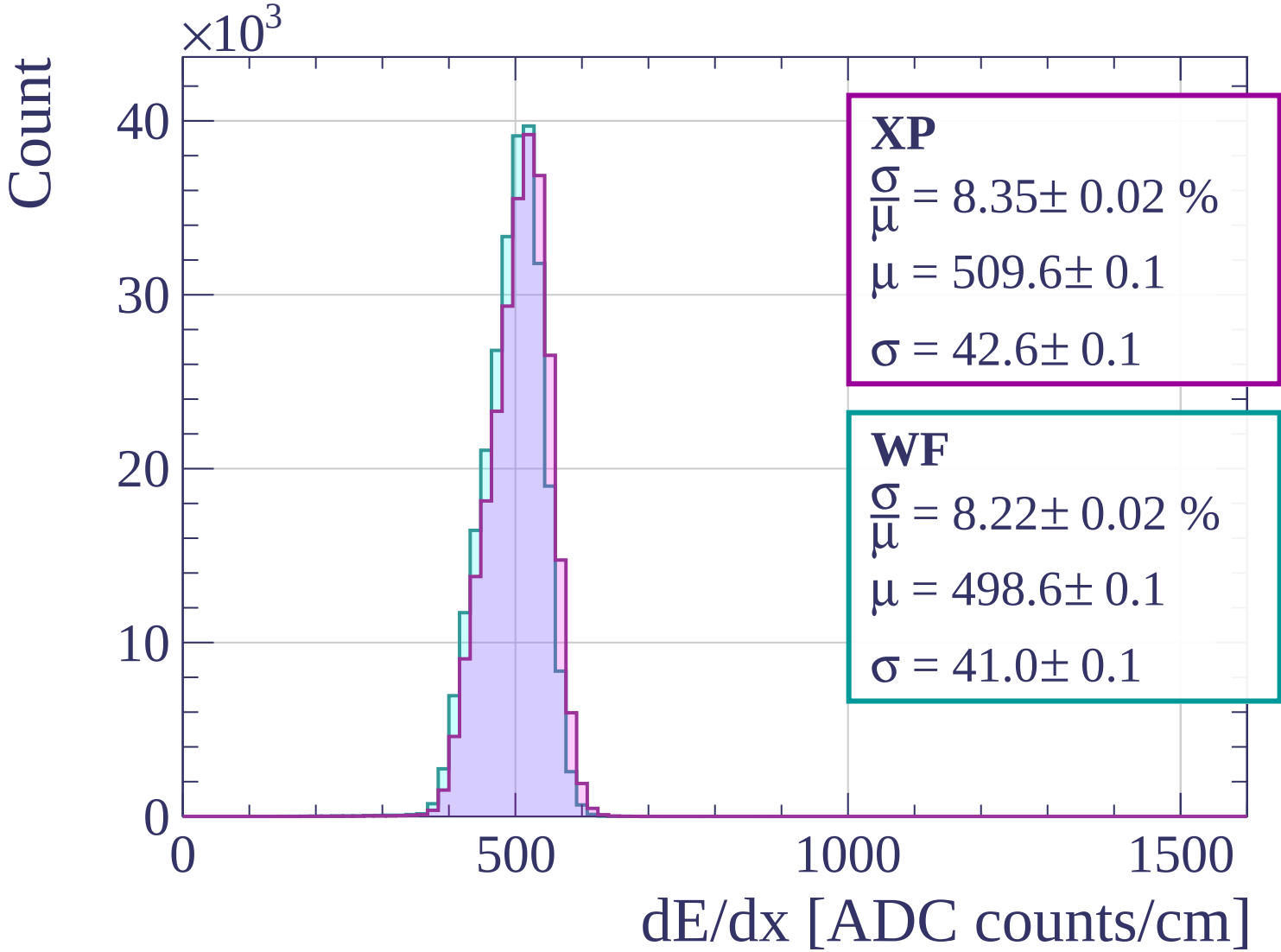
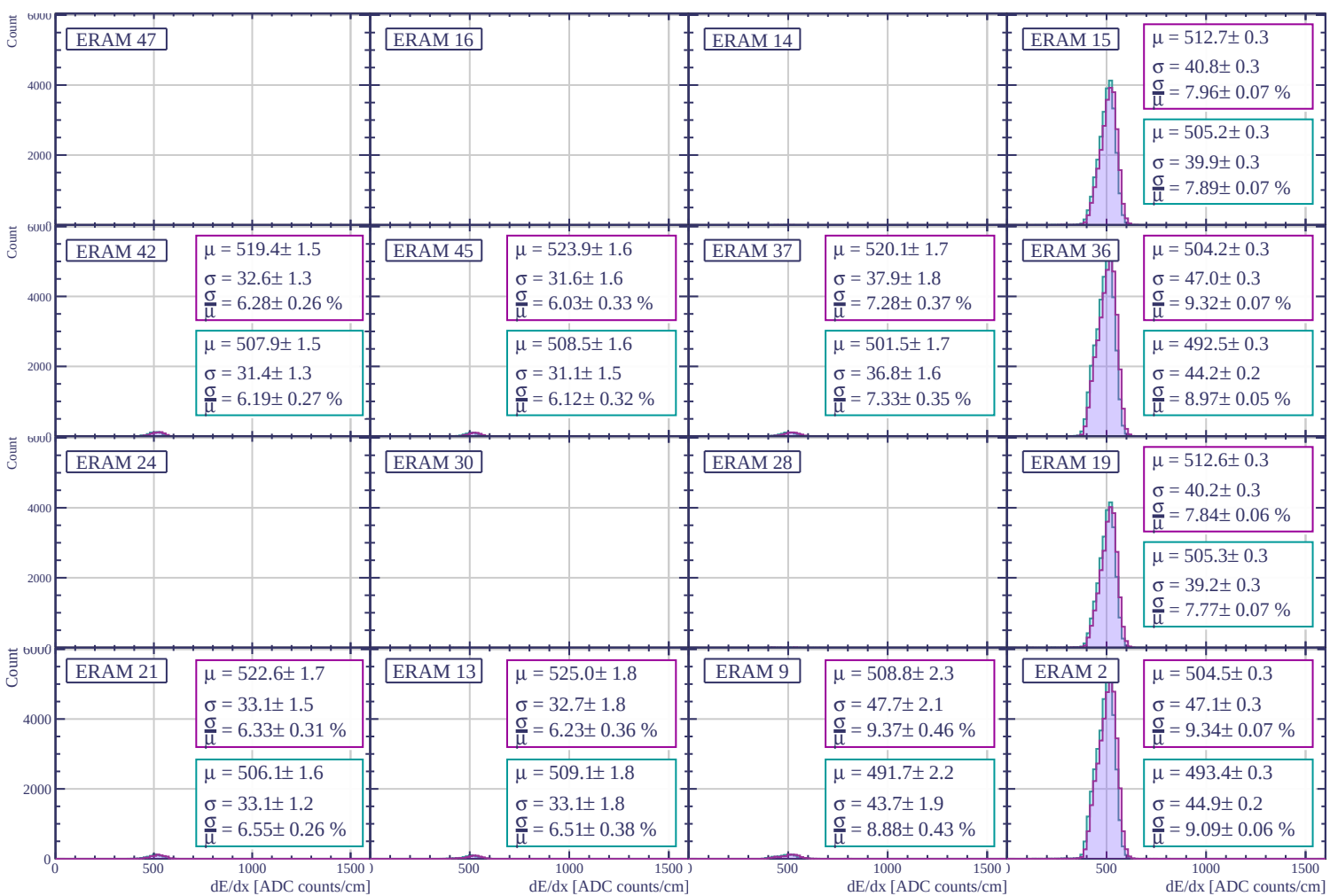
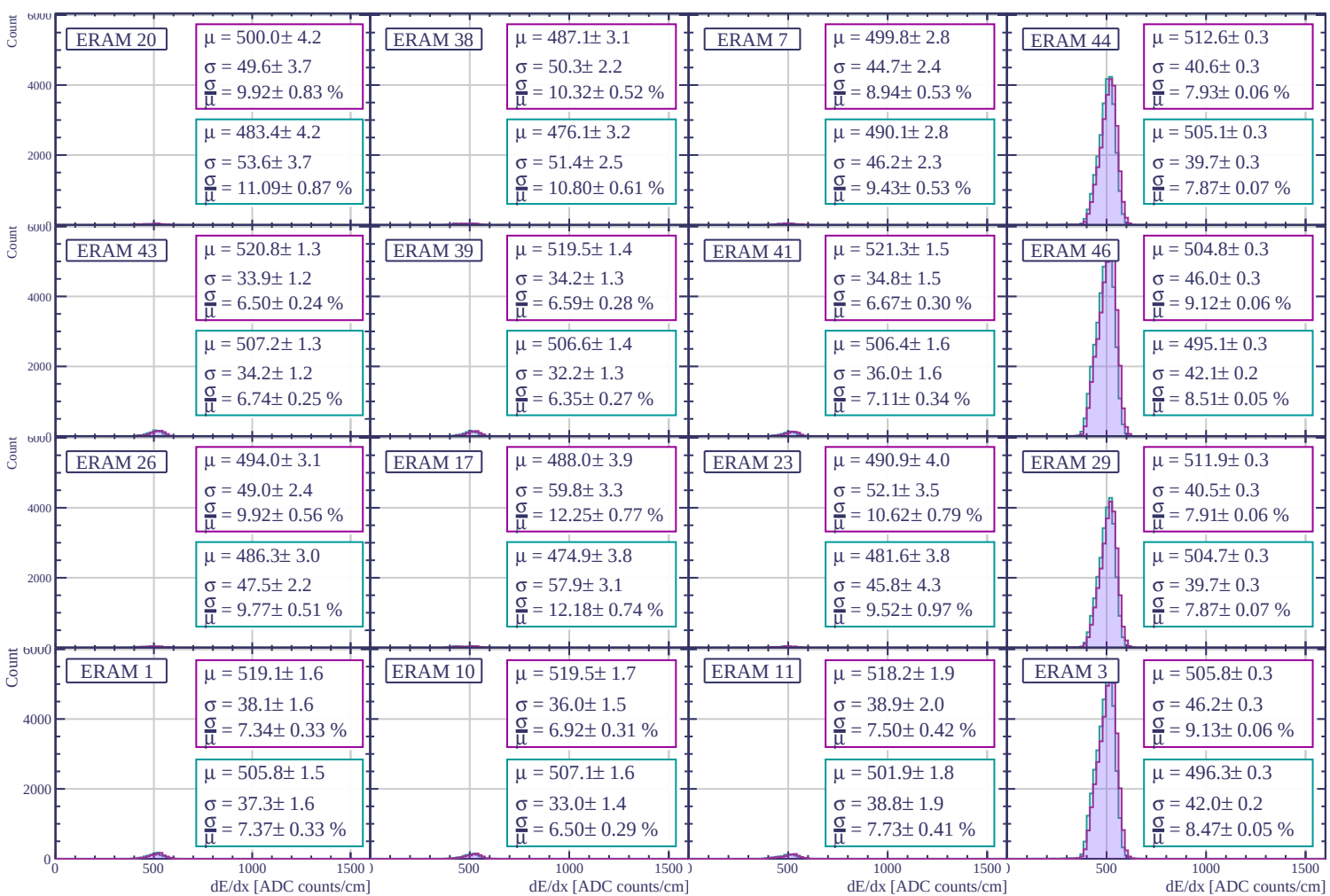
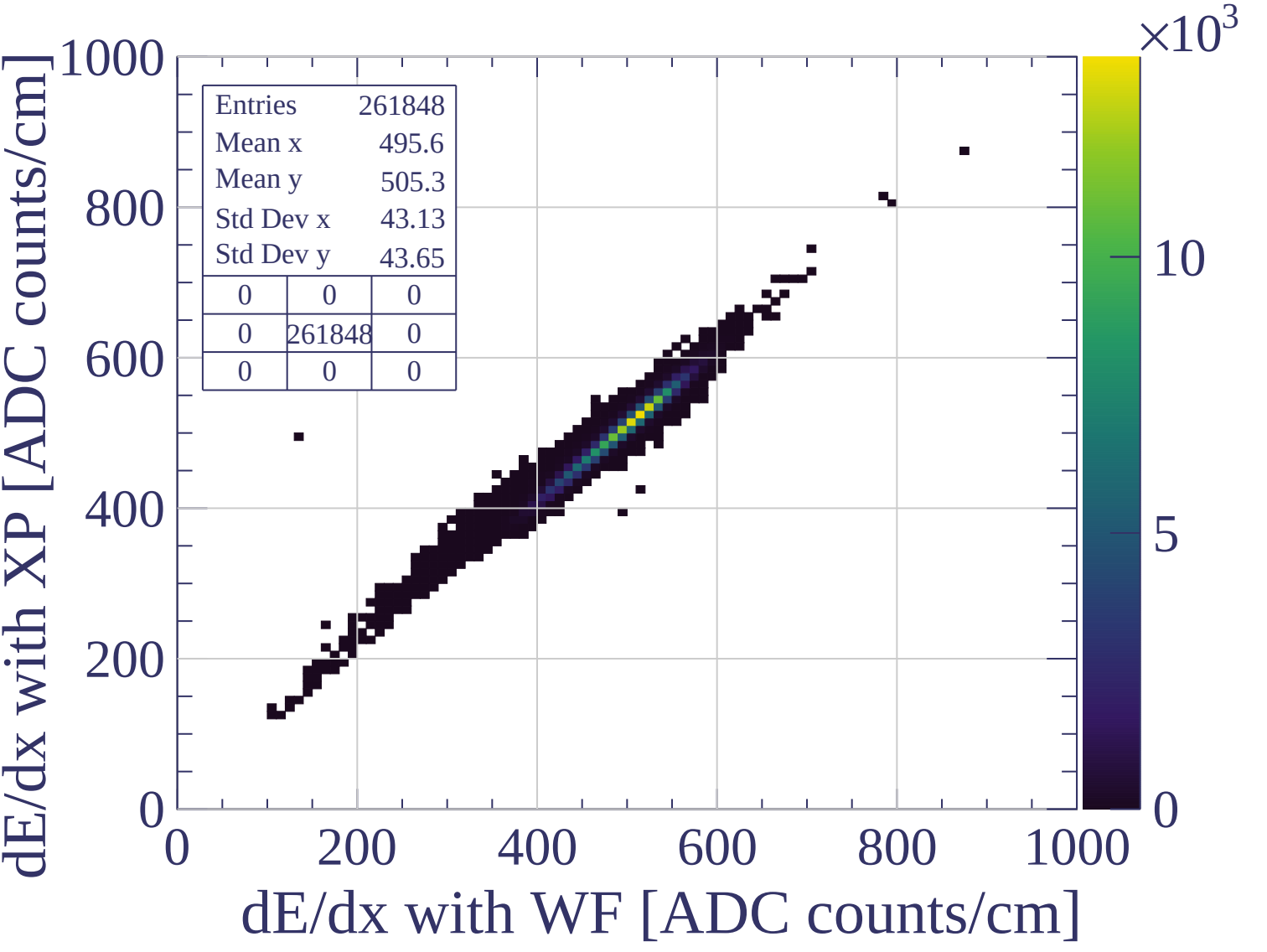
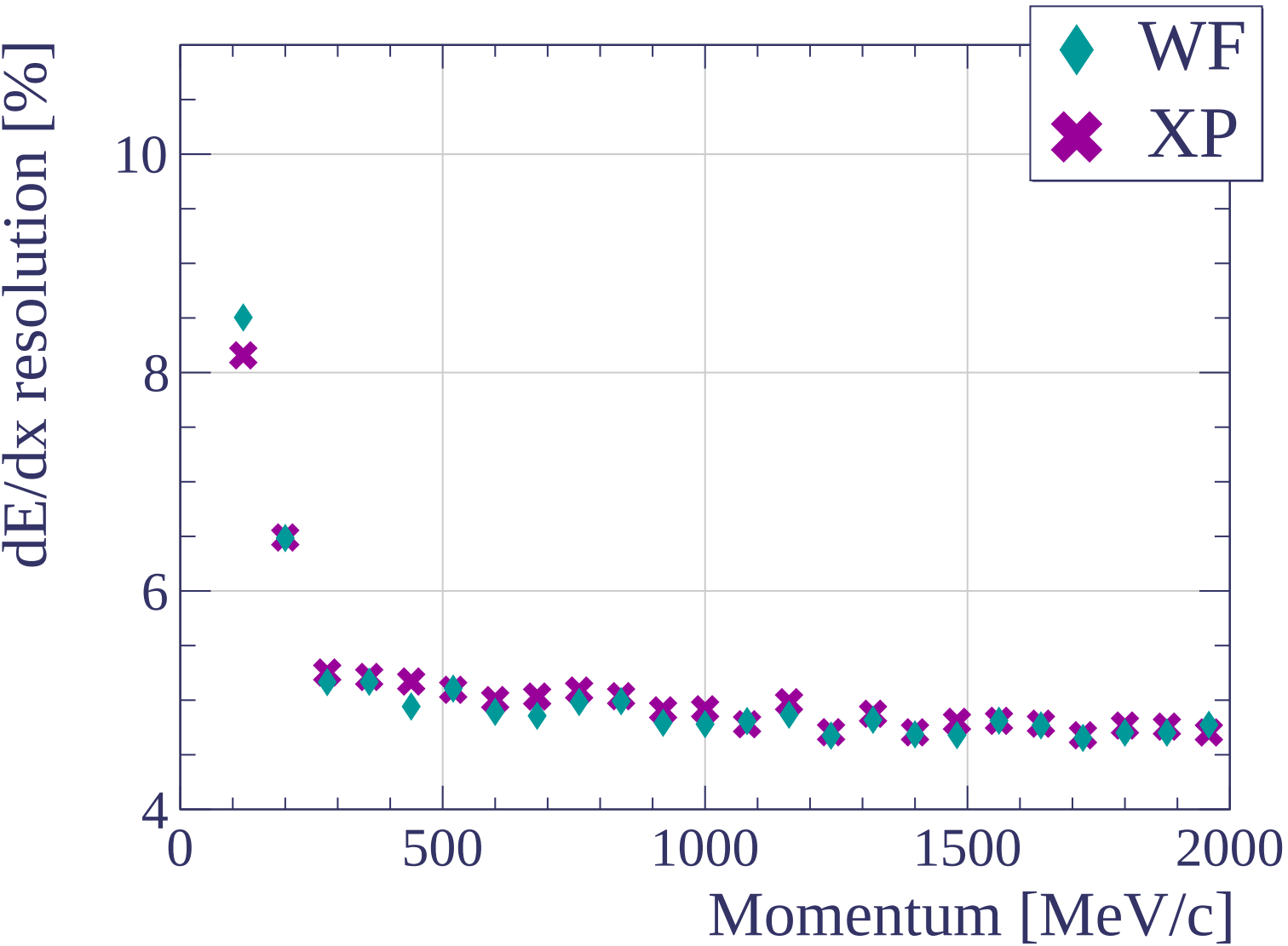


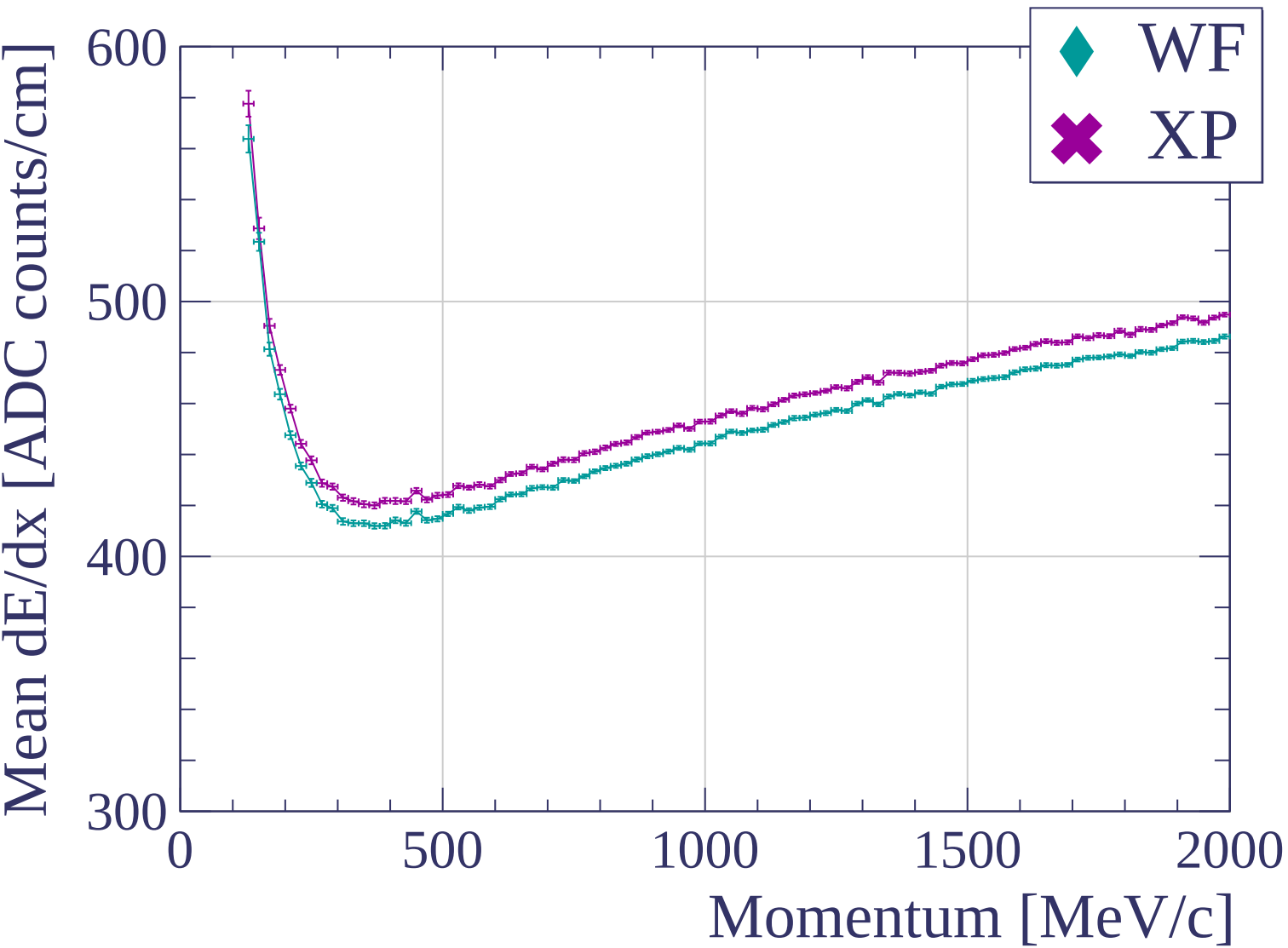


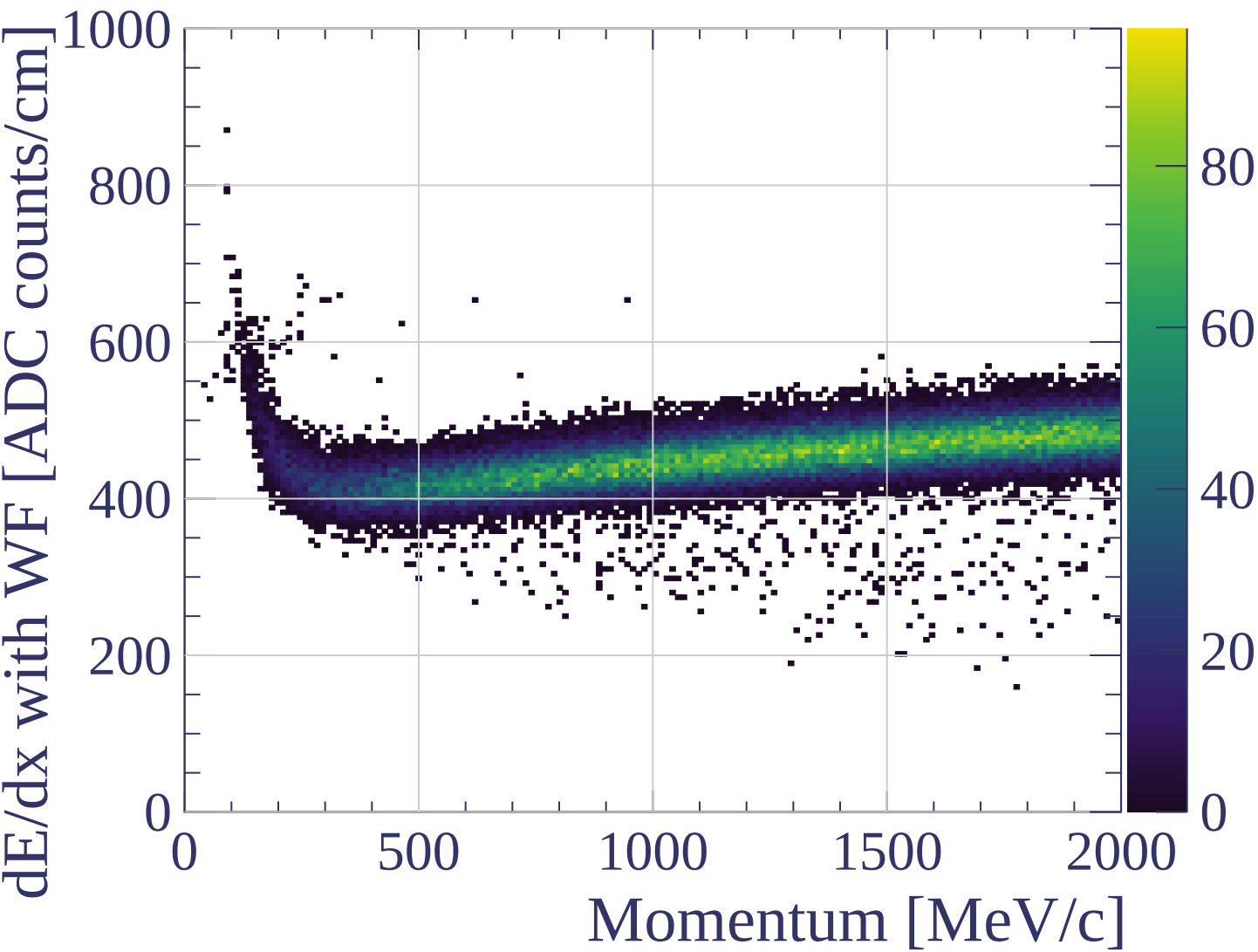


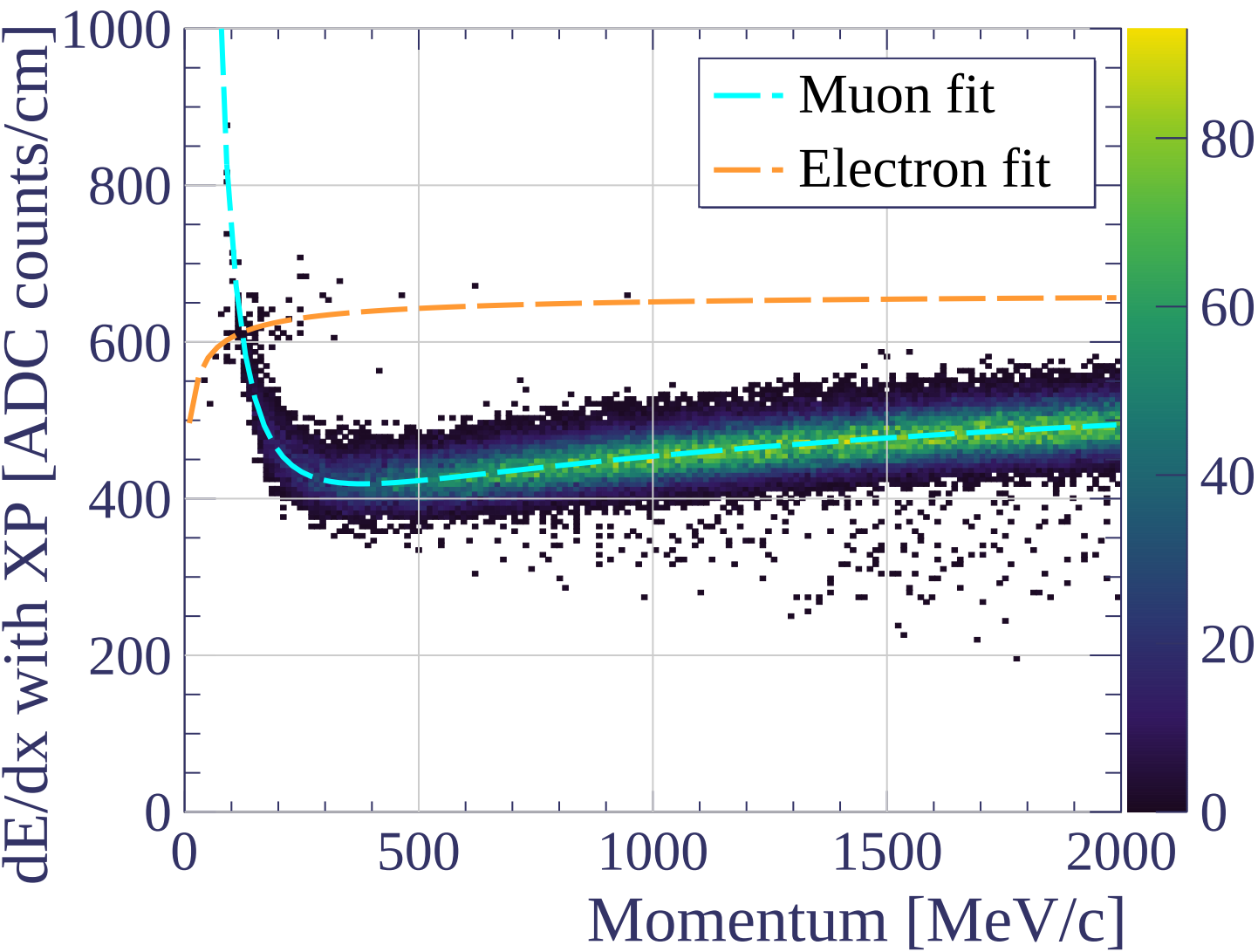




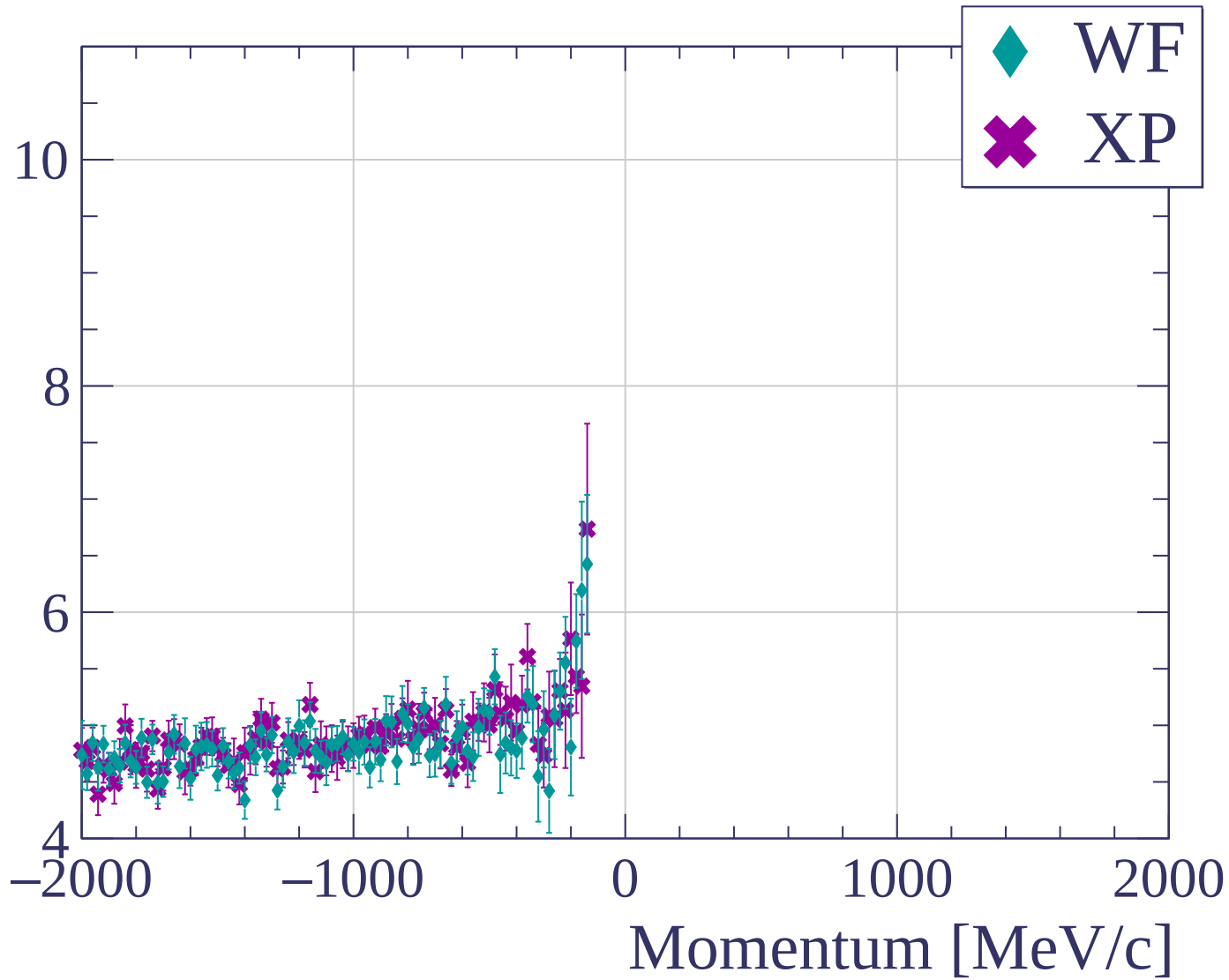






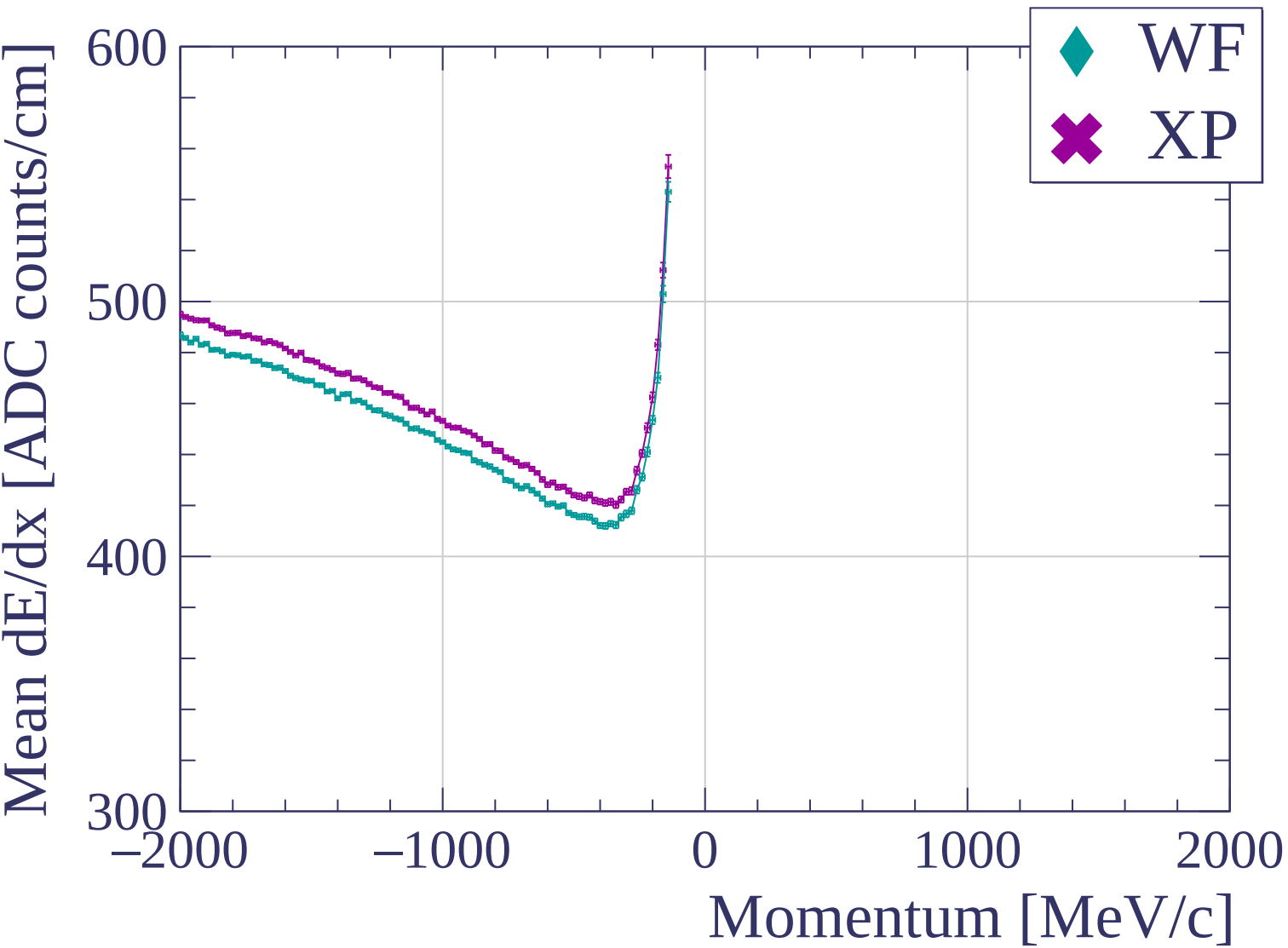


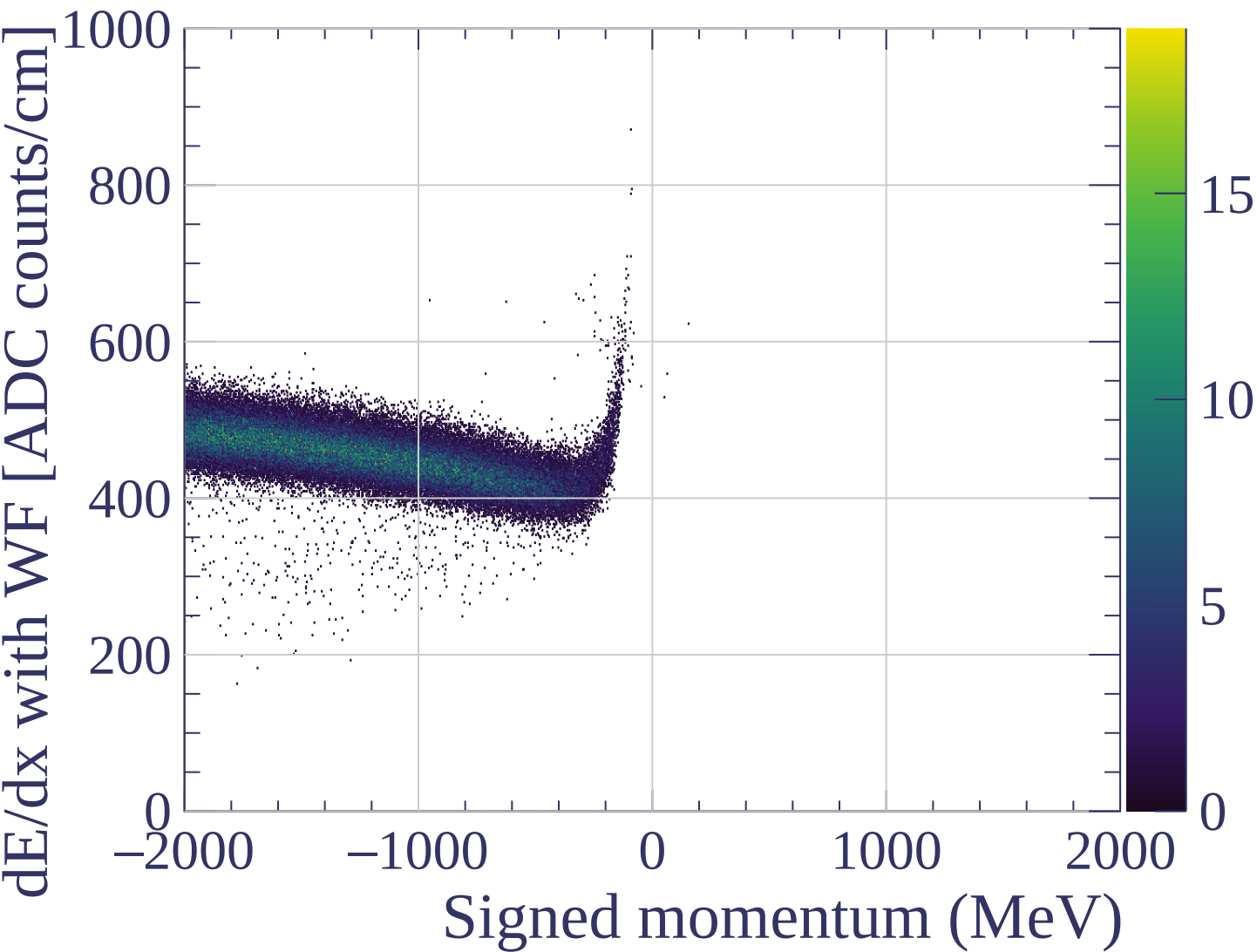
dE/dx resolution [%]

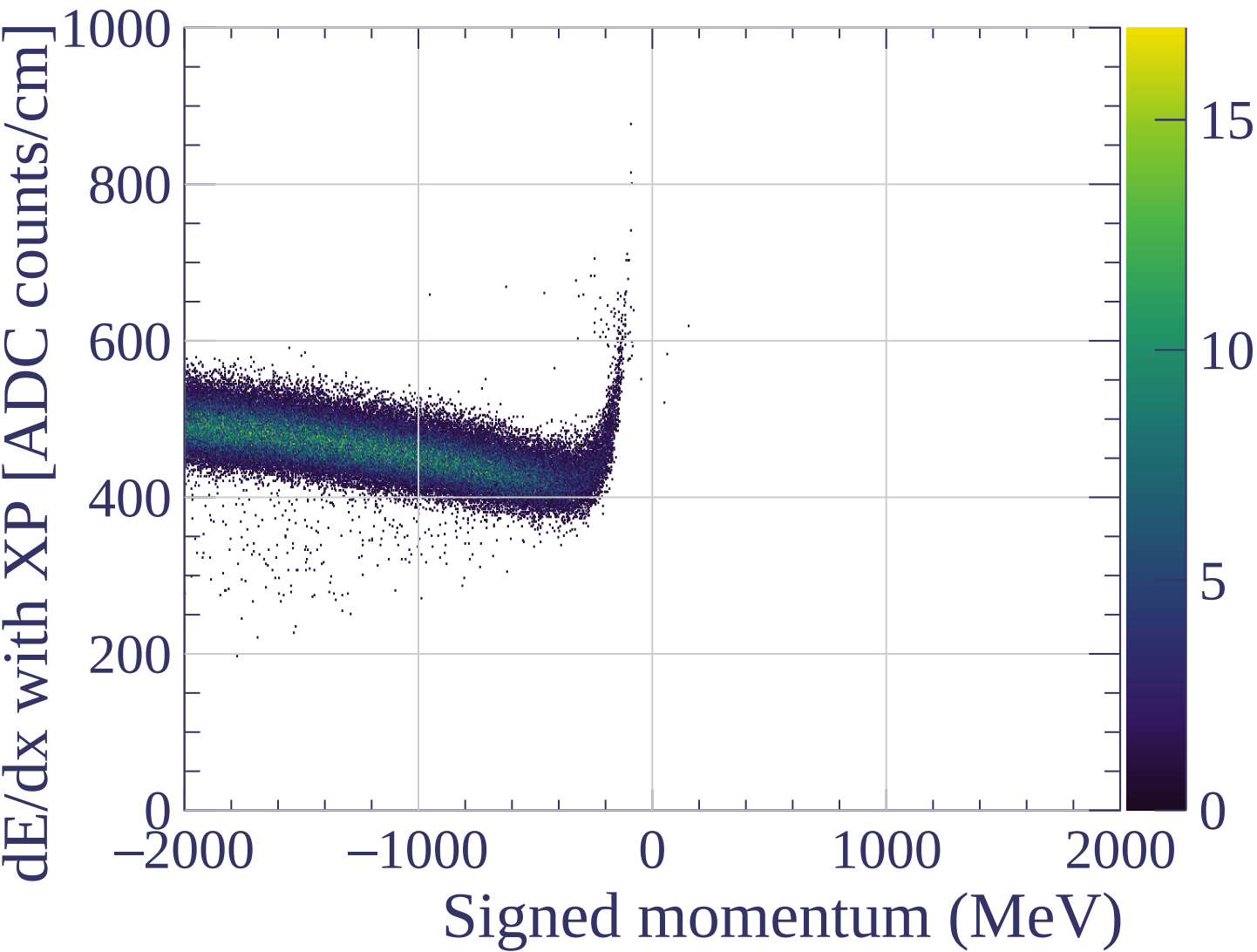


WF
XP

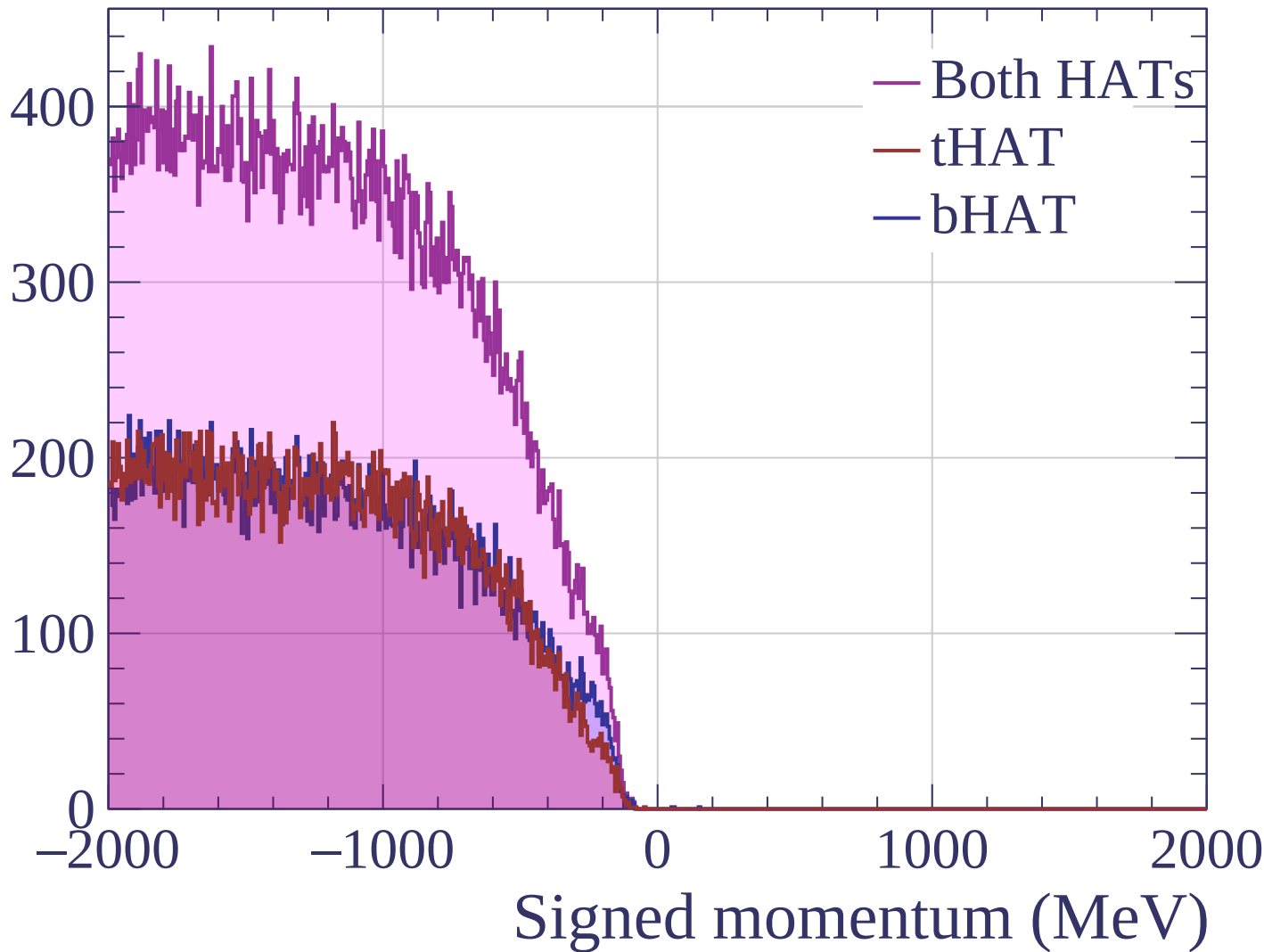
Momentum [MeV/c]

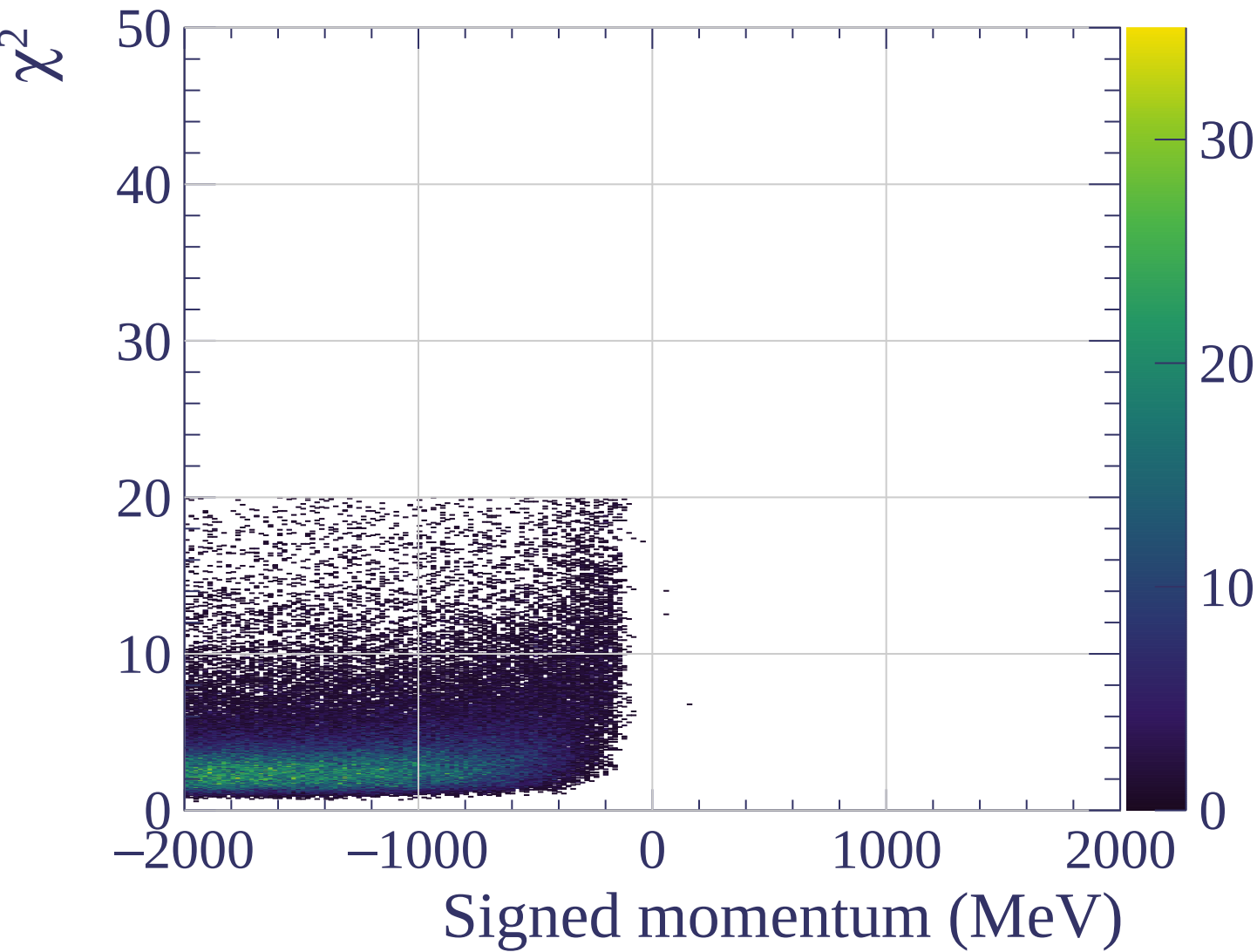




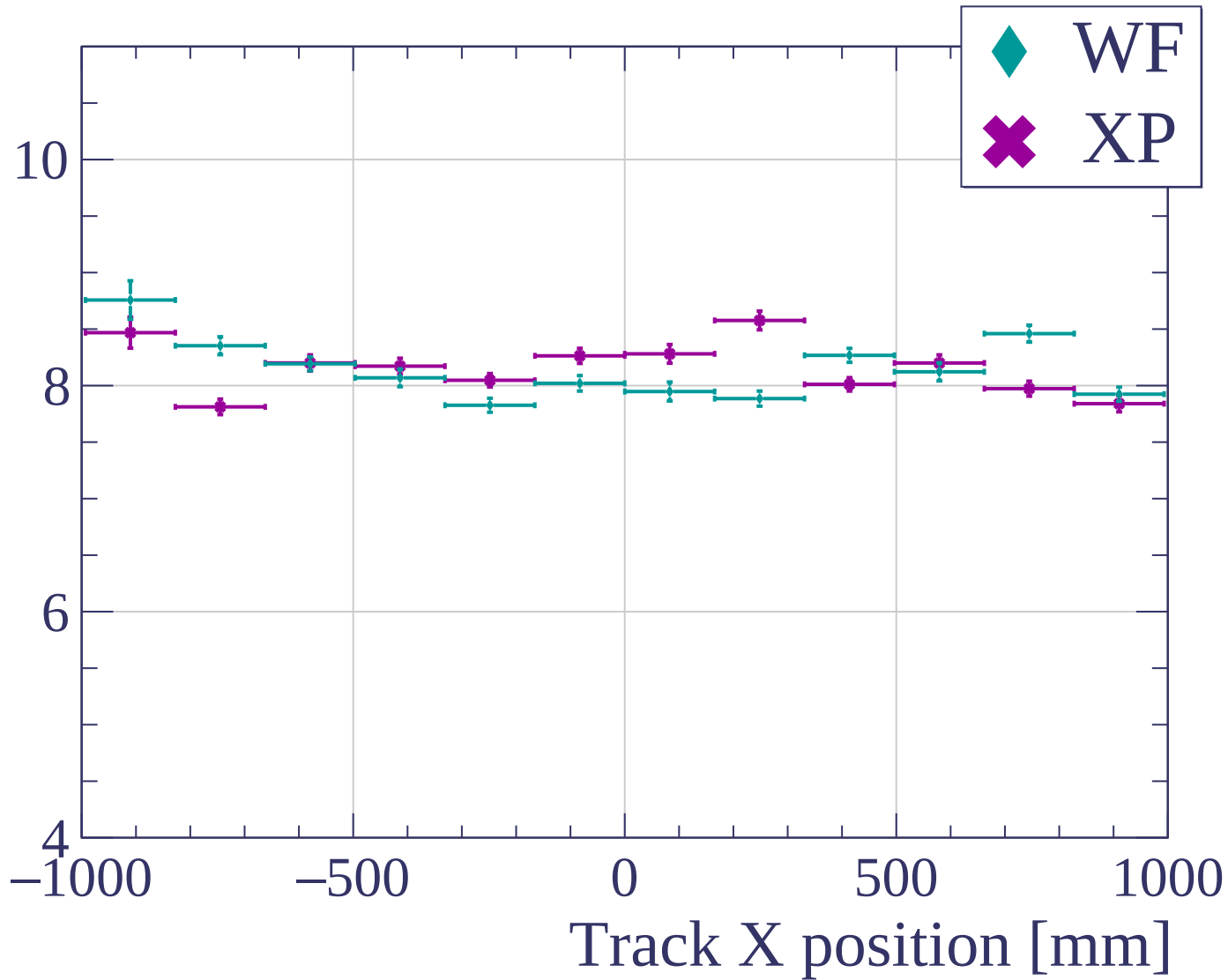


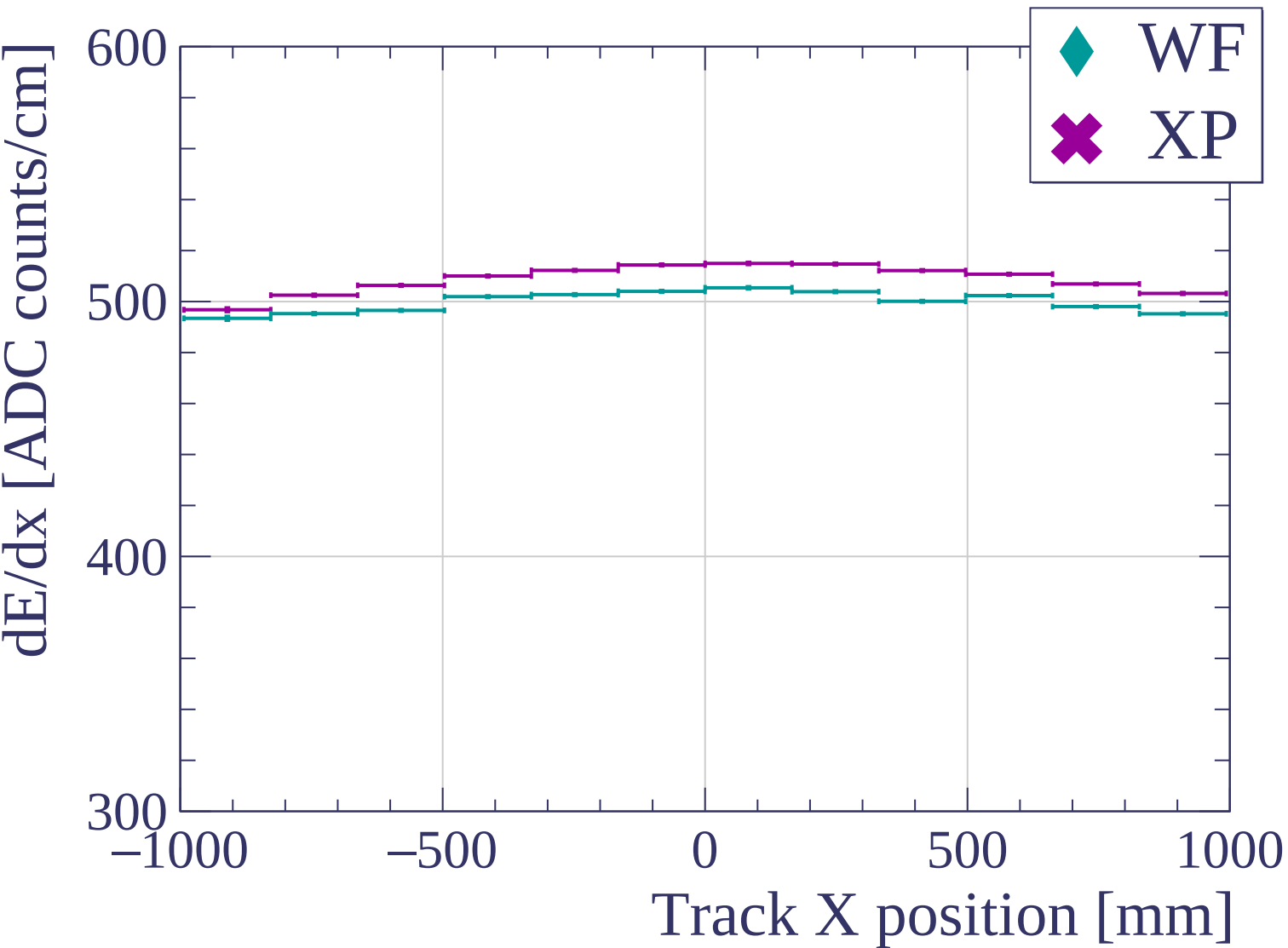
Count

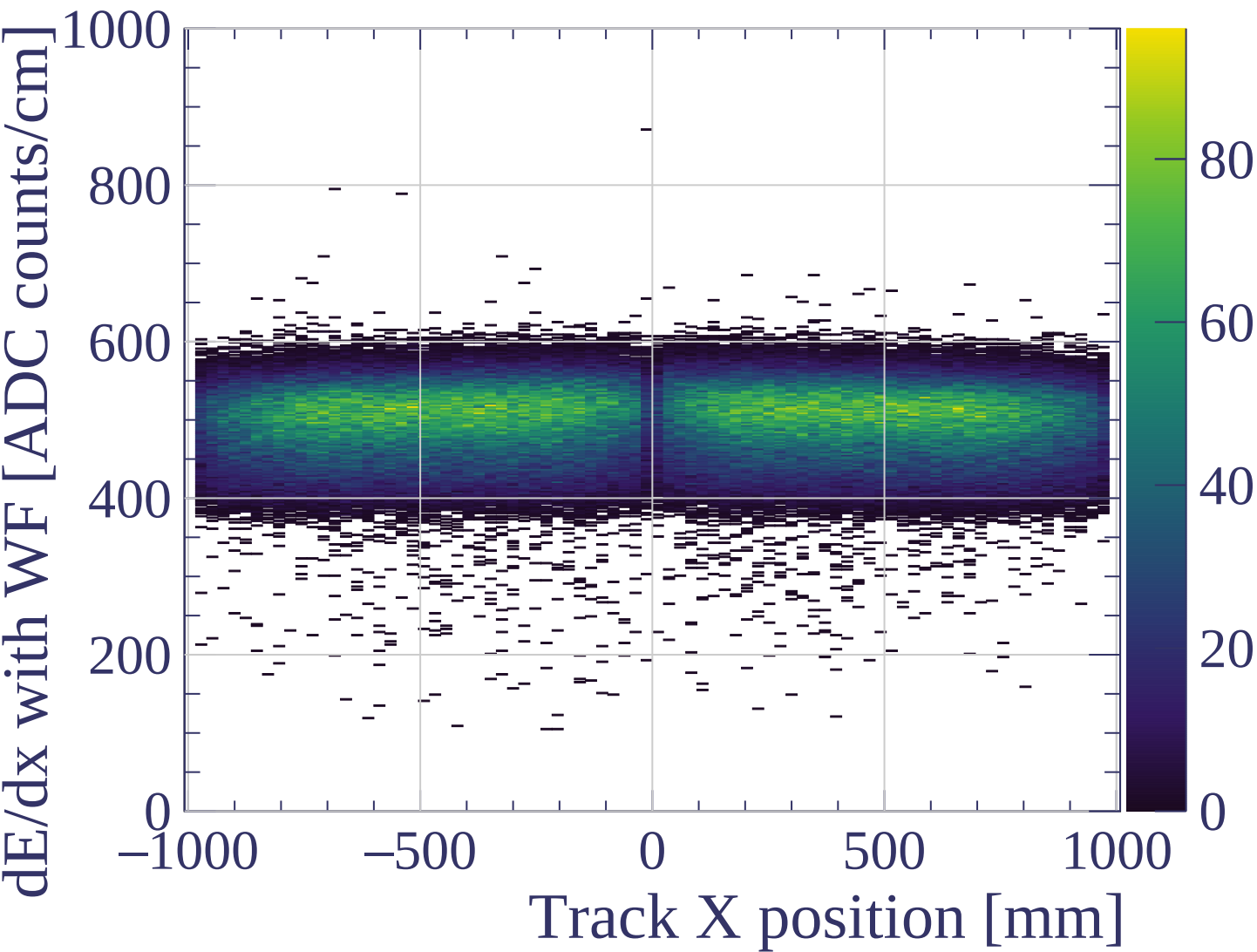


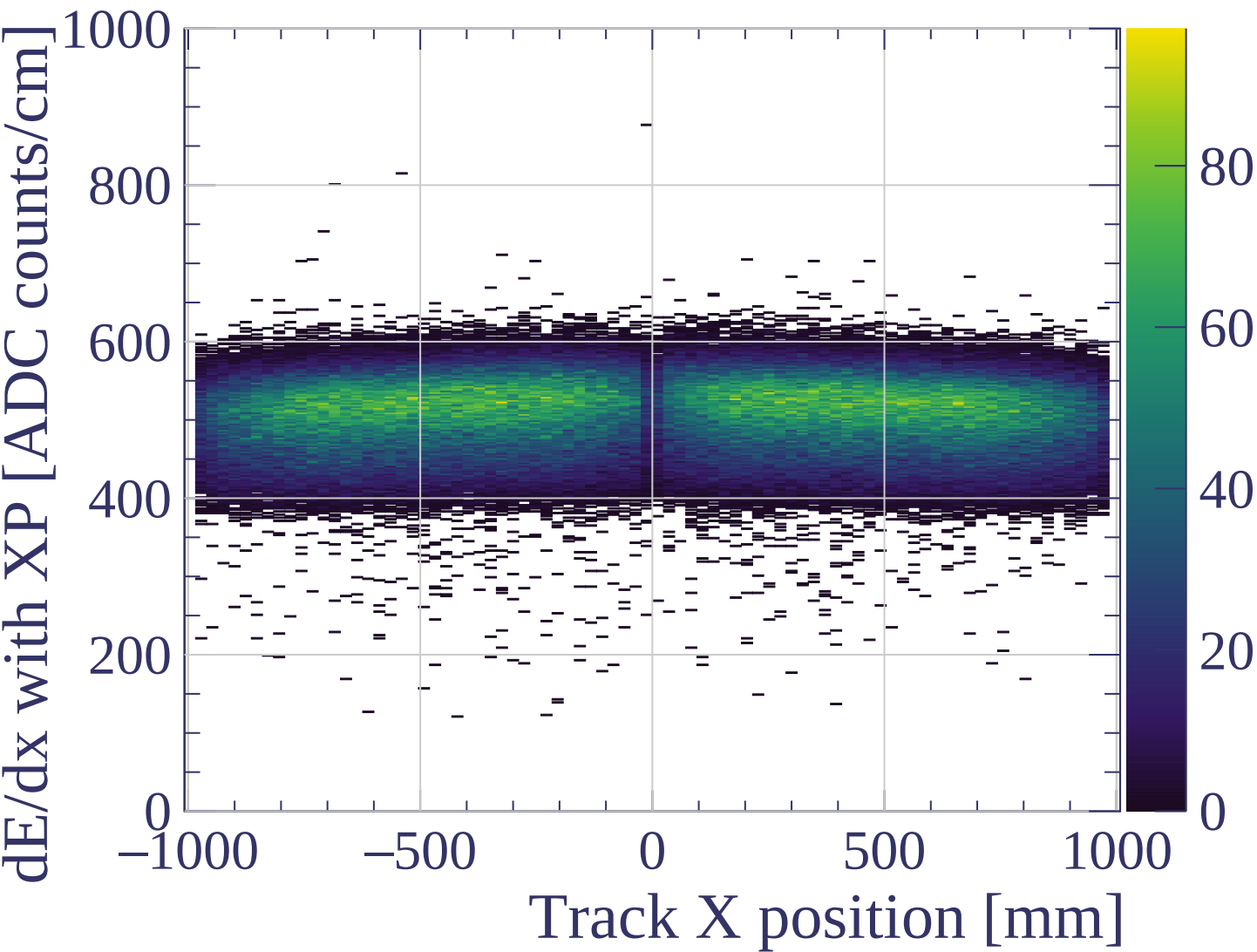


dE/dx resolution [%]

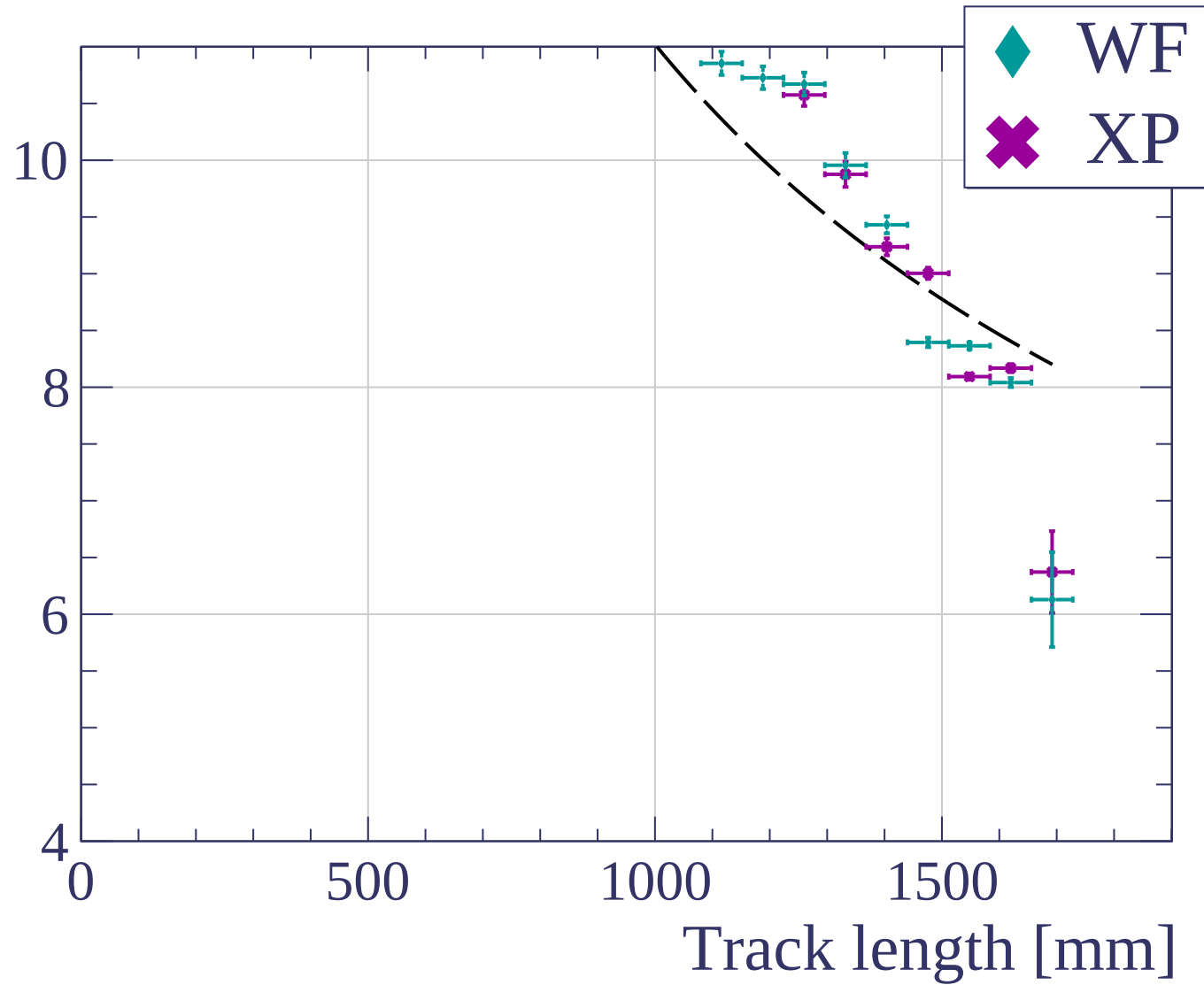


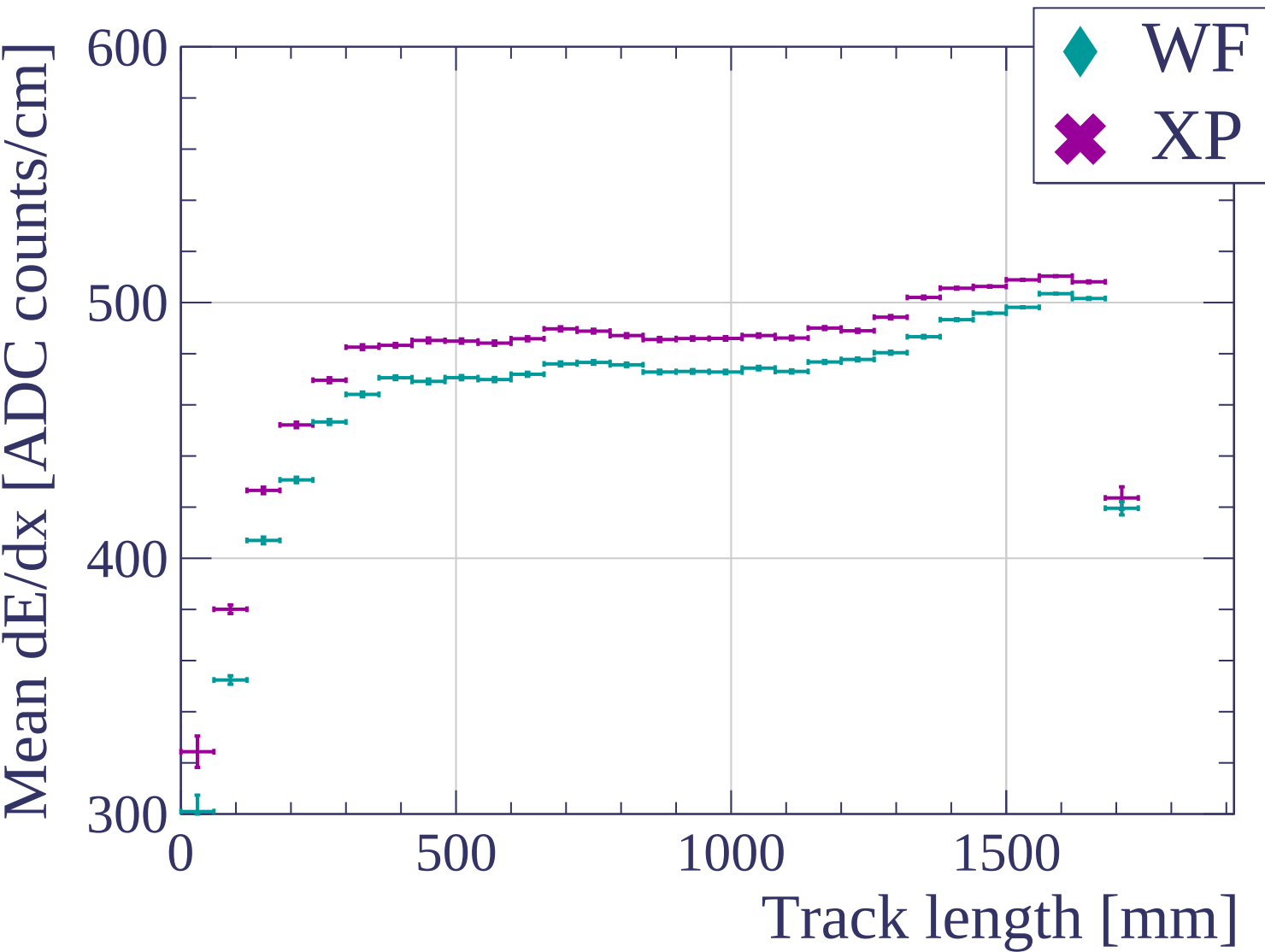


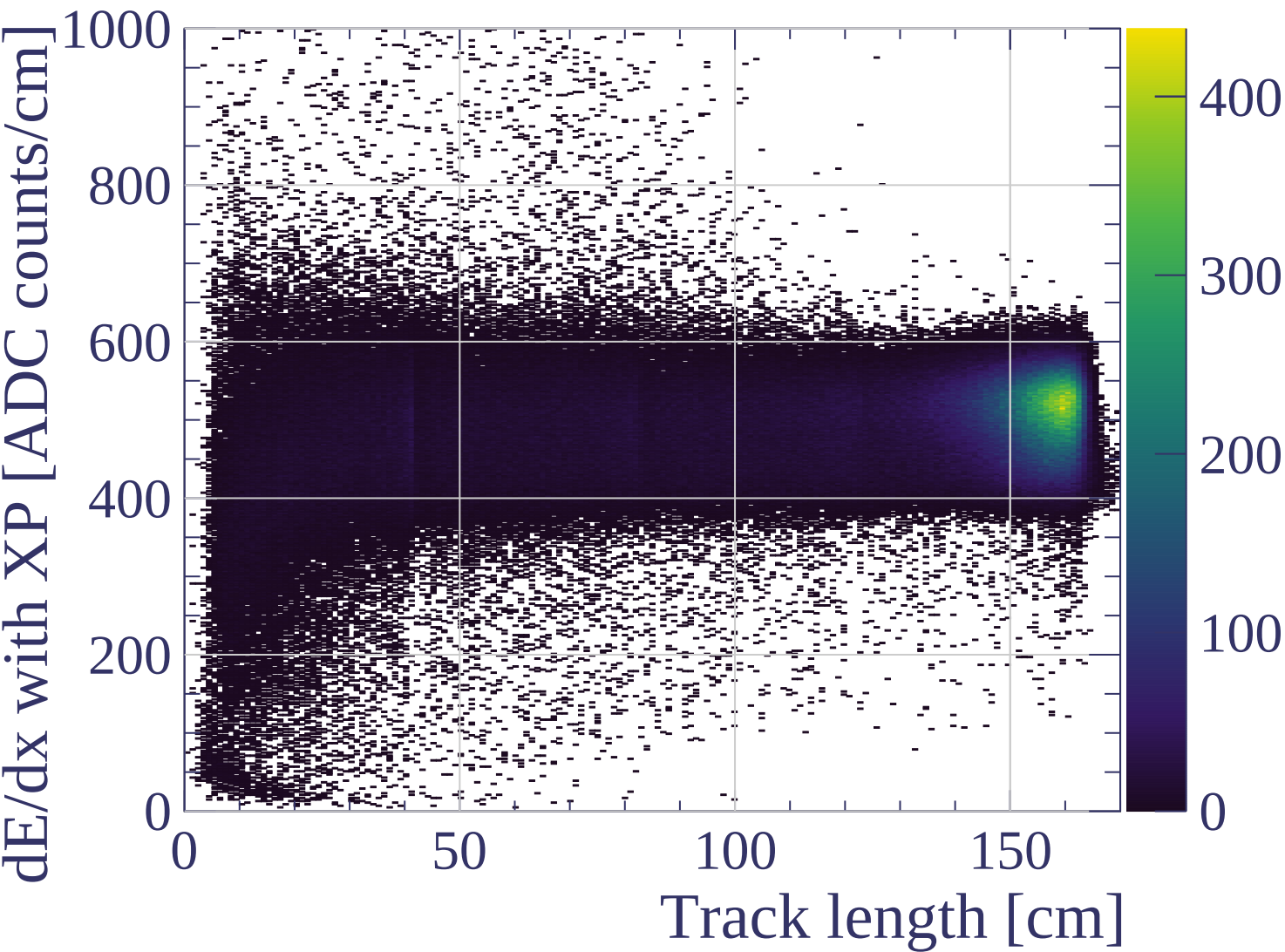


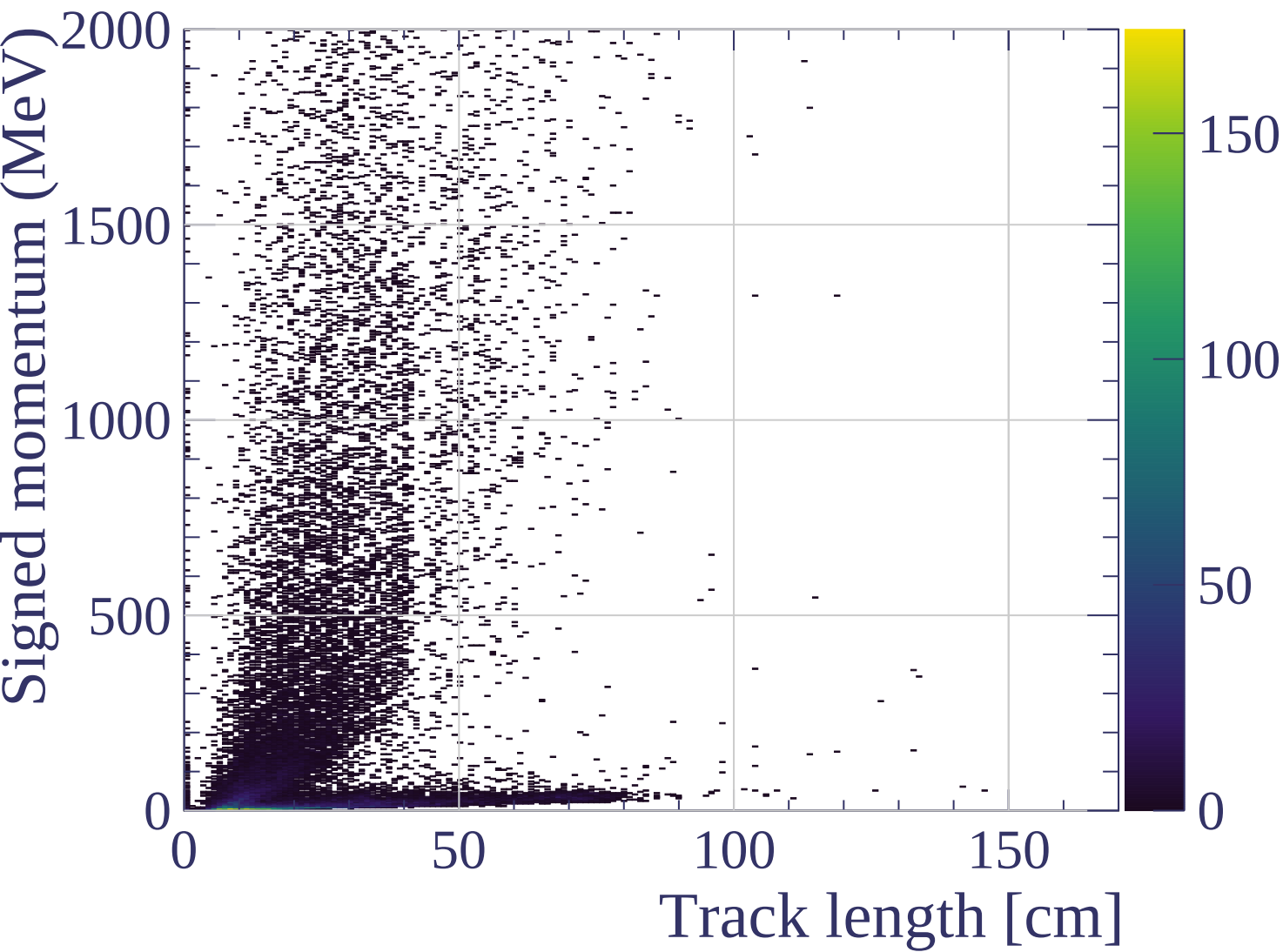


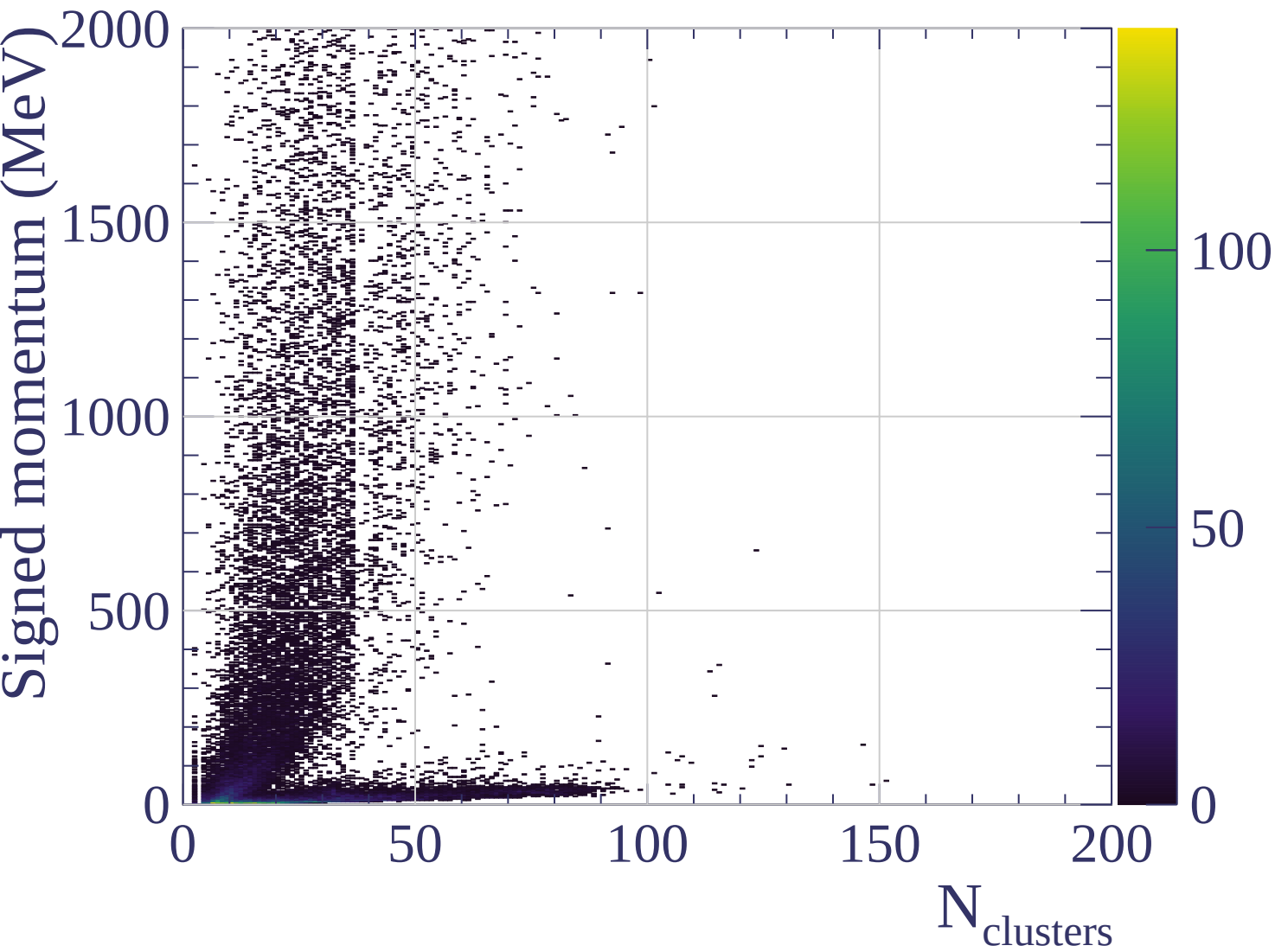
dE/dx resolution [%]



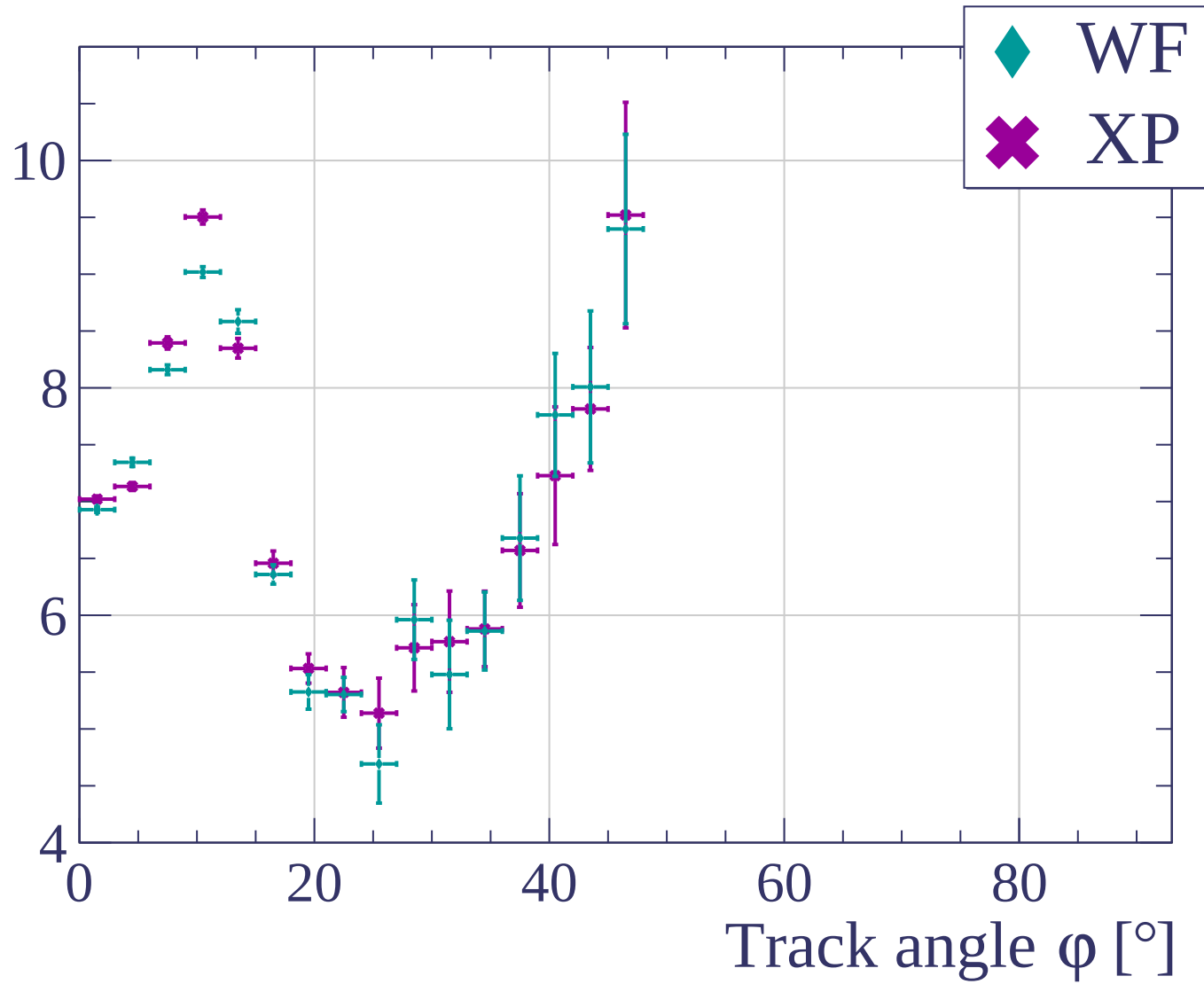


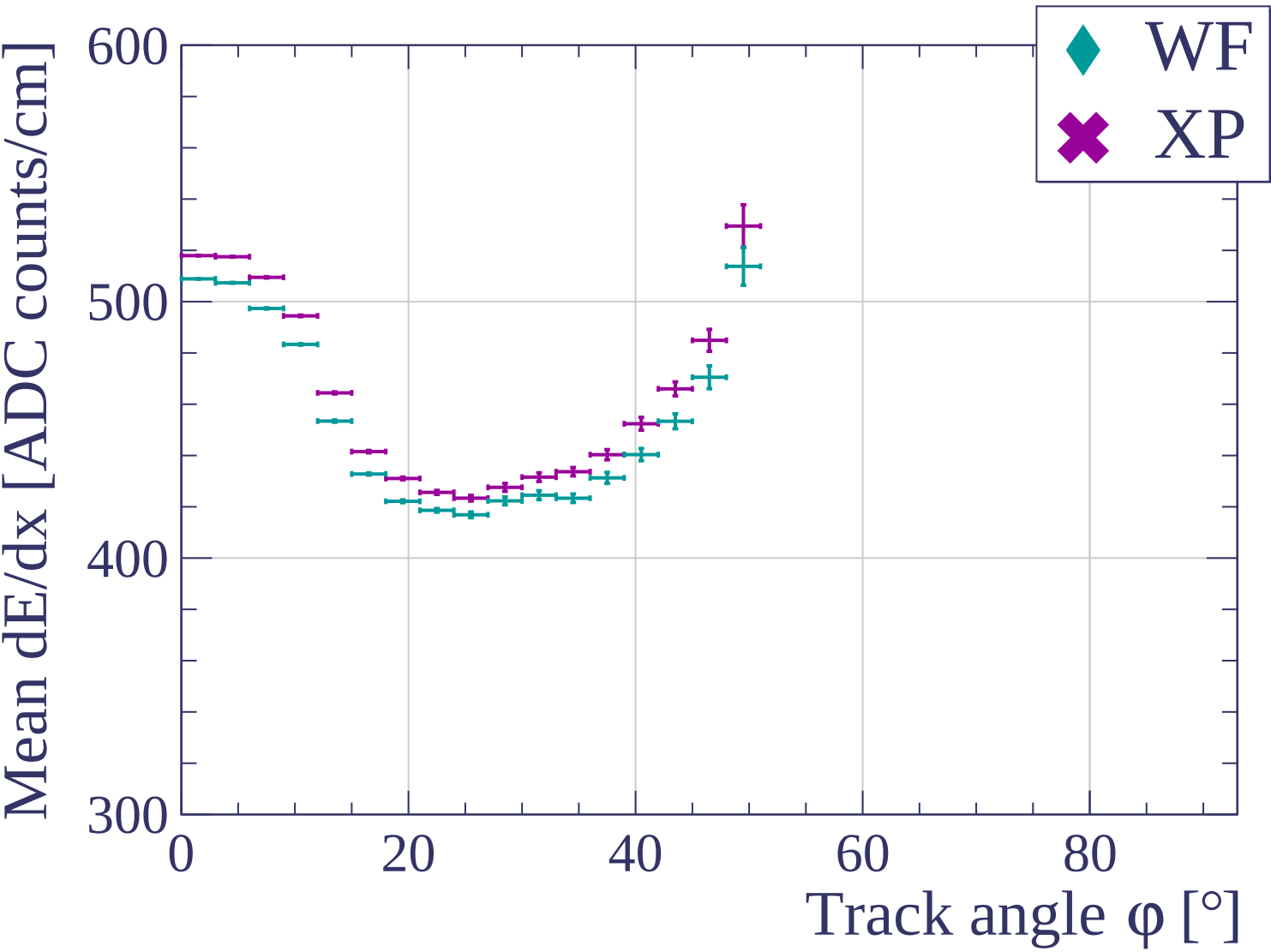




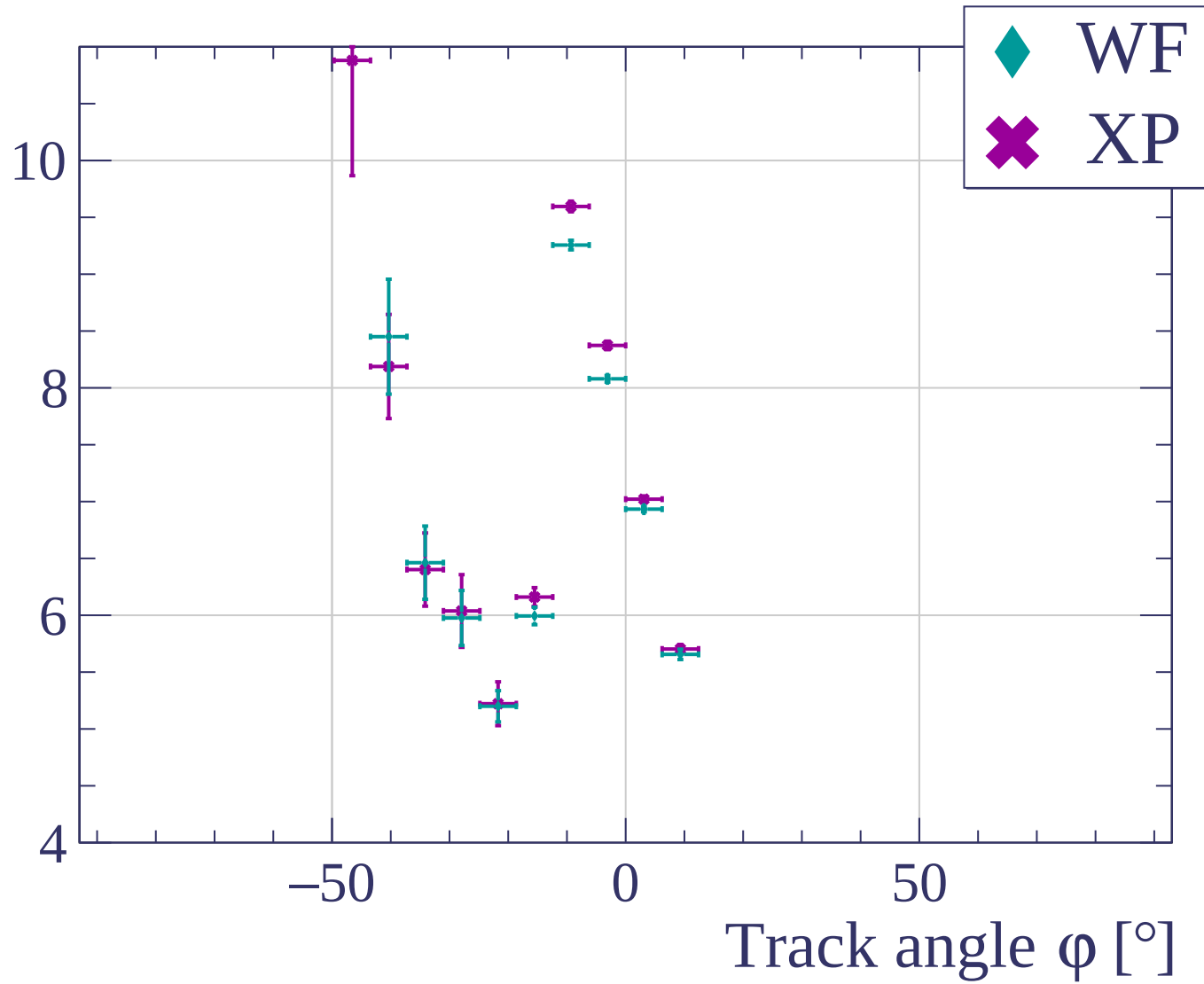


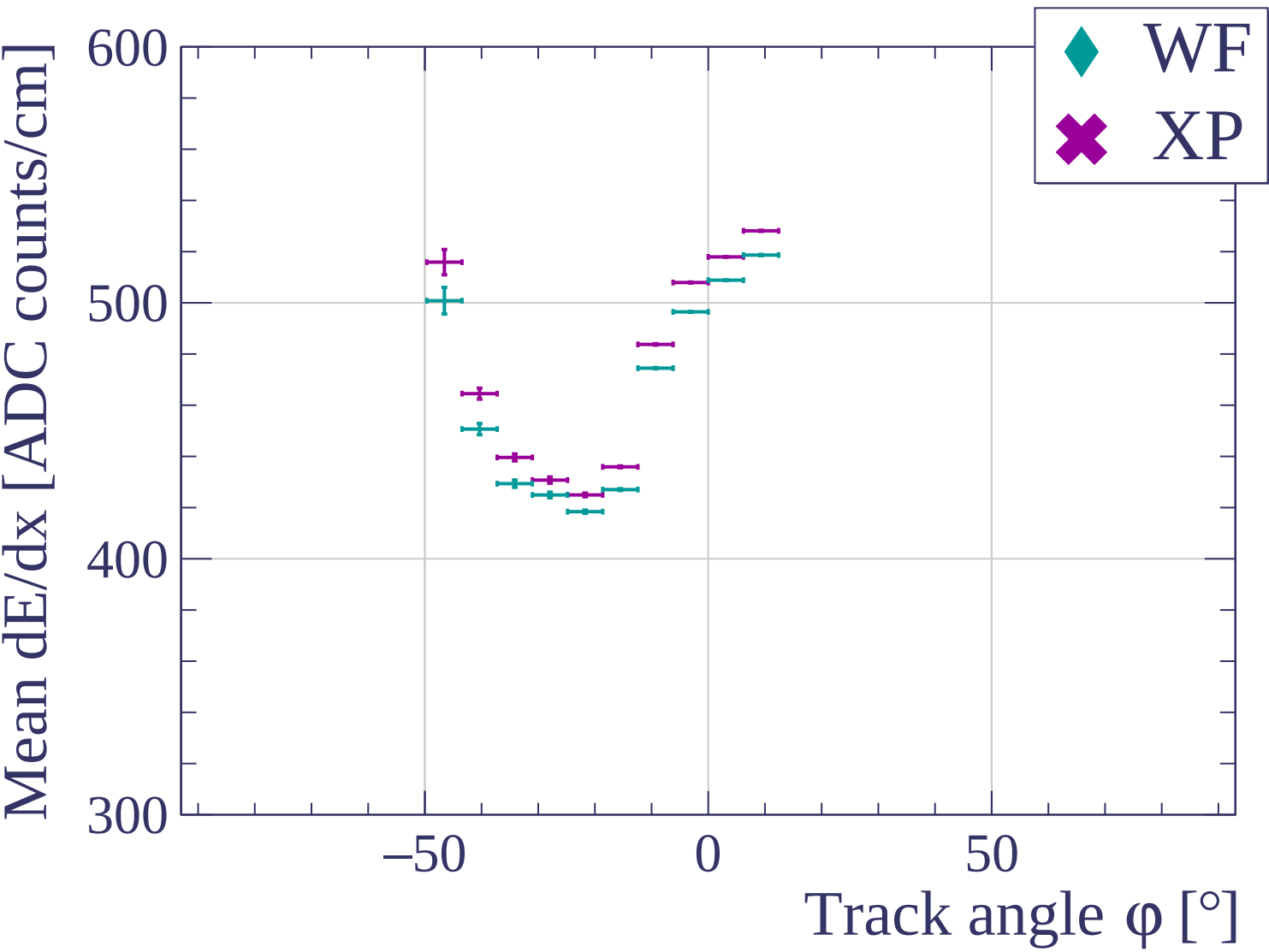
dE/dx resolution [%]

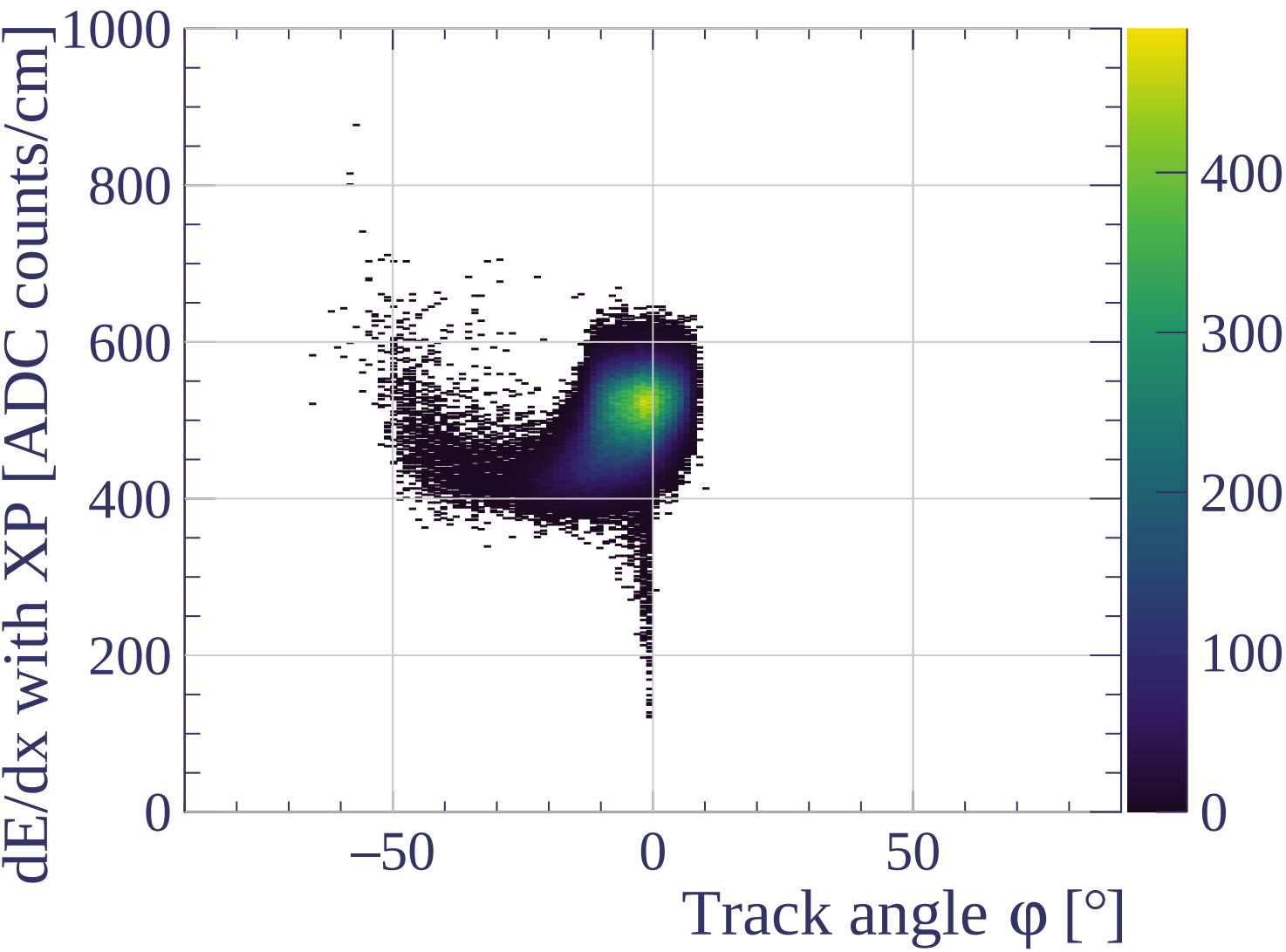




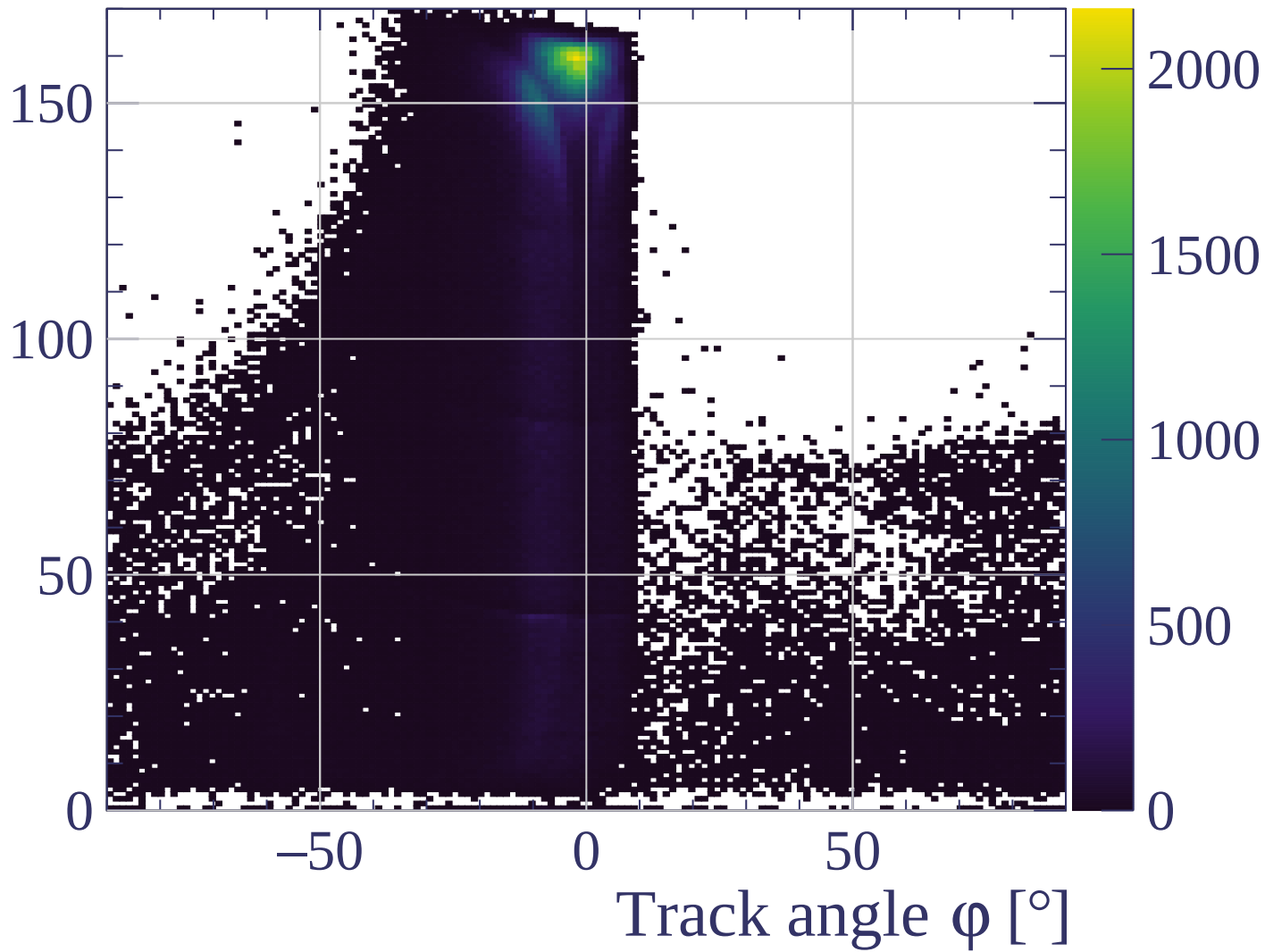
dE/dx resolution [%]

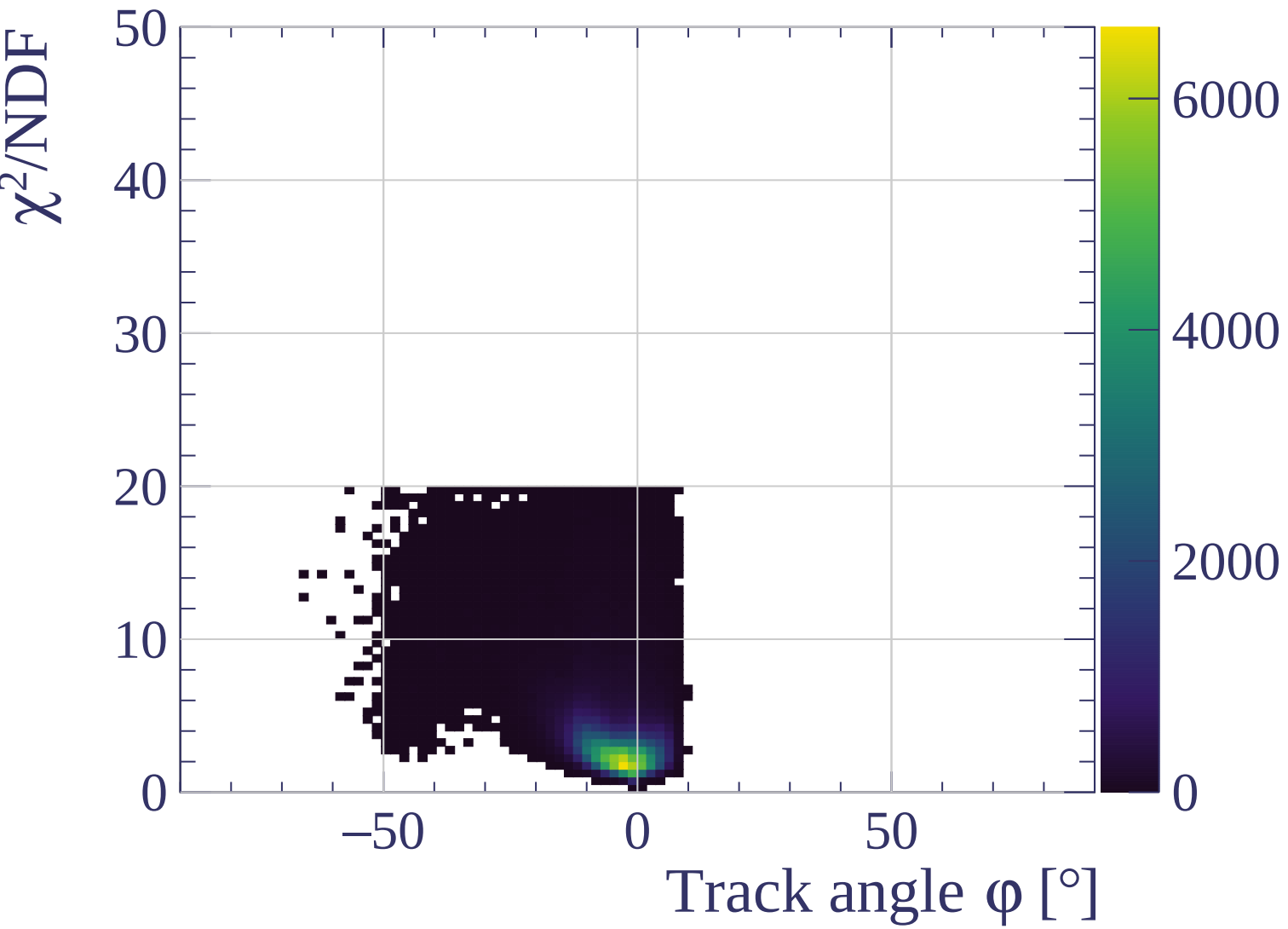


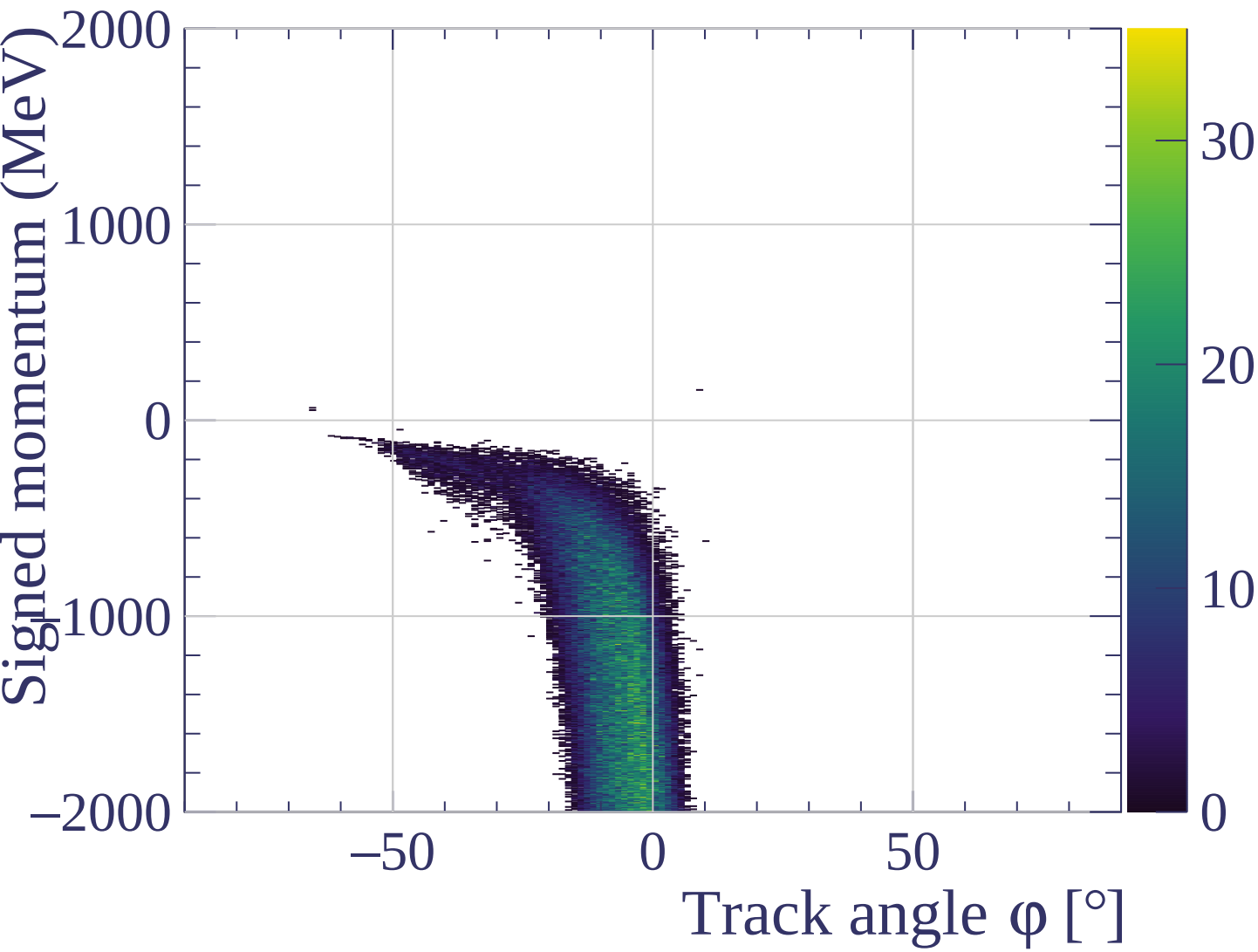


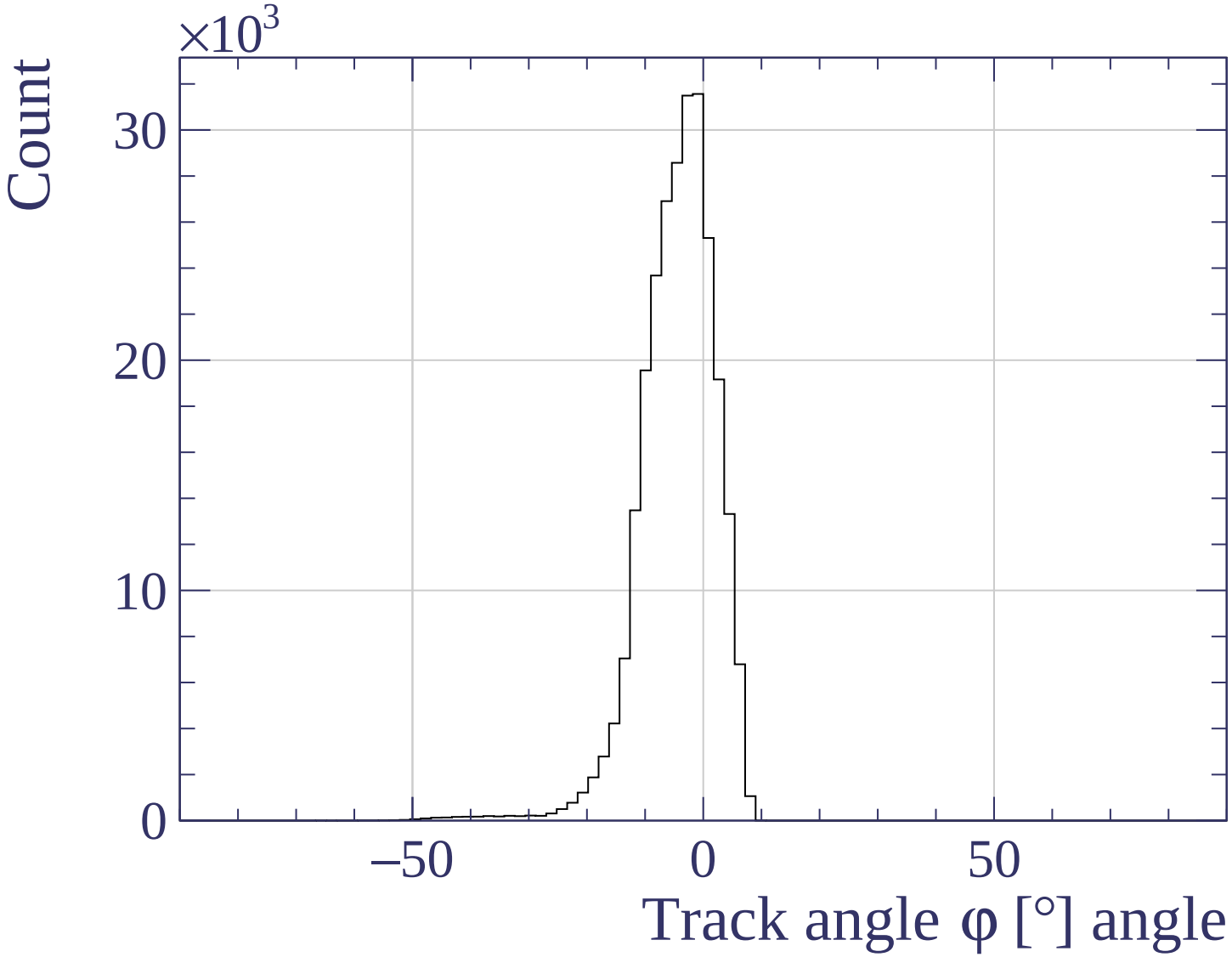


Track length [cm]

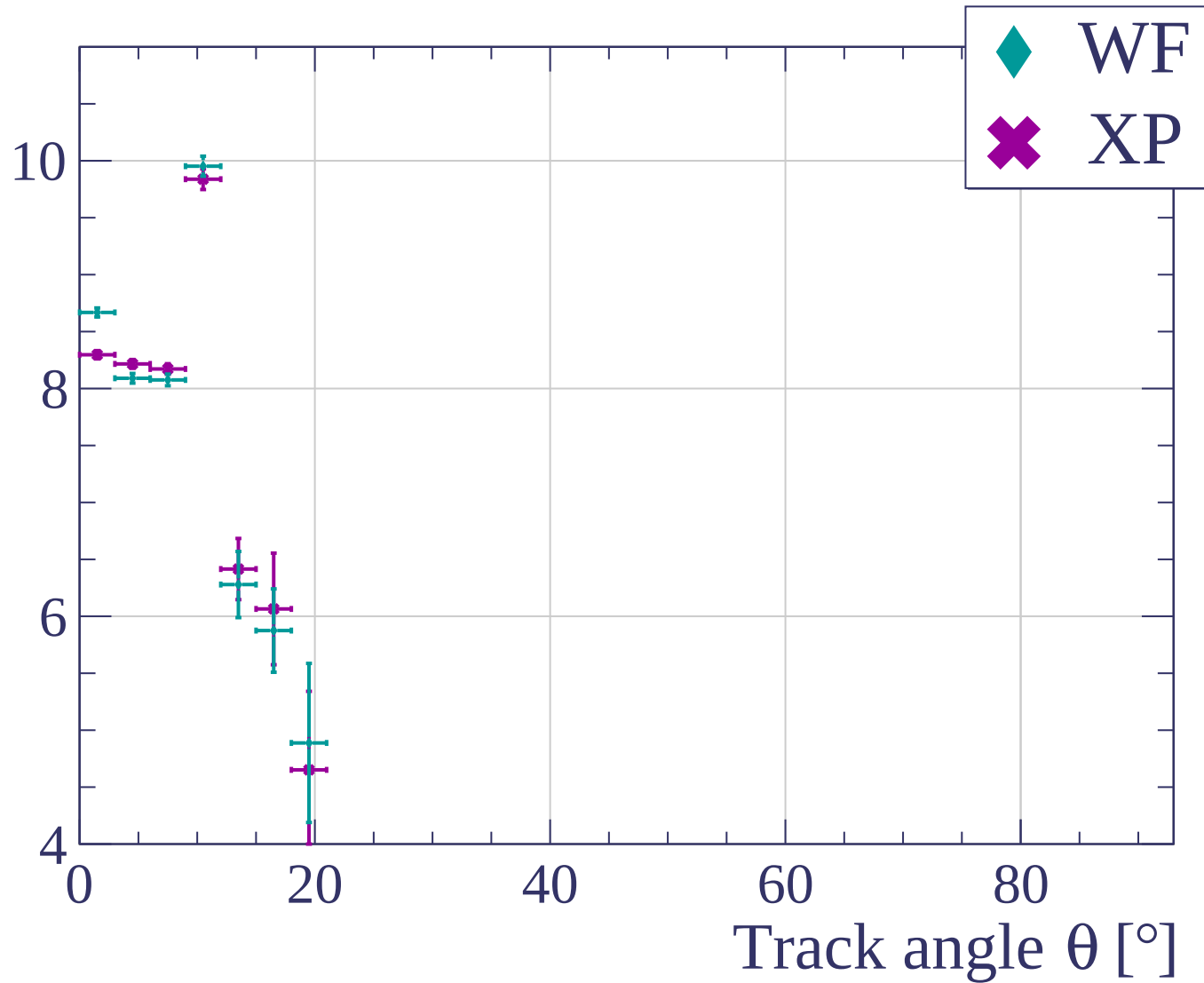


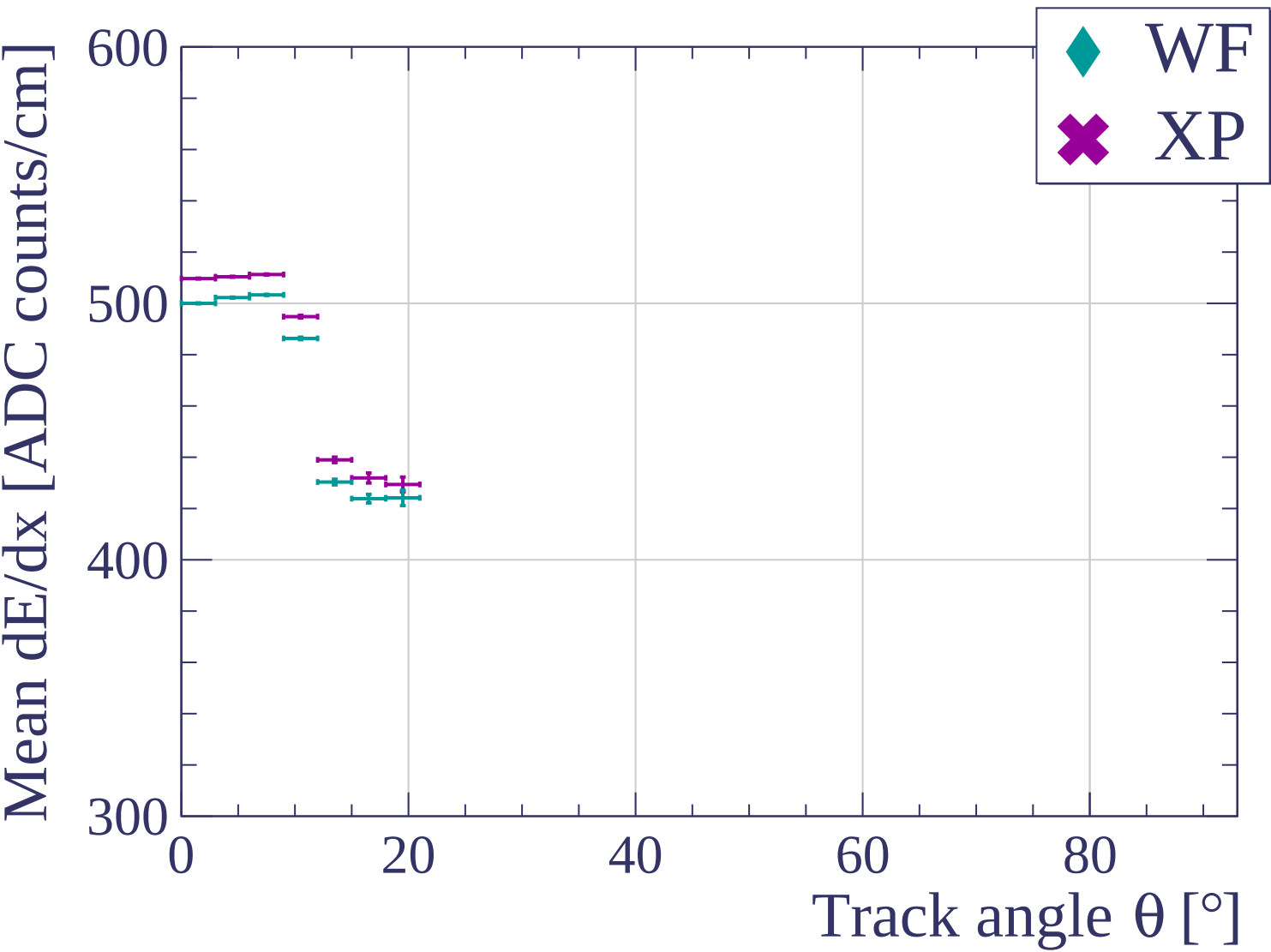


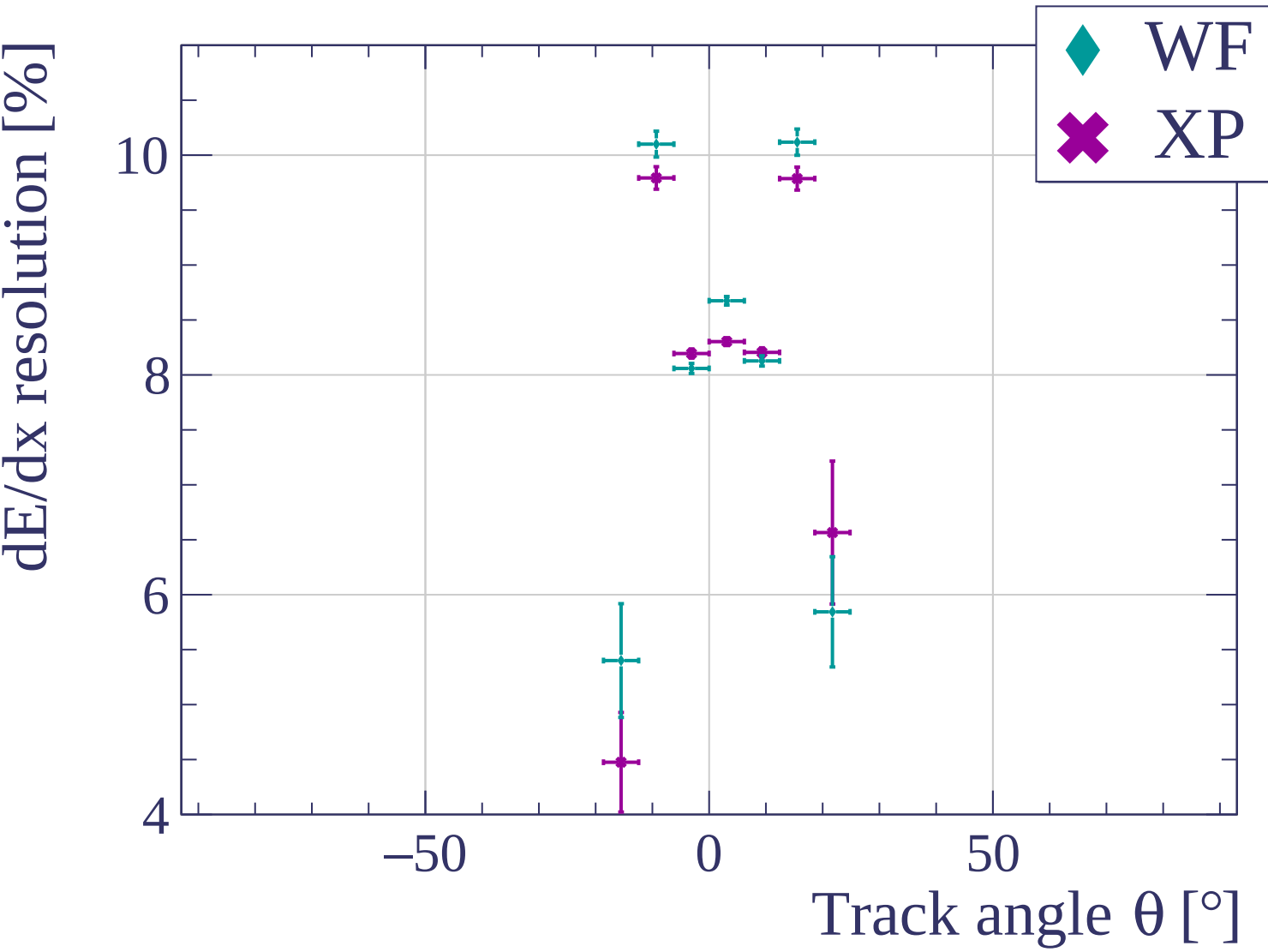


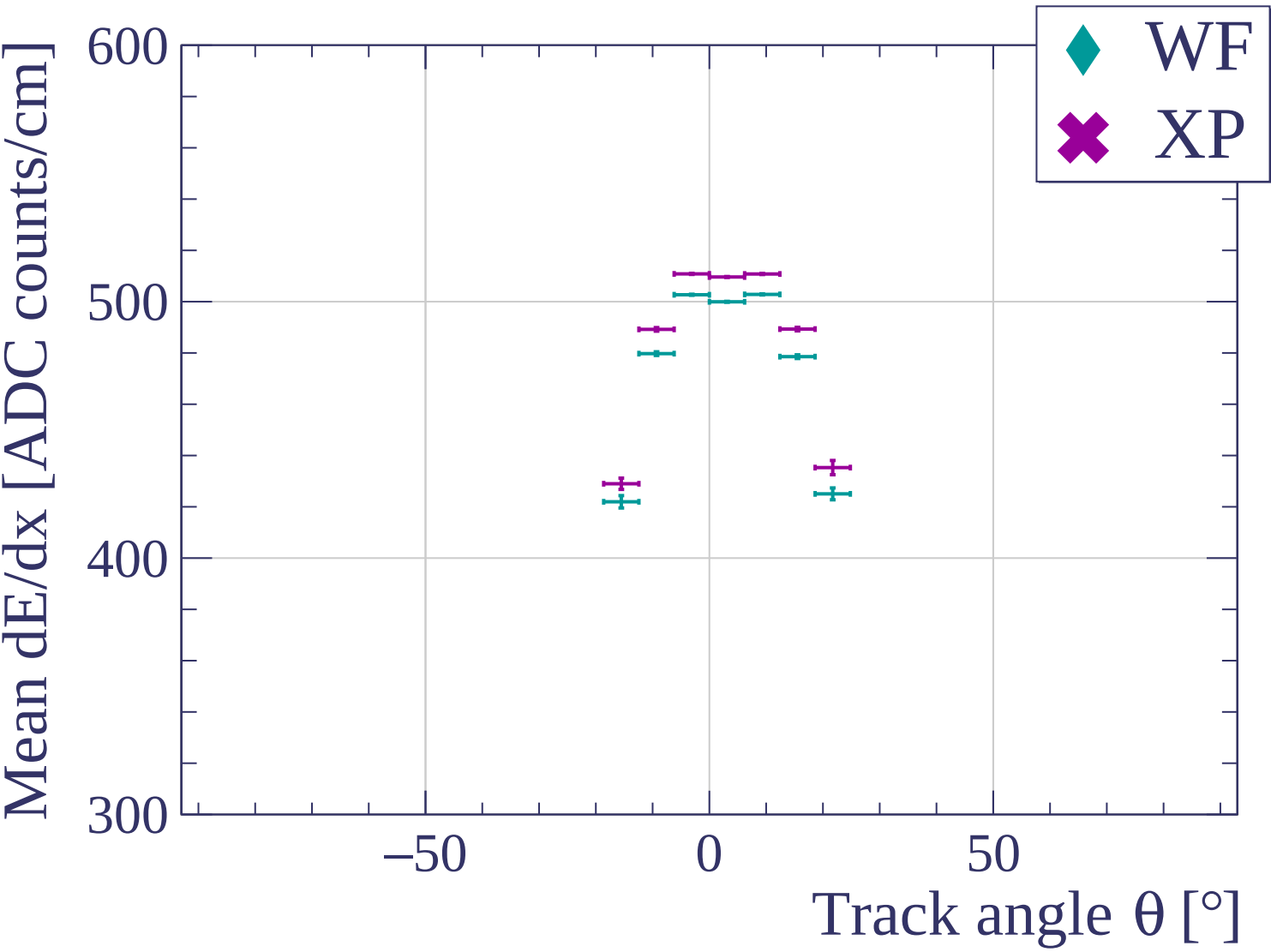


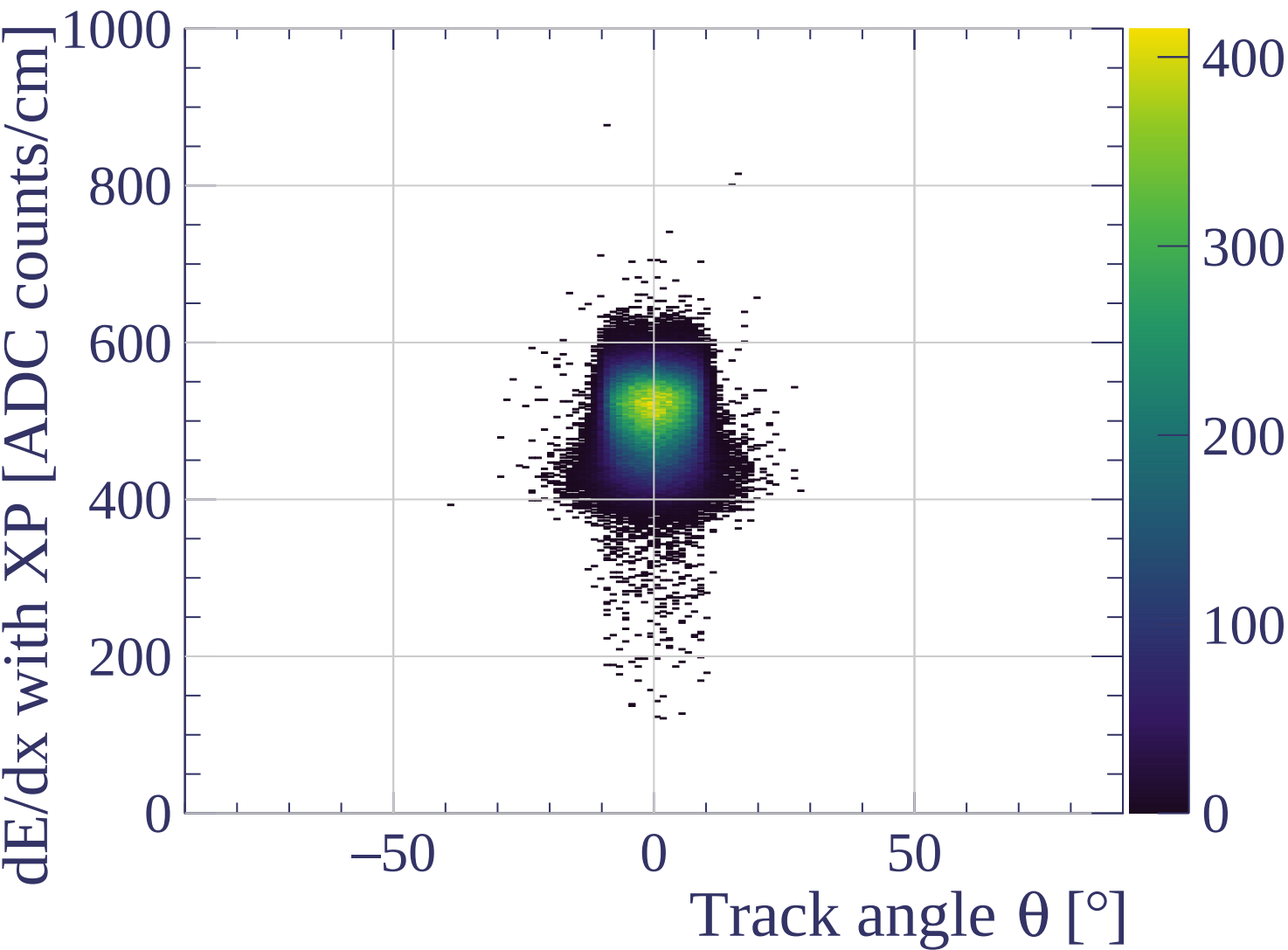
dE/dx resolution [%]



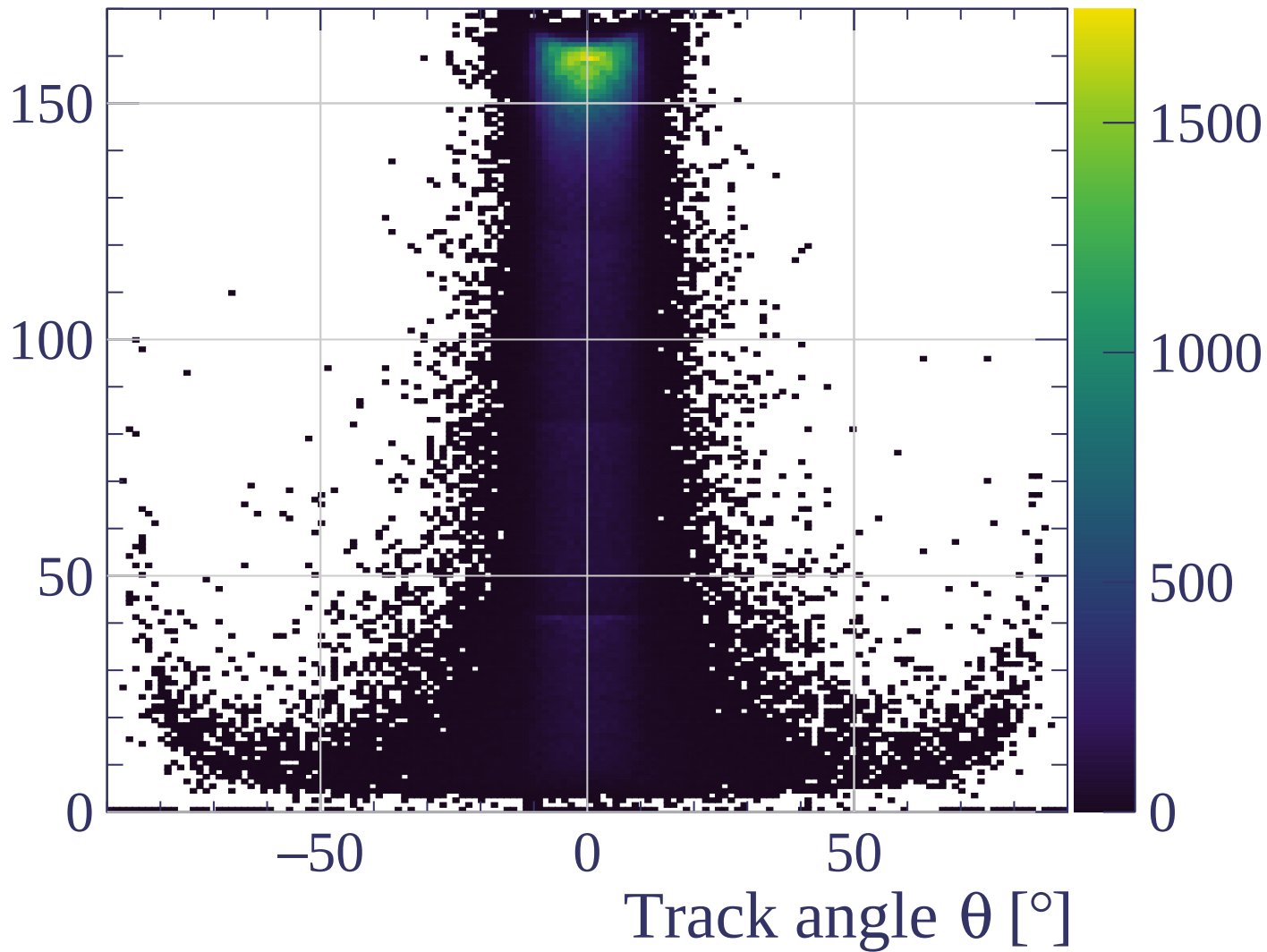


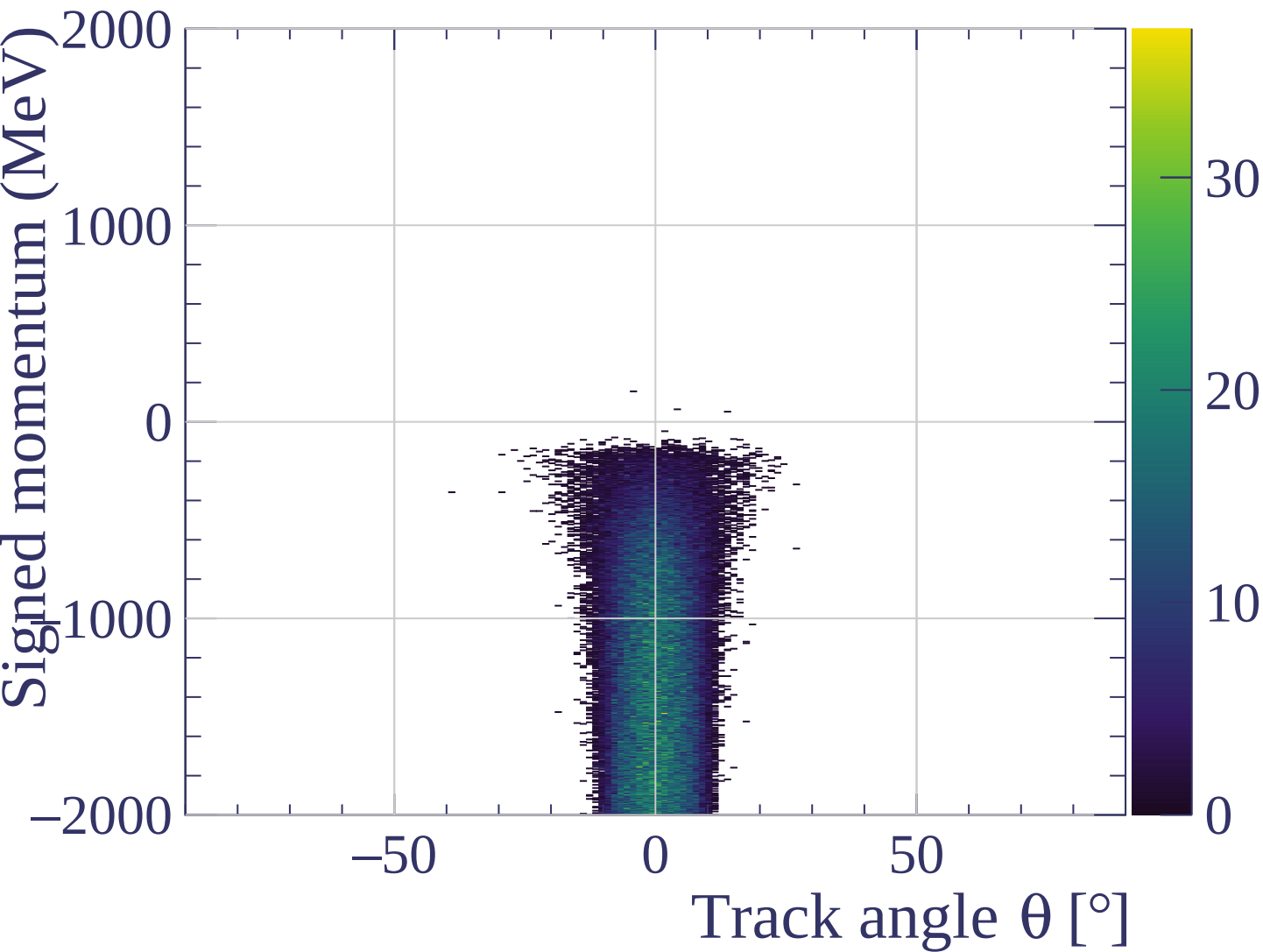


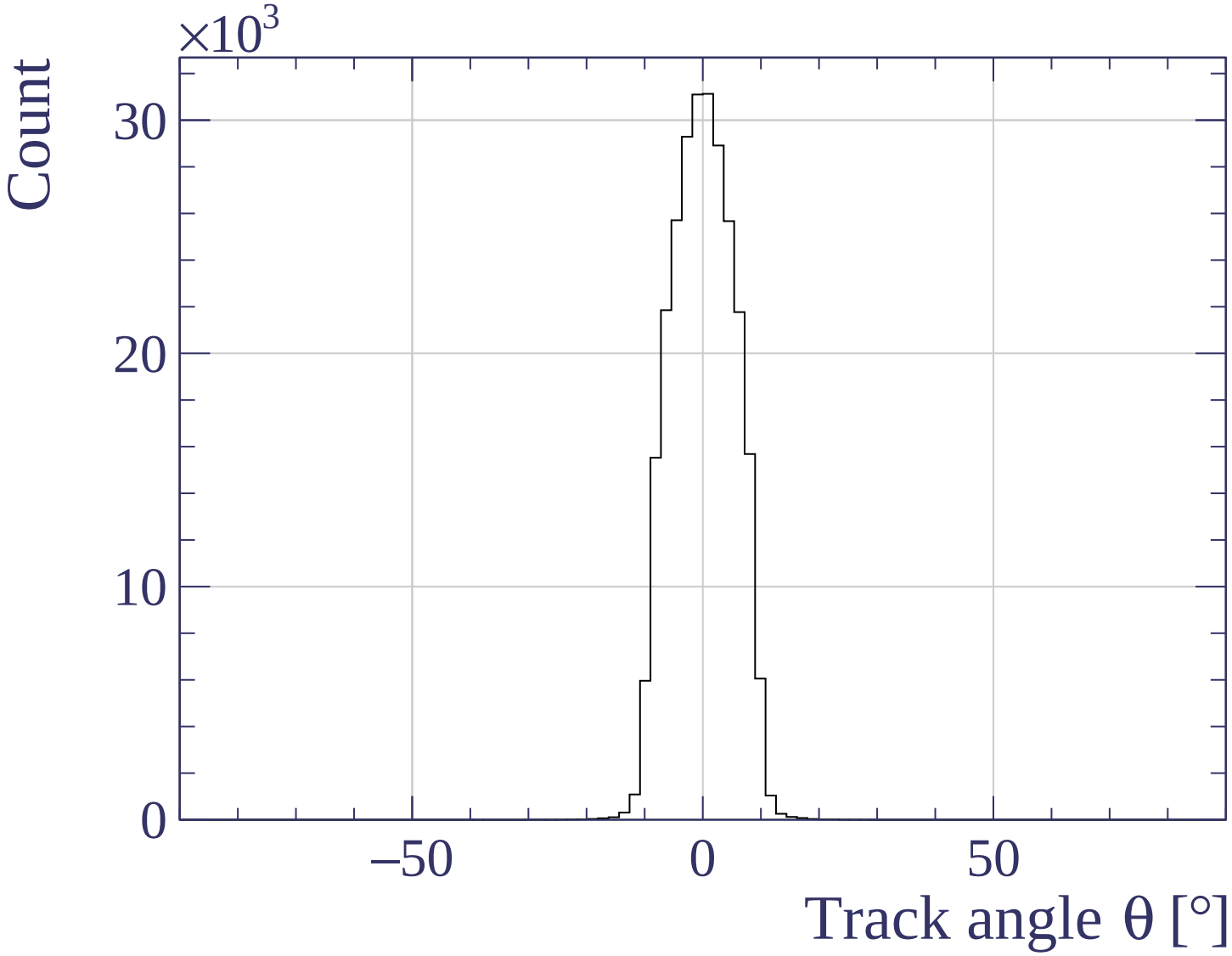




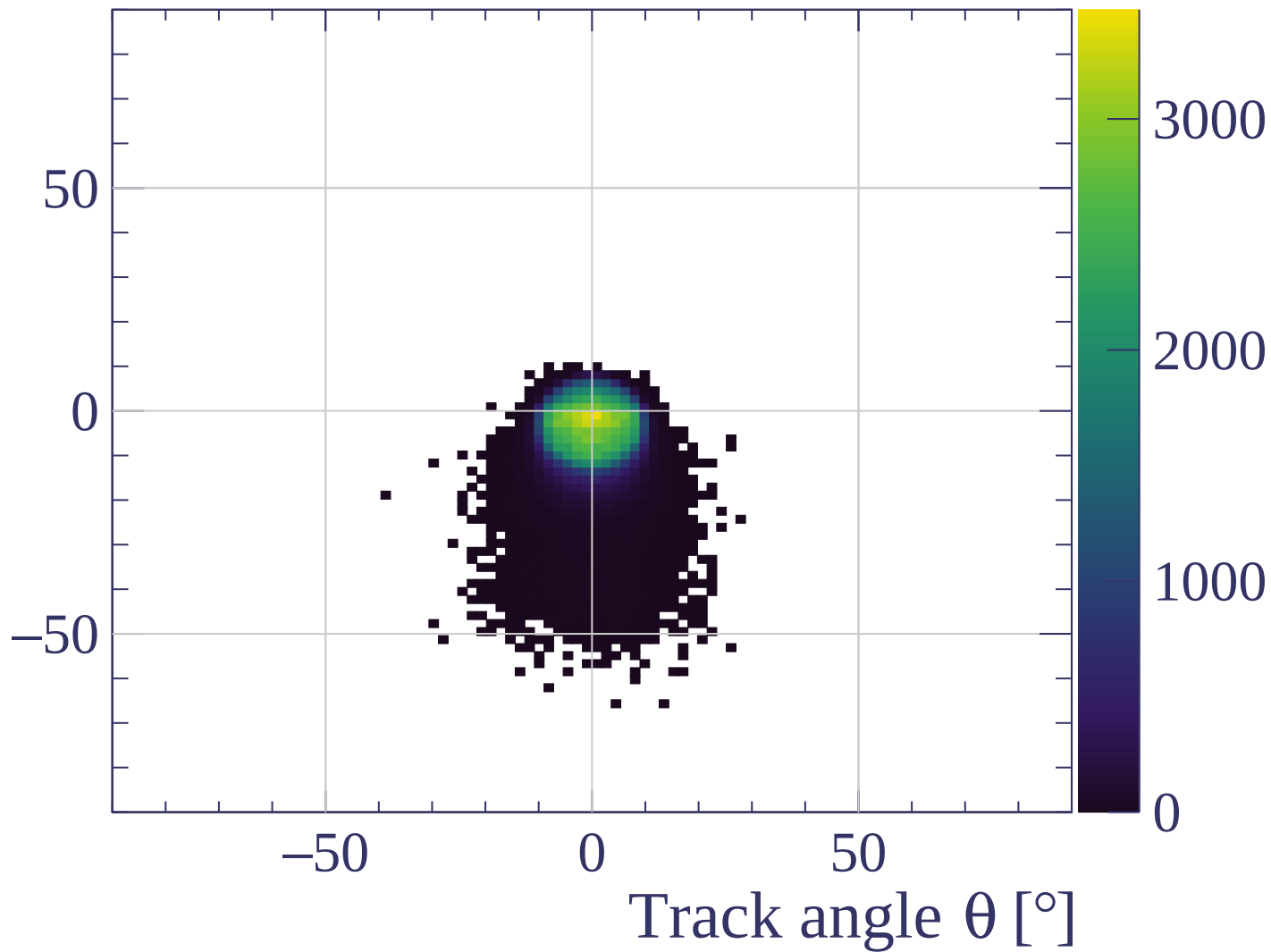
Track length [cm]



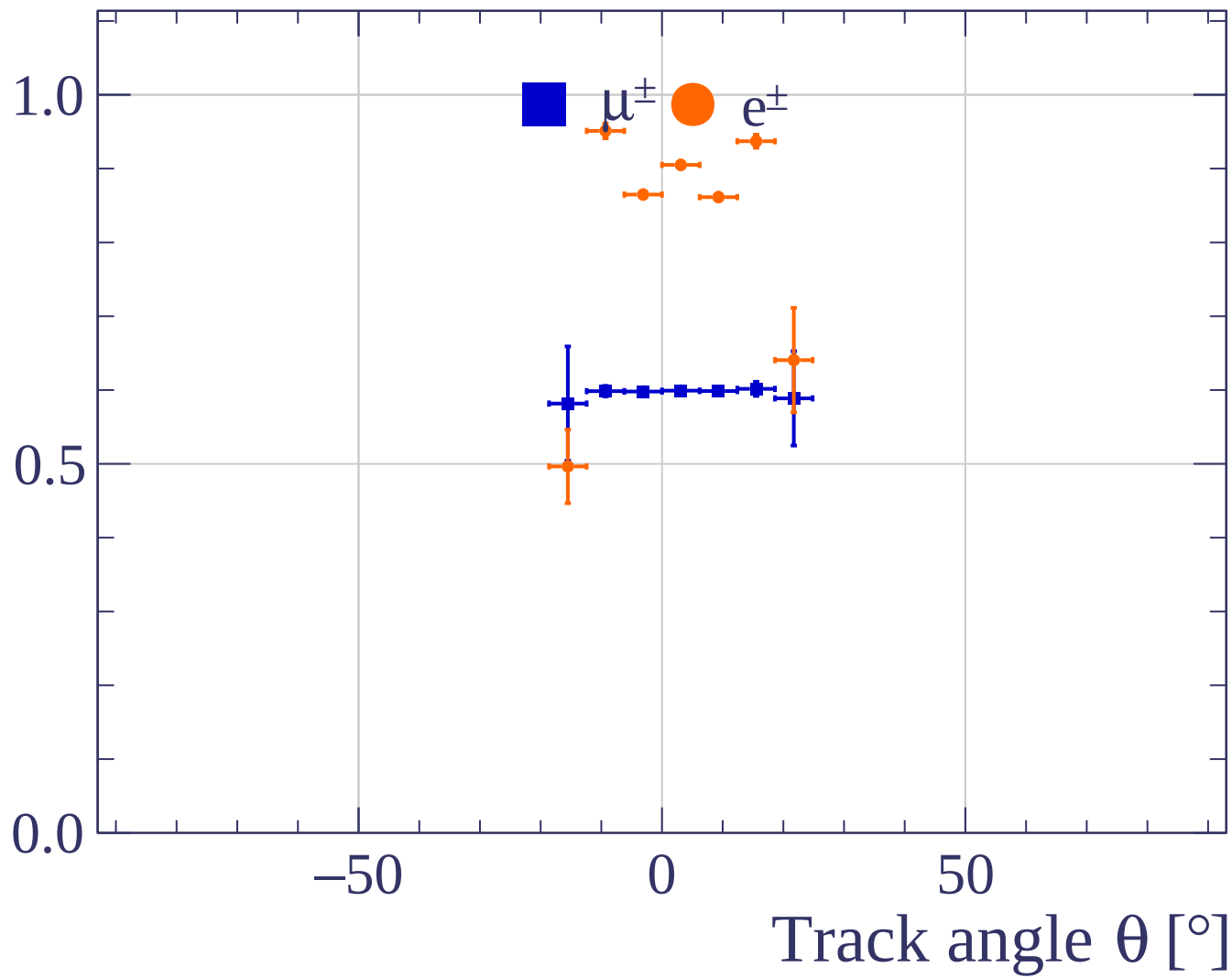




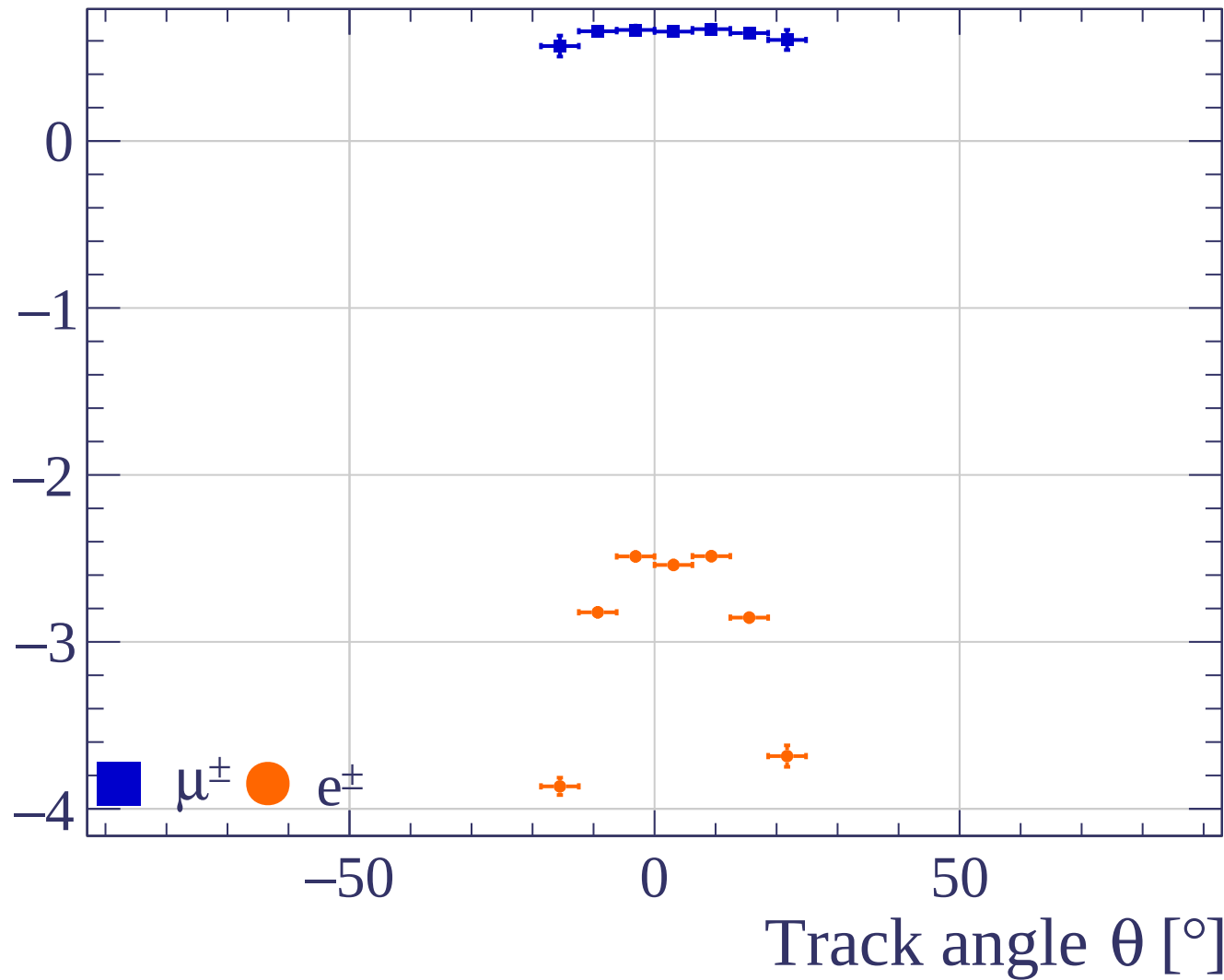
Track angle ϕ [°]



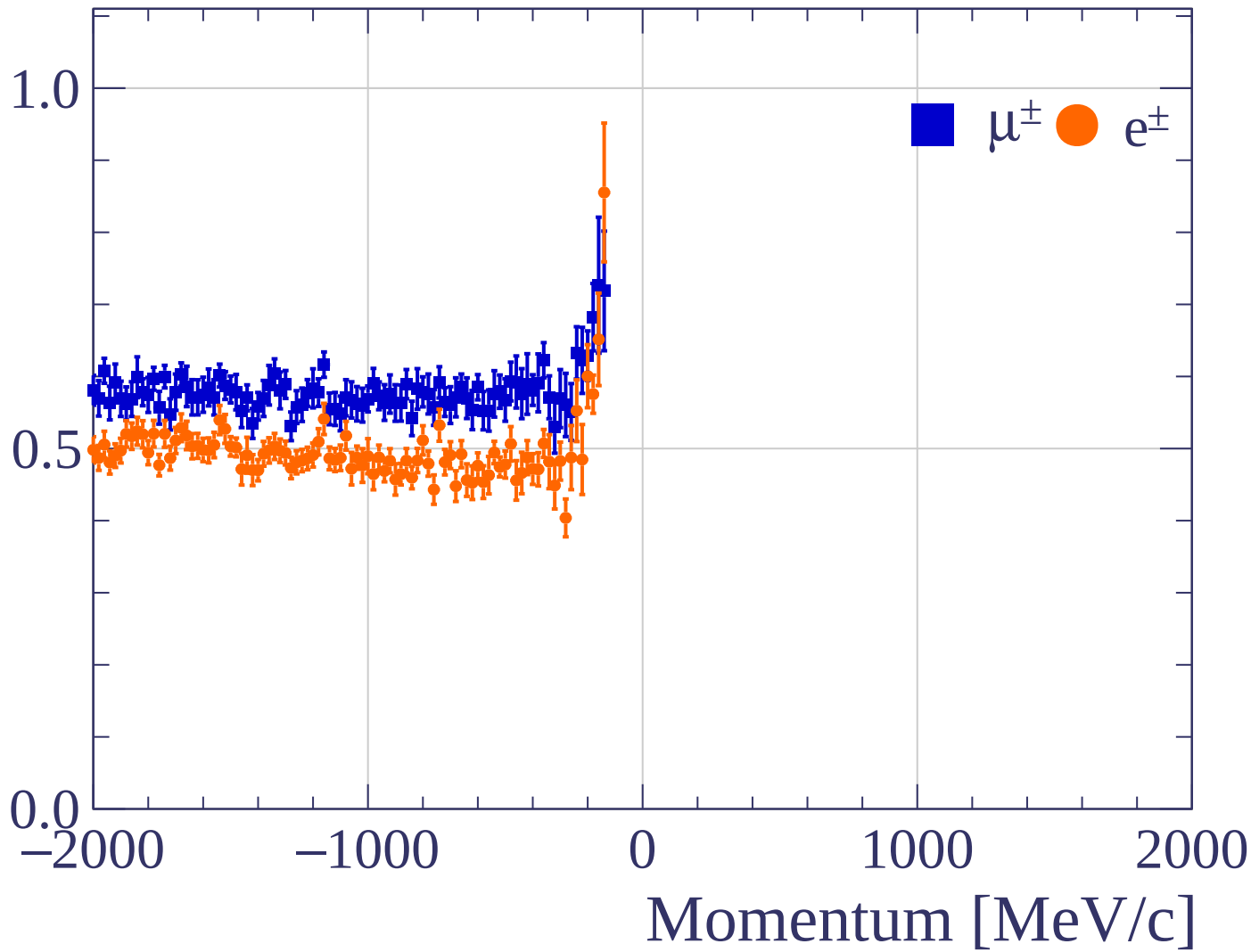
Pulls standard deviation



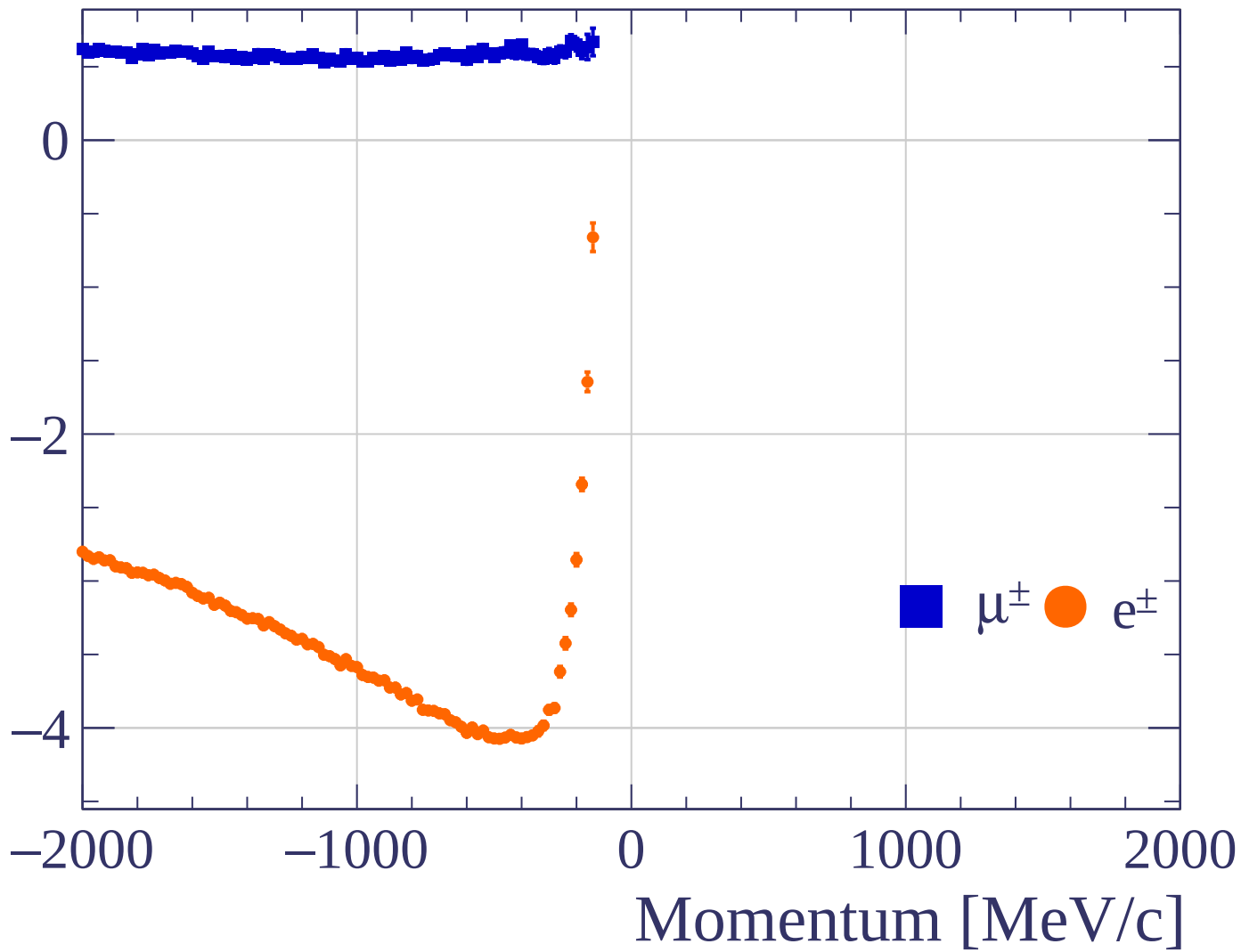
Pulls mean

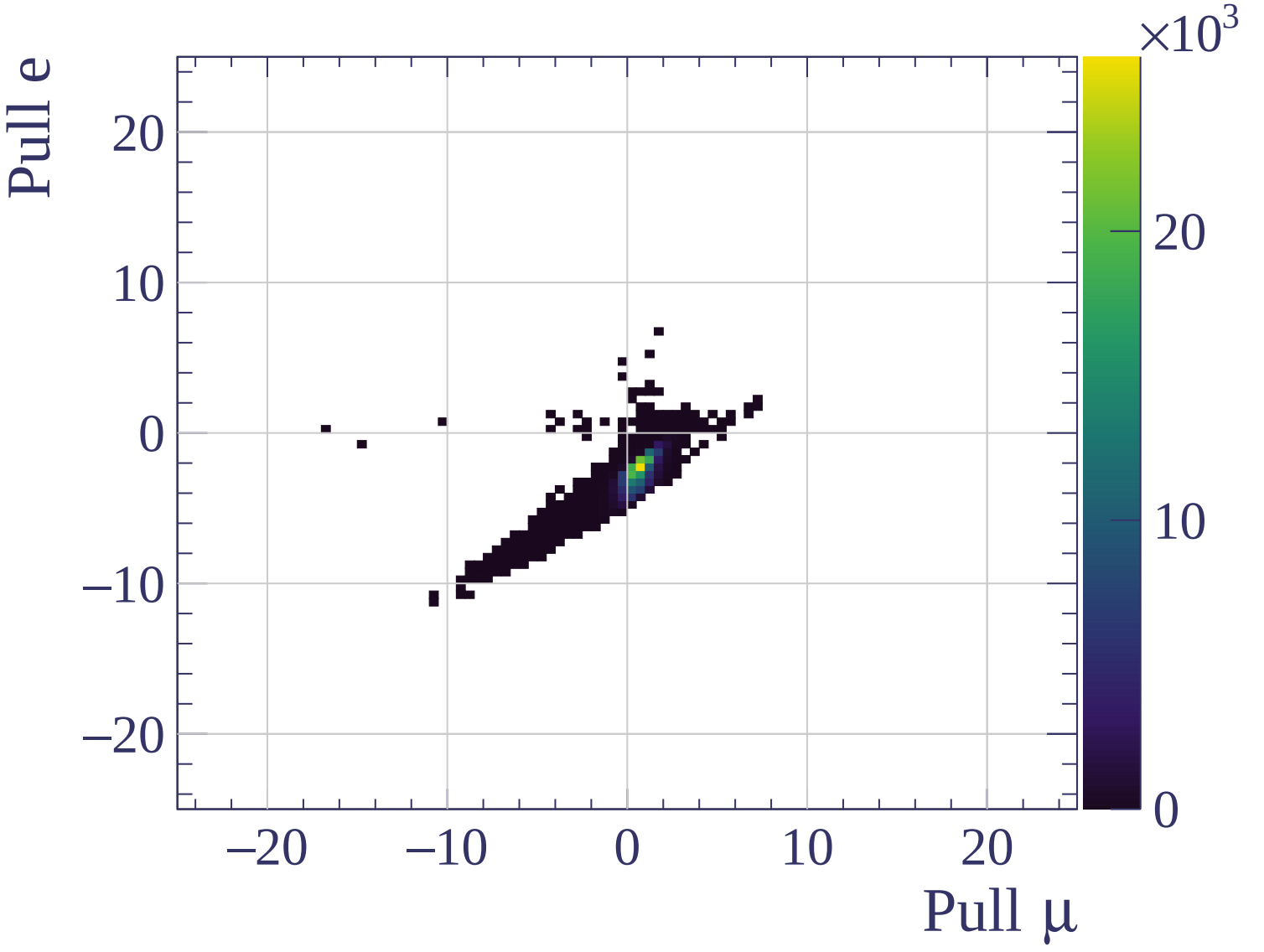


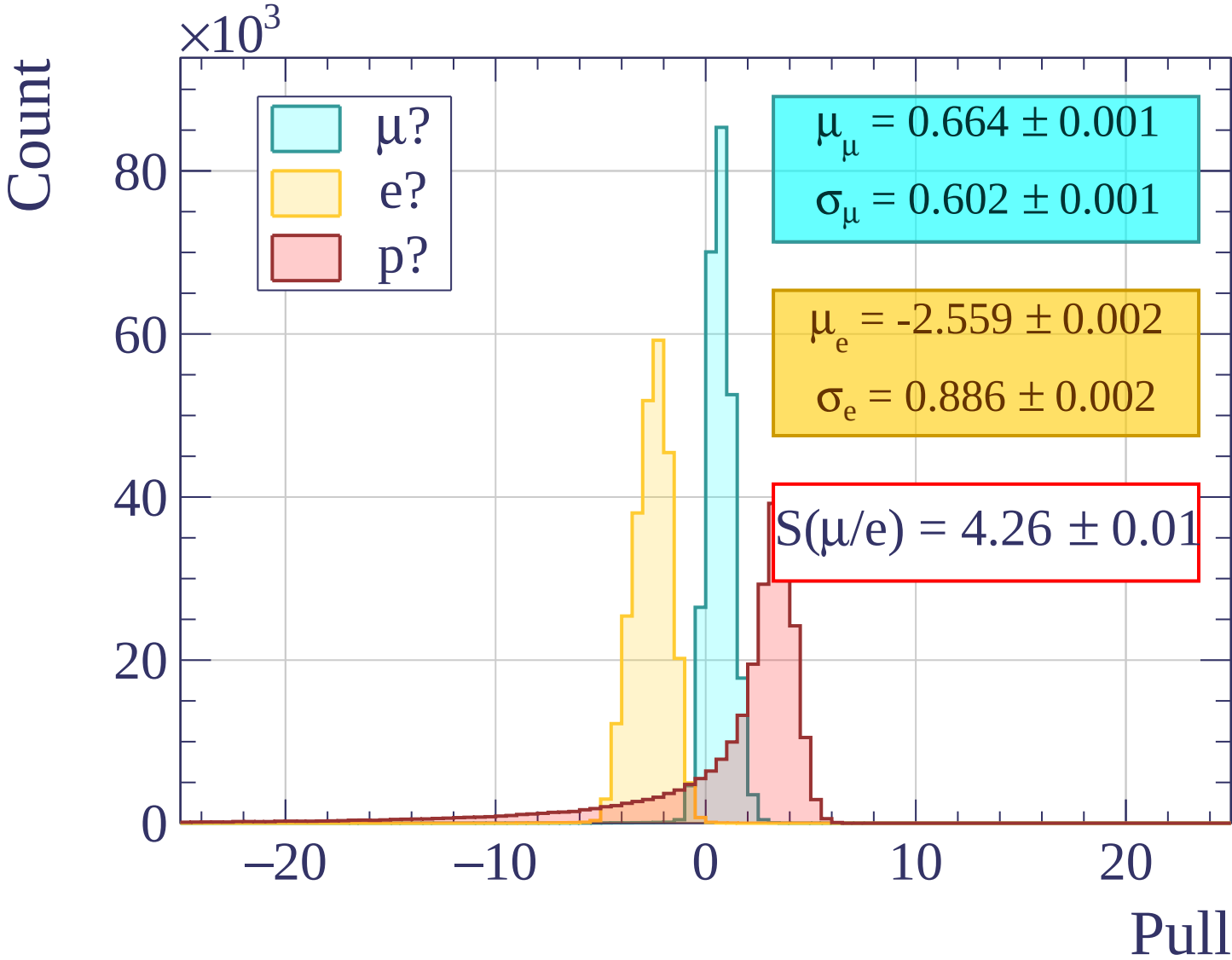
Pulls standard deviation

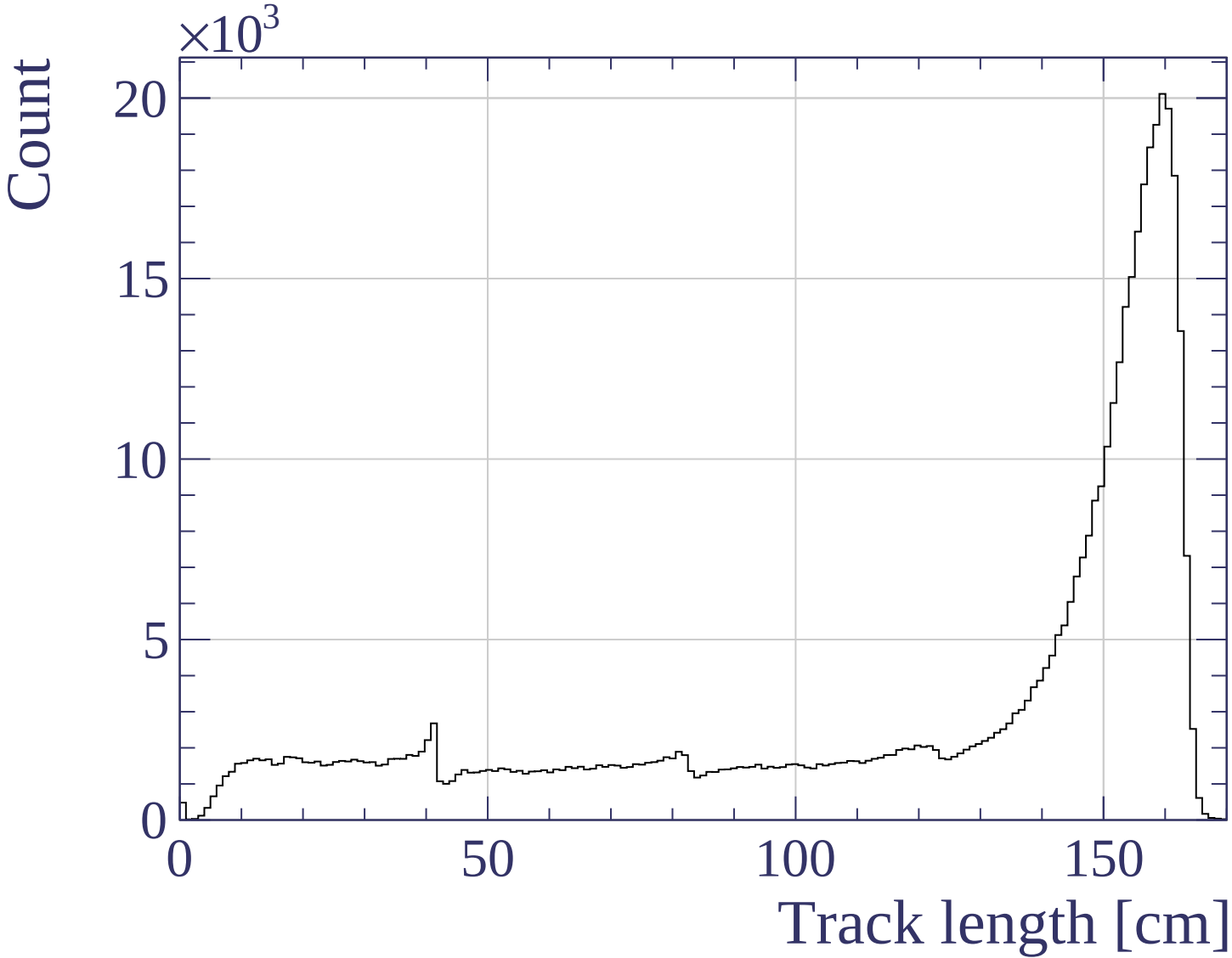


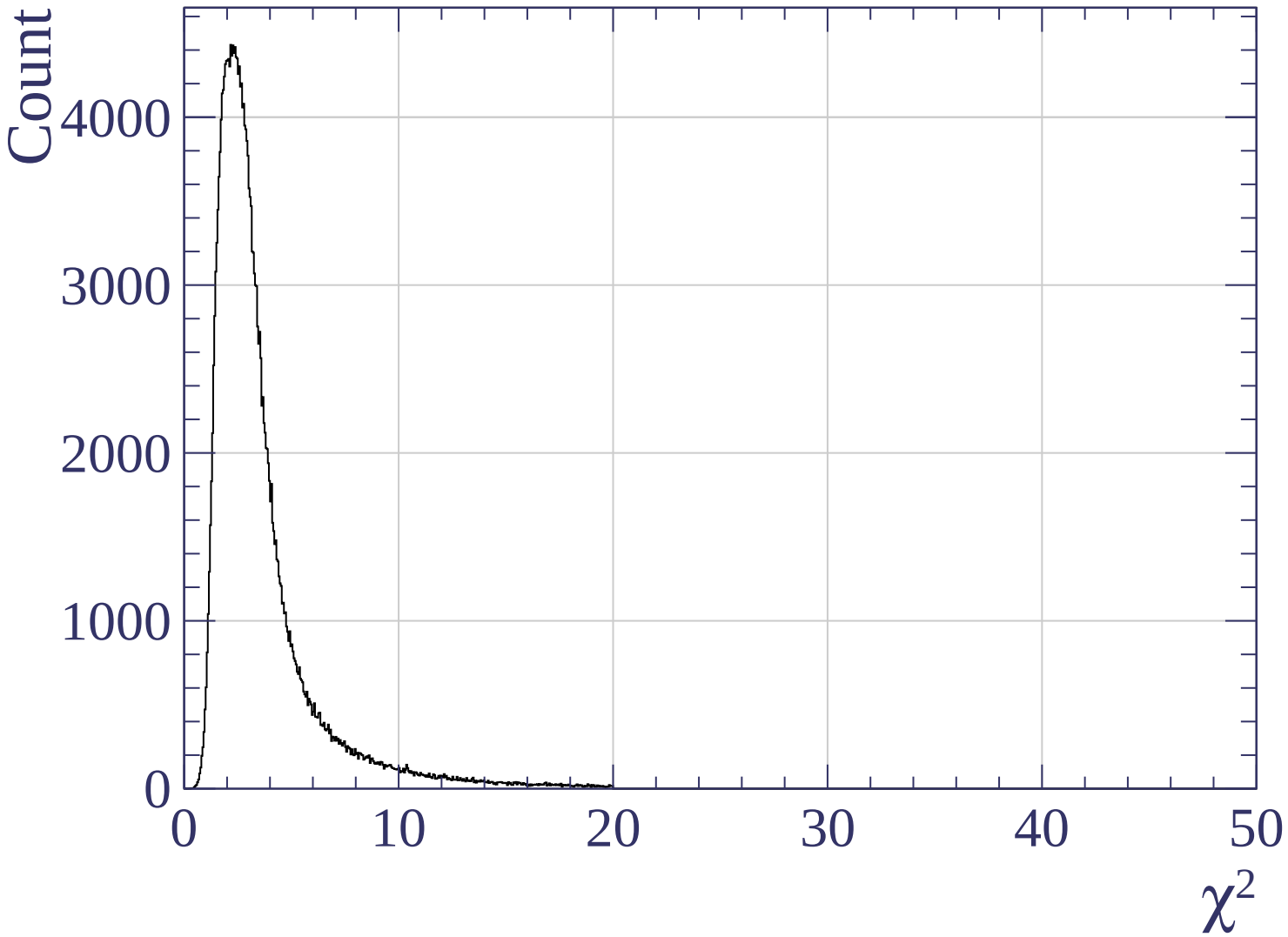
Pulls mean

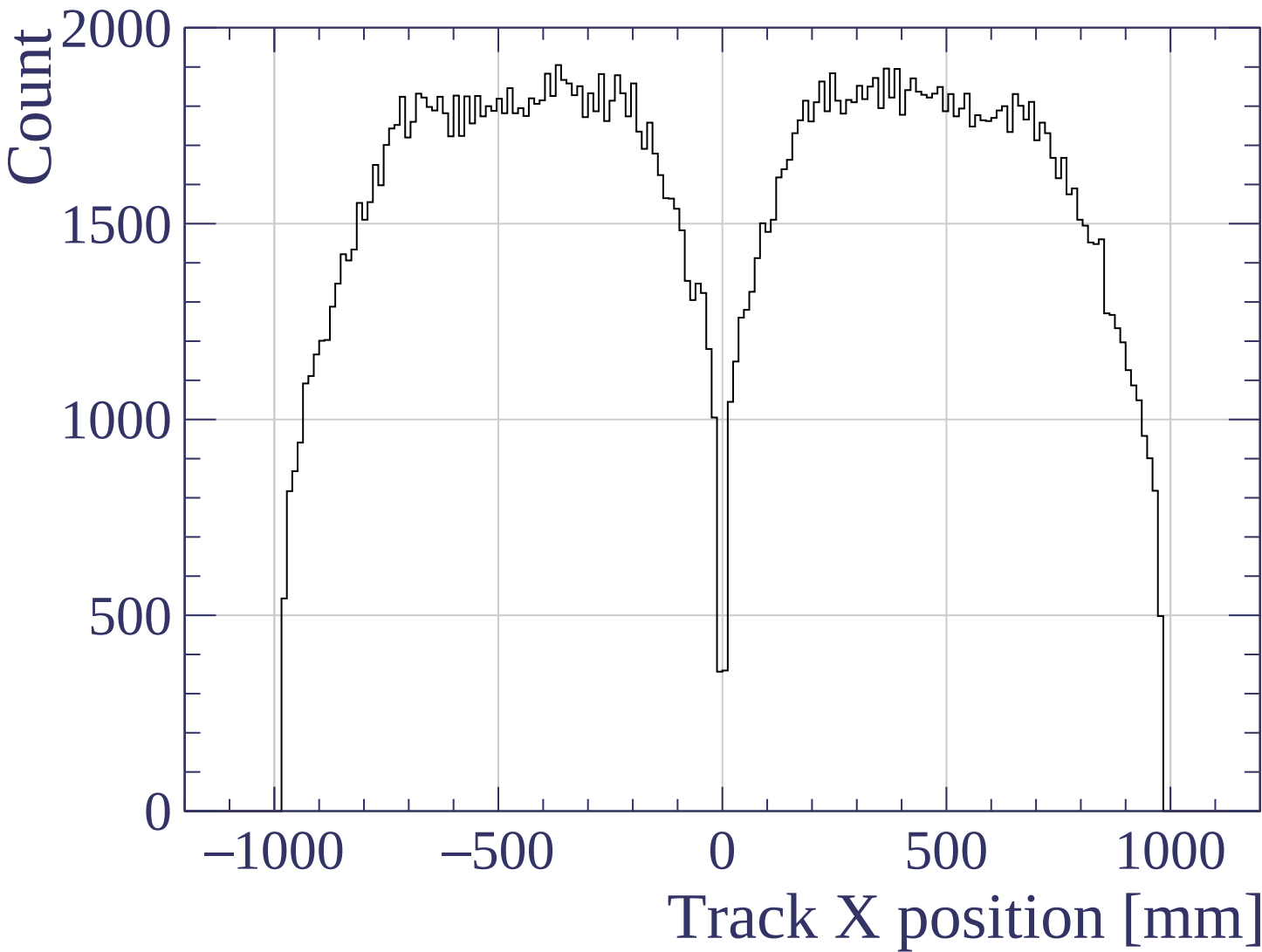


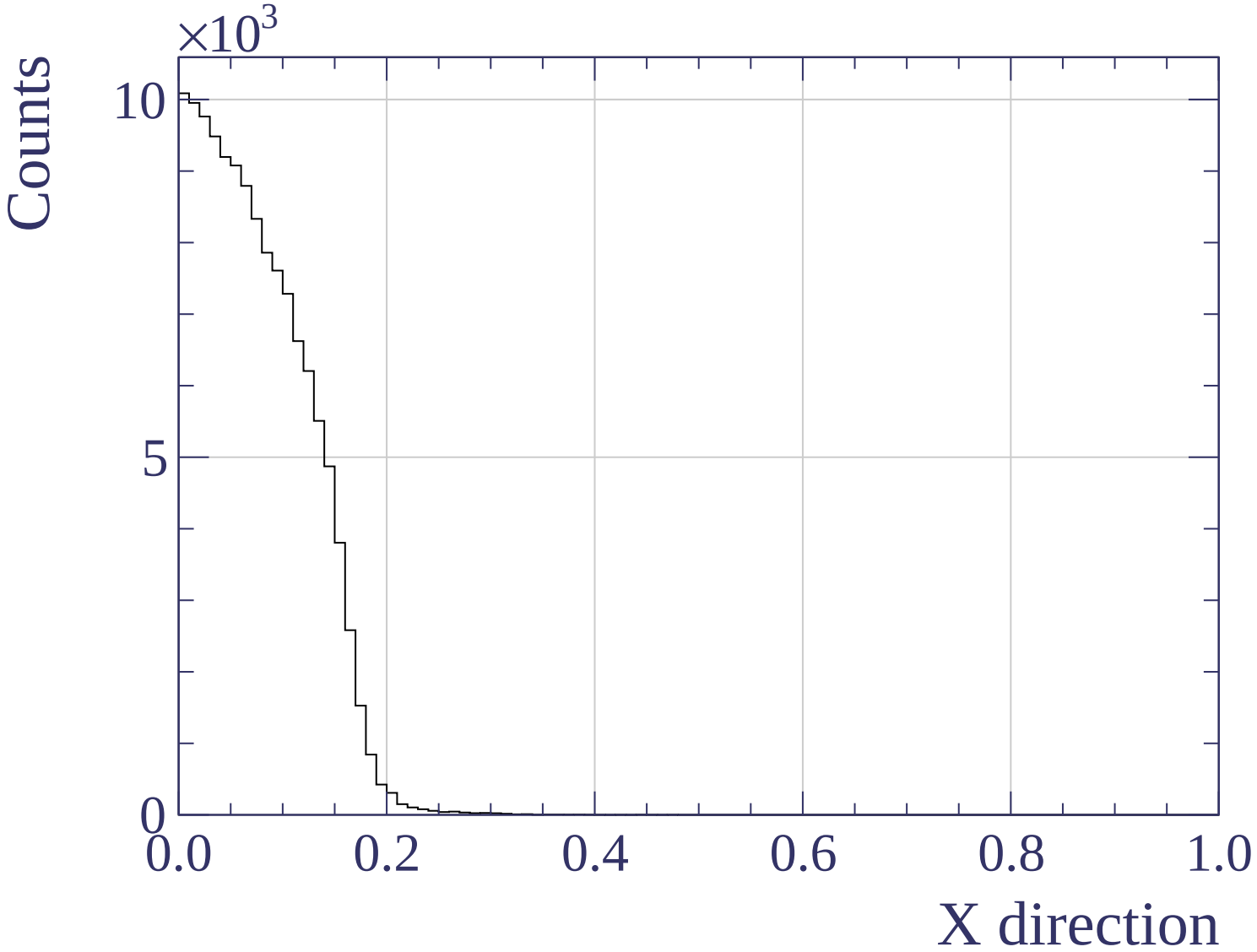


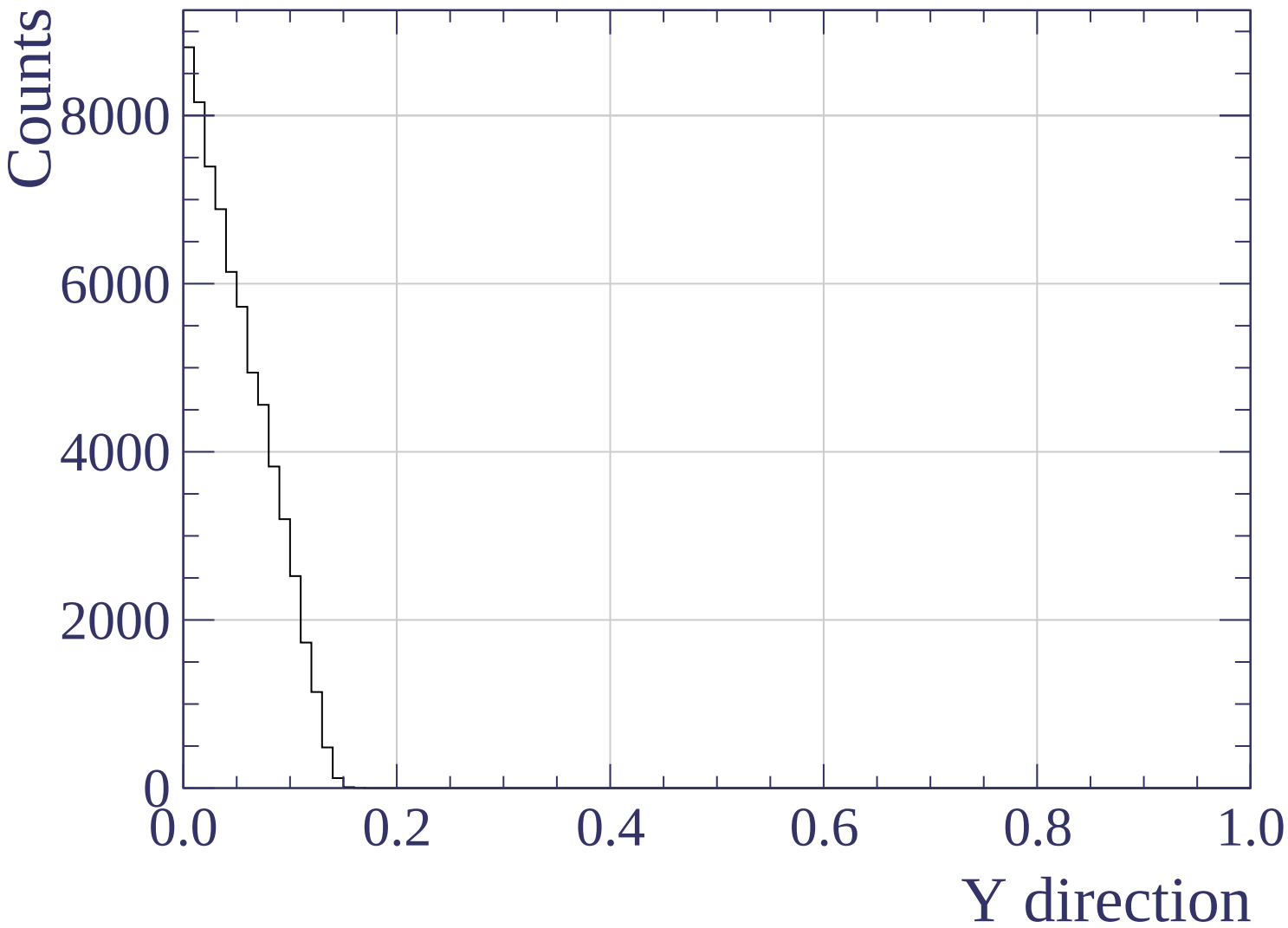


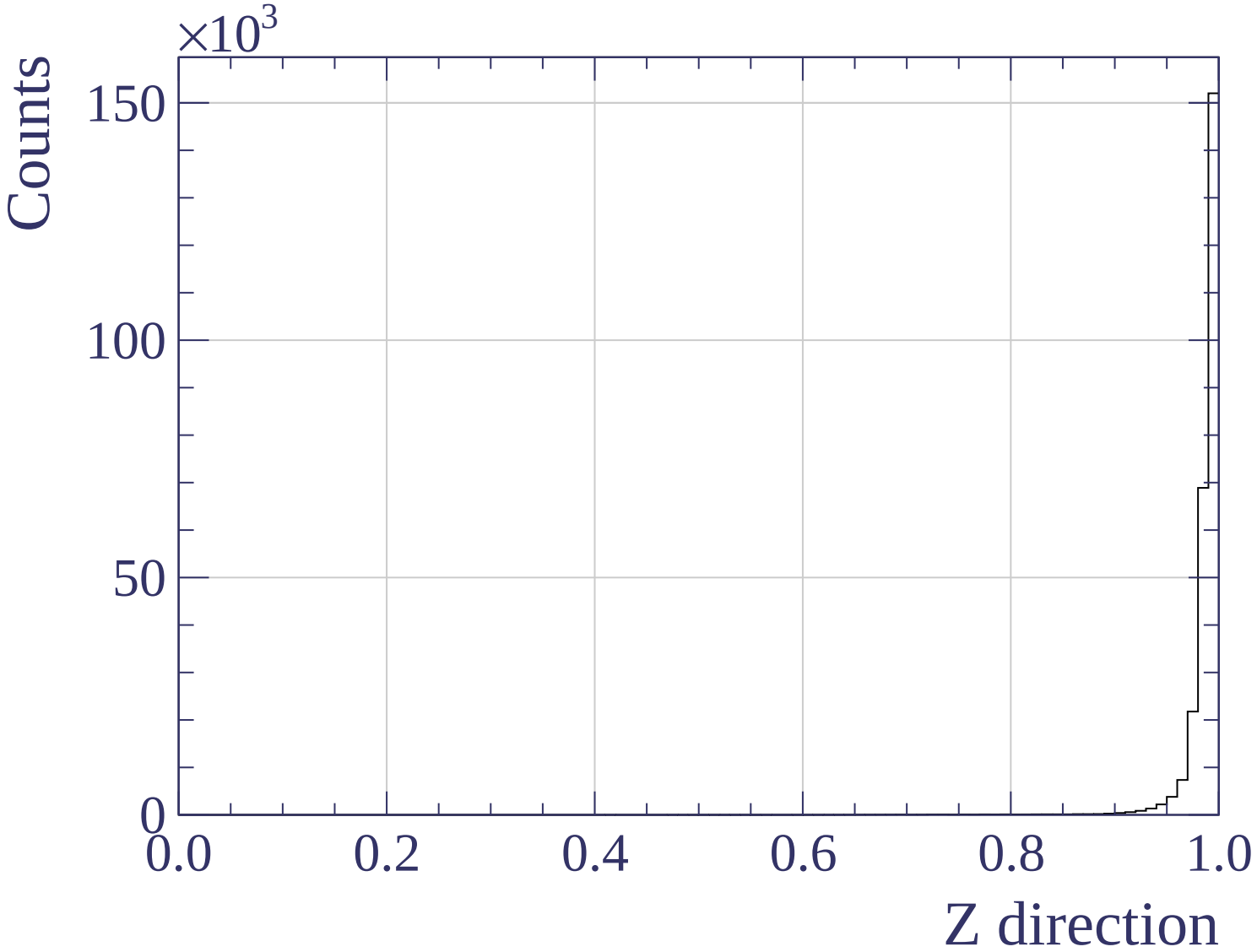


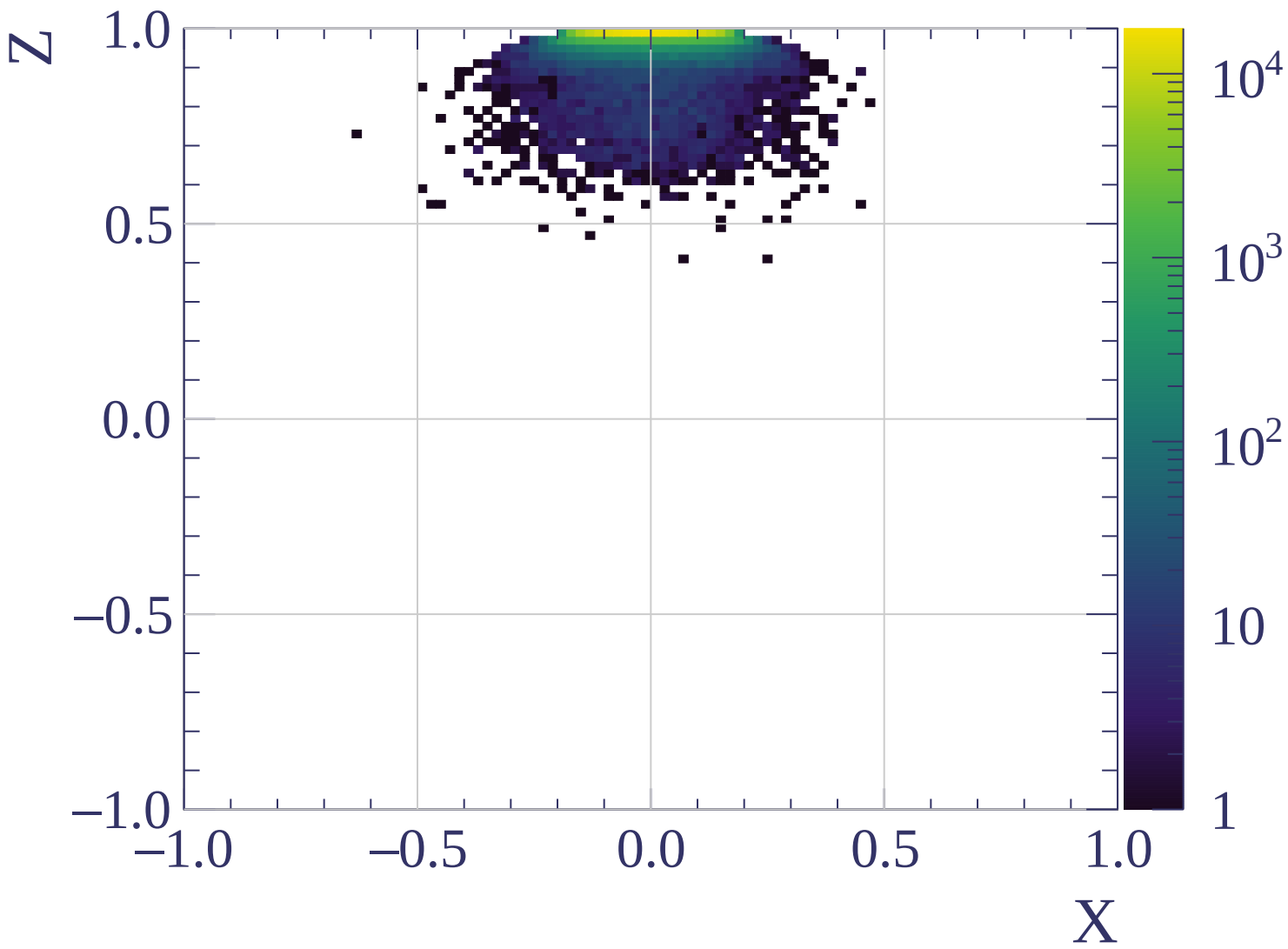


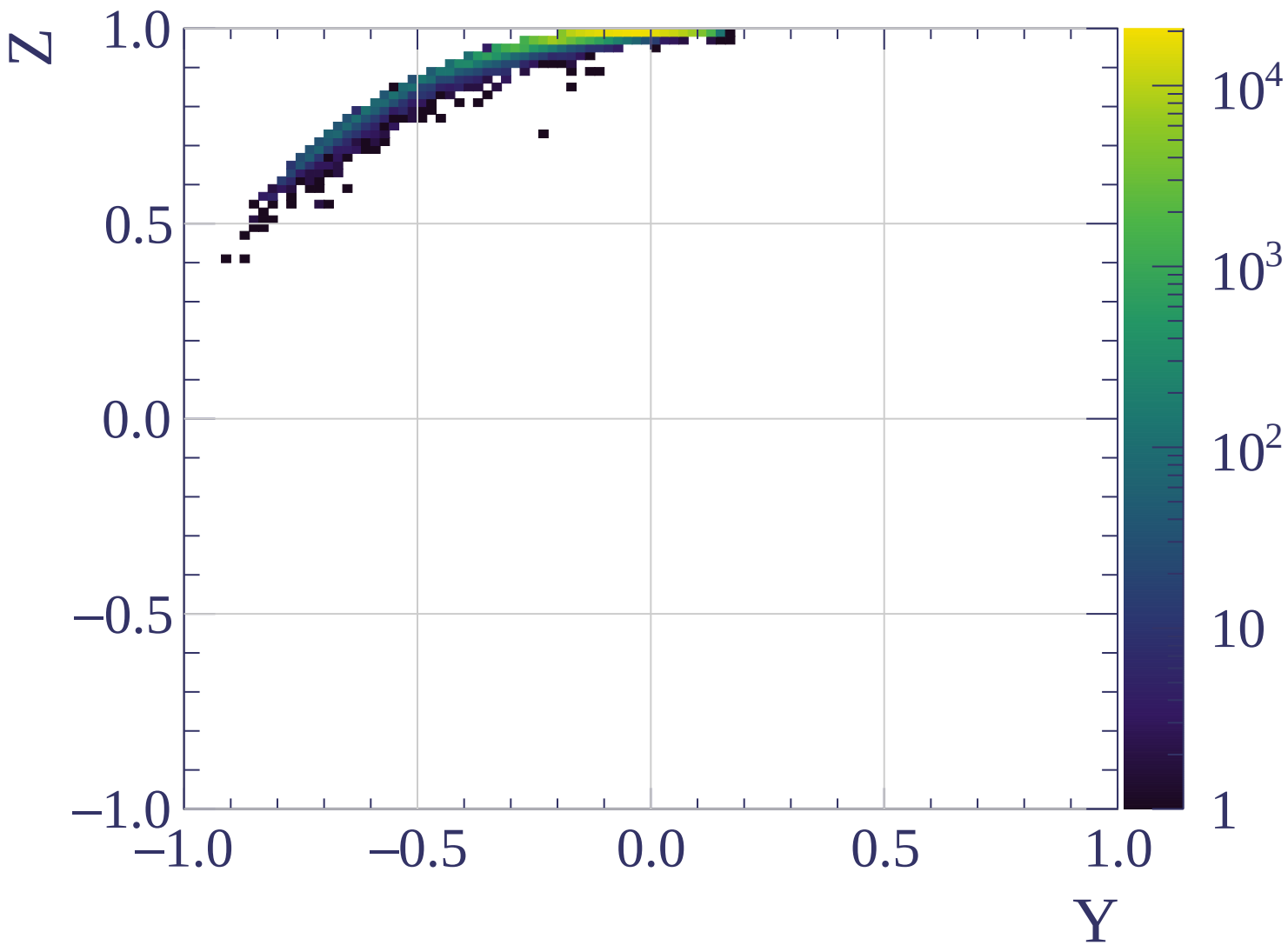


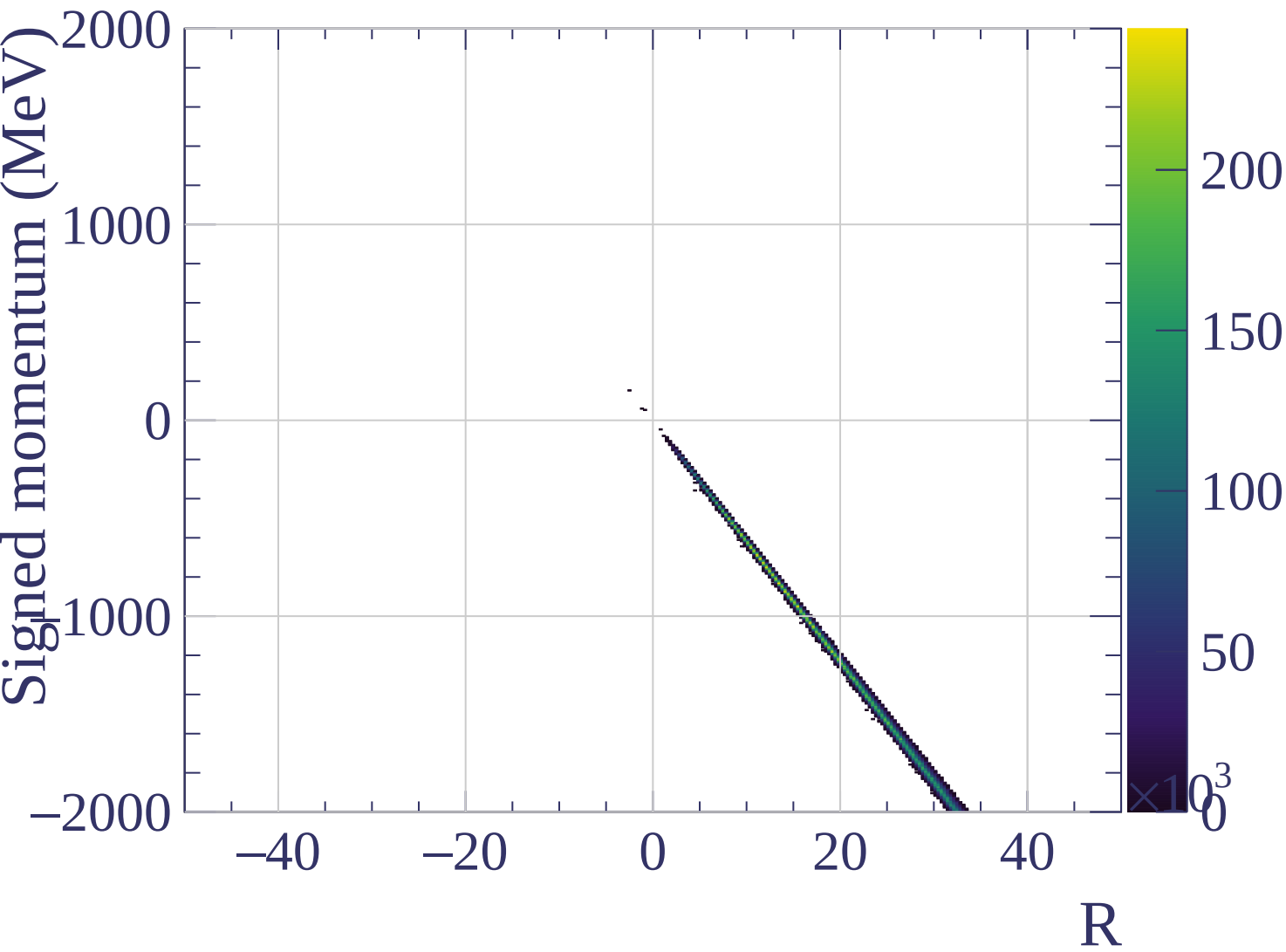


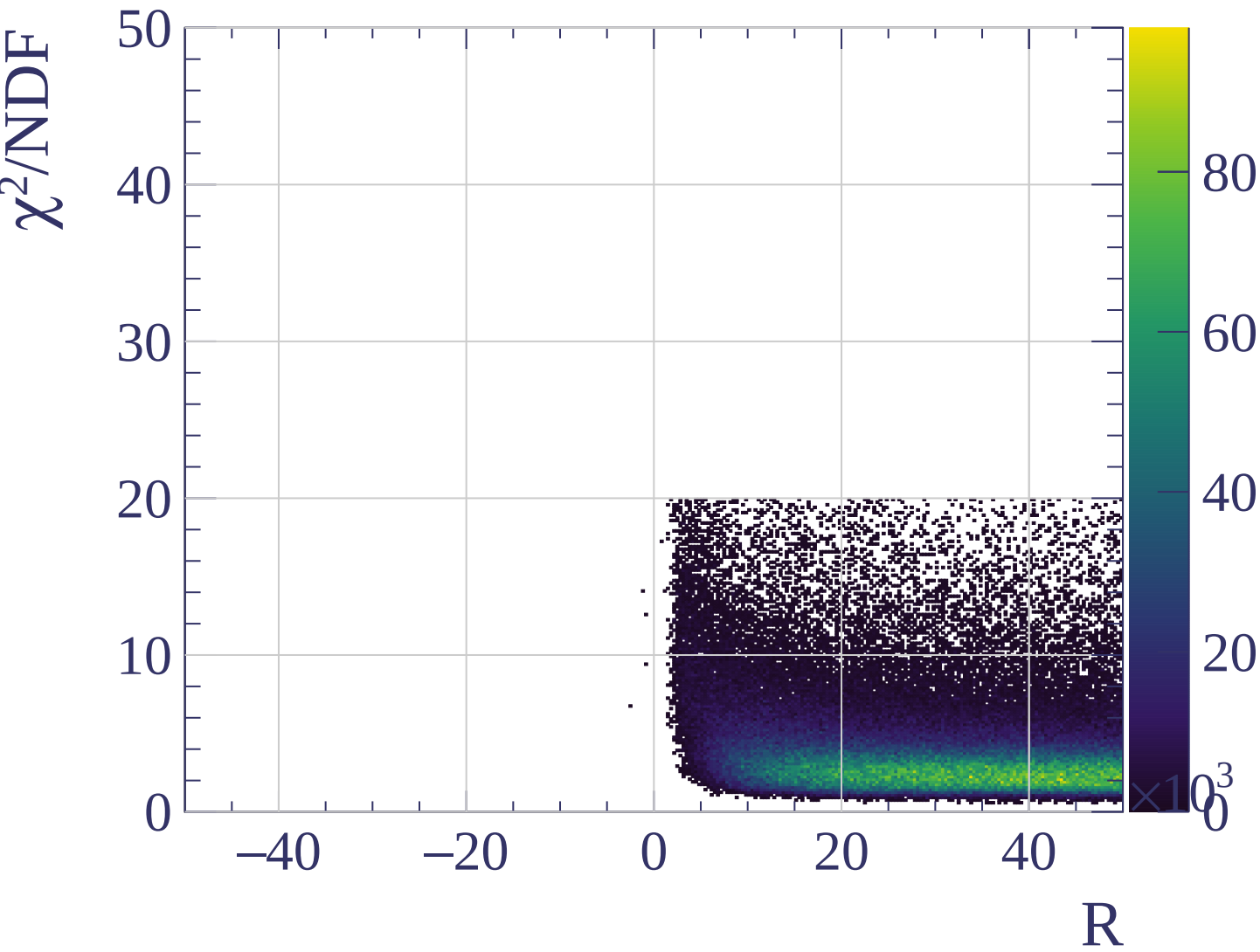




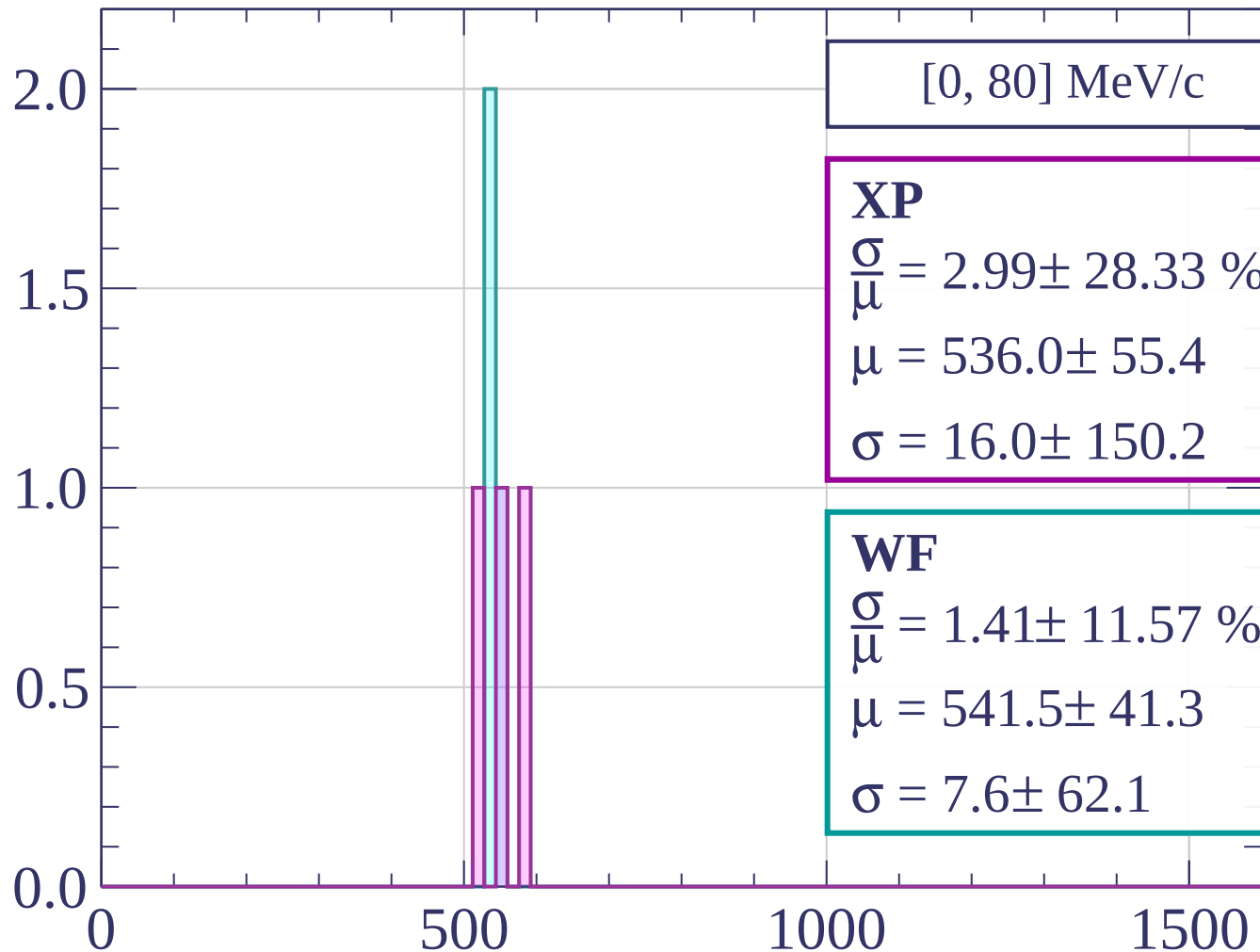








Count



[0, 80] MeV/c

XP

$$\frac{\sigma}{\mu} = 2.99 \pm 28.33 \%$$

$$\mu = 536.0 \pm 55.4$$

$$\sigma = 16.0 \pm 150.2$$

WF

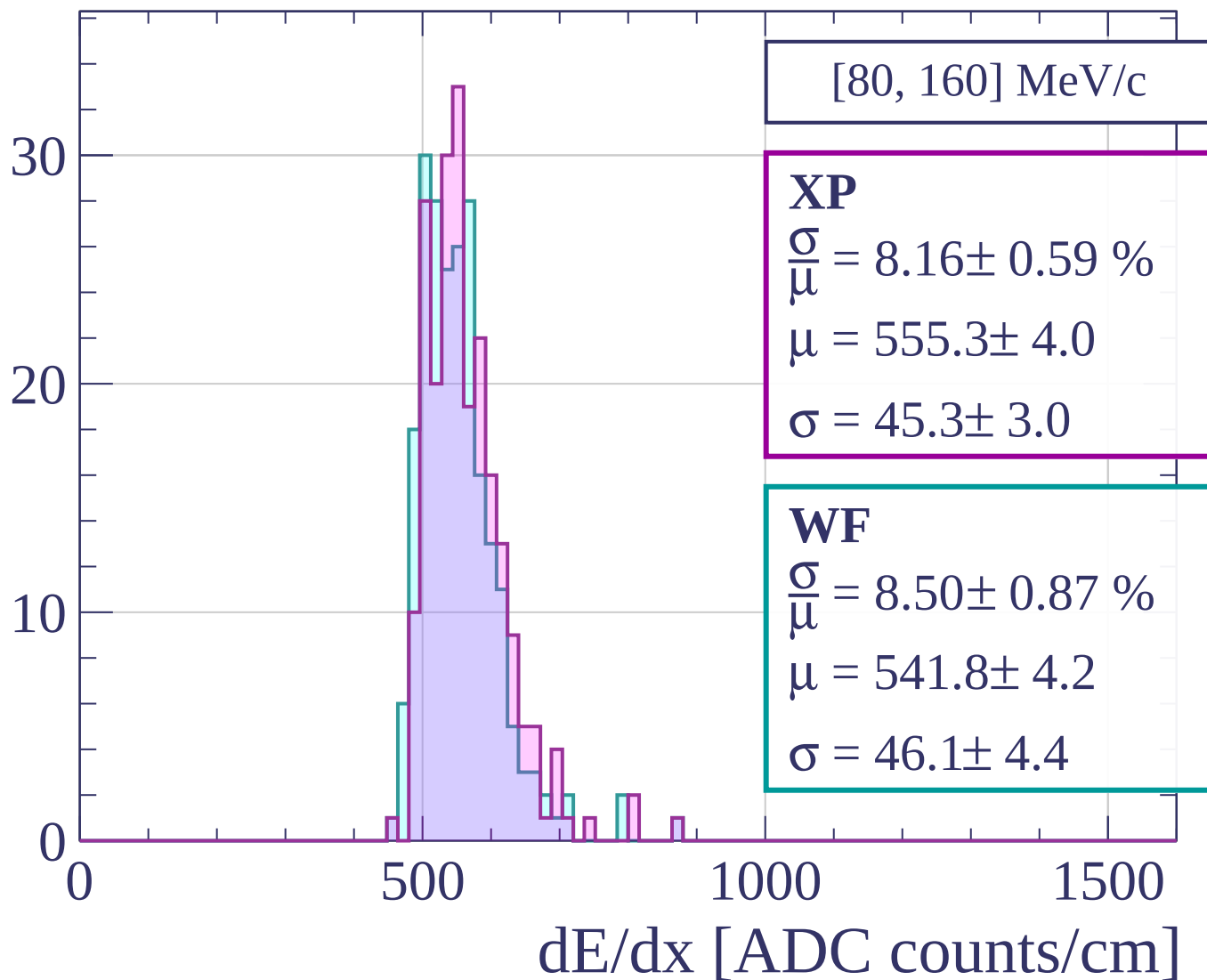
$$\frac{\sigma}{\mu} = 1.41 \pm 11.57 \%$$

$$\mu = 541.5 \pm 41.3$$

$$\sigma = 7.6 \pm 62.1$$

dE/dx [ADC counts/cm]

Count



Count

[160, 240] MeV/c

XP

$$\frac{\sigma}{\mu} = 6.49 \pm 0.23 \%$$

$$\mu = 462.0 \pm 1.1$$

$$\sigma = 30.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.48 \pm 0.28 \%$$

$$\mu = 451.7 \pm 1.1$$

$$\sigma = 29.3 \pm 1.2$$

200

100

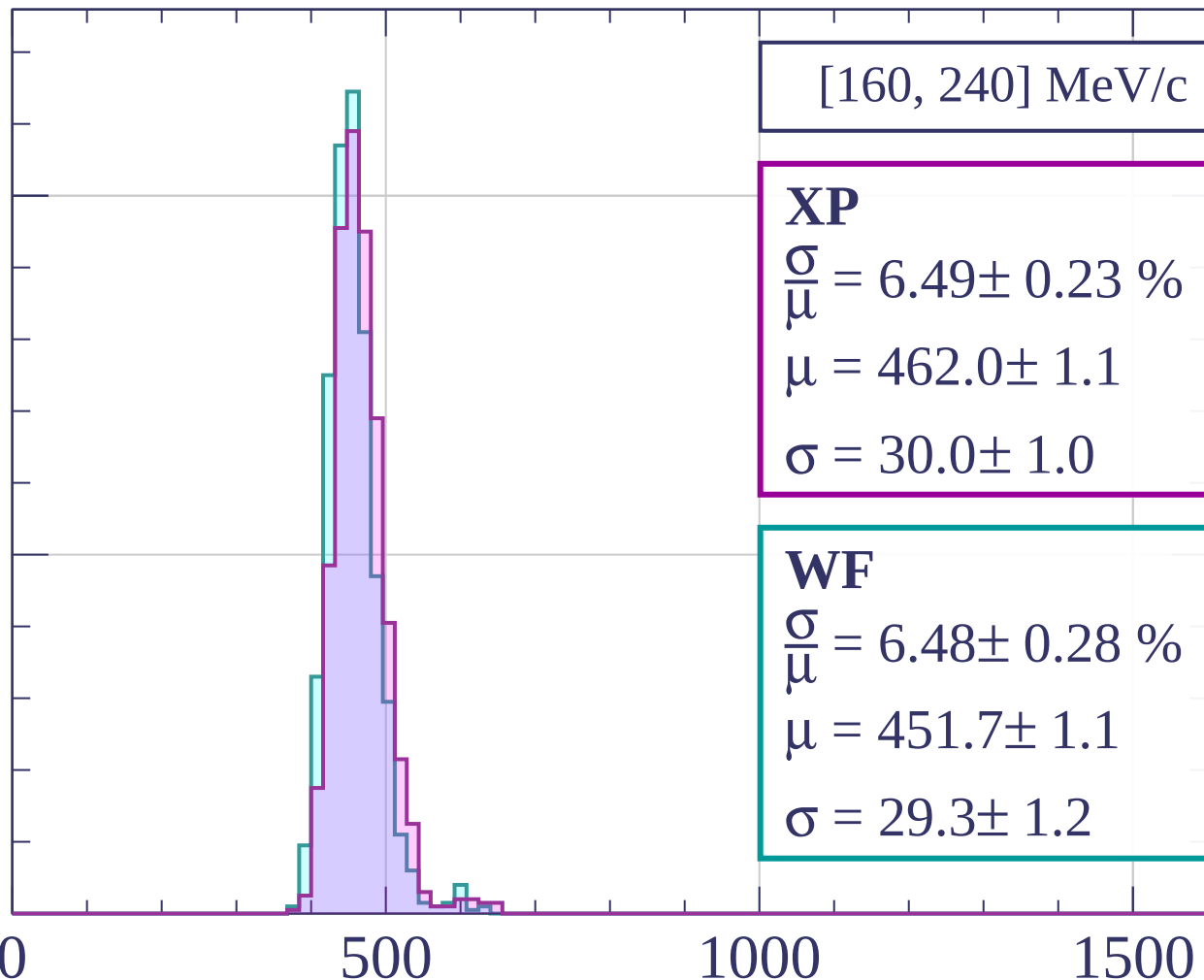
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[240, 320] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.25 \pm 0.16 \%$$

$$\mu = 429.1 \pm 0.7$$

$$\sigma = 22.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.17 \pm 0.19 \%$$

$$\mu = 420.4 \pm 0.7$$

$$\sigma = 21.7 \pm 0.8$$

400

300

200

100

0

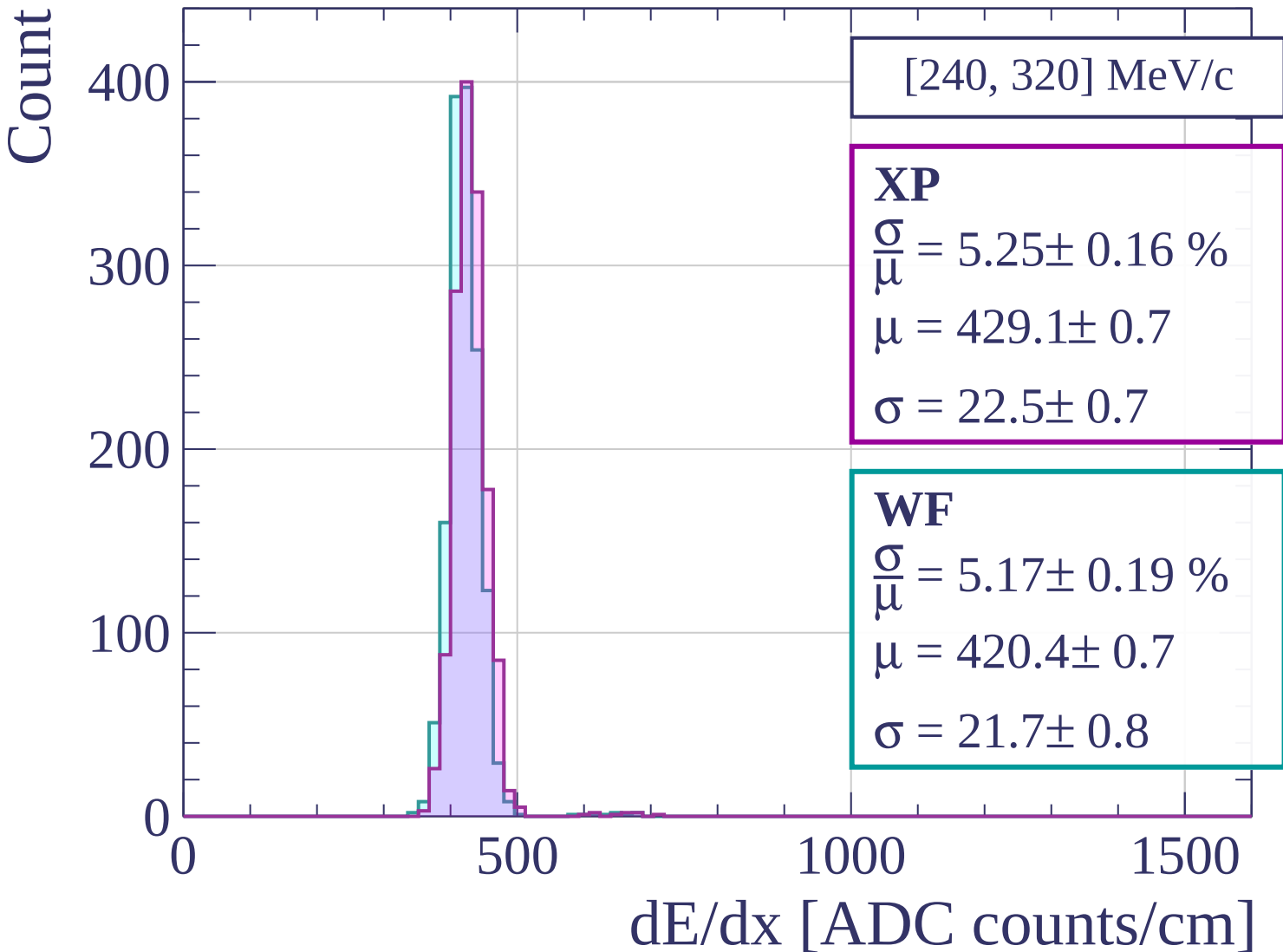
0

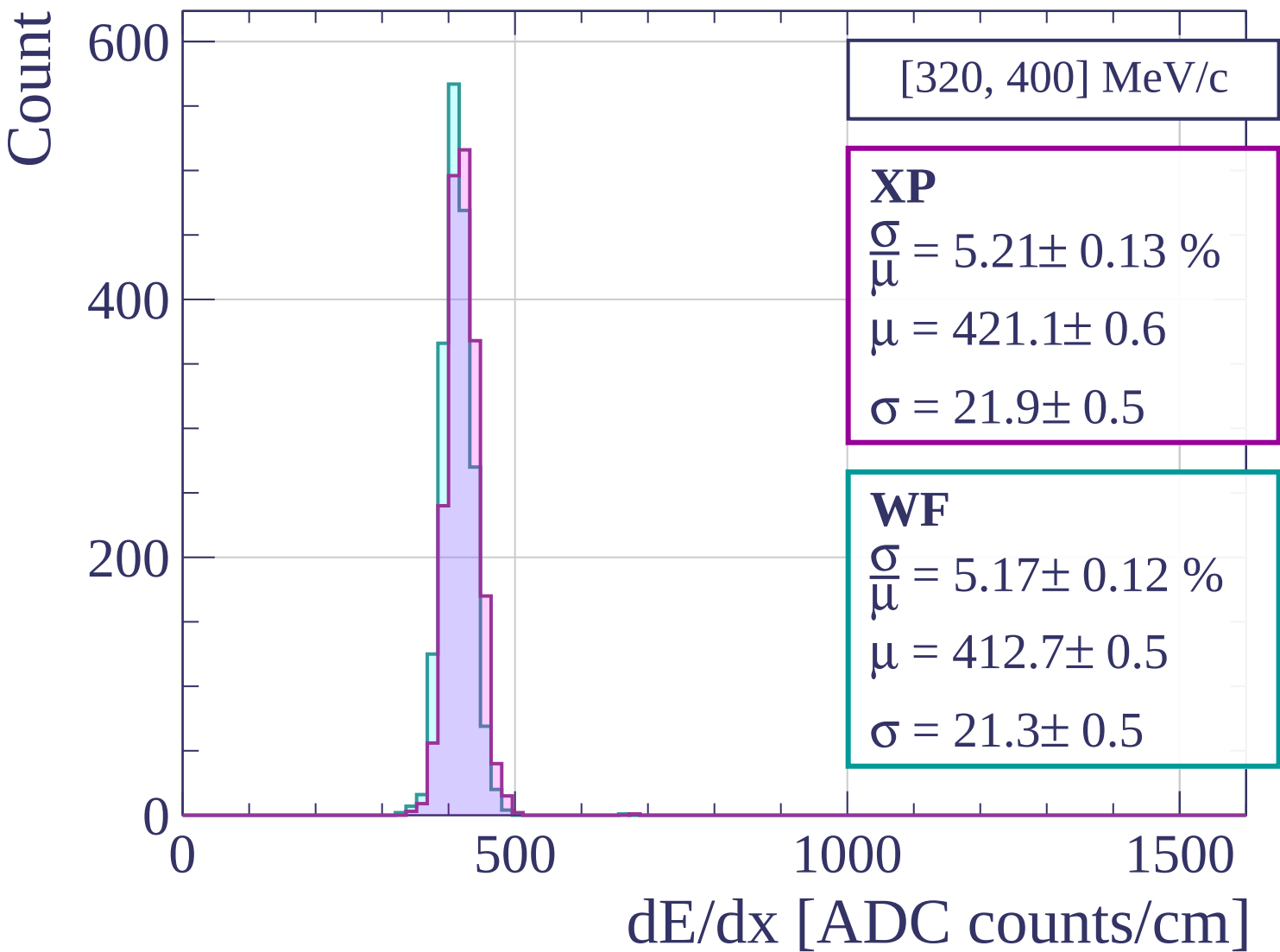
500

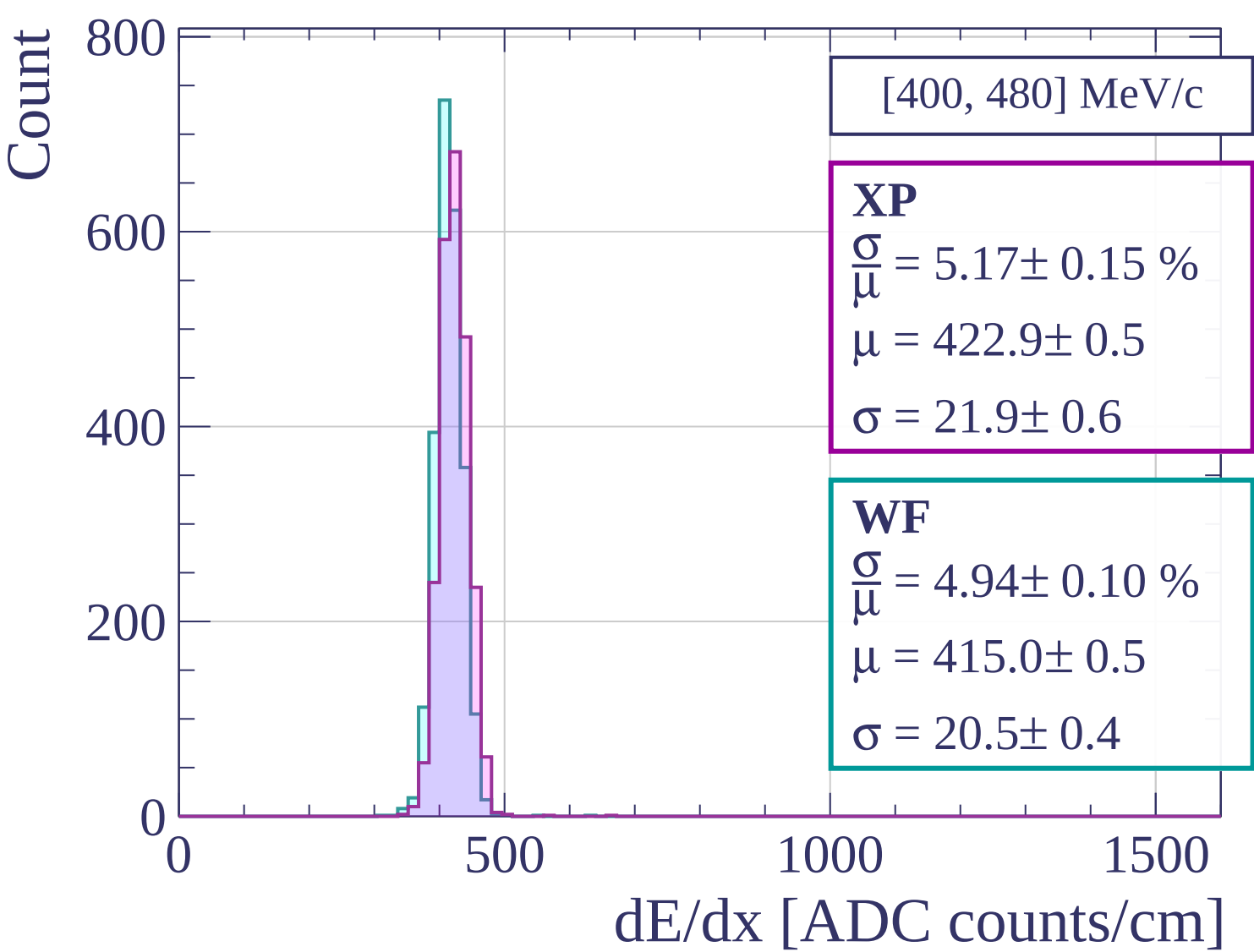
1000

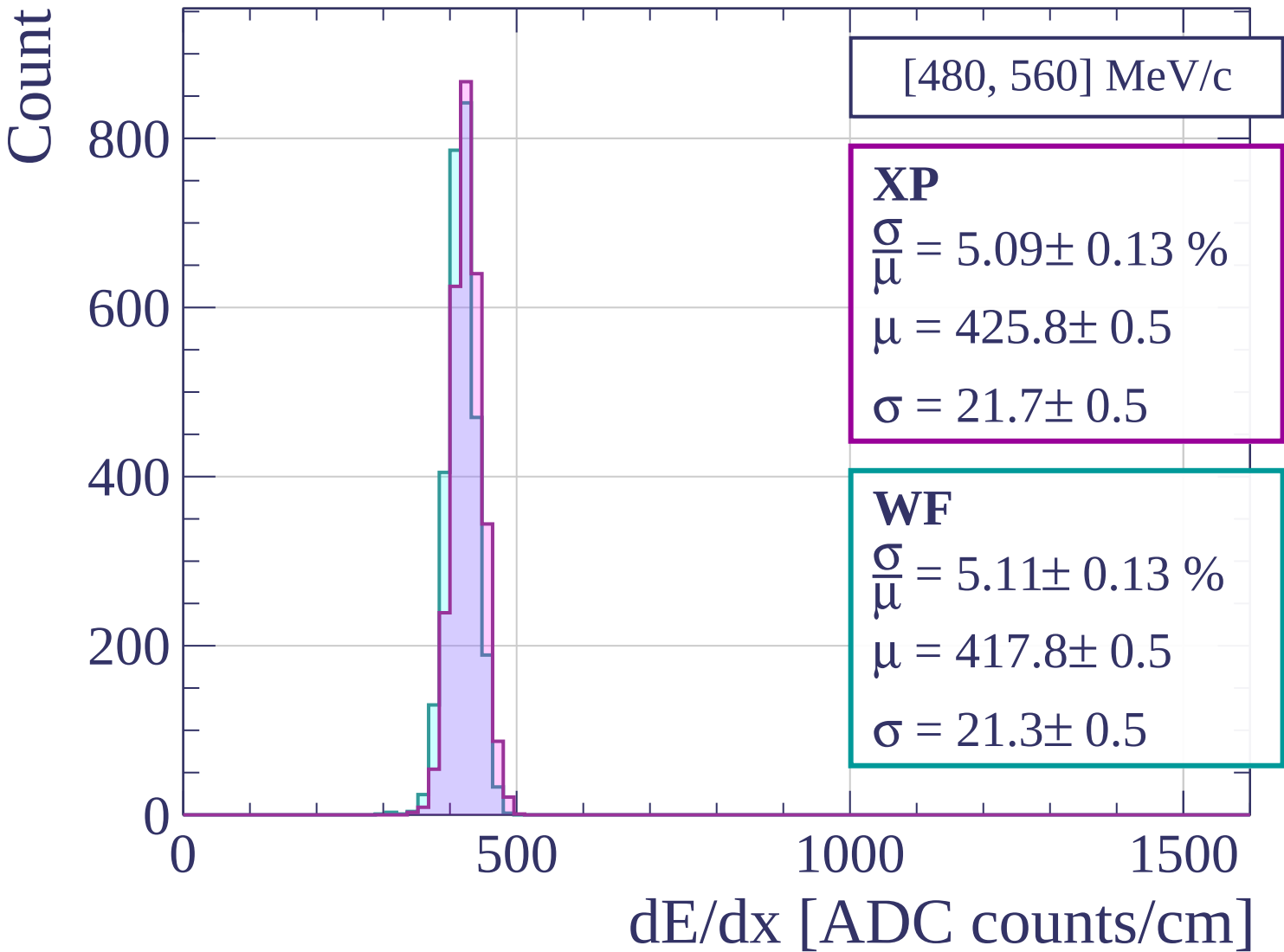
1500

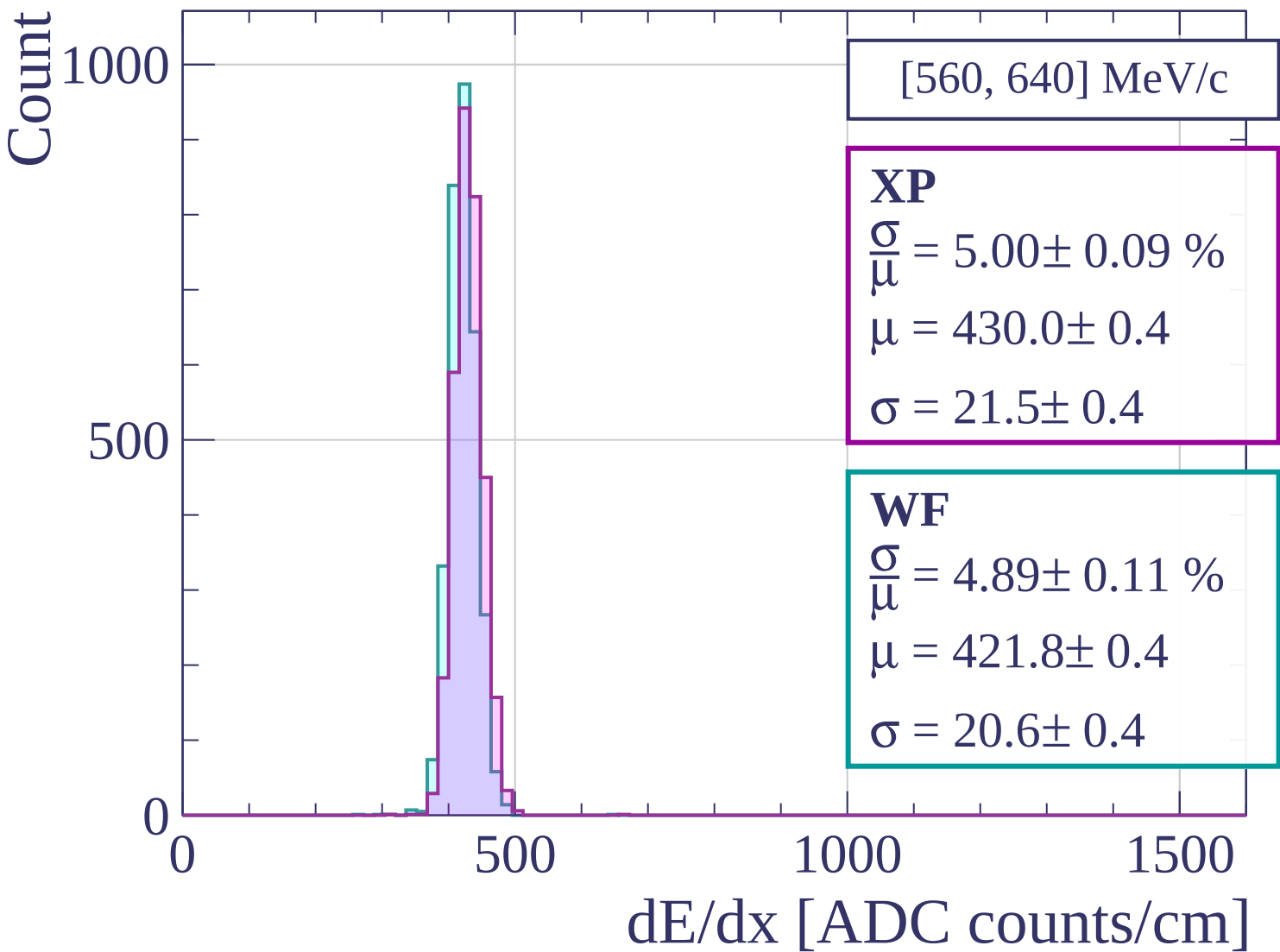
dE/dx [ADC counts/cm]

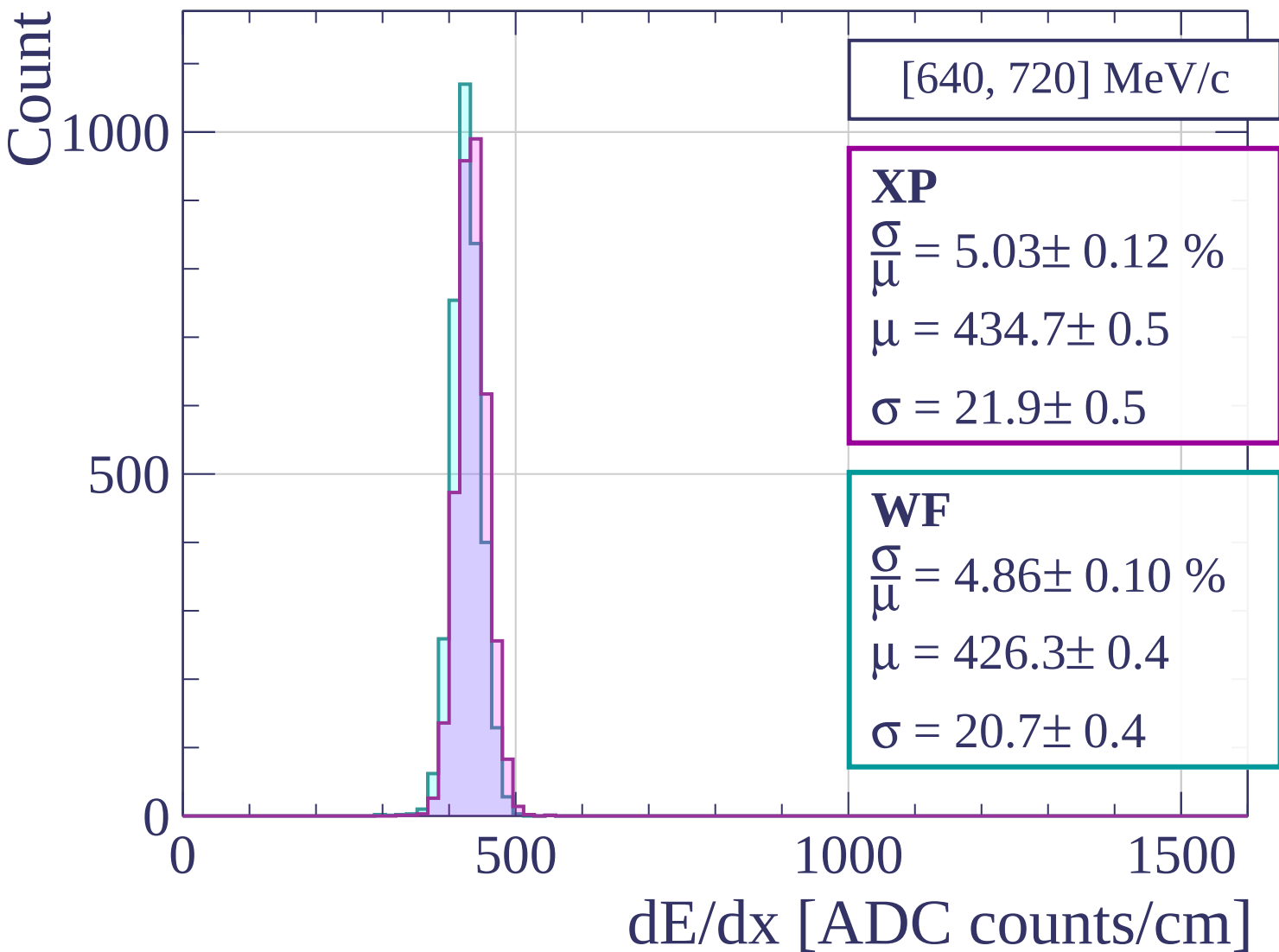


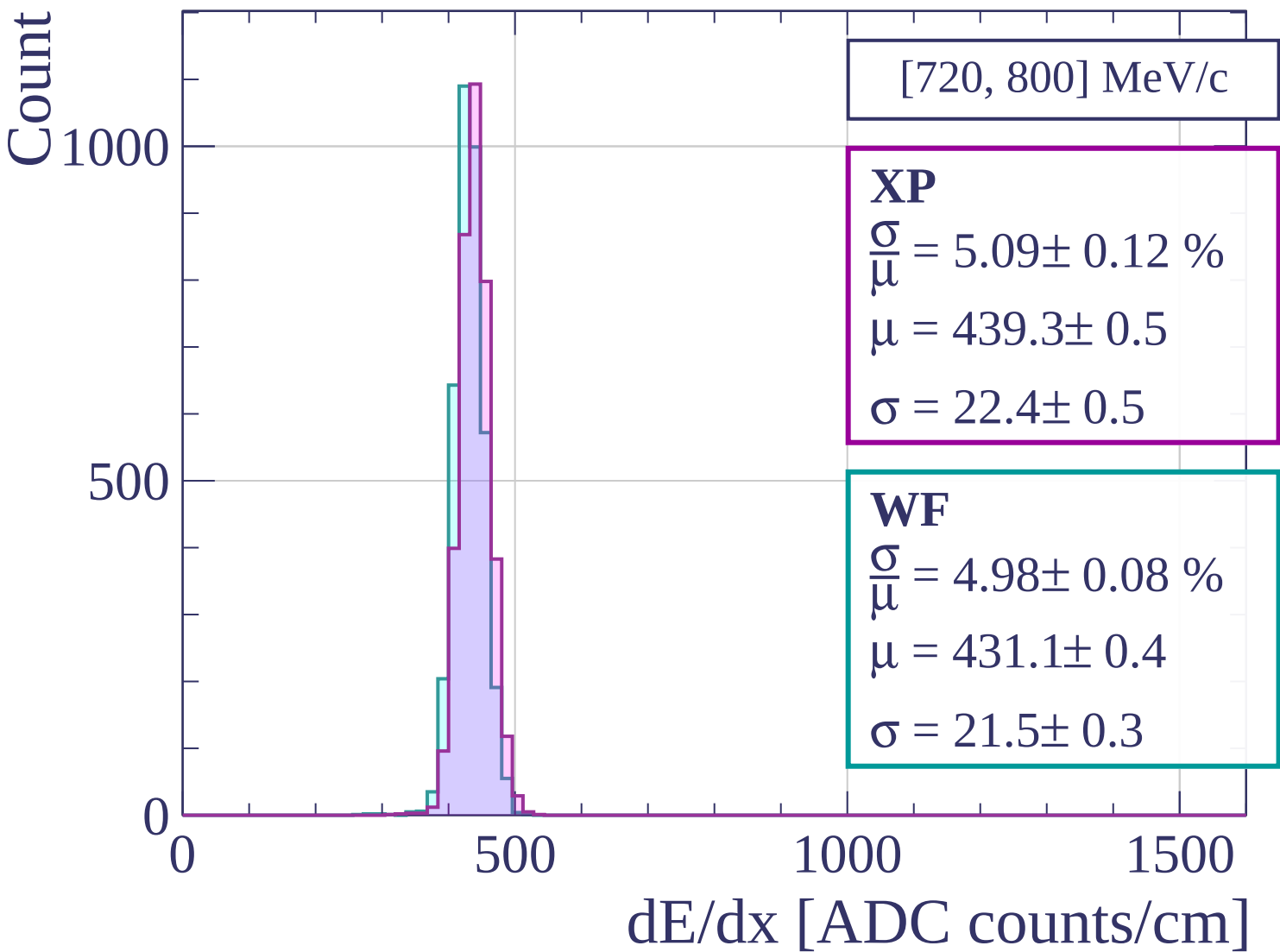


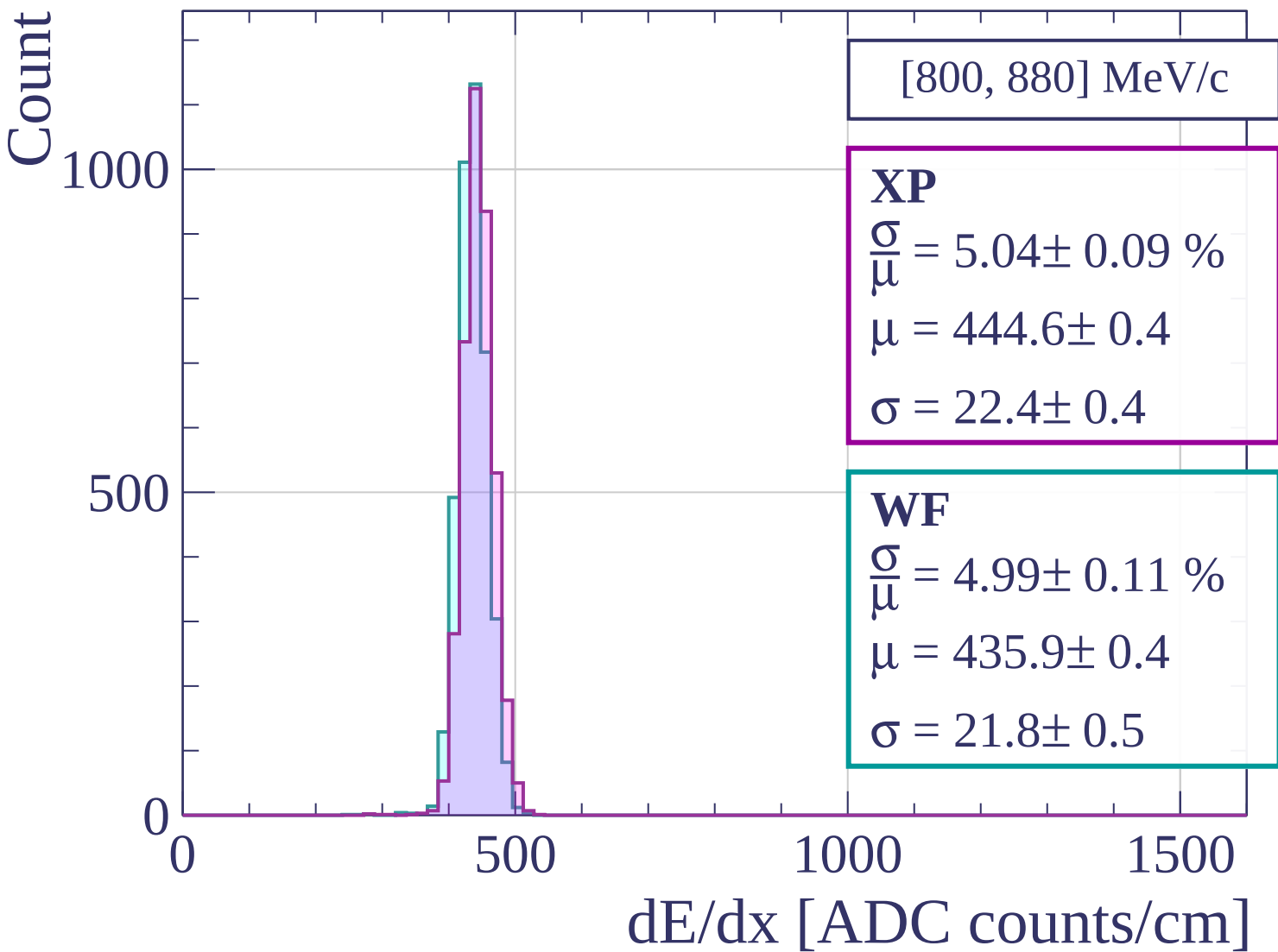


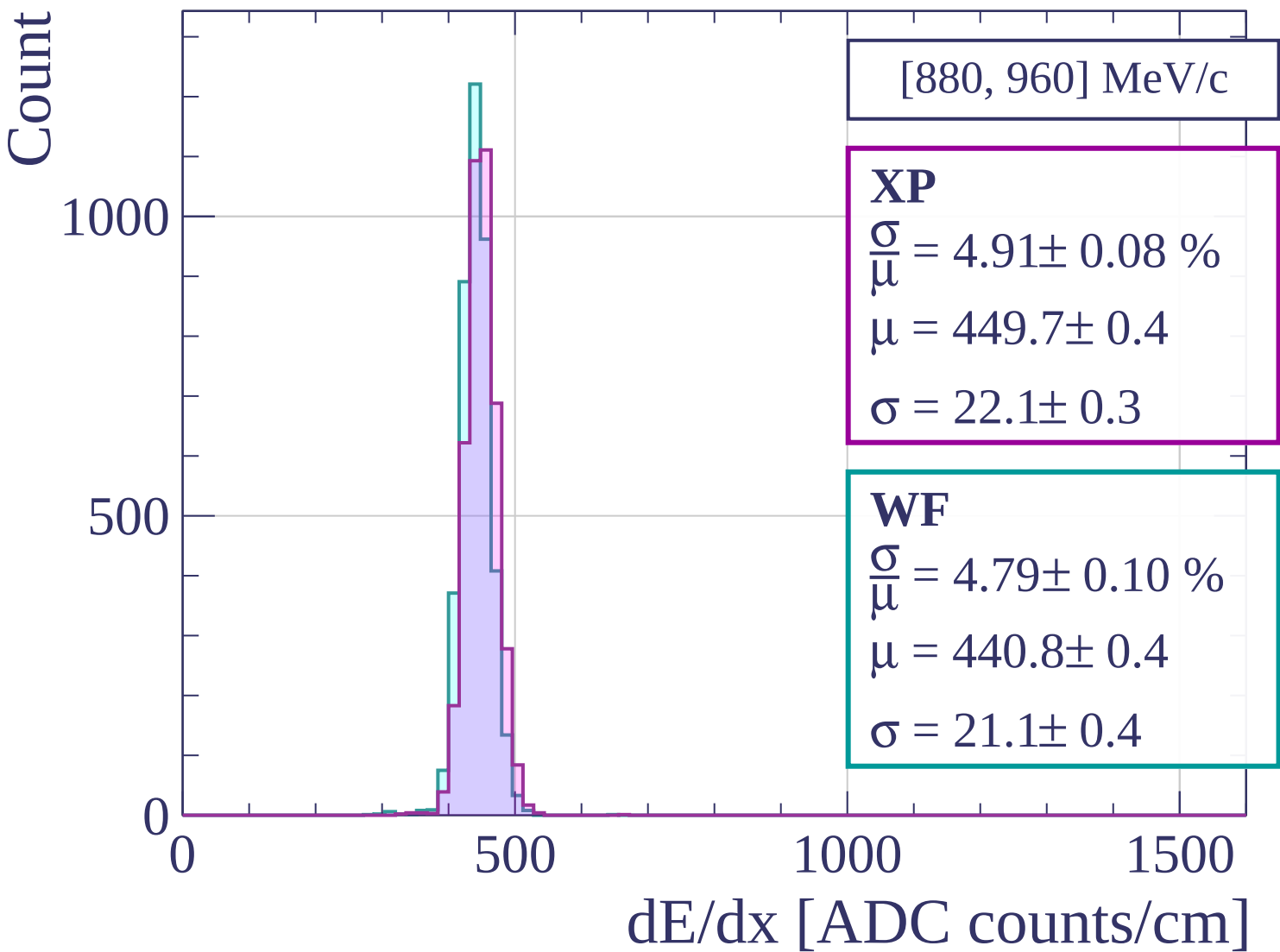


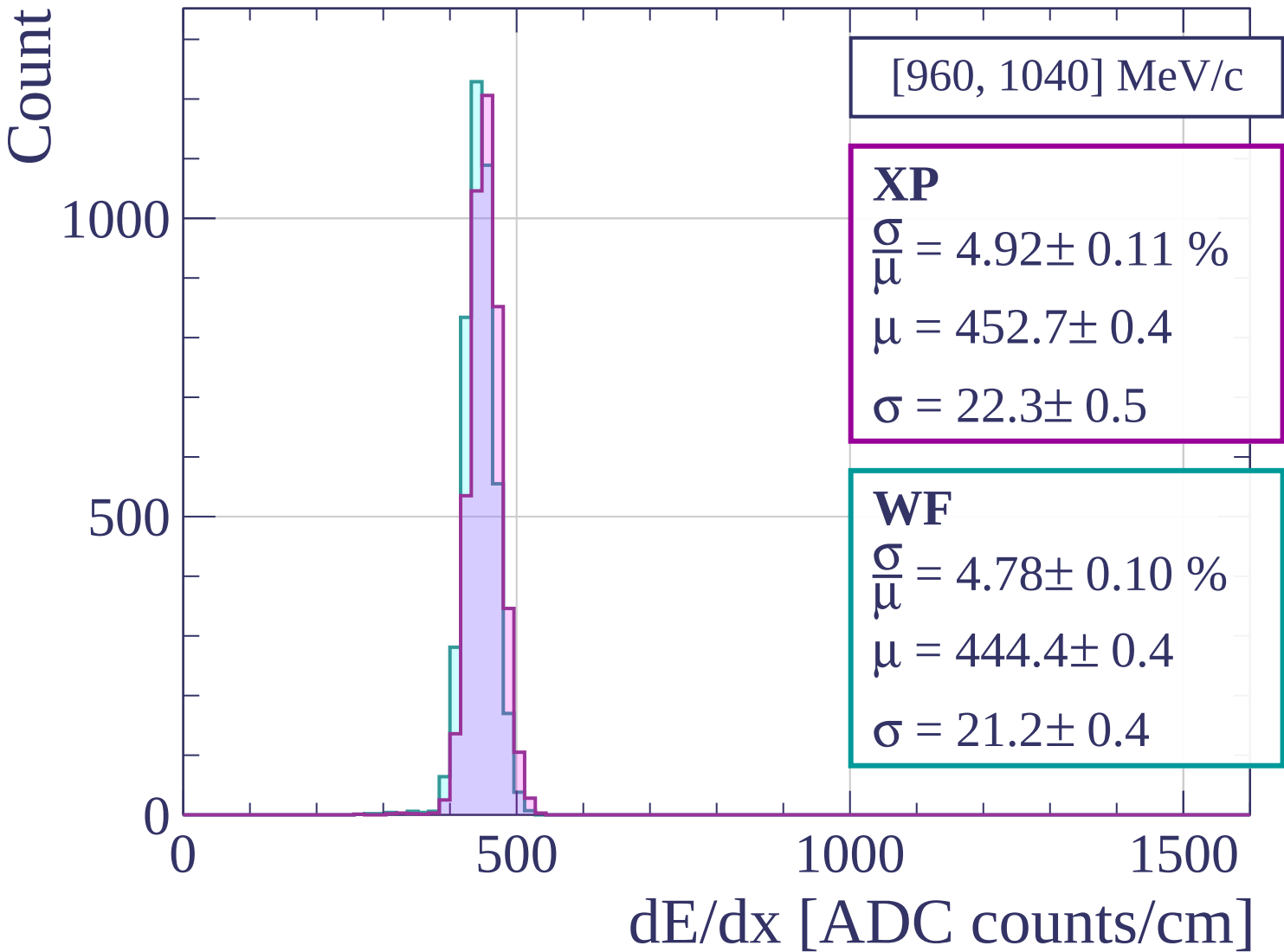


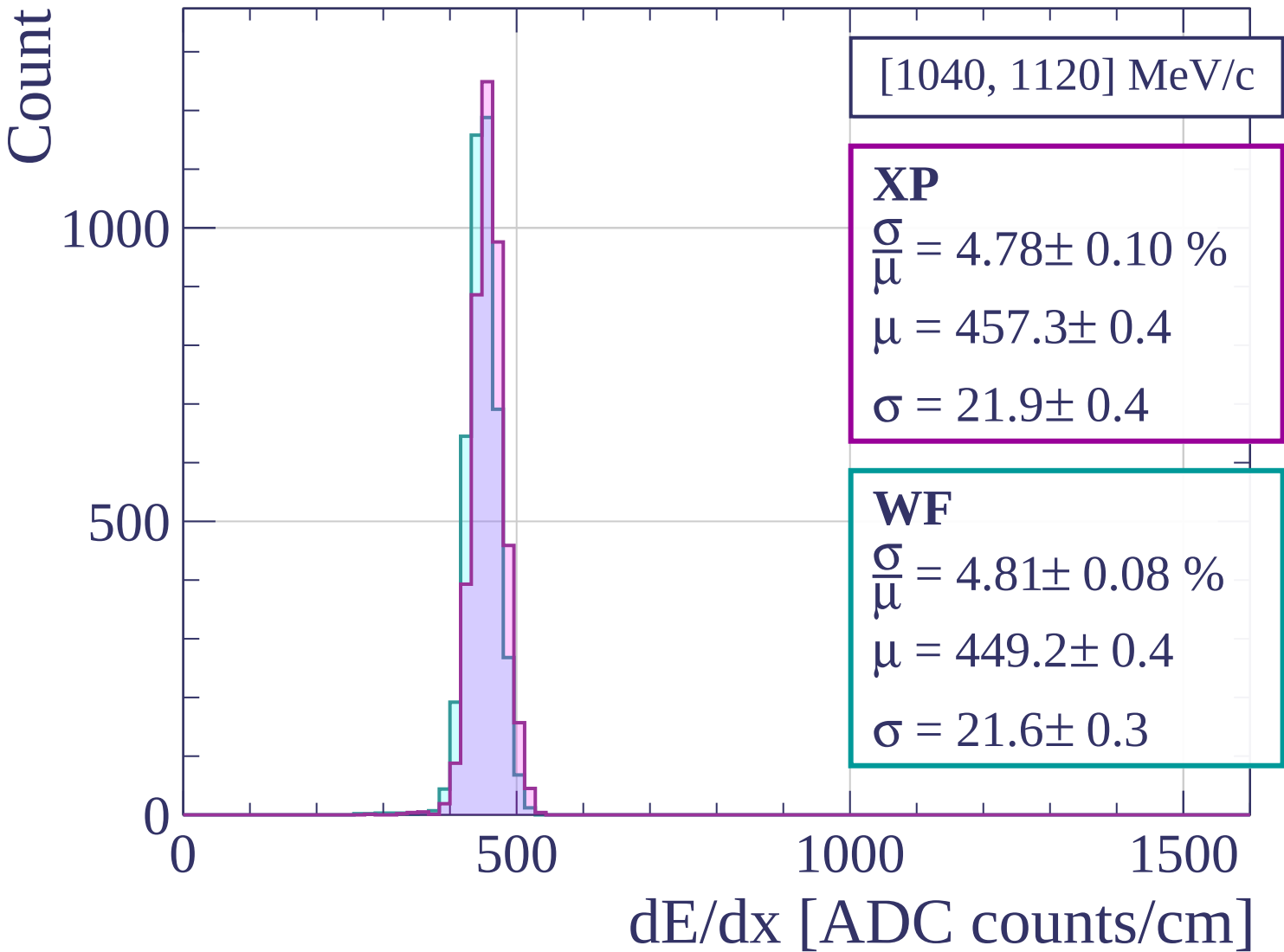


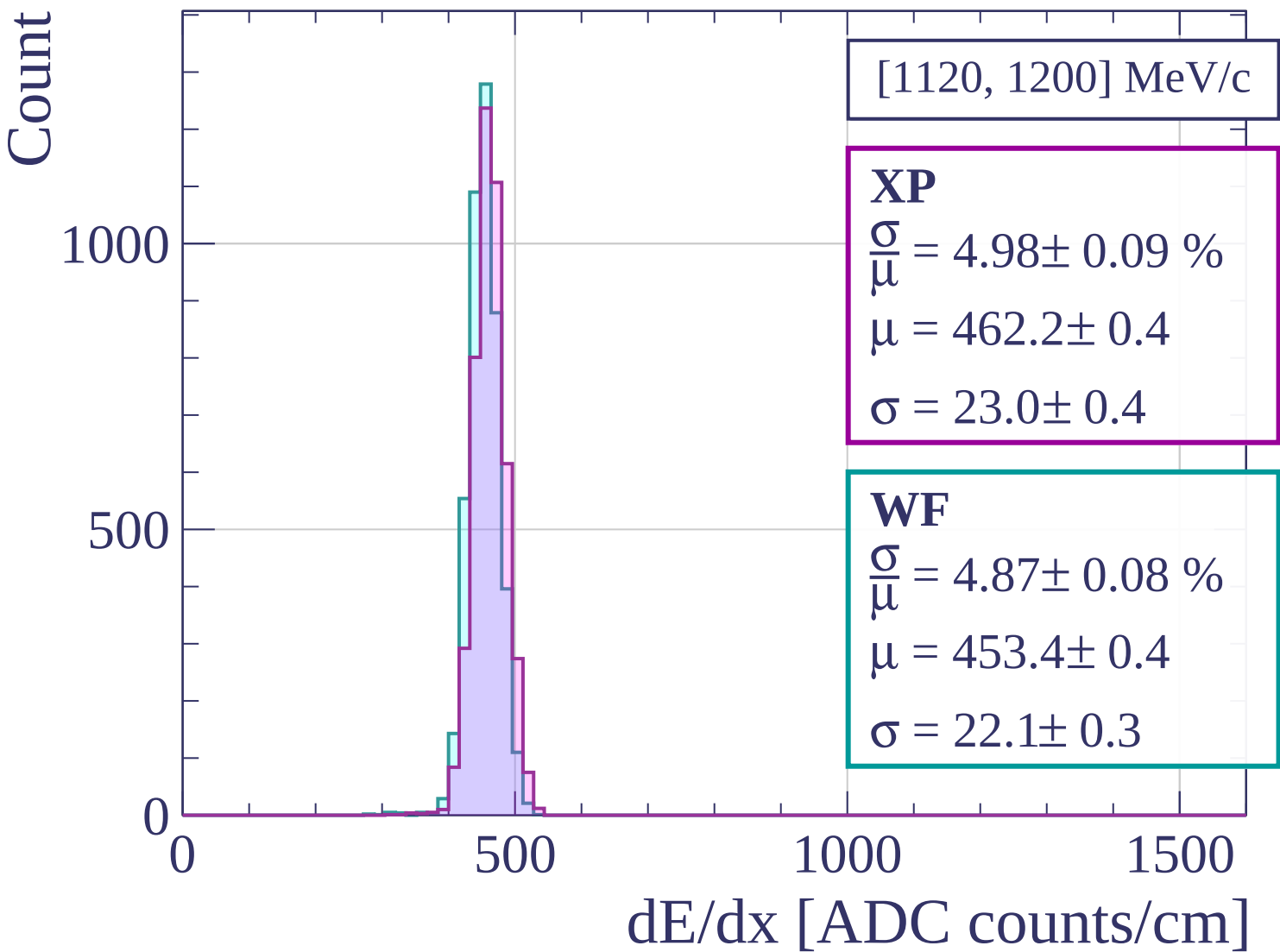


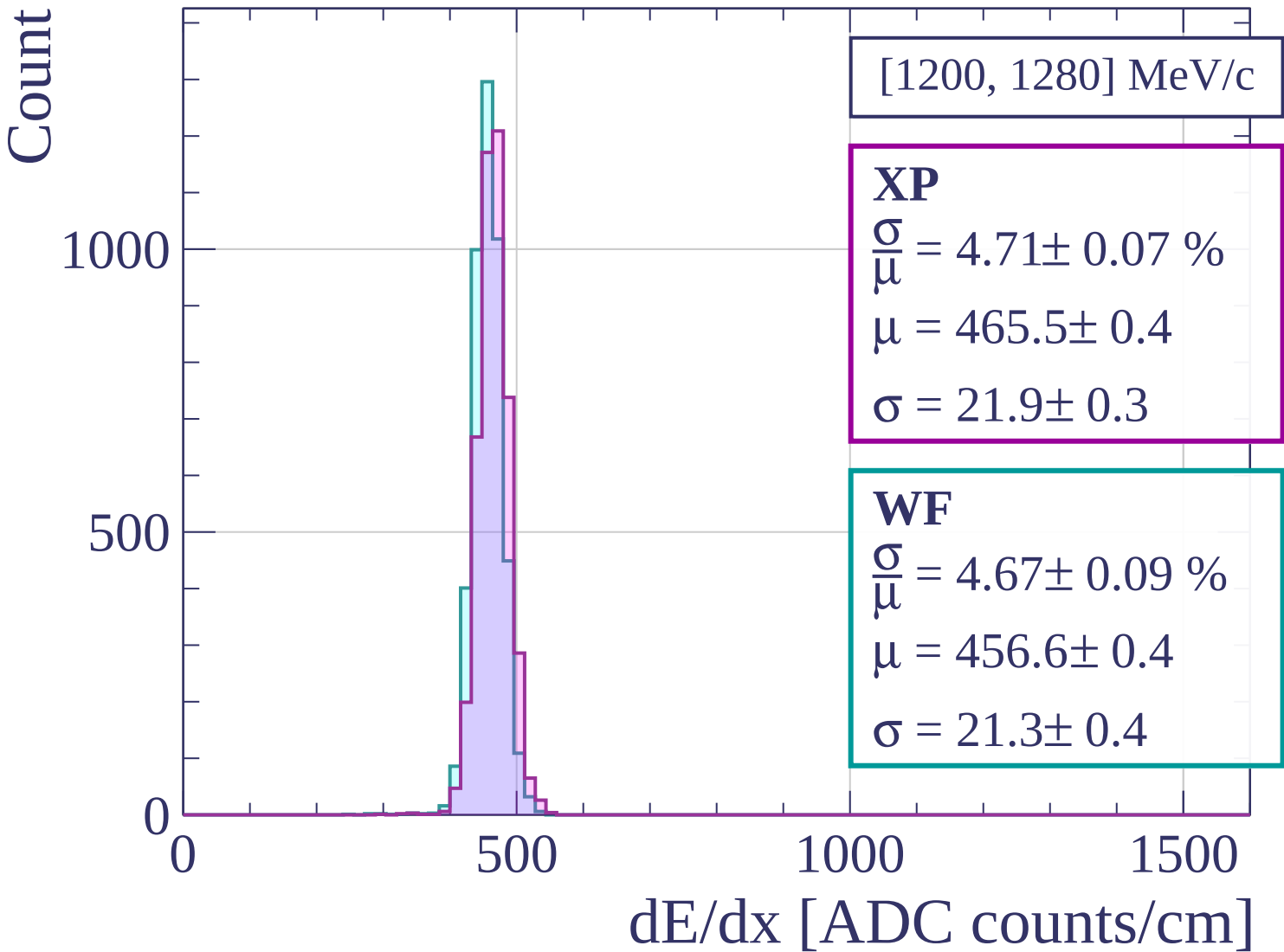


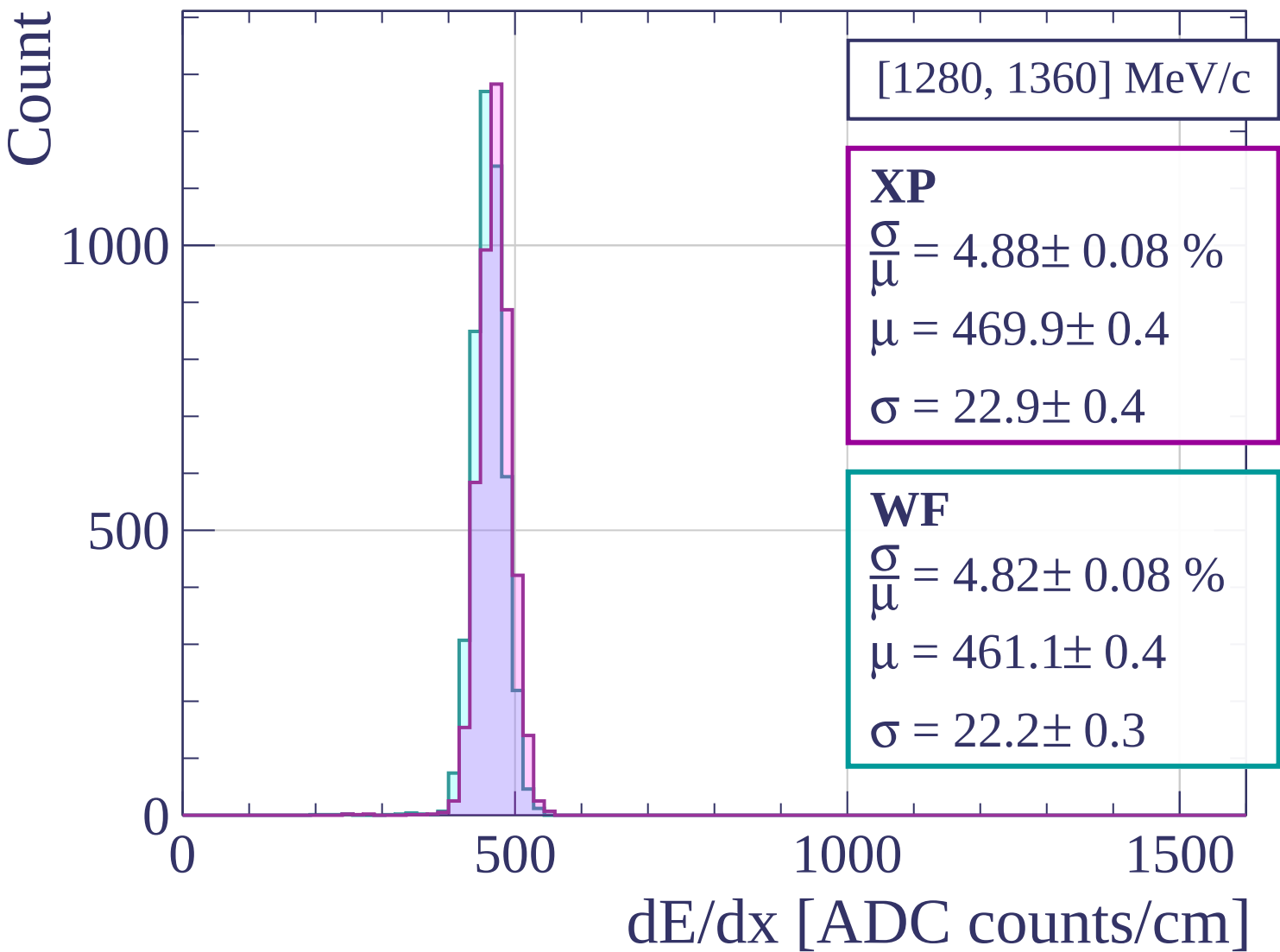




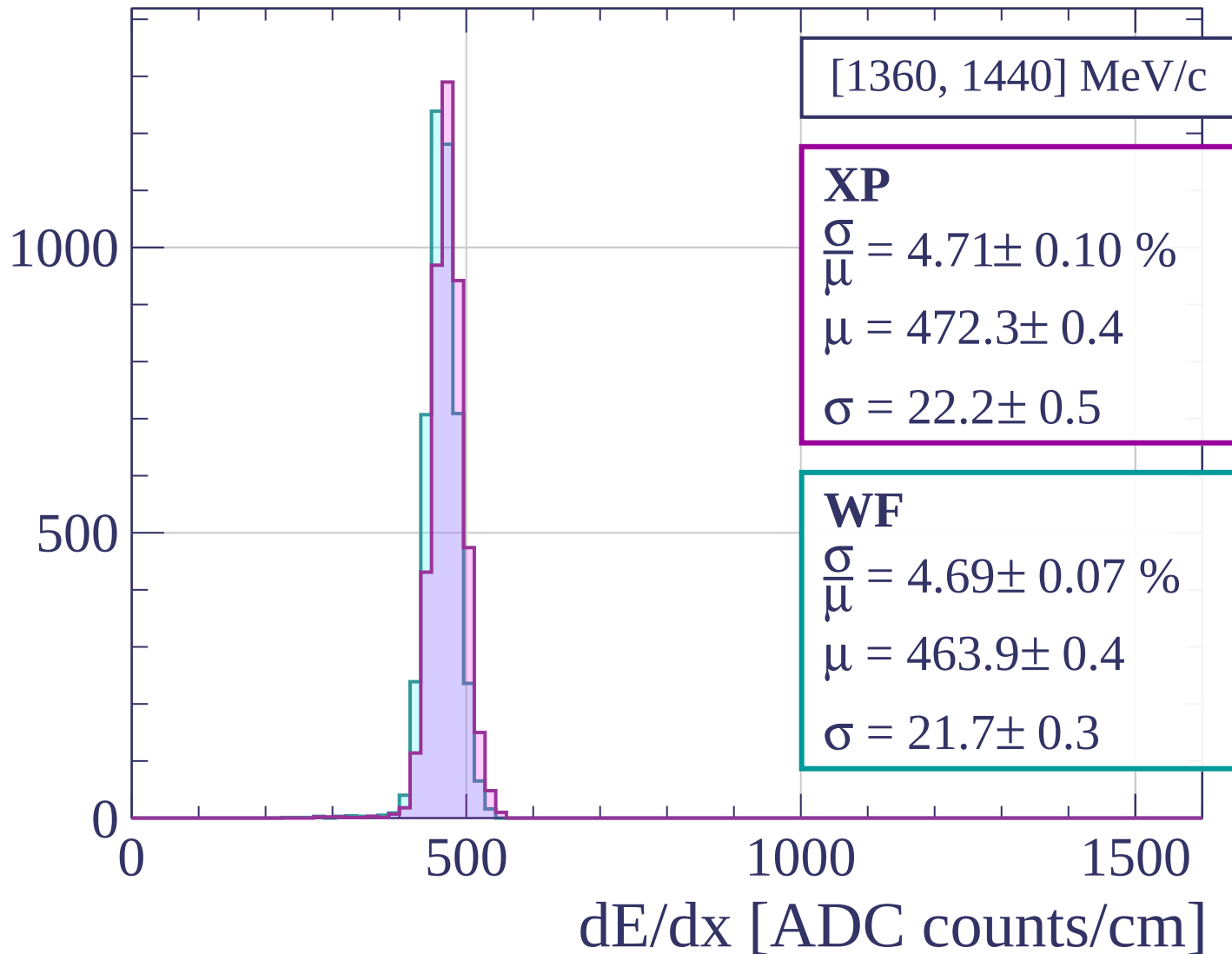


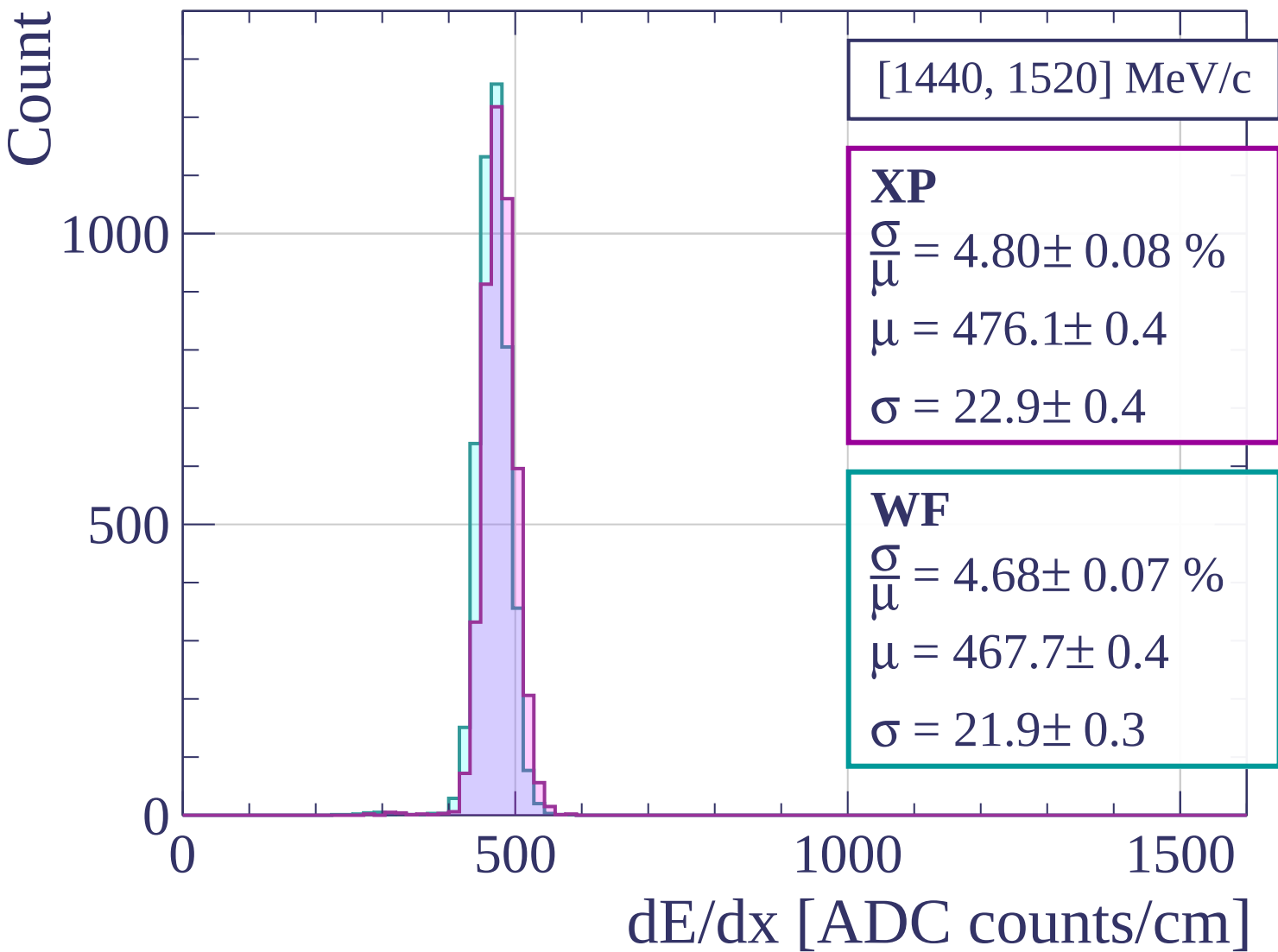


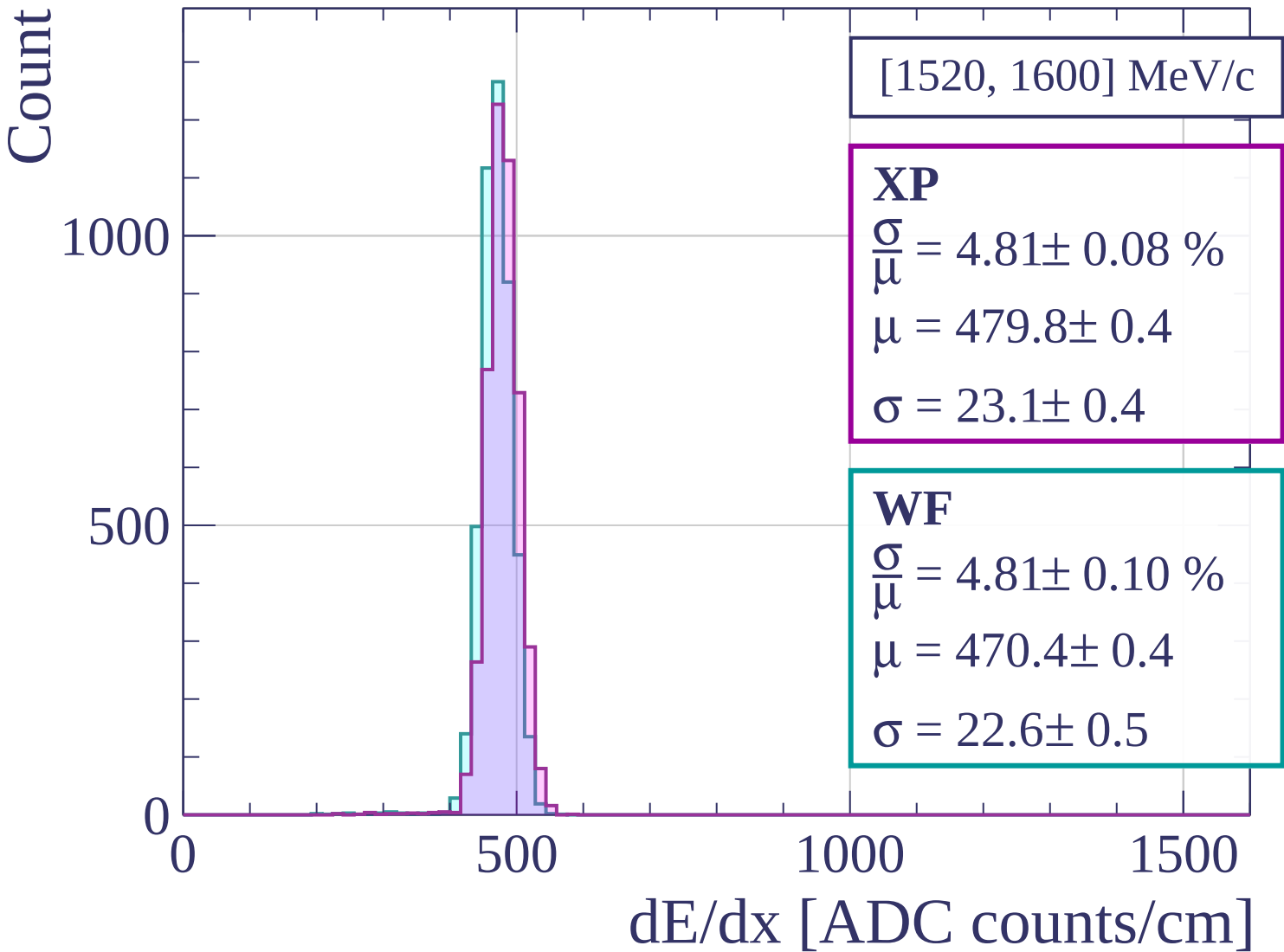


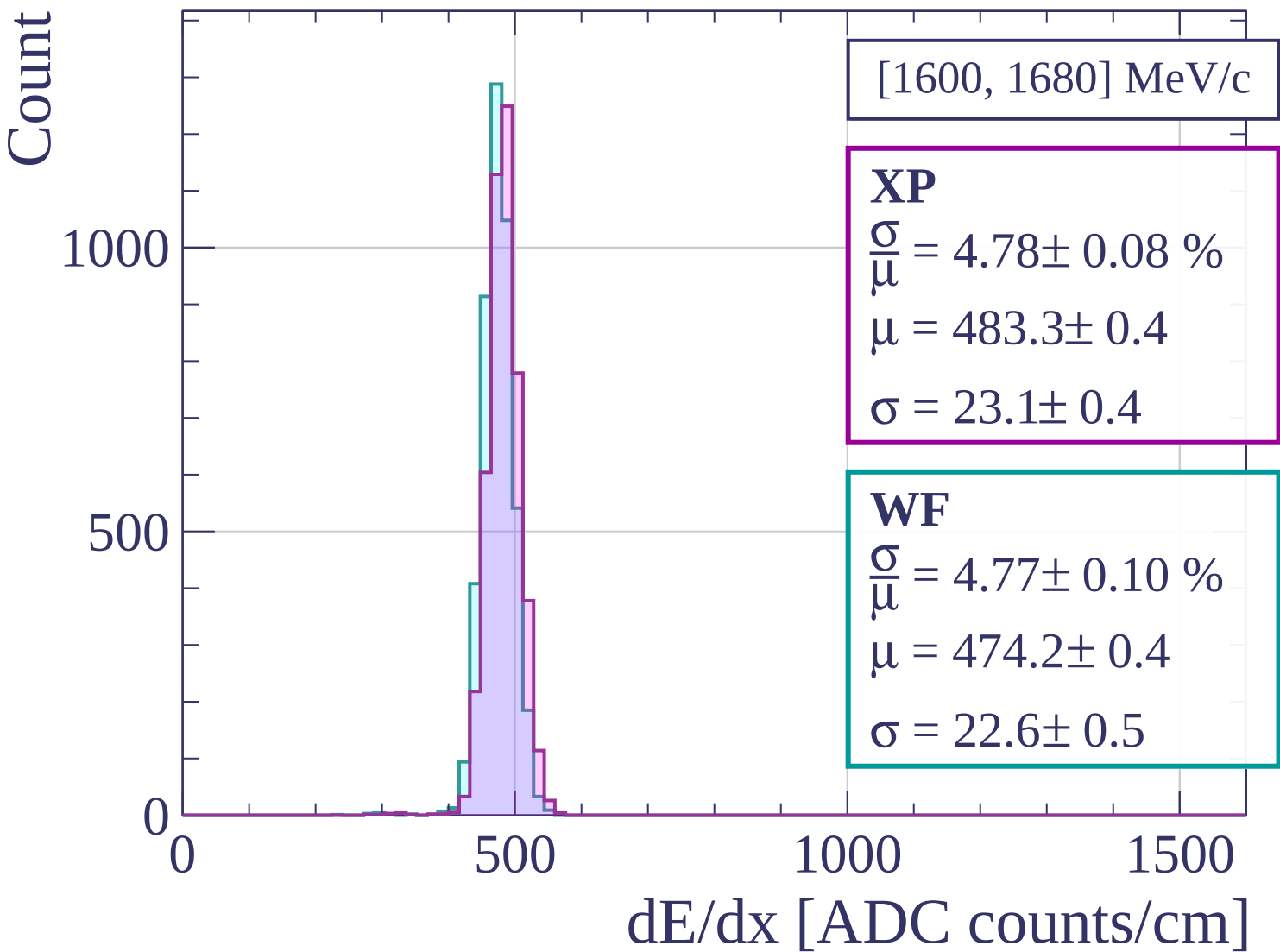


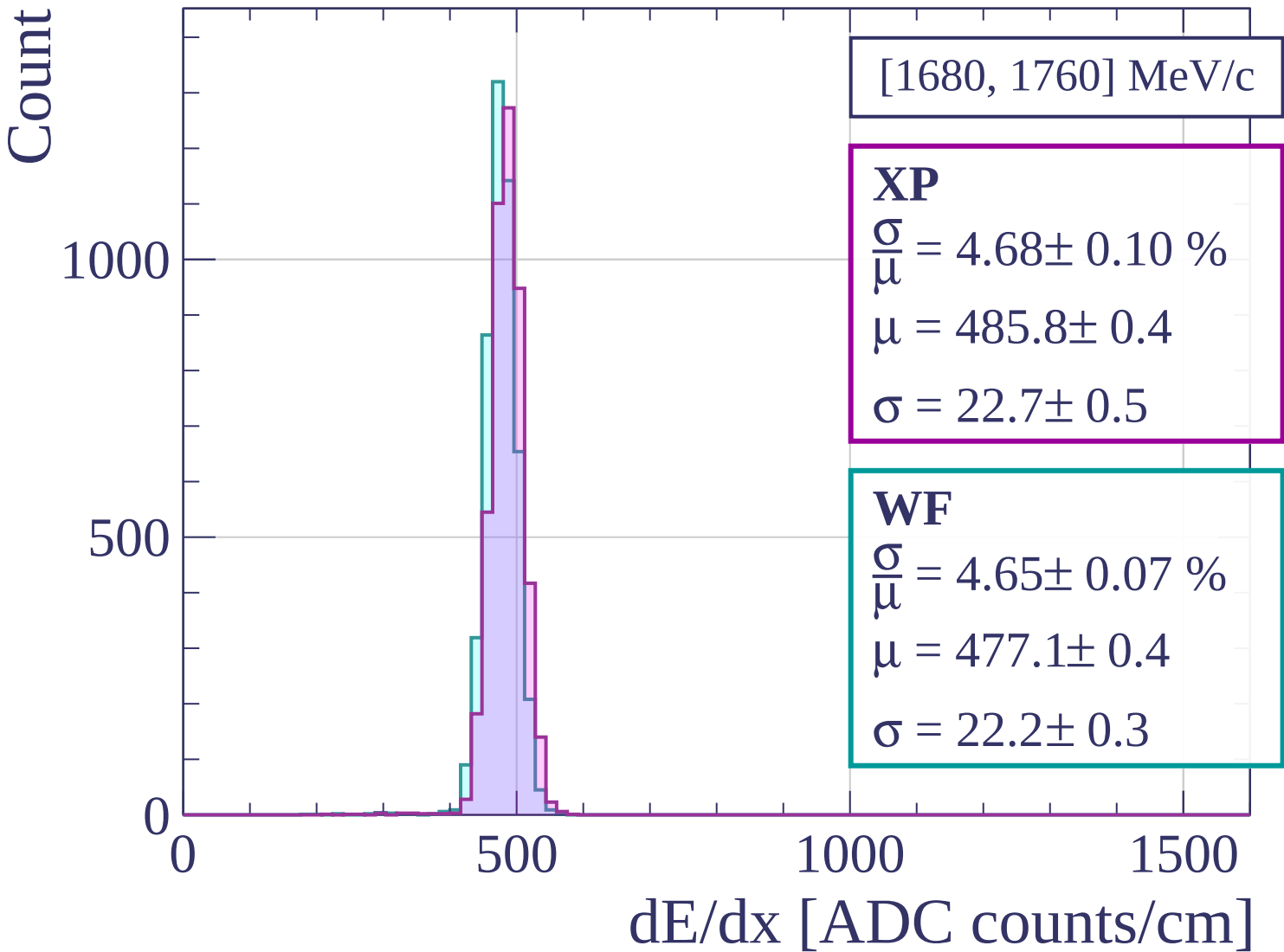
Count

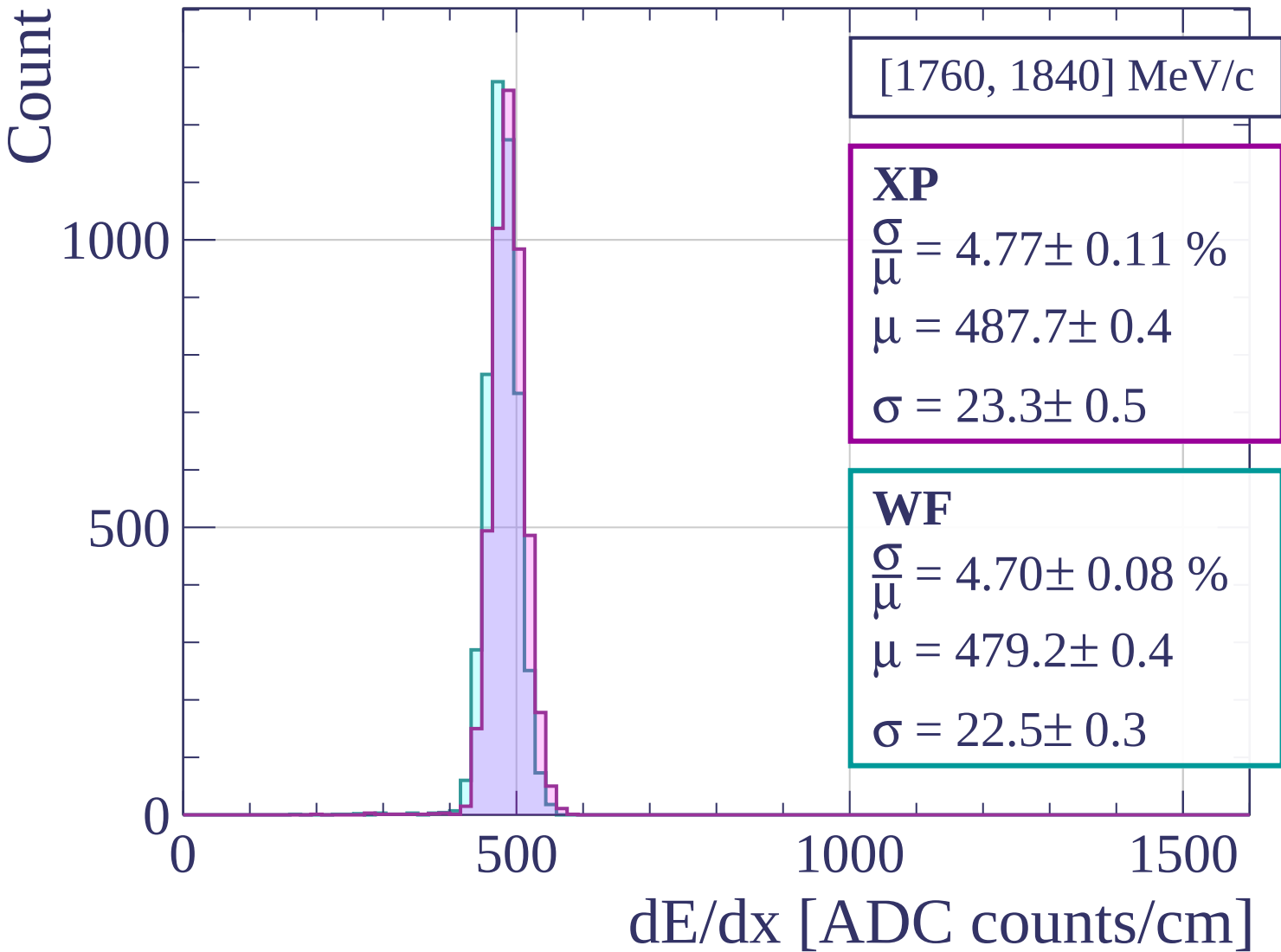




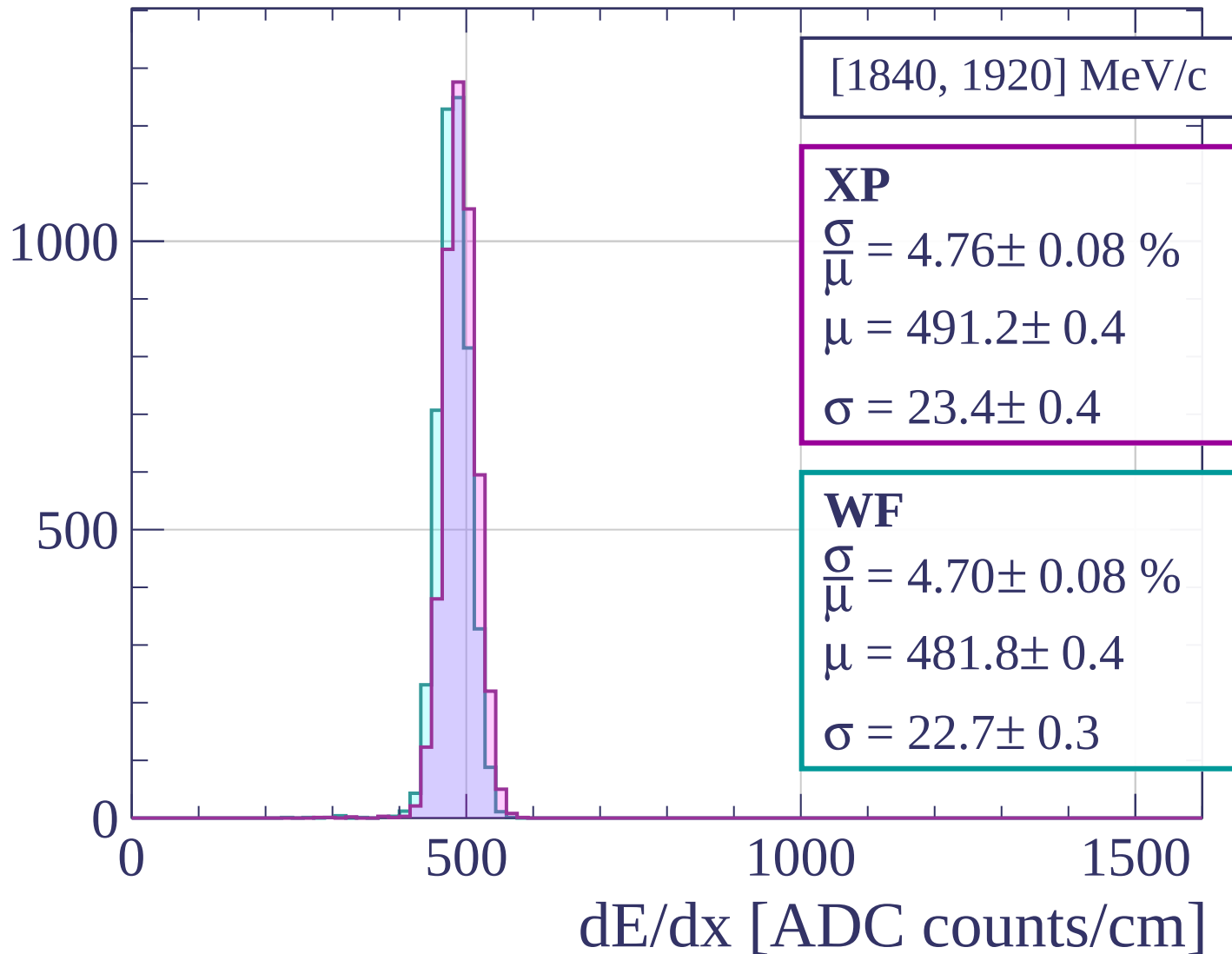


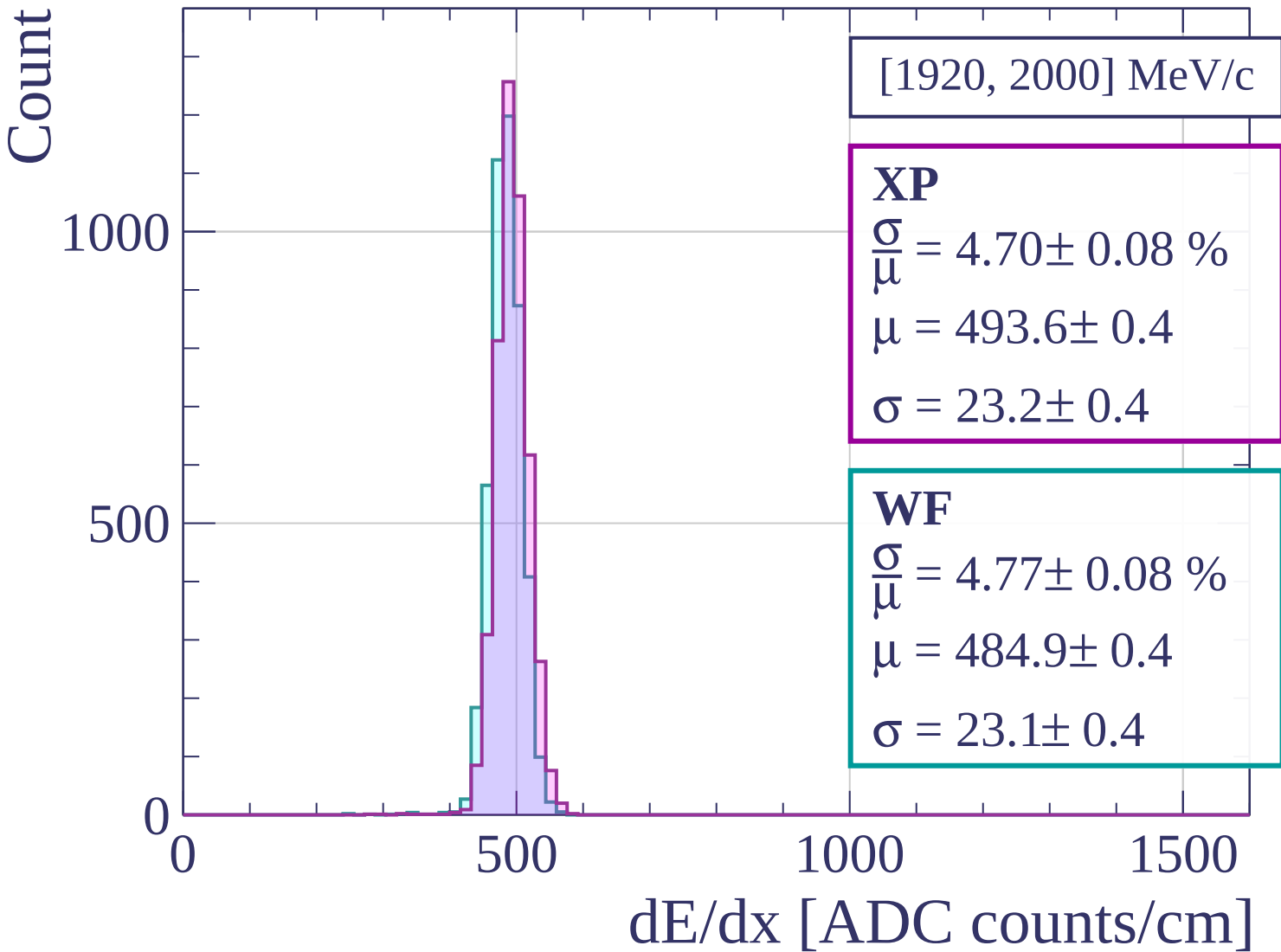


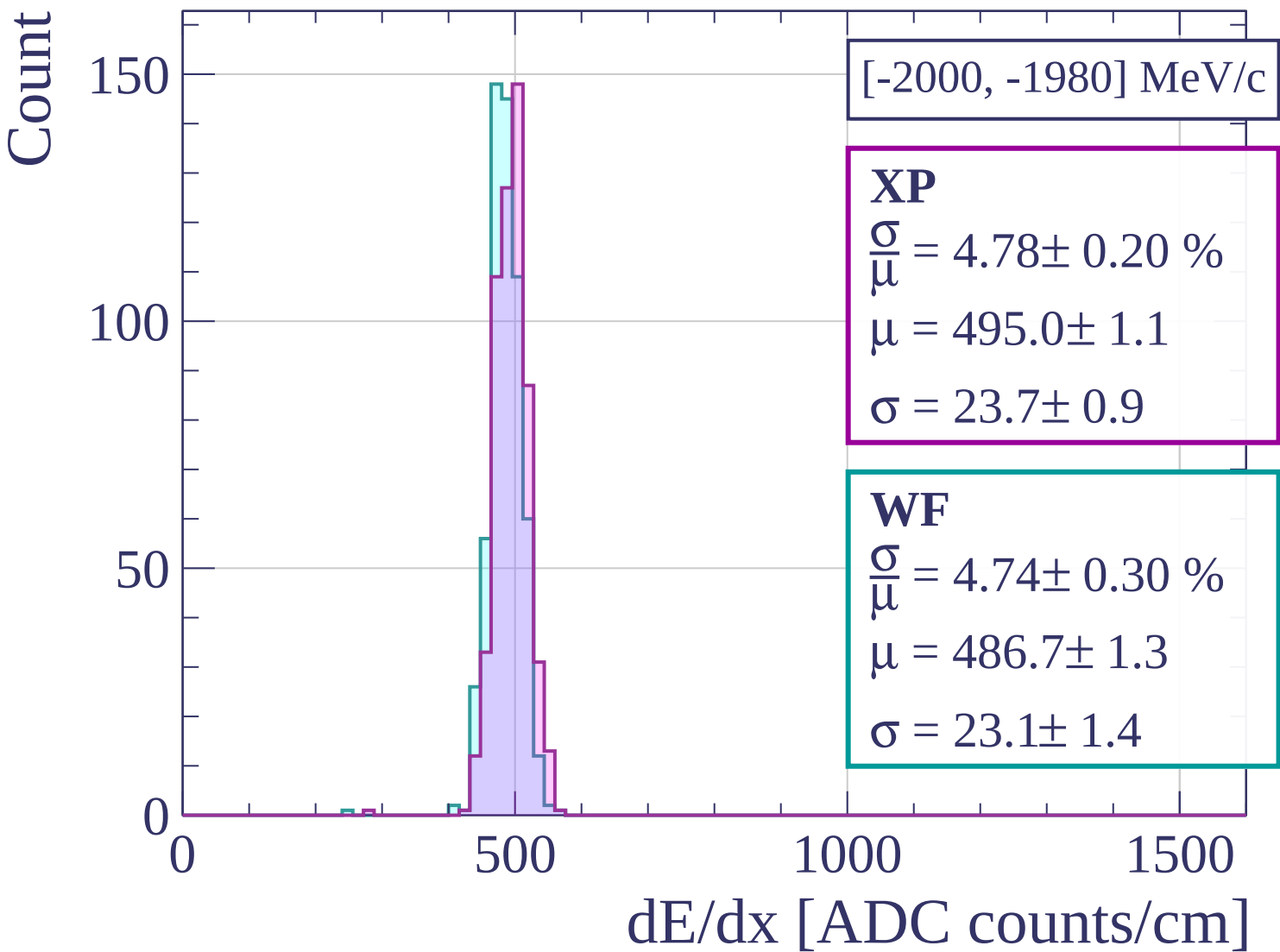


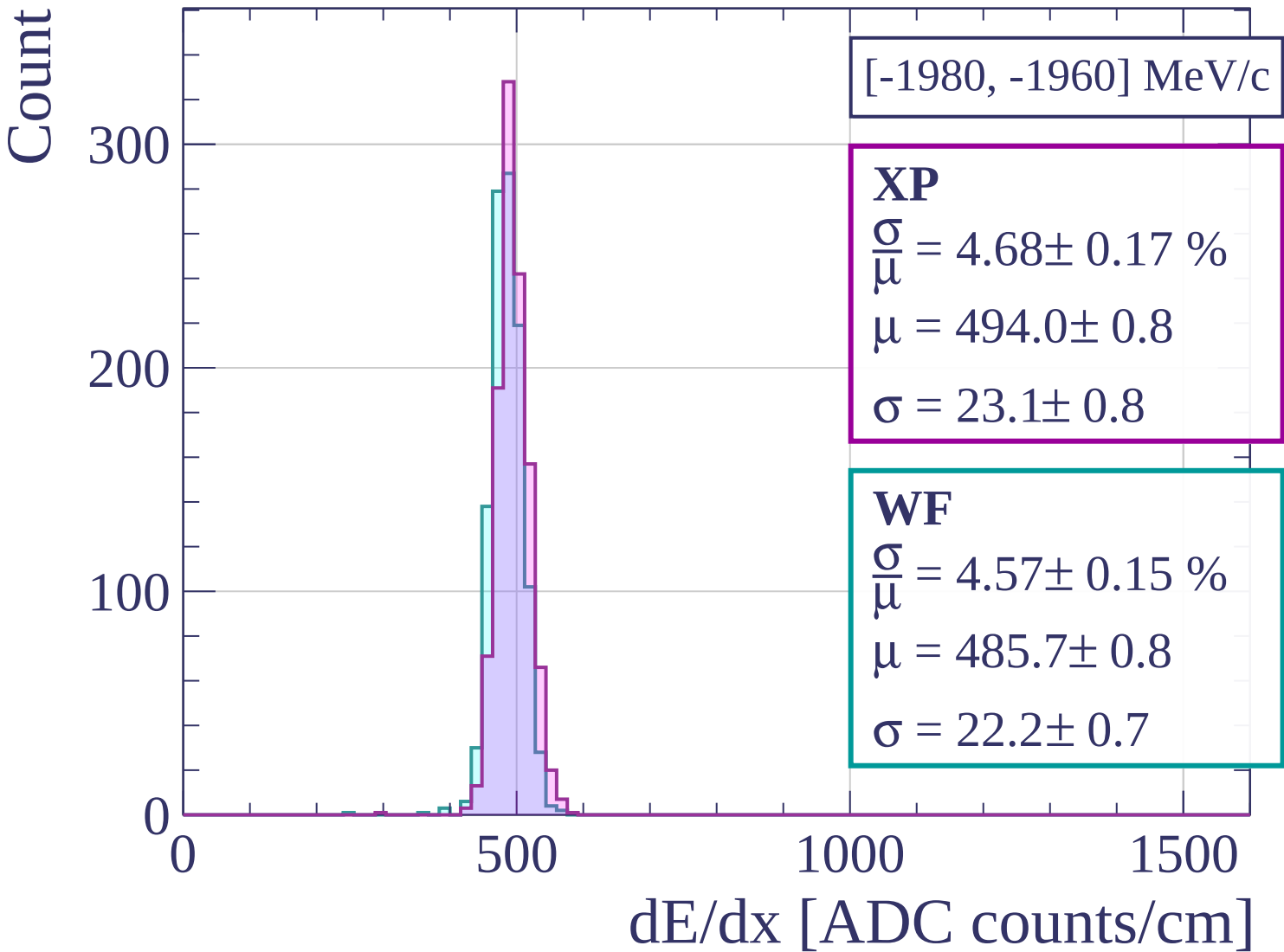


Count

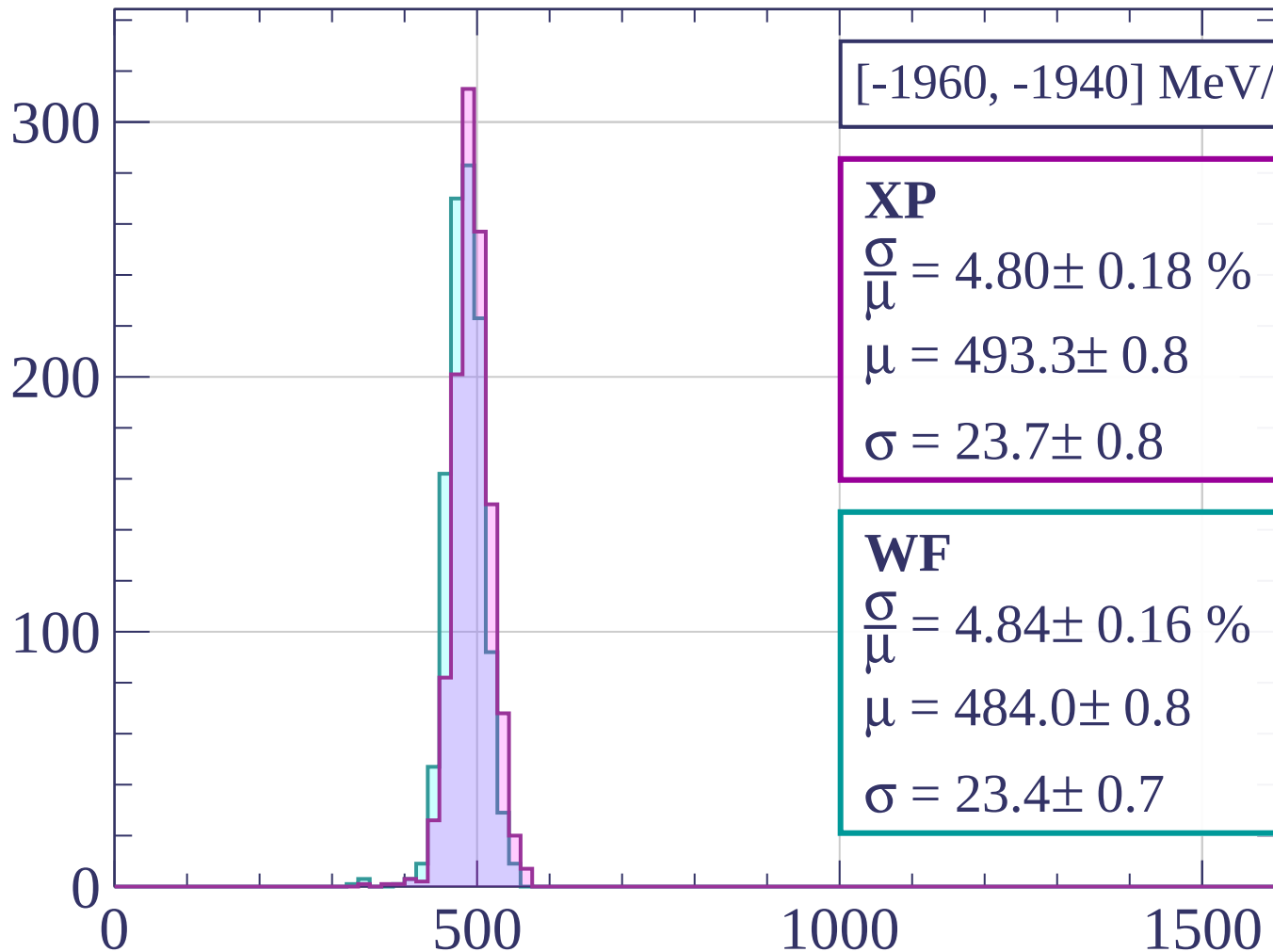




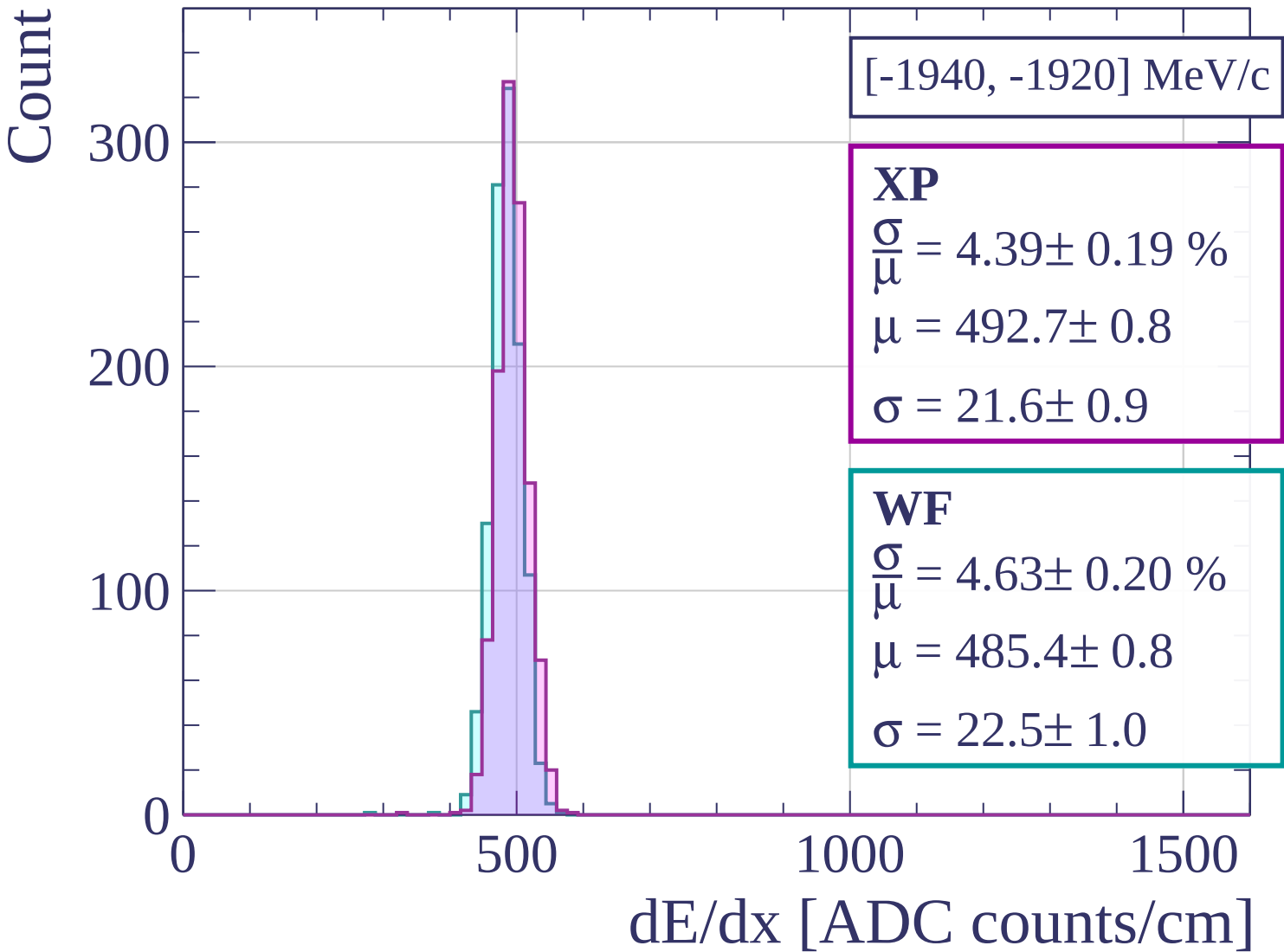


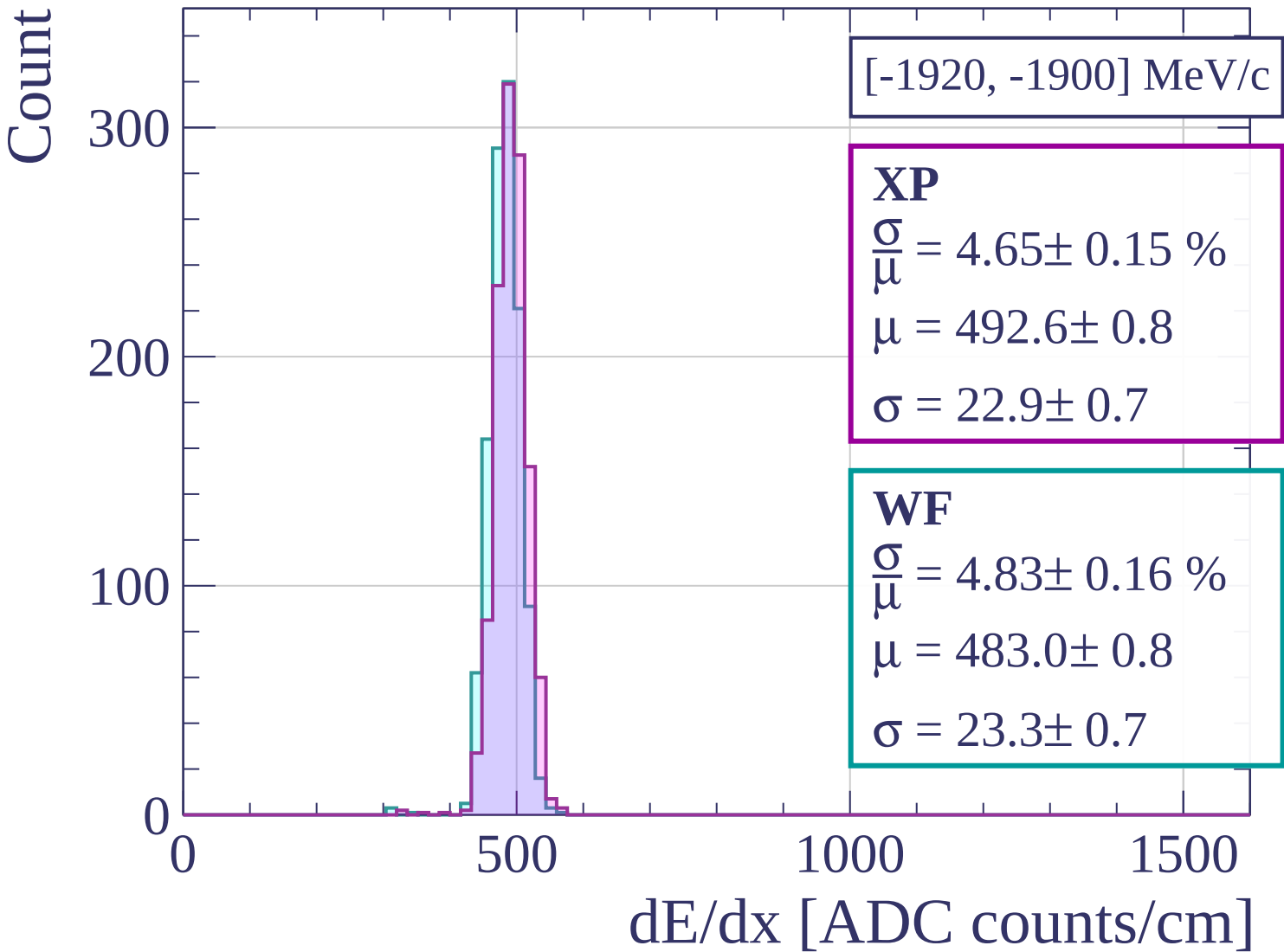


Count



dE/dx [ADC counts/cm]





Count

[-1900, -1880] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.61 \pm 0.15 \%$$

$$\mu = 492.6 \pm 0.8$$

$$\sigma = 22.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.61 \pm 0.14 \%$$

$$\mu = 483.5 \pm 0.7$$

$$\sigma = 22.3 \pm 0.7$$

300

200

100

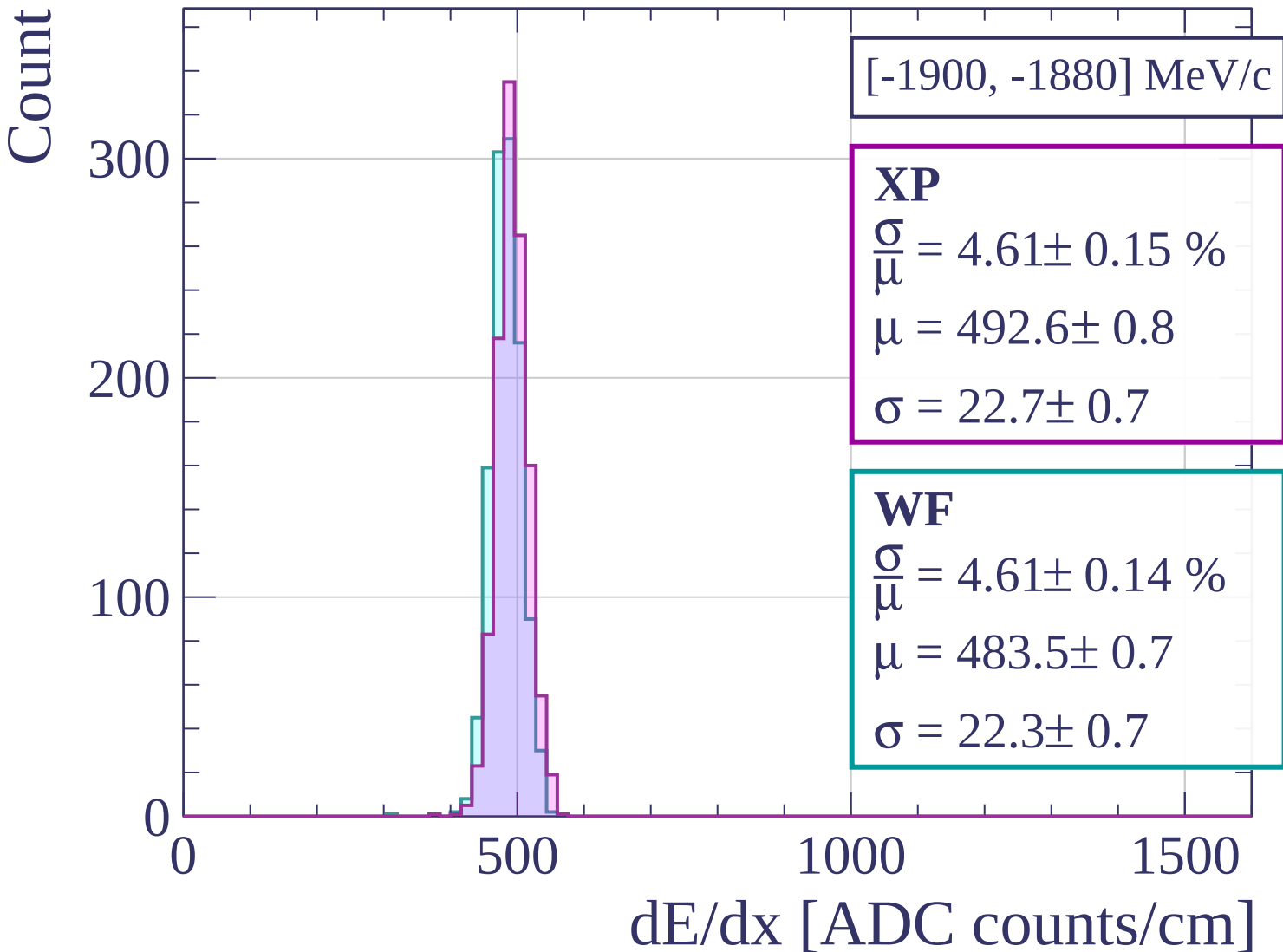
0

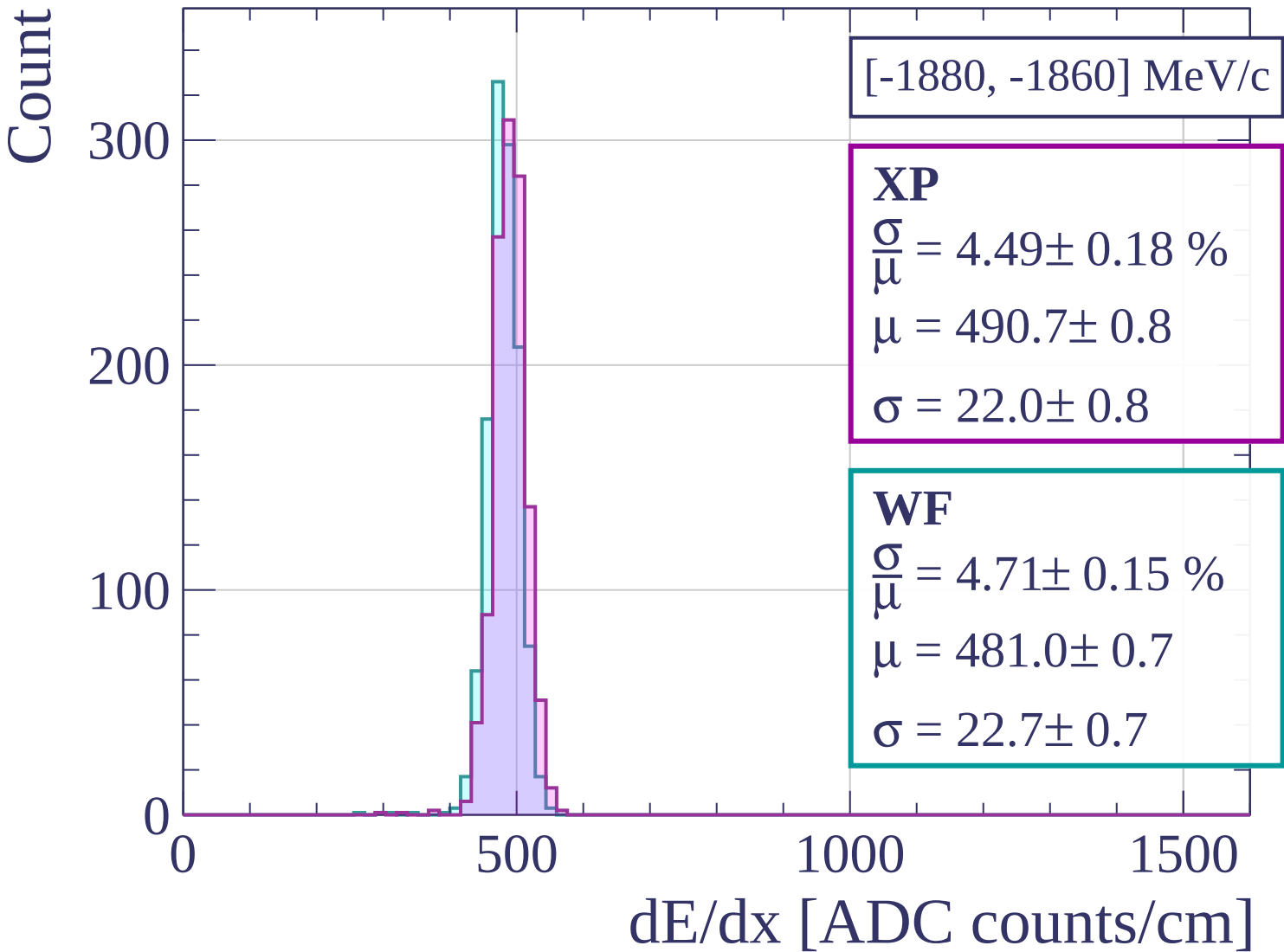
500

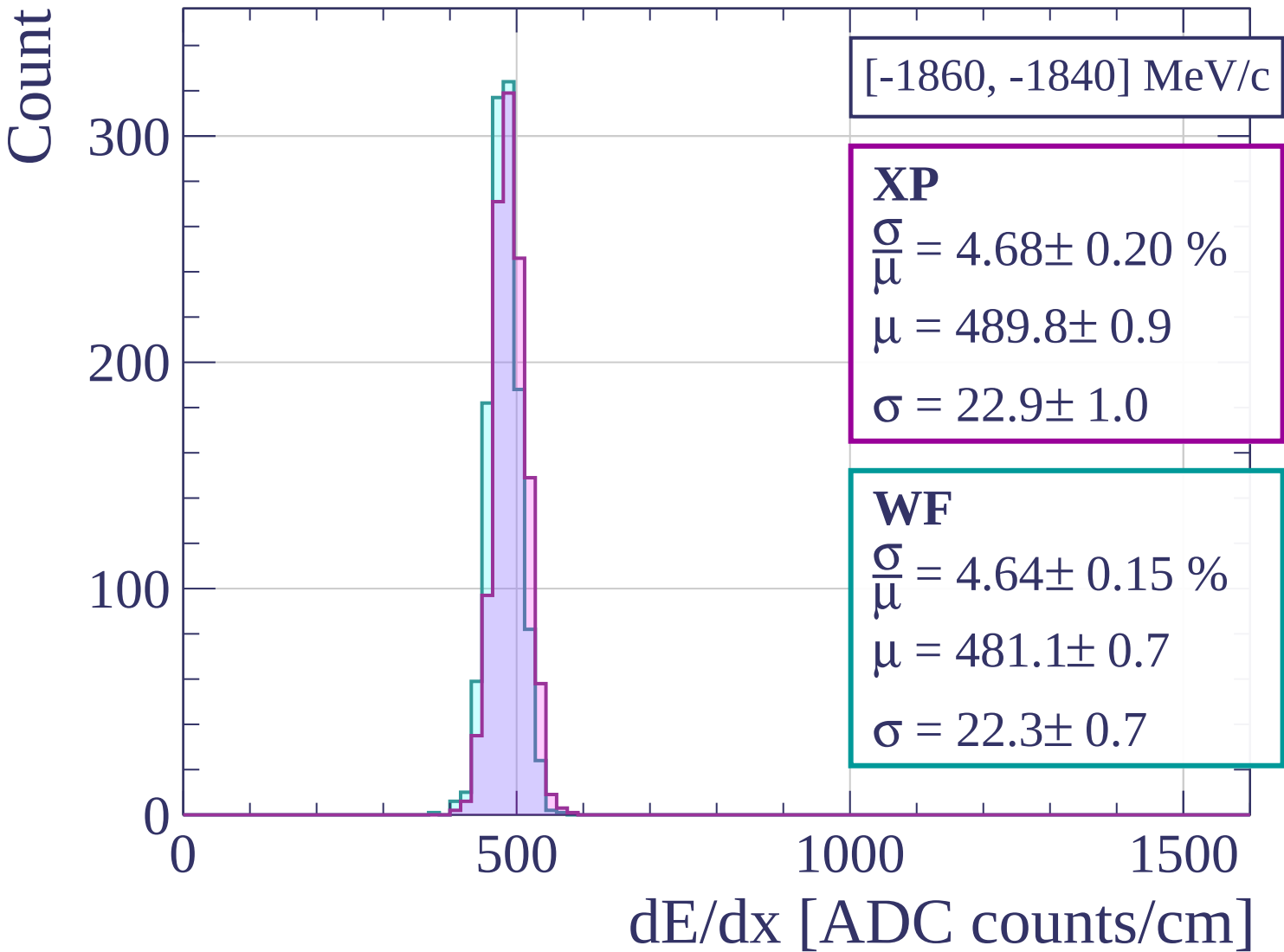
1000

1500

dE/dx [ADC counts/cm]







Count

[-1840, -1820] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.99 \pm 0.19 \%$$

$$\mu = 489.3 \pm 0.9$$

$$\sigma = 24.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.84 \pm 0.16 \%$$

$$\mu = 480.5 \pm 0.8$$

$$\sigma = 23.3 \pm 0.7$$

300

200

100

0

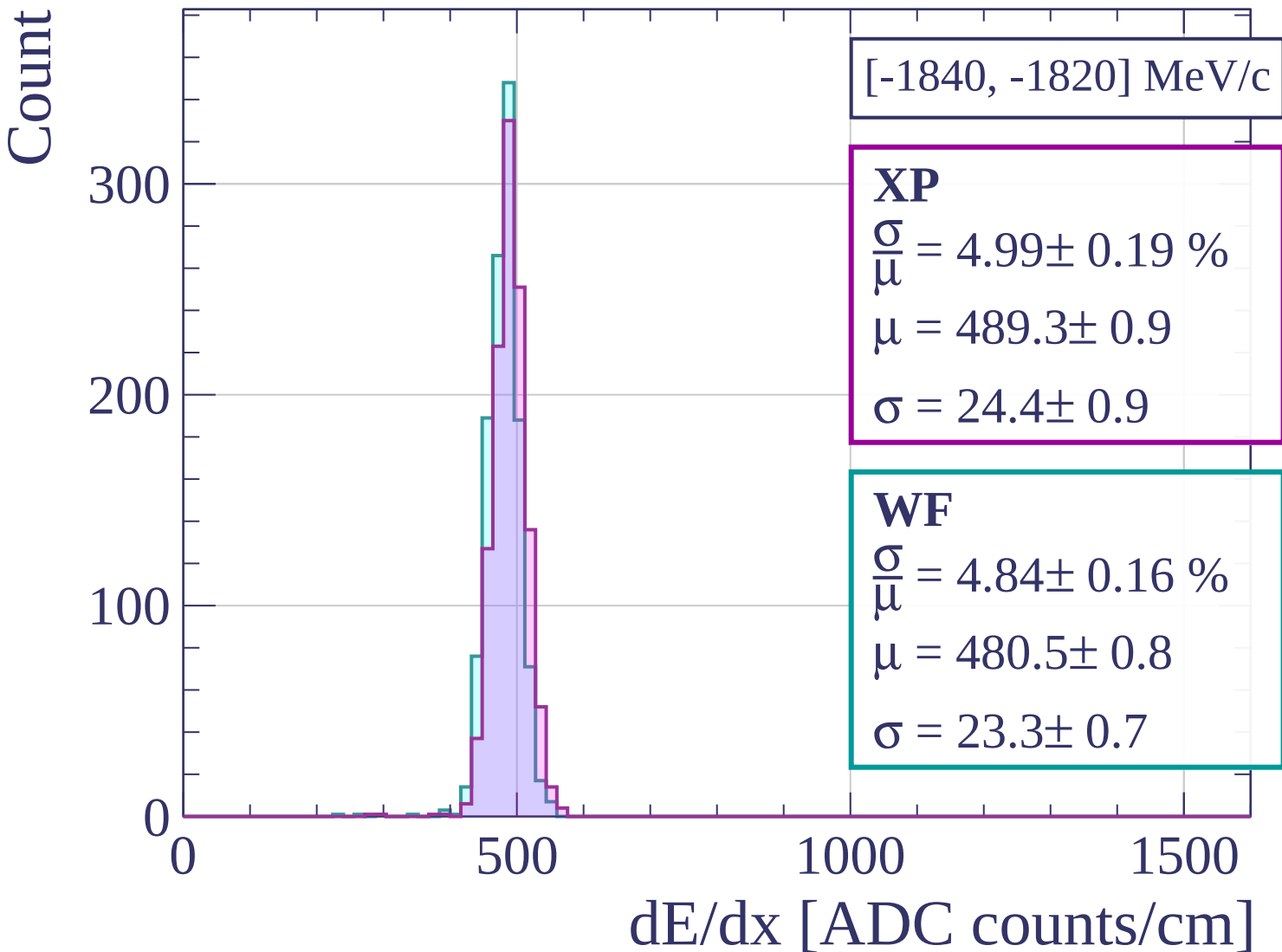
0

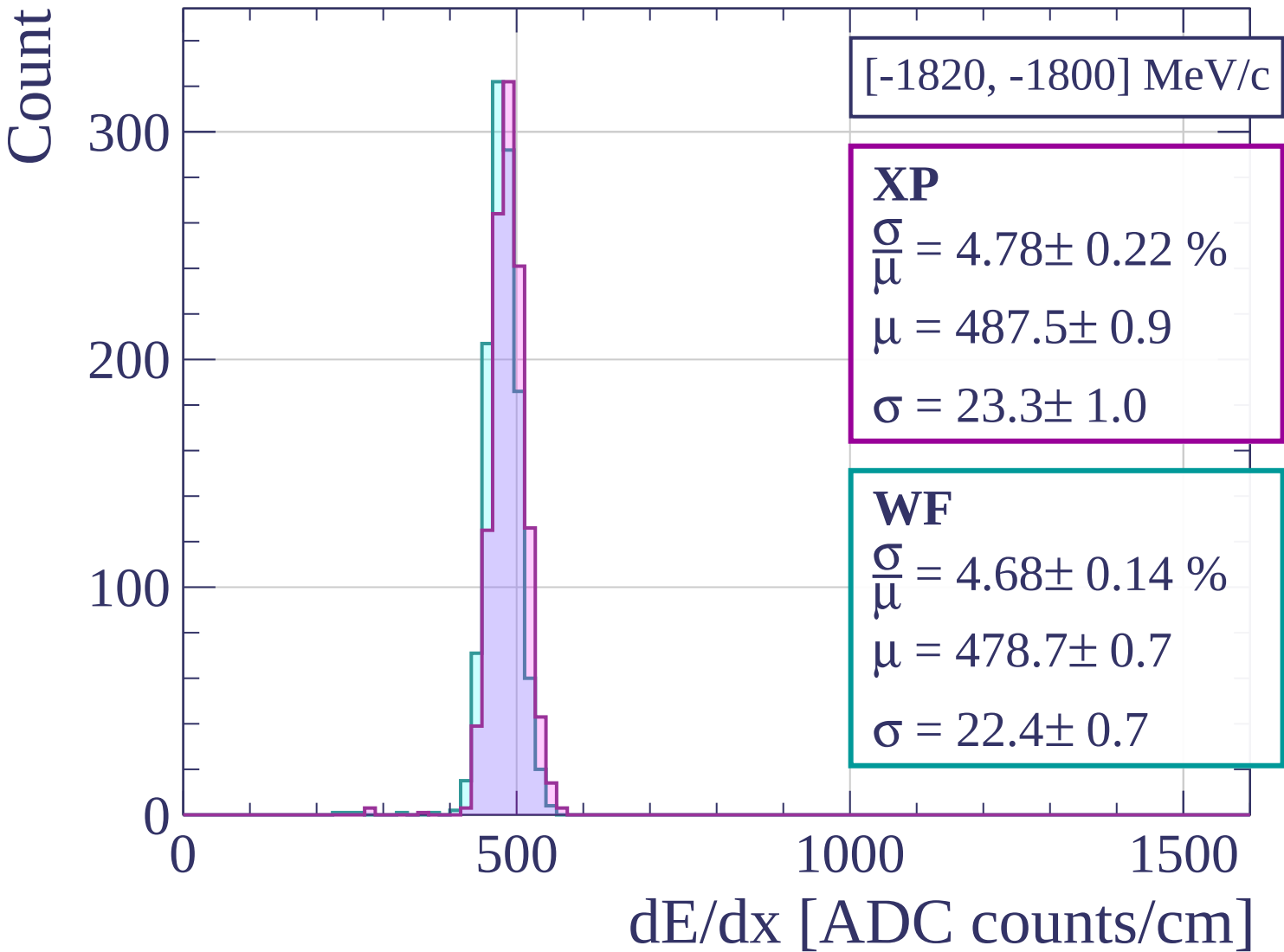
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-1800, -1780] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.65 \pm 0.20 \%$$

$$\mu = 487.7 \pm 0.8$$

$$\sigma = 22.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.63 \pm 0.15 \%$$

$$\mu = 479.2 \pm 0.7$$

$$\sigma = 22.2 \pm 0.7$$

300

200

100

0

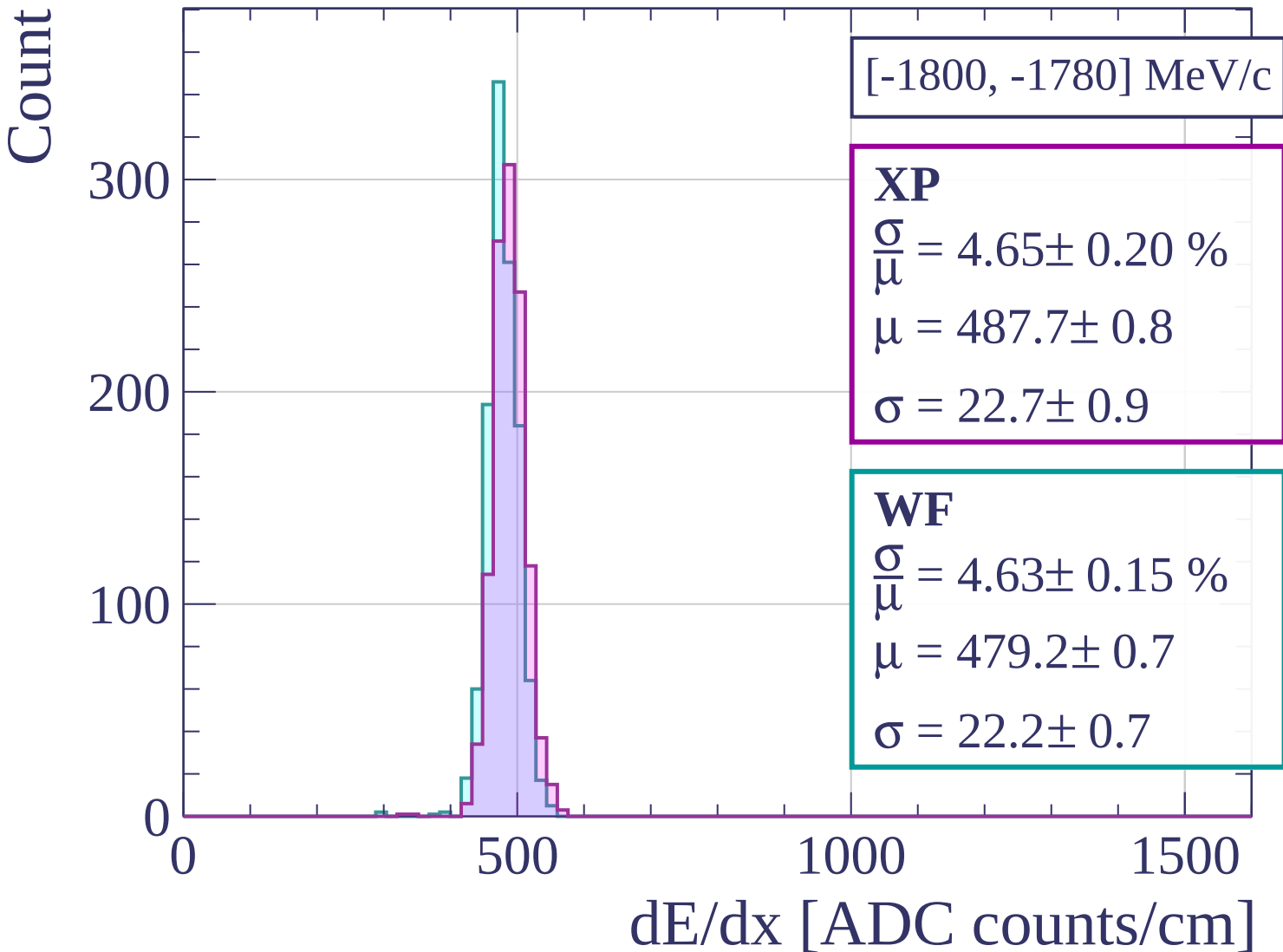
0

500

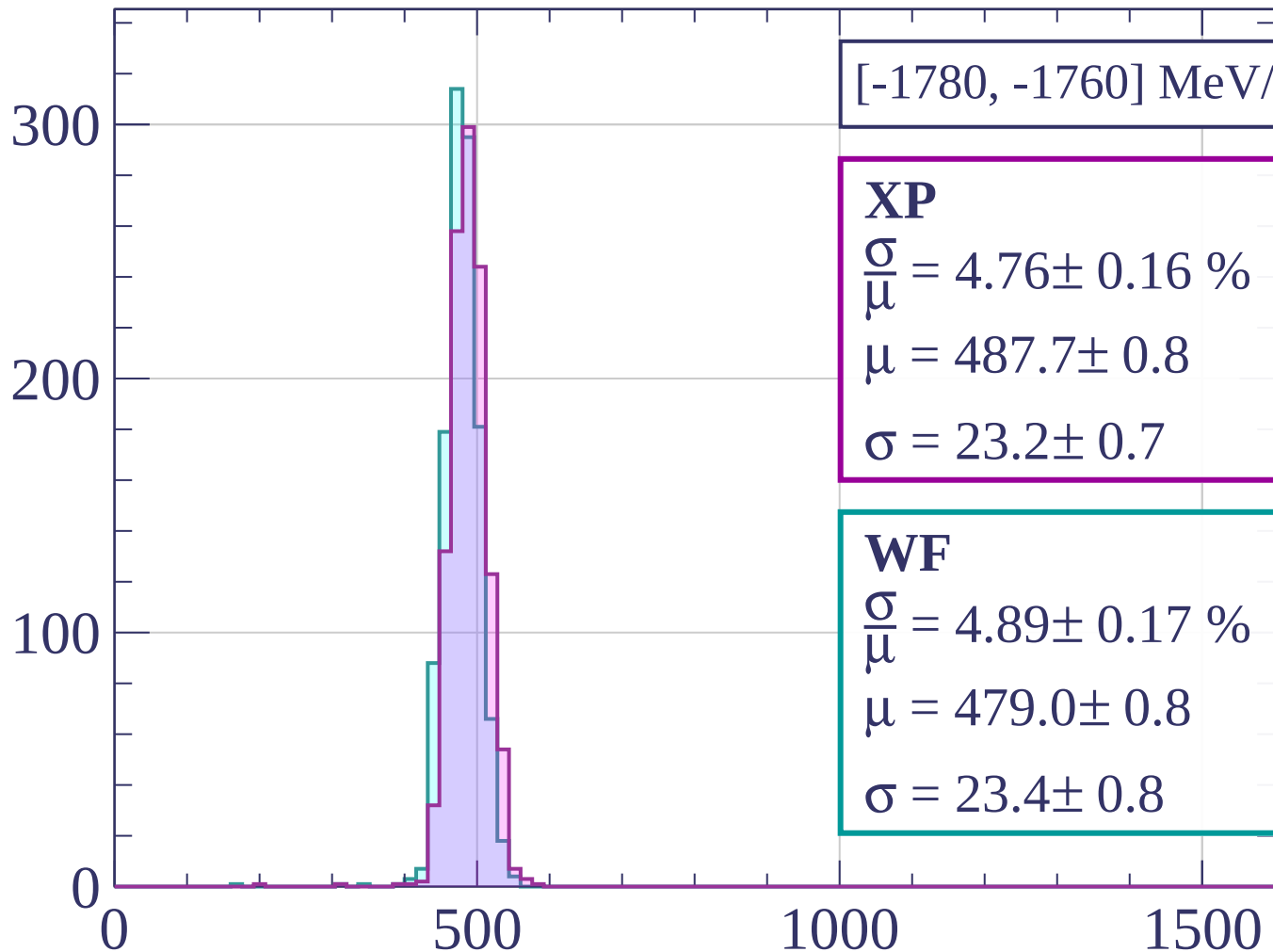
1000

1500

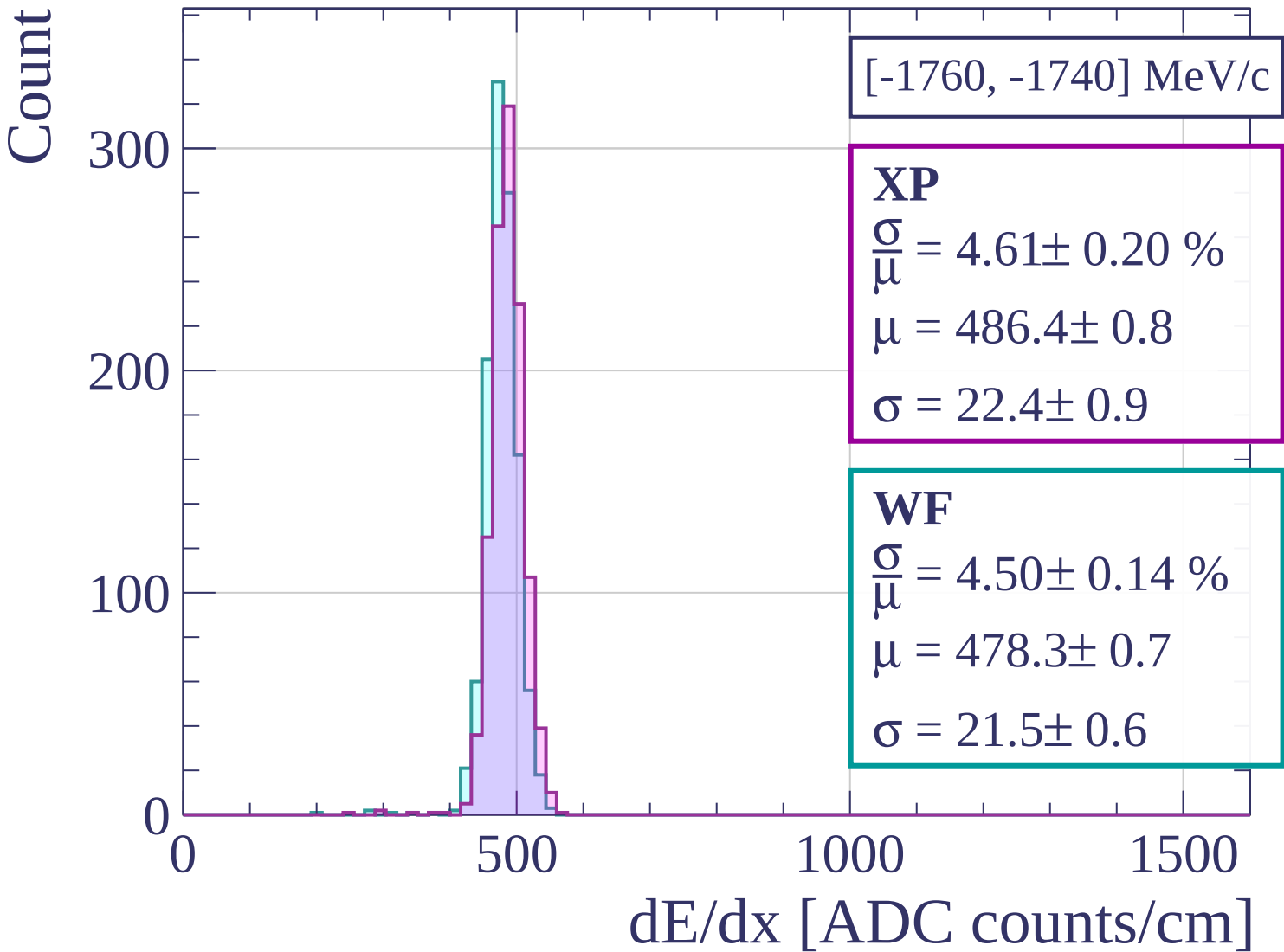
dE/dx [ADC counts/cm]



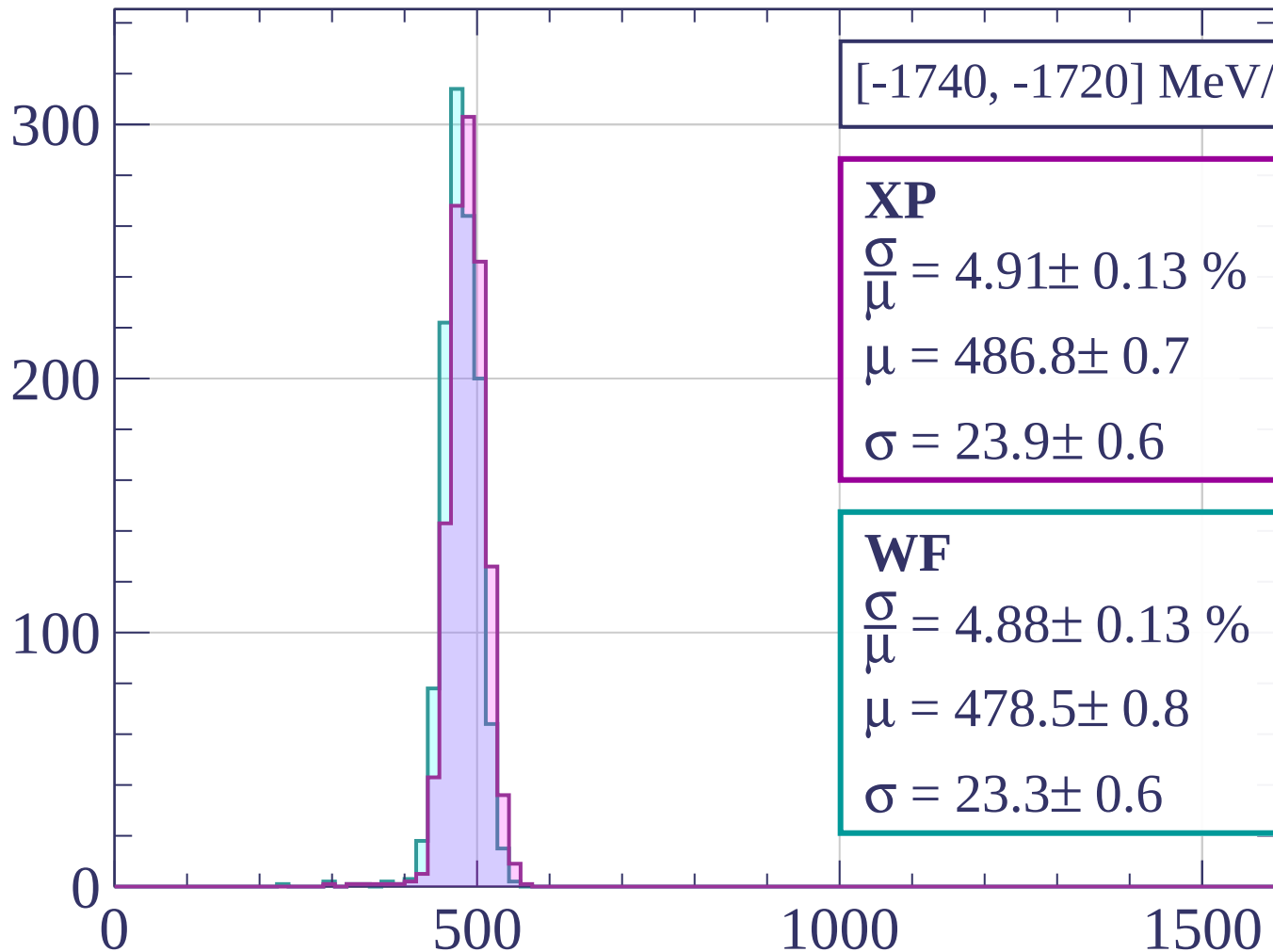
Count



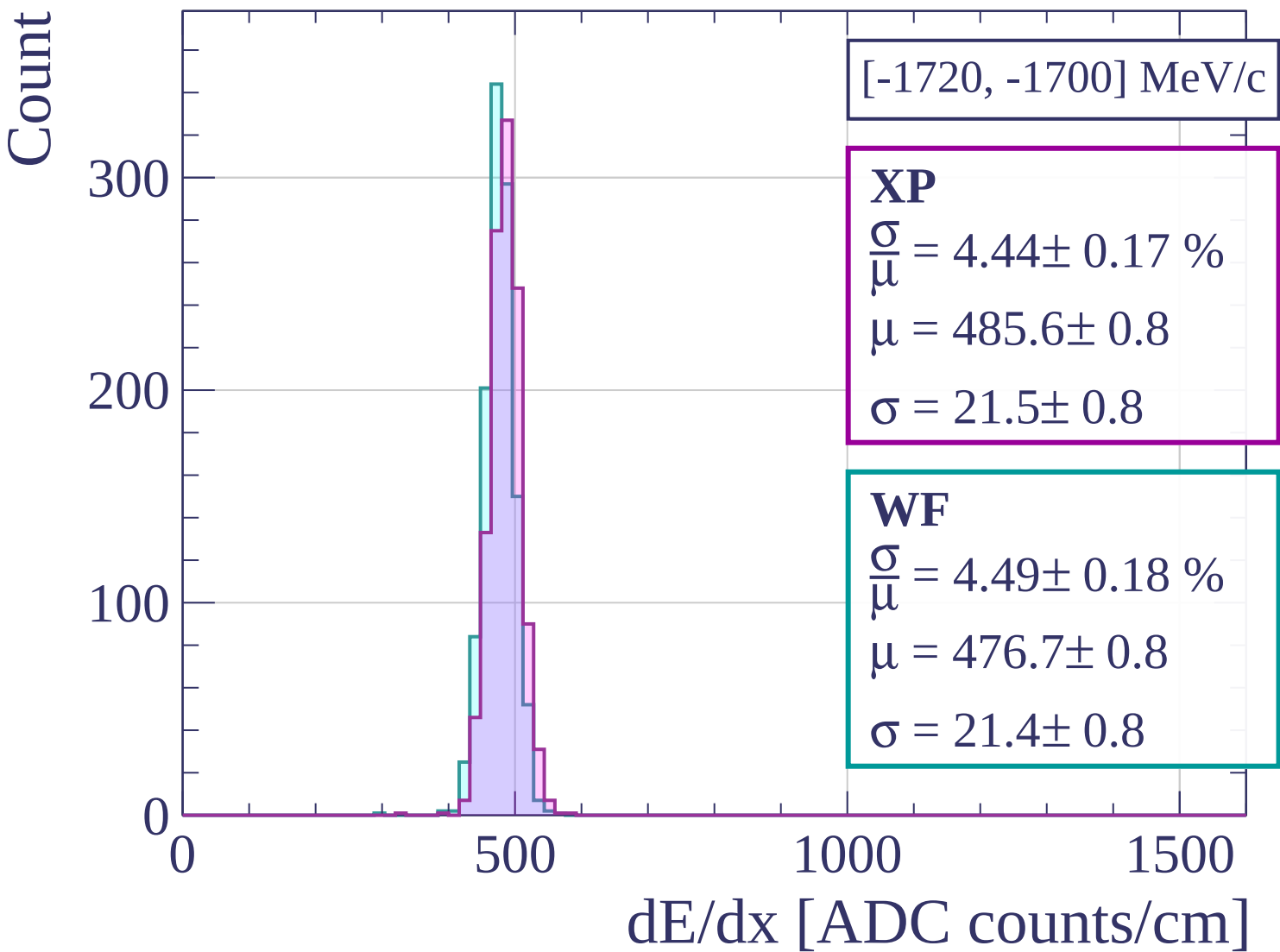
dE/dx [ADC counts/cm]

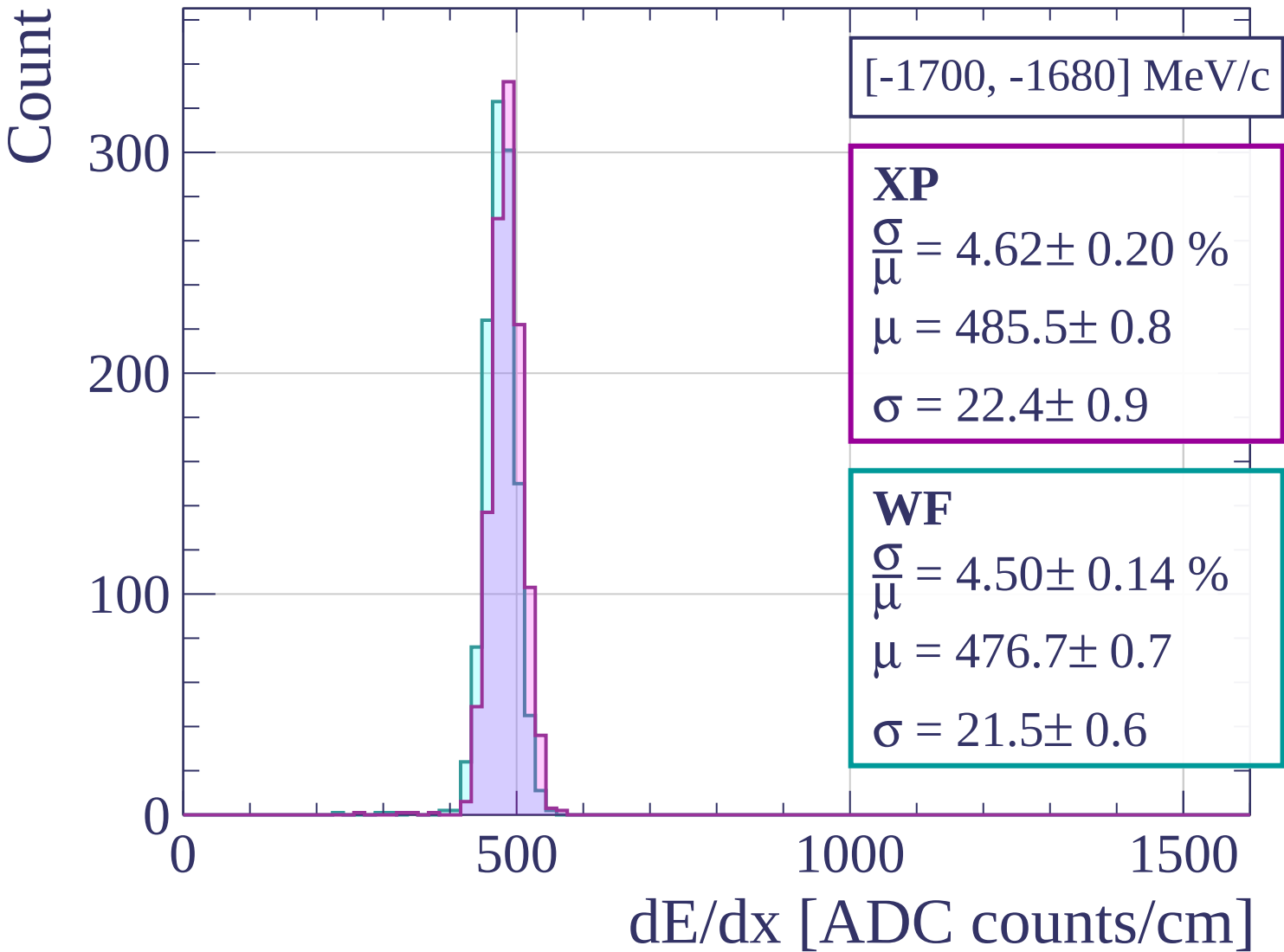


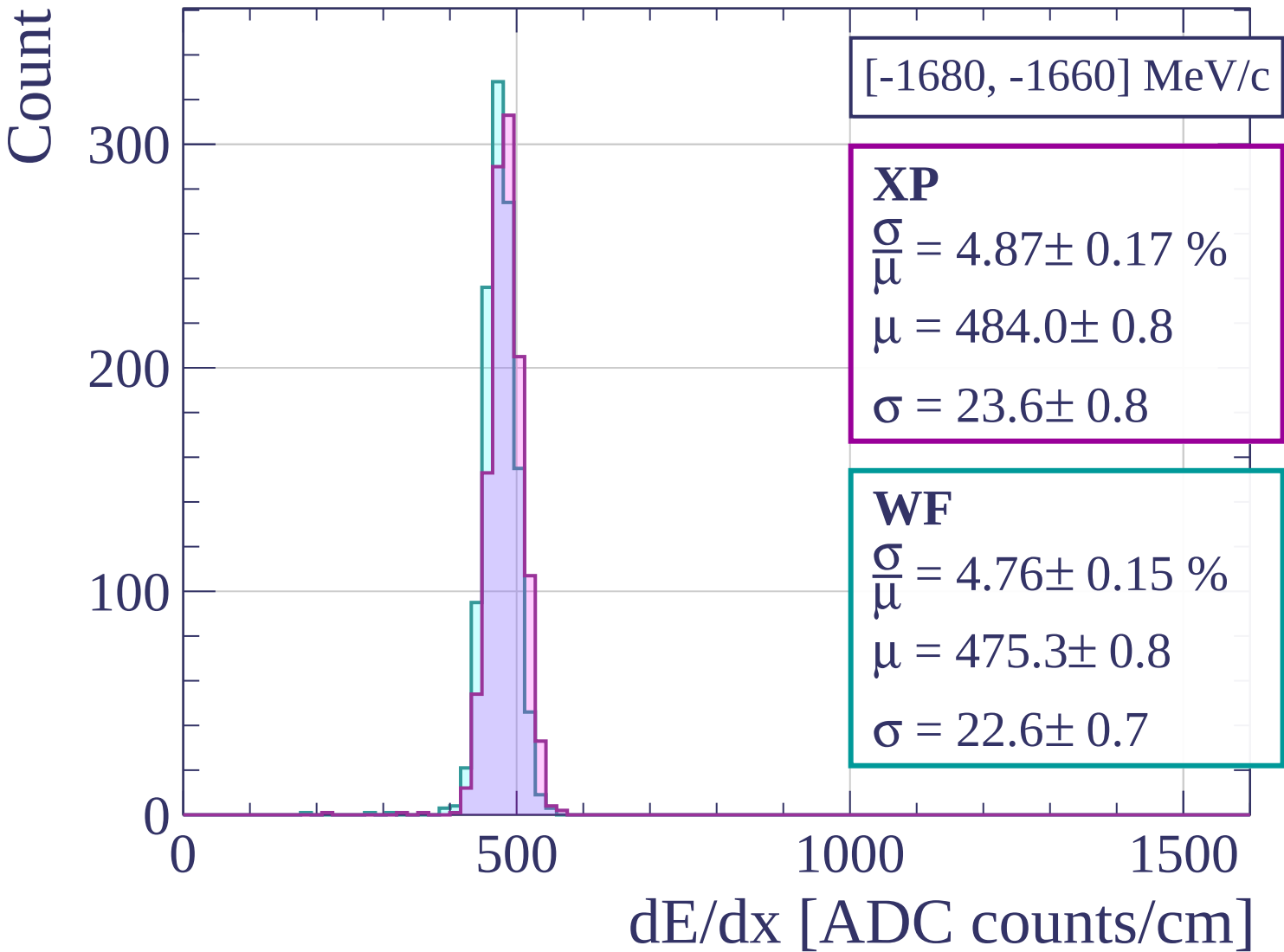
Count



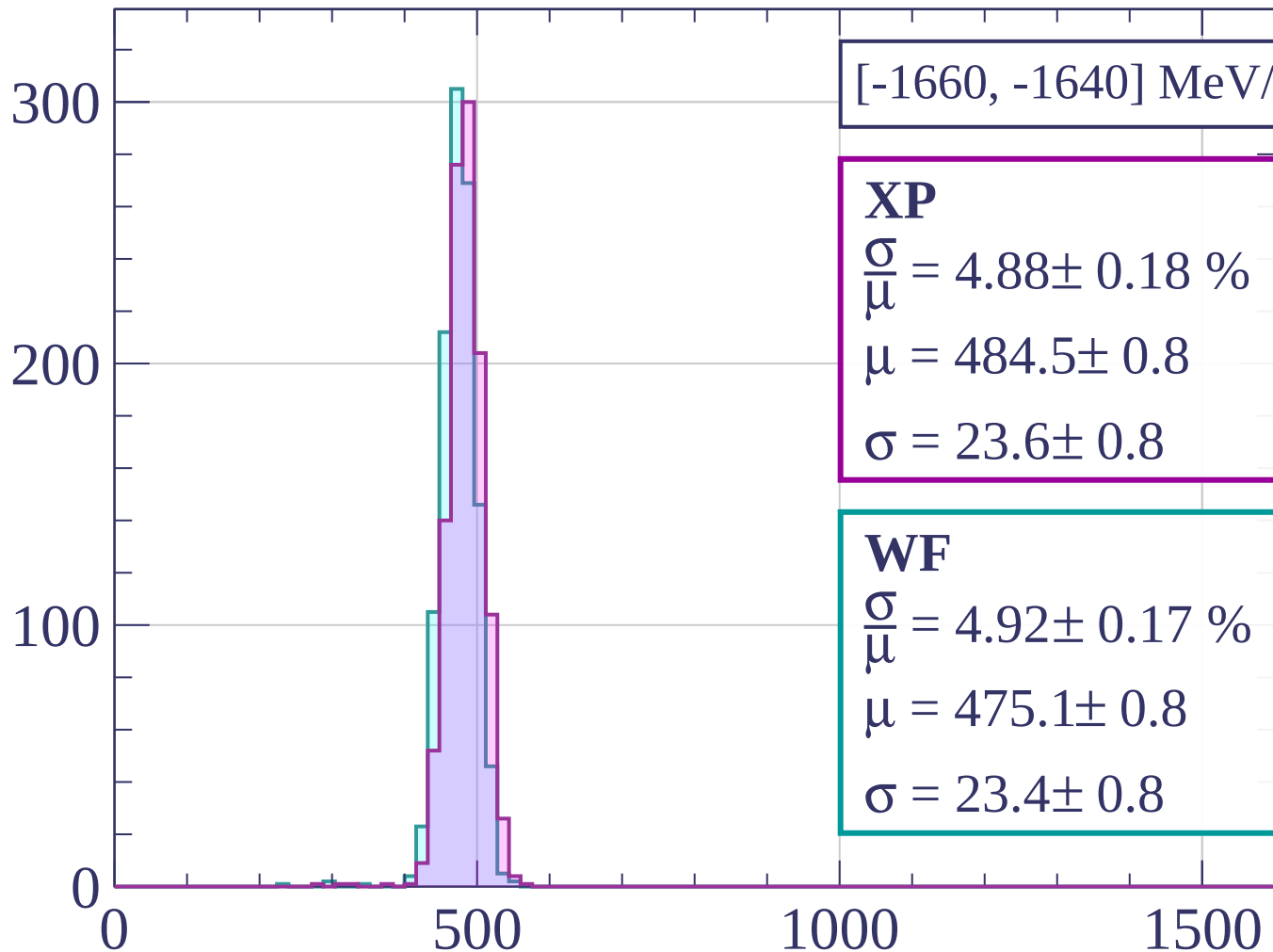
dE/dx [ADC counts/cm]



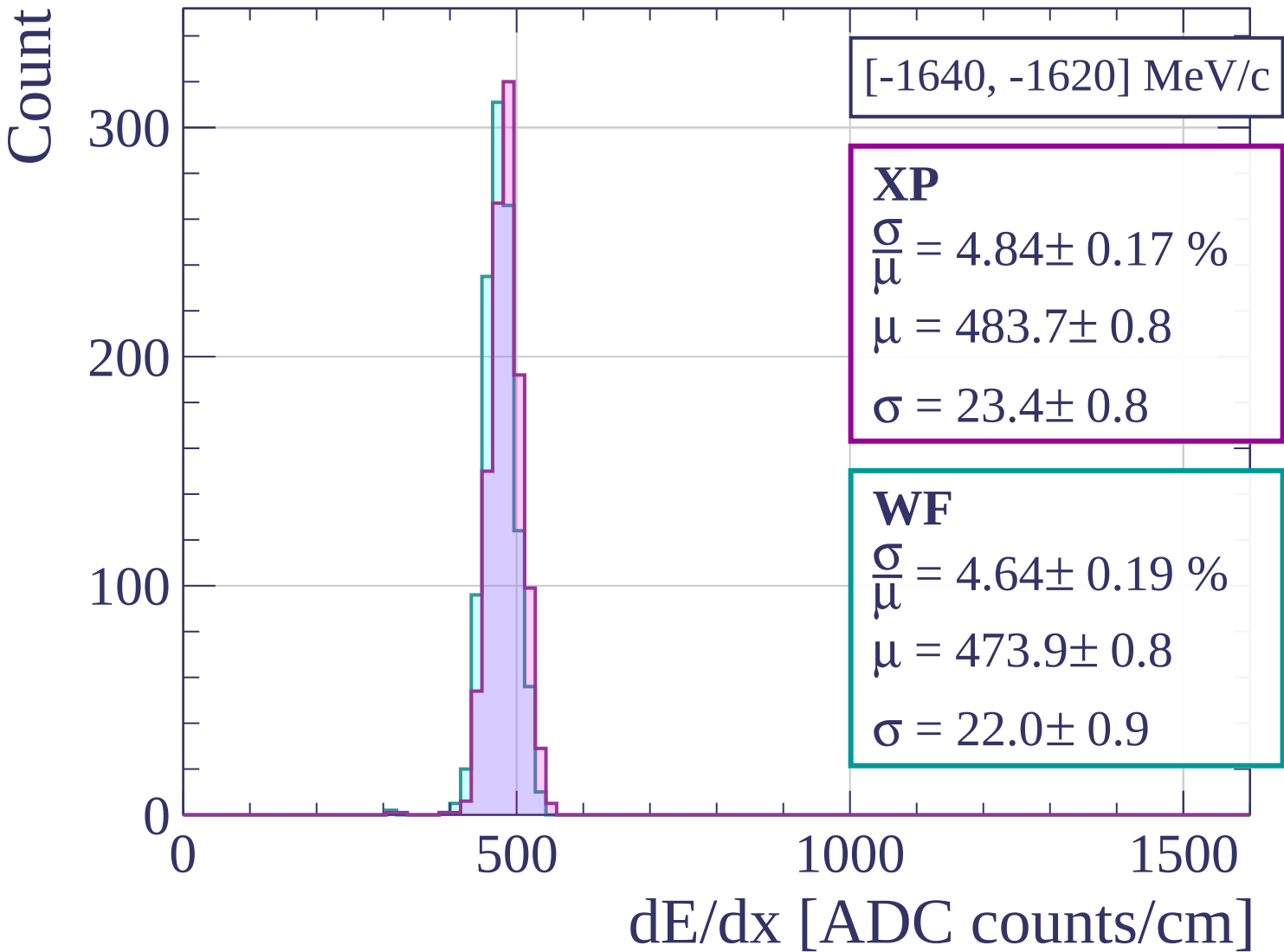


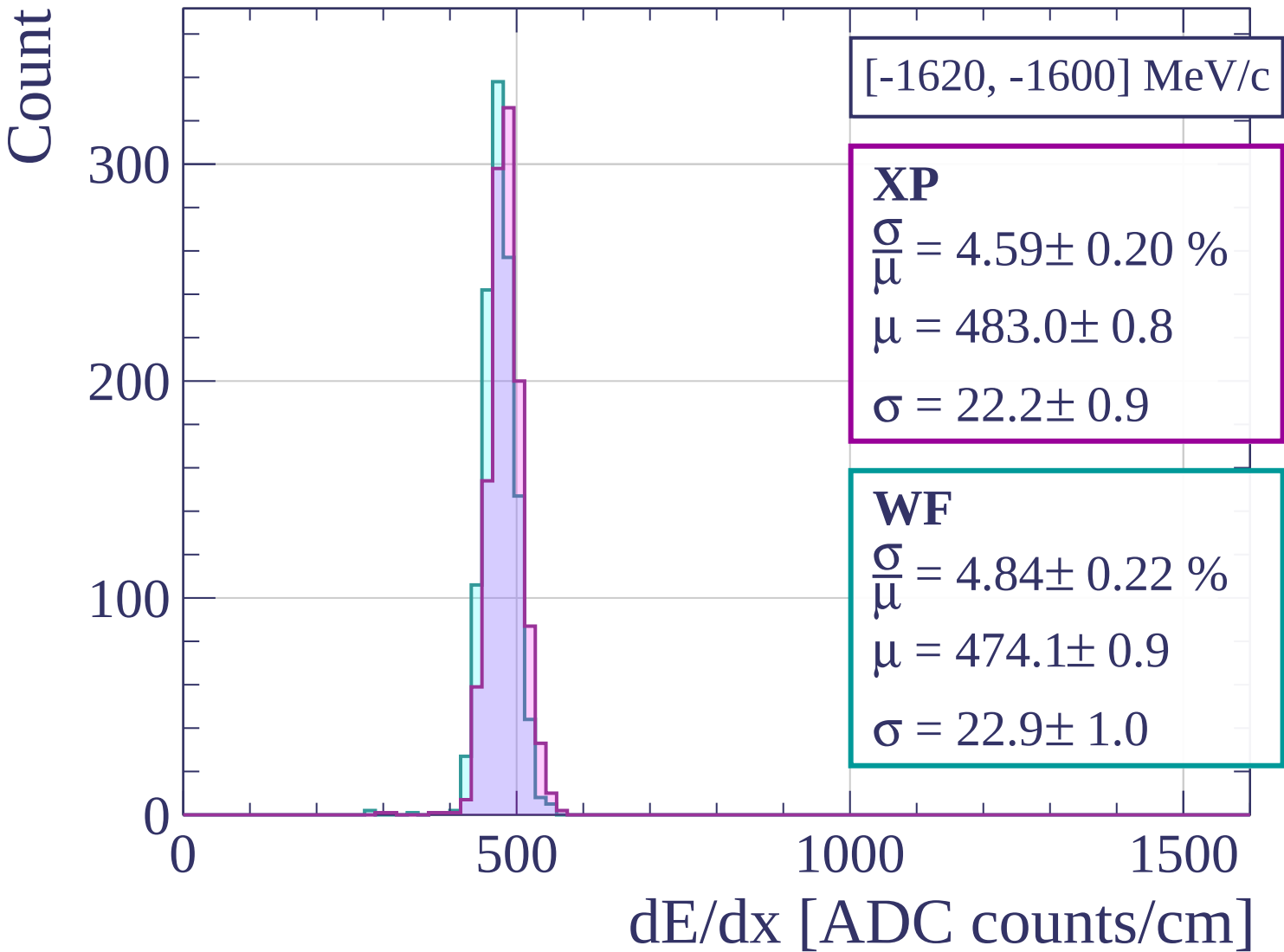


Count



dE/dx [ADC counts/cm]





Count

[-1600, -1580] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.61 \pm 0.14 \%$$

$$\mu = 481.6 \pm 0.8$$

$$\sigma = 22.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.53 \pm 0.19 \%$$

$$\mu = 472.8 \pm 0.8$$

$$\sigma = 21.4 \pm 0.9$$

300

200

100

0

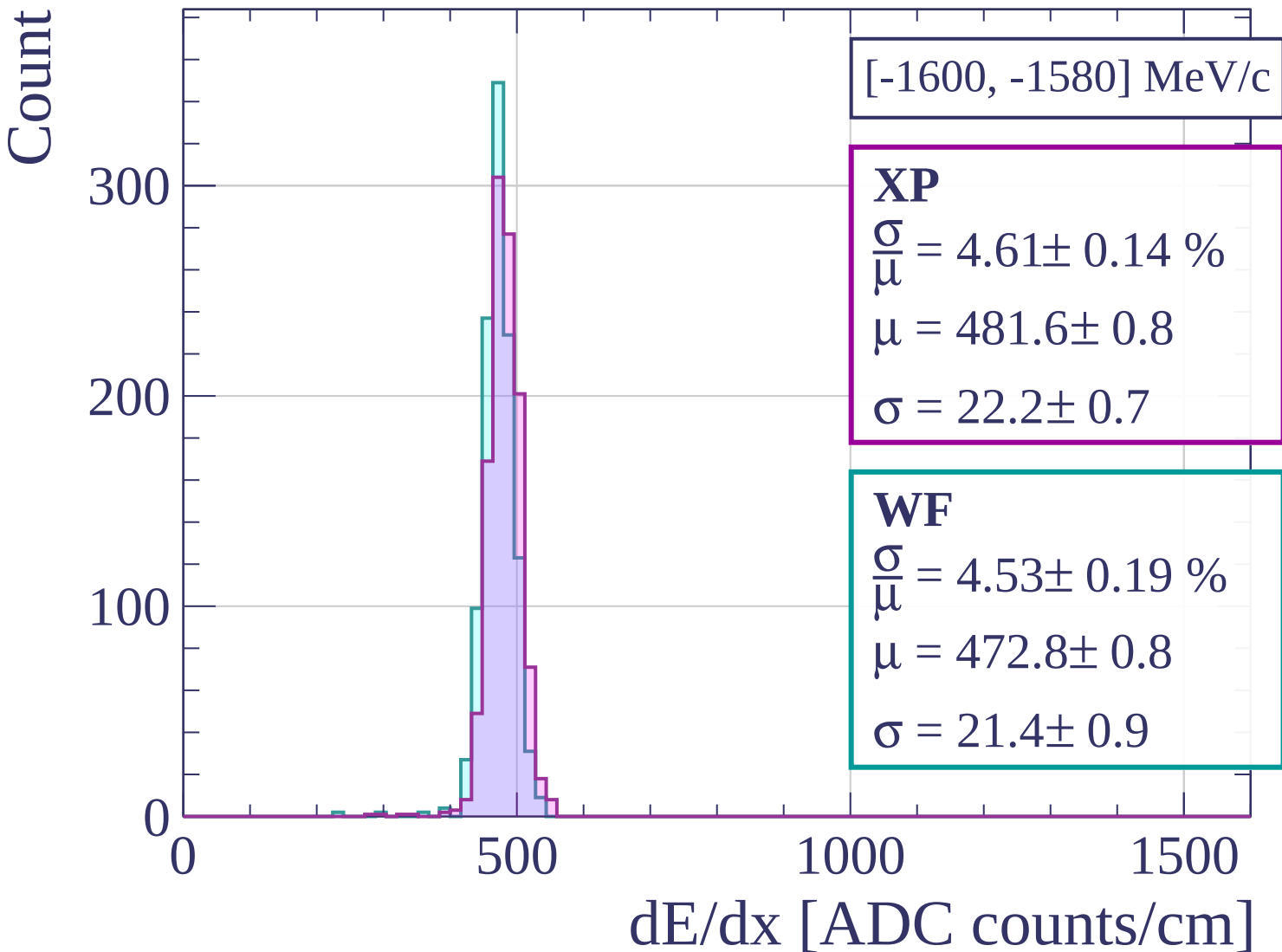
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1580, -1560] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.71 \pm 0.15 \%$$

$$\mu = 480.2 \pm 0.8$$

$$\sigma = 22.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.79 \pm 0.21 \%$$

$$\mu = 470.9 \pm 0.8$$

$$\sigma = 22.5 \pm 0.9$$

300

200

100

0

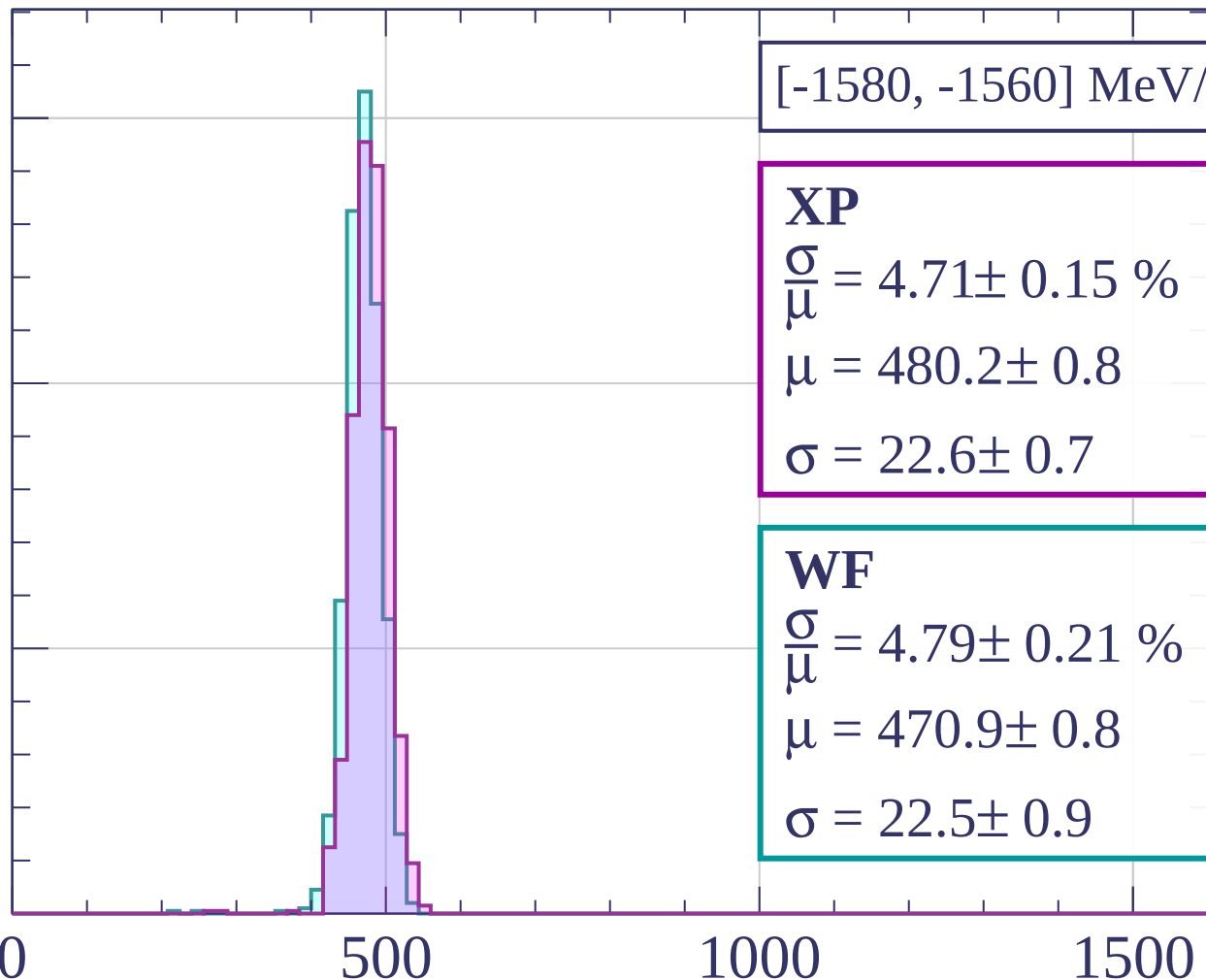
0

500

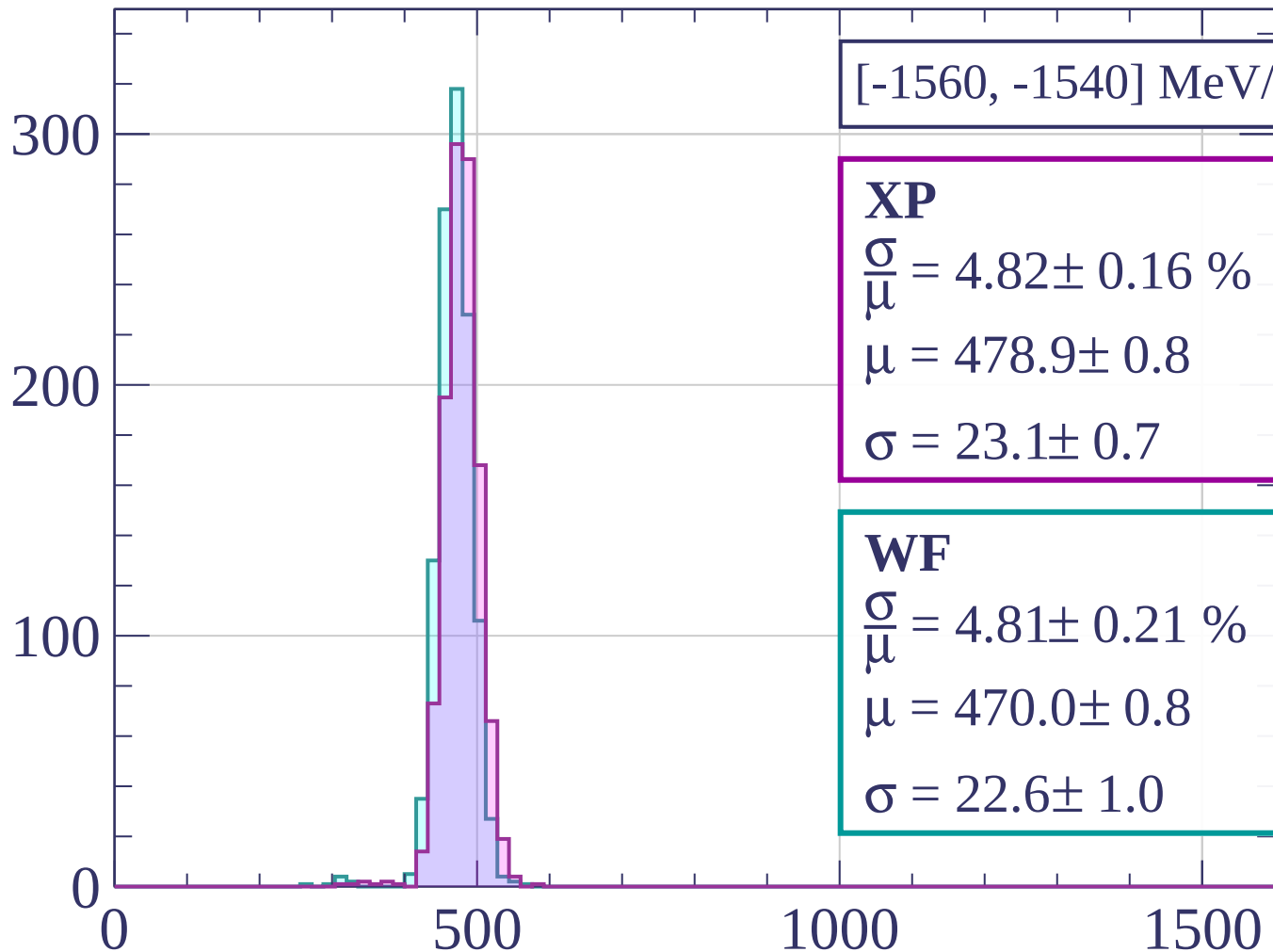
1000

1500

dE/dx [ADC counts/cm]



Count



[-1560, -1540] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.82 \pm 0.16 \%$$

$$\mu = 478.9 \pm 0.8$$

$$\sigma = 23.1 \pm 0.7$$

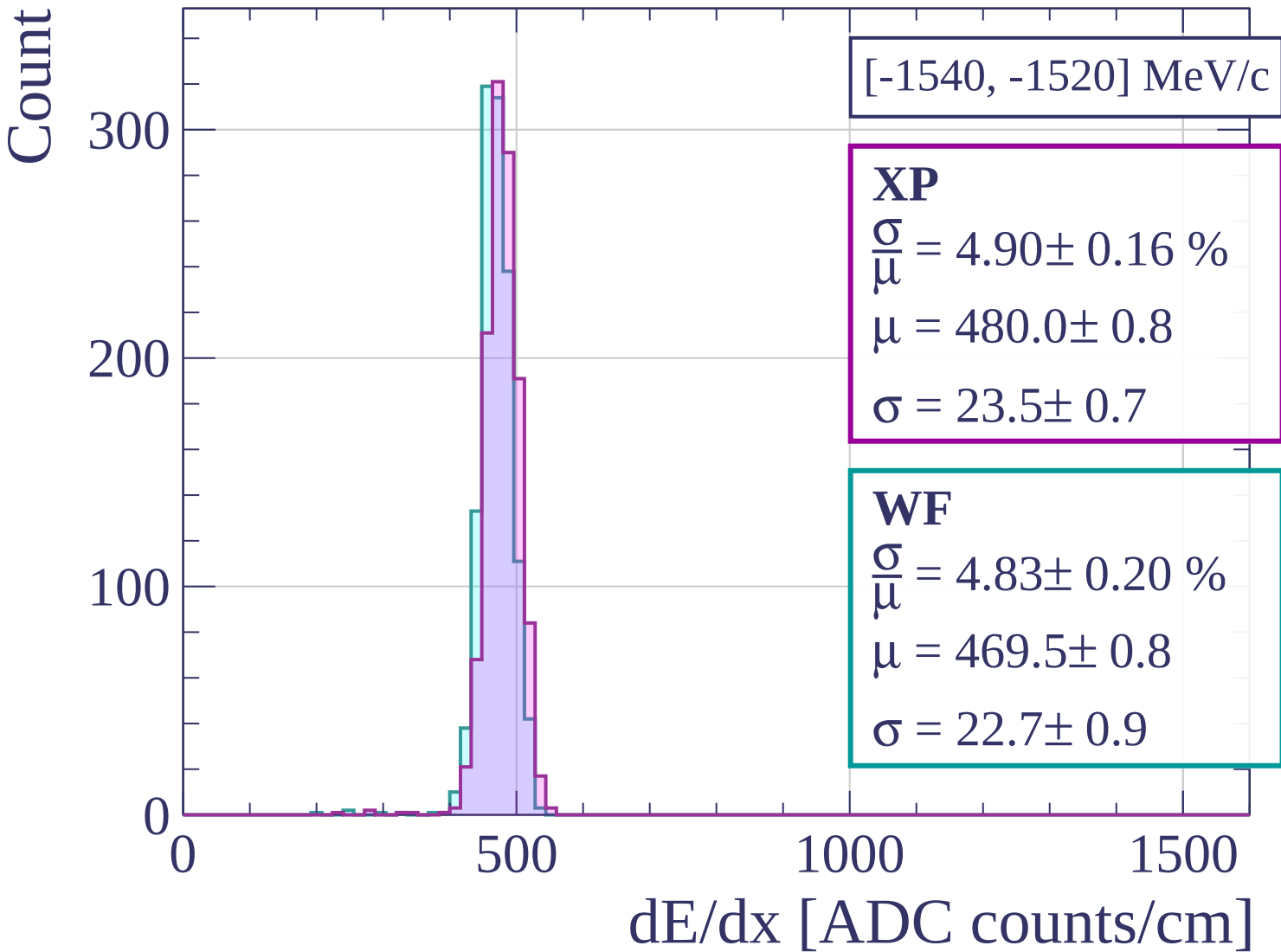
WF

$$\frac{\sigma}{\mu} = 4.81 \pm 0.21 \%$$

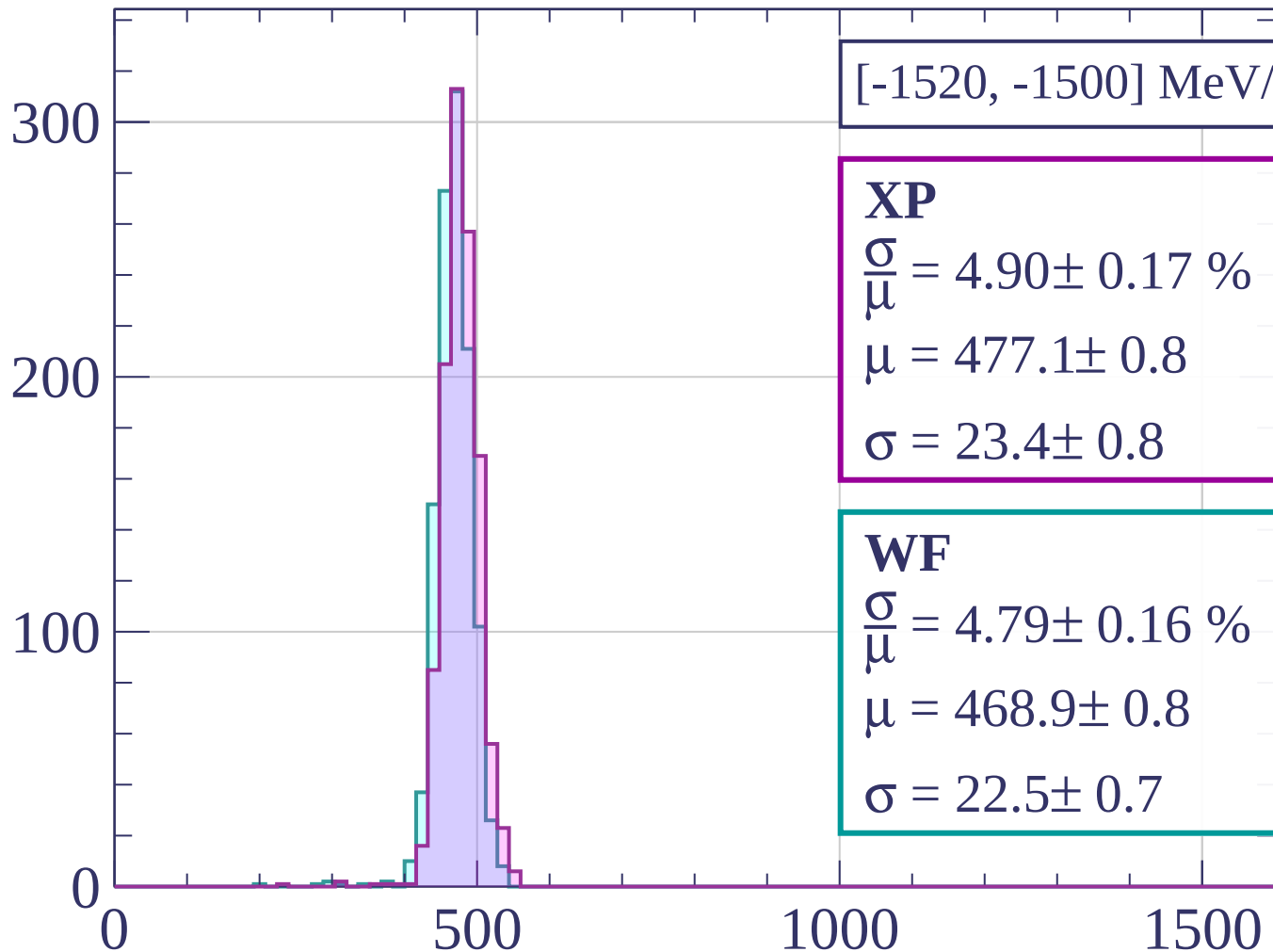
$$\mu = 470.0 \pm 0.8$$

$$\sigma = 22.6 \pm 1.0$$

dE/dx [ADC counts/cm]

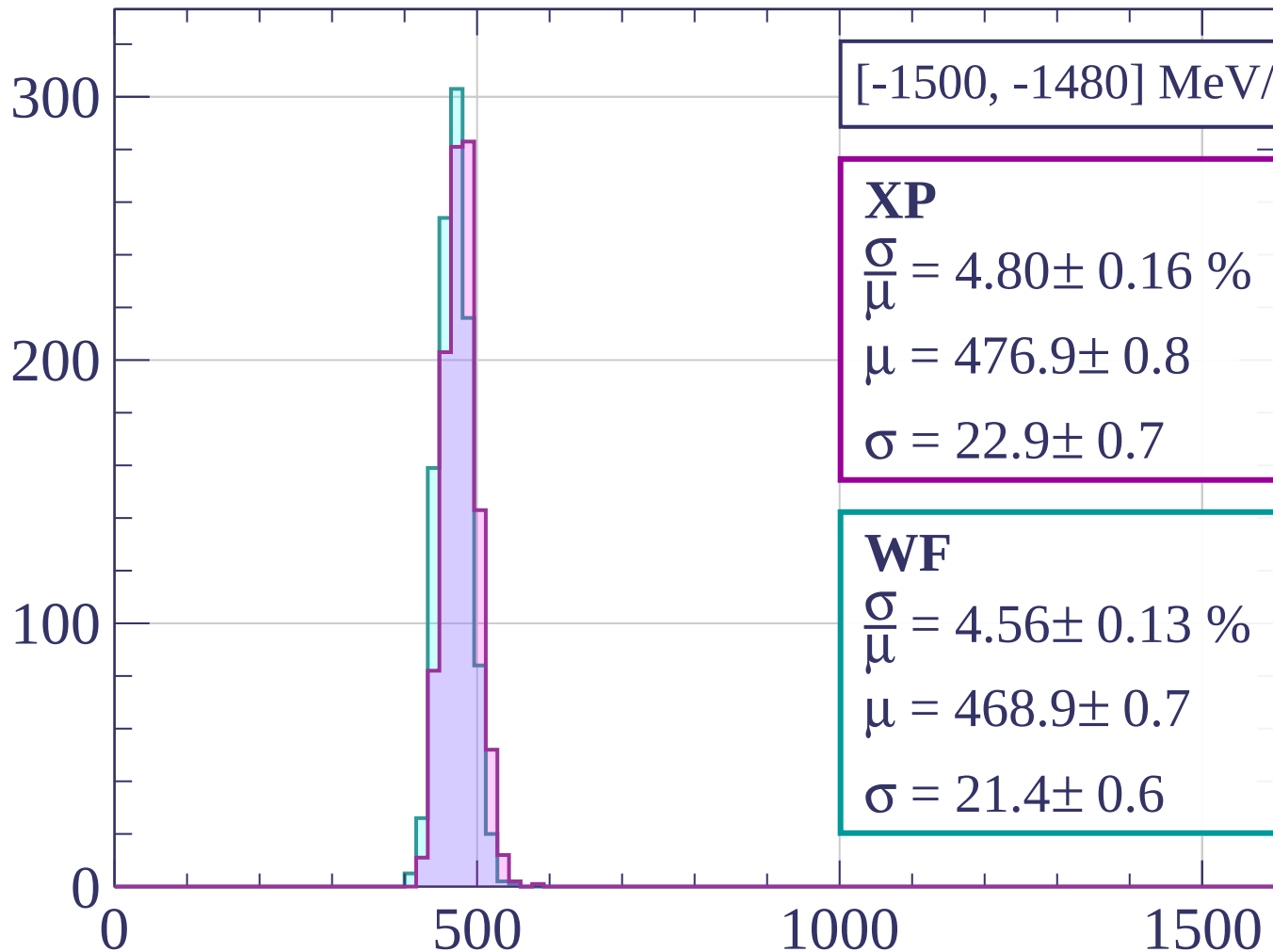


Count



dE/dx [ADC counts/cm]

Count



[-1500, -1480] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.80 \pm 0.16 \%$$

$$\mu = 476.9 \pm 0.8$$

$$\sigma = 22.9 \pm 0.7$$

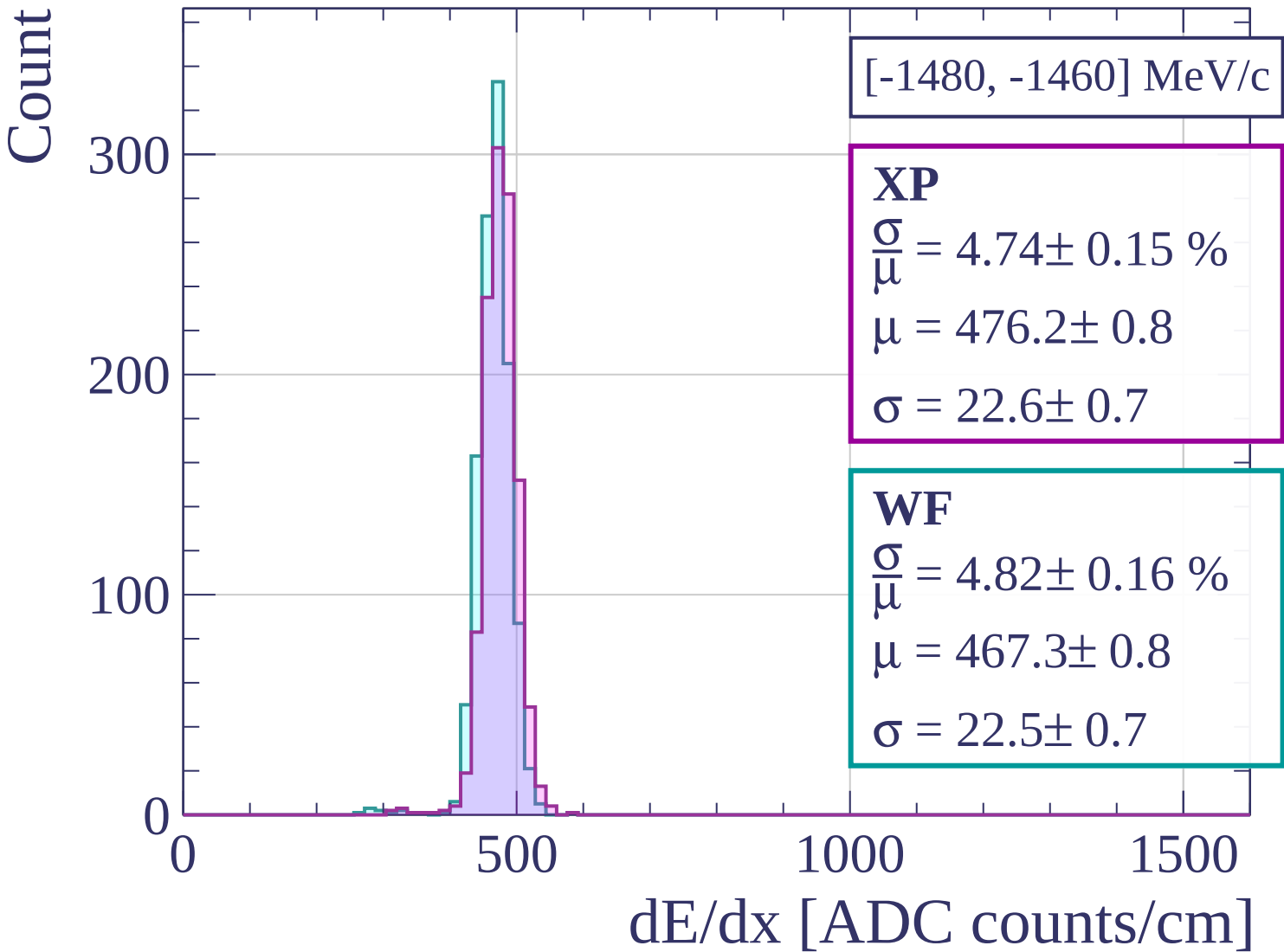
WF

$$\frac{\sigma}{\mu} = 4.56 \pm 0.13 \%$$

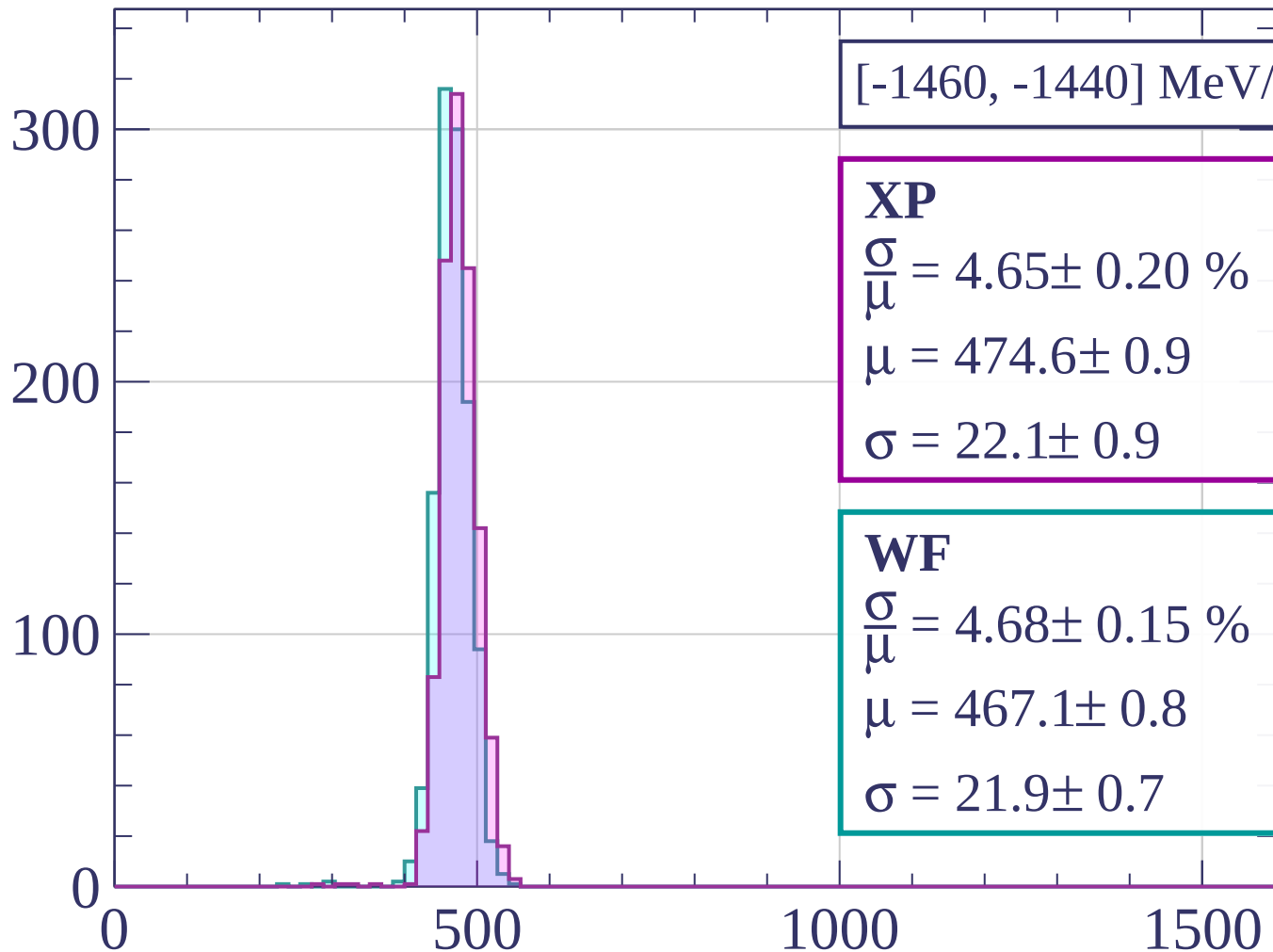
$$\mu = 468.9 \pm 0.7$$

$$\sigma = 21.4 \pm 0.6$$

dE/dx [ADC counts/cm]



Count



[-1460, -1440] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.65 \pm 0.20 \%$$

$$\mu = 474.6 \pm 0.9$$

$$\sigma = 22.1 \pm 0.9$$

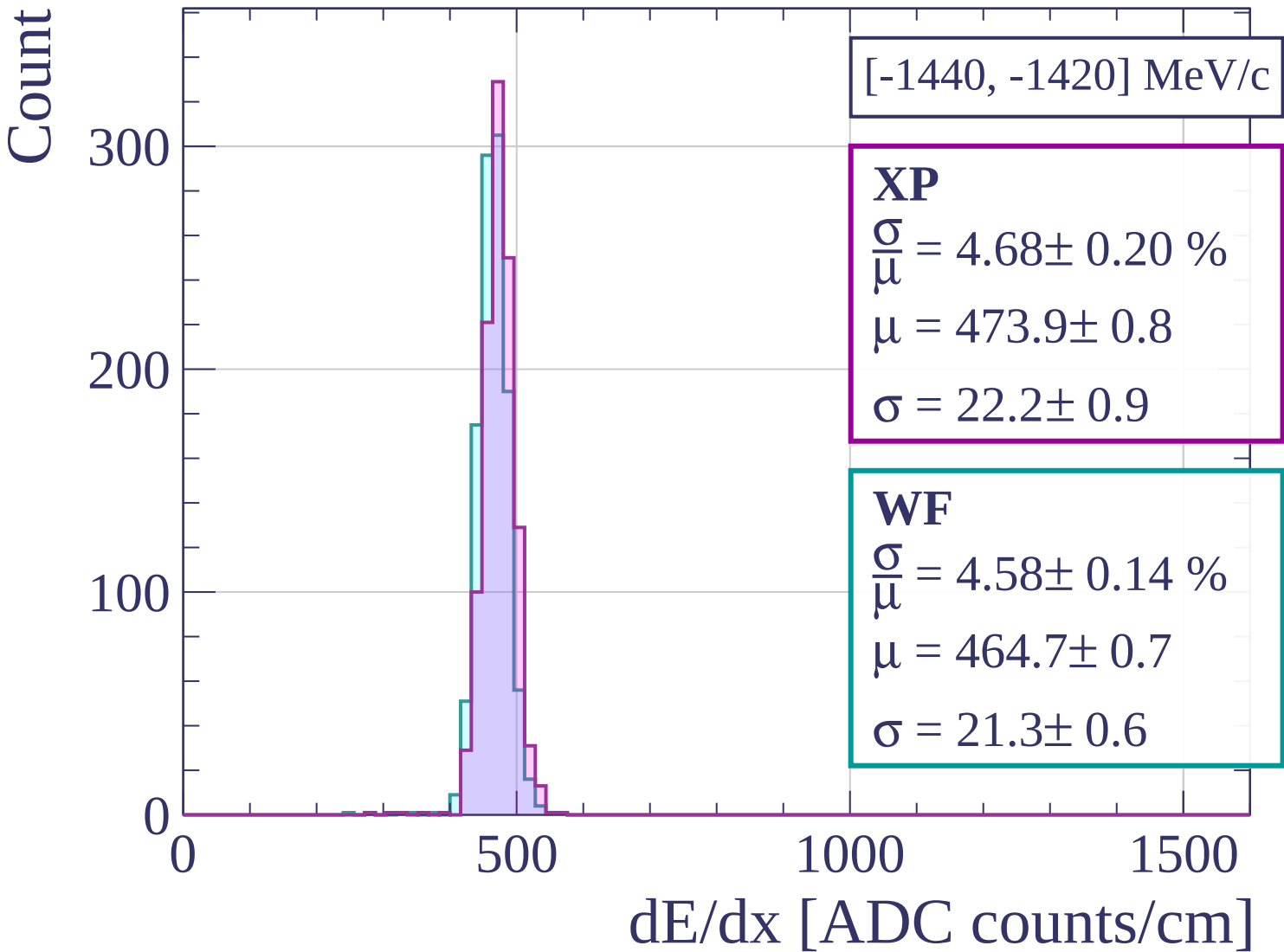
WF

$$\frac{\sigma}{\mu} = 4.68 \pm 0.15 \%$$

$$\mu = 467.1 \pm 0.8$$

$$\sigma = 21.9 \pm 0.7$$

dE/dx [ADC counts/cm]



Count

[-1420, -1400] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.48 \pm 0.18 \%$$

$$\mu = 473.2 \pm 0.8$$

$$\sigma = 21.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.62 \pm 0.14 \%$$

$$\mu = 465.0 \pm 0.7$$

$$\sigma = 21.5 \pm 0.6$$

300

200

100

0

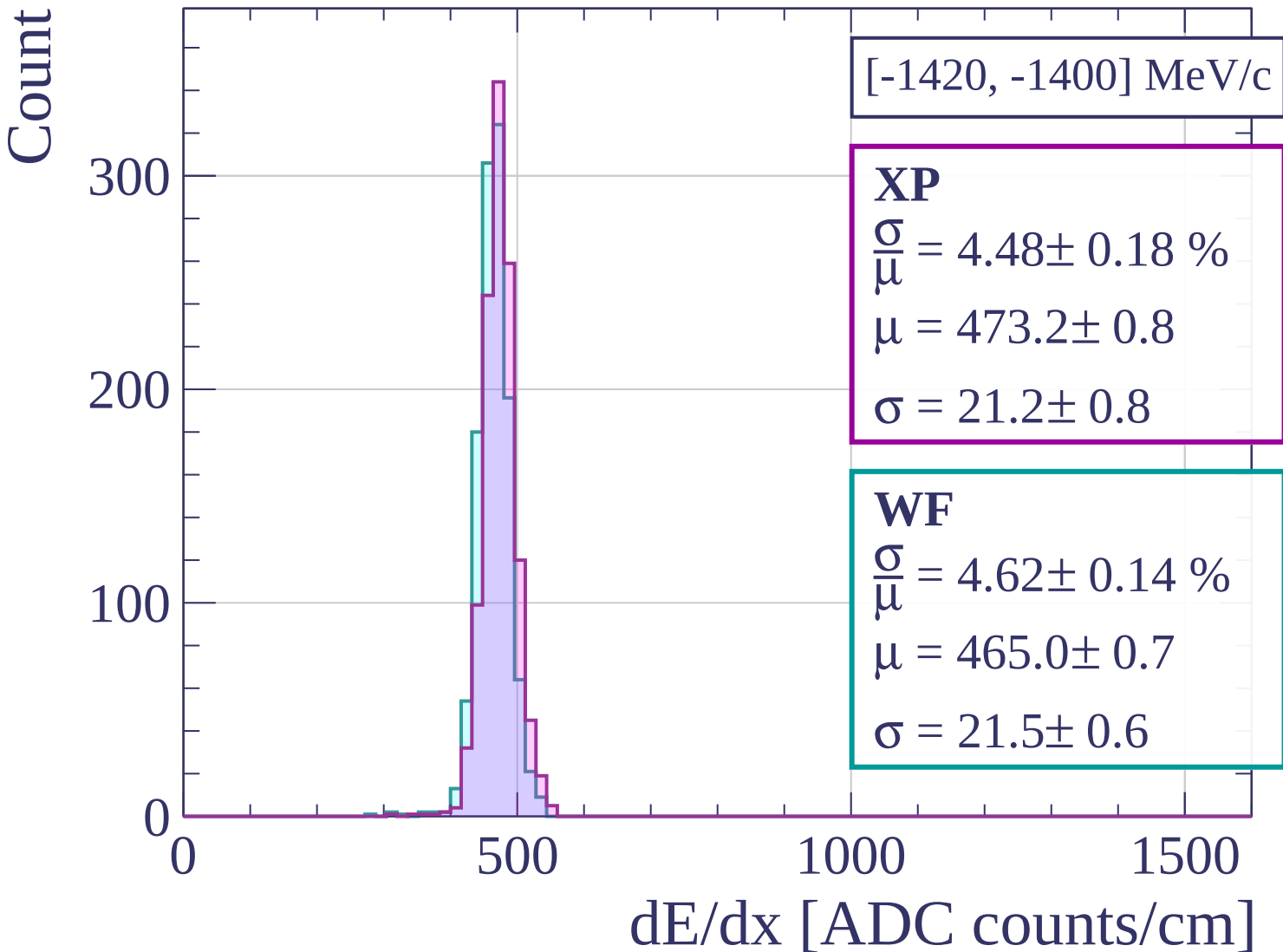
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-1400, -1380] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.76 \pm 0.22 \%$$

$$\mu = 471.8 \pm 0.9$$

$$\sigma = 22.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 4.34 \pm 0.17 \%$$

$$\mu = 462.0 \pm 0.8$$

$$\sigma = 20.0 \pm 0.7$$

300

200

100

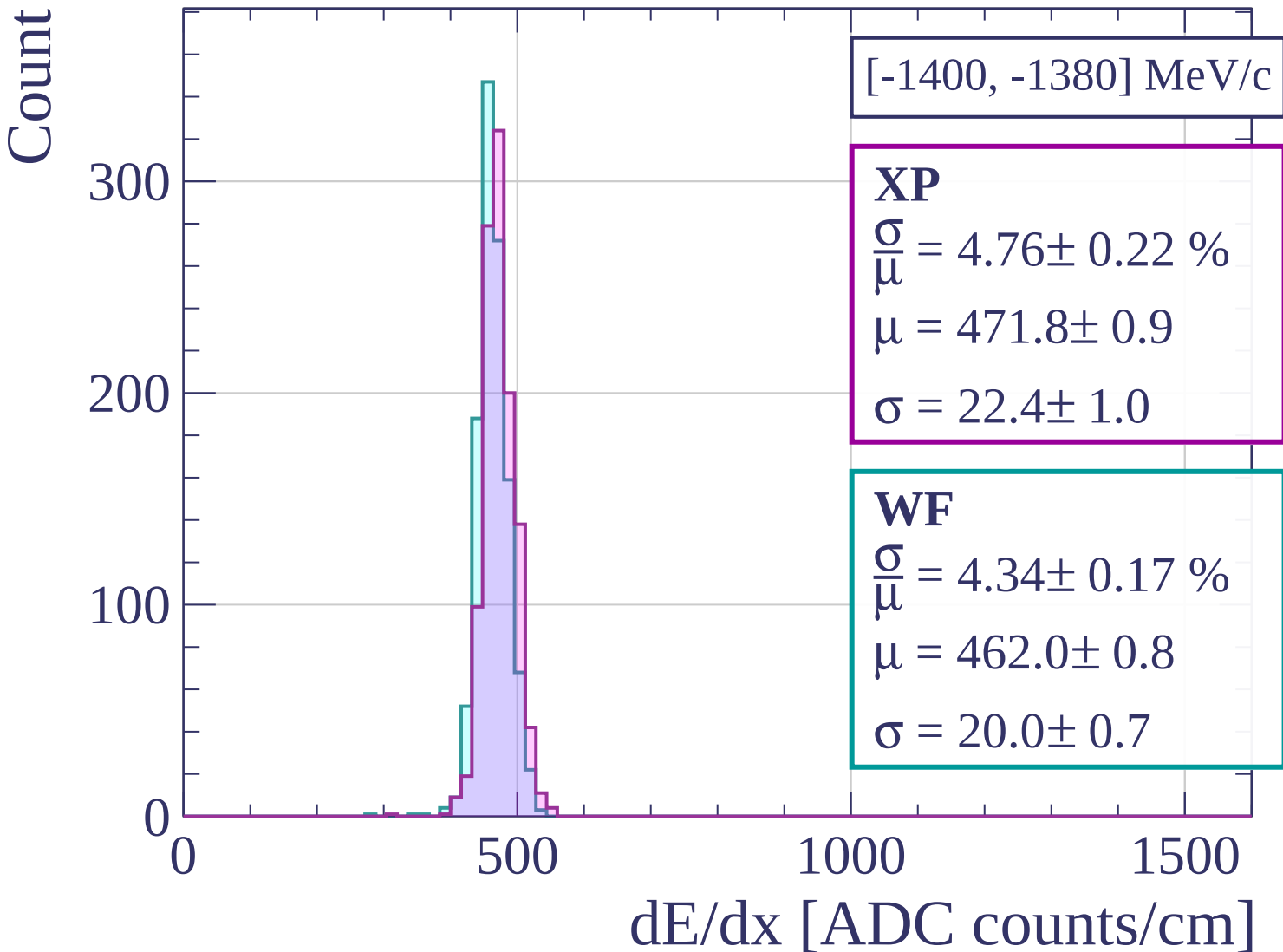
0

500

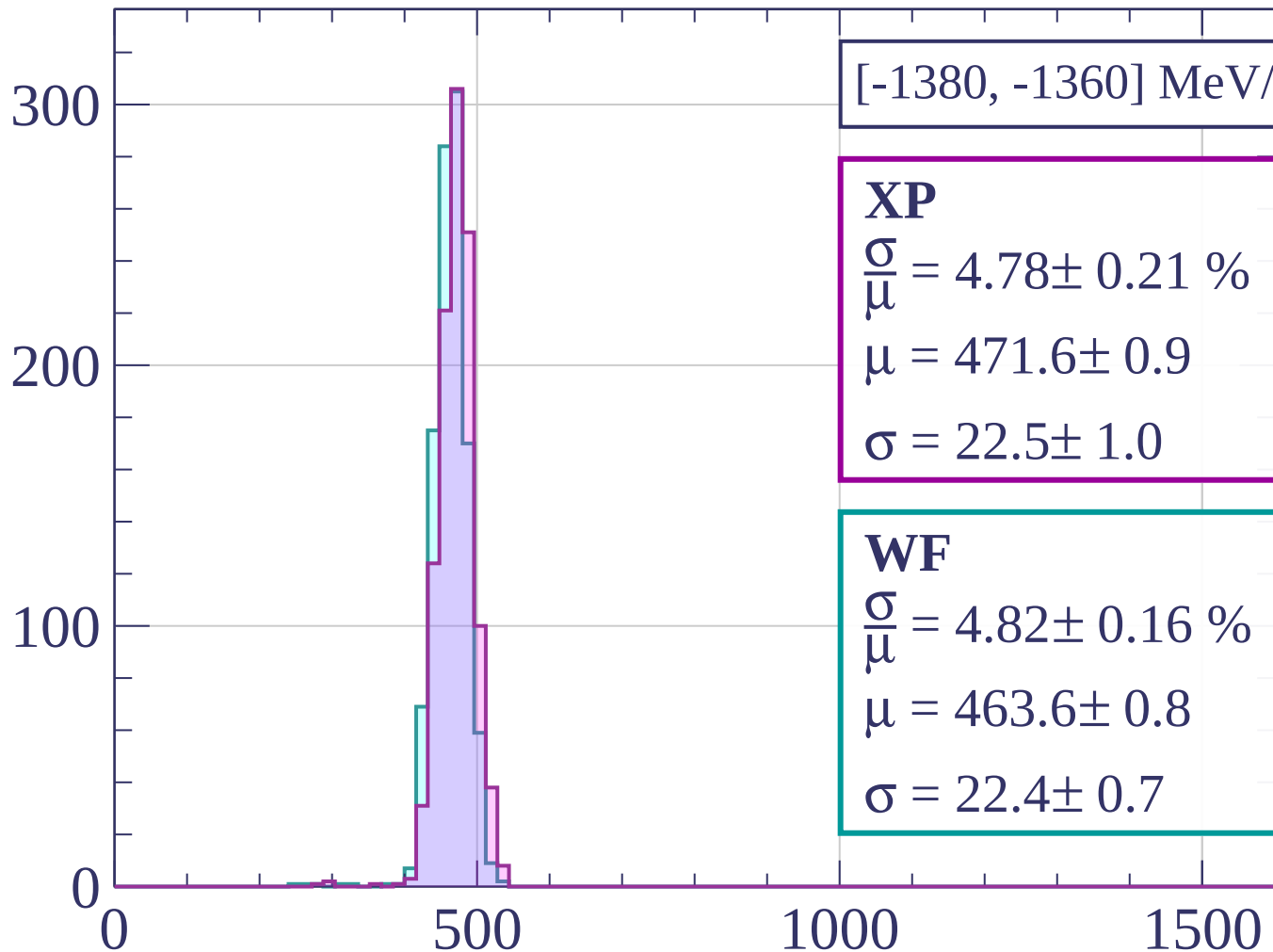
1000

1500

dE/dx [ADC counts/cm]

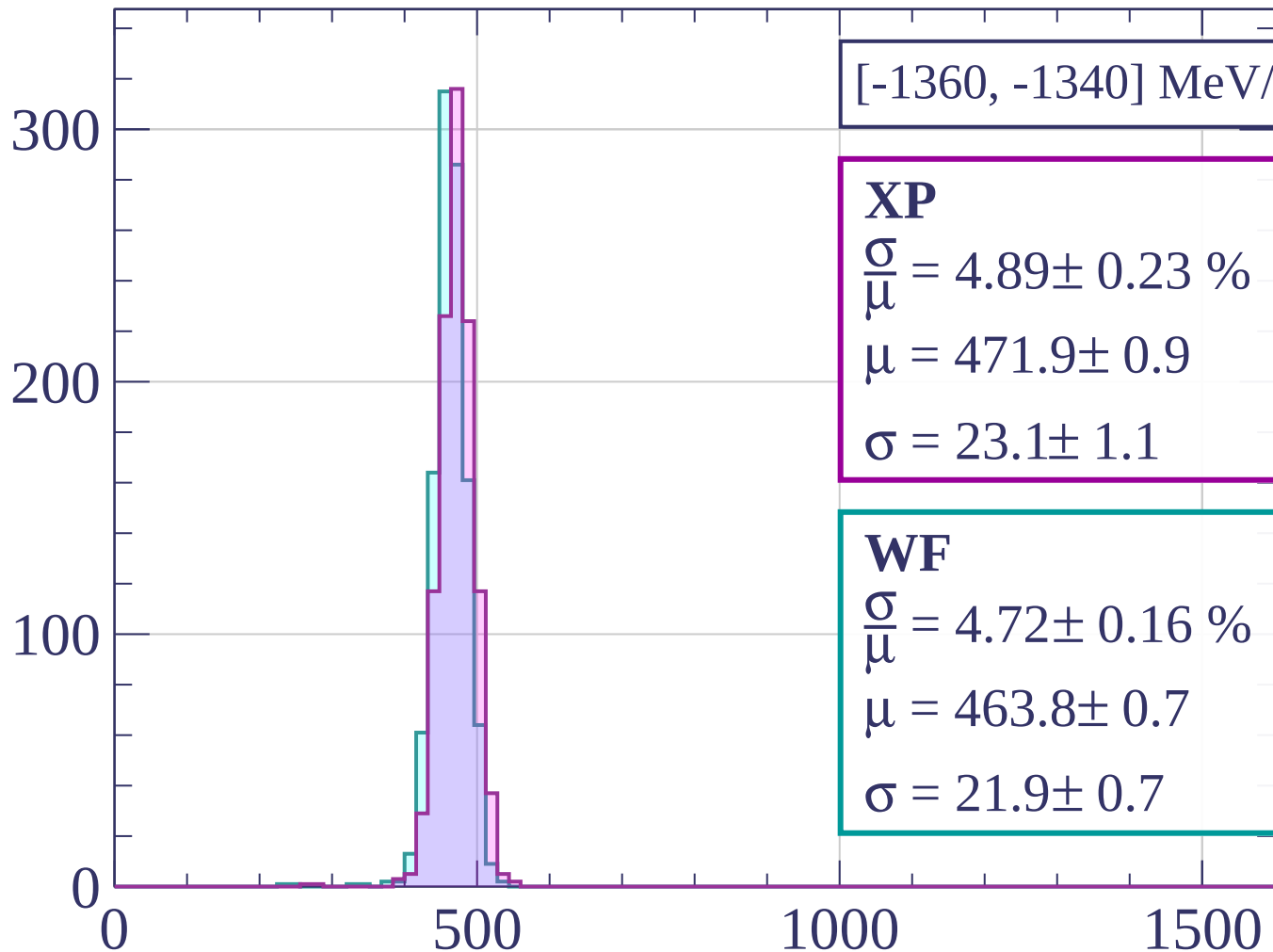


Count



dE/dx [ADC counts/cm]

Count



$[-1360, -1340]$ MeV/c

XP

$$\frac{\sigma}{\mu} = 4.89 \pm 0.23 \%$$

$$\mu = 471.9 \pm 0.9$$

$$\sigma = 23.1 \pm 1.1$$

WF

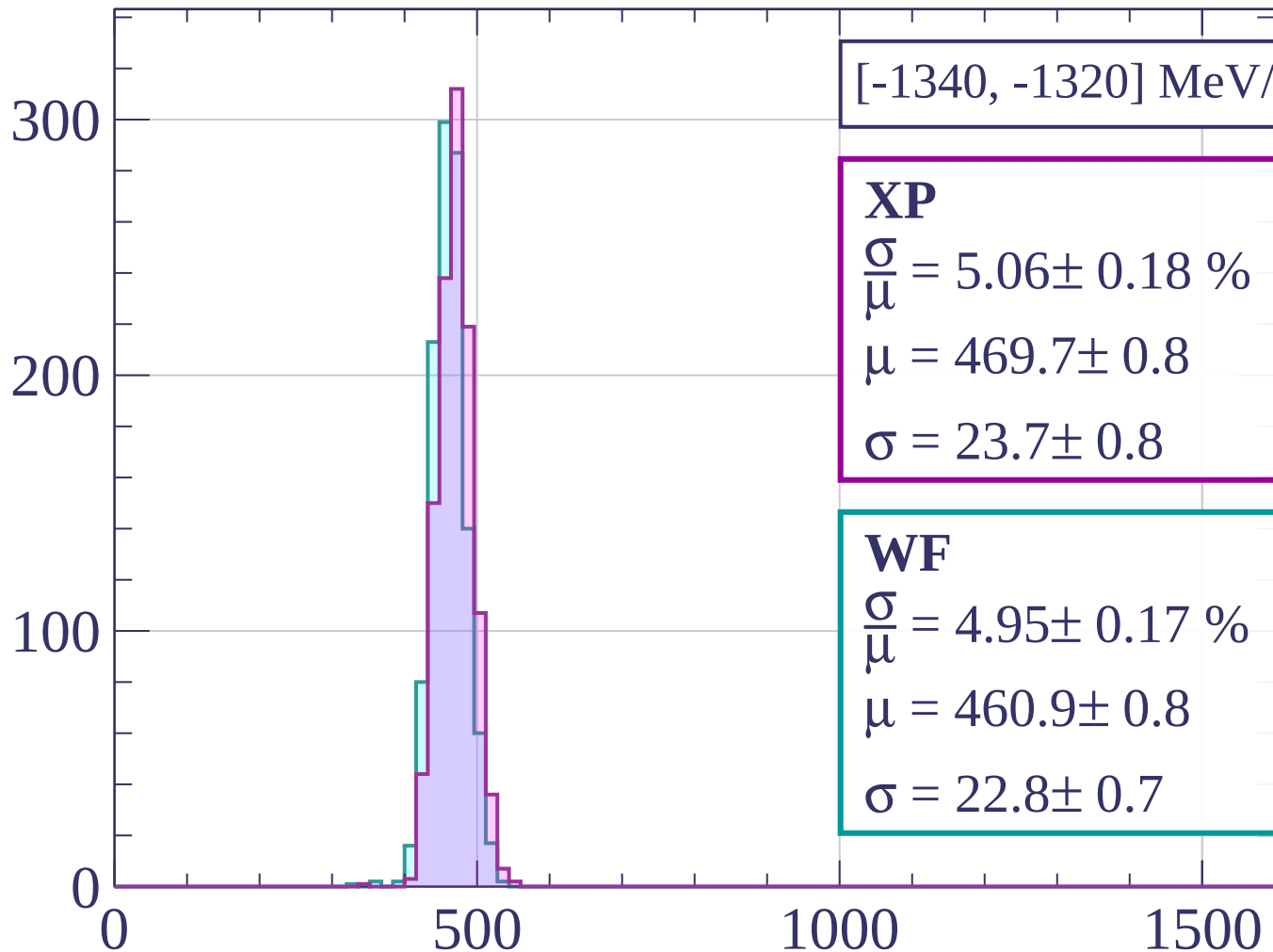
$$\frac{\sigma}{\mu} = 4.72 \pm 0.16 \%$$

$$\mu = 463.8 \pm 0.7$$

$$\sigma = 21.9 \pm 0.7$$

dE/dx [ADC counts/cm]

Count



dE/dx [ADC counts/cm]

Count

[-1320, -1300] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.85 \pm 0.22 \%$$

$$\mu = 469.9 \pm 0.8$$

$$\sigma = 22.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 4.74 \pm 0.15 \%$$

$$\mu = 461.2 \pm 0.7$$

$$\sigma = 21.9 \pm 0.6$$

300

200

100

0

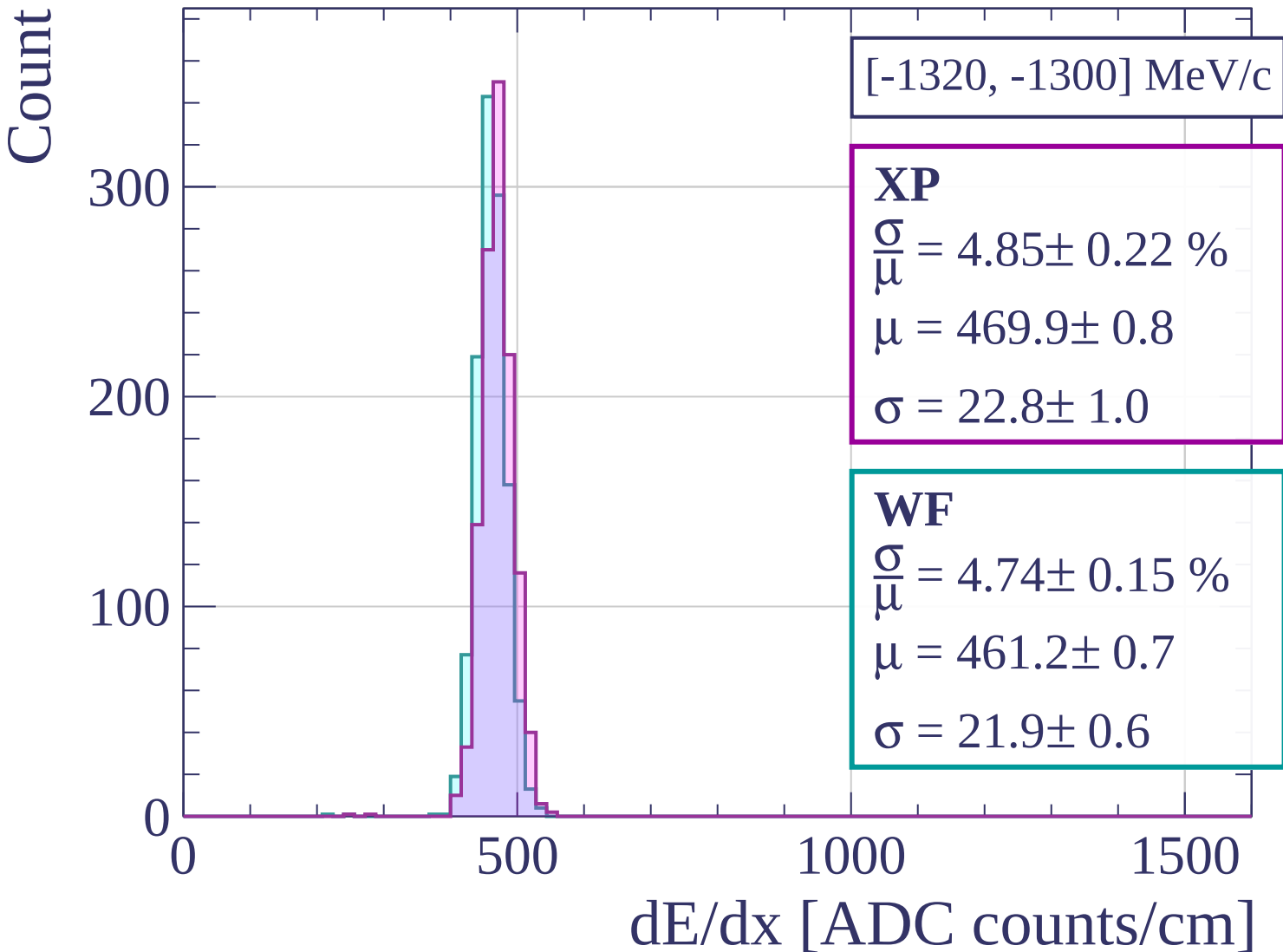
0

500

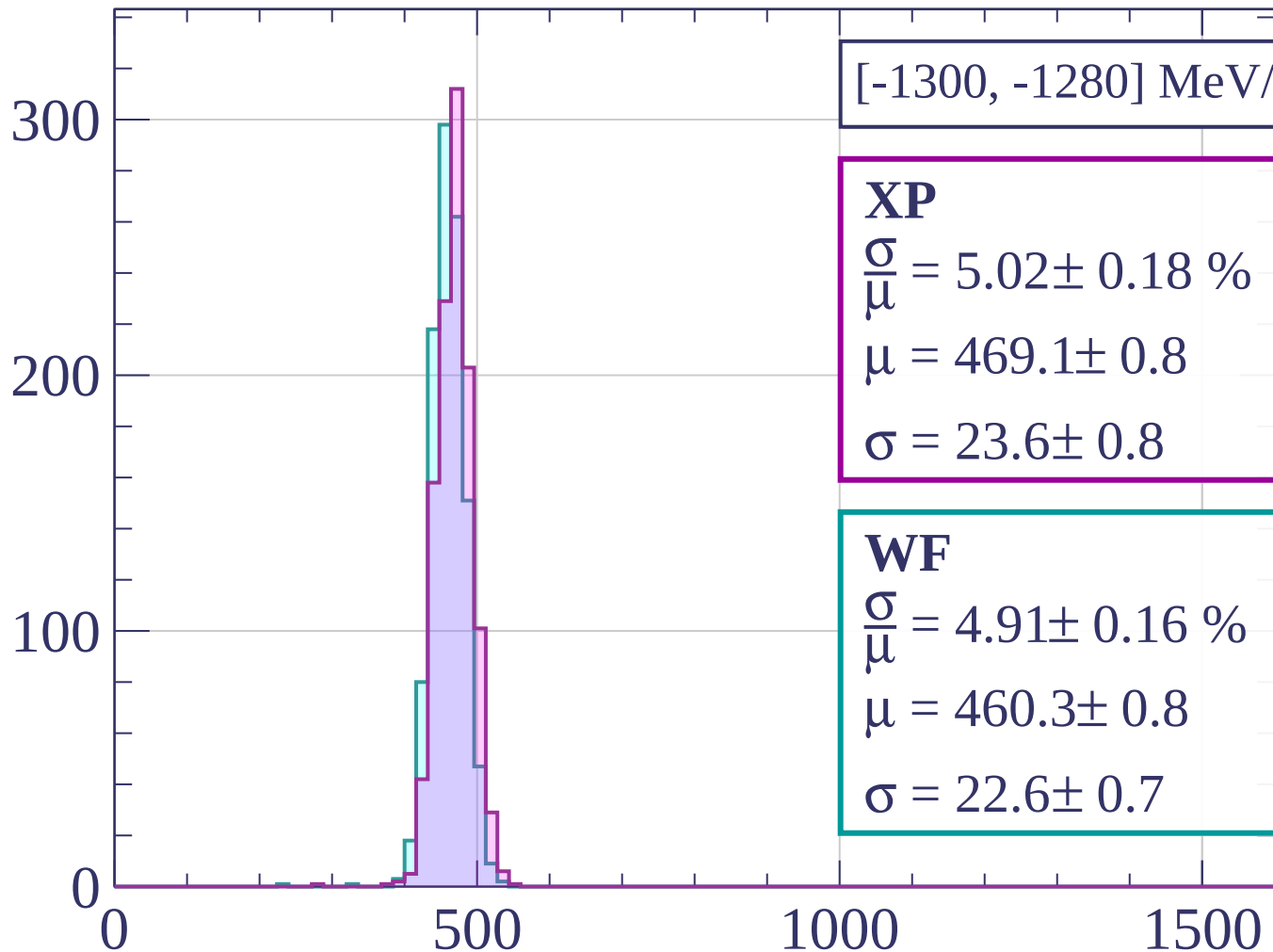
1000

1500

dE/dx [ADC counts/cm]



Count



[-1300, -1280] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.02 \pm 0.18 \%$$

$$\mu = 469.1 \pm 0.8$$

$$\sigma = 23.6 \pm 0.8$$

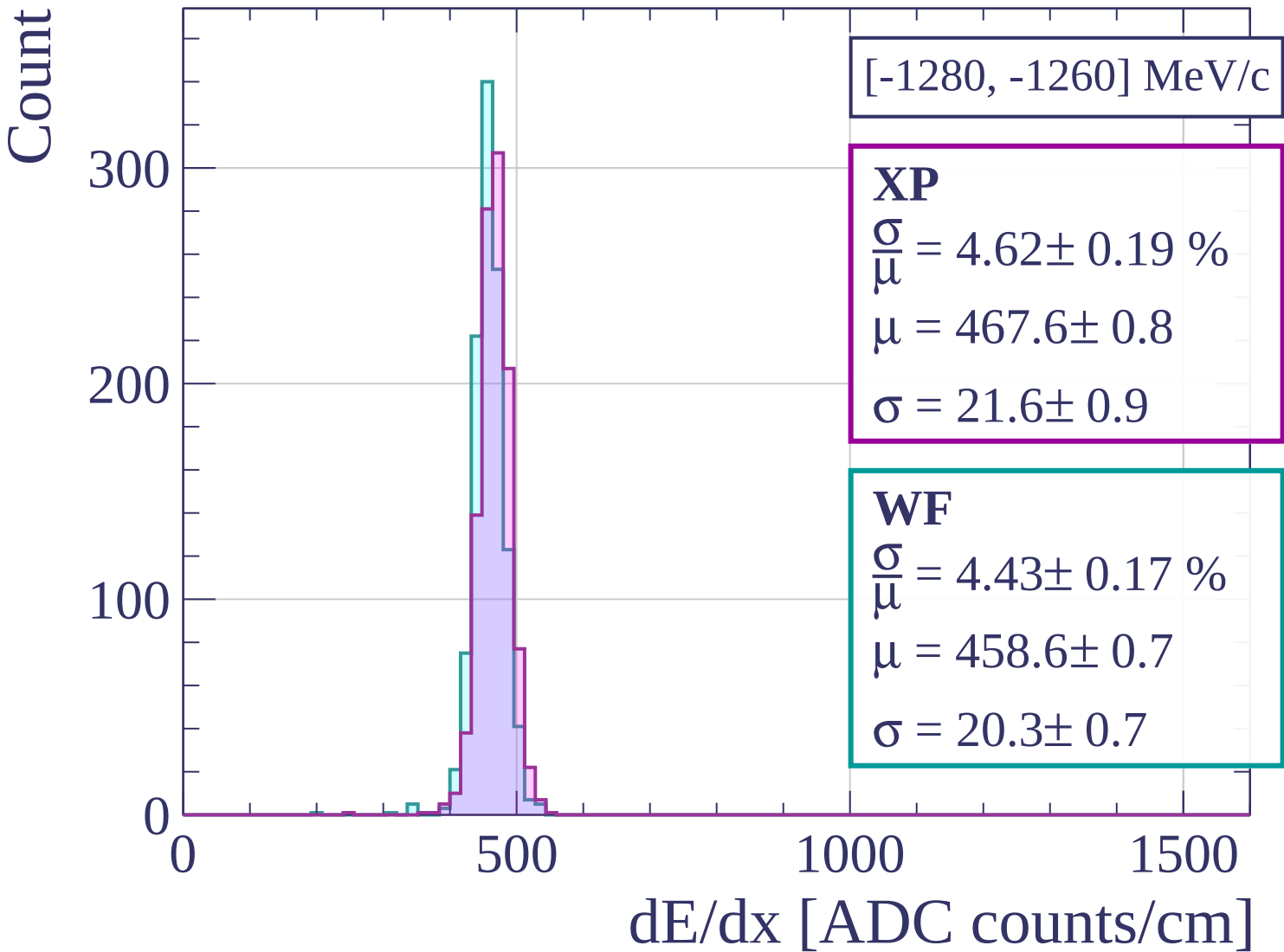
WF

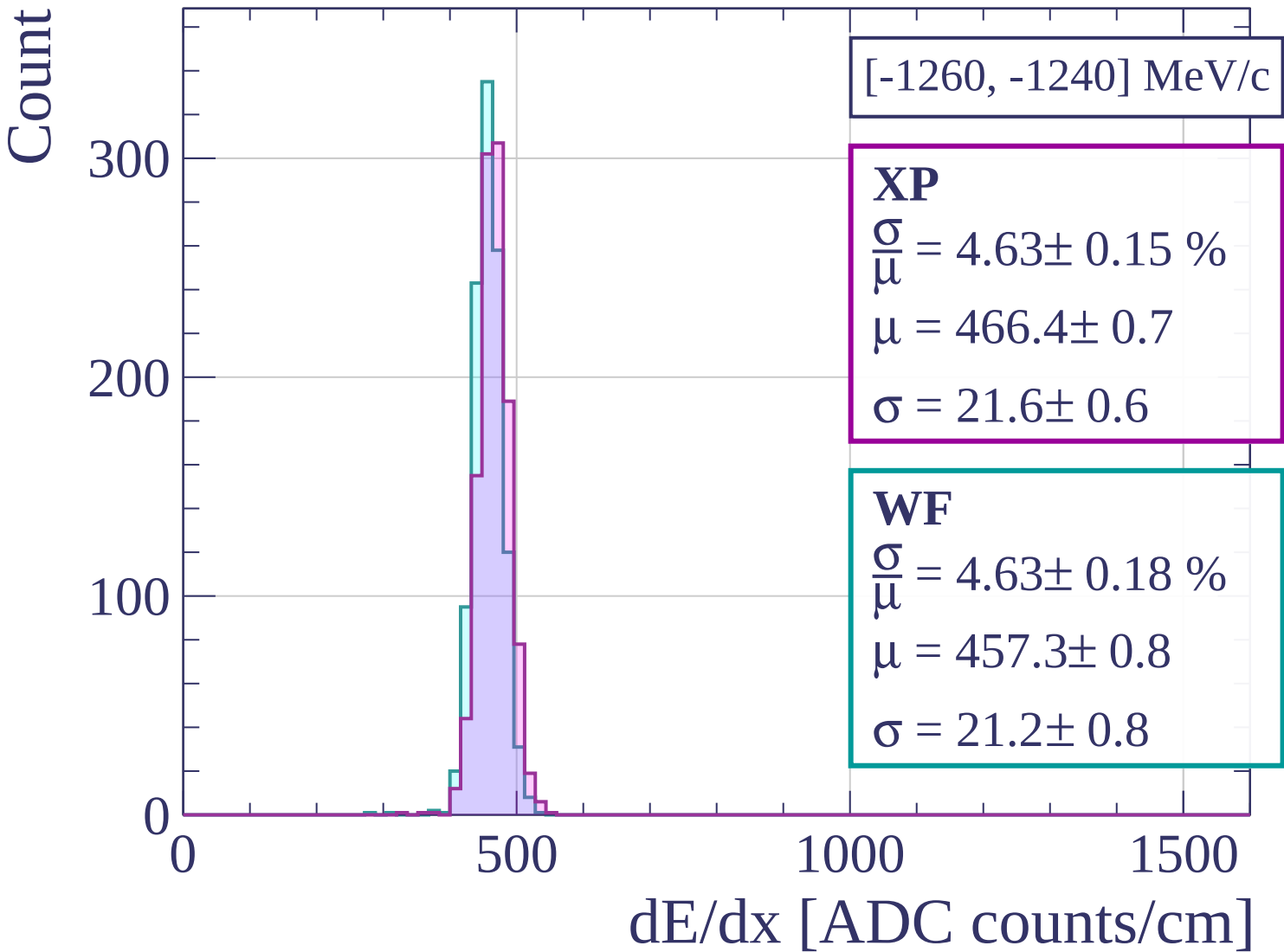
$$\frac{\sigma}{\mu} = 4.91 \pm 0.16 \%$$

$$\mu = 460.3 \pm 0.8$$

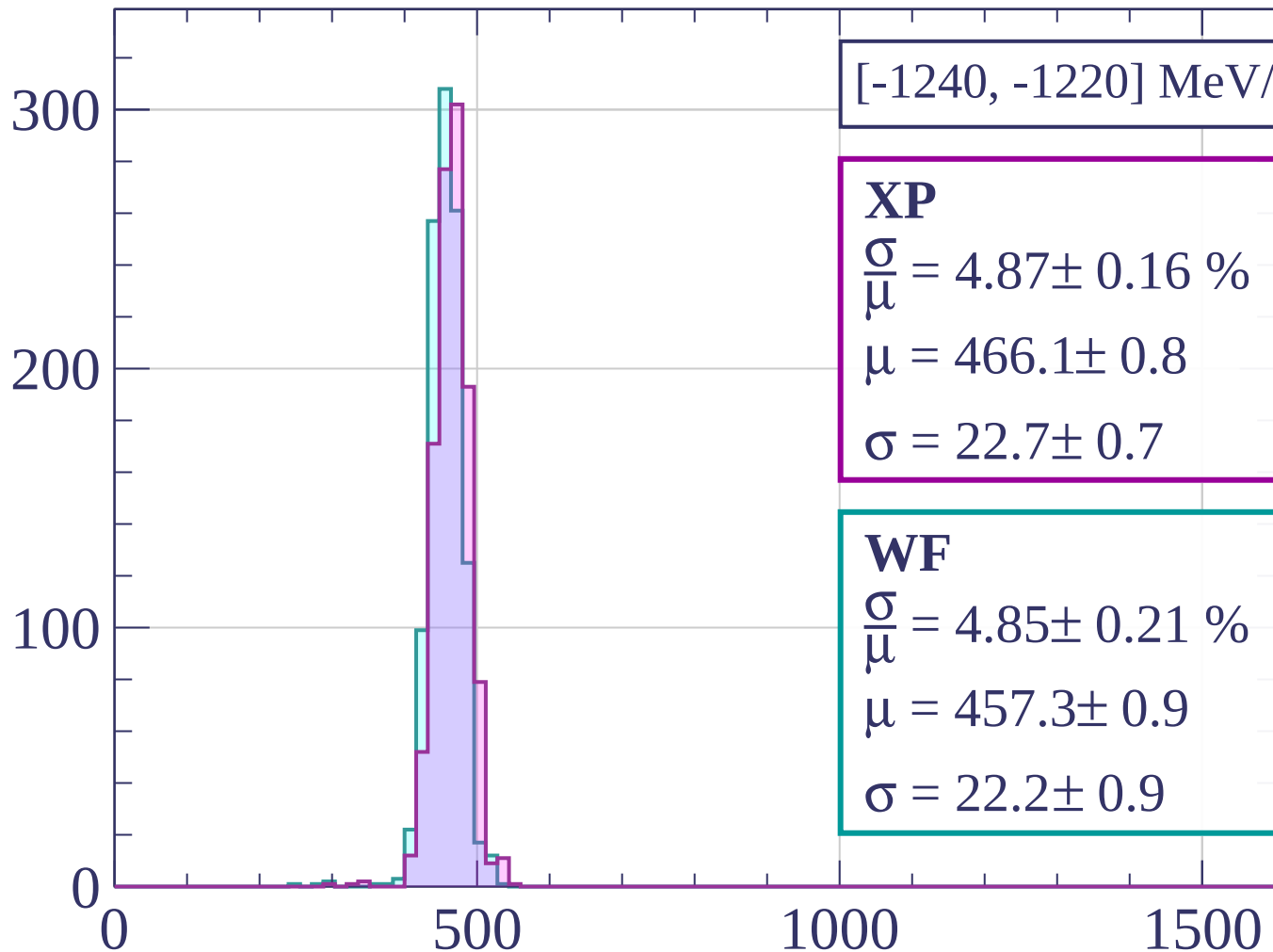
$$\sigma = 22.6 \pm 0.7$$

dE/dx [ADC counts/cm]



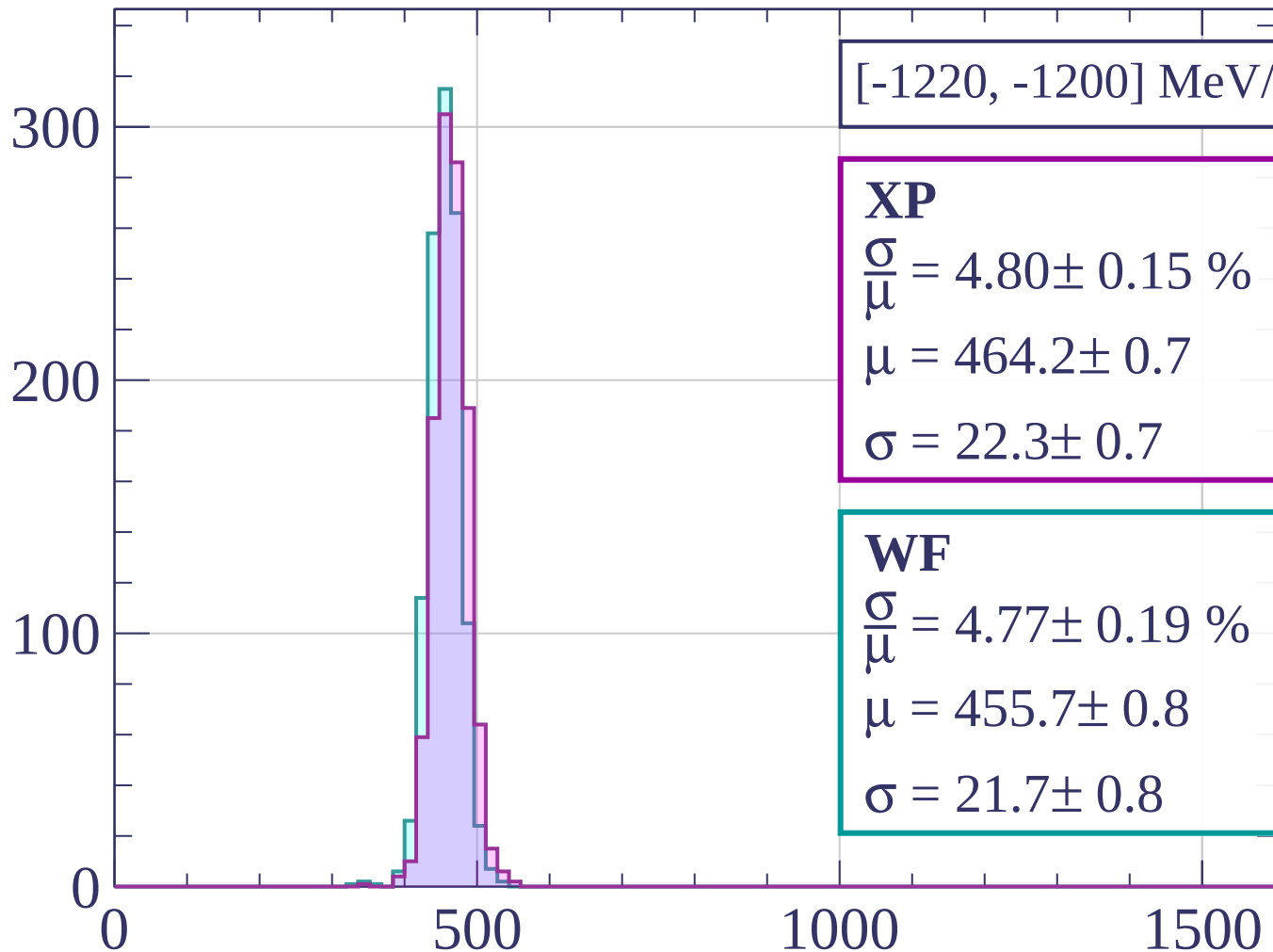


Count

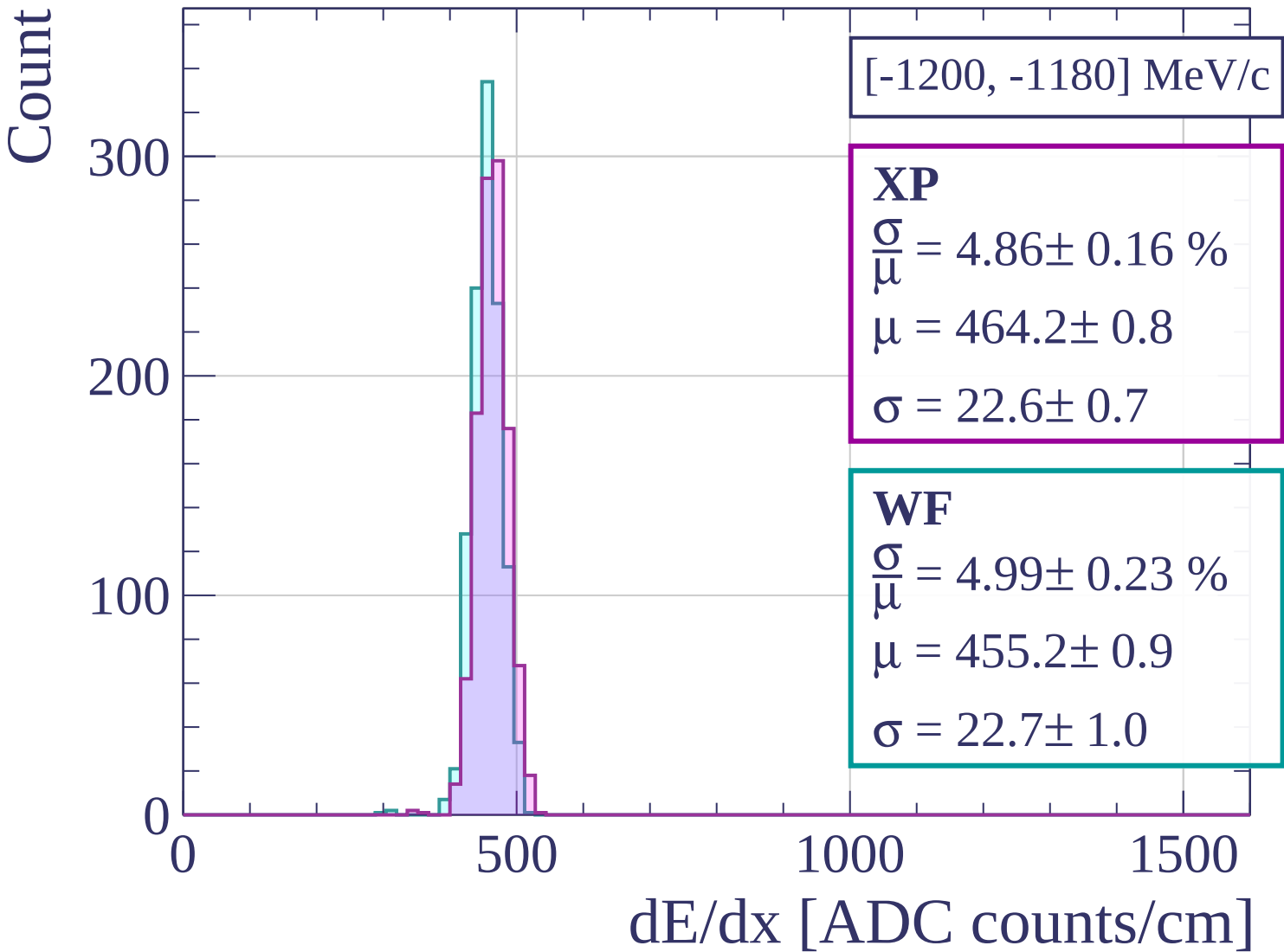


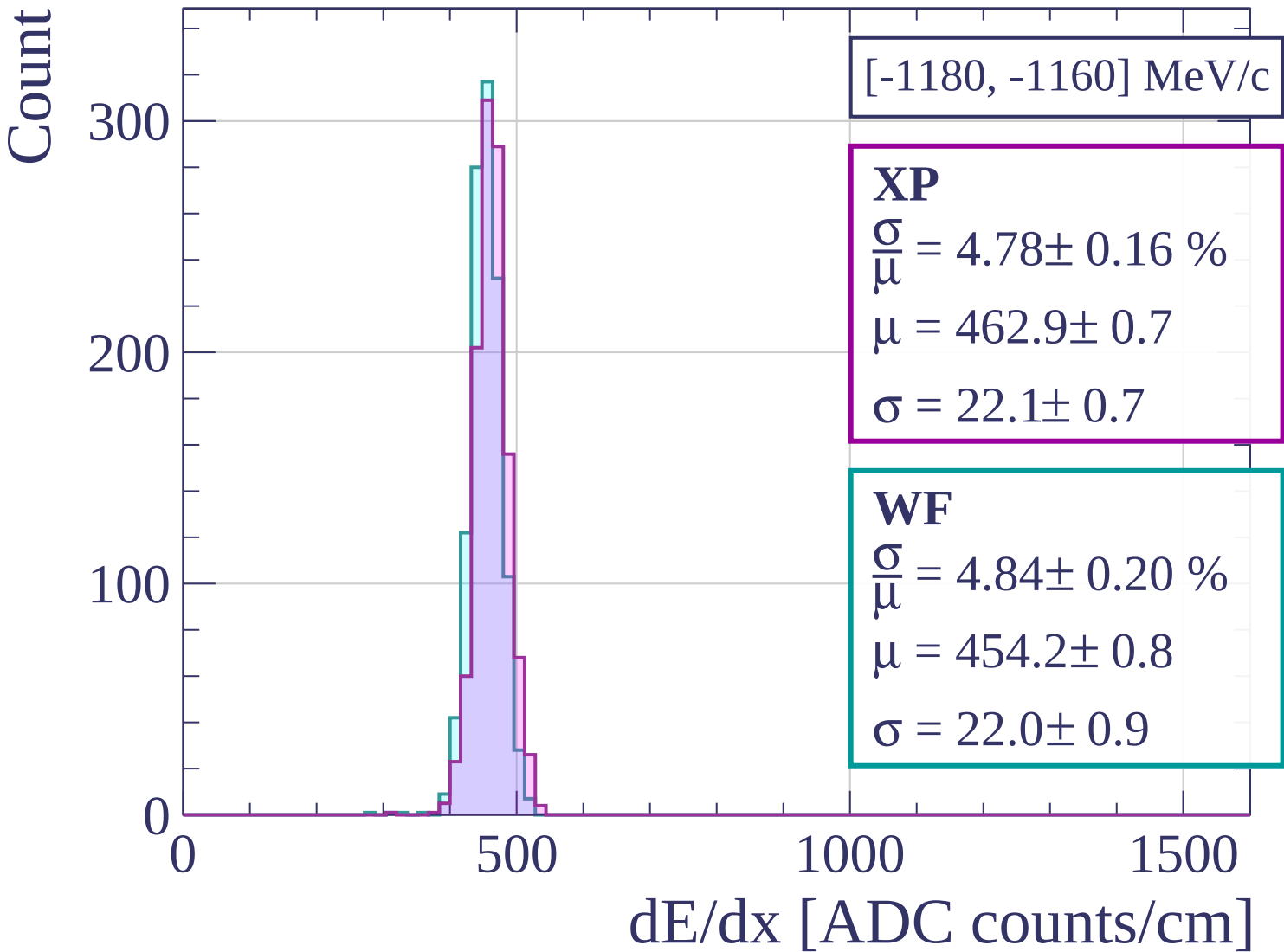
dE/dx [ADC counts/cm]

Count

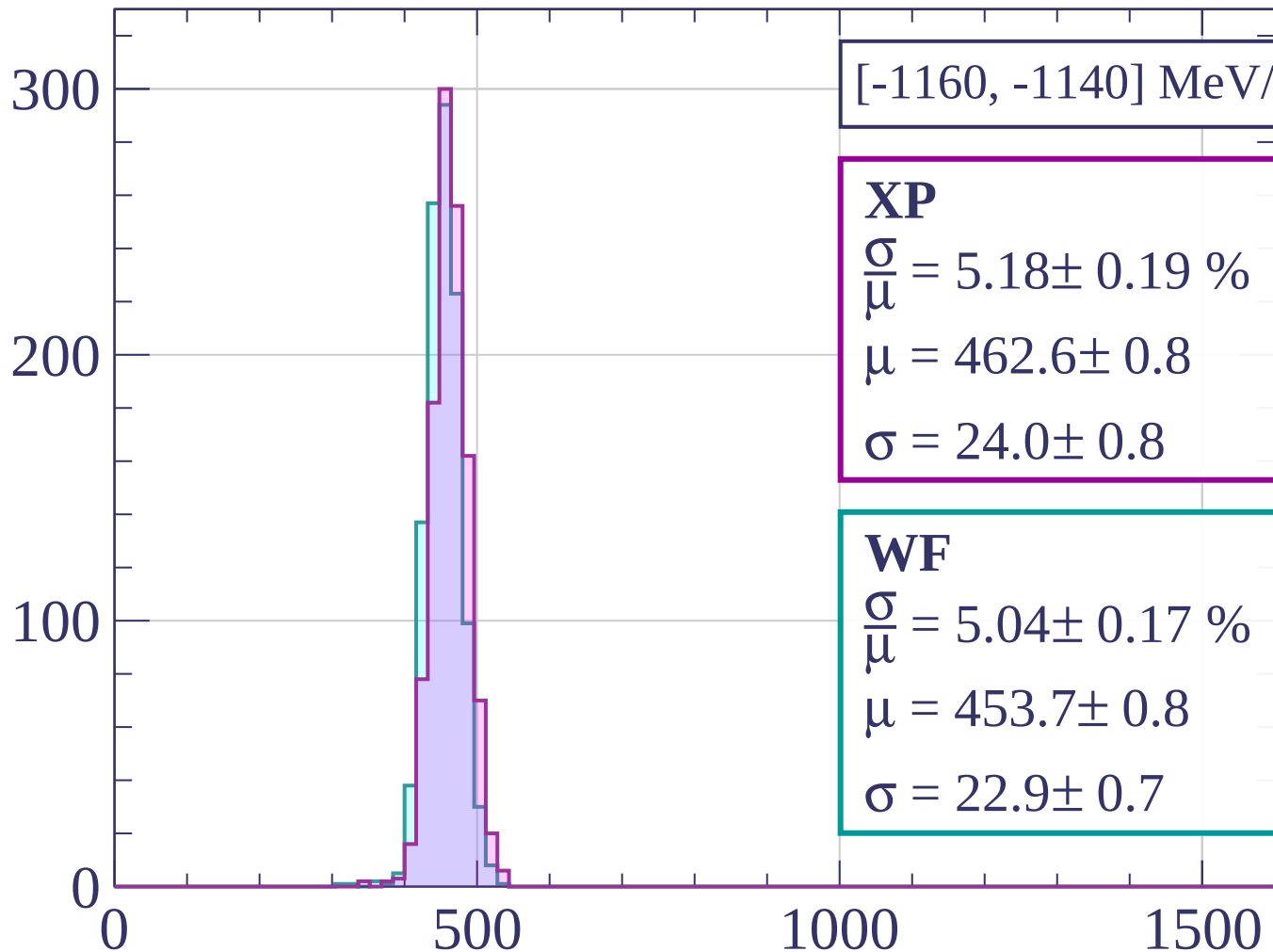


dE/dx [ADC counts/cm]





Count



$[-1160, -1140] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 5.18 \pm 0.19 \%$$

$$\mu = 462.6 \pm 0.8$$

$$\sigma = 24.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.04 \pm 0.17 \%$$

$$\mu = 453.7 \pm 0.8$$

$$\sigma = 22.9 \pm 0.7$$

dE/dx [ADC counts/cm]

Count

[-1140, -1120] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.59 \pm 0.18 \%$$

$$\mu = 460.3 \pm 0.8$$

$$\sigma = 21.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 4.77 \pm 0.19 \%$$

$$\mu = 452.1 \pm 0.8$$

$$\sigma = 21.6 \pm 0.8$$

300

200

100

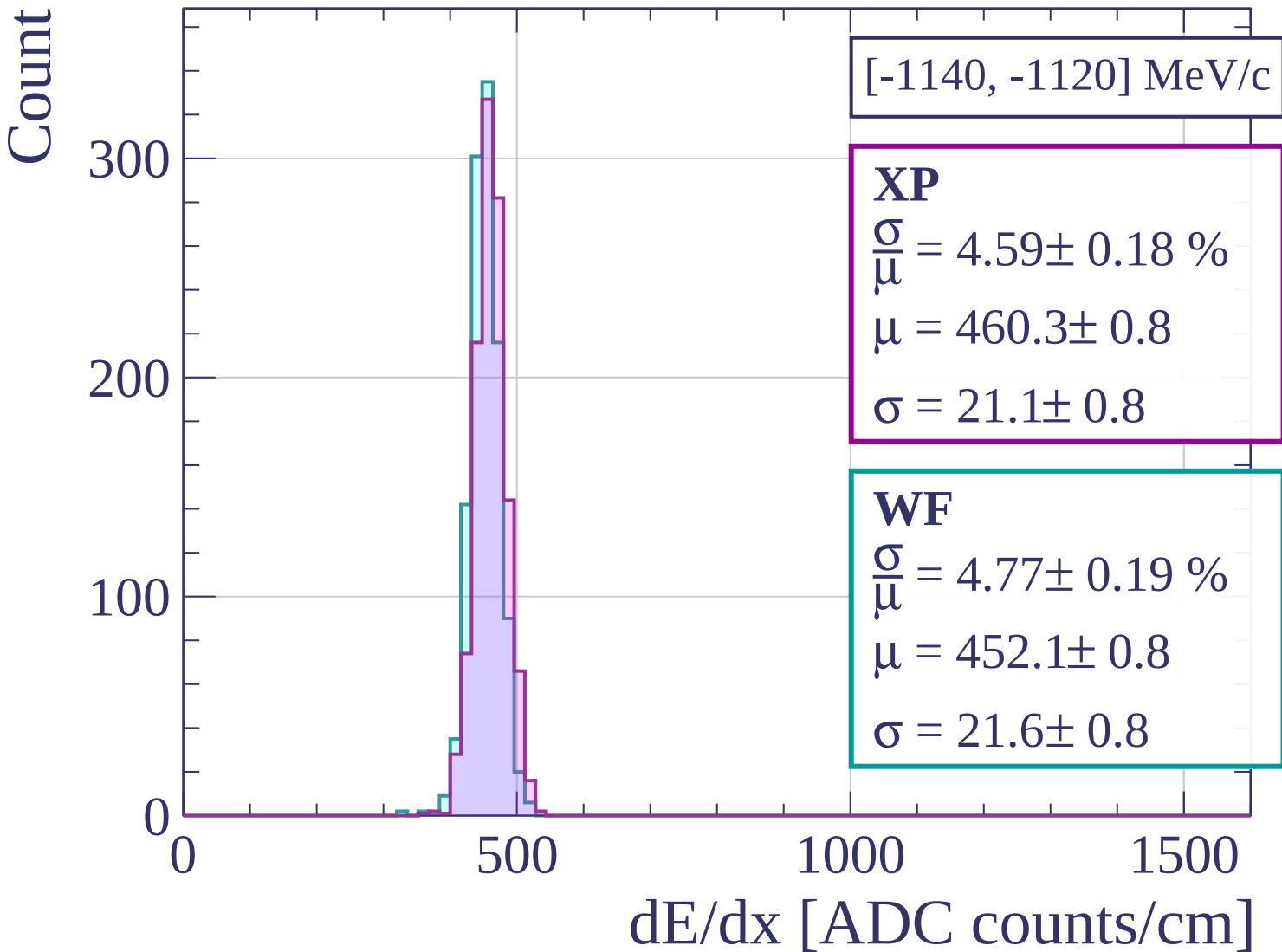
0

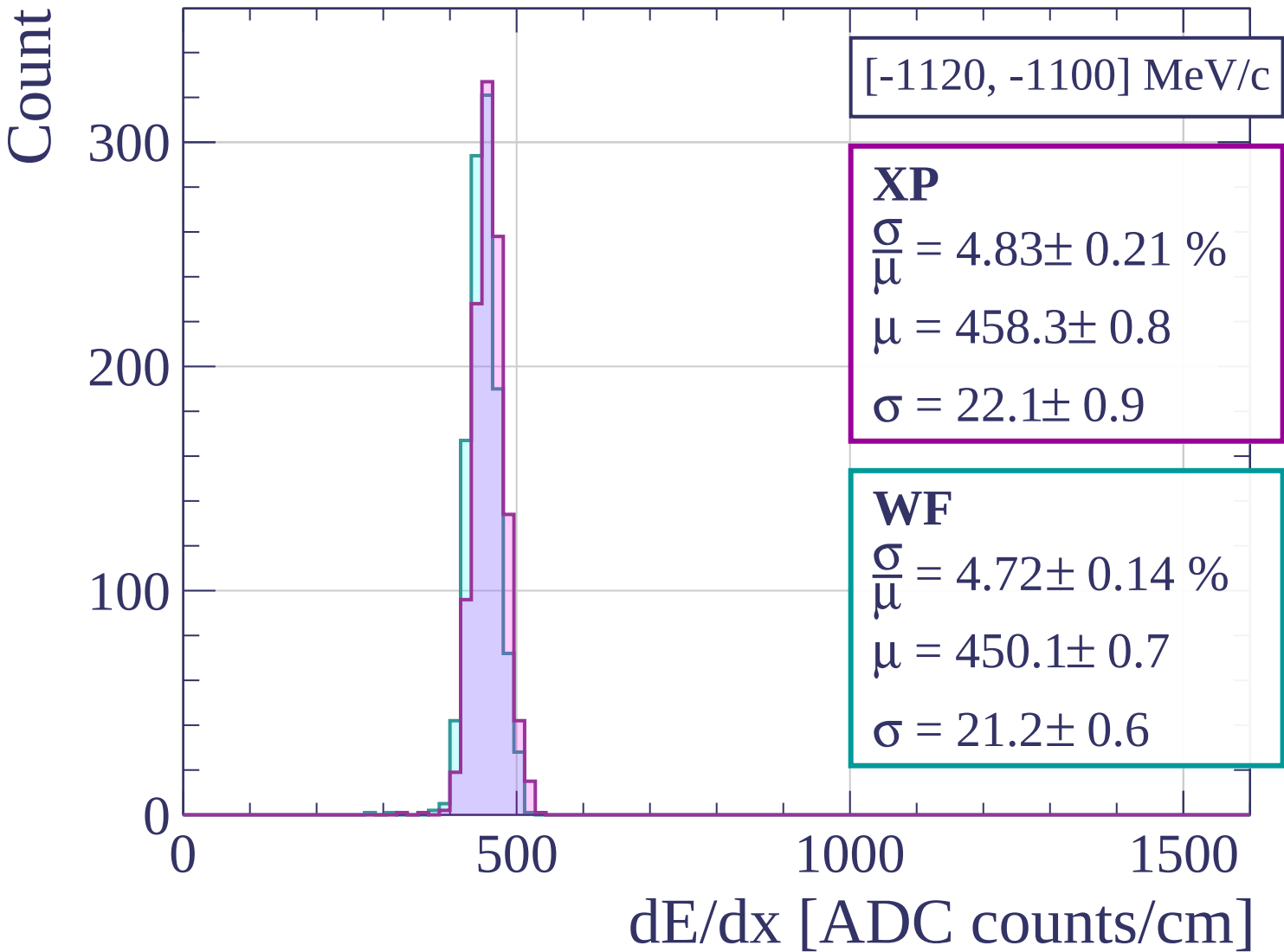
500

1000

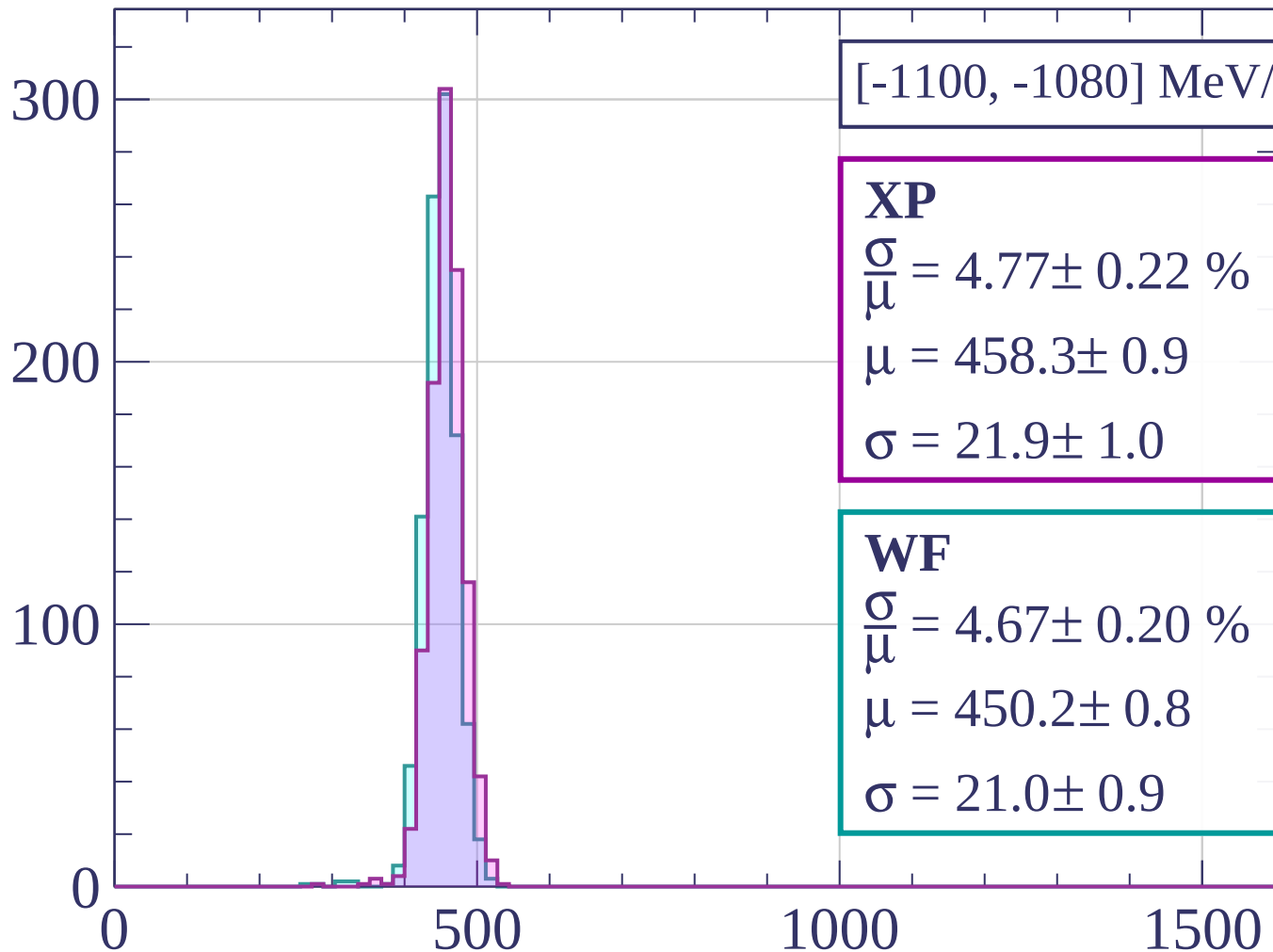
1500

dE/dx [ADC counts/cm]



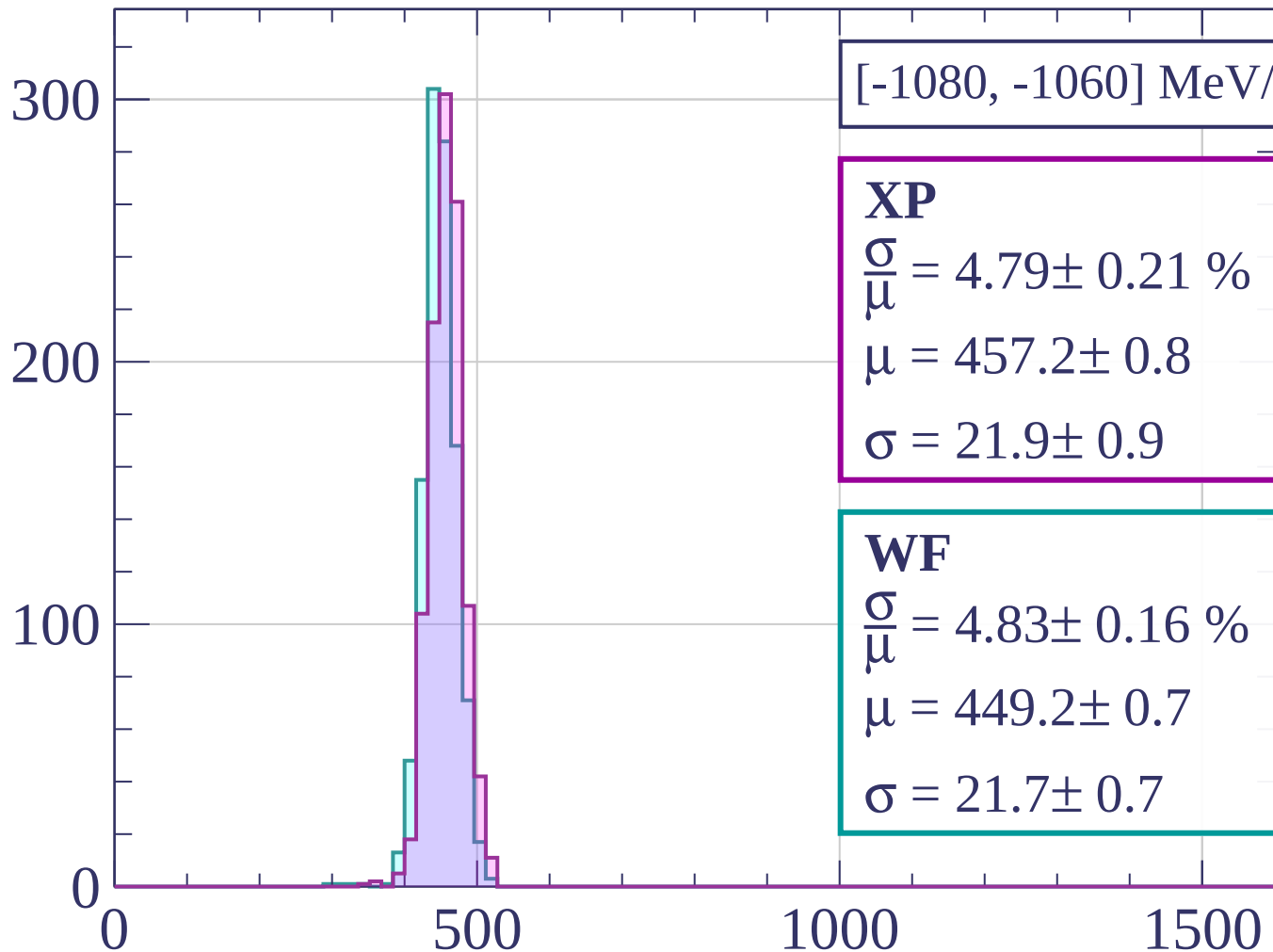


Count



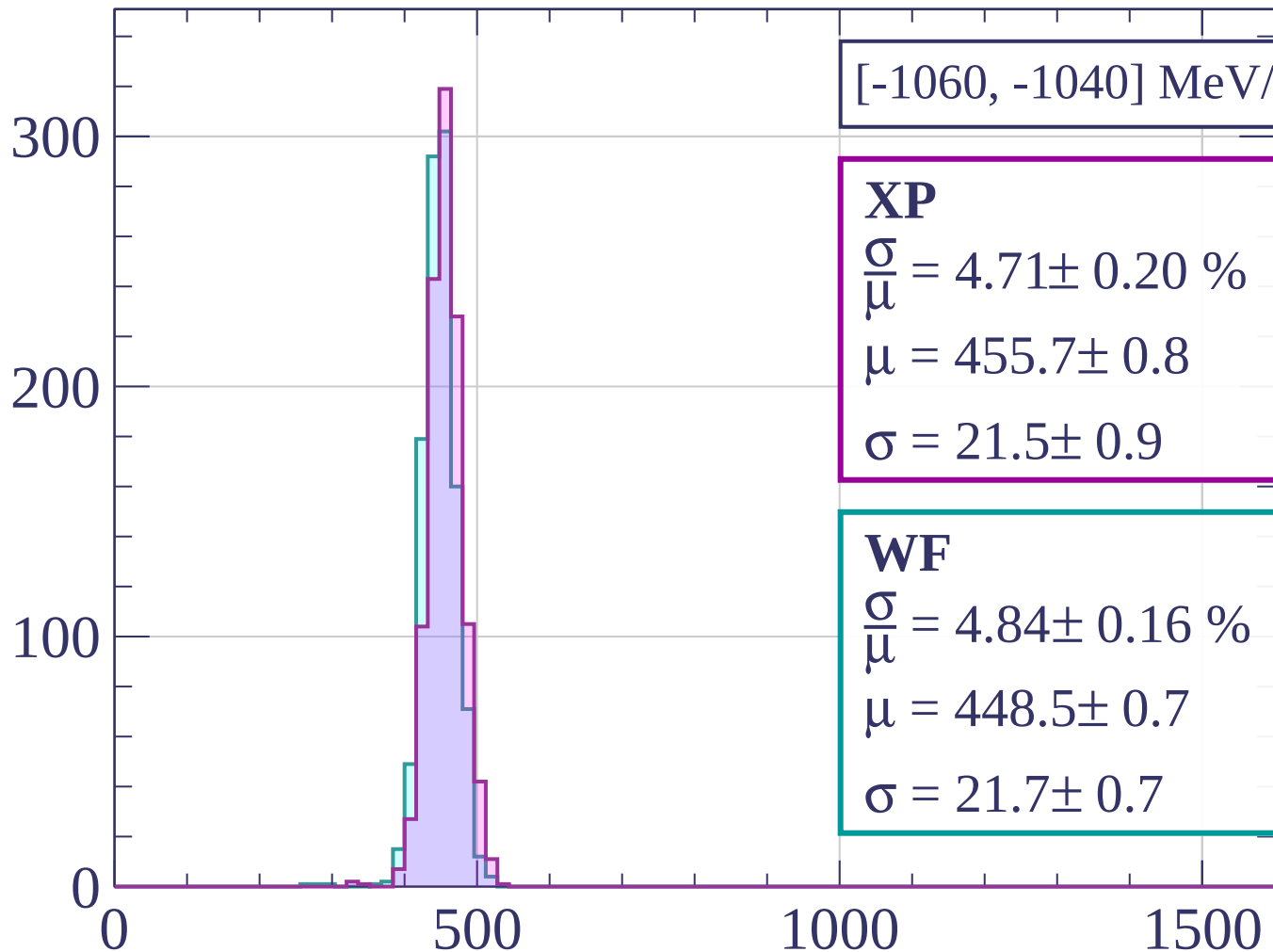
dE/dx [ADC counts/cm]

Count

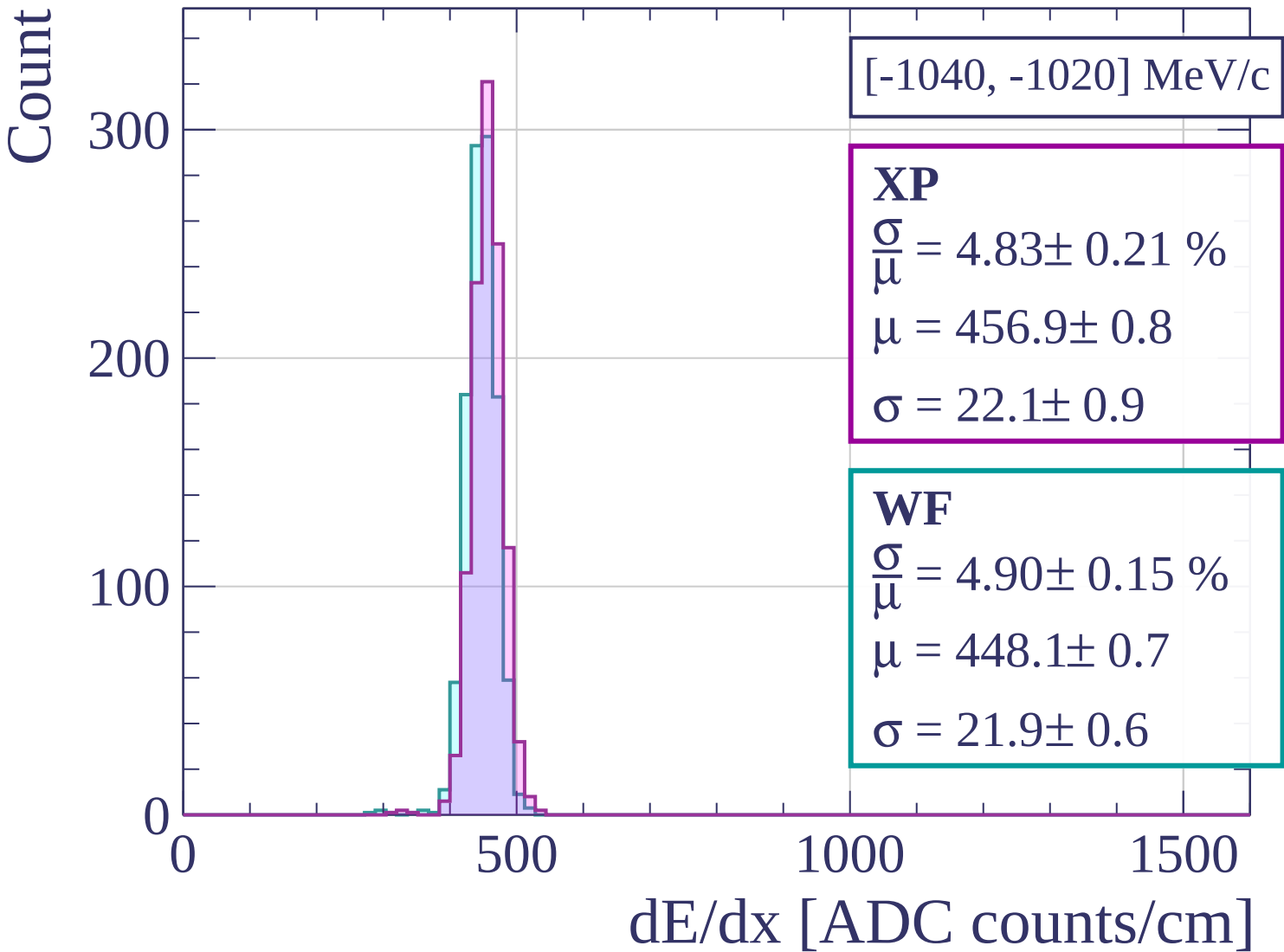


dE/dx [ADC counts/cm]

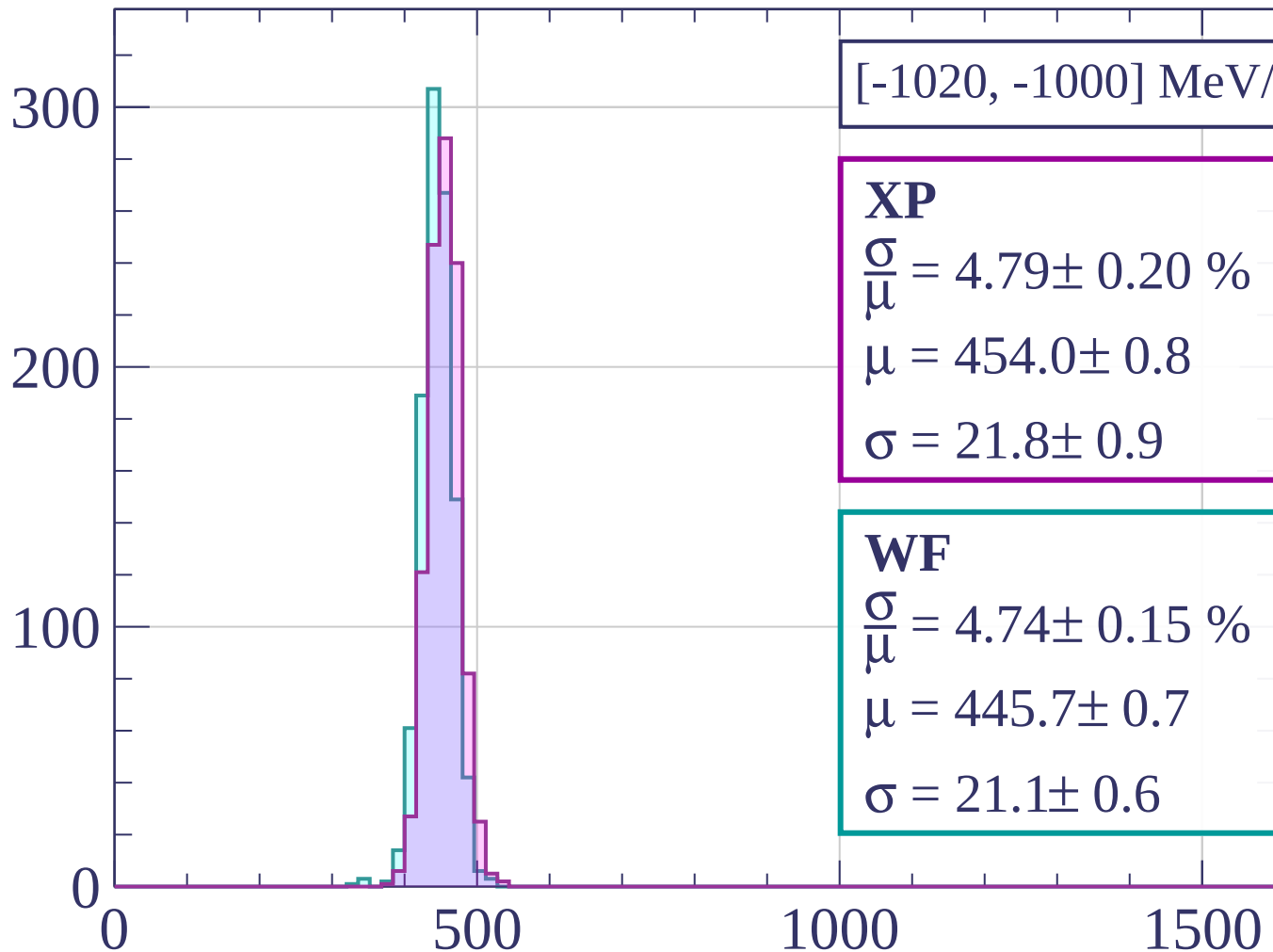
Count



dE/dx [ADC counts/cm]



Count



dE/dx [ADC counts/cm]

Count

[-1000, -980] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.82 \pm 0.20 \%$$

$$\mu = 453.2 \pm 0.8$$

$$\sigma = 21.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.84 \pm 0.15 \%$$

$$\mu = 444.8 \pm 0.7$$

$$\sigma = 21.5 \pm 0.6$$

300

200

100

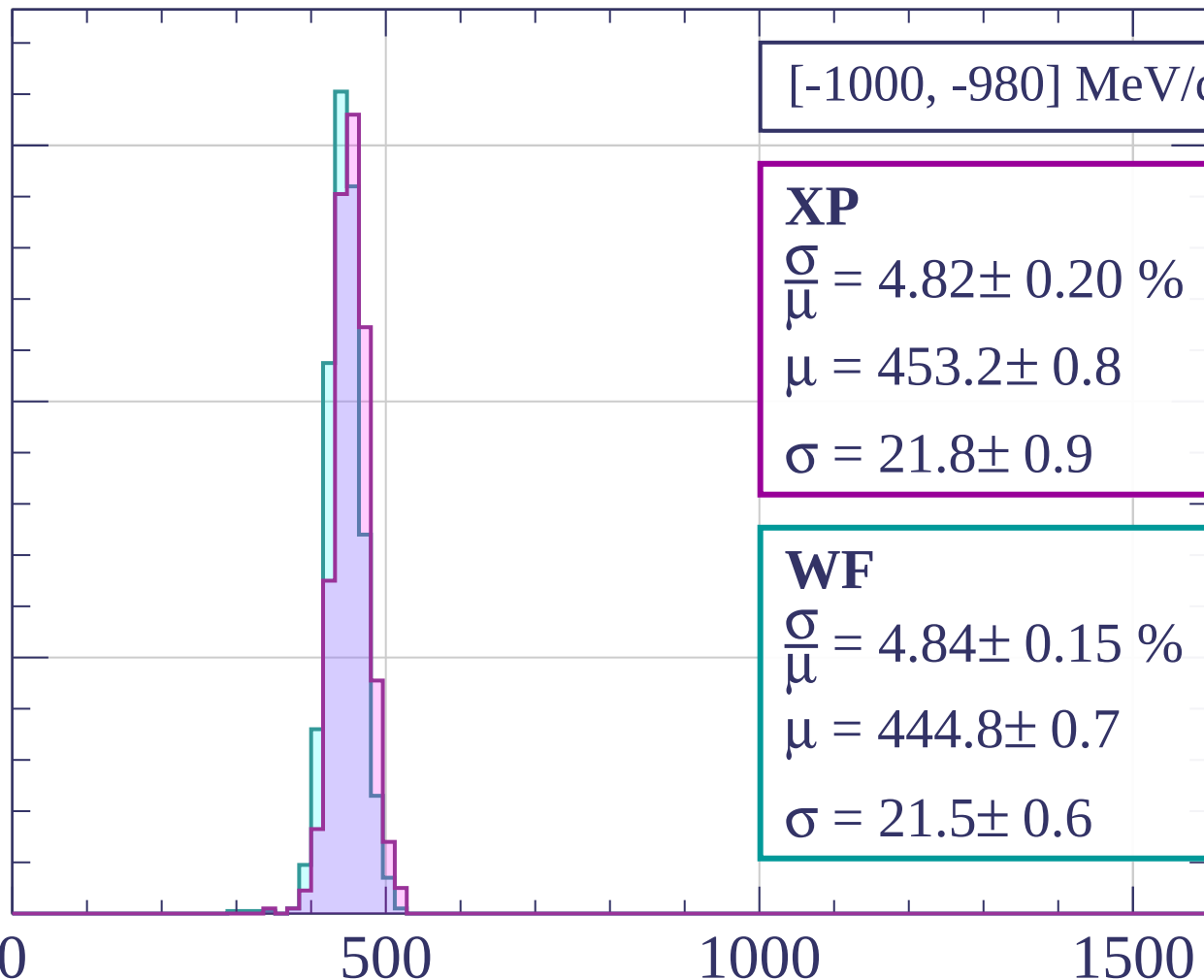
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-980, -960] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.91 \pm 0.16 \%$$

$$\mu = 451.3 \pm 0.8$$

$$\sigma = 22.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.77 \pm 0.19 \%$$

$$\mu = 443.2 \pm 0.8$$

$$\sigma = 21.1 \pm 0.8$$

300

200

100

0

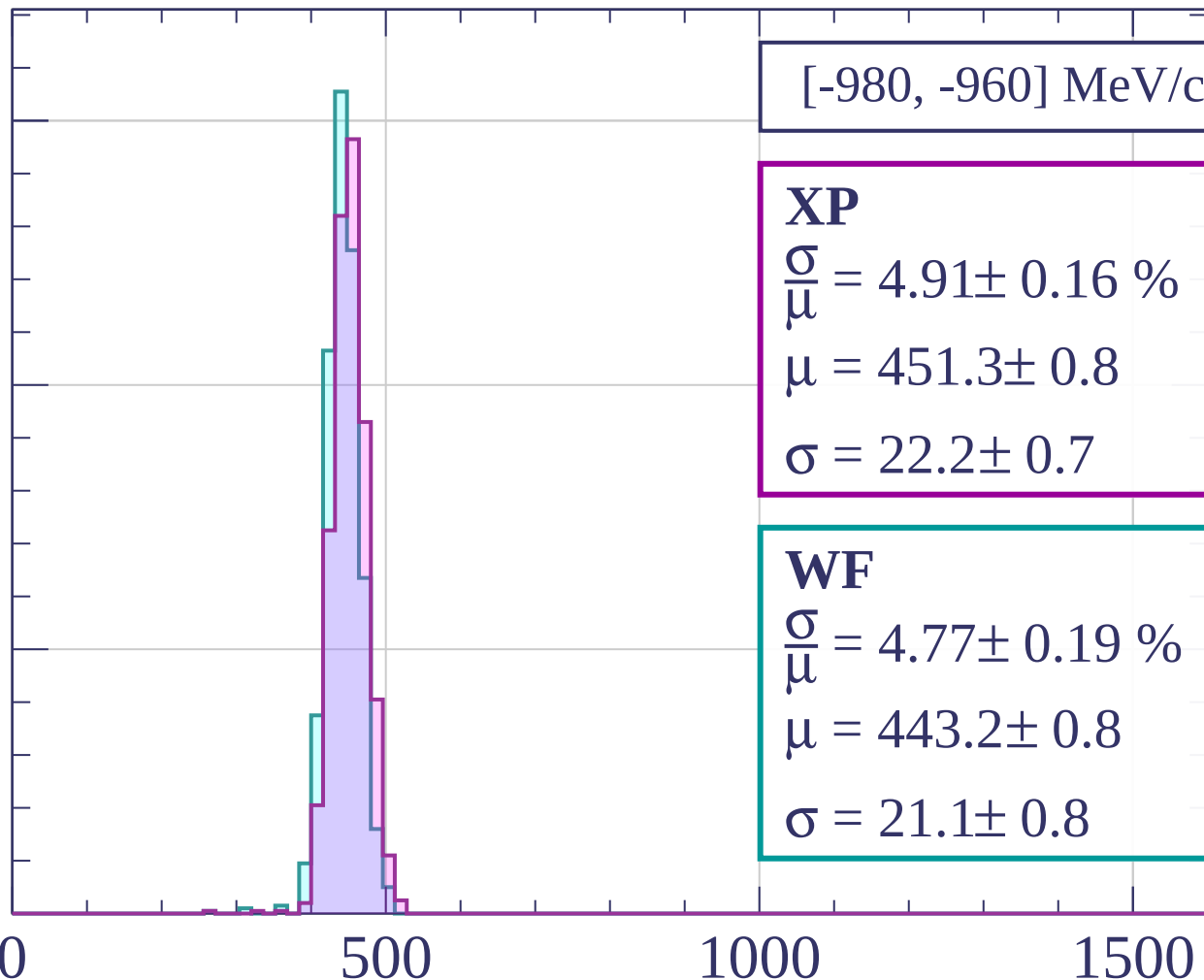
0

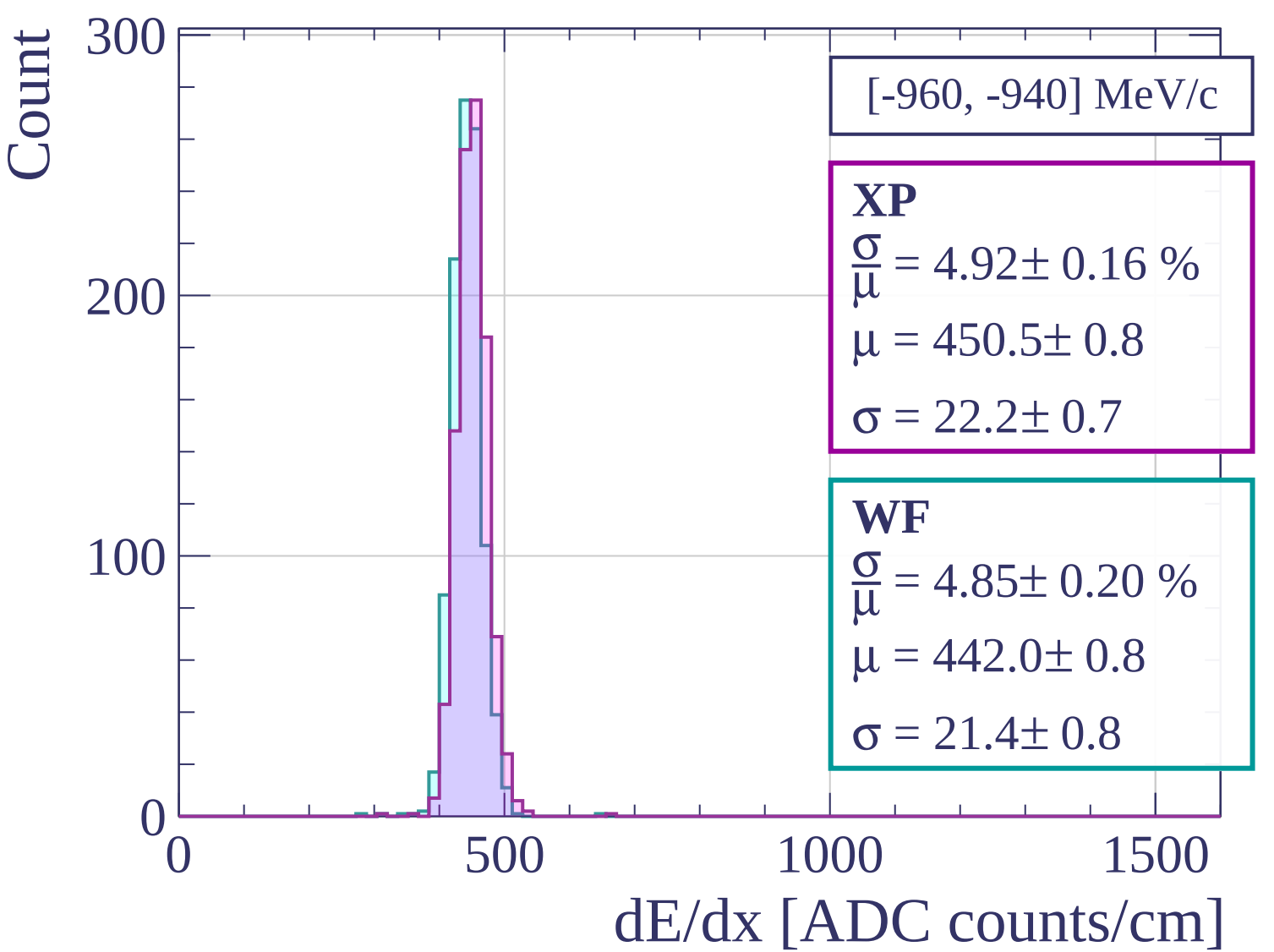
500

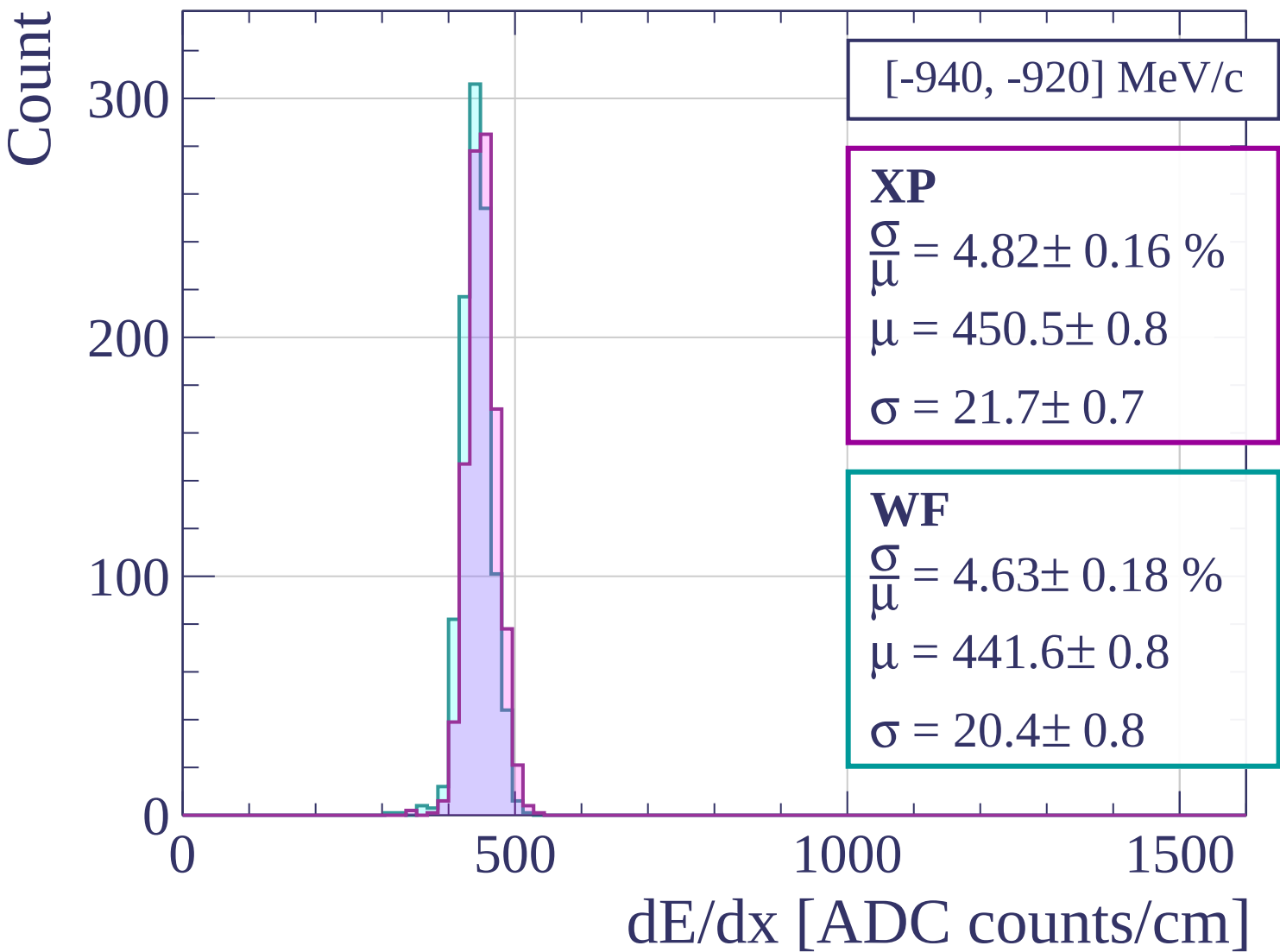
1000

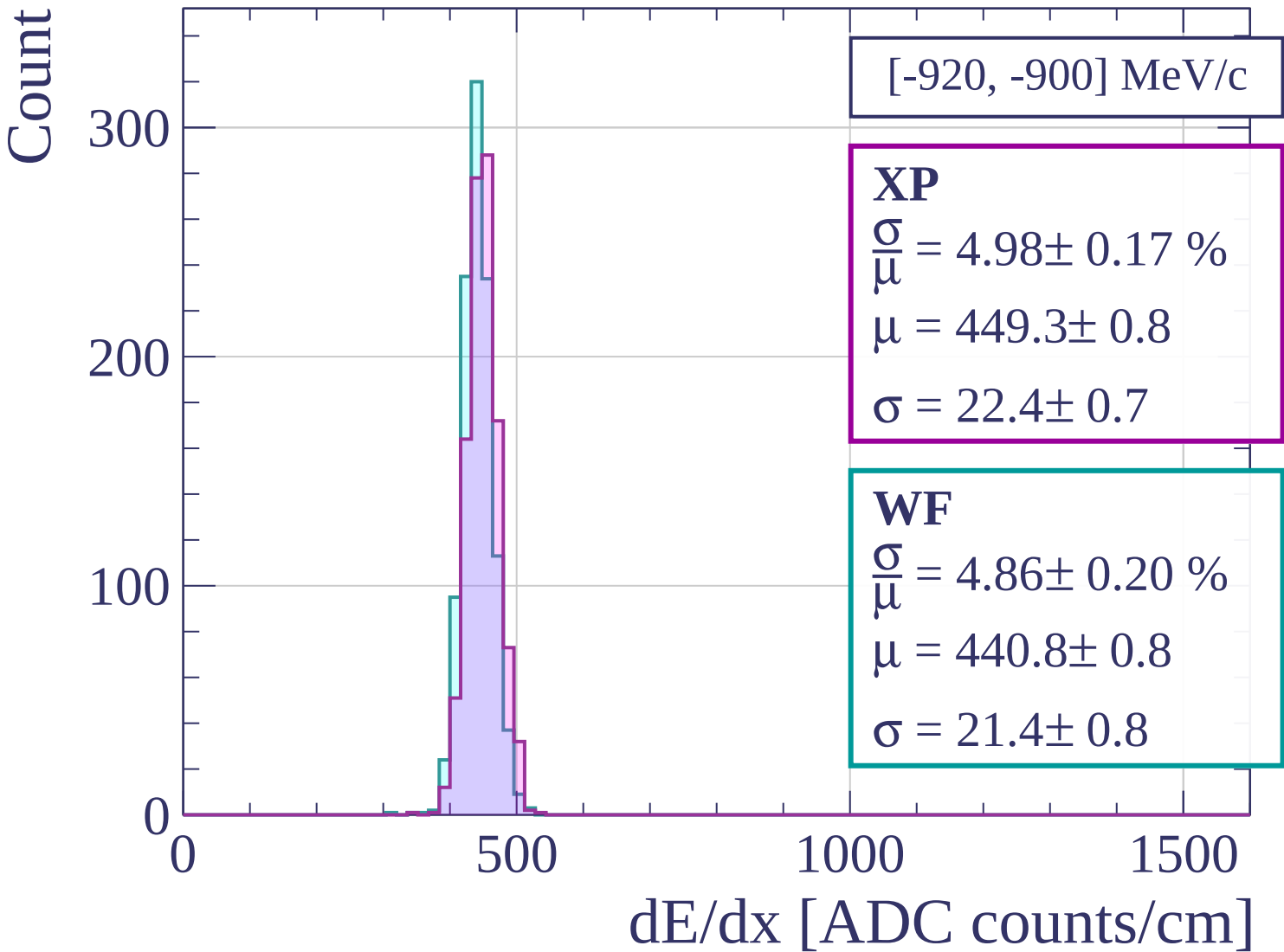
1500

dE/dx [ADC counts/cm]









Count

[-900, -880] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.81 \pm 0.16 \%$$

$$\mu = 448.8 \pm 0.7$$

$$\sigma = 21.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.70 \pm 0.19 \%$$

$$\mu = 440.5 \pm 0.8$$

$$\sigma = 20.7 \pm 0.8$$

300

200

100

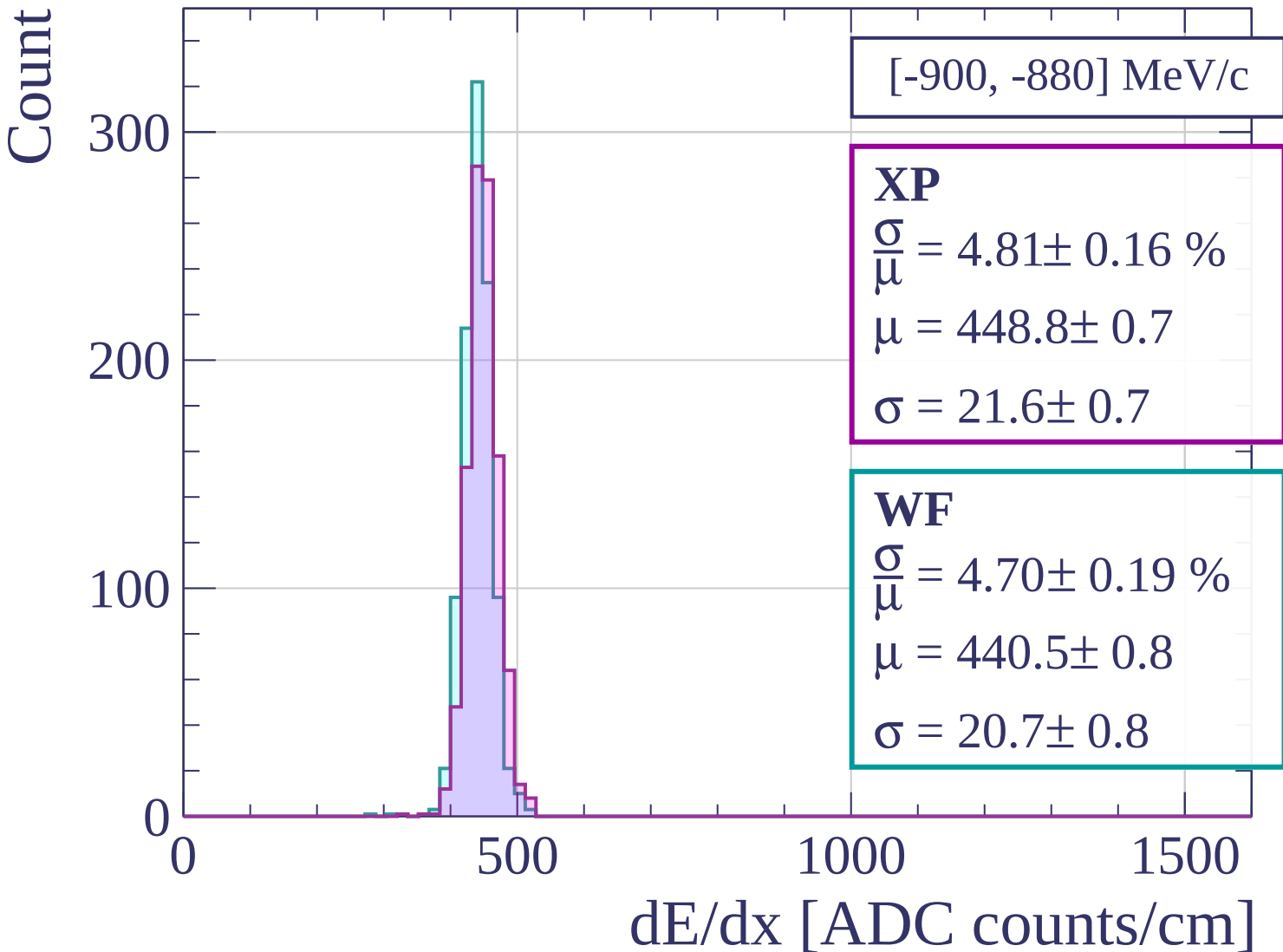
0

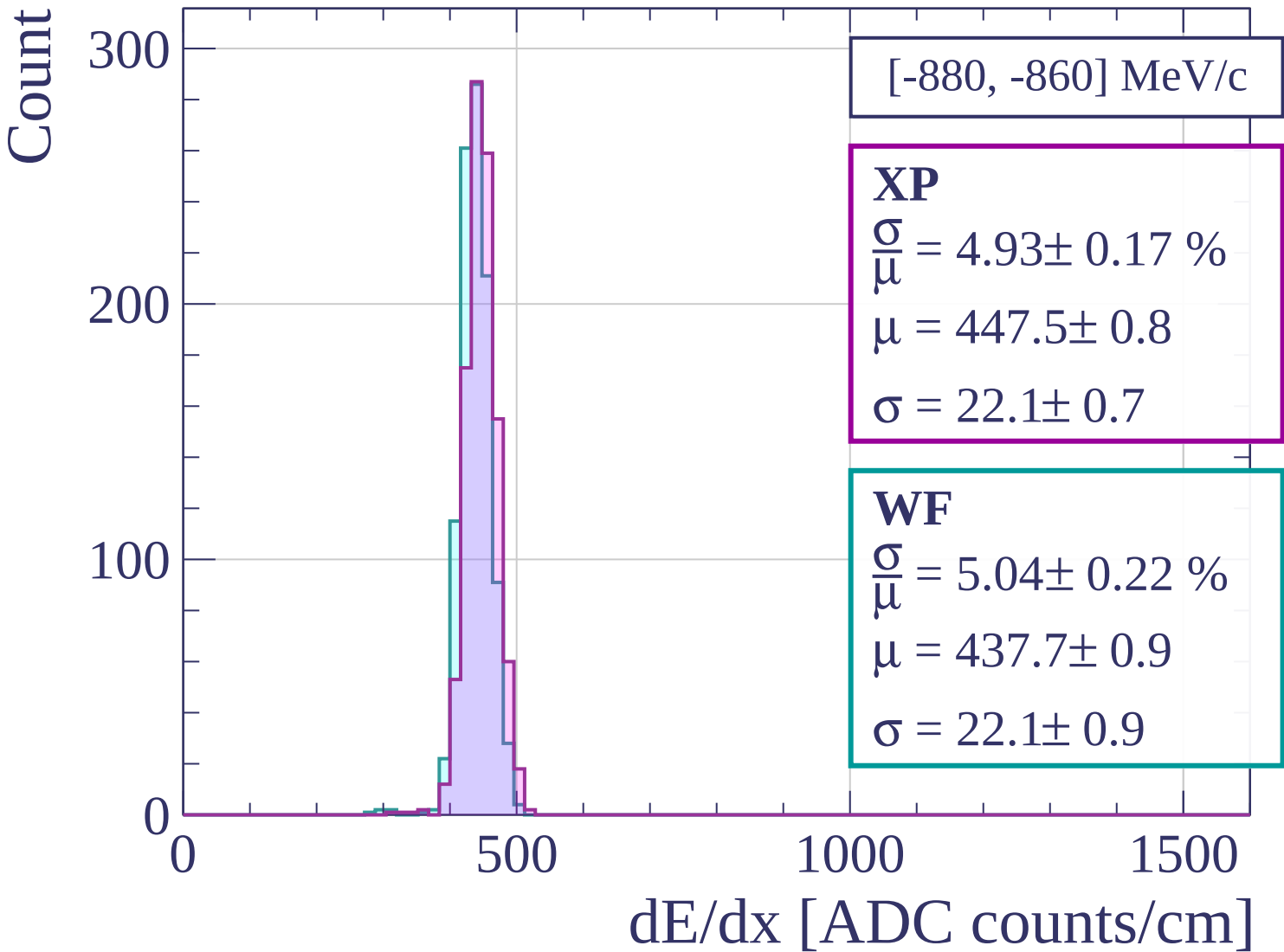
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-860, -840] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.02 \pm 0.17 \%$$

$$\mu = 446.1 \pm 0.8$$

$$\sigma = 22.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.03 \pm 0.23 \%$$

$$\mu = 437.0 \pm 0.9$$

$$\sigma = 22.0 \pm 0.9$$

200

100

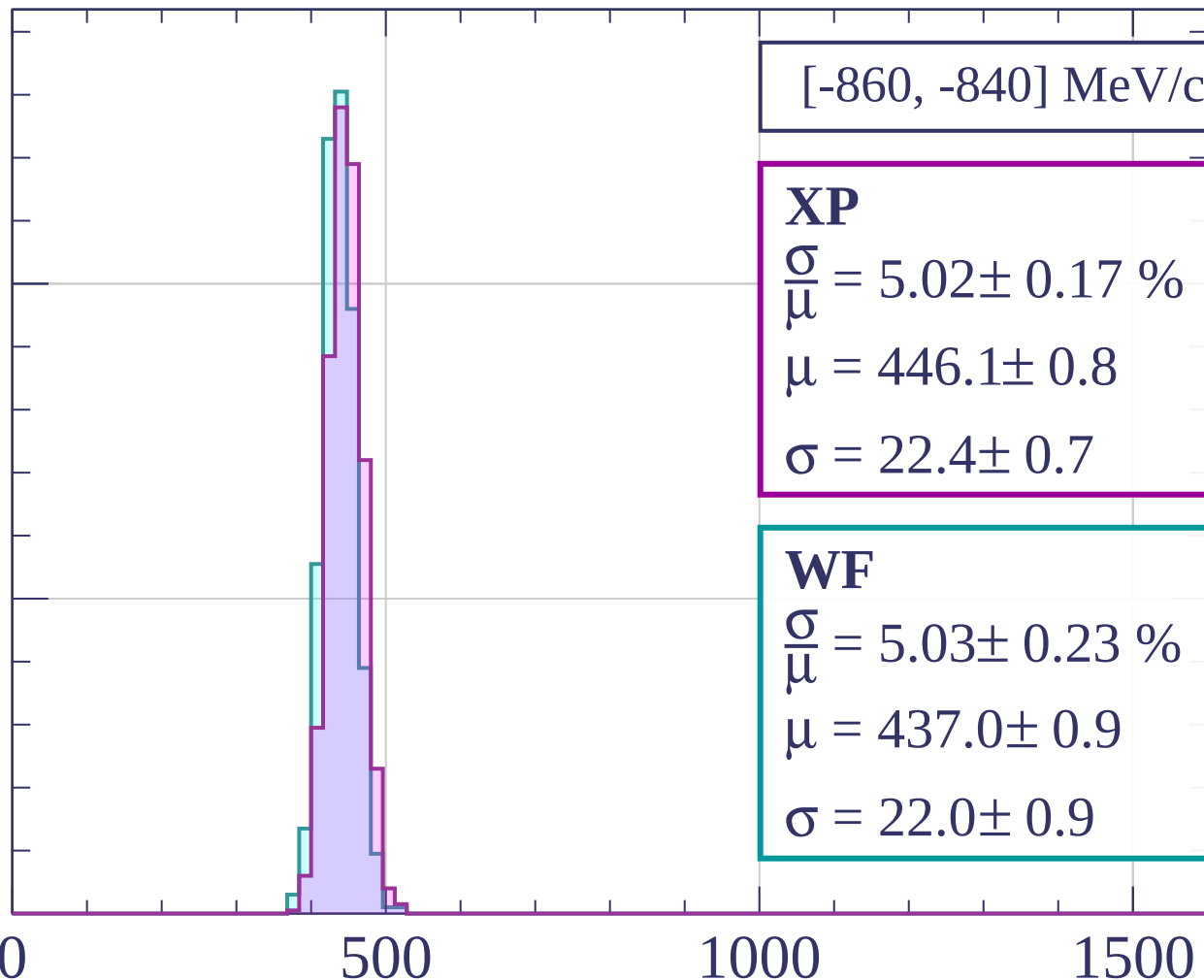
0

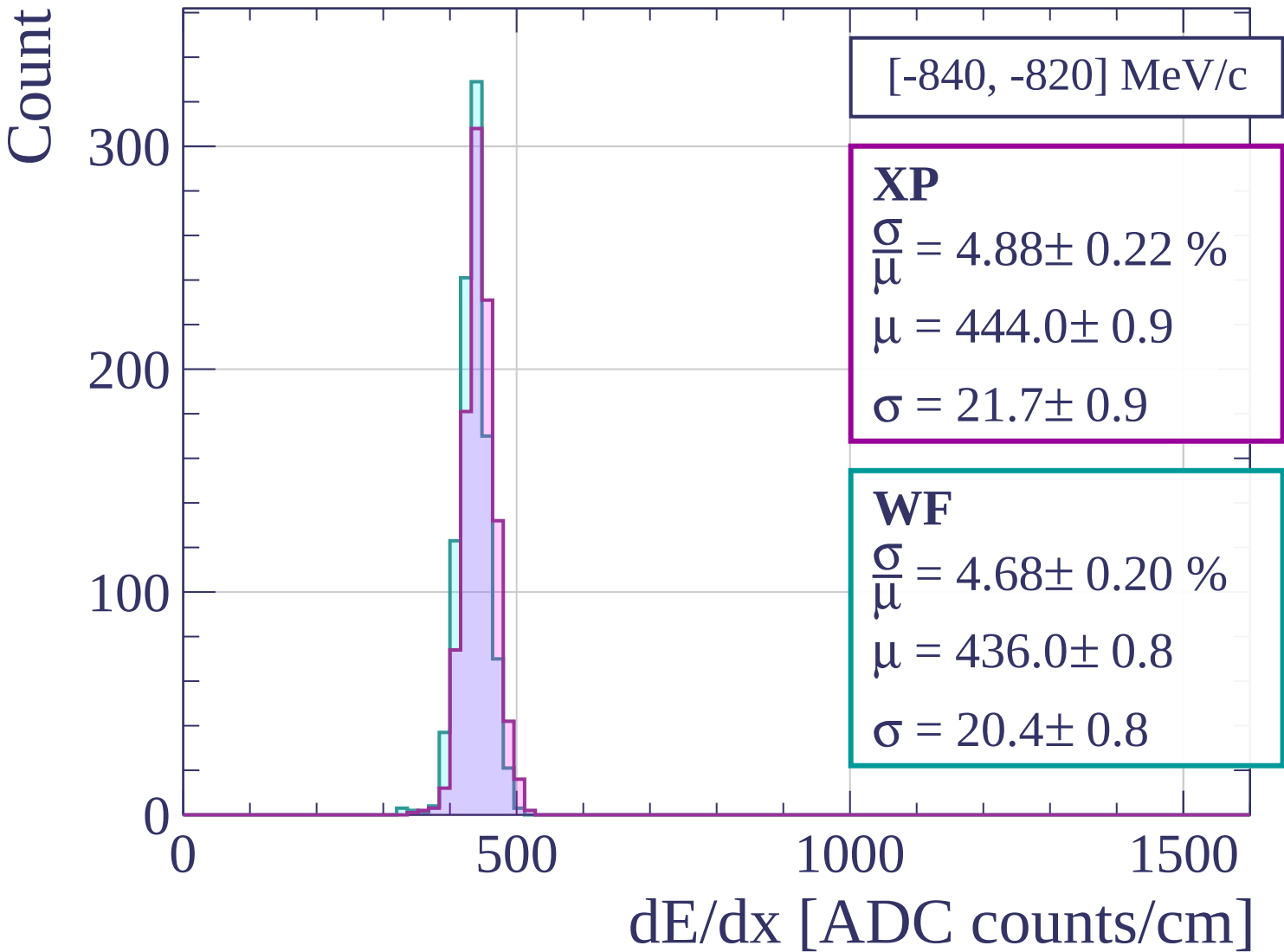
500

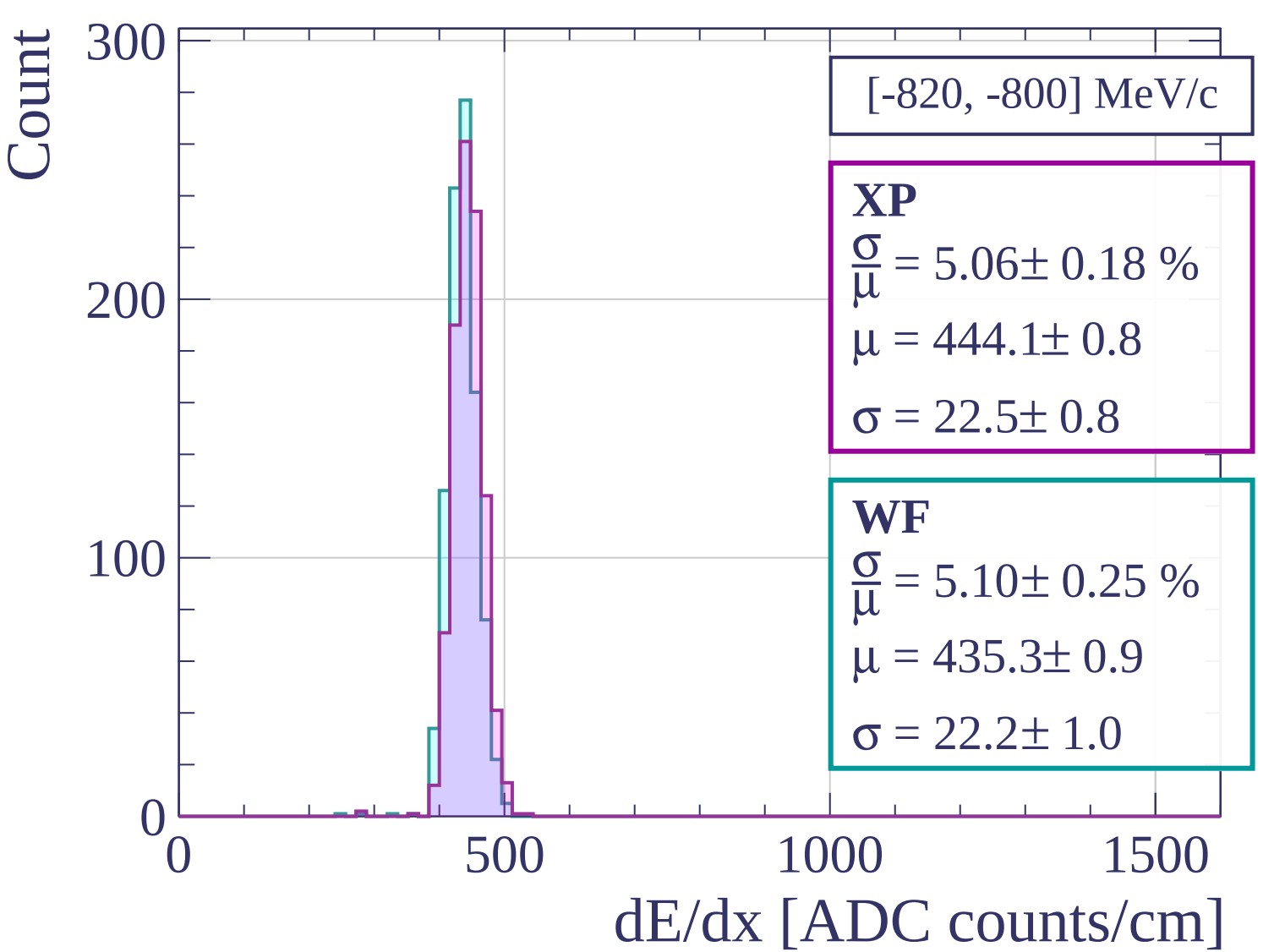
1000

1500

dE/dx [ADC counts/cm]







Count

[-800, -780] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.14 \pm 0.25 \%$$

$$\mu = 441.5 \pm 0.9$$

$$\sigma = 22.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 5.02 \pm 0.17 \%$$

$$\mu = 434.0 \pm 0.8$$

$$\sigma = 21.8 \pm 0.7$$

200

100

0

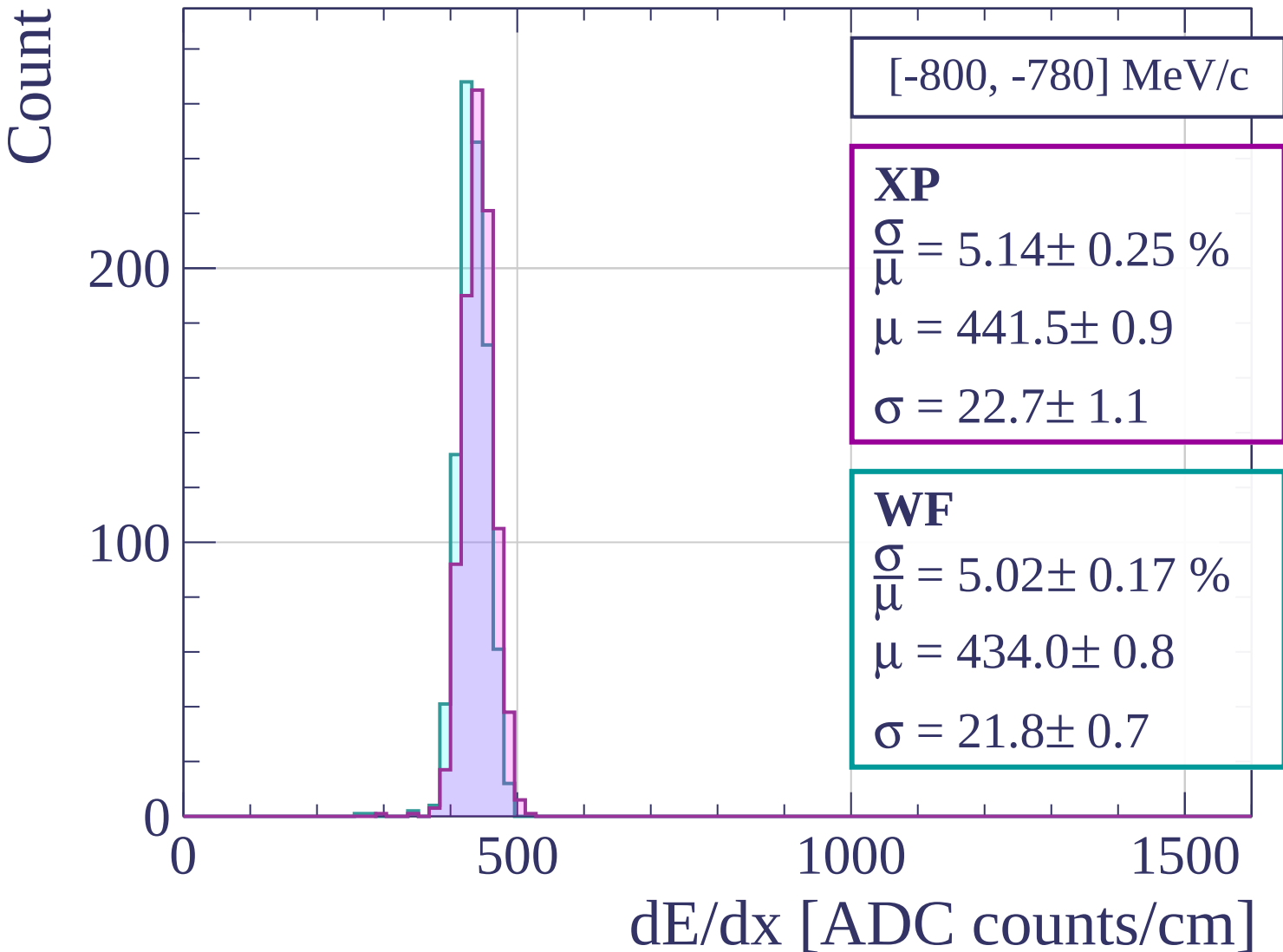
0

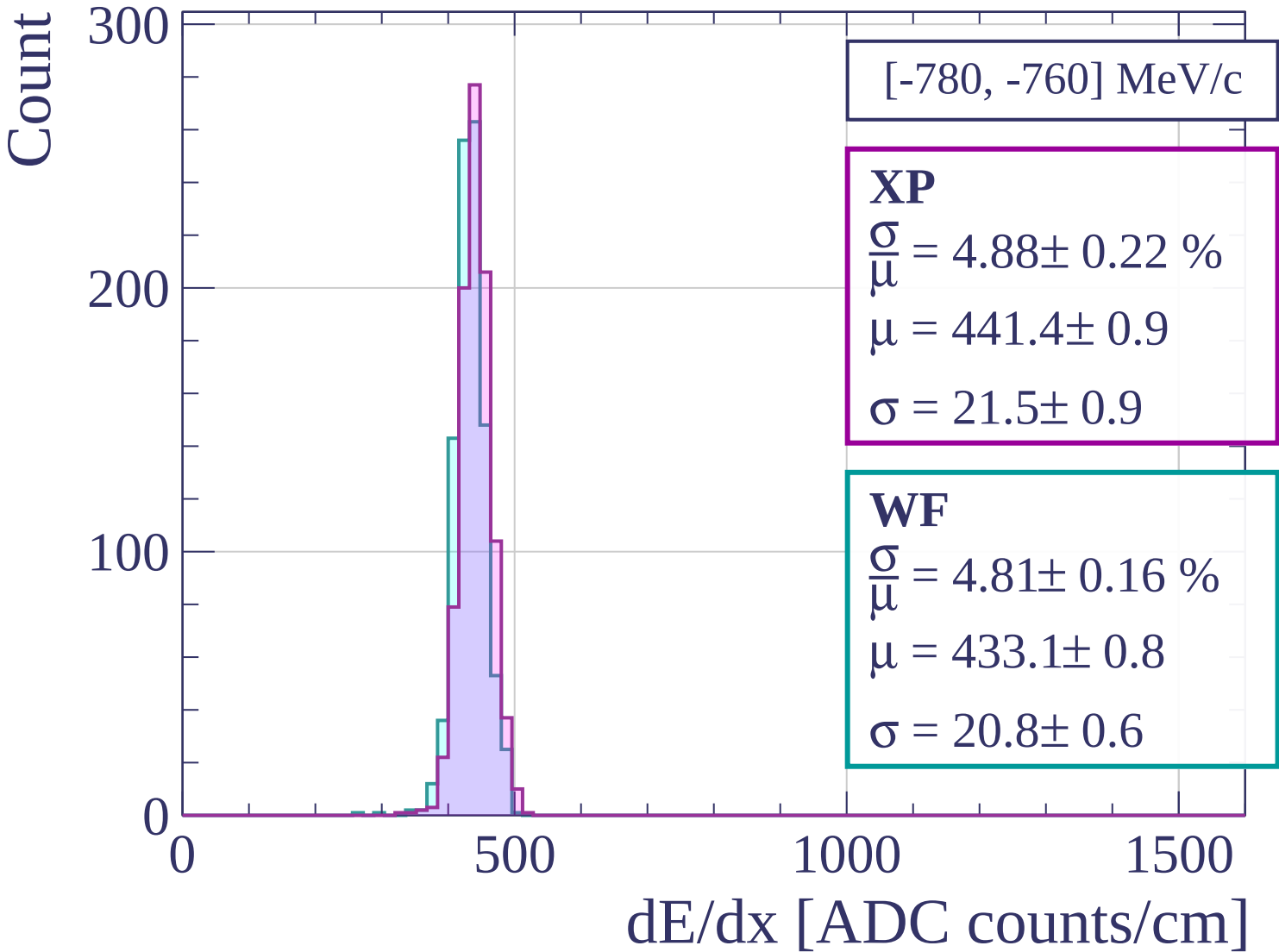
500

1000

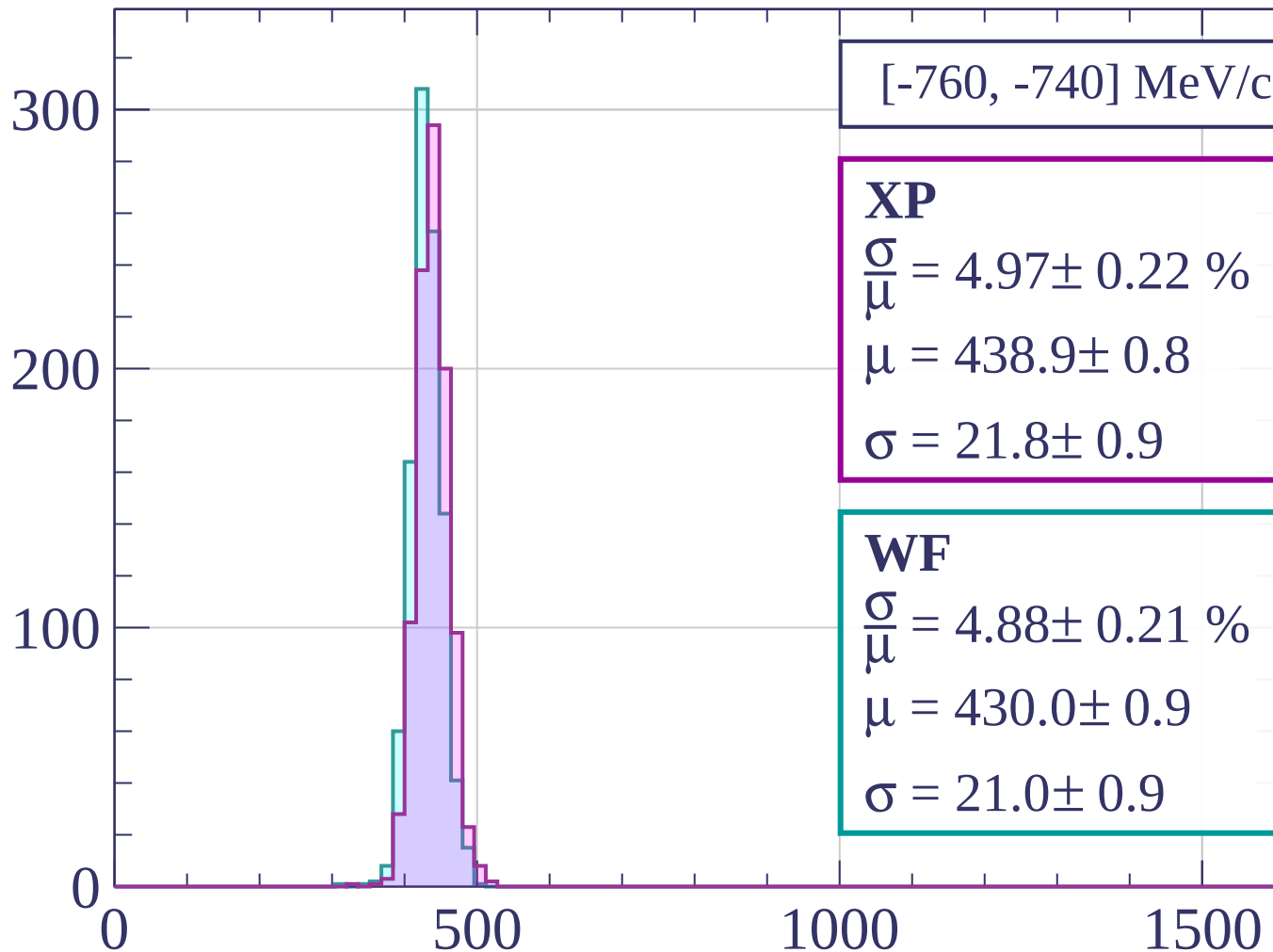
1500

dE/dx [ADC counts/cm]





Count



dE/dx [ADC counts/cm]

Count

[-740, -720] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.11 \pm 0.18 \%$$

$$\mu = 438.1 \pm 0.9$$

$$\sigma = 22.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.16 \pm 0.18 \%$$

$$\mu = 429.6 \pm 0.8$$

$$\sigma = 22.1 \pm 0.7$$

200

100

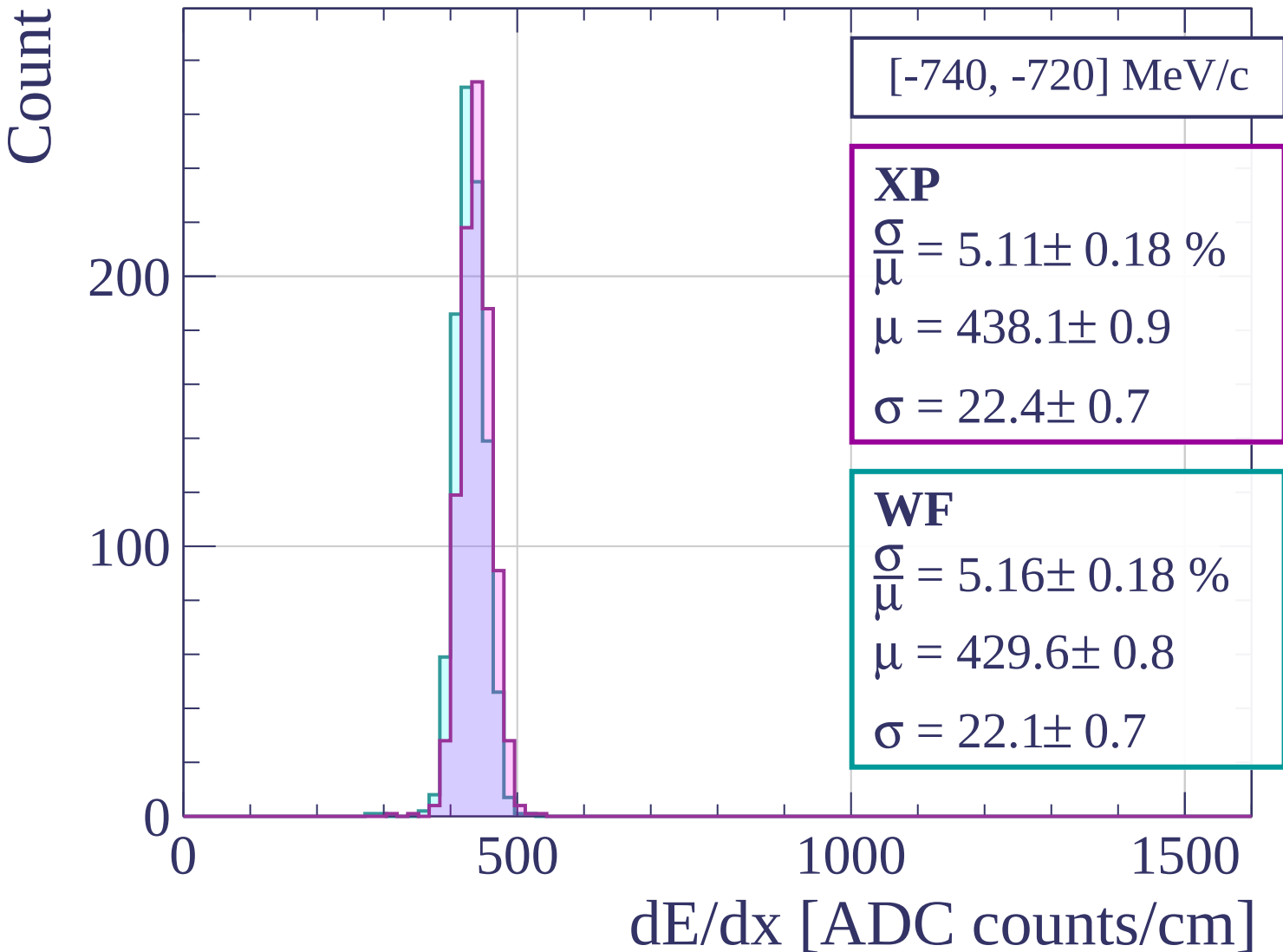
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-720, -700] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.96 \pm 0.17 \%$$

$$\mu = 437.0 \pm 0.8$$

$$\sigma = 21.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.73 \pm 0.19 \%$$

$$\mu = 427.8 \pm 0.8$$

$$\sigma = 20.2 \pm 0.8$$

200

100

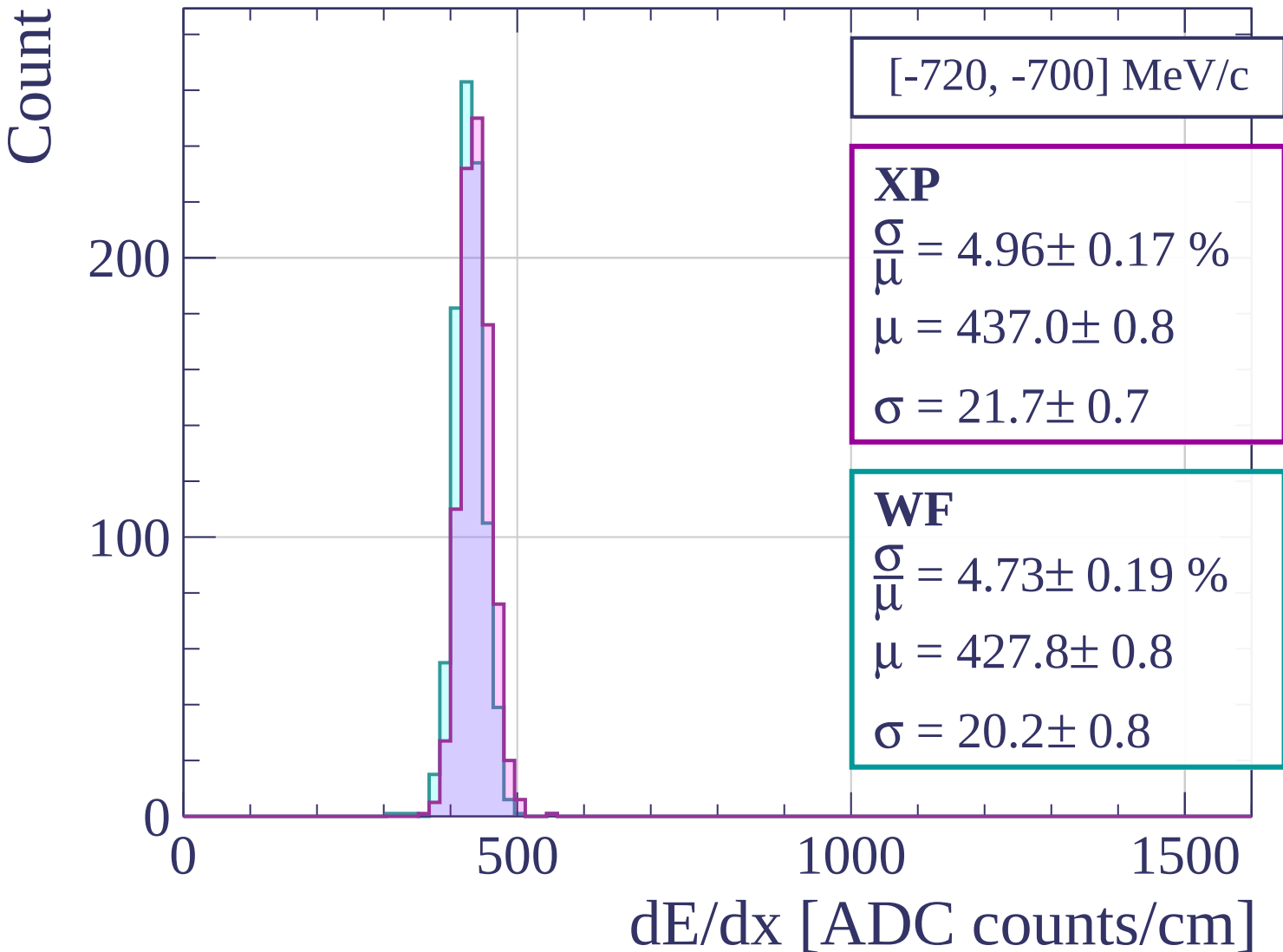
0

500

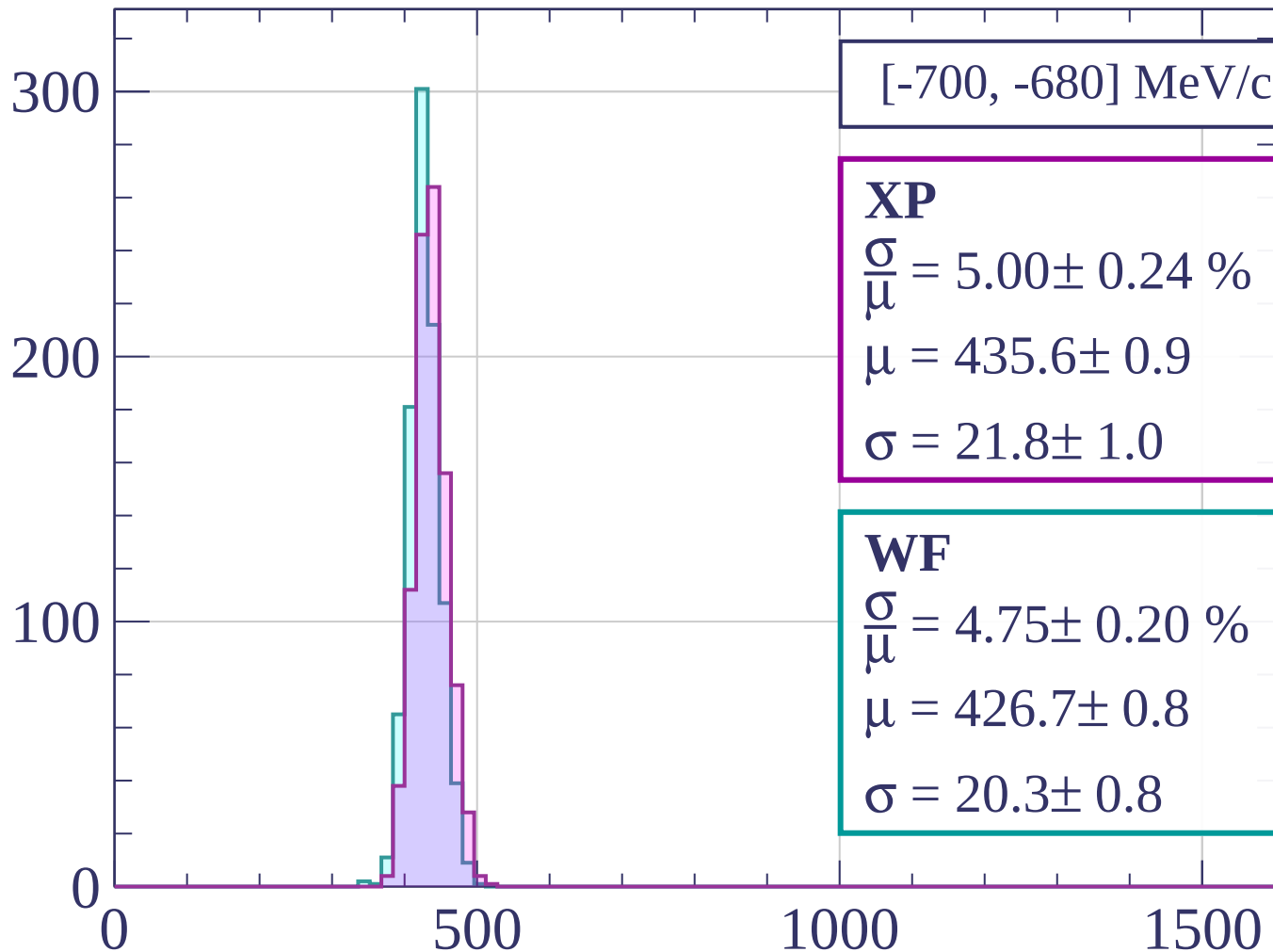
1000

1500

dE/dx [ADC counts/cm]



Count



[-700, -680] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.00 \pm 0.24 \%$$

$$\mu = 435.6 \pm 0.9$$

$$\sigma = 21.8 \pm 1.0$$

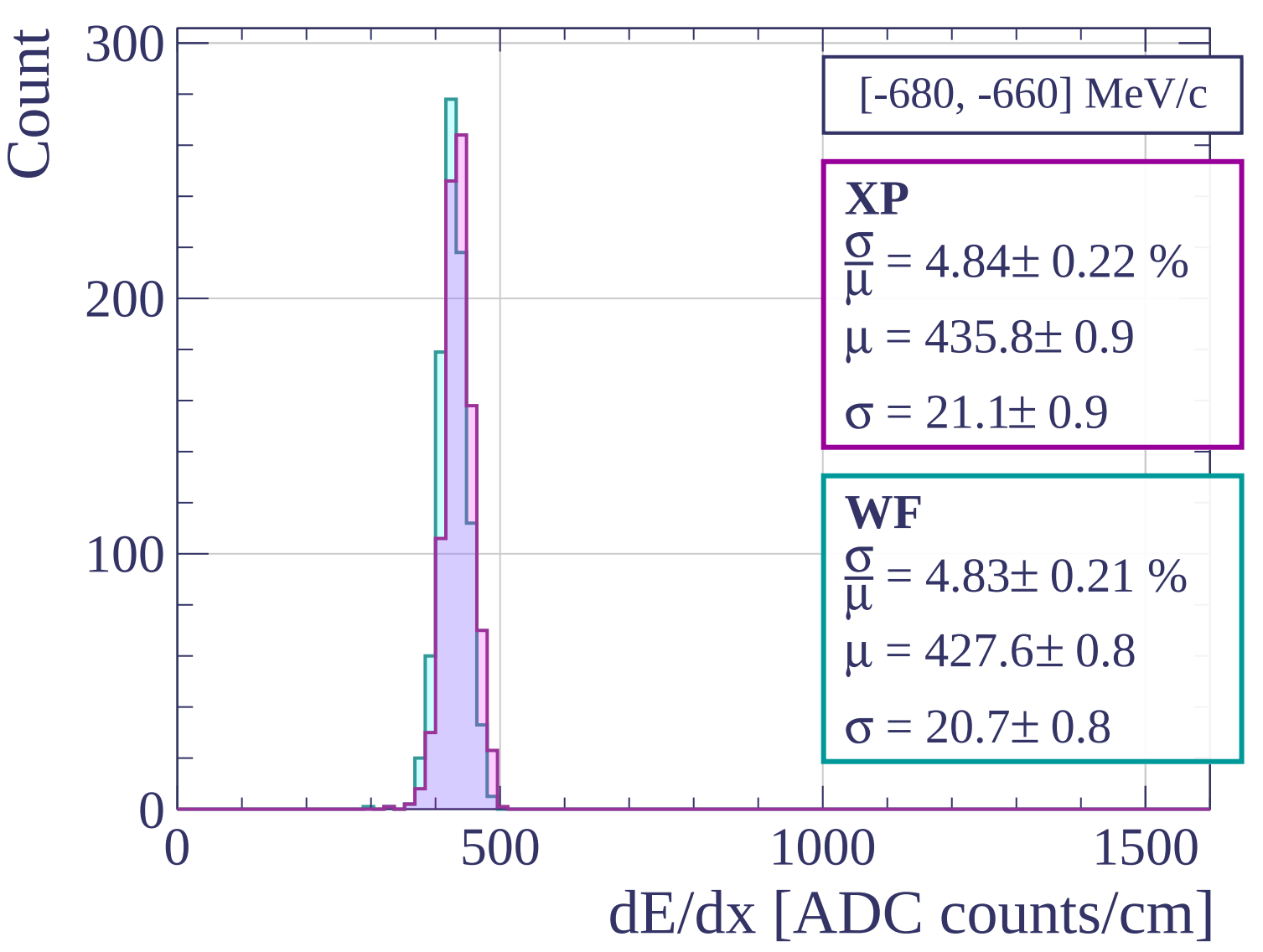
WF

$$\frac{\sigma}{\mu} = 4.75 \pm 0.20 \%$$

$$\mu = 426.7 \pm 0.8$$

$$\sigma = 20.3 \pm 0.8$$

dE/dx [ADC counts/cm]



Count

[-660, -640] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.14 \pm 0.19 \%$$

$$\mu = 434.4 \pm 0.9$$

$$\sigma = 22.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.18 \pm 0.25 \%$$

$$\mu = 426.0 \pm 0.9$$

$$\sigma = 22.1 \pm 1.0$$

200

100

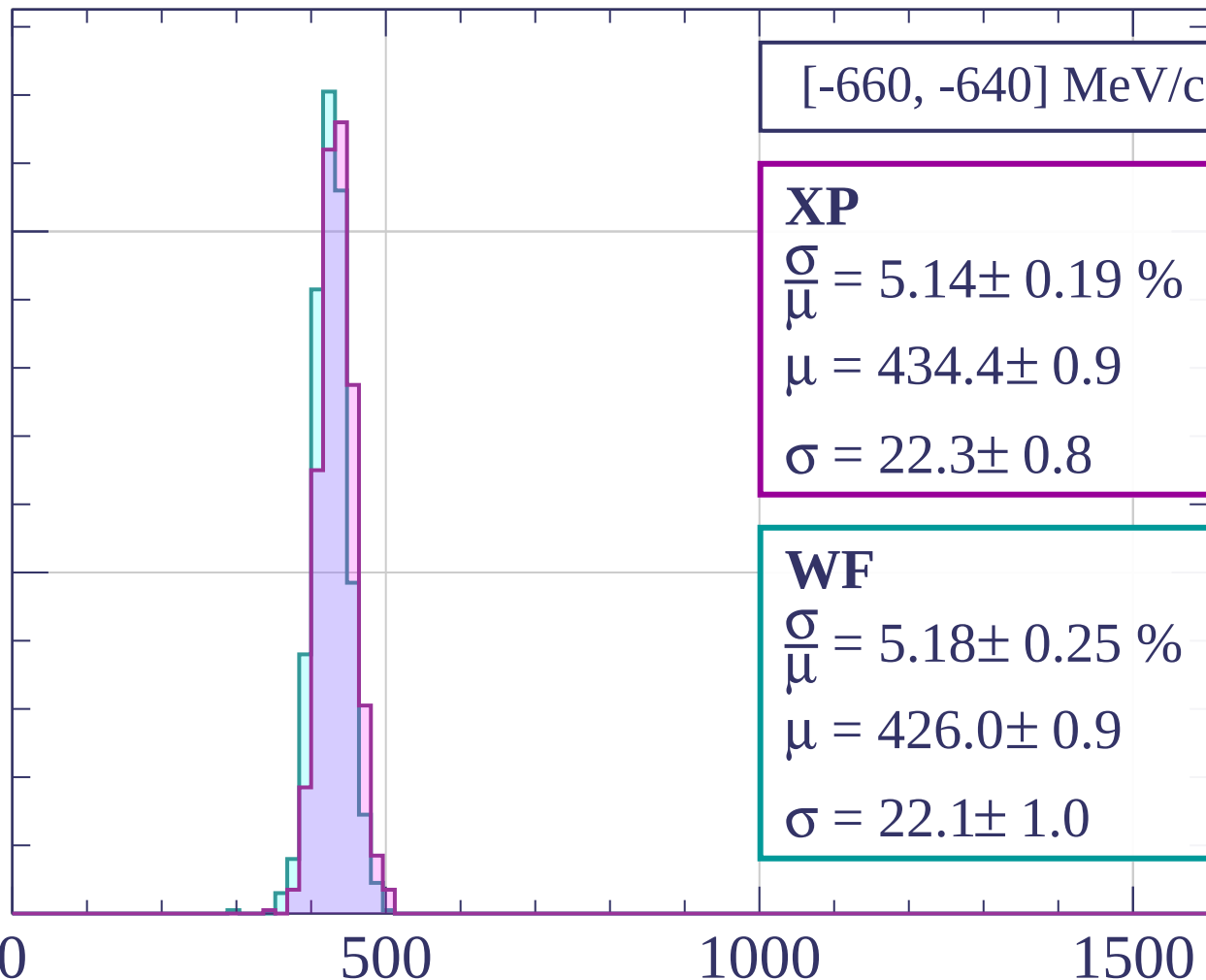
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-640, -620] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.60 \pm 0.14 \%$$

$$\mu = 432.8 \pm 0.7$$

$$\sigma = 19.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 4.67 \pm 0.18 \%$$

$$\mu = 424.6 \pm 0.8$$

$$\sigma = 19.8 \pm 0.7$$

200

100

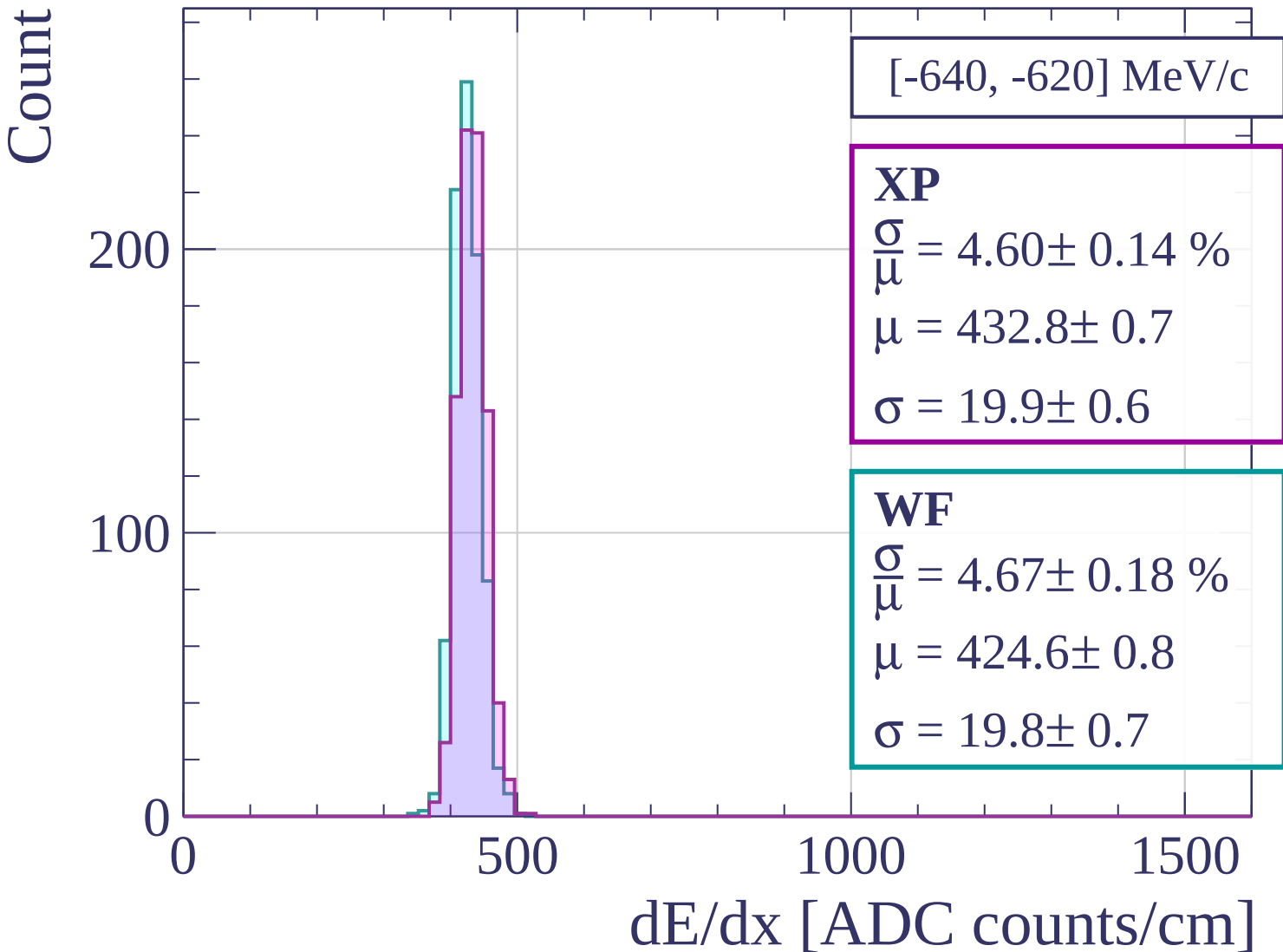
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-620, -600] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.81 \pm 0.23 \%$$

$$\mu = 430.2 \pm 1.0$$

$$\sigma = 20.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.90 \pm 0.22 \%$$

$$\mu = 422.7 \pm 0.9$$

$$\sigma = 20.7 \pm 0.9$$

200

100

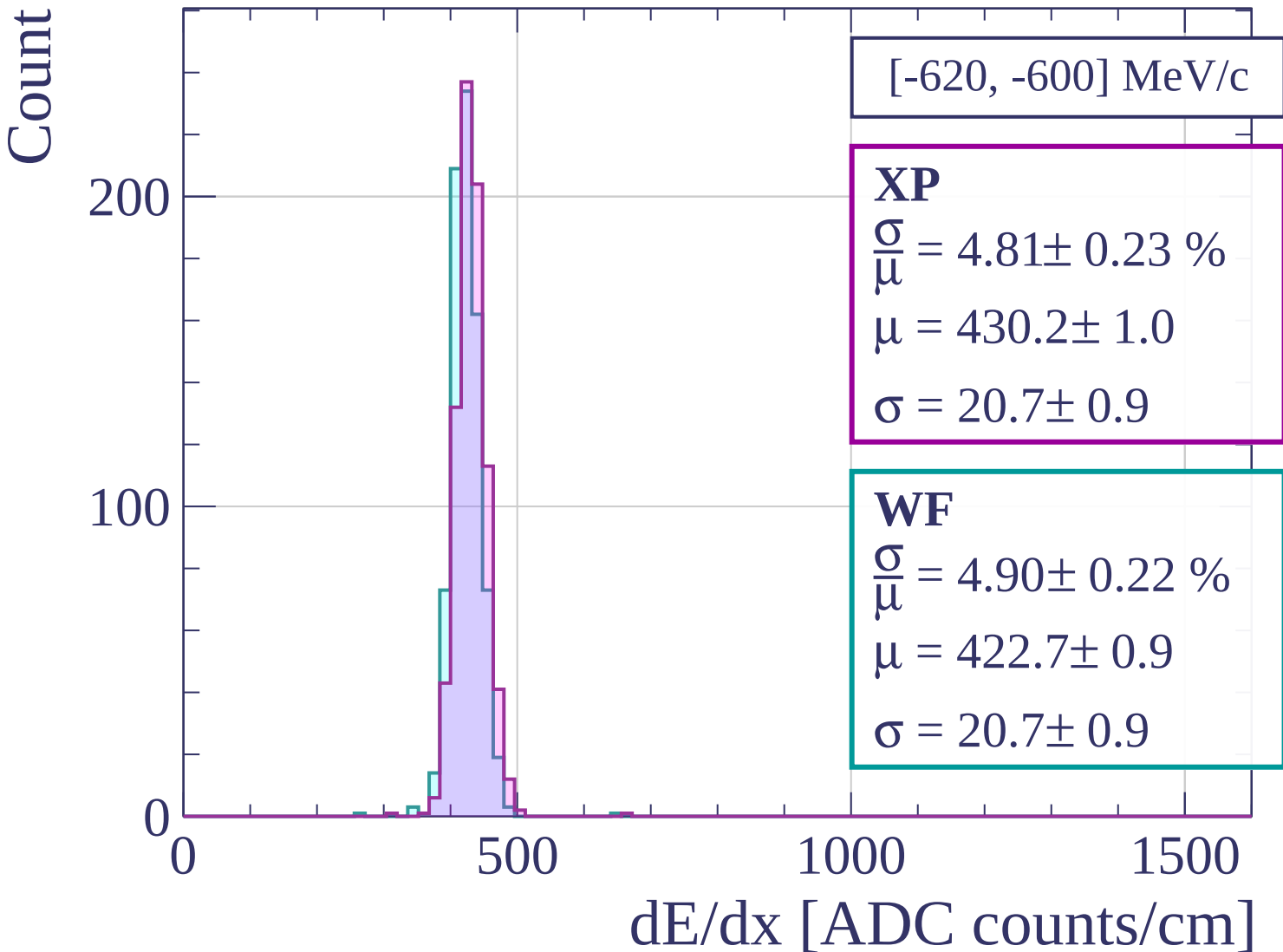
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-600, -580] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.96 \pm 0.23 \%$$

$$\mu = 428.1 \pm 1.0$$

$$\sigma = 21.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.99 \pm 0.24 \%$$

$$\mu = 420.5 \pm 0.9$$

$$\sigma = 21.0 \pm 1.0$$

200

100

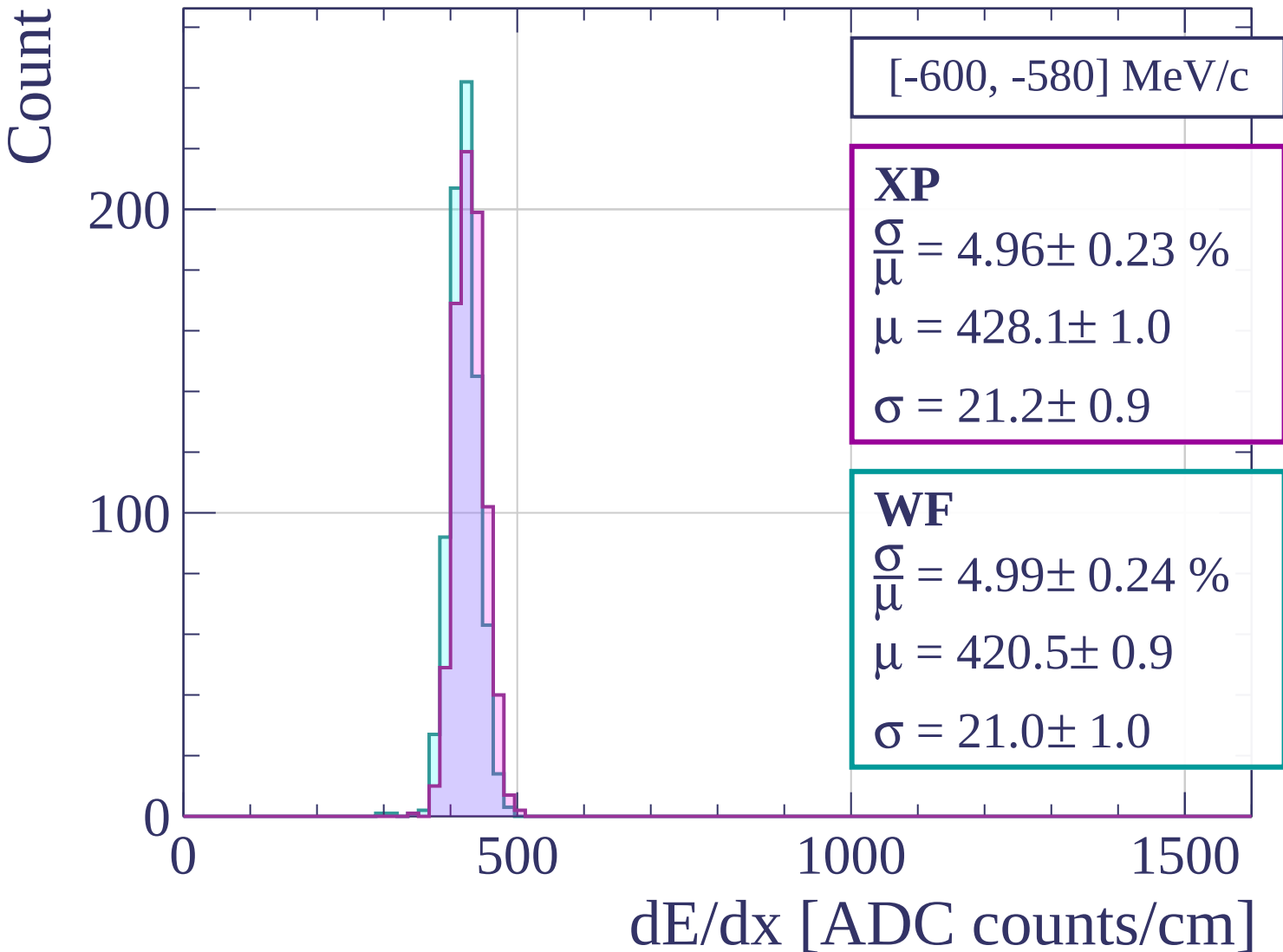
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-580, -560] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.67 \pm 0.21 \%$$

$$\mu = 429.0 \pm 0.8$$

$$\sigma = 20.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.77 \pm 0.21 \%$$

$$\mu = 420.8 \pm 0.8$$

$$\sigma = 20.1 \pm 0.8$$

200

100

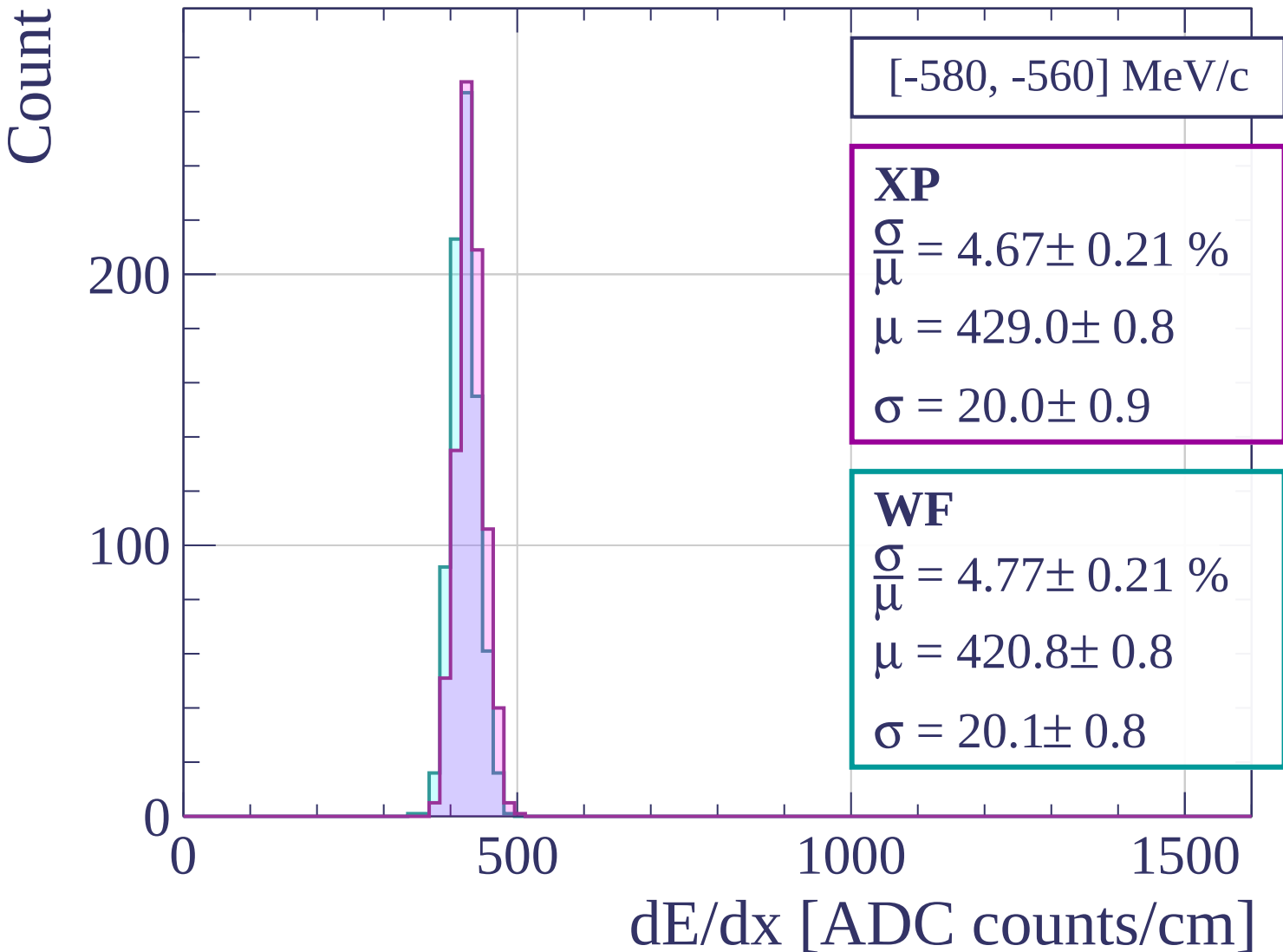
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-560, -540] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.26 \%$$

$$\mu = 427.2 \pm 1.0$$

$$\sigma = 21.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 4.72 \pm 0.21 \%$$

$$\mu = 419.6 \pm 0.9$$

$$\sigma = 19.8 \pm 0.9$$

200

150

100

50

0

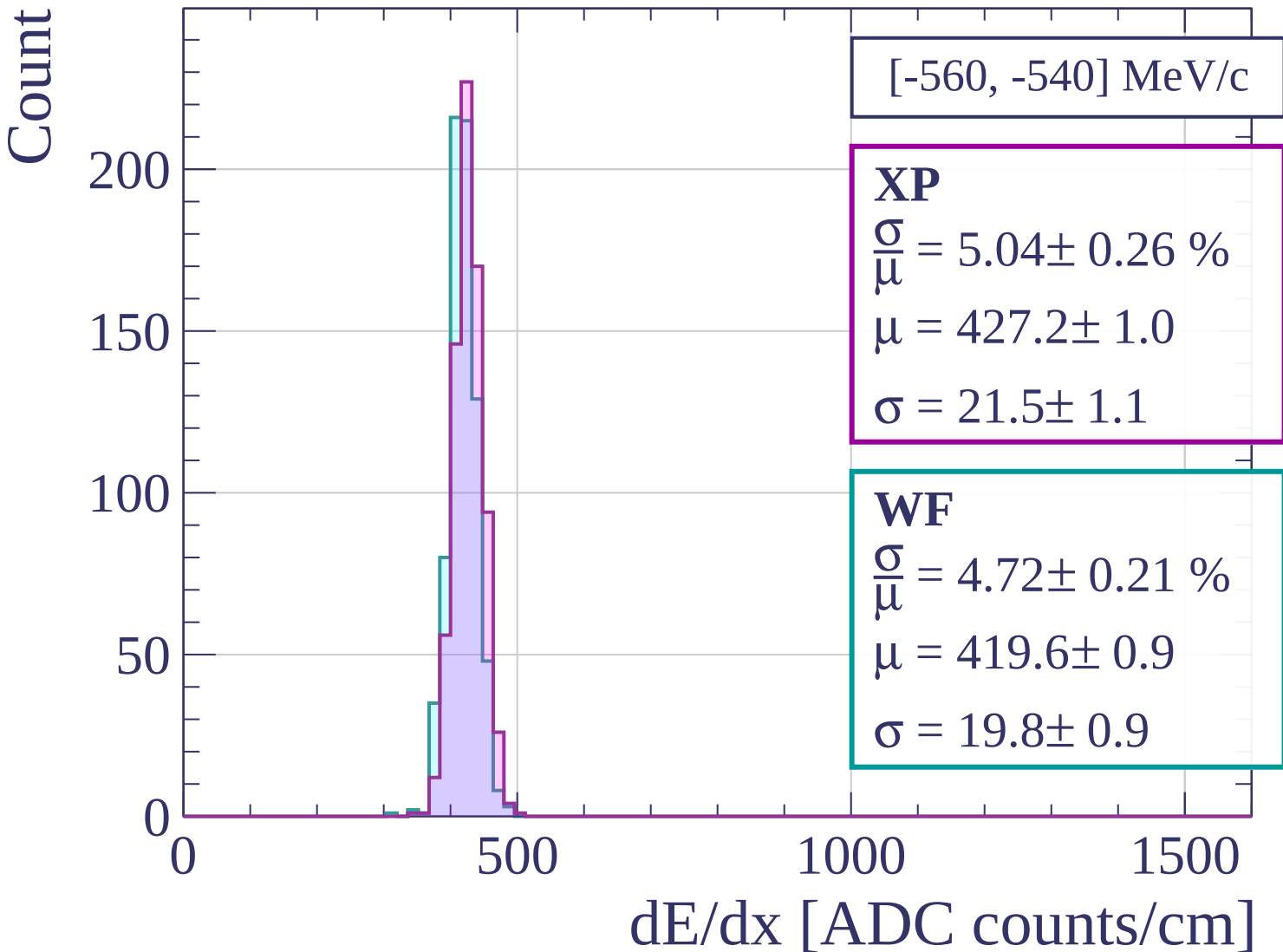
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

[-540, -520] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.03 \pm 0.16 \%$$

$$\mu = 427.4 \pm 0.9$$

$$\sigma = 21.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.98 \pm 0.25 \%$$

$$\mu = 420.0 \pm 0.9$$

$$\sigma = 20.9 \pm 1.0$$

200

150

100

50

0

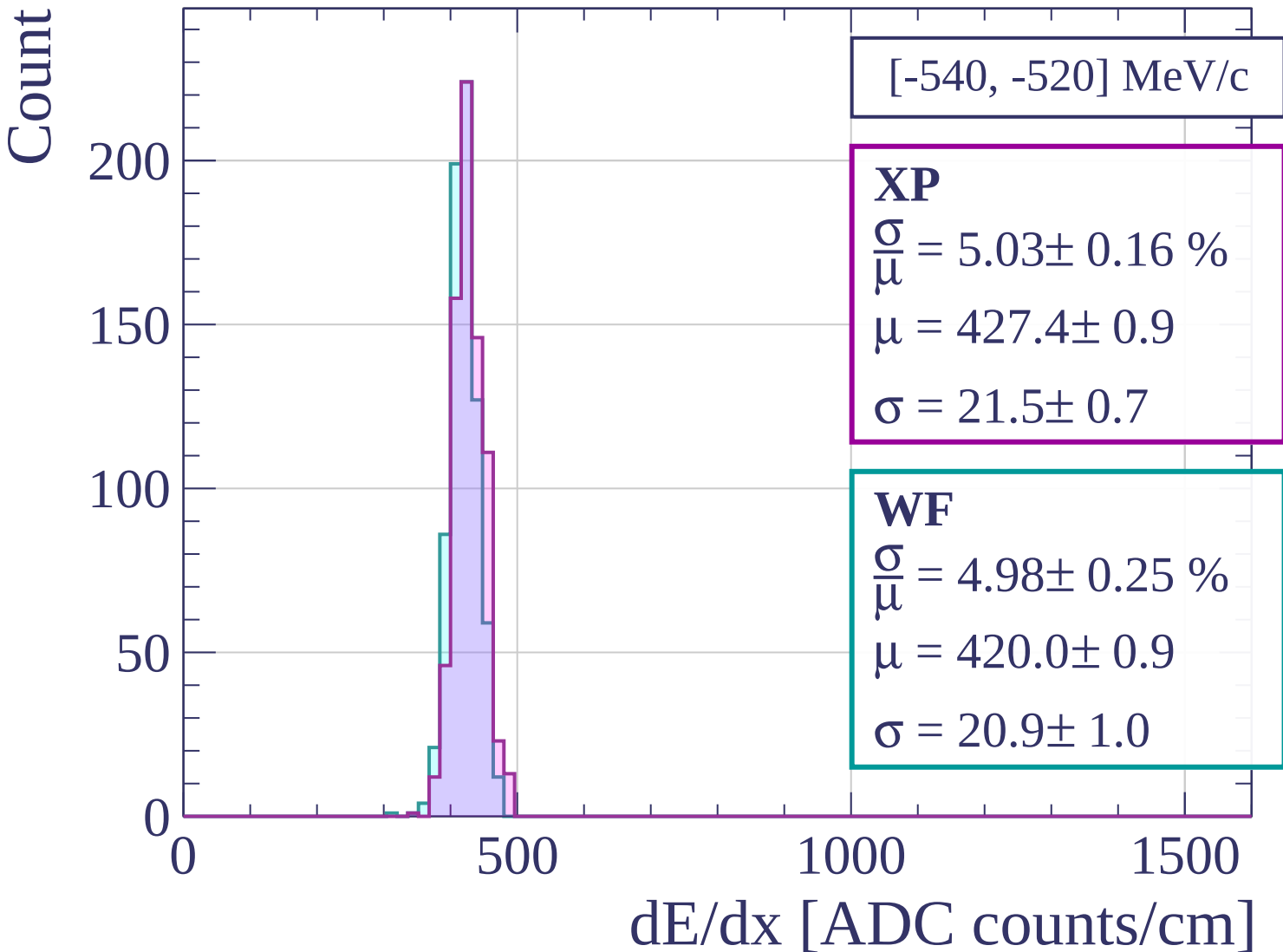
0

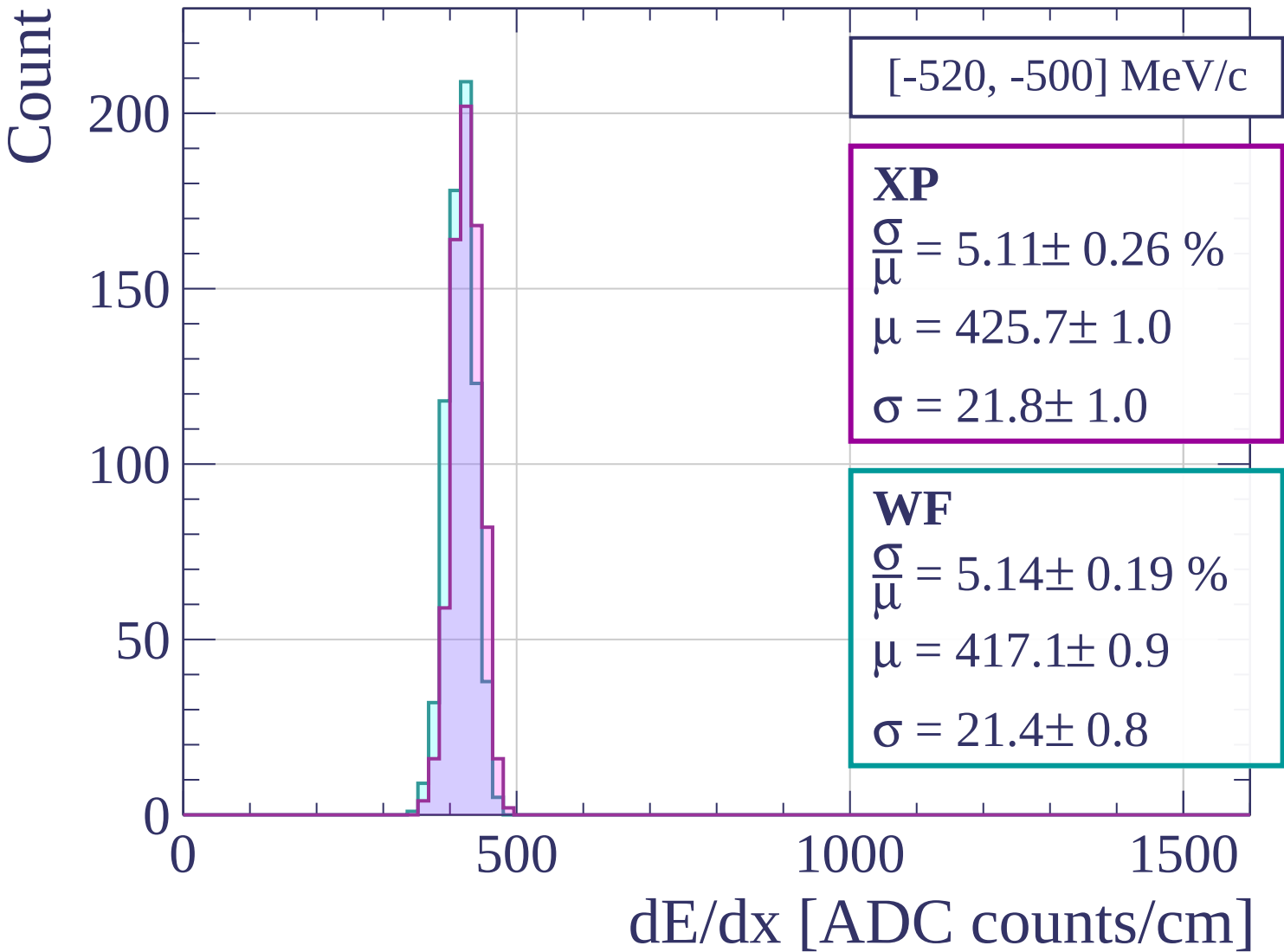
500

1000

1500

dE/dx [ADC counts/cm]





Count

[-500, -480] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.00 \pm 0.24 \%$$

$$\mu = 424.1 \pm 0.9$$

$$\sigma = 21.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 5.11 \pm 0.20 \%$$

$$\mu = 416.2 \pm 0.9$$

$$\sigma = 21.3 \pm 0.8$$

200

150

100

50

0

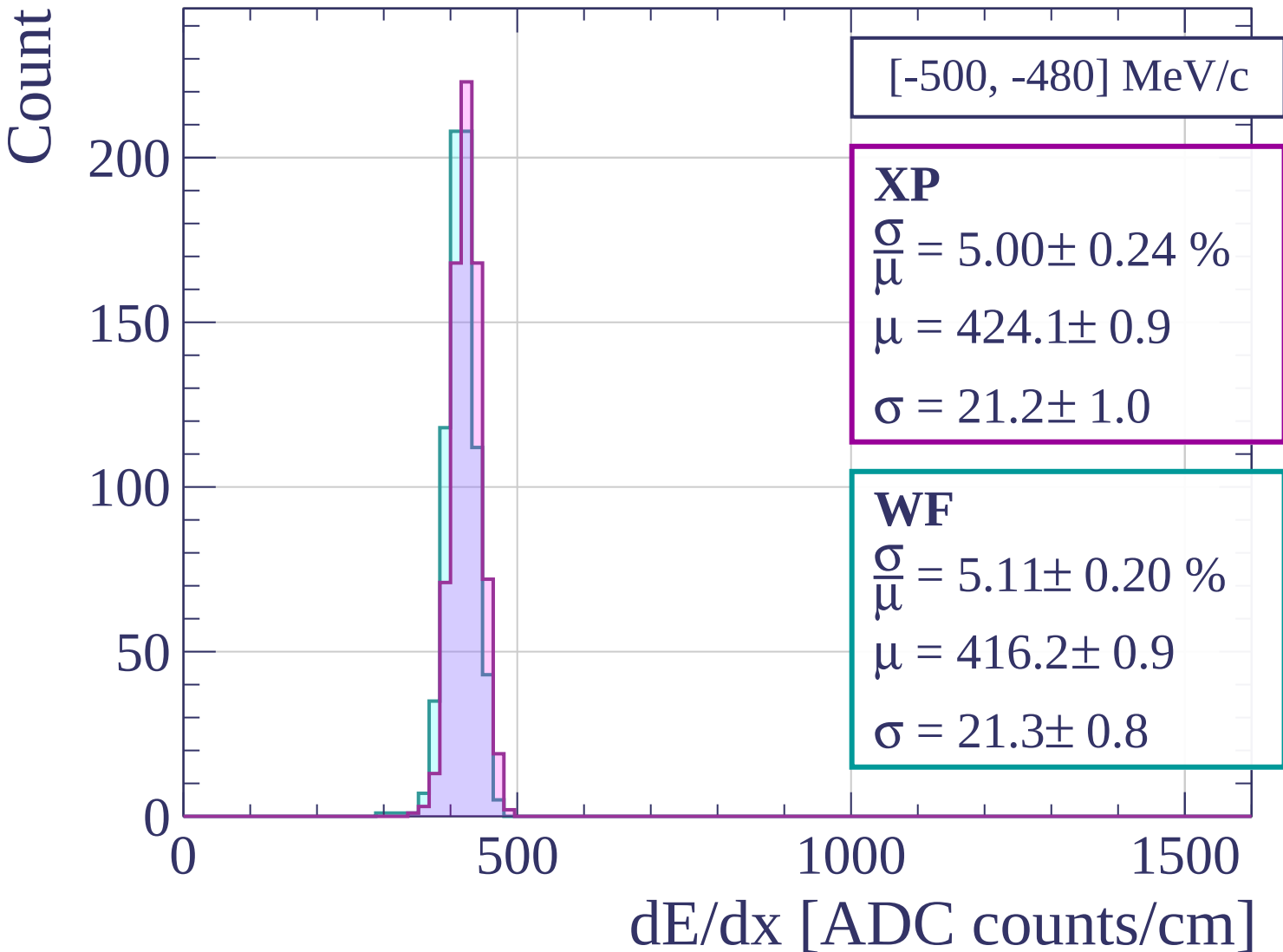
0

500

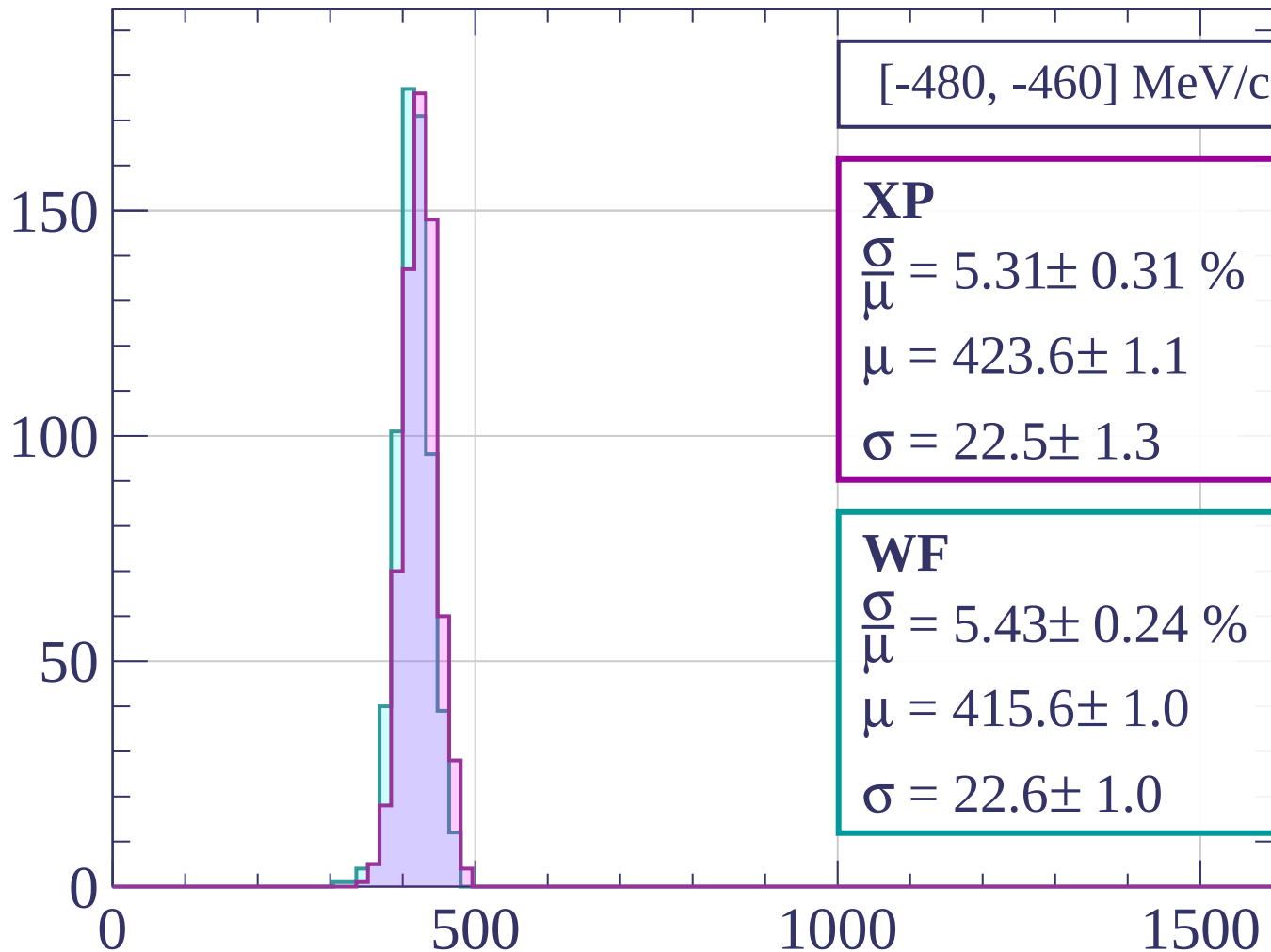
1000

1500

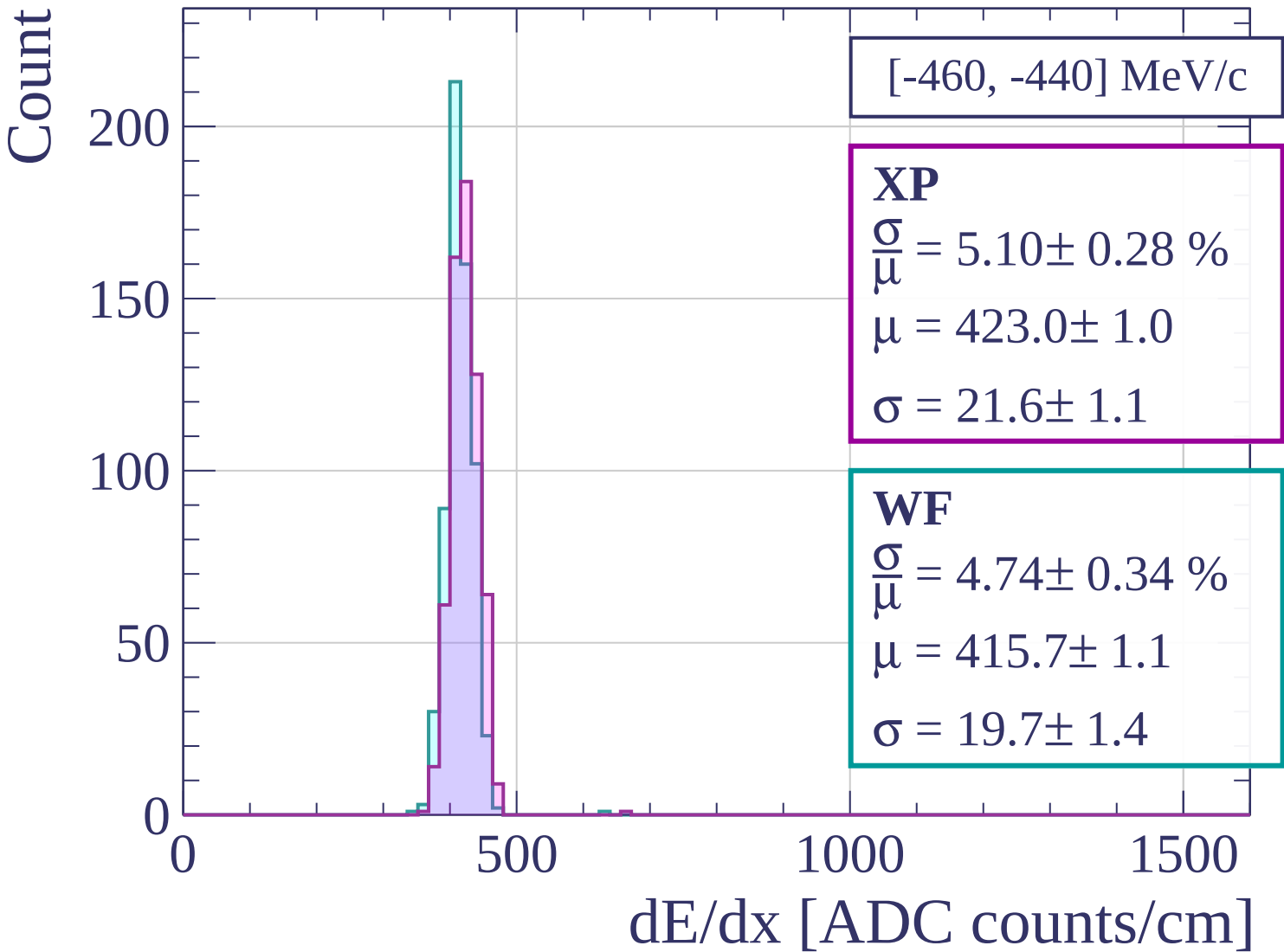
dE/dx [ADC counts/cm]



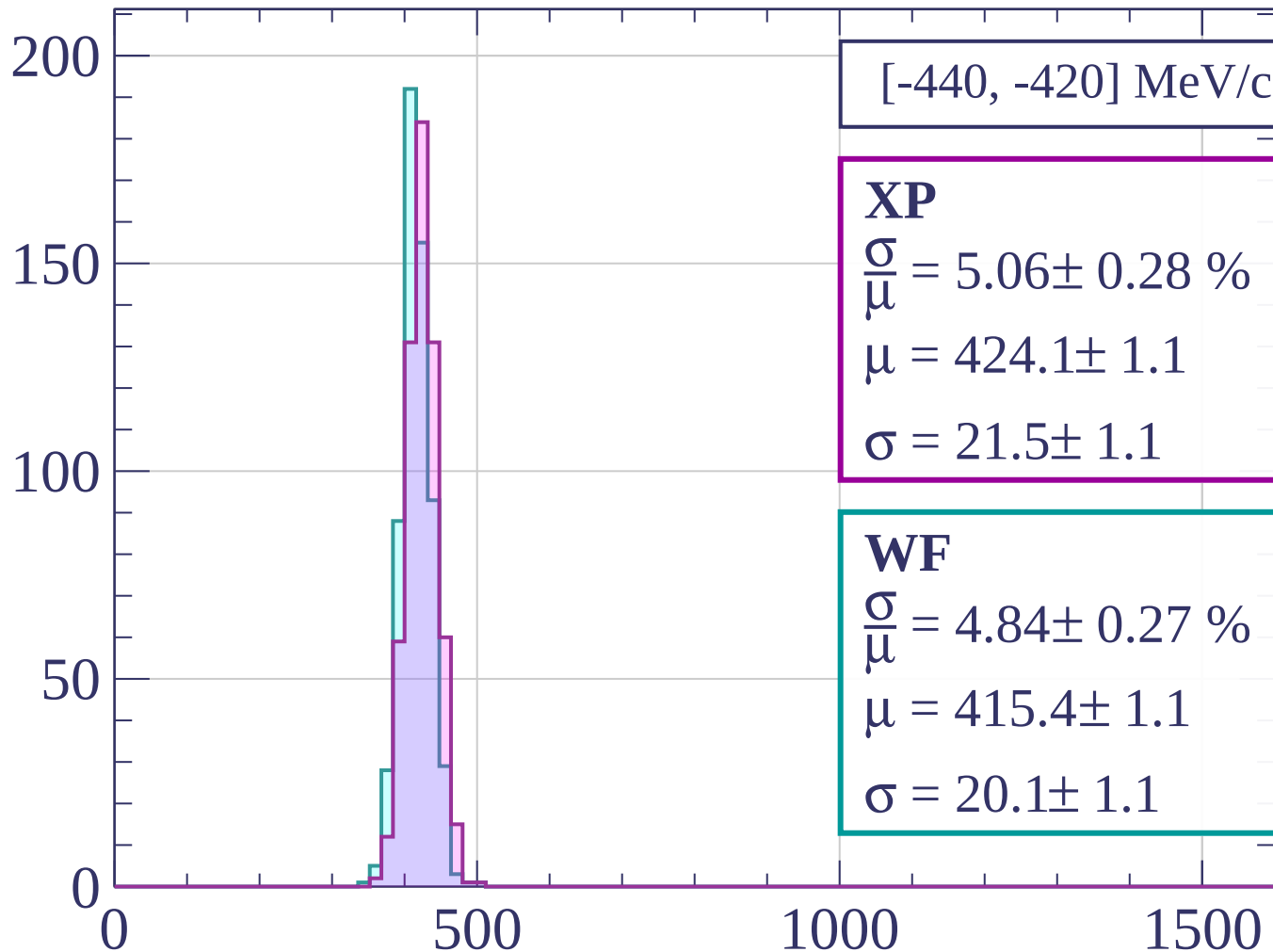
Count



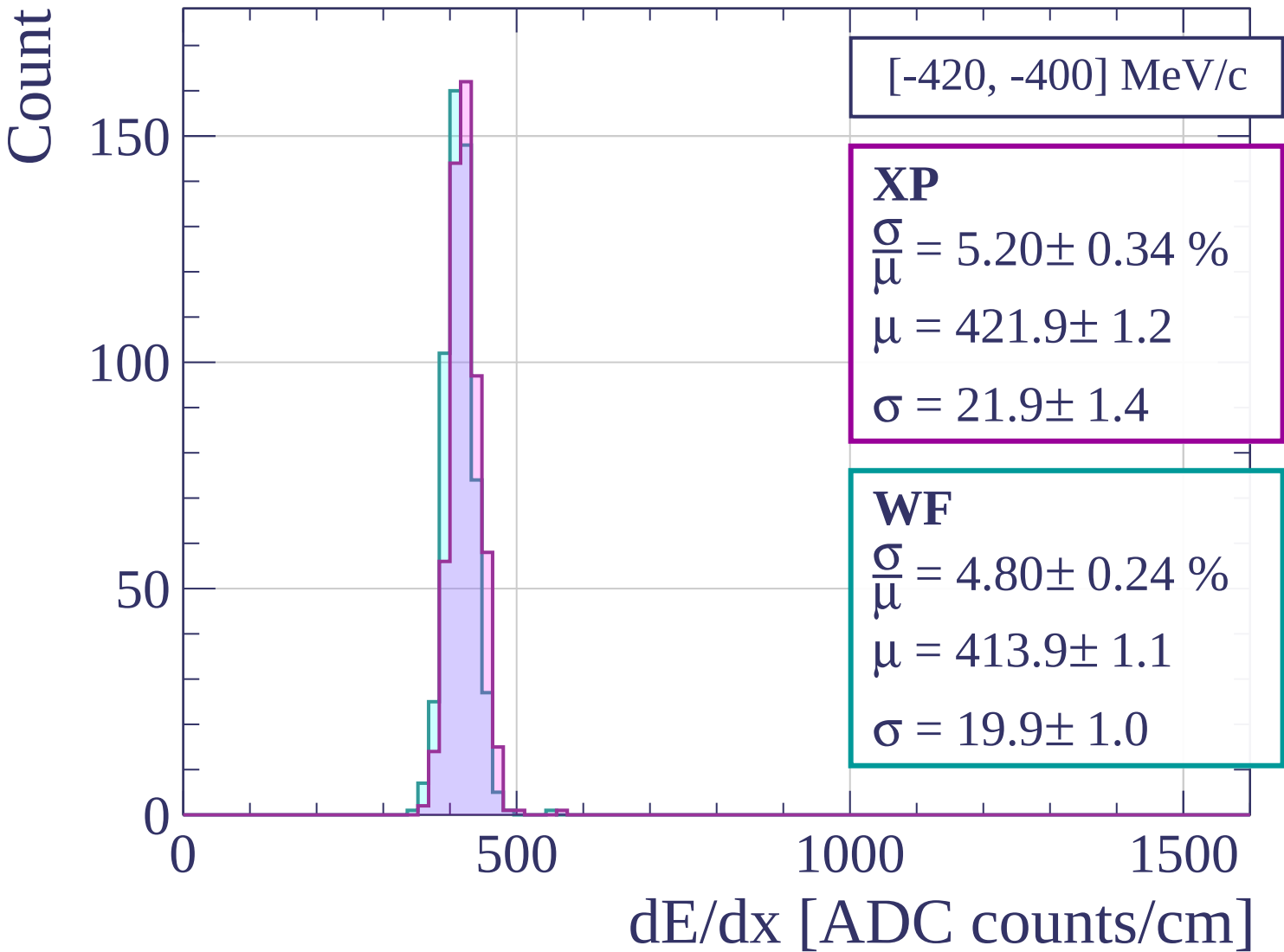
dE/dx [ADC counts/cm]



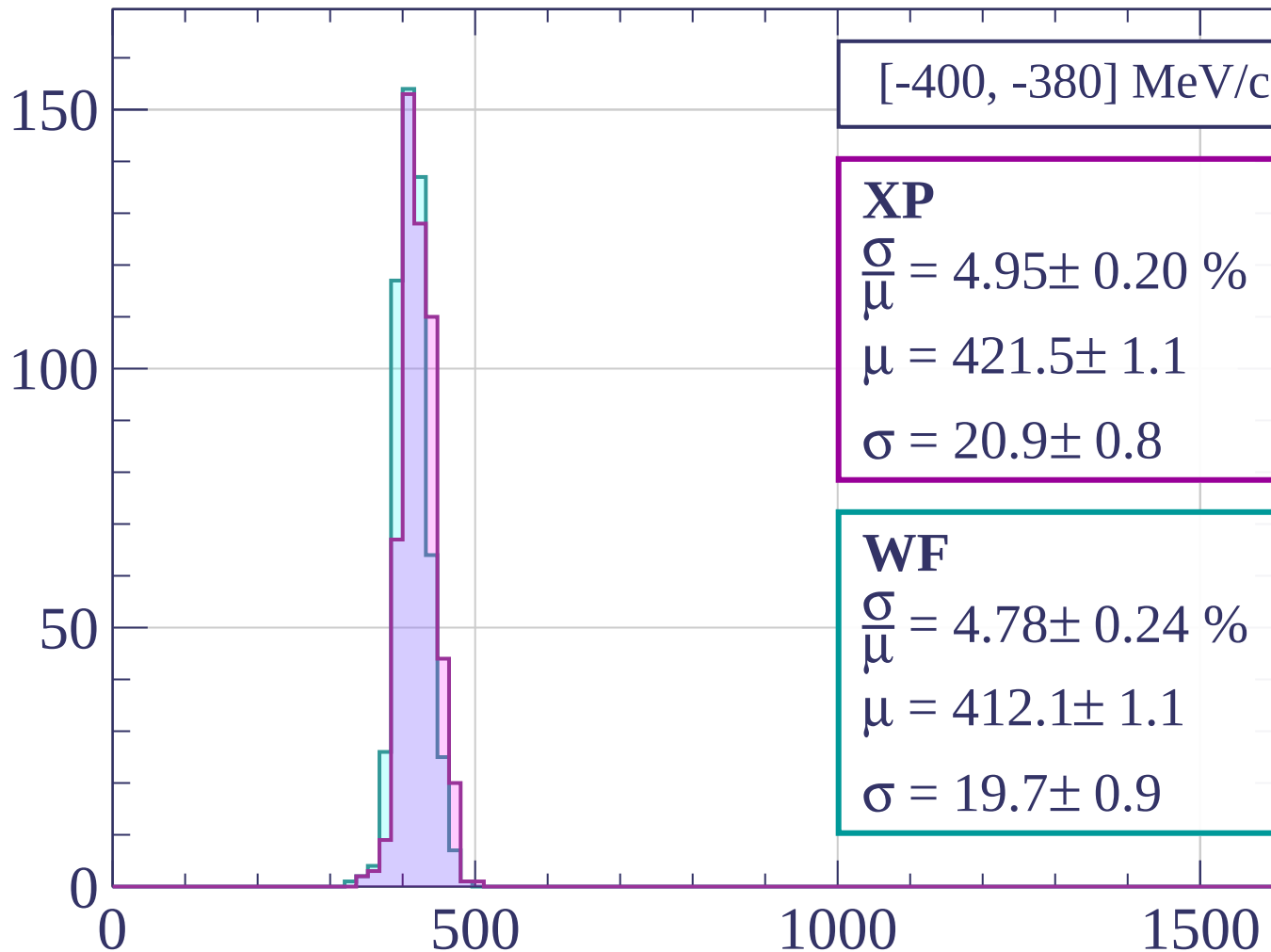
Count



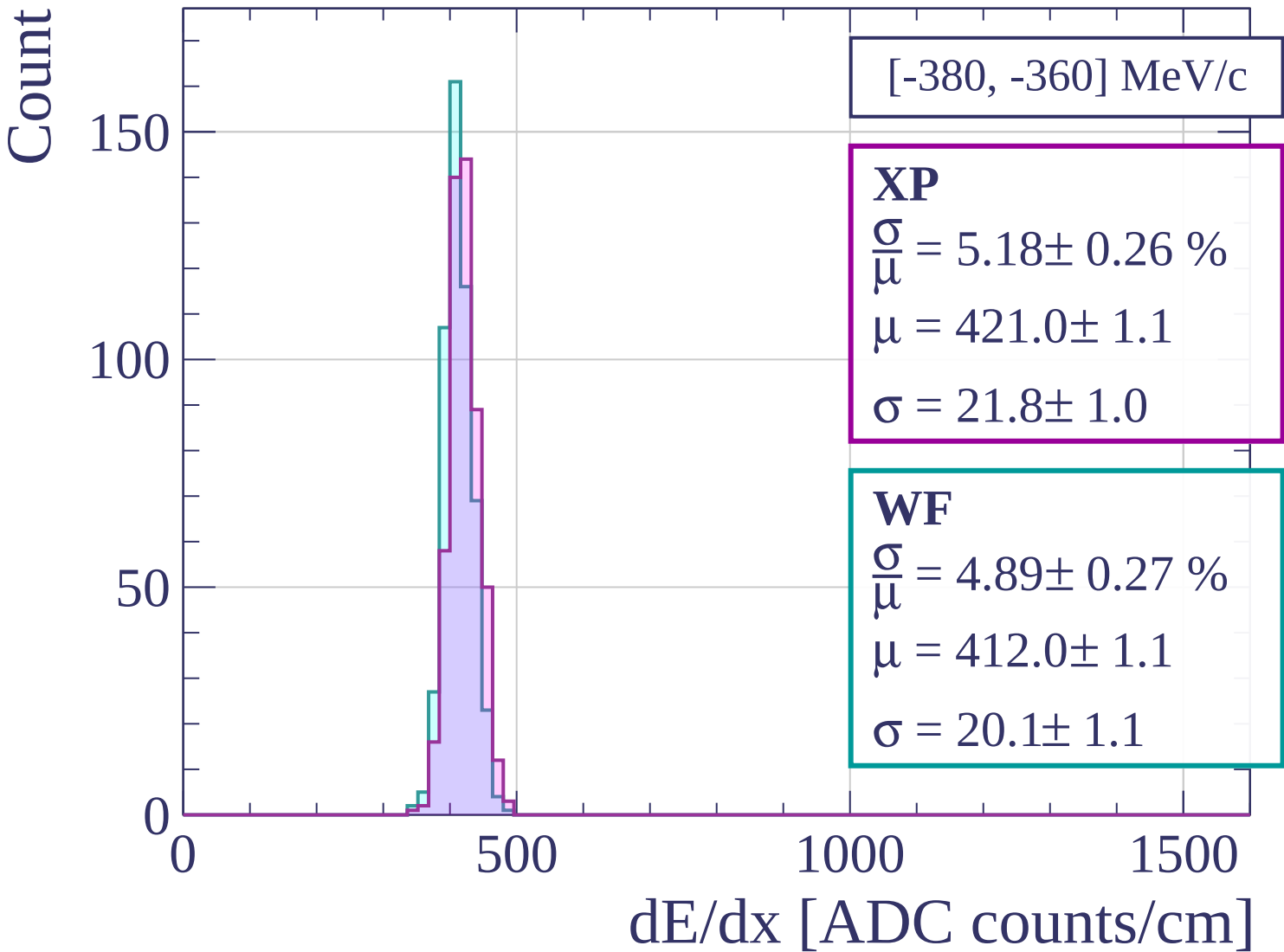
dE/dx [ADC counts/cm]

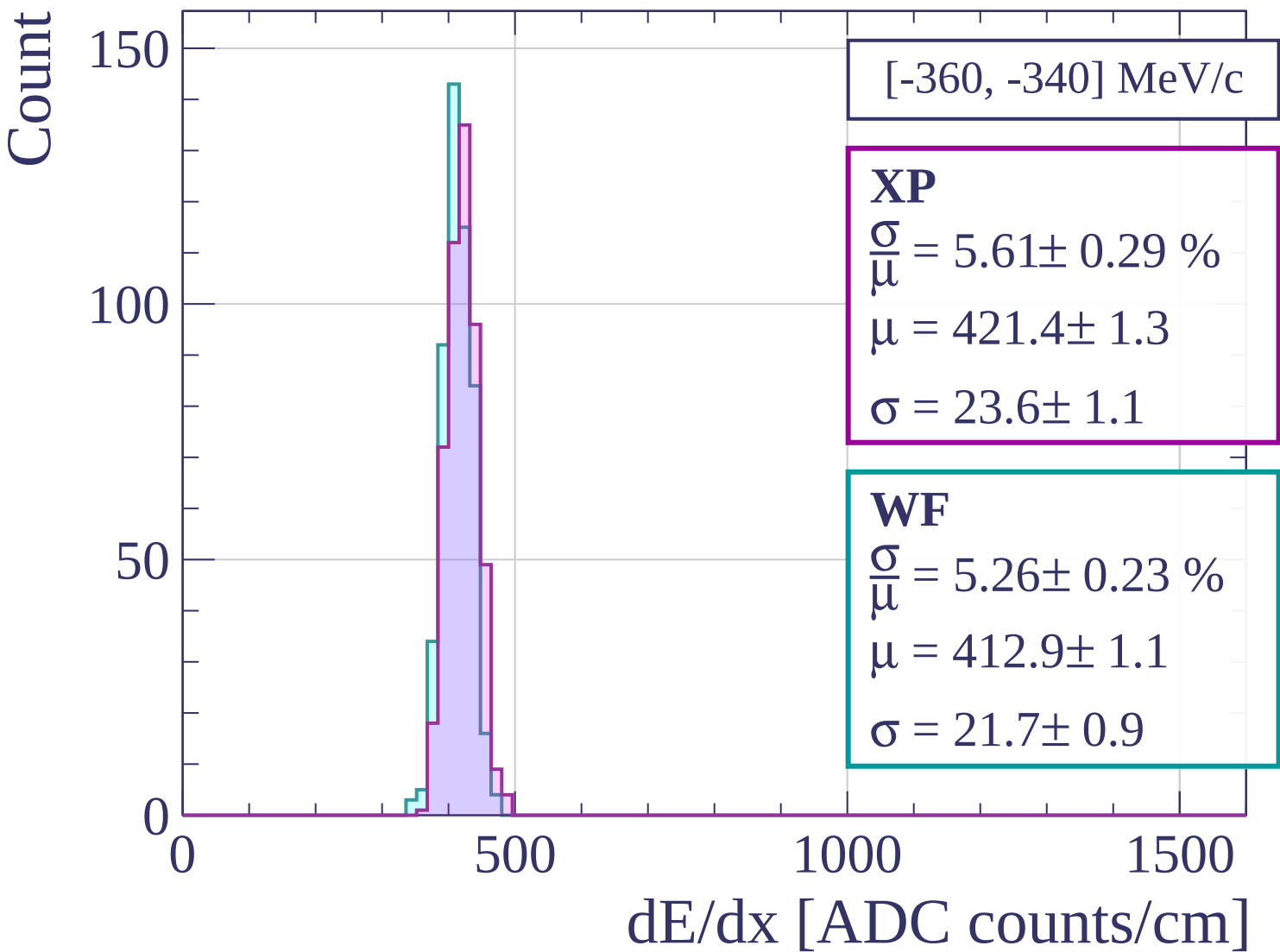


Count



dE/dx [ADC counts/cm]





Count

[-340, -320] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.21 \pm 0.35 \%$$

$$\mu = 420.3 \pm 1.3$$

$$\sigma = 21.9 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 5.19 \pm 0.34 \%$$

$$\mu = 412.3 \pm 1.3$$

$$\sigma = 21.4 \pm 1.3$$

100

50

0

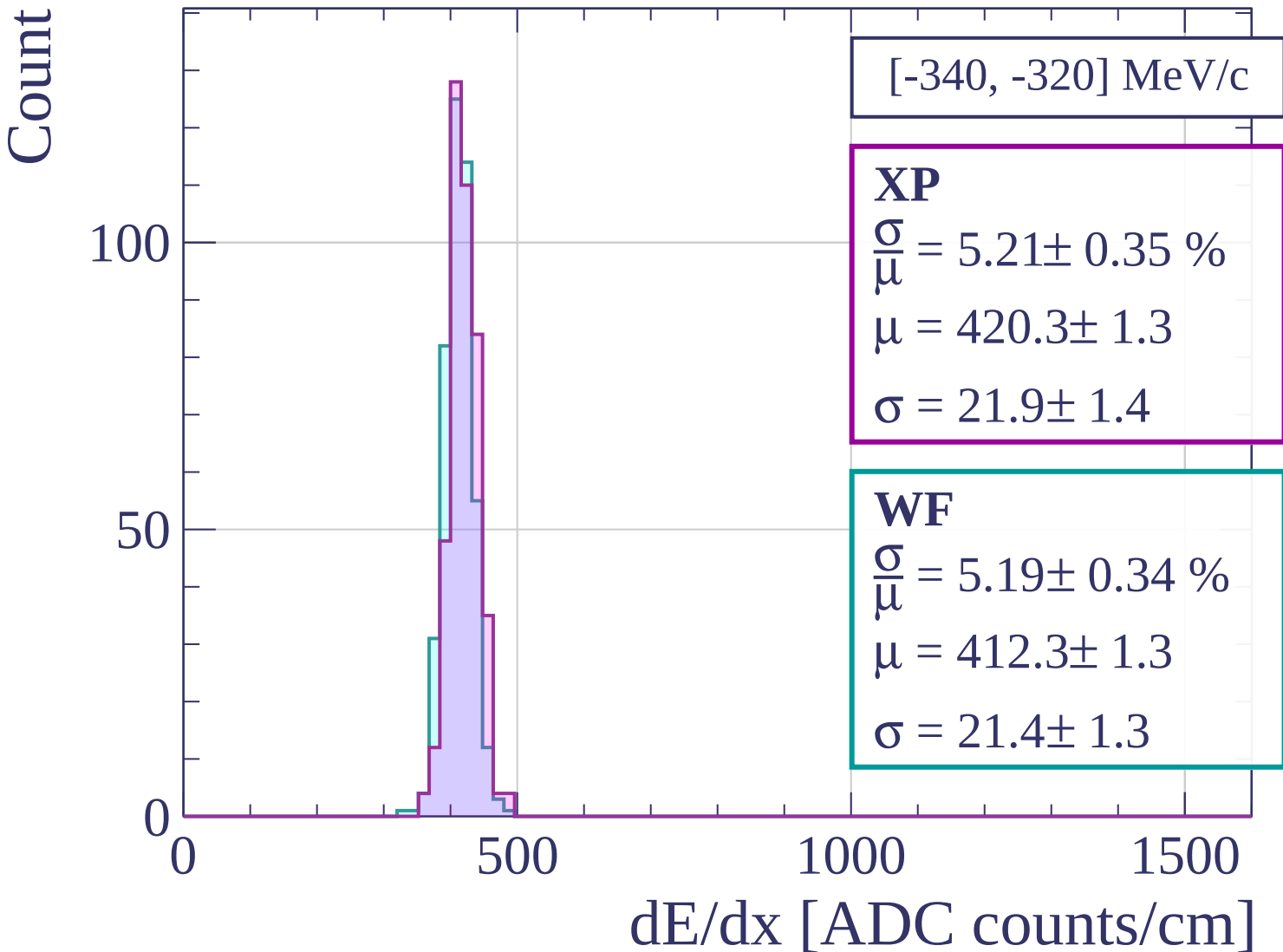
0

500

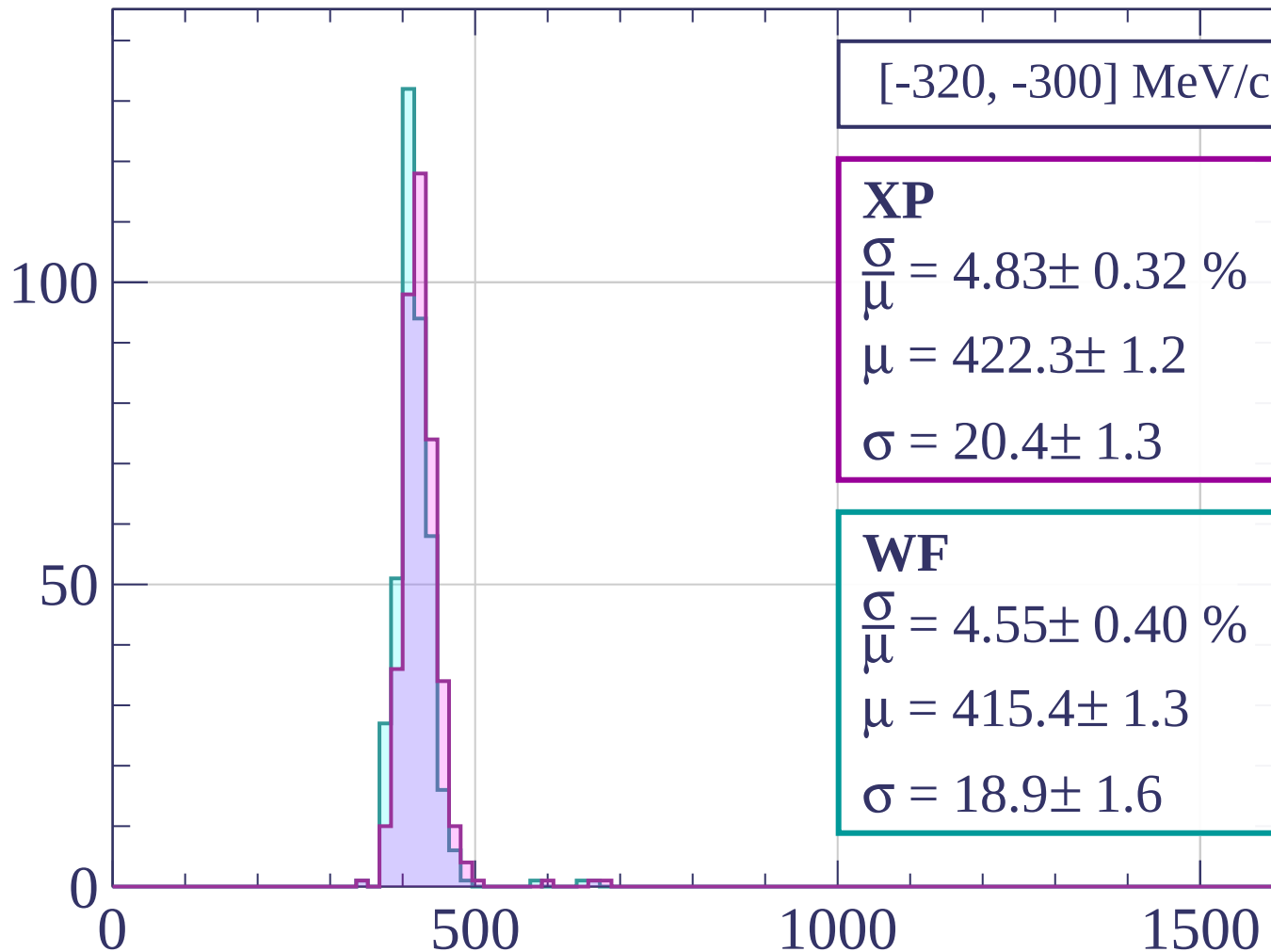
1000

1500

dE/dx [ADC counts/cm]



Count



[-320, -300] MeV/c

XP

$$\frac{\sigma}{\mu} = 4.83 \pm 0.32 \%$$

$$\mu = 422.3 \pm 1.2$$

$$\sigma = 20.4 \pm 1.3$$

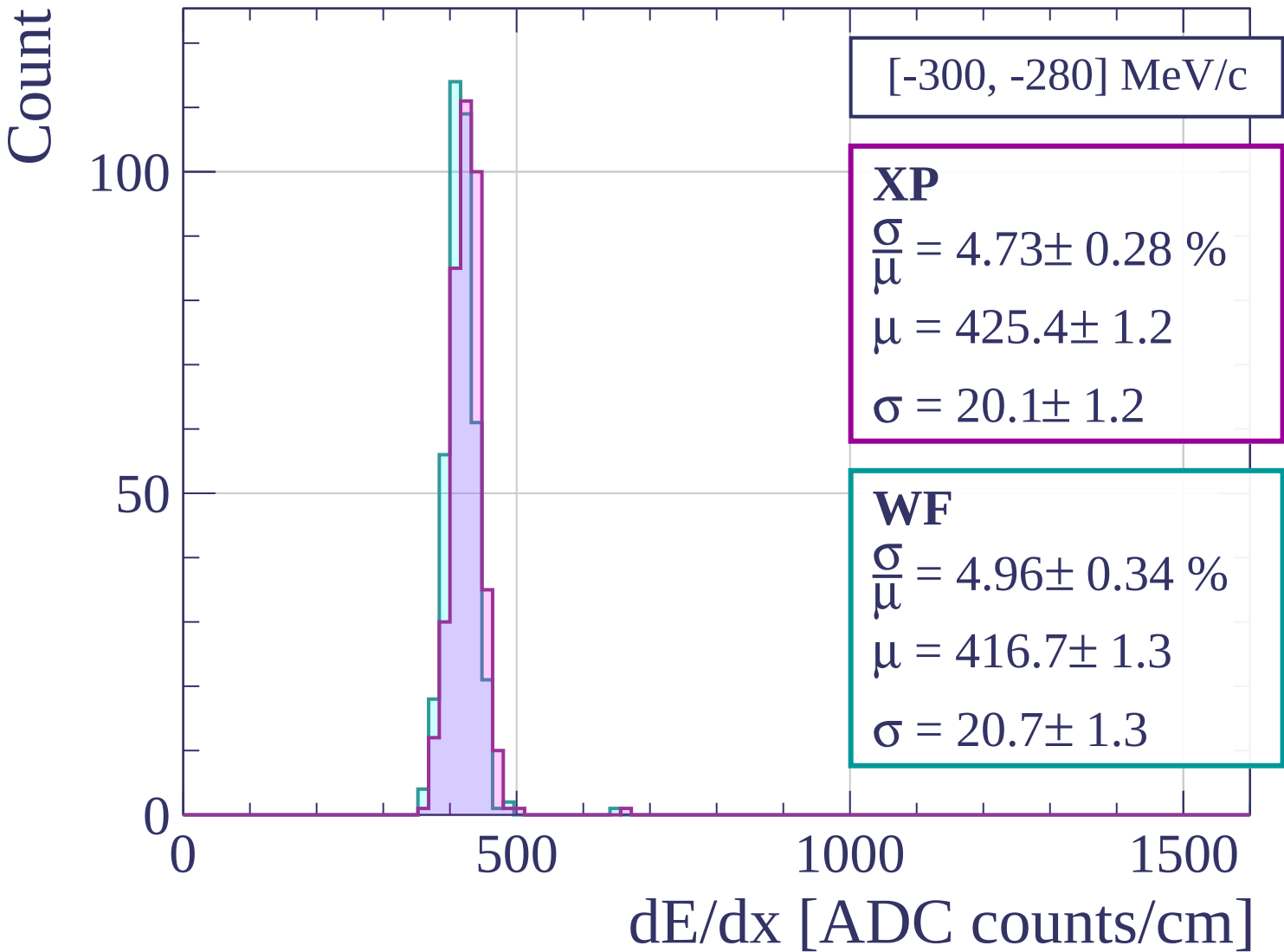
WF

$$\frac{\sigma}{\mu} = 4.55 \pm 0.40 \%$$

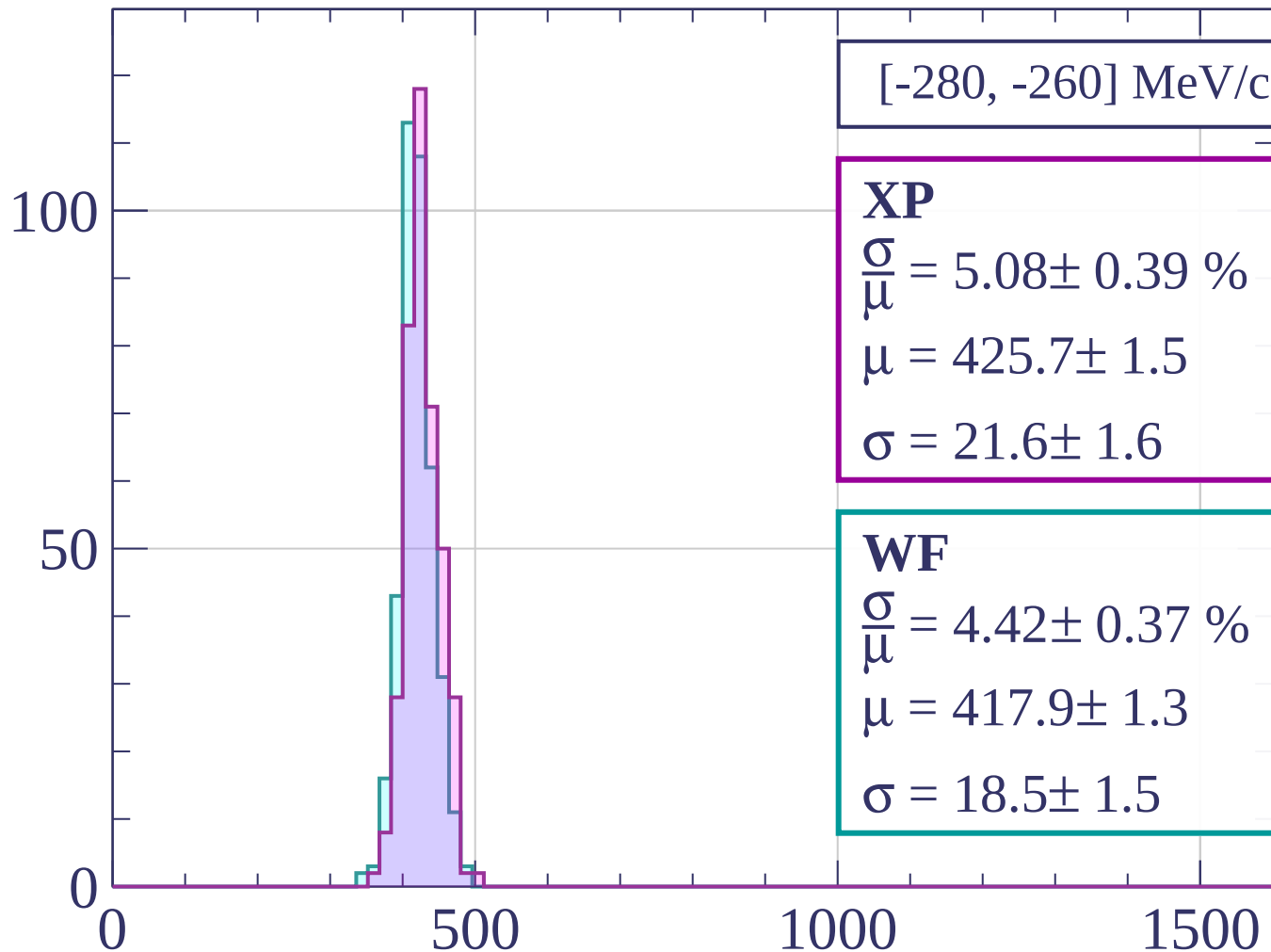
$$\mu = 415.4 \pm 1.3$$

$$\sigma = 18.9 \pm 1.6$$

dE/dx [ADC counts/cm]



Count



$[-280, -260] \text{ MeV/c}$

XP

$$\frac{\sigma}{\mu} = 5.08 \pm 0.39 \%$$

$$\mu = 425.7 \pm 1.5$$

$$\sigma = 21.6 \pm 1.6$$

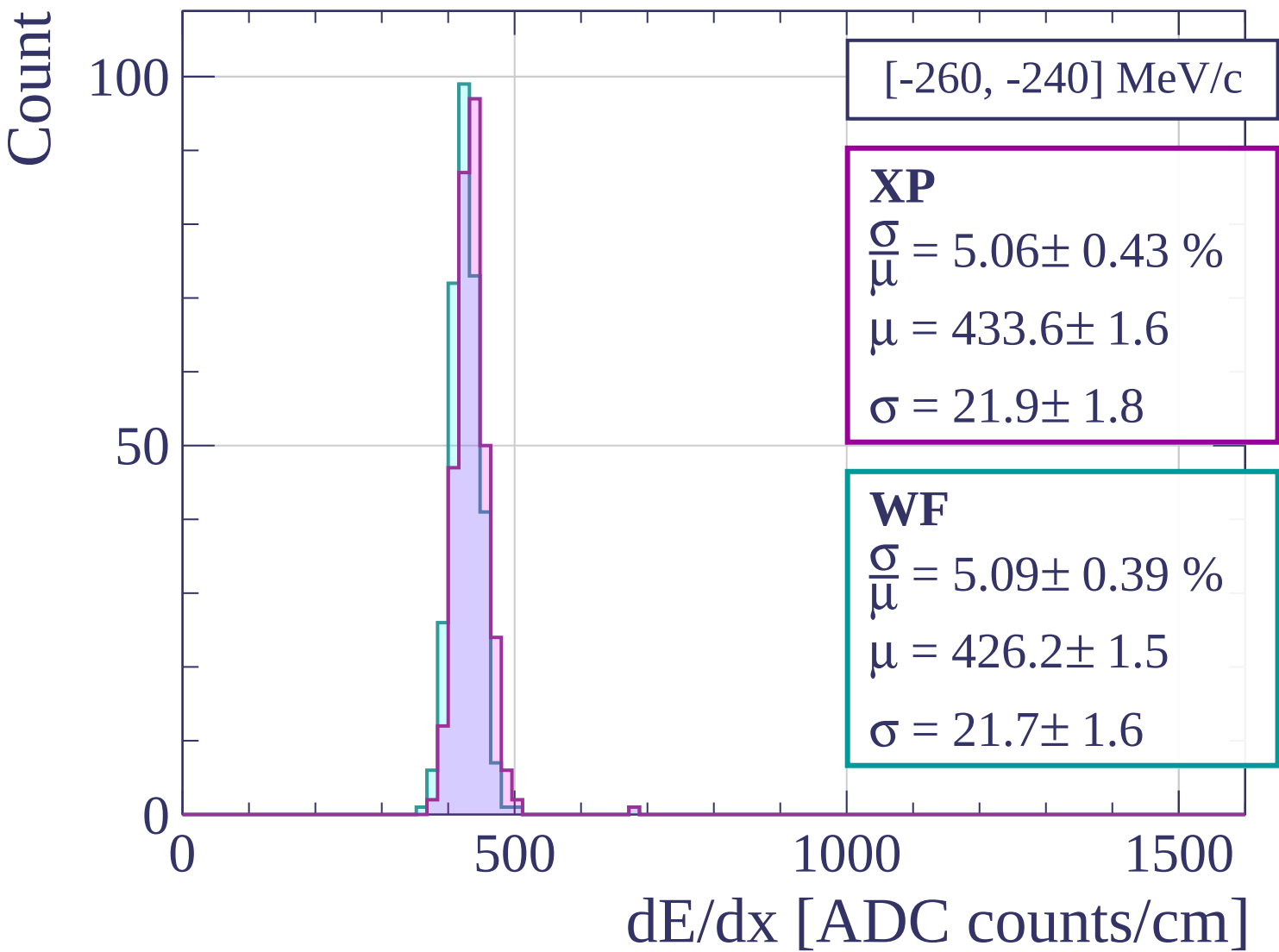
WF

$$\frac{\sigma}{\mu} = 4.42 \pm 0.37 \%$$

$$\mu = 417.9 \pm 1.3$$

$$\sigma = 18.5 \pm 1.5$$

dE/dx [ADC counts/cm]



Count

[-240, -220] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.31 \pm 0.28 \%$$

$$\mu = 440.4 \pm 1.4$$

$$\sigma = 23.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 5.30 \pm 0.34 \%$$

$$\mu = 431.2 \pm 1.5$$

$$\sigma = 22.9 \pm 1.4$$

80

60

40

20

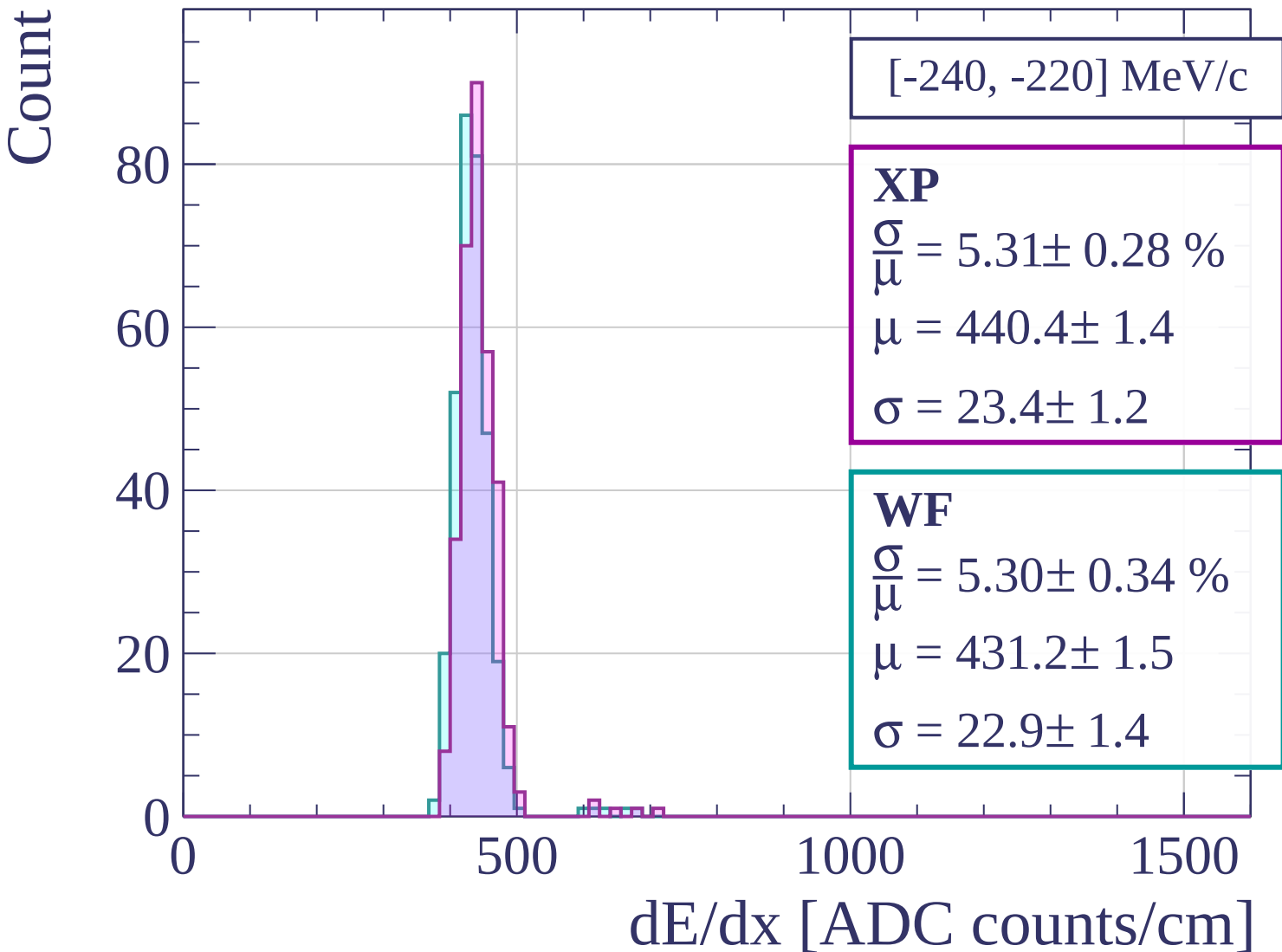
0

500

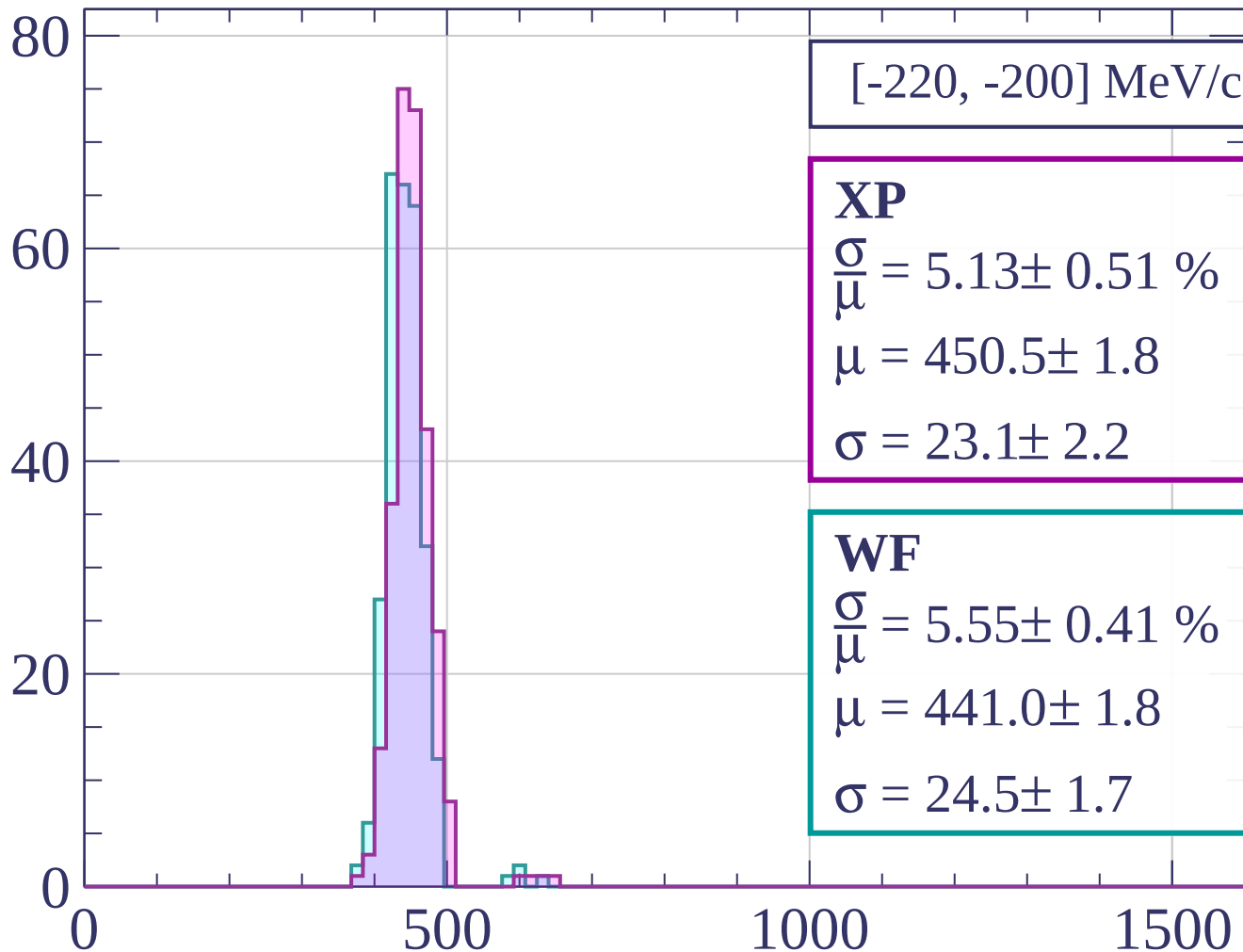
1000

1500

dE/dx [ADC counts/cm]

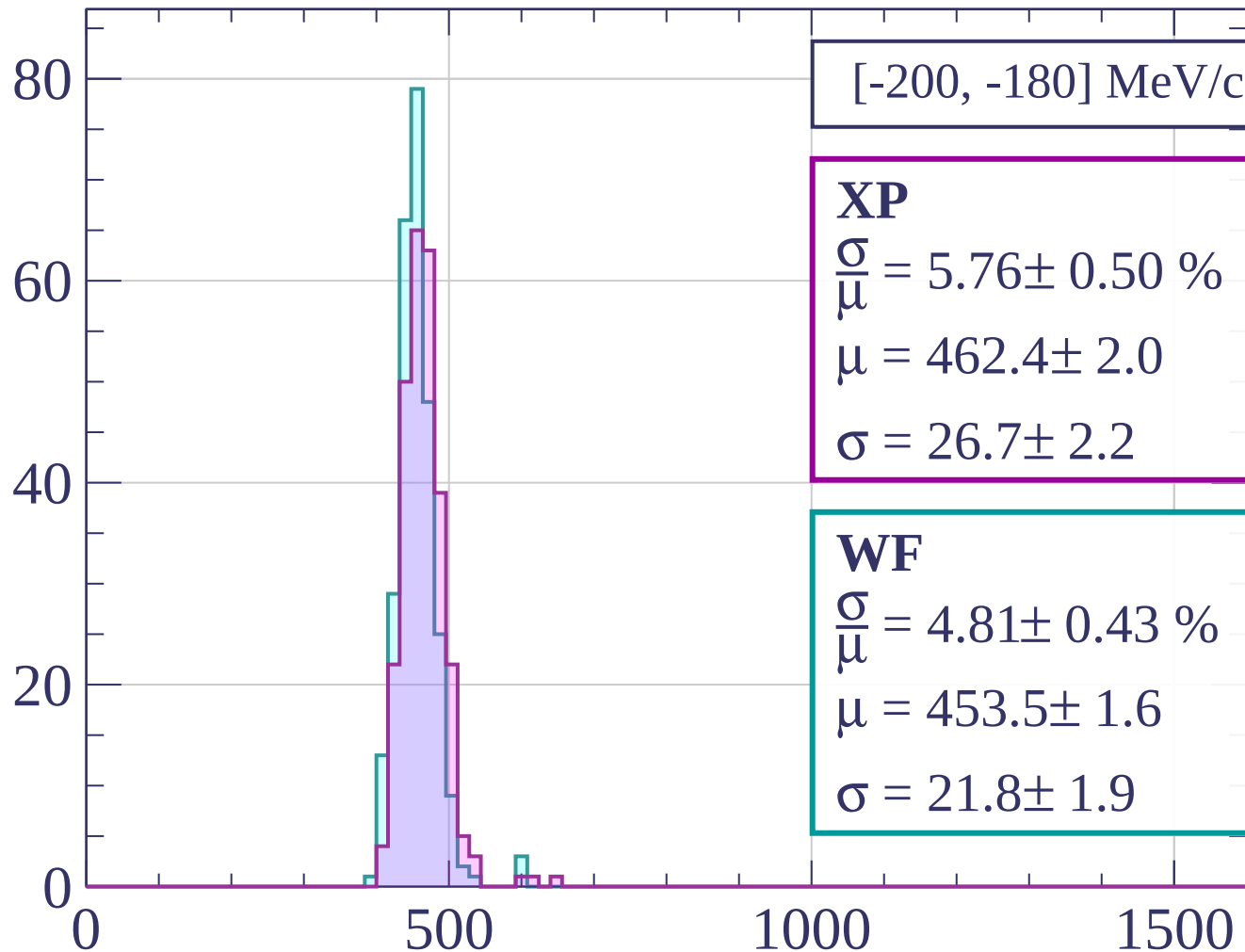


Count



dE/dx [ADC counts/cm]

Count



[-200, -180] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.76 \pm 0.50 \%$$

$$\mu = 462.4 \pm 2.0$$

$$\sigma = 26.7 \pm 2.2$$

WF

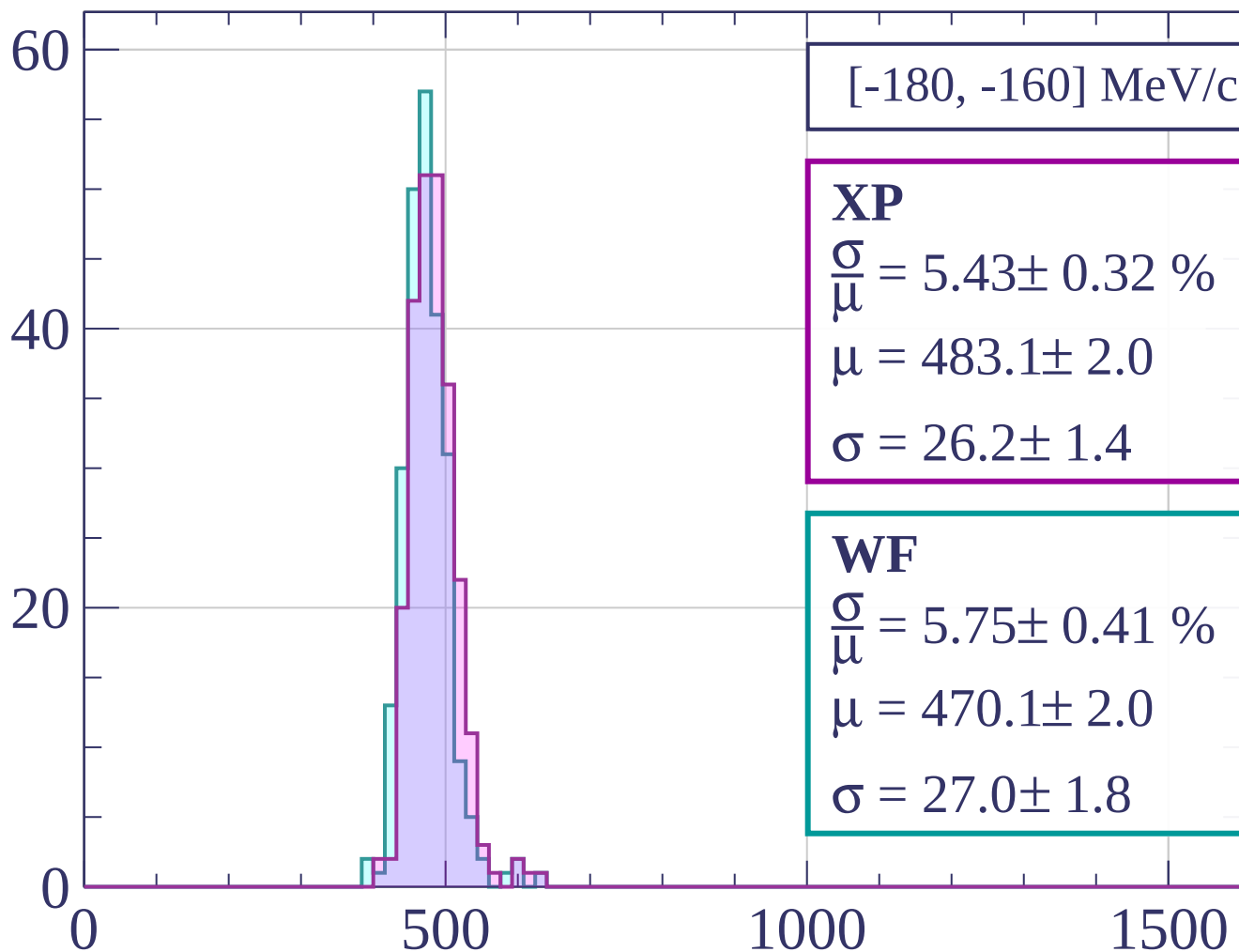
$$\frac{\sigma}{\mu} = 4.81 \pm 0.43 \%$$

$$\mu = 453.5 \pm 1.6$$

$$\sigma = 21.8 \pm 1.9$$

dE/dx [ADC counts/cm]

Count



[-180, -160] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.43 \pm 0.32 \%$$

$$\mu = 483.1 \pm 2.0$$

$$\sigma = 26.2 \pm 1.4$$

WF

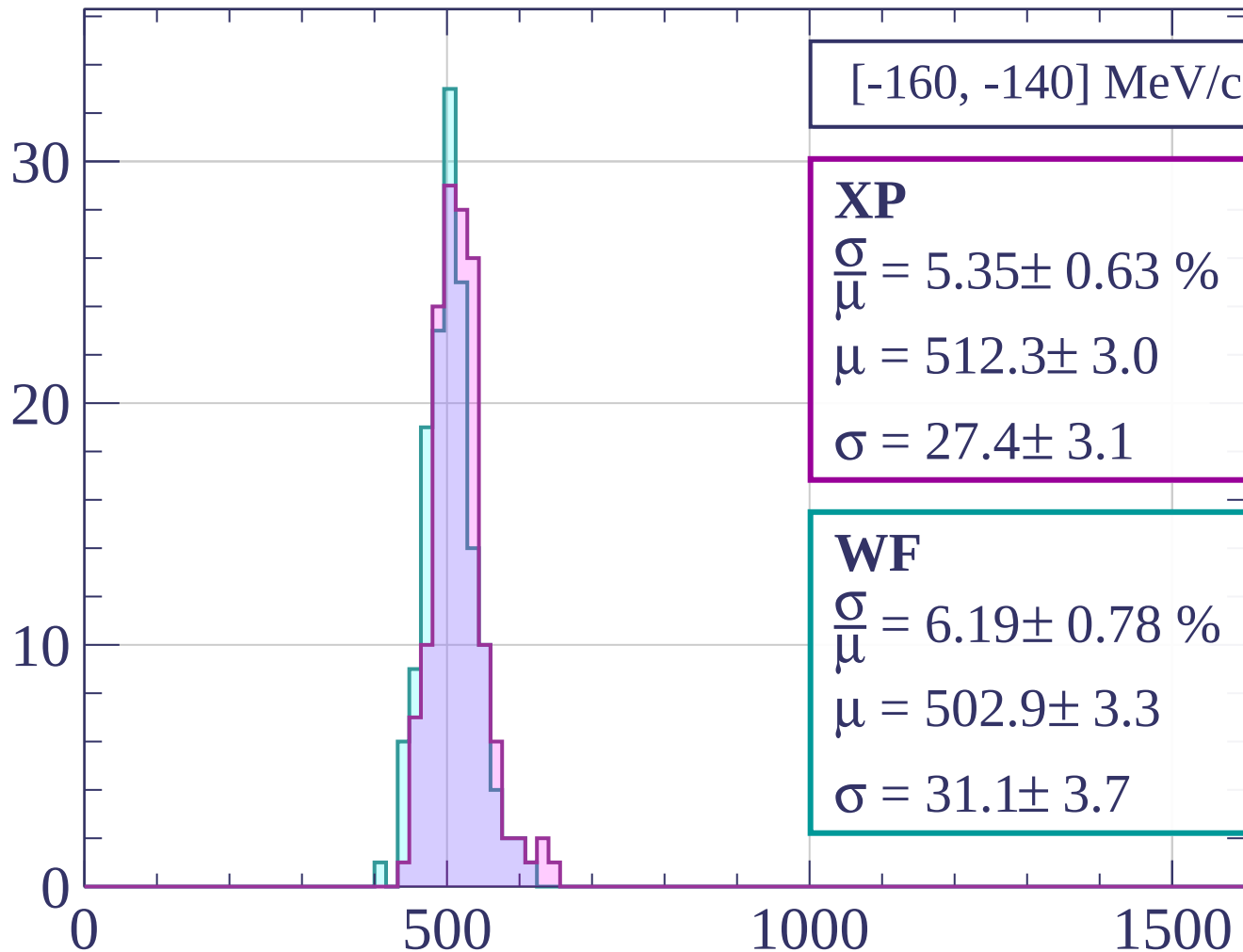
$$\frac{\sigma}{\mu} = 5.75 \pm 0.41 \%$$

$$\mu = 470.1 \pm 2.0$$

$$\sigma = 27.0 \pm 1.8$$

dE/dx [ADC counts/cm]

Count



[-160, -140] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.35 \pm 0.63 \%$$

$$\mu = 512.3 \pm 3.0$$

$$\sigma = 27.4 \pm 3.1$$

WF

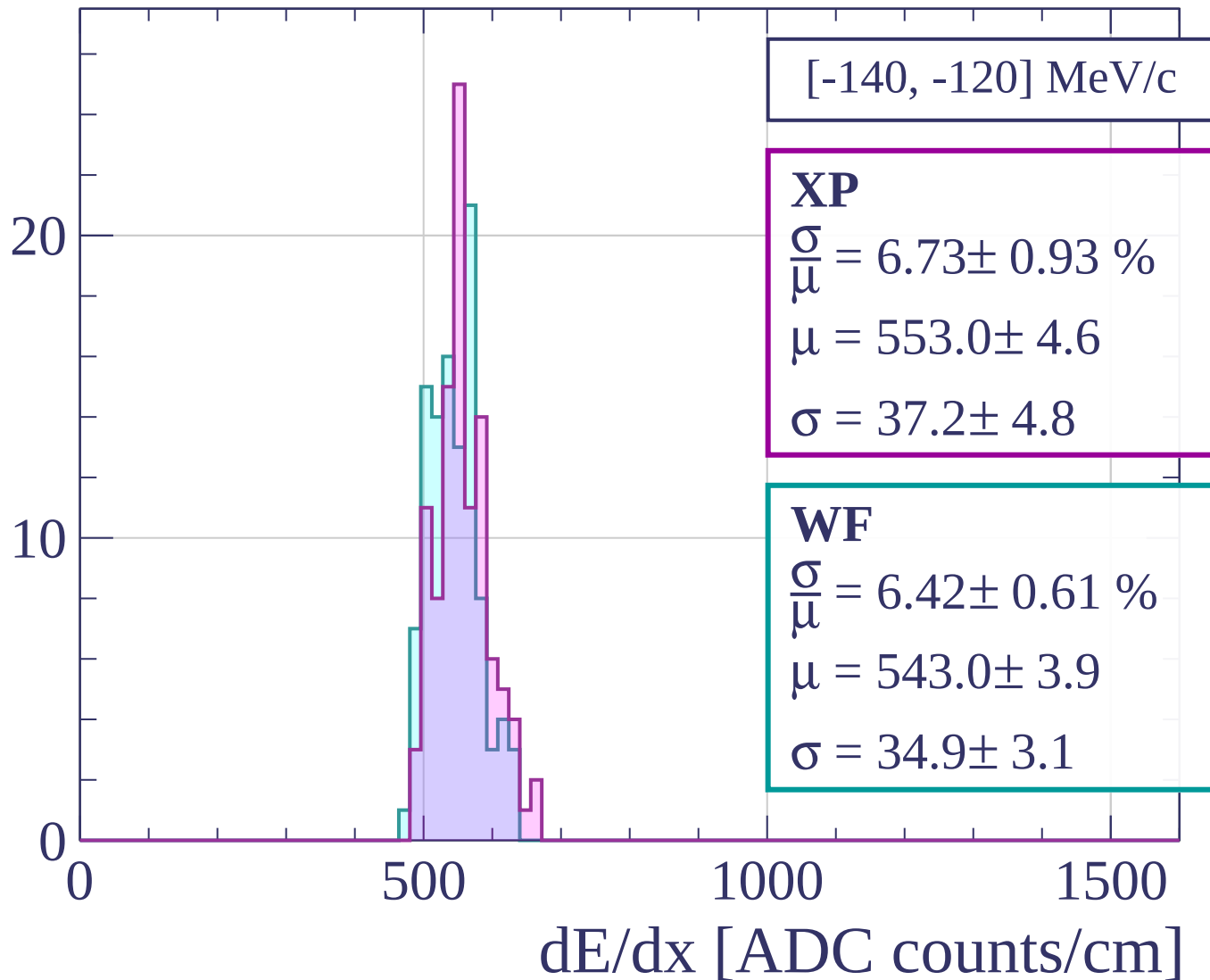
$$\frac{\sigma}{\mu} = 6.19 \pm 0.78 \%$$

$$\mu = 502.9 \pm 3.3$$

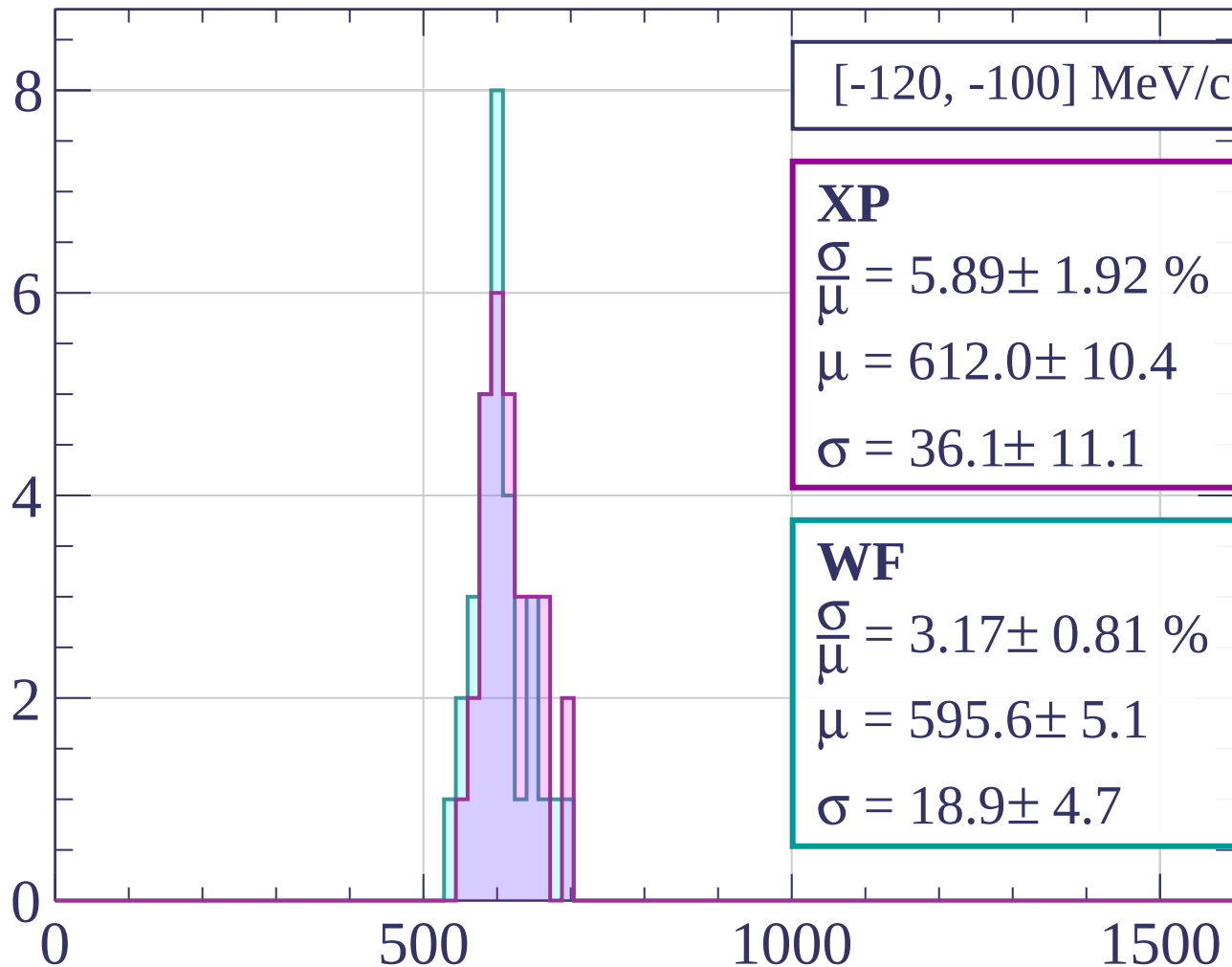
$$\sigma = 31.1 \pm 3.7$$

dE/dx [ADC counts/cm]

Count



Count



[-120, -100] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.89 \pm 1.92 \%$$

$$\mu = 612.0 \pm 10.4$$

$$\sigma = 36.1 \pm 11.1$$

WF

$$\frac{\sigma}{\mu} = 3.17 \pm 0.81 \%$$

$$\mu = 595.6 \pm 5.1$$

$$\sigma = 18.9 \pm 4.7$$

dE/dx [ADC counts/cm]

Count

[-100, -80] MeV/c

XP

$$\frac{\sigma}{\mu} = 32.29 \pm 90.74 \%$$

$$\mu = 608.8 \pm 307.6$$

$$\sigma = 196.6 \pm 453.1$$

WF

$$\frac{\sigma}{\mu} = 40.93 \pm 96.38 \%$$

$$\mu = 664.0 \pm 343.0$$

$$\sigma = 271.8 \pm 499.6$$

2.0
1.5
1.0
0.5
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

[-80, -60] MeV/c

XP

$$\frac{\sigma}{\mu} = 5.09 \pm 6.76 \%$$

$$\mu = 610.0 \pm 19.5$$

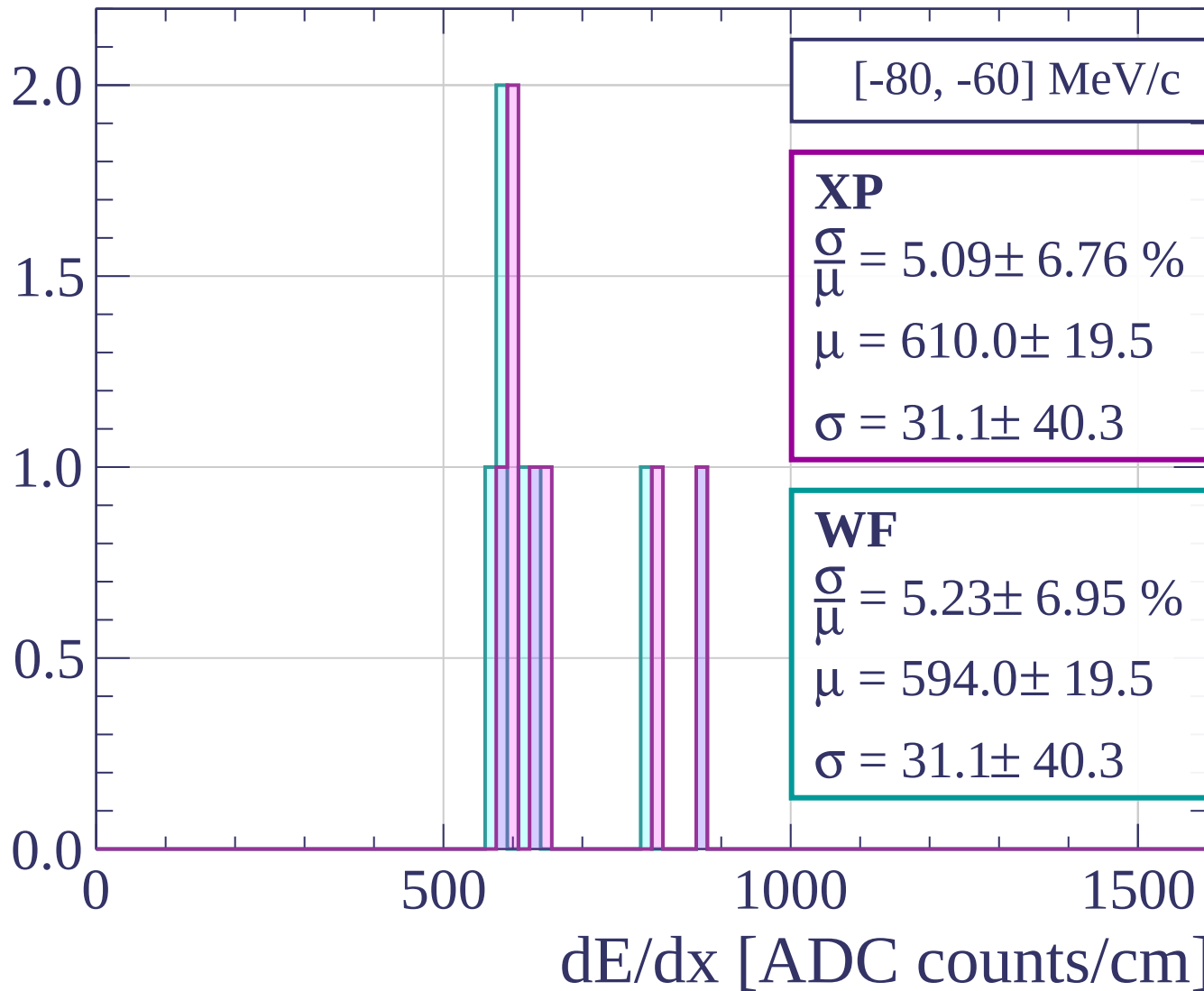
$$\sigma = 31.1 \pm 40.3$$

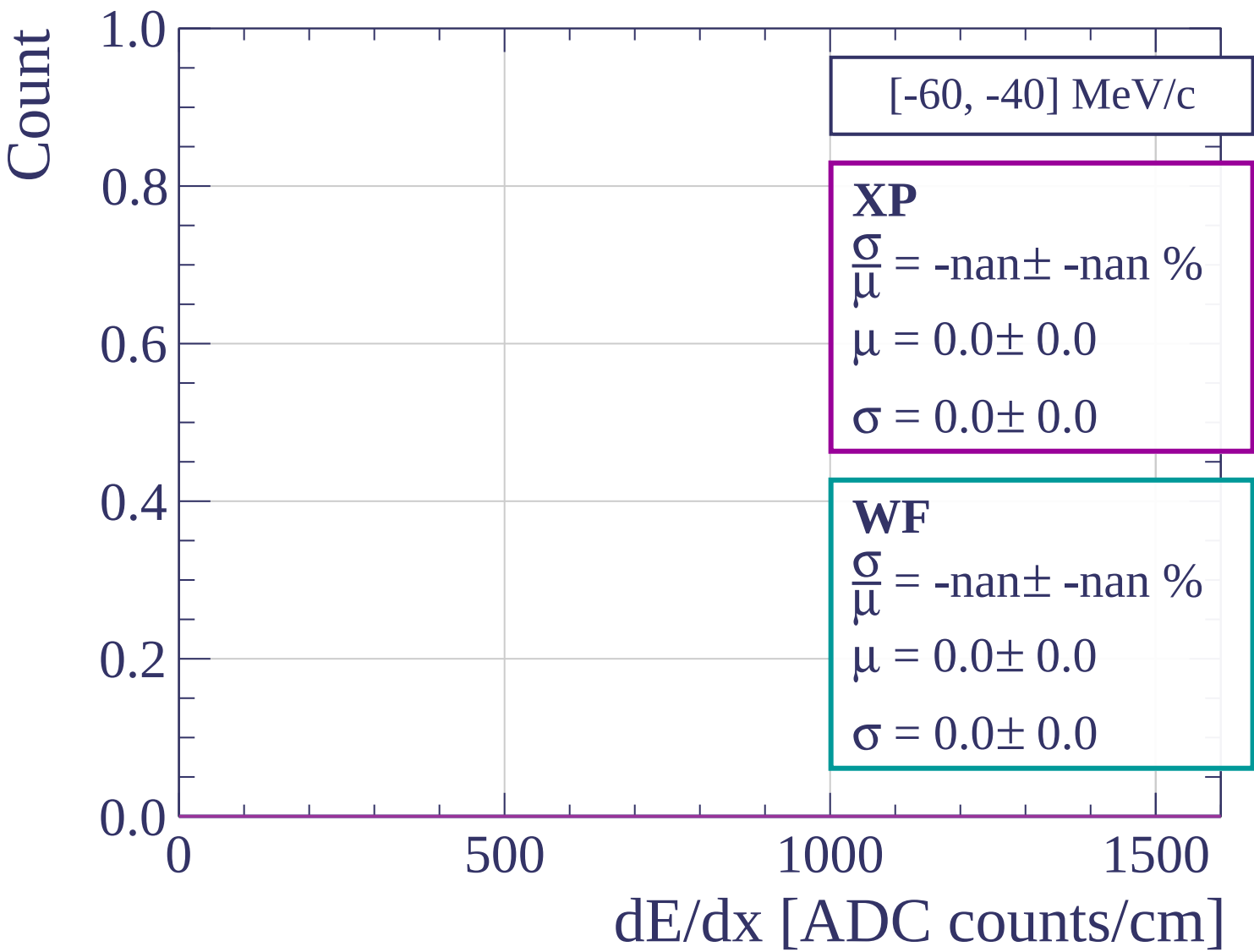
WF

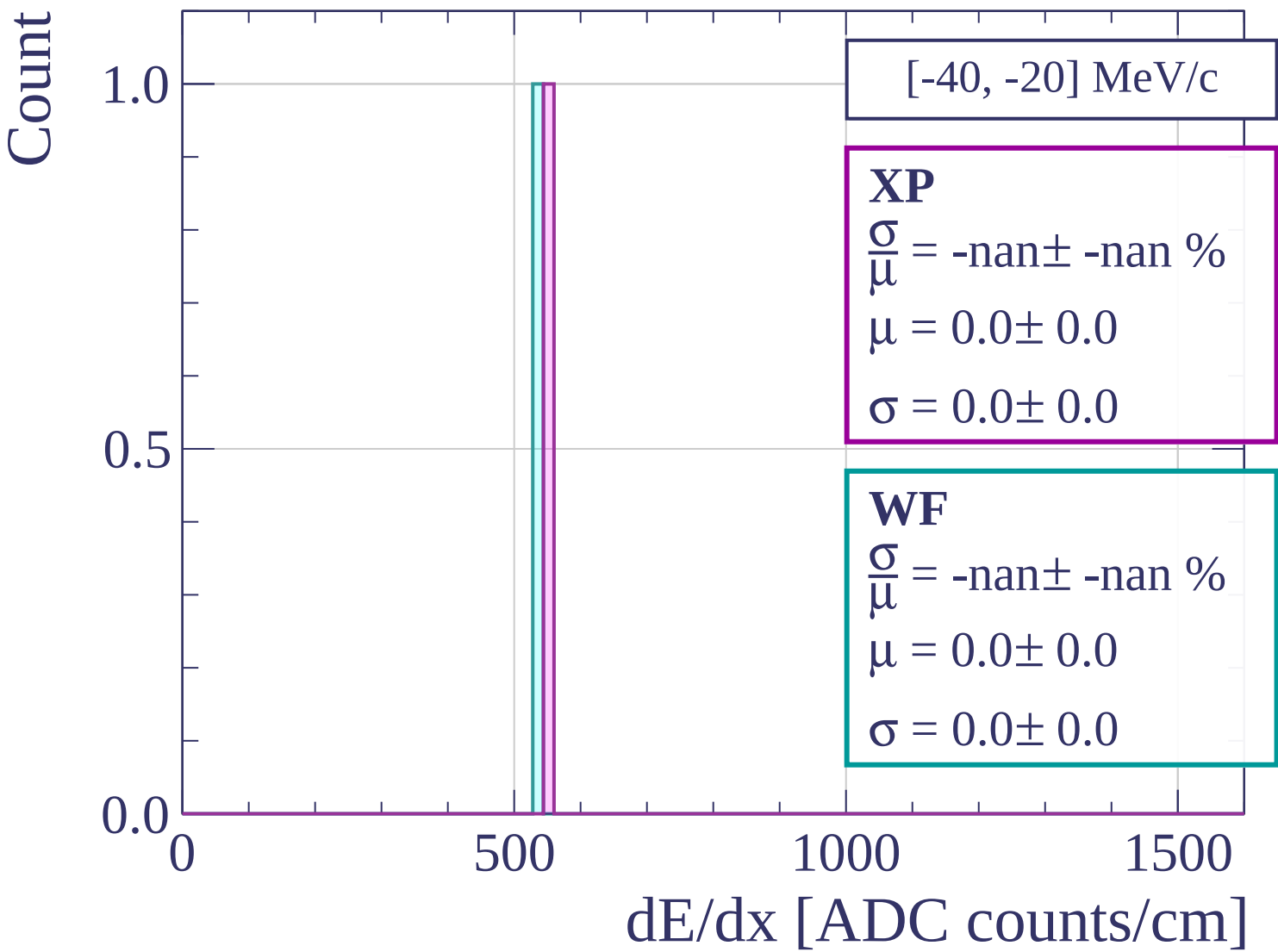
$$\frac{\sigma}{\mu} = 5.23 \pm 6.95 \%$$

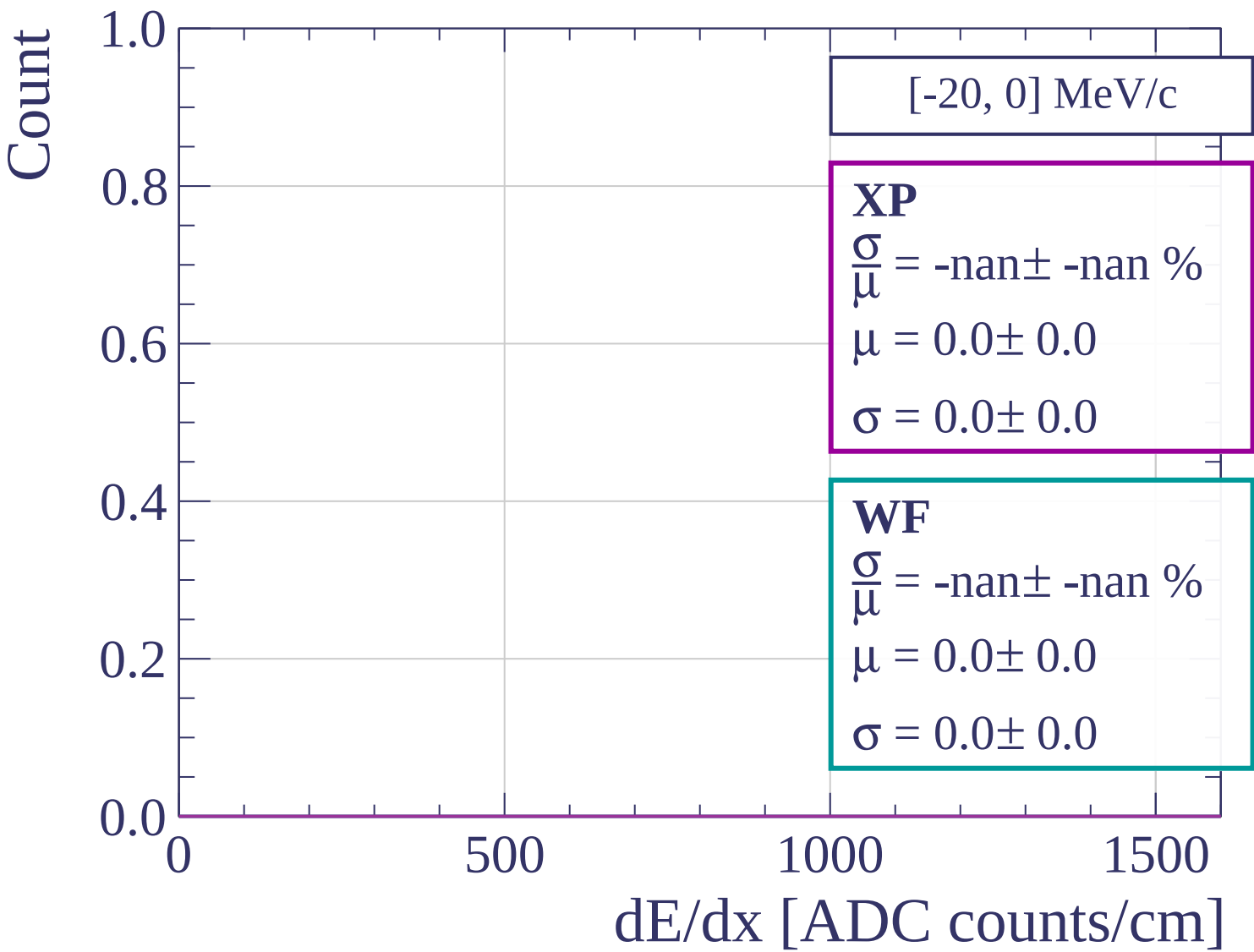
$$\mu = 594.0 \pm 19.5$$

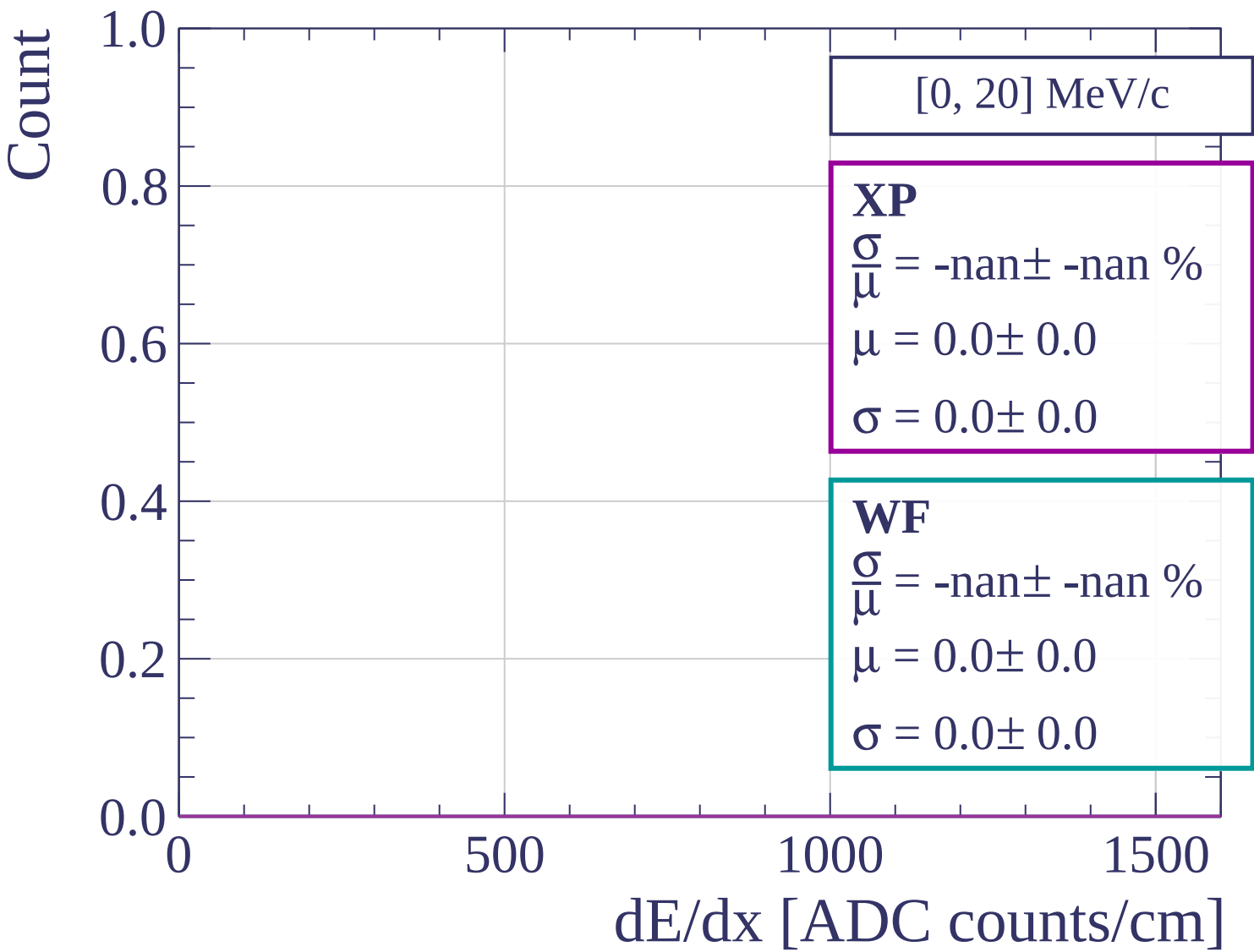
$$\sigma = 31.1 \pm 40.3$$

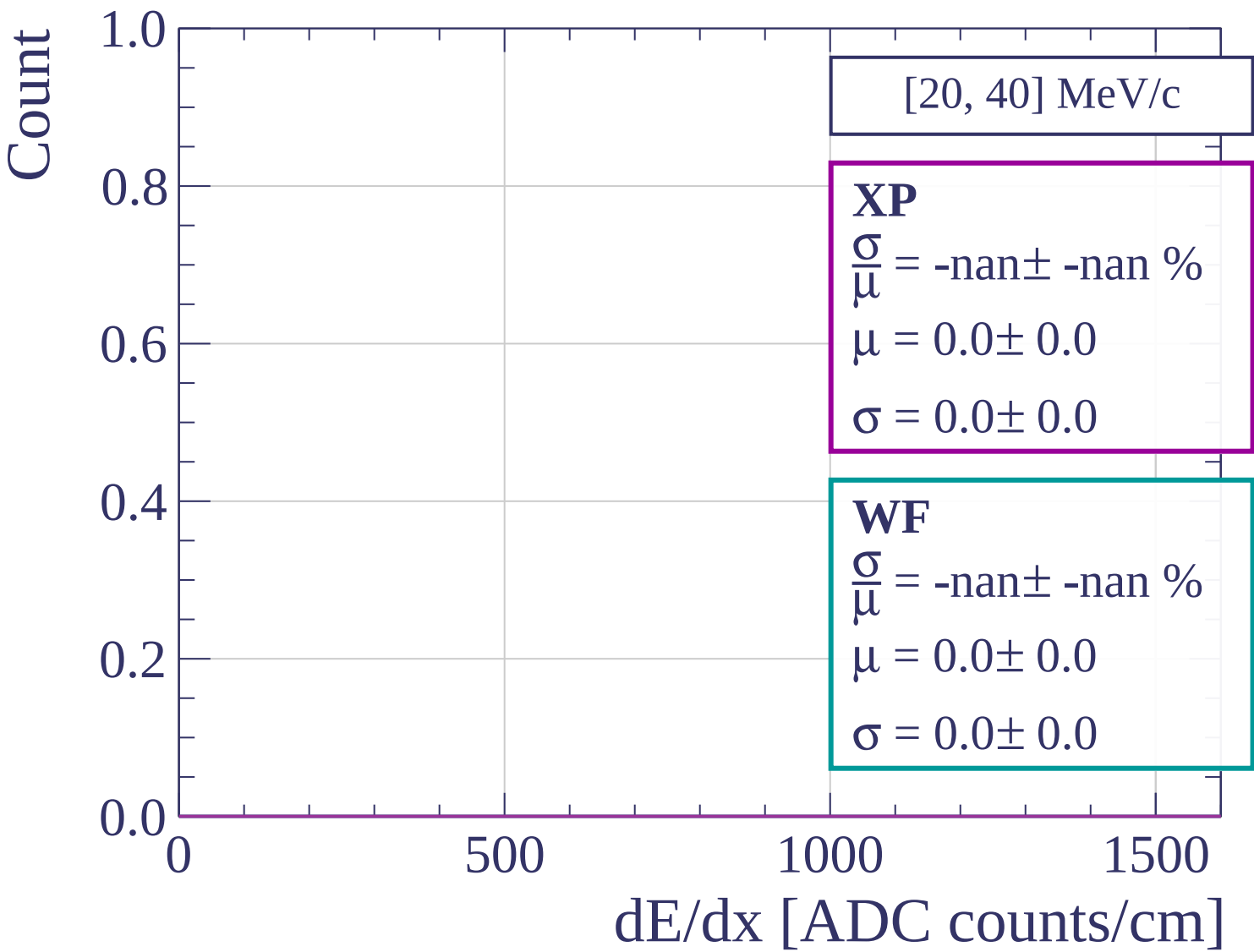


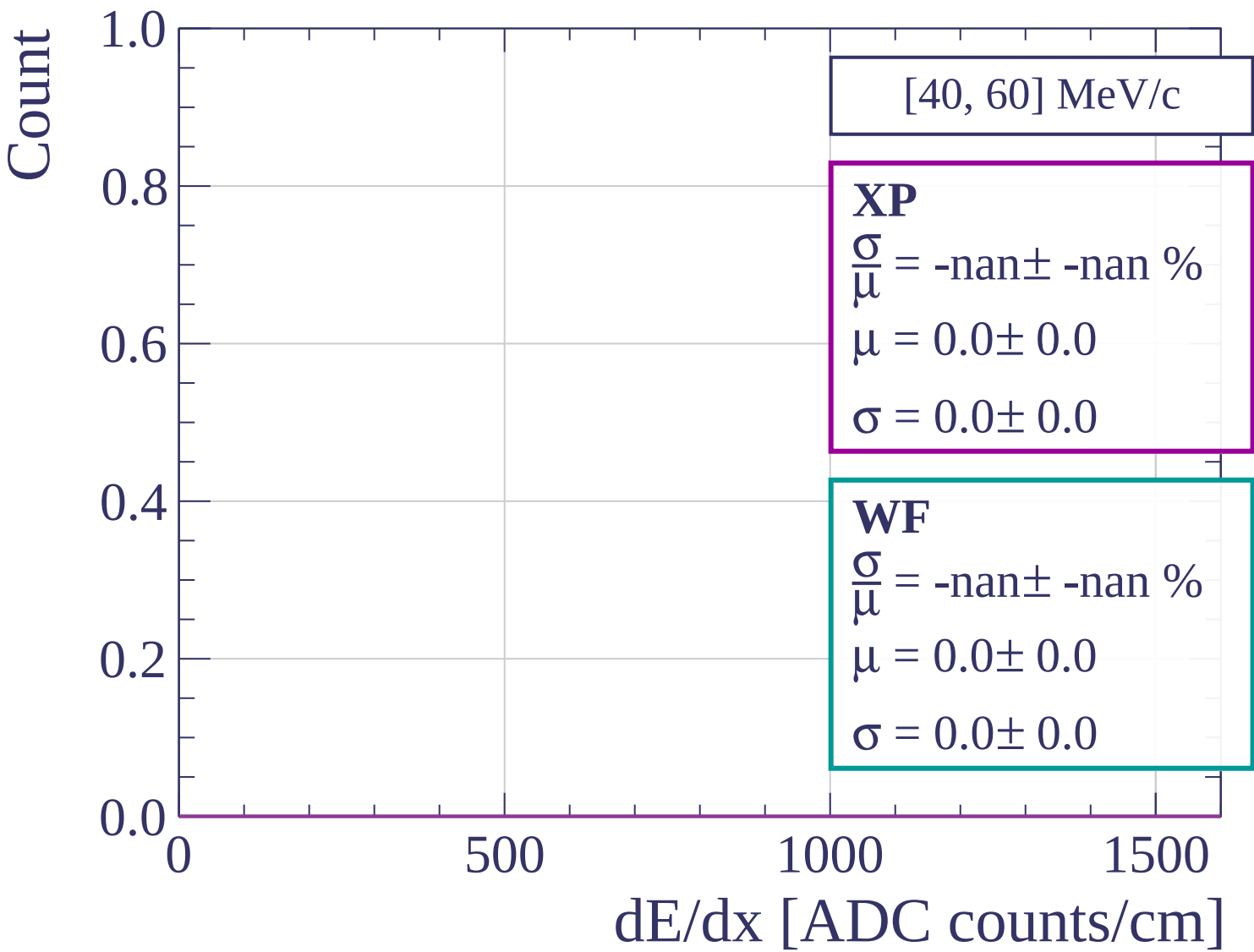




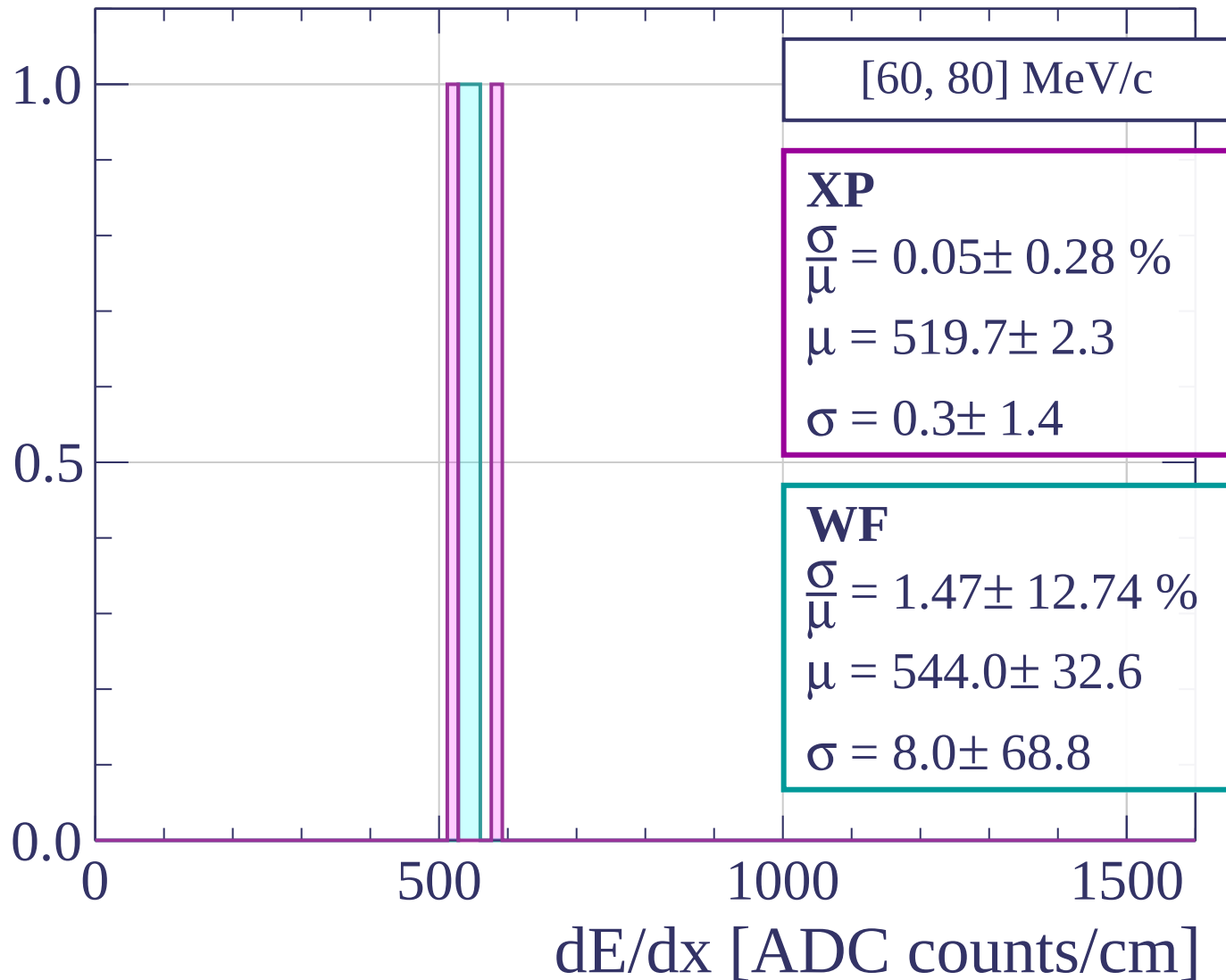


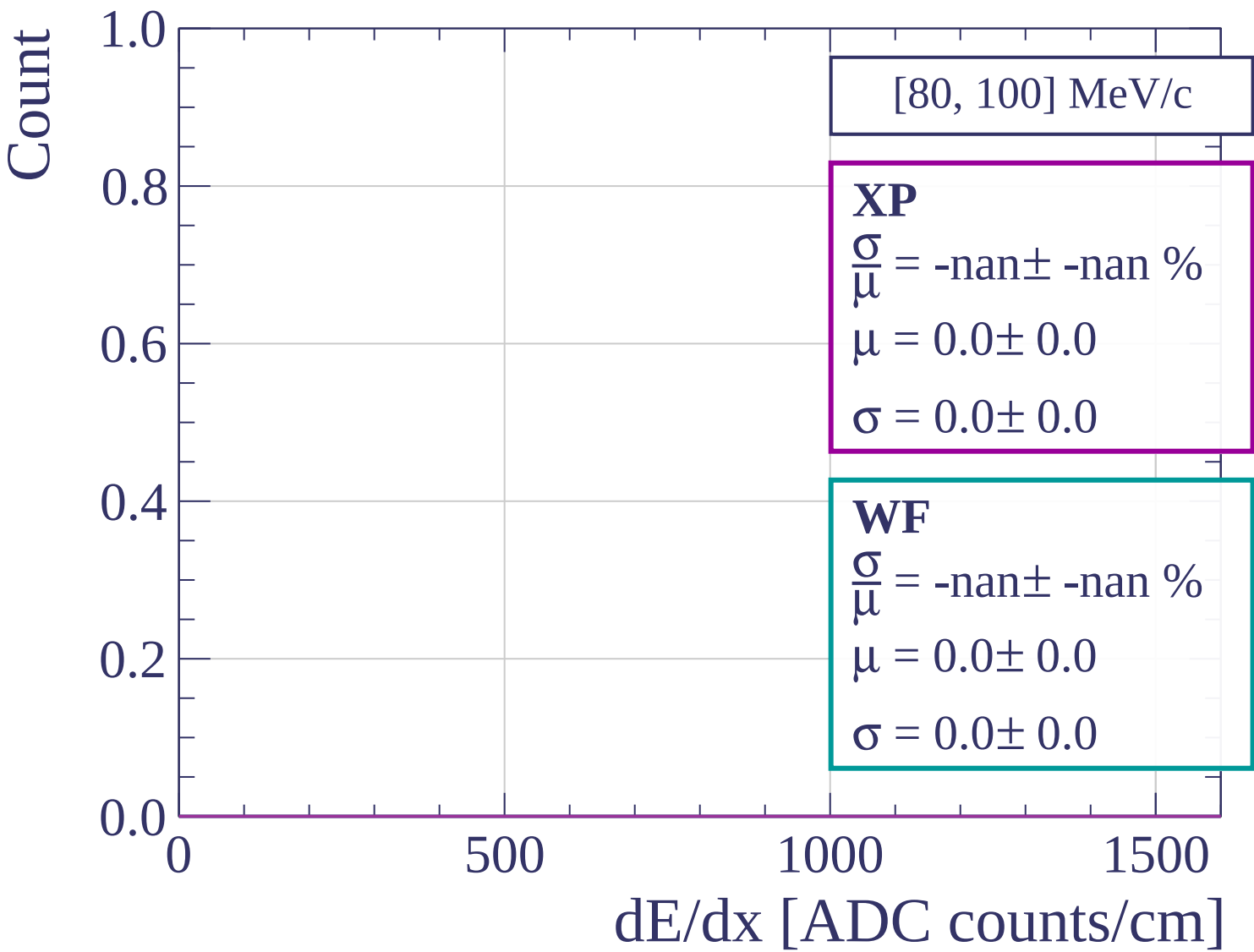


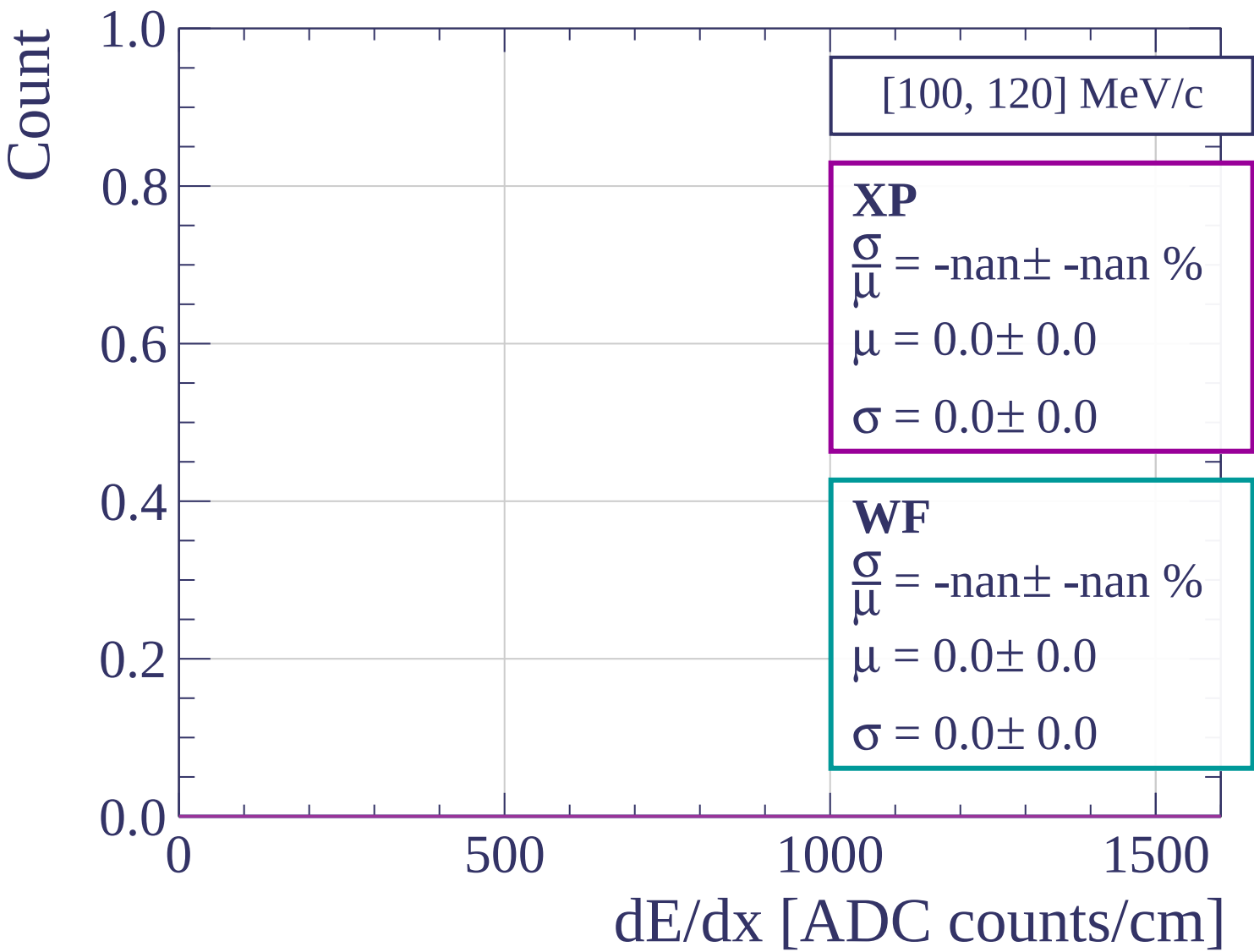


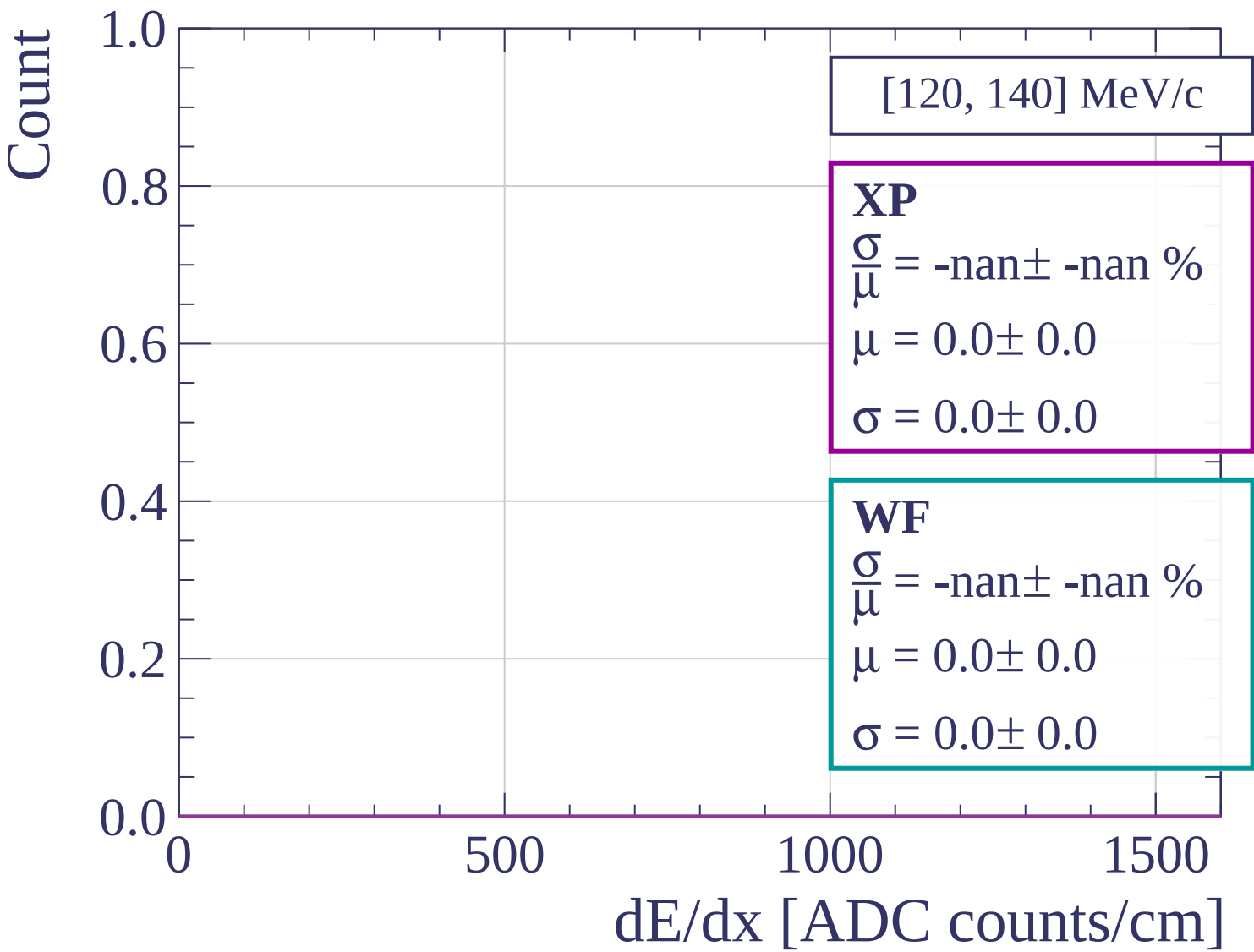


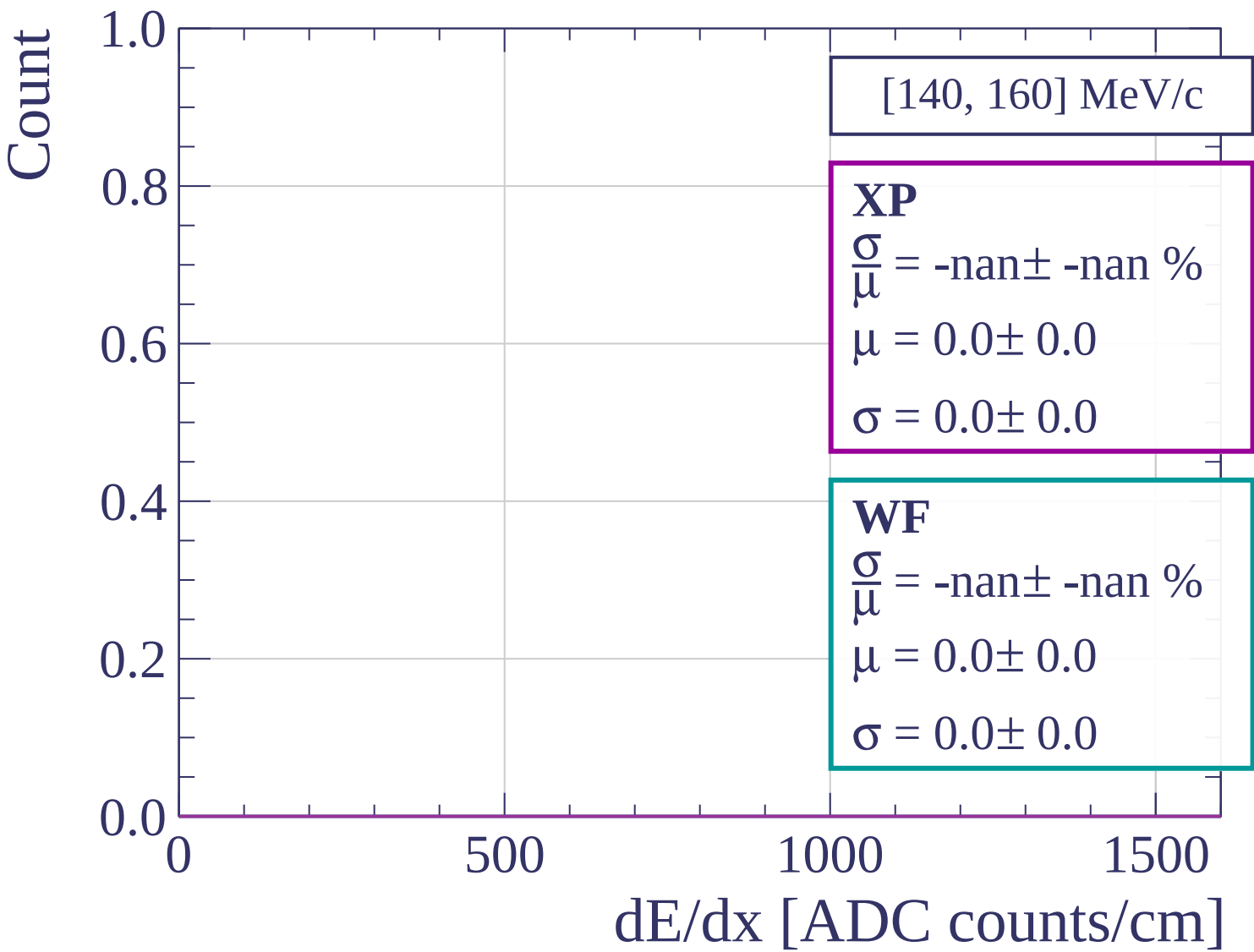
Count











Count

[160, 180] MeV/c

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.5

0.0

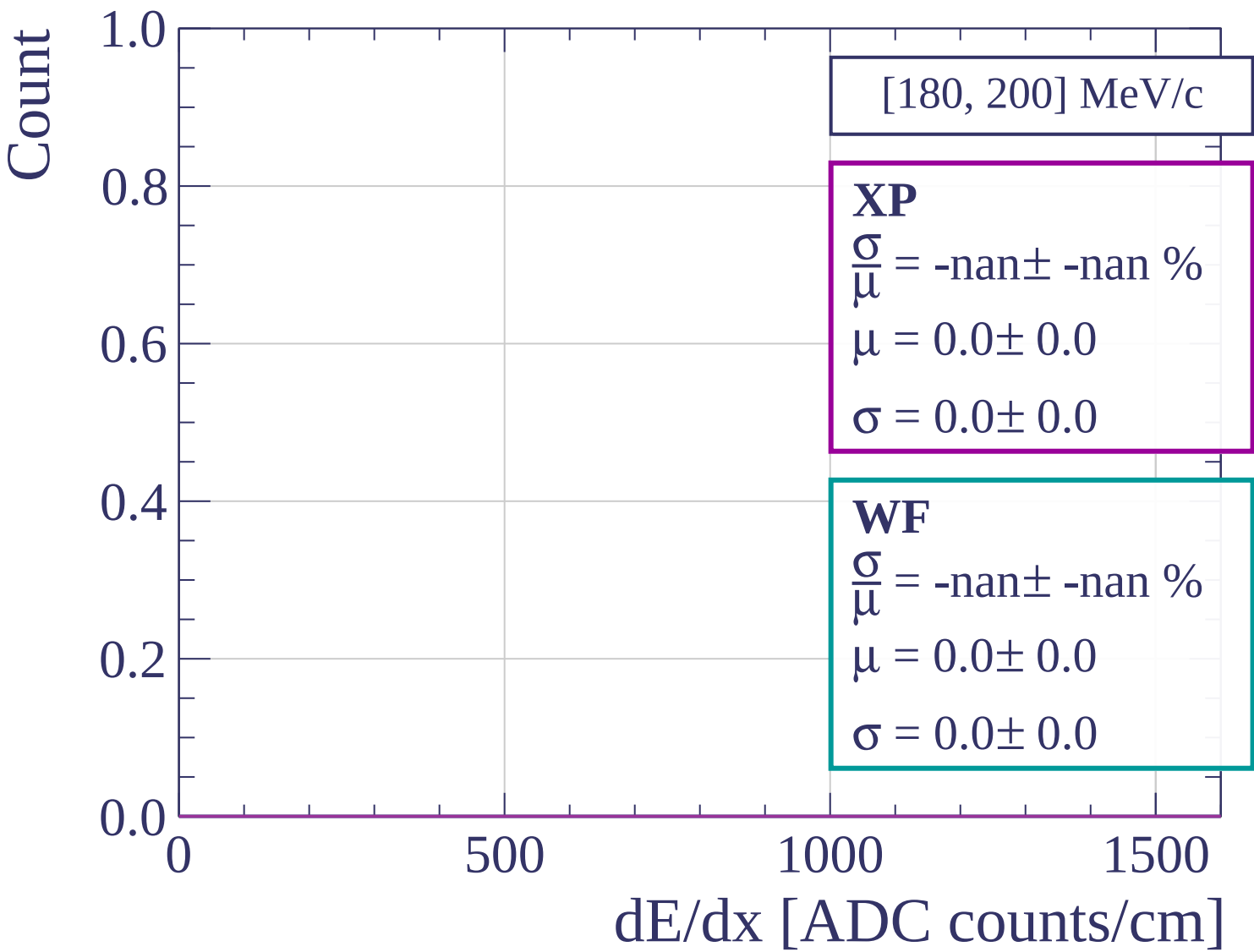
0

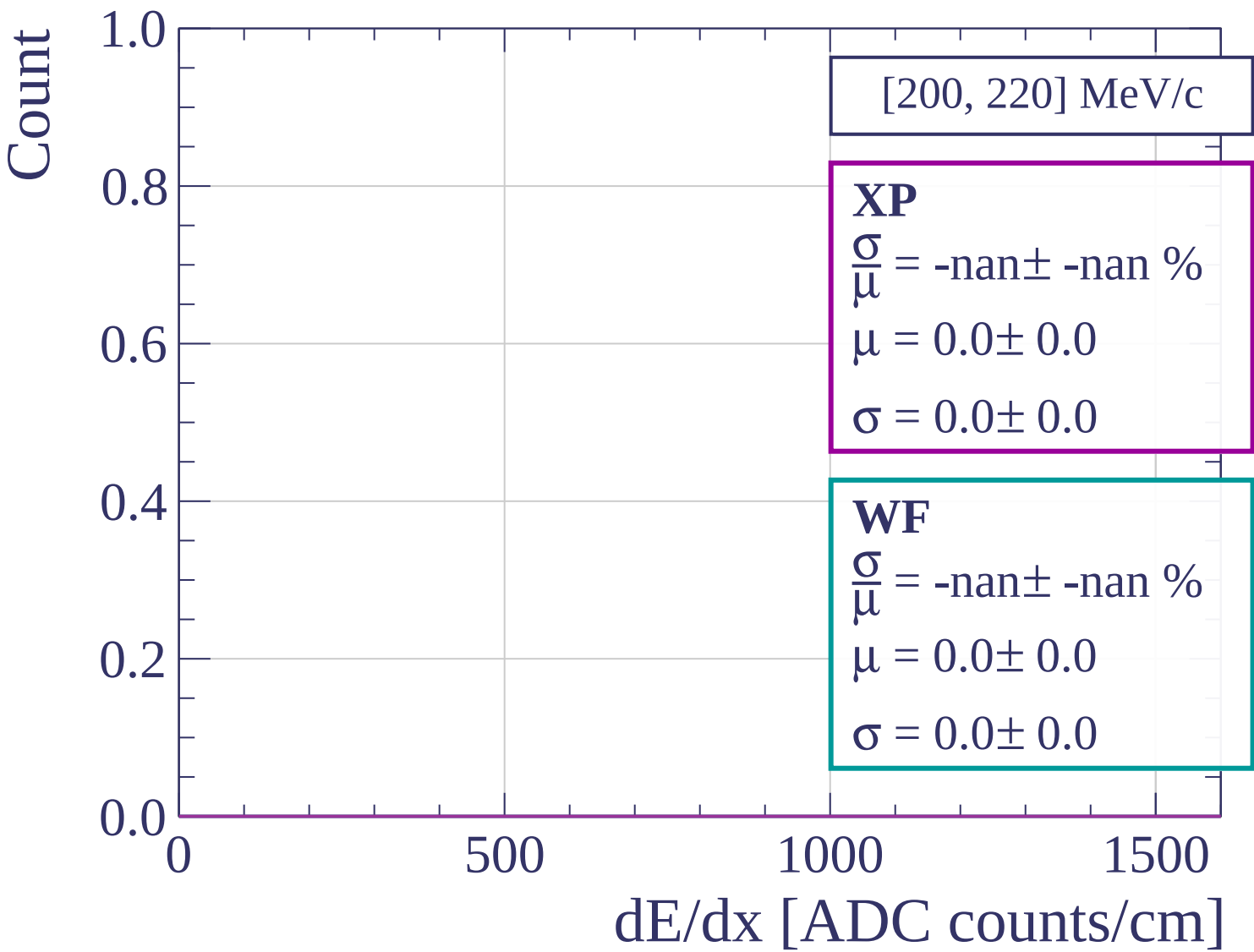
500

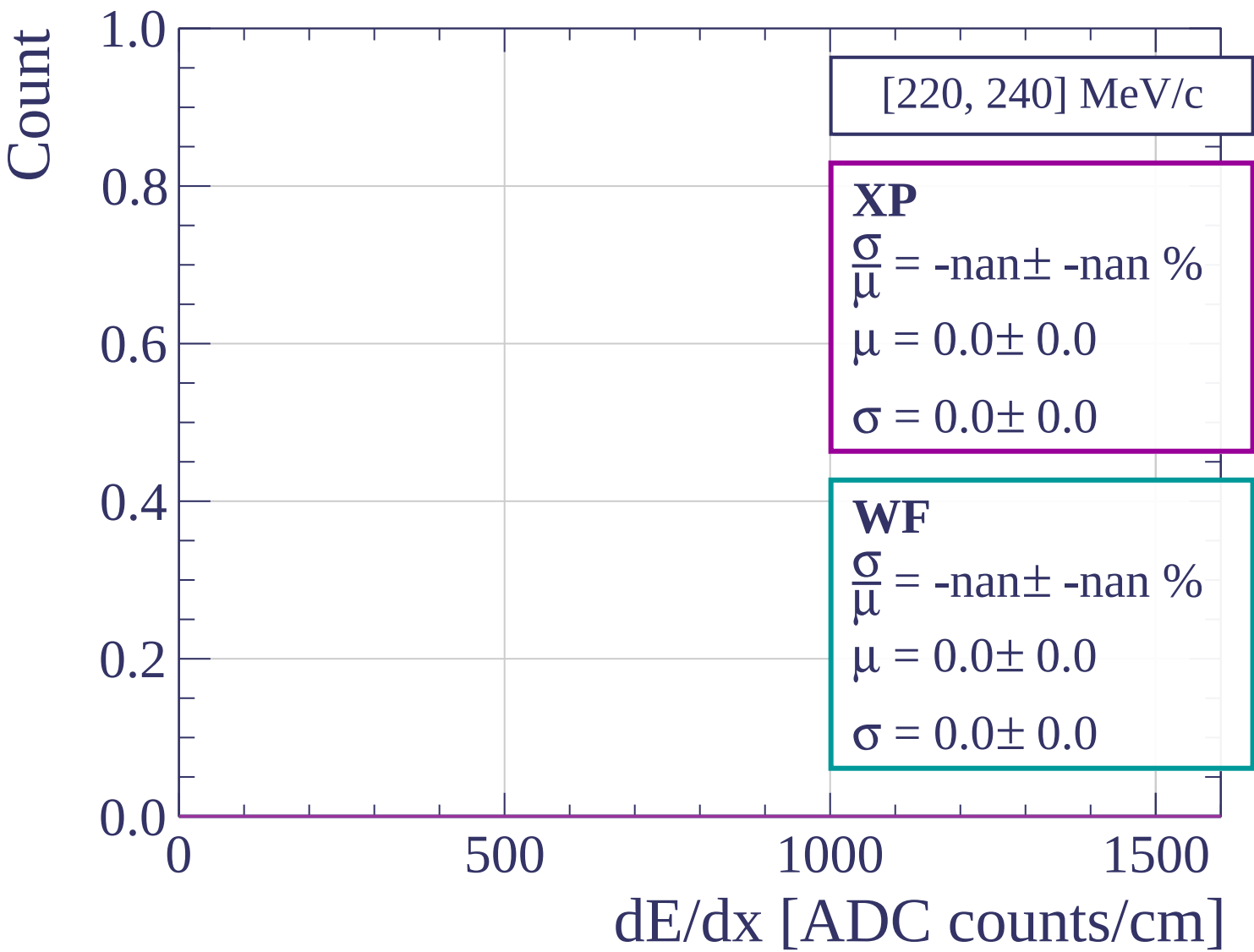
1000

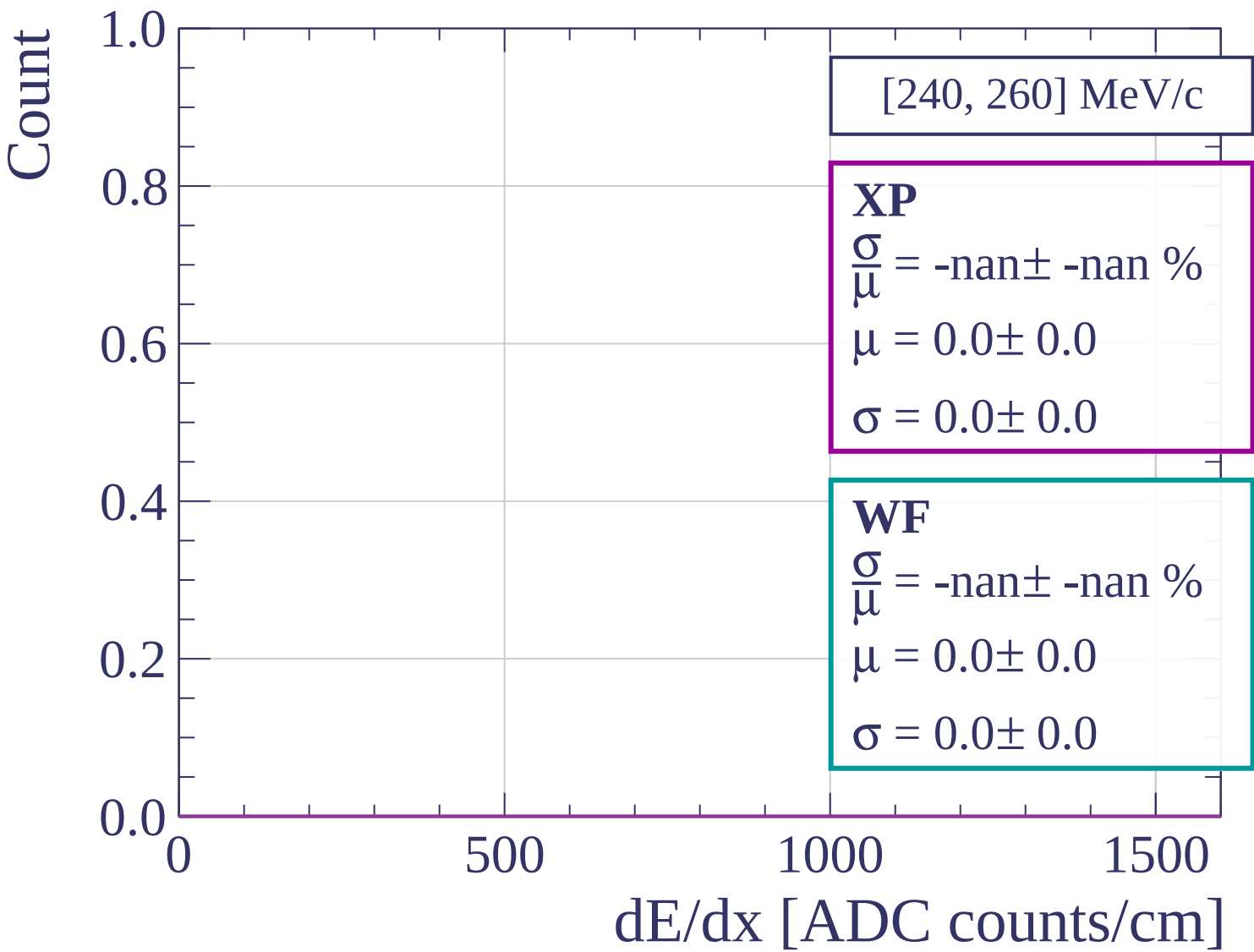
1500

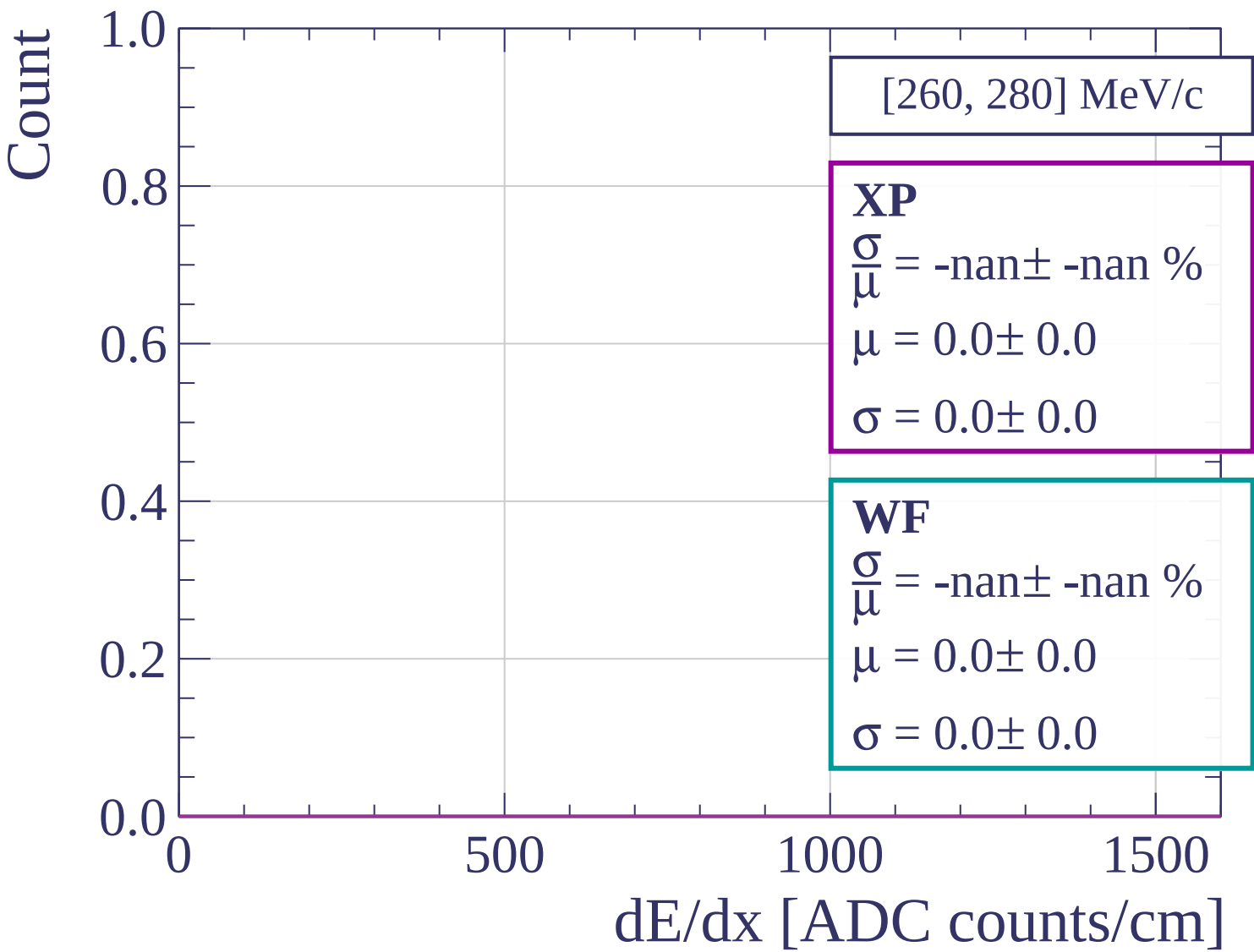
dE/dx [ADC counts/cm]

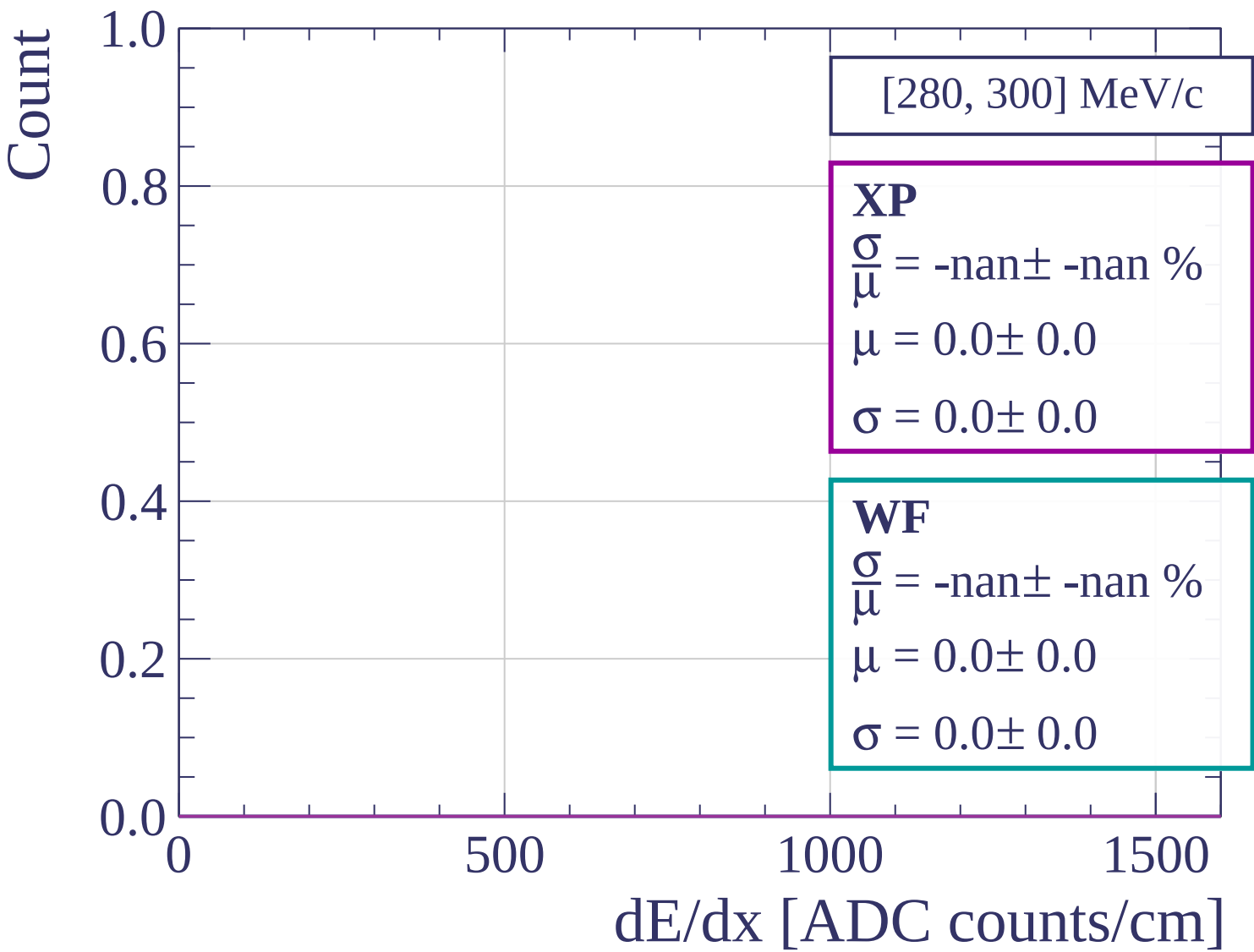


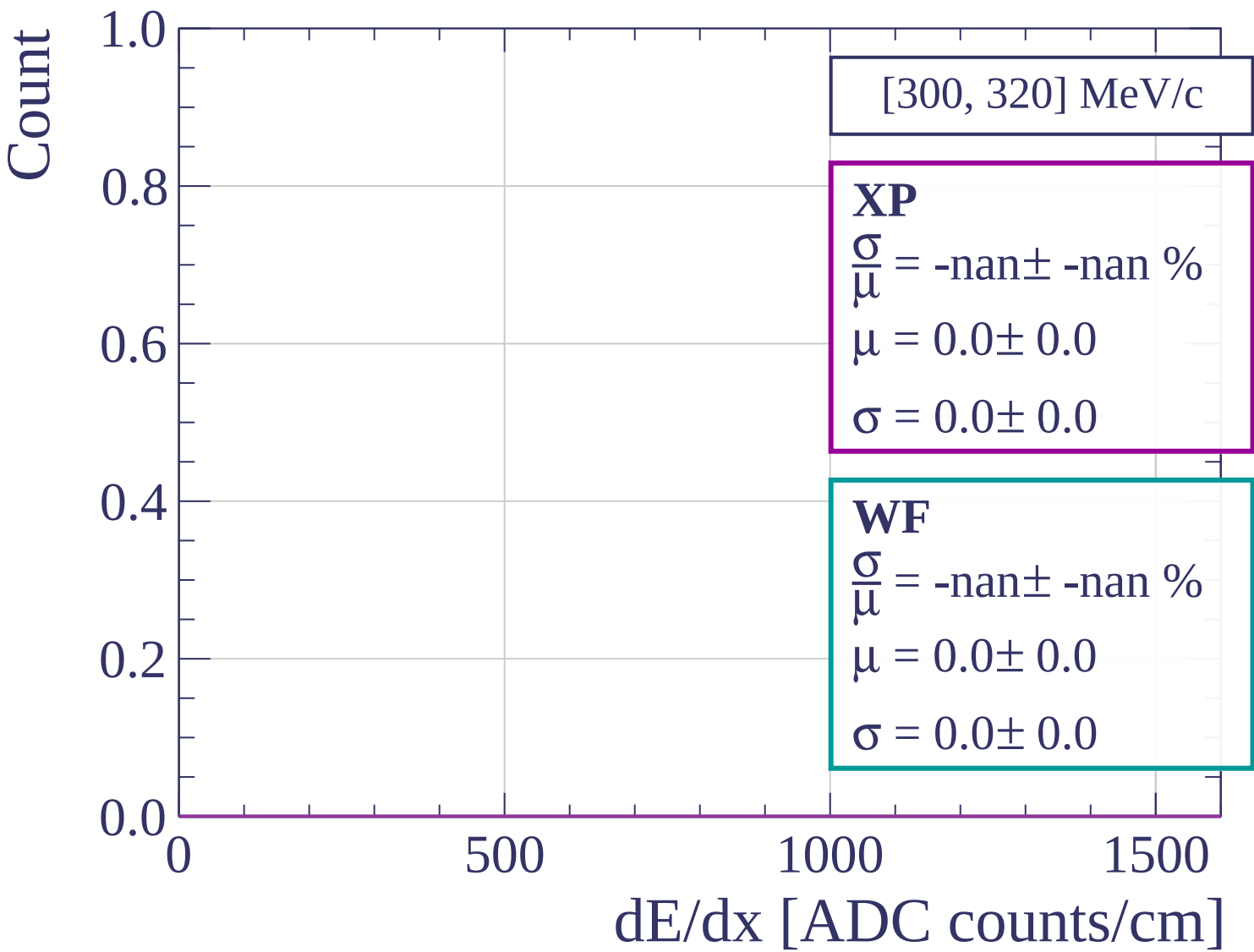


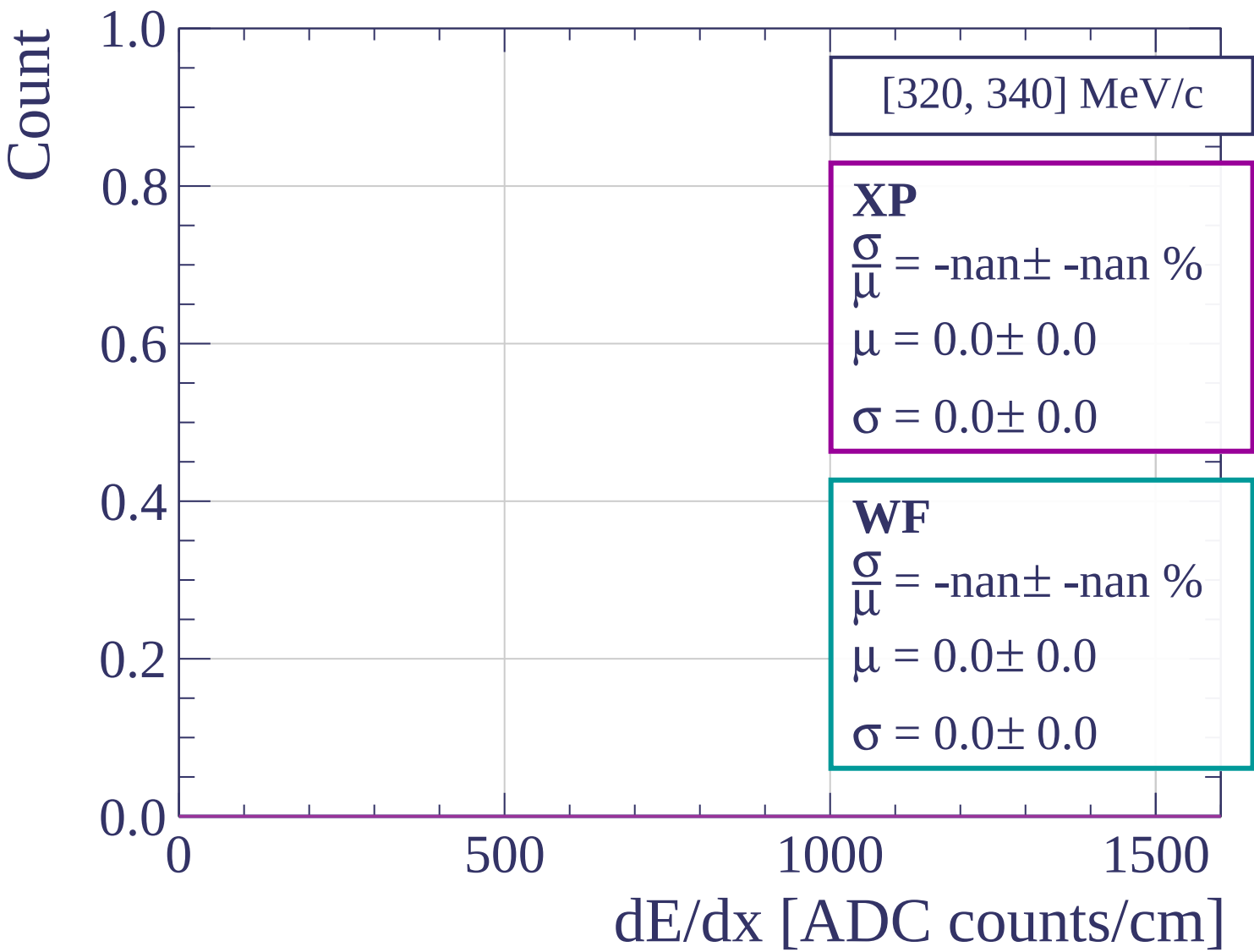


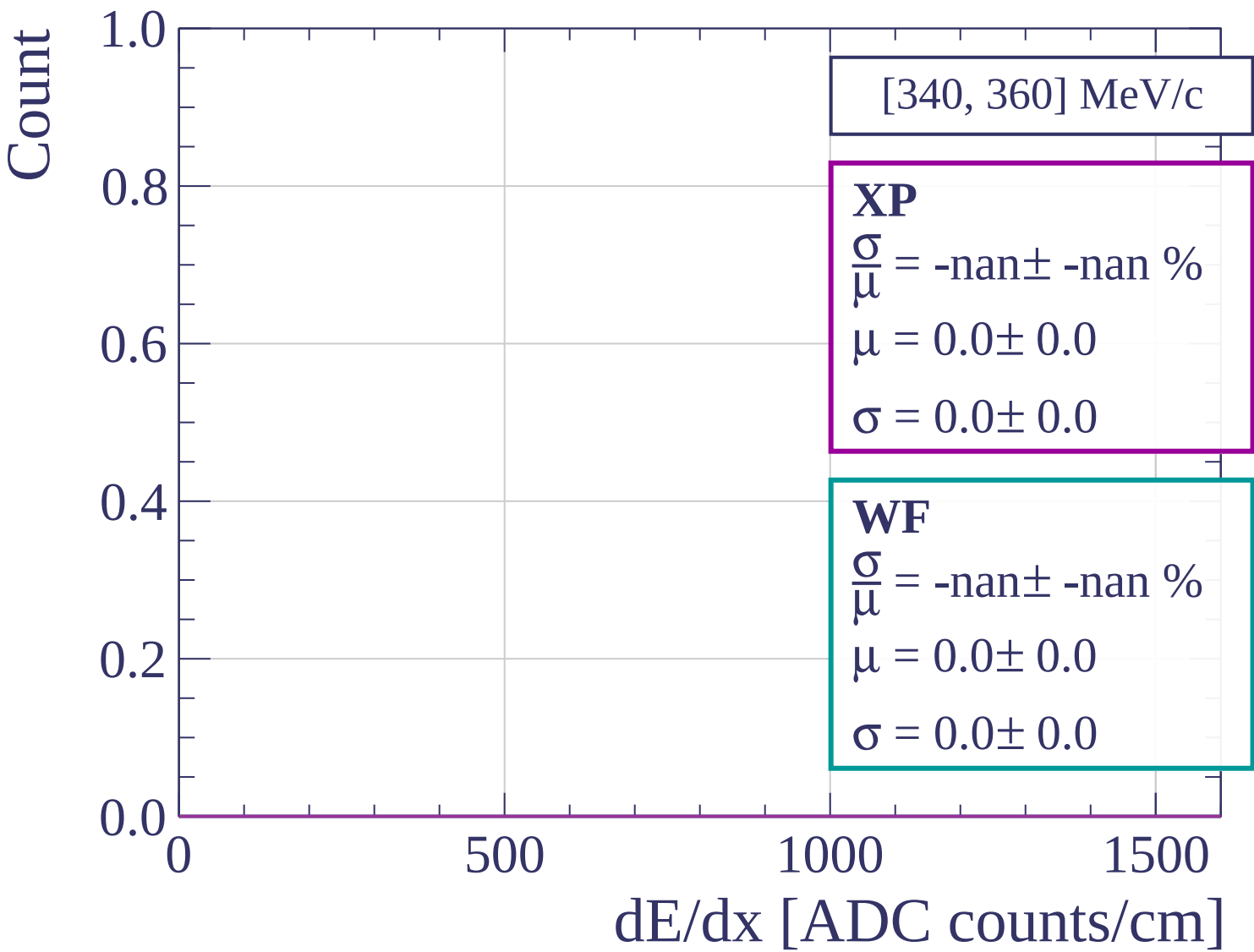


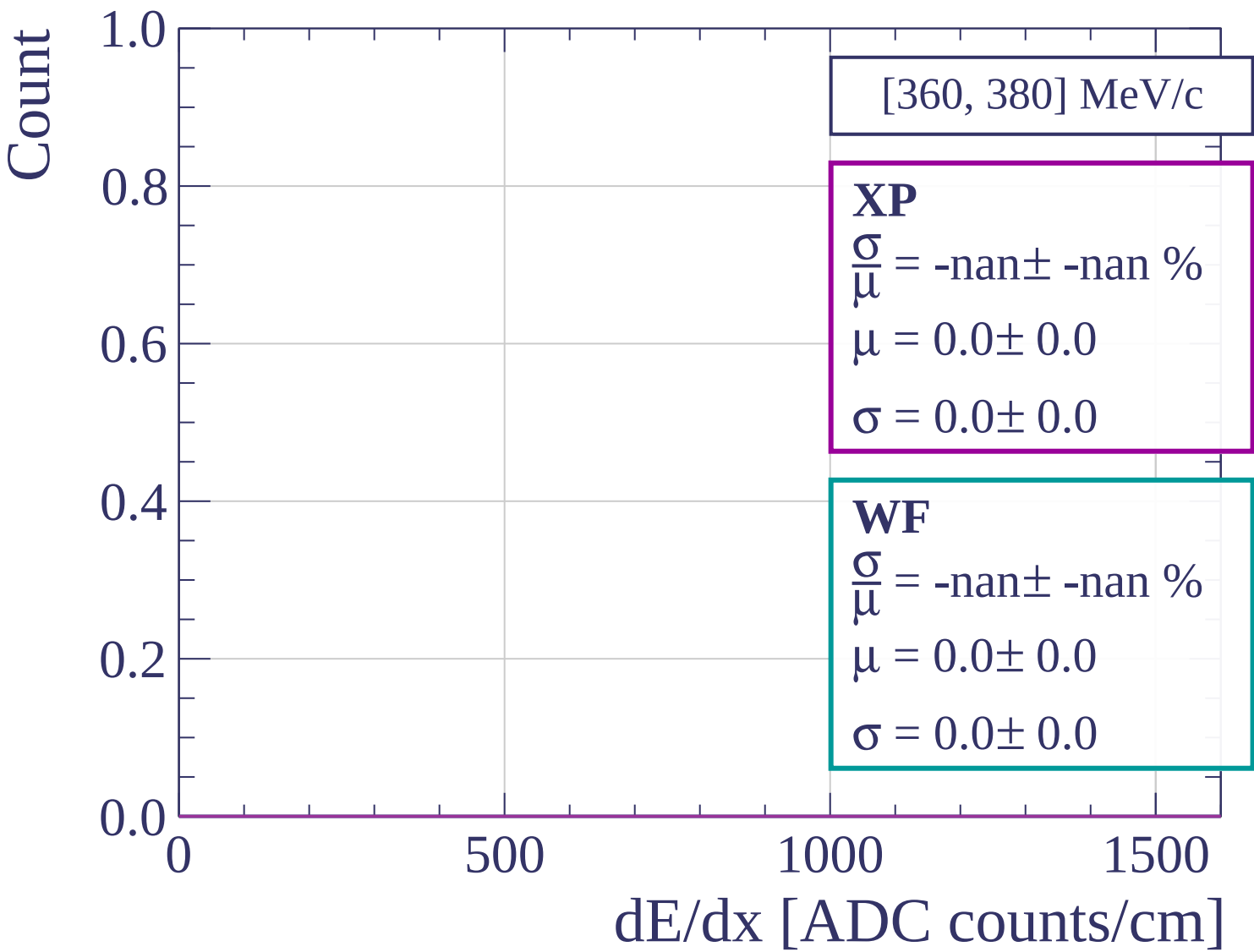


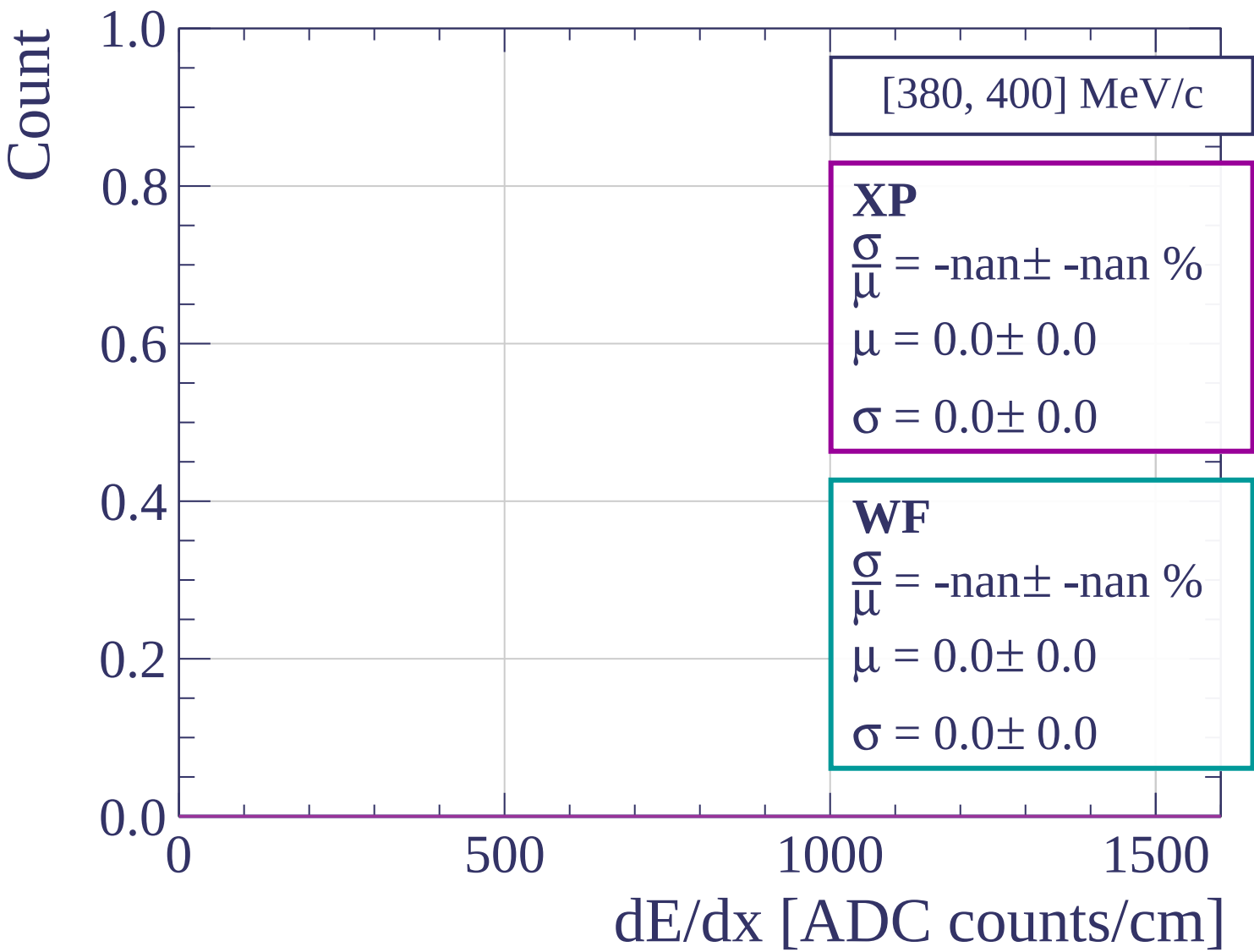


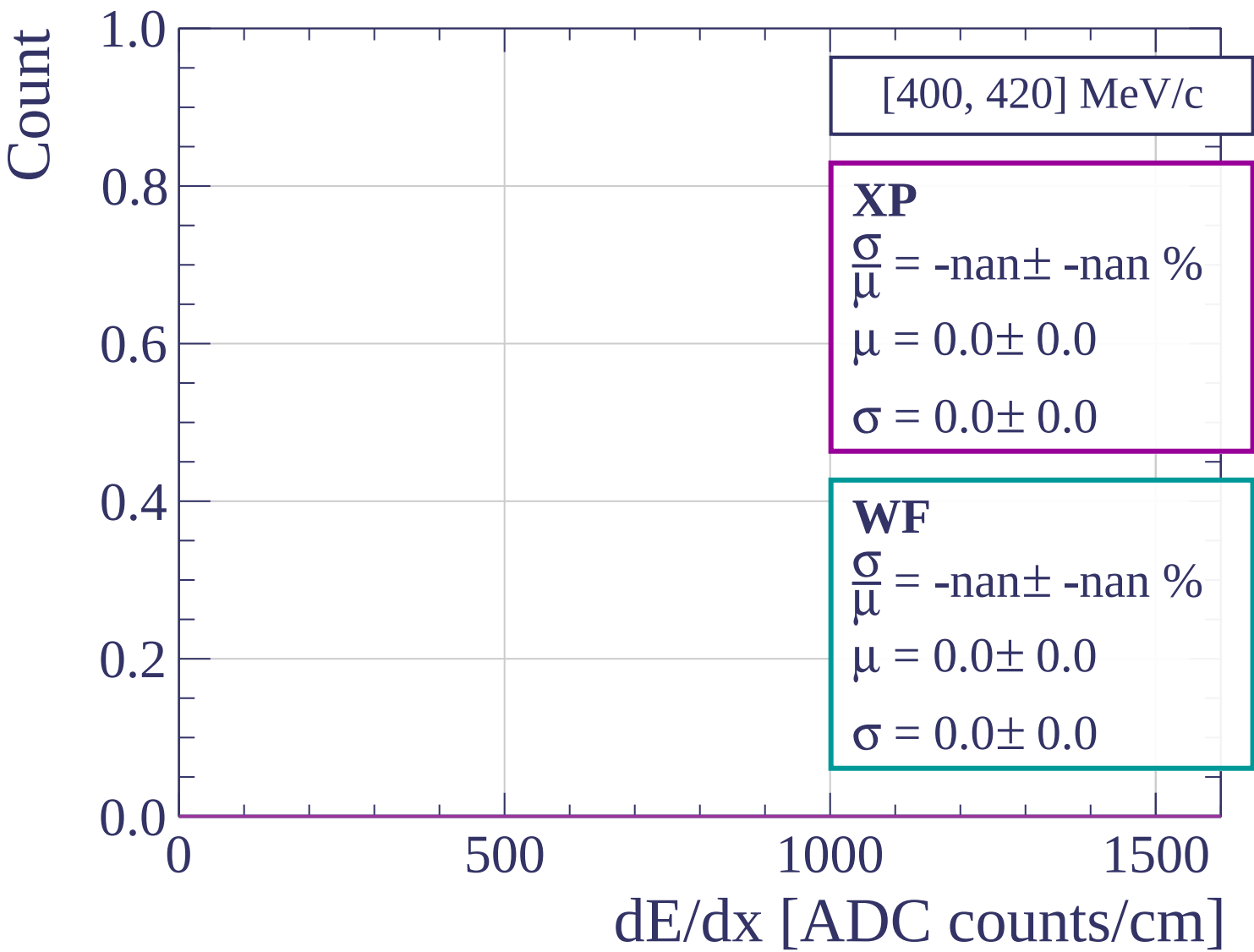


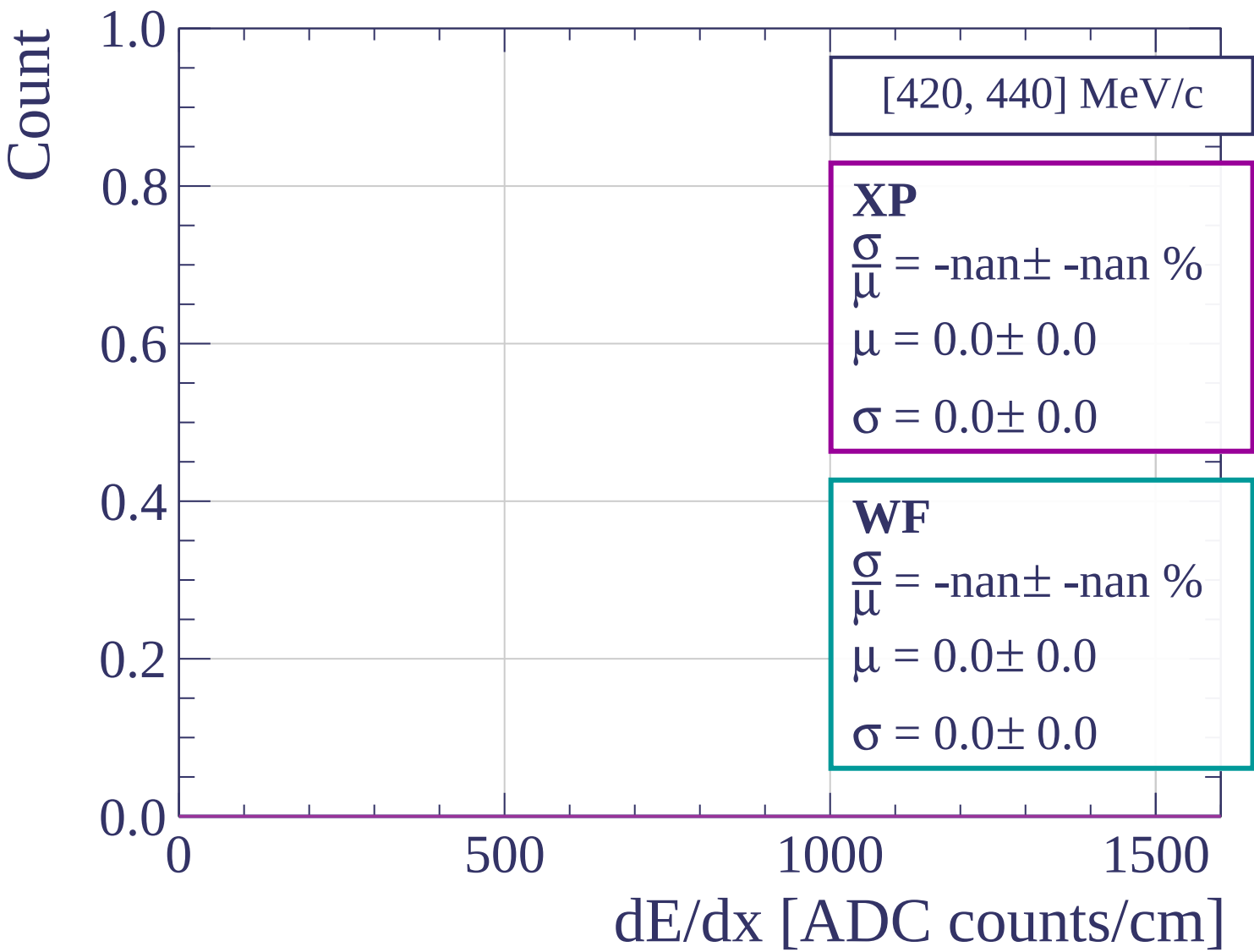


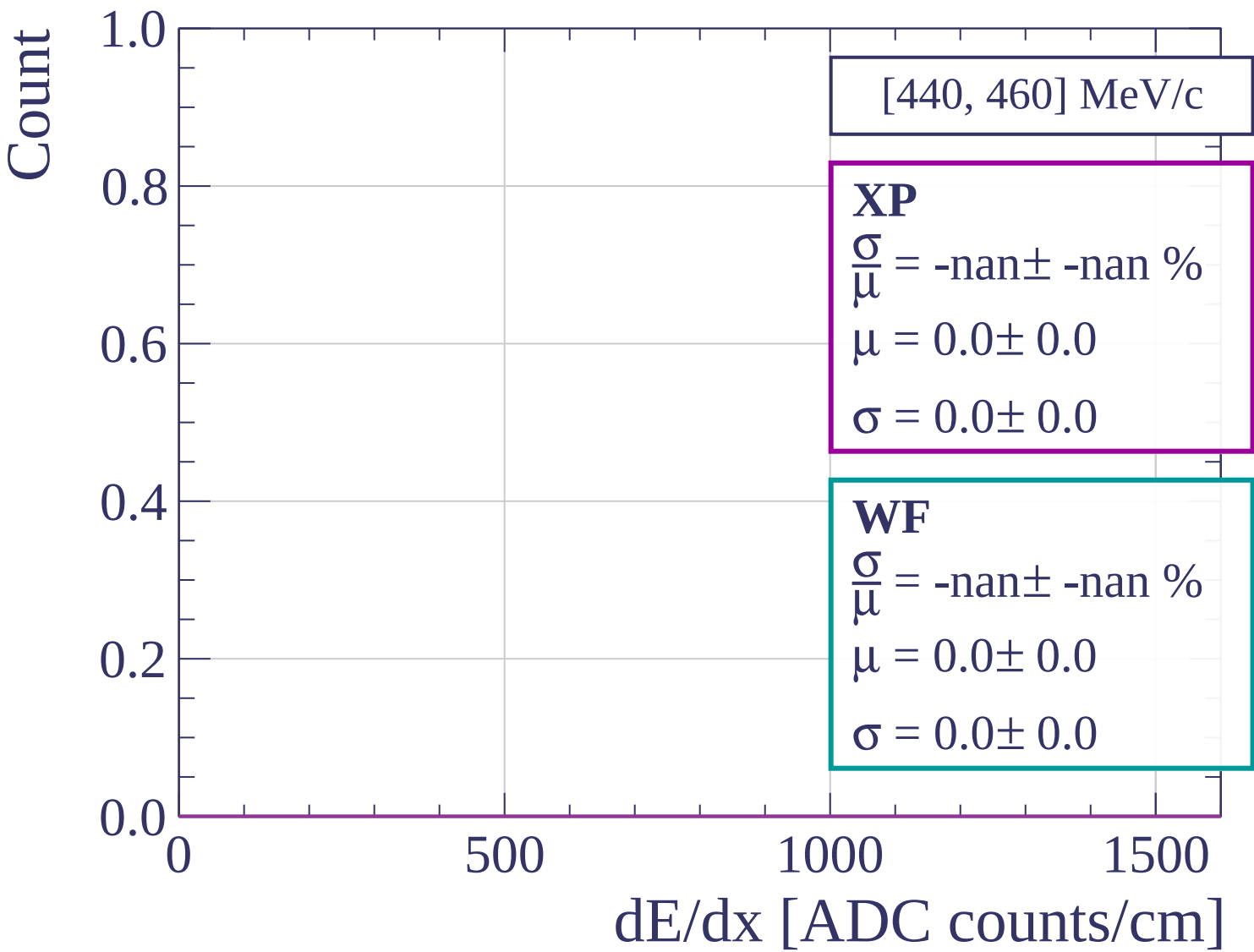


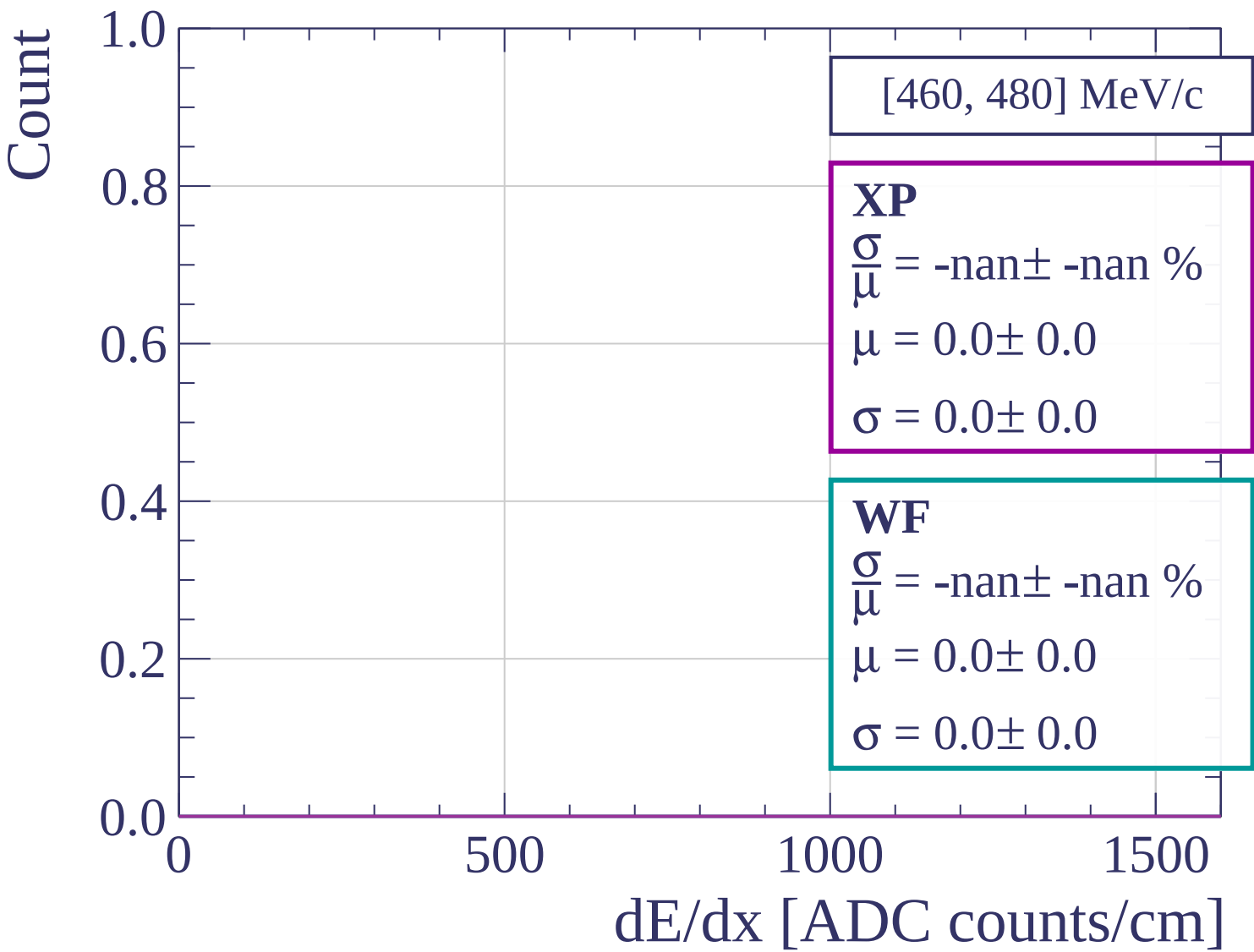


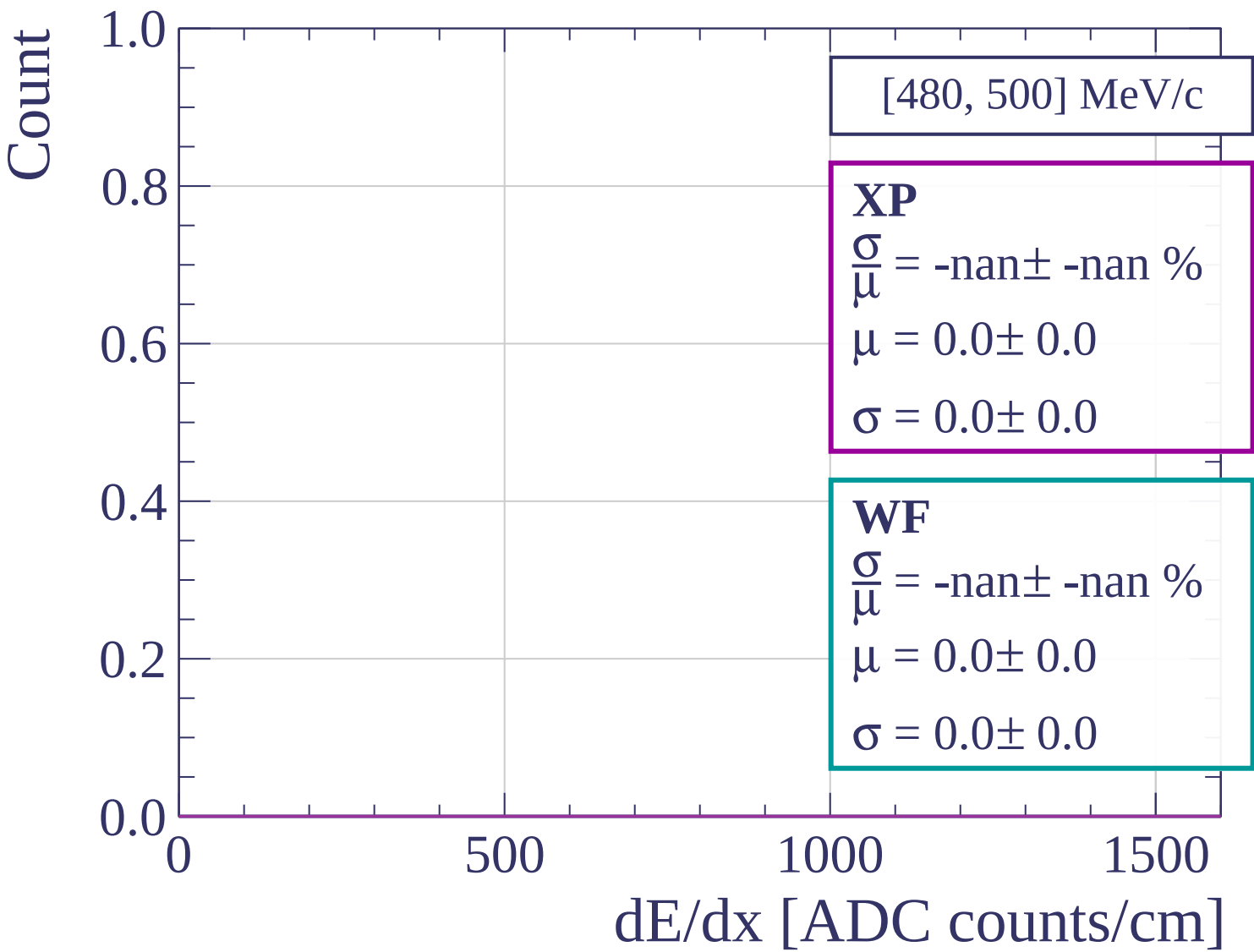


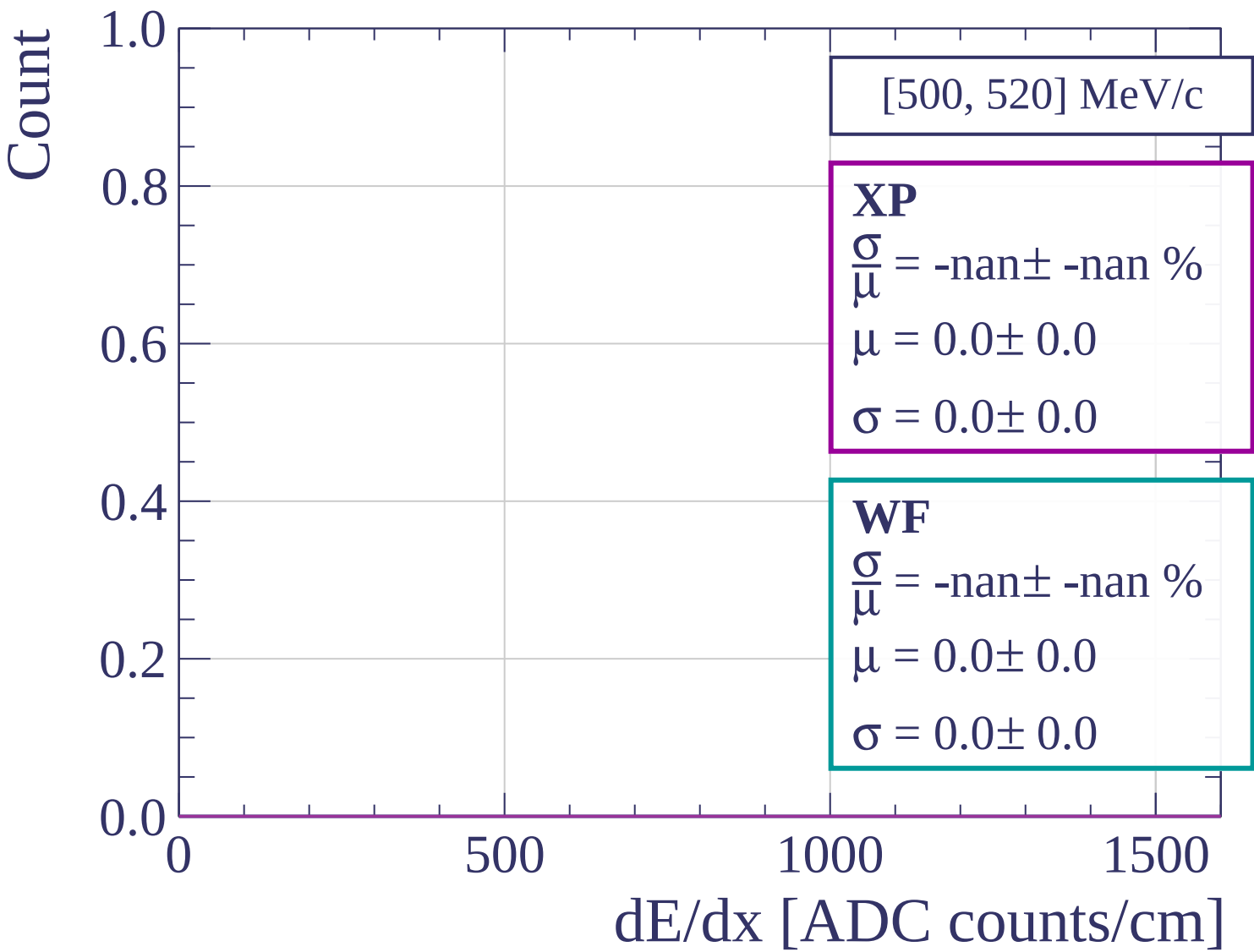


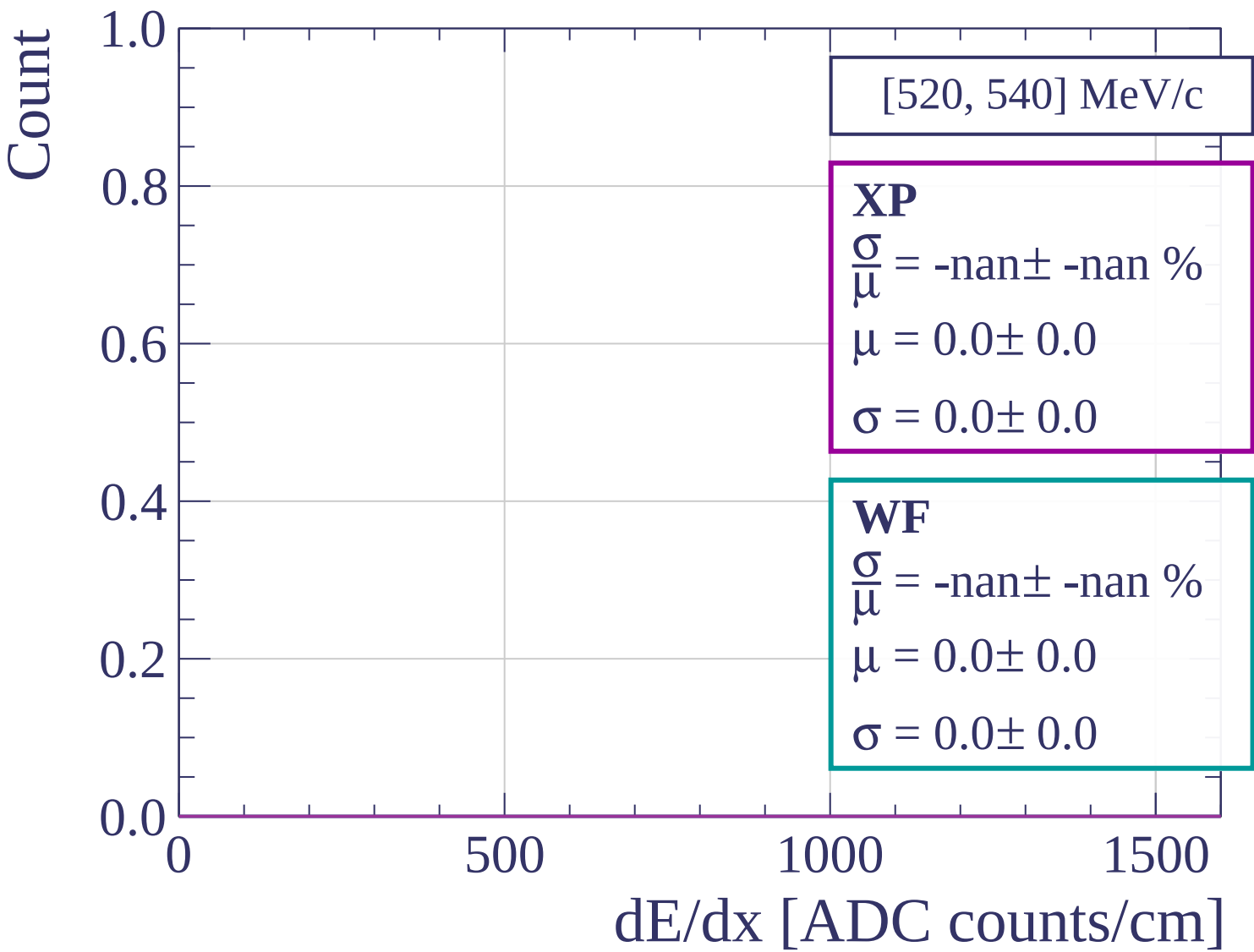


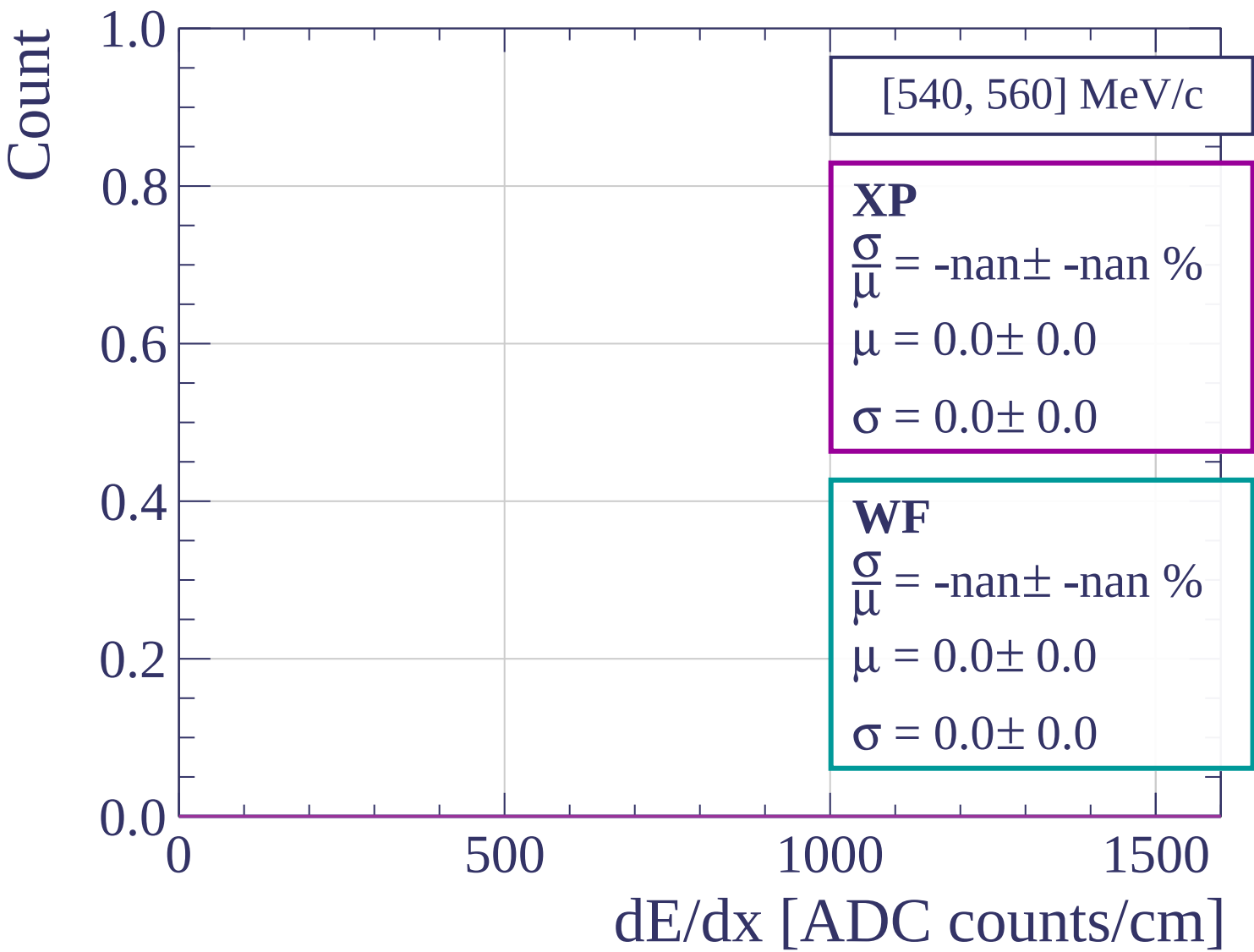


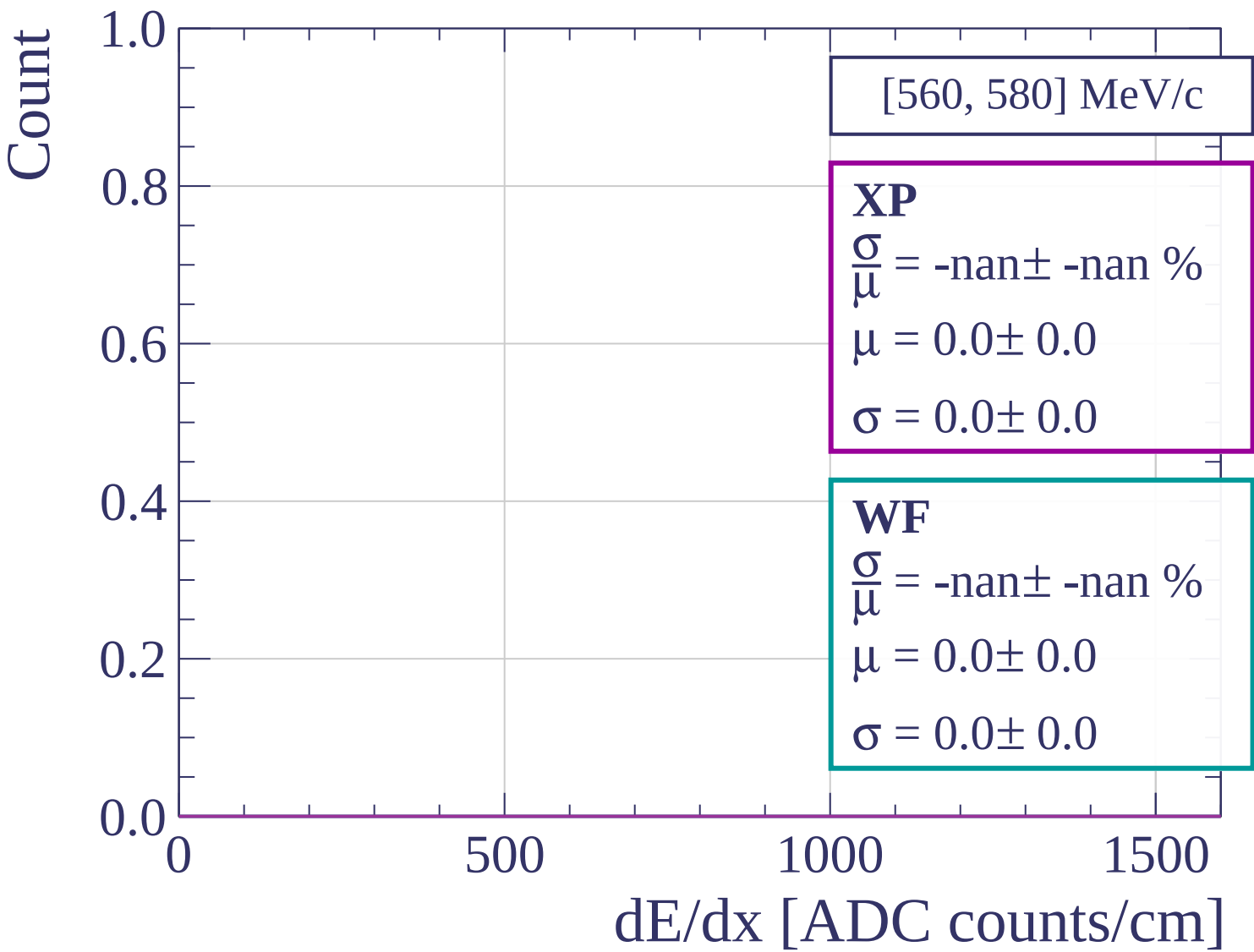


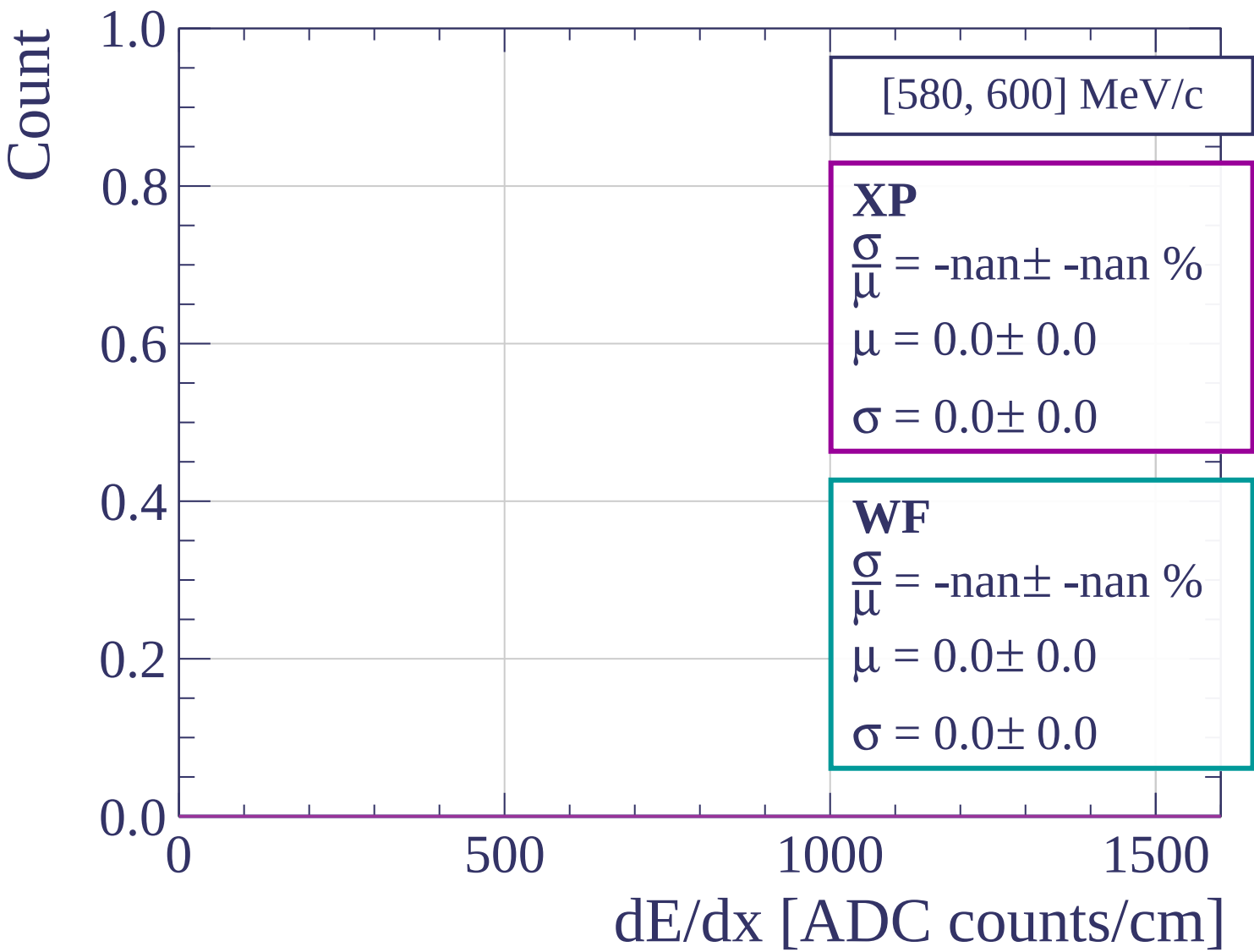


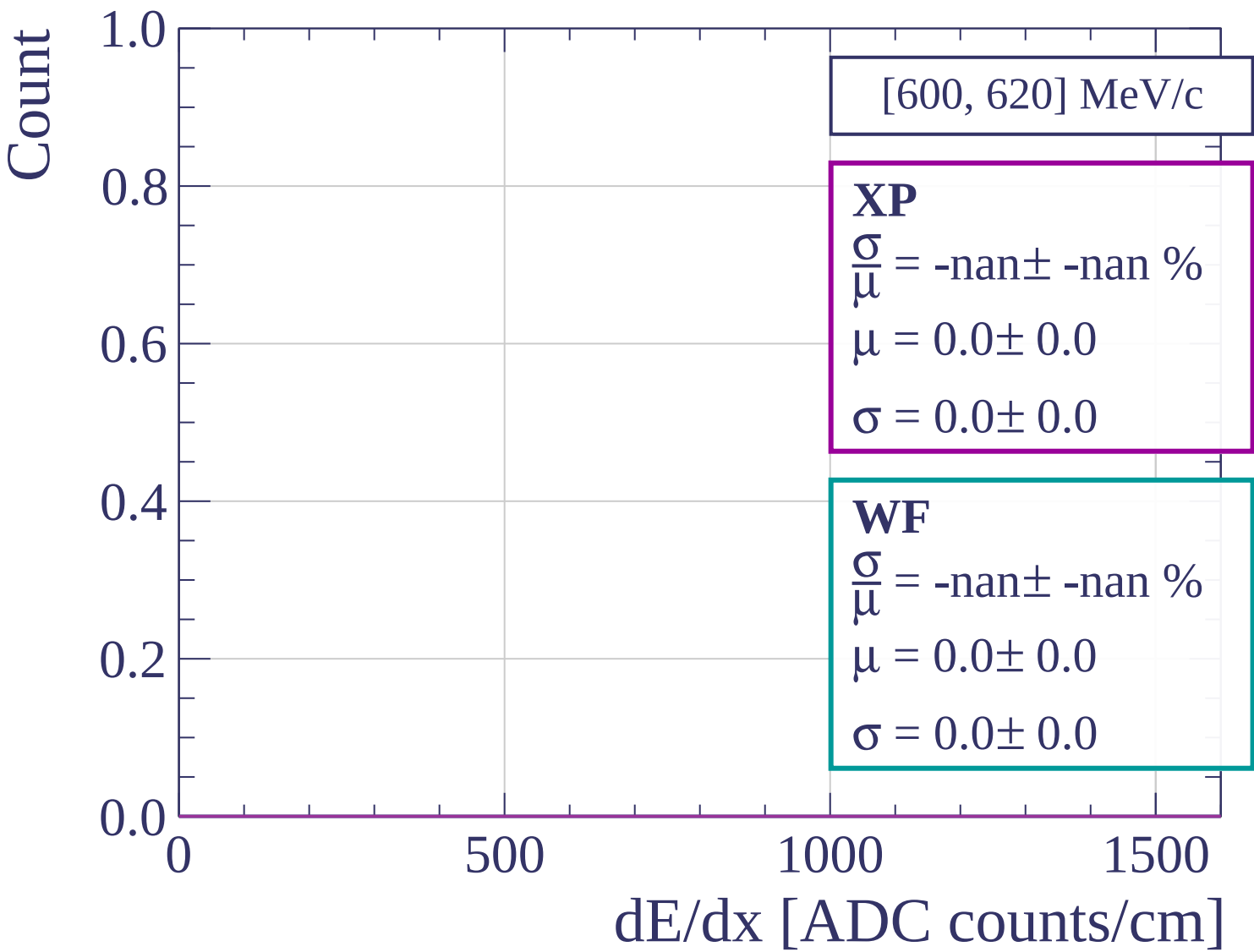


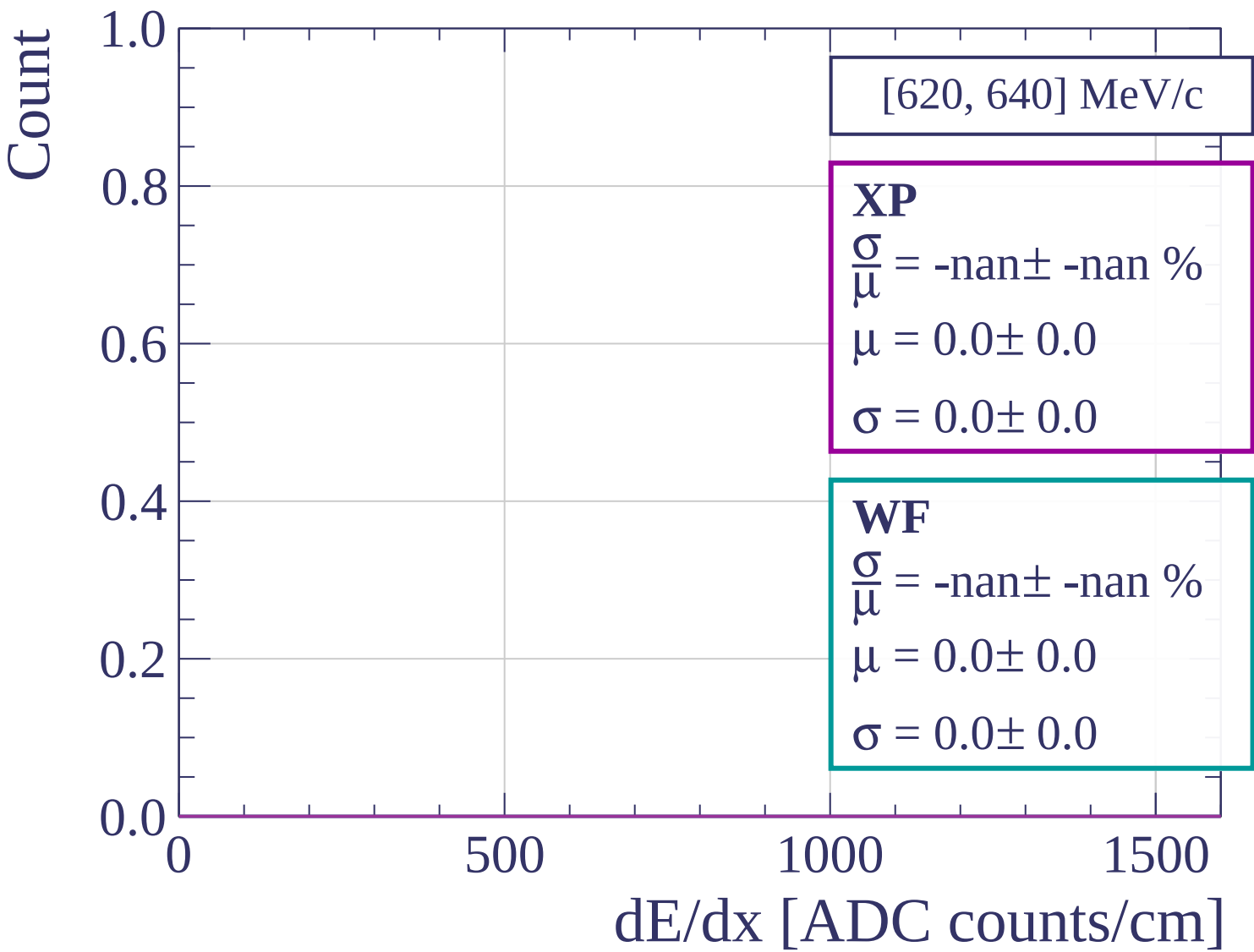


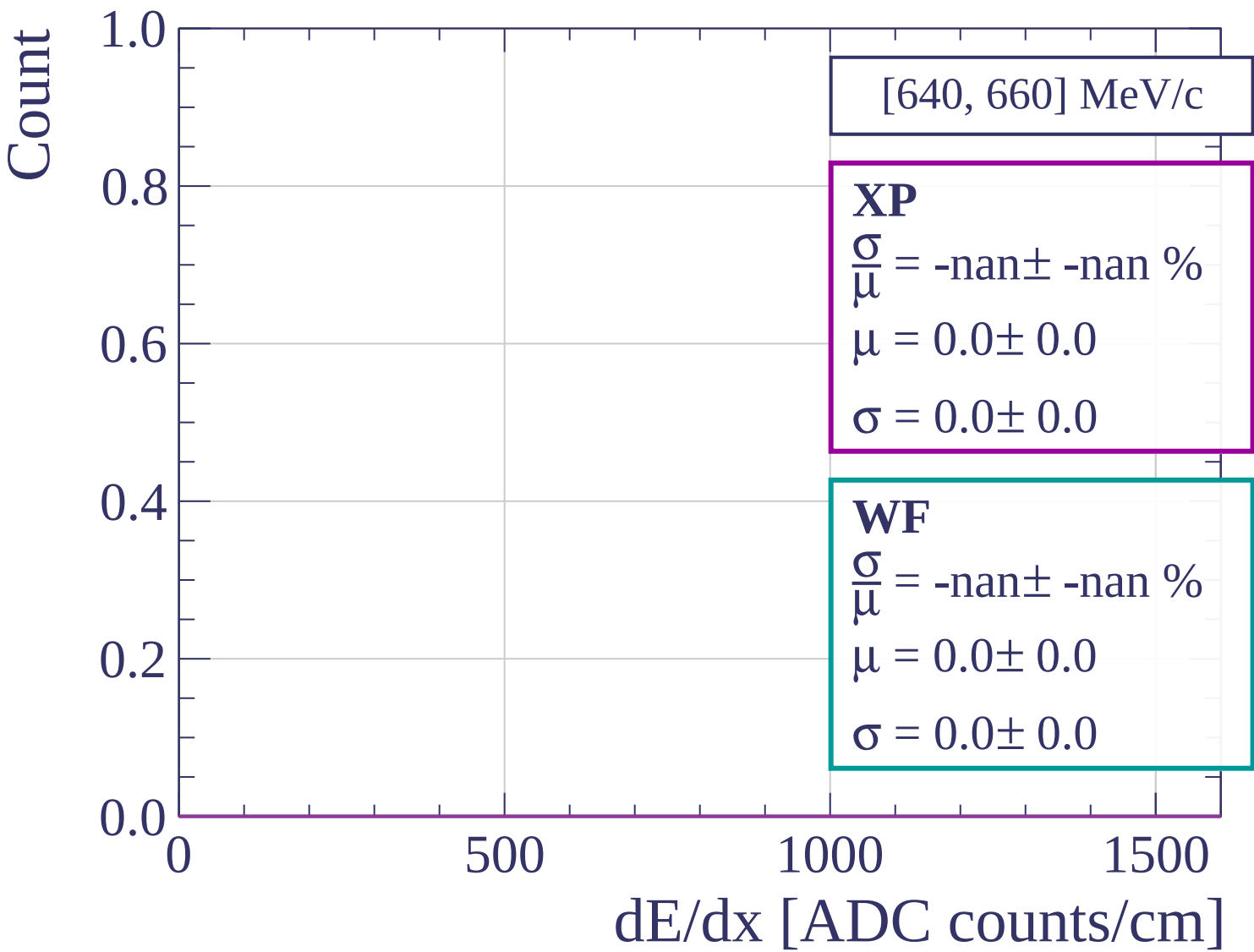


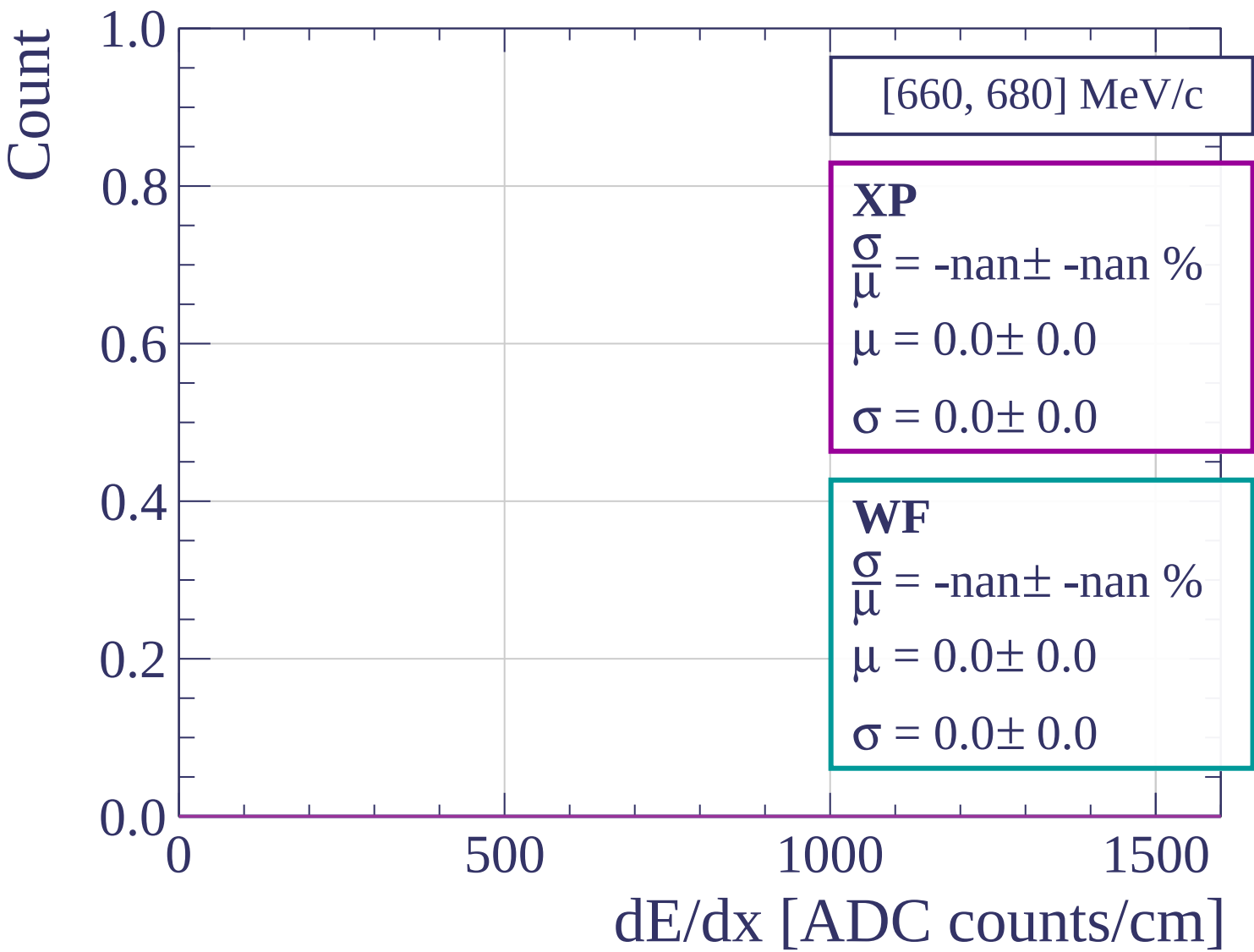


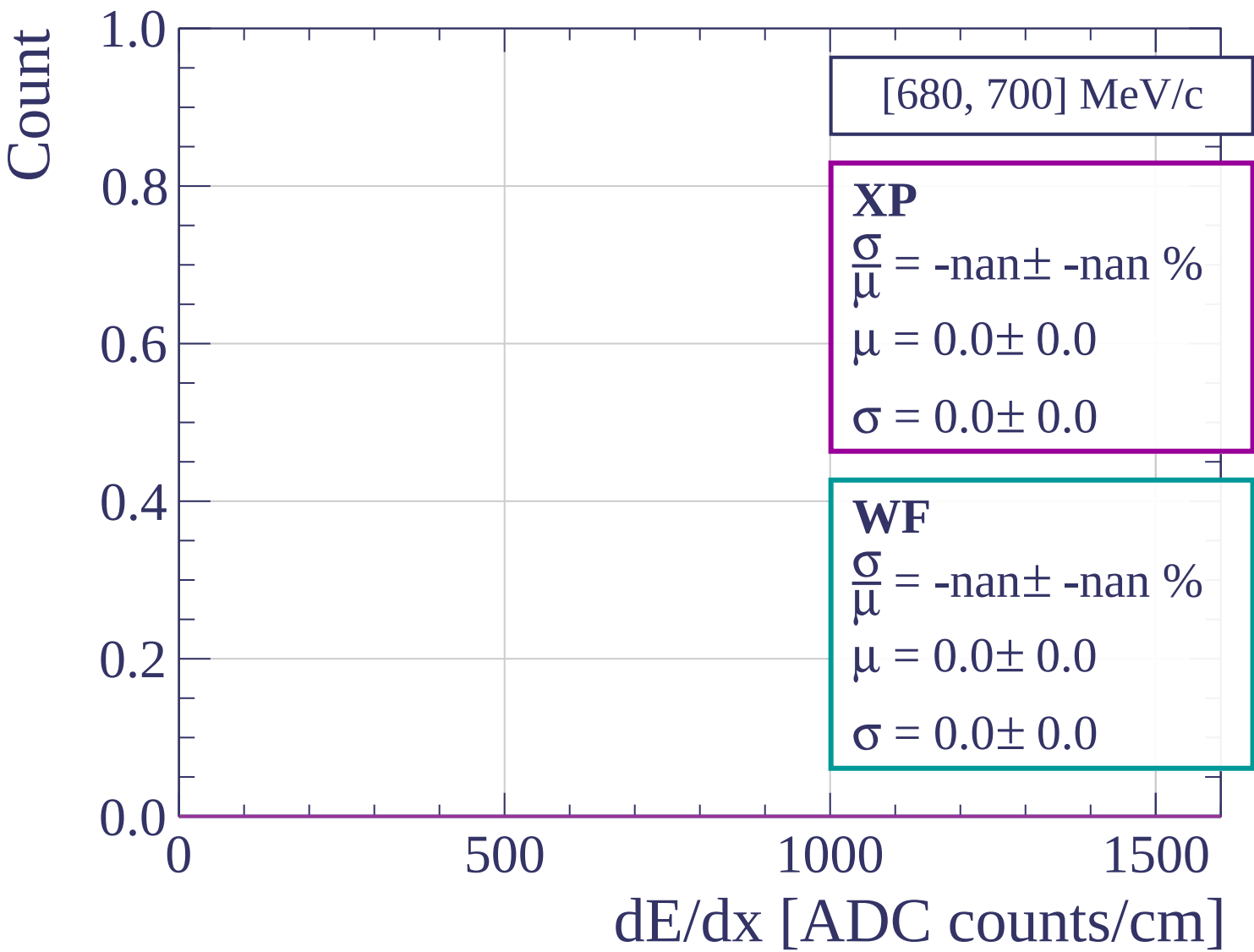


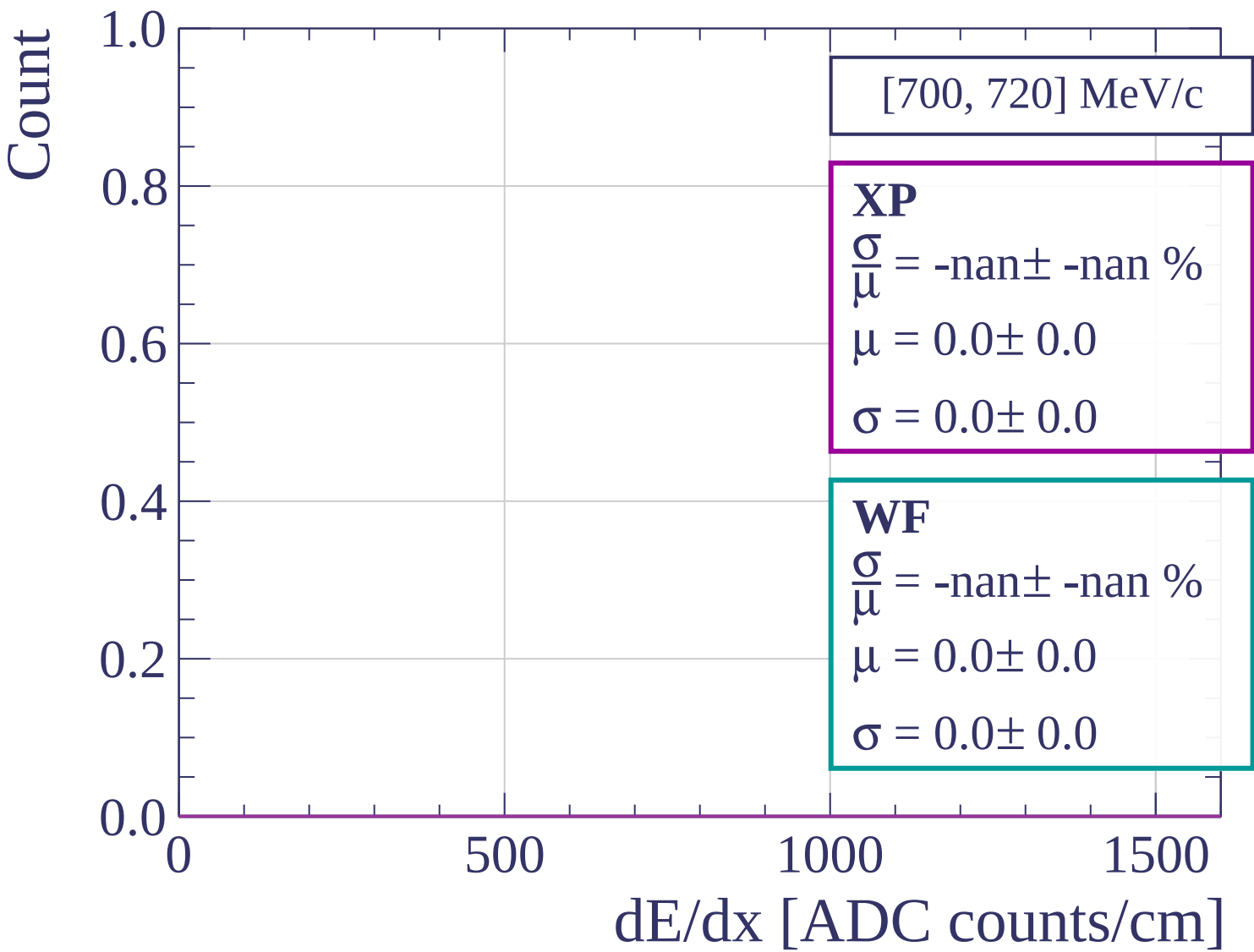


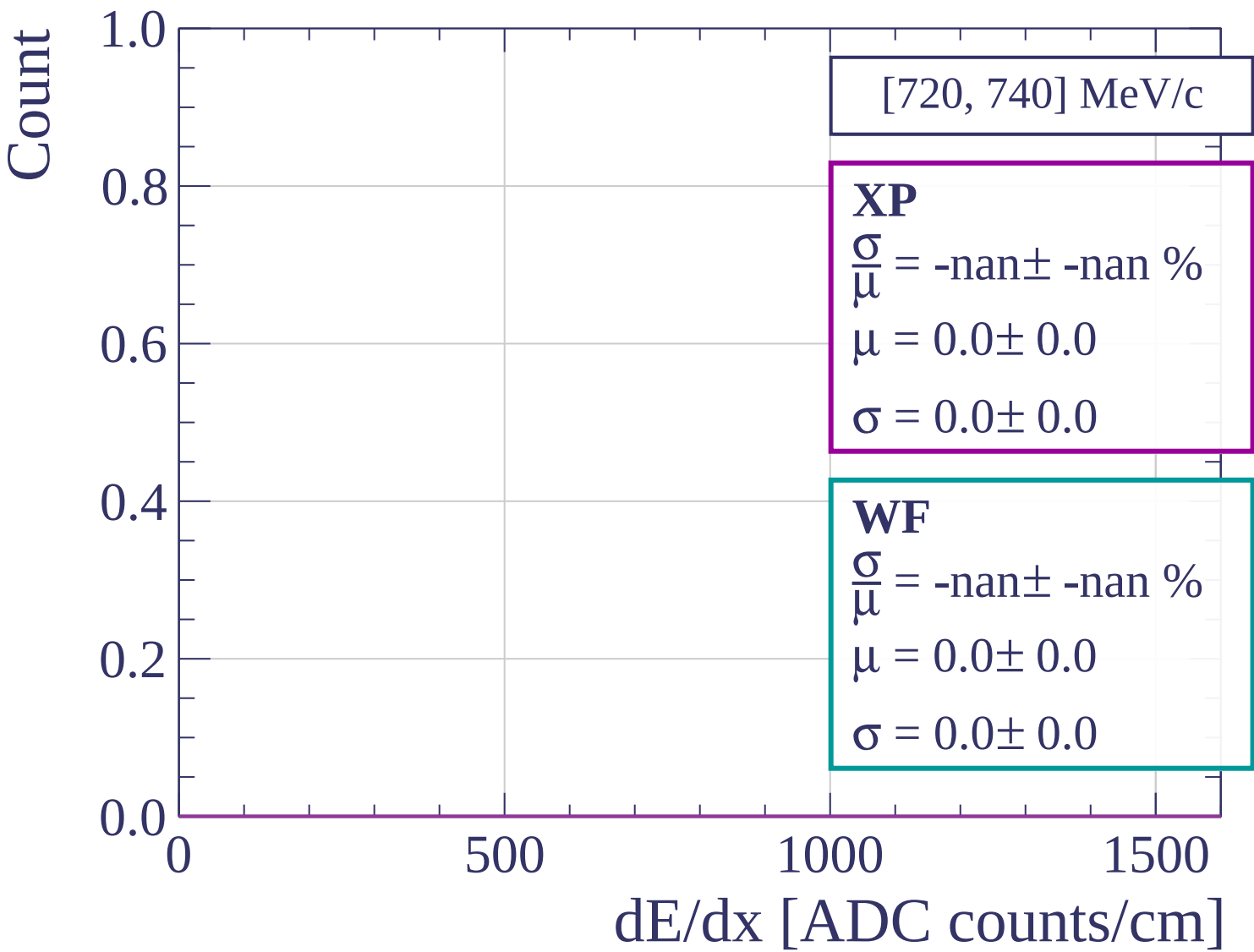


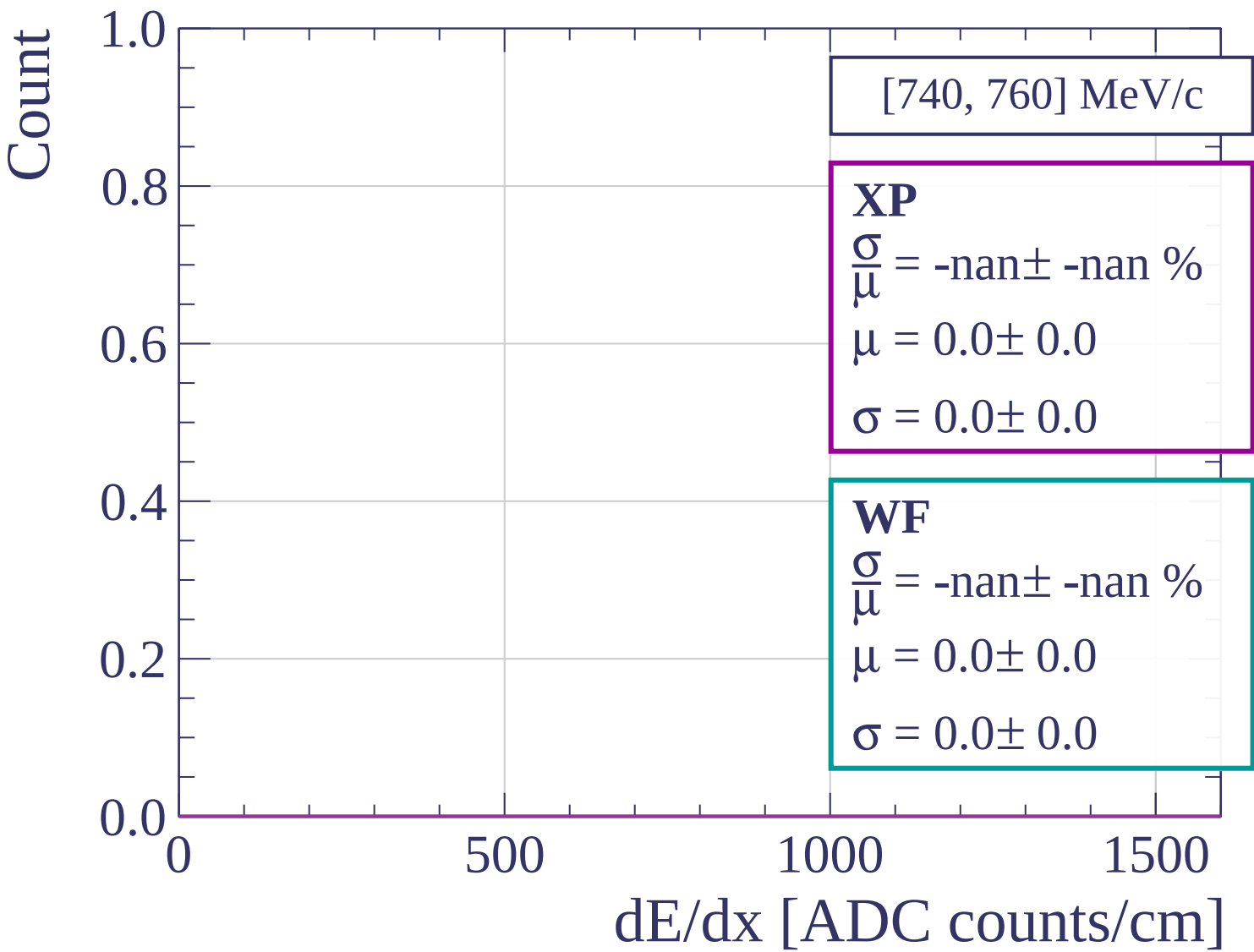


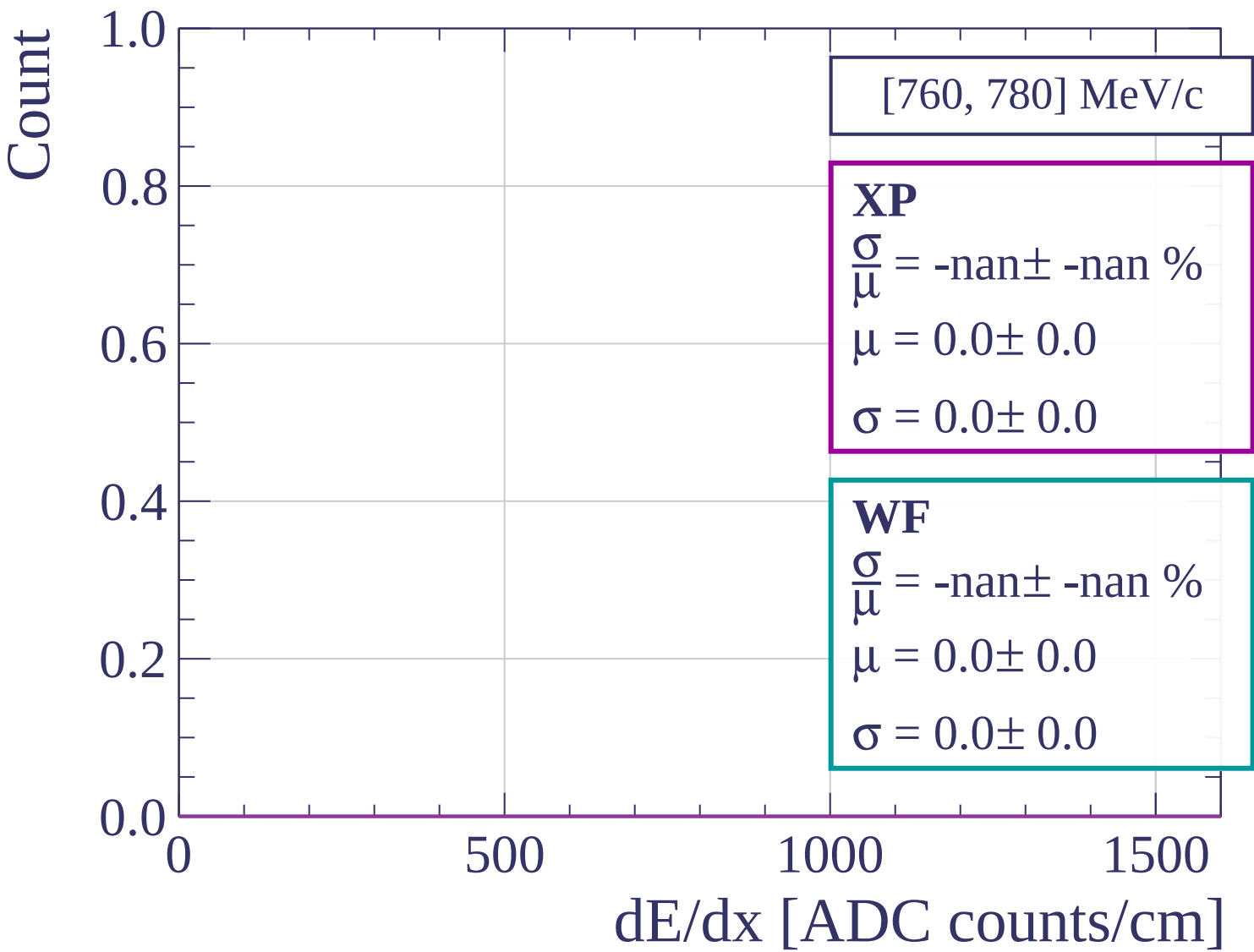


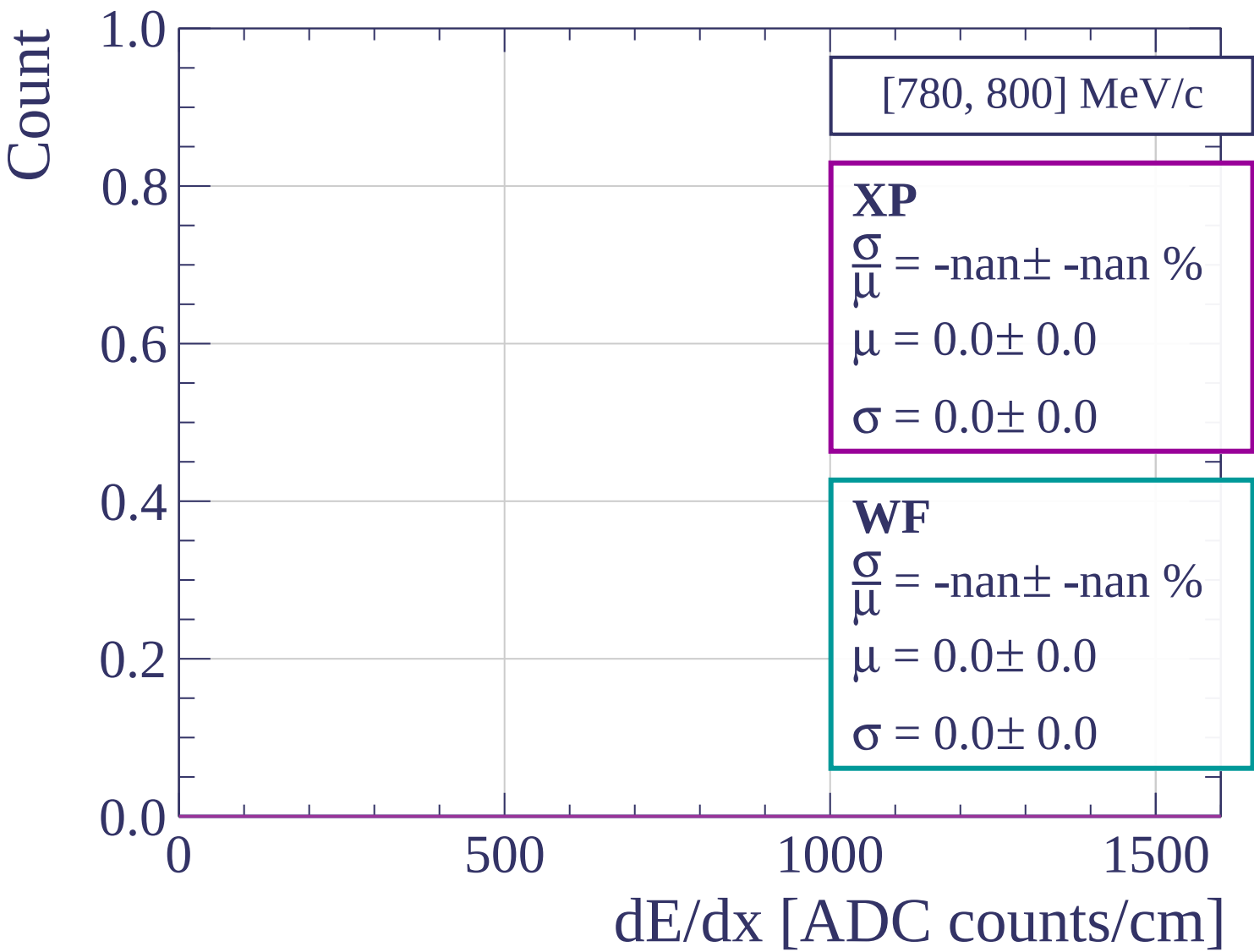


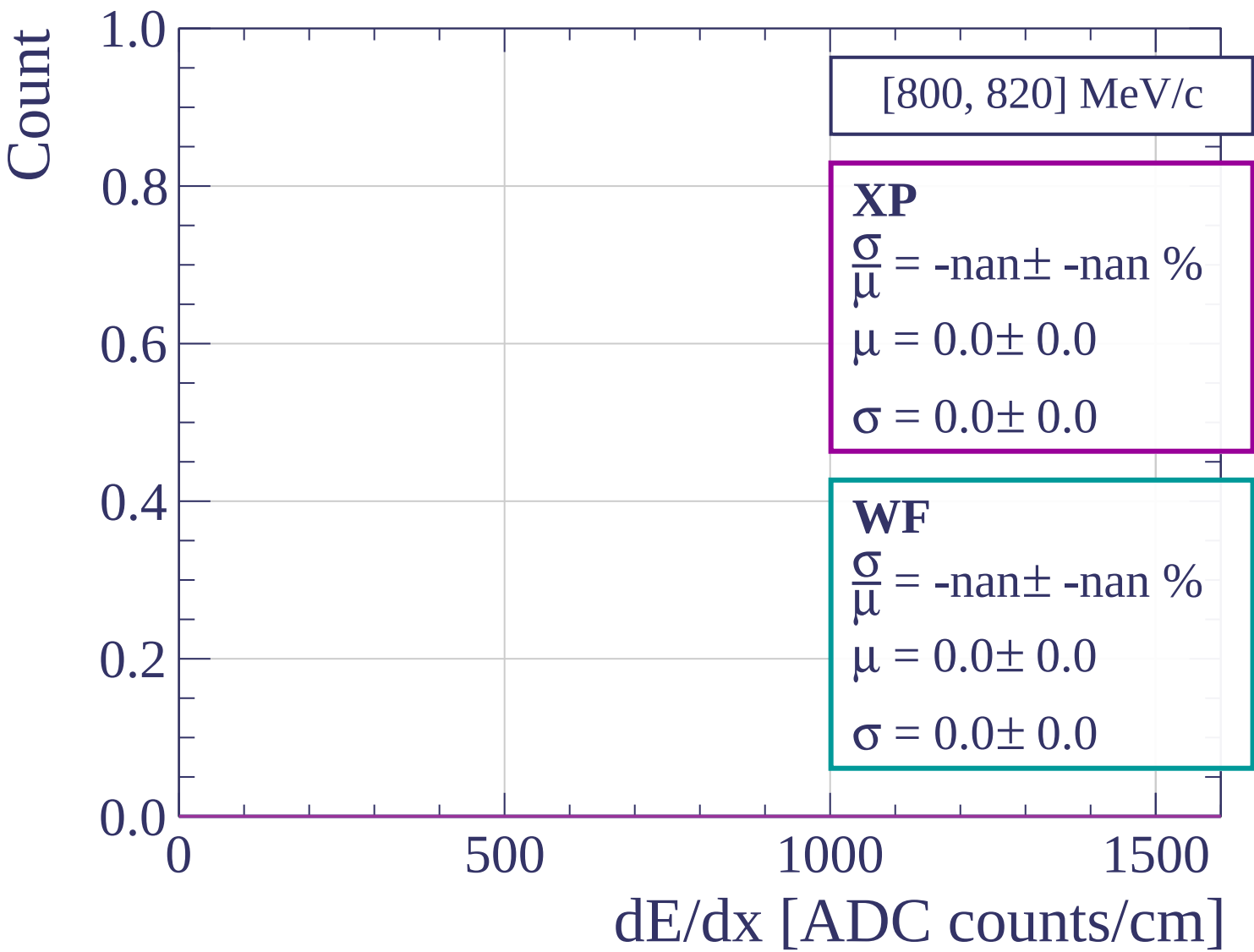


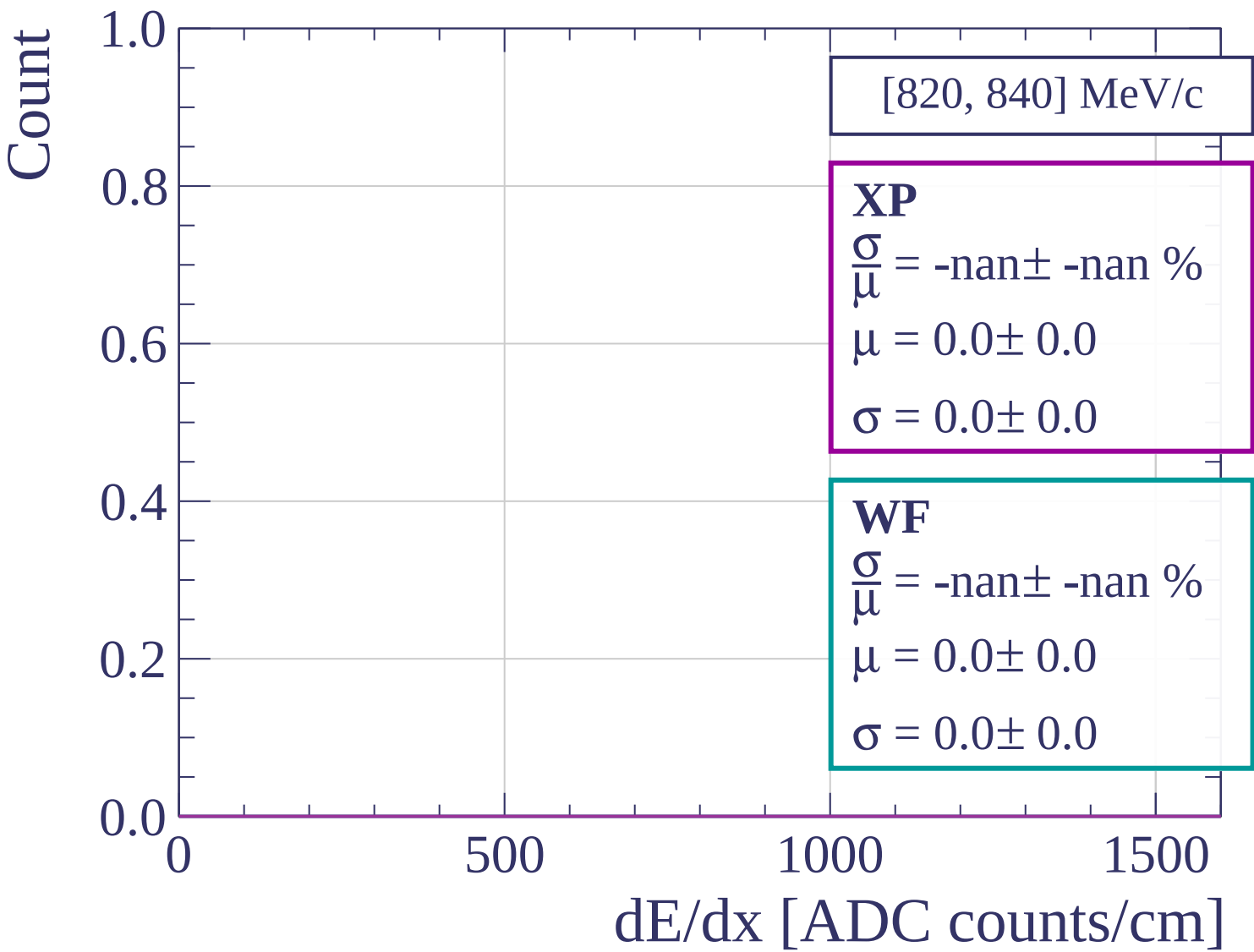


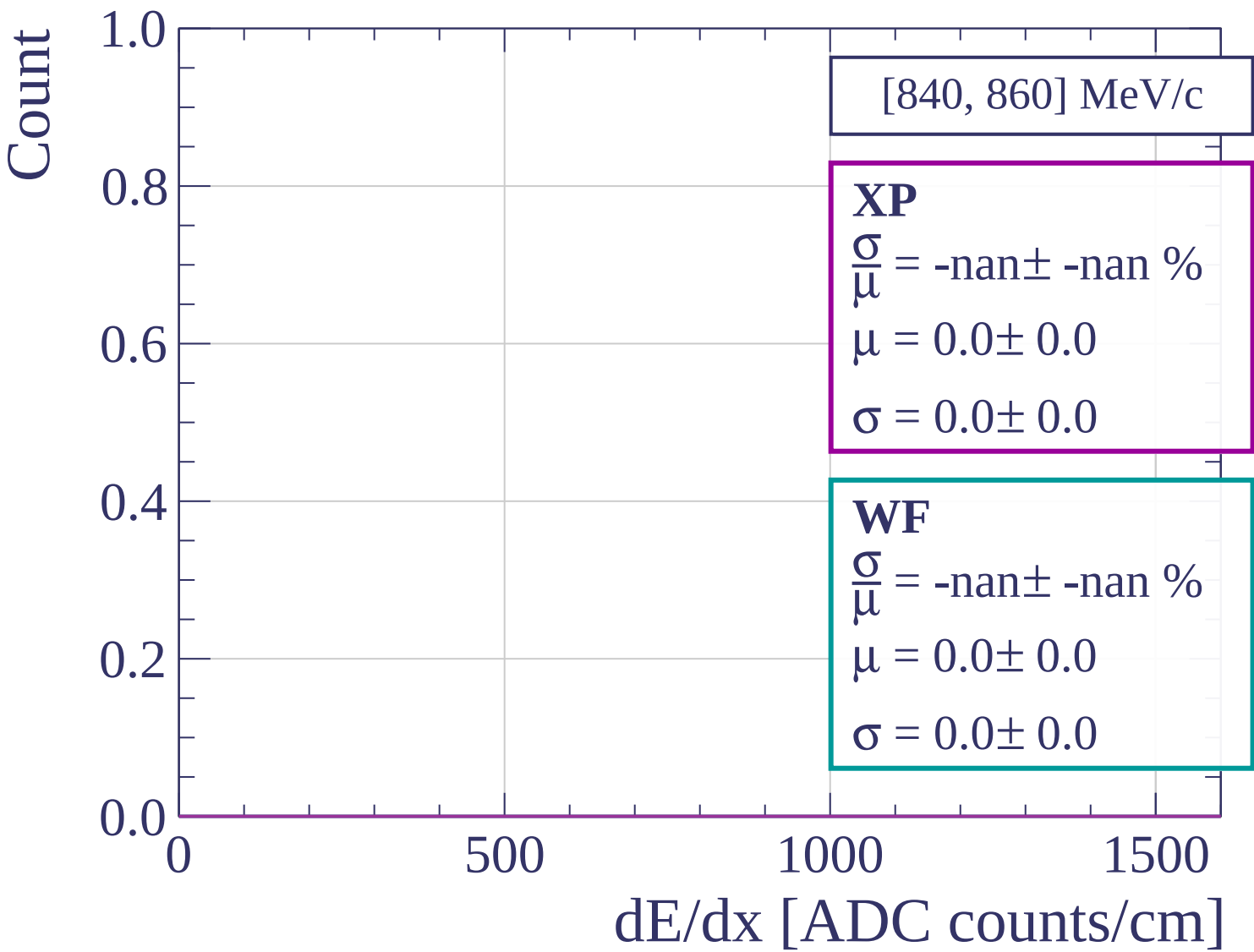


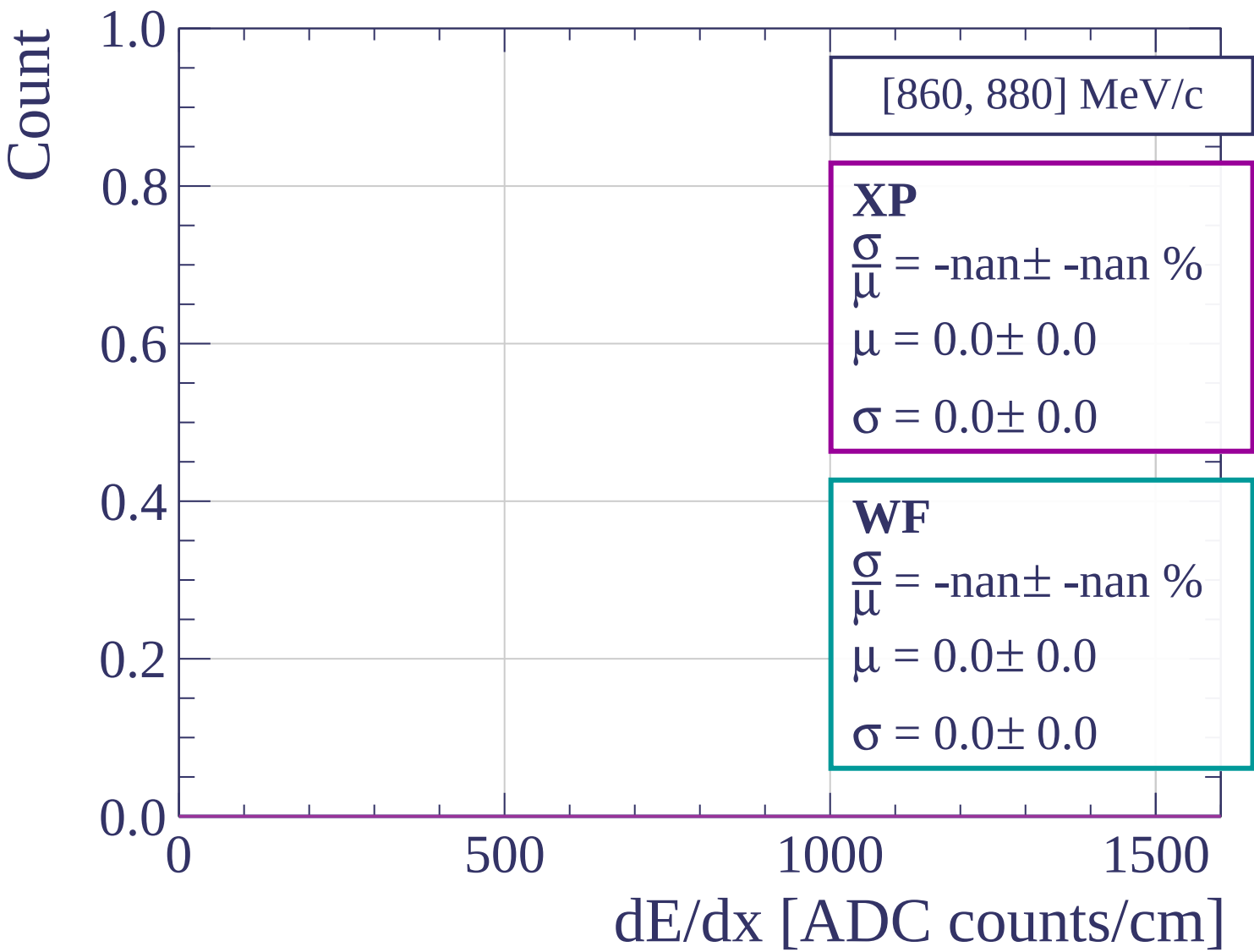


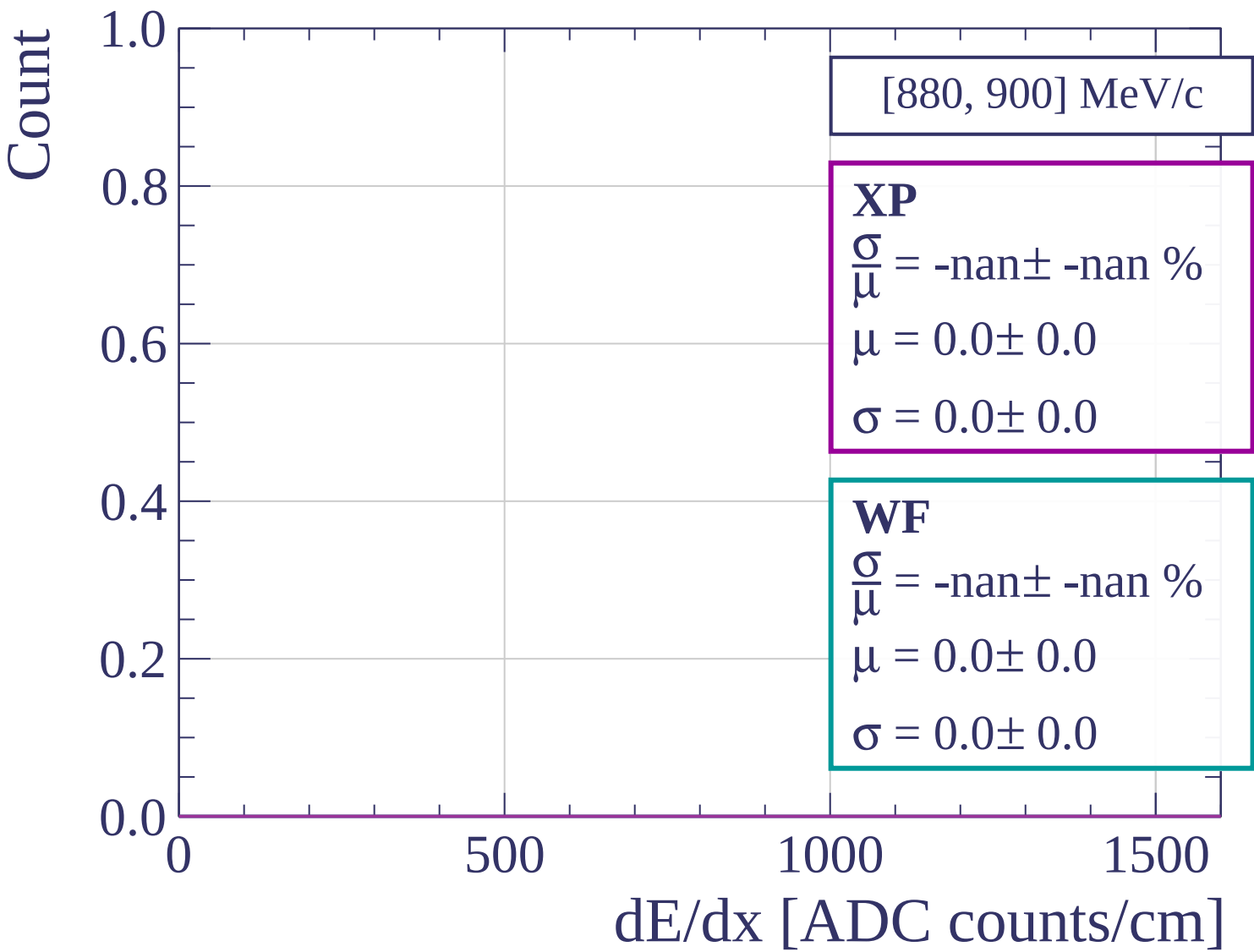


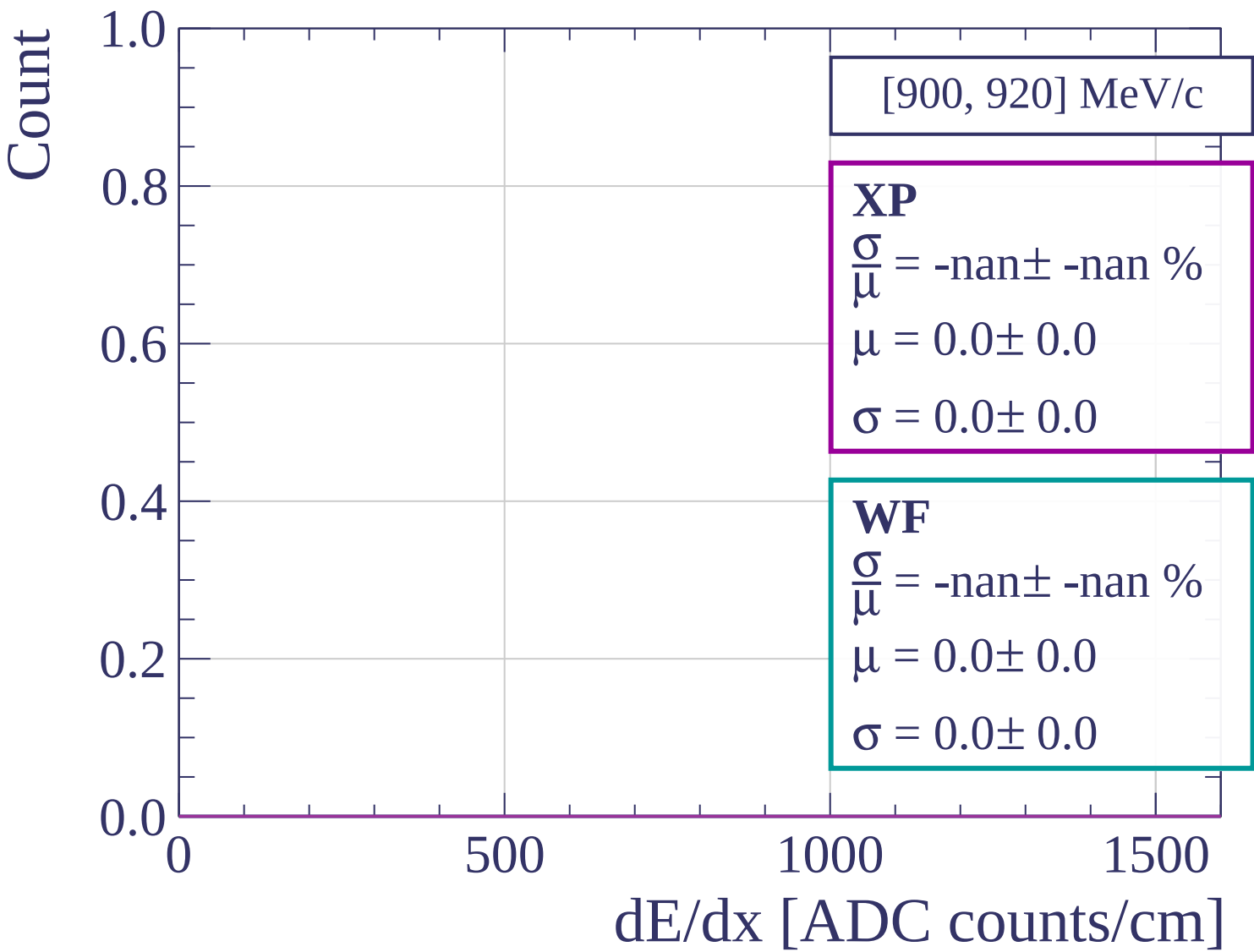


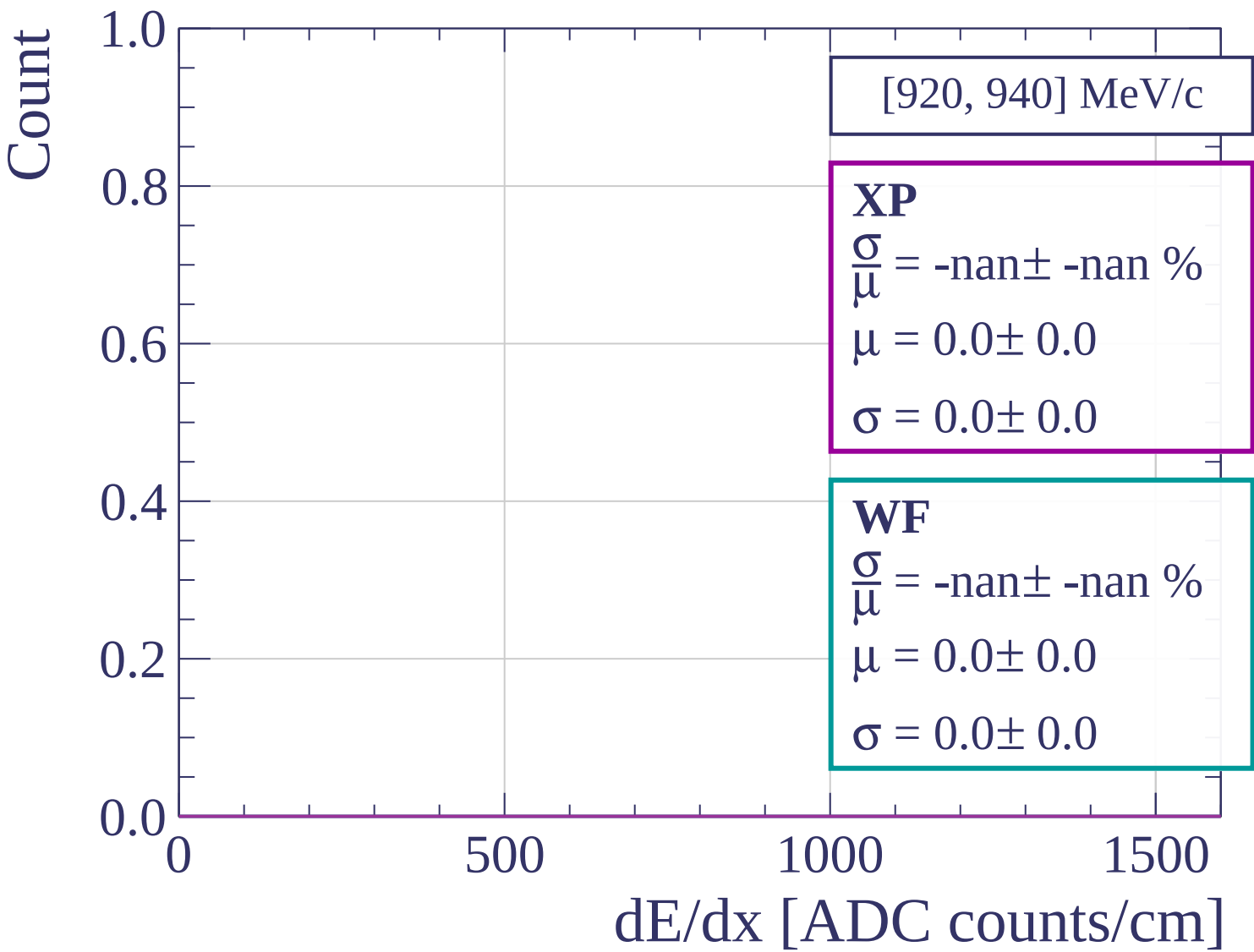


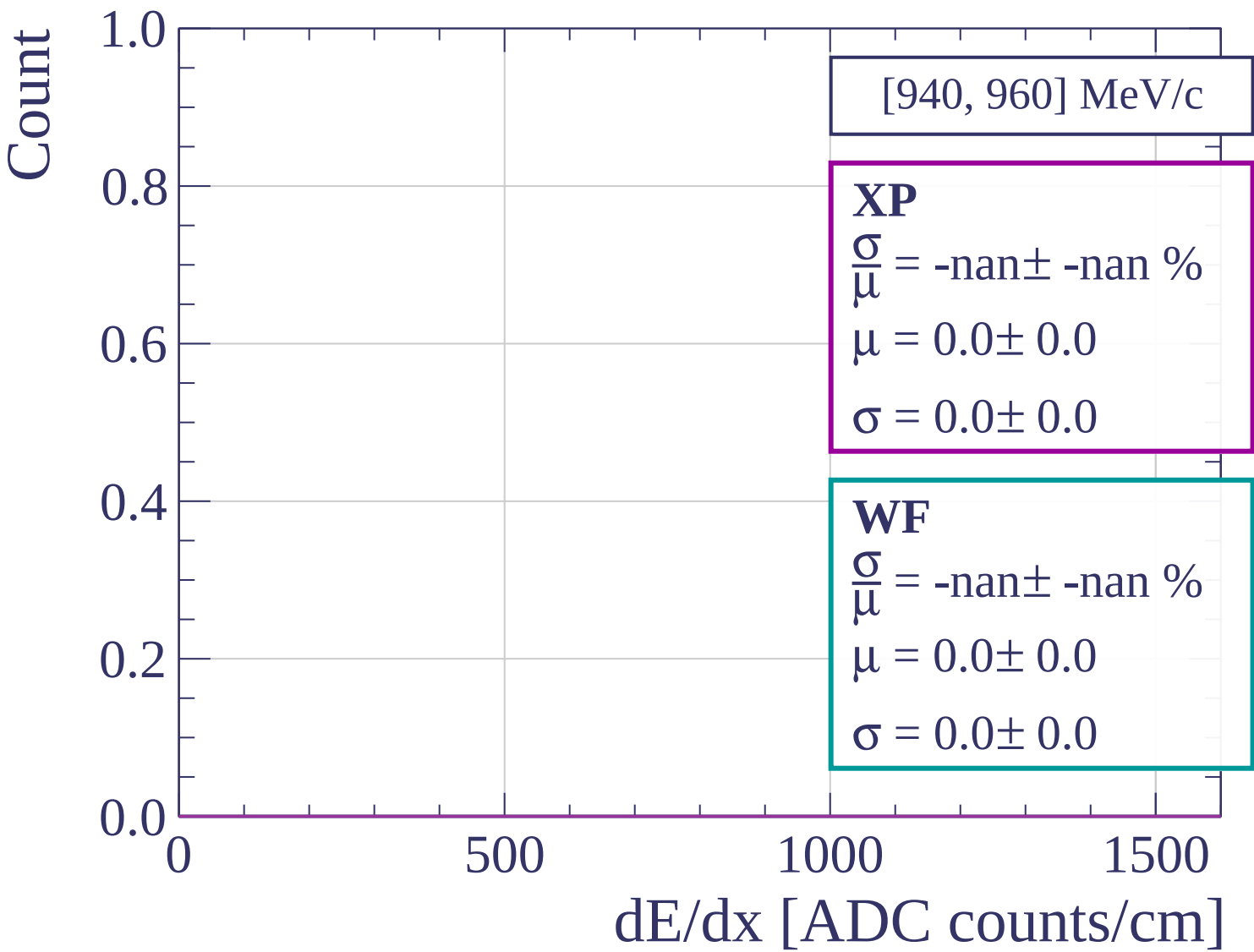


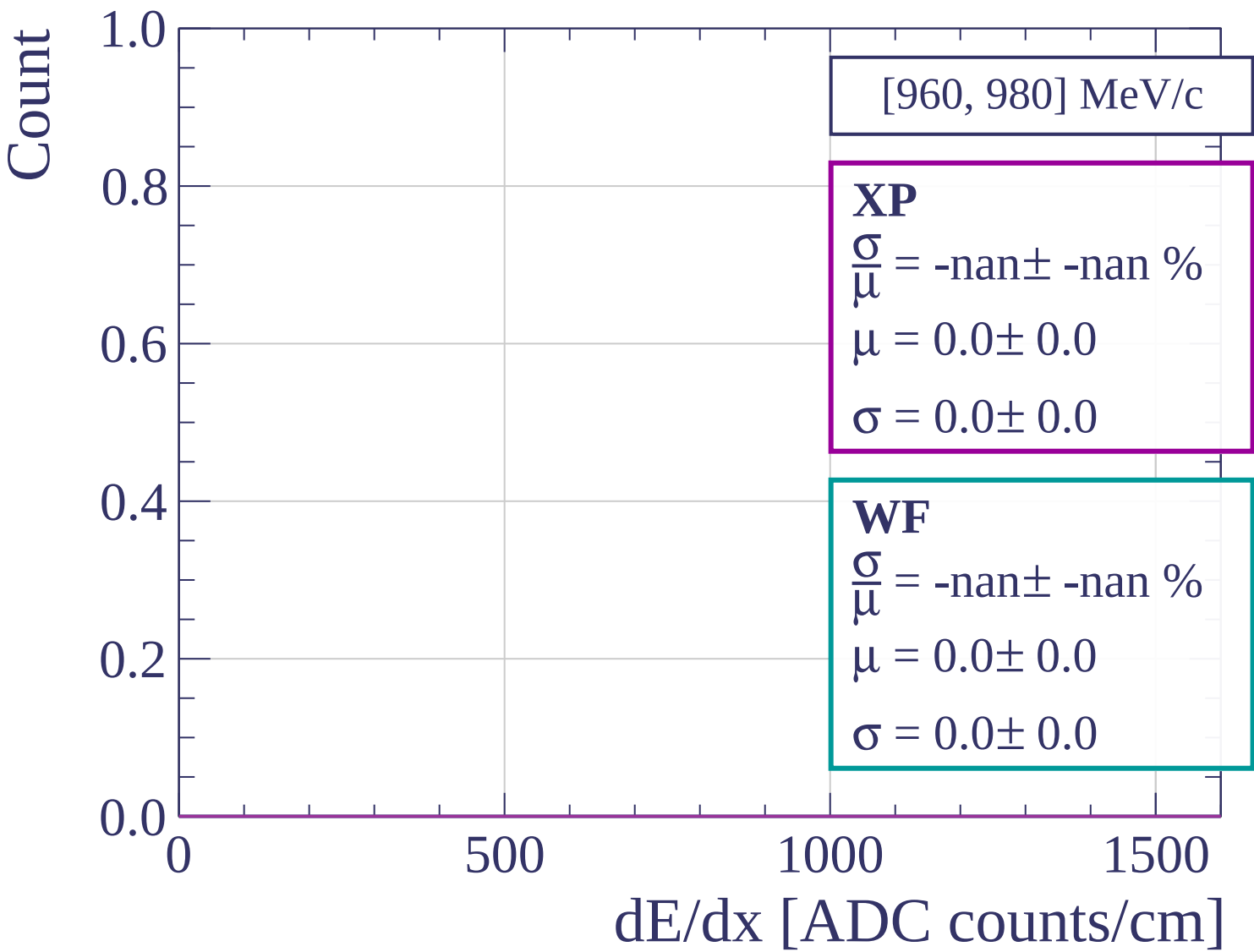


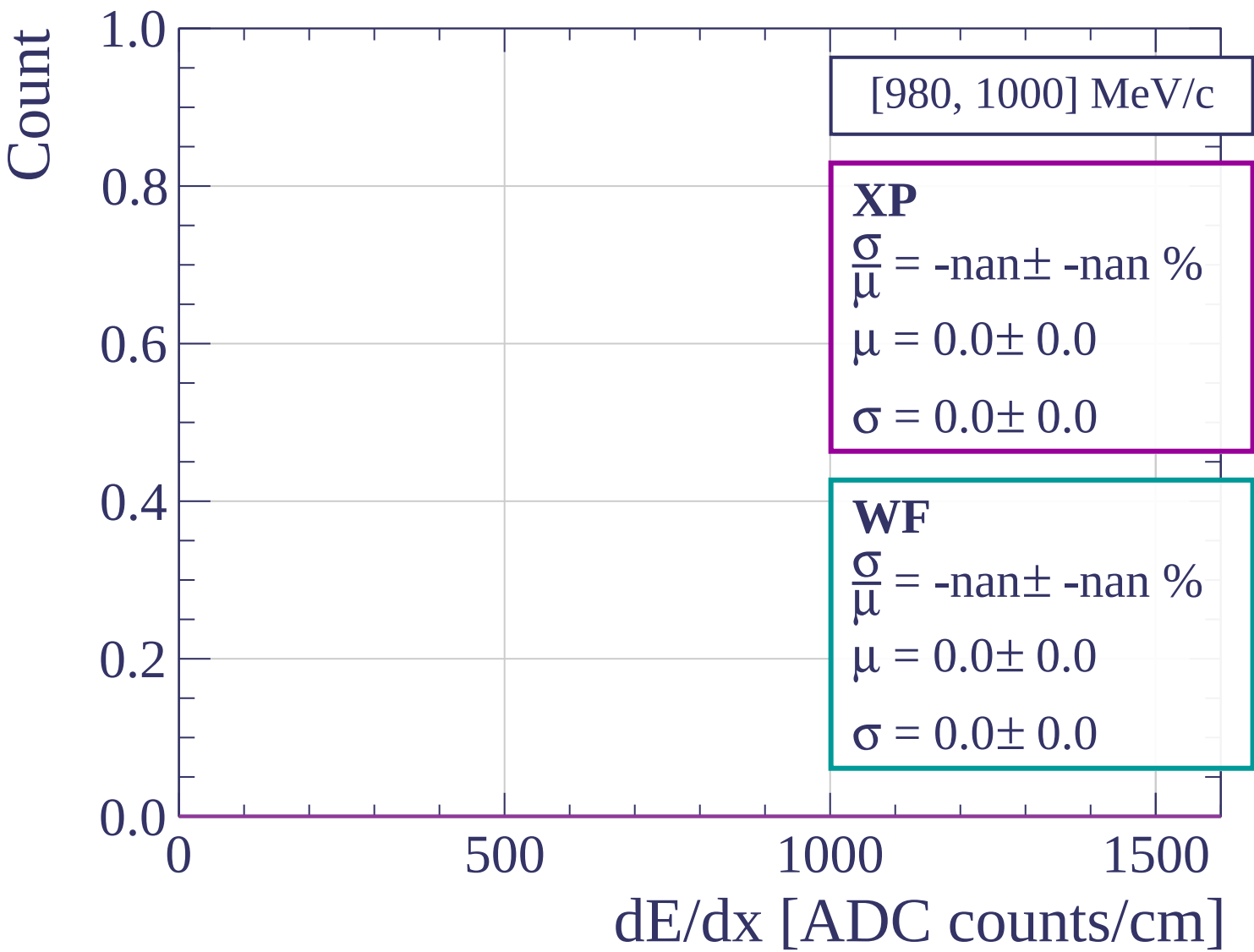


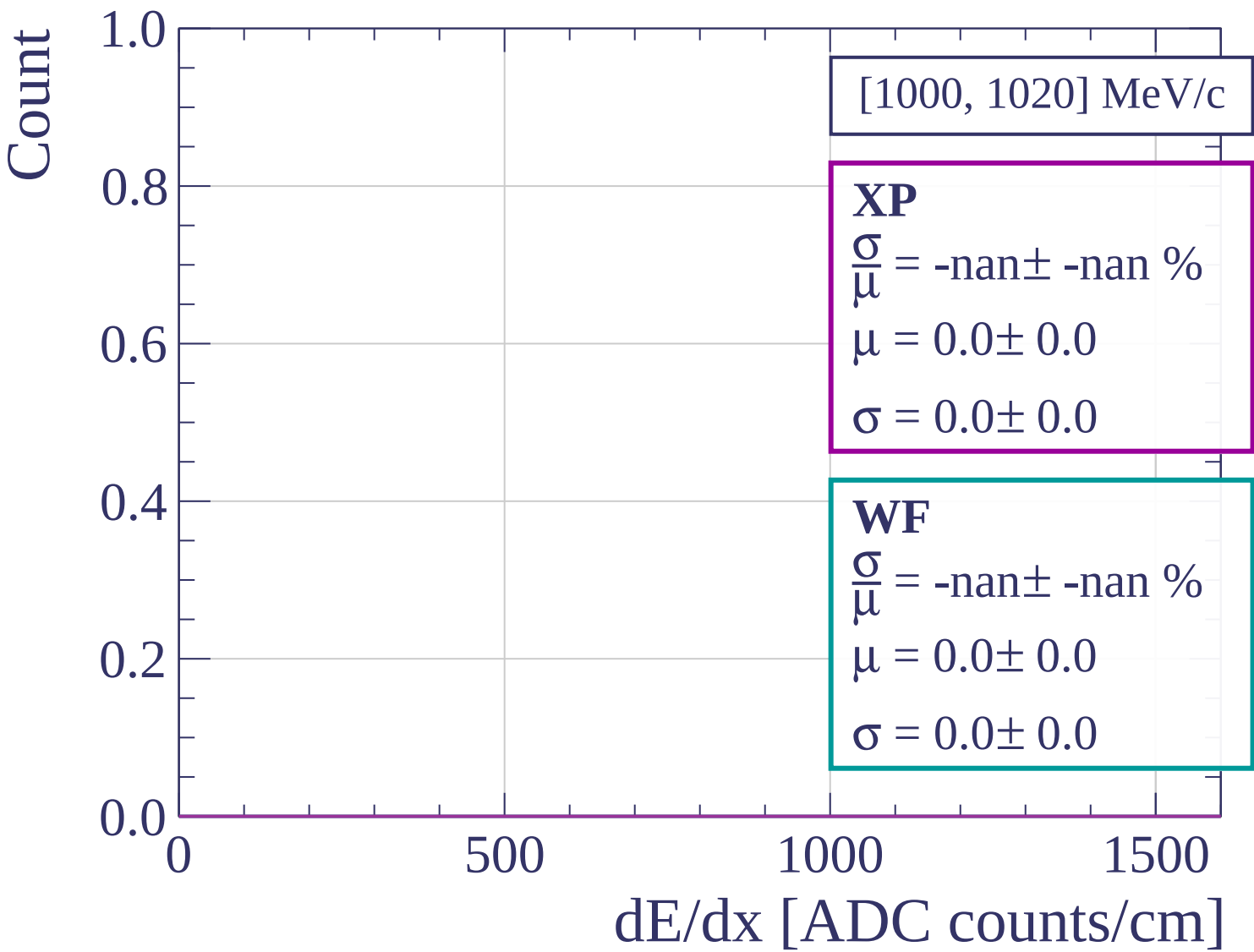


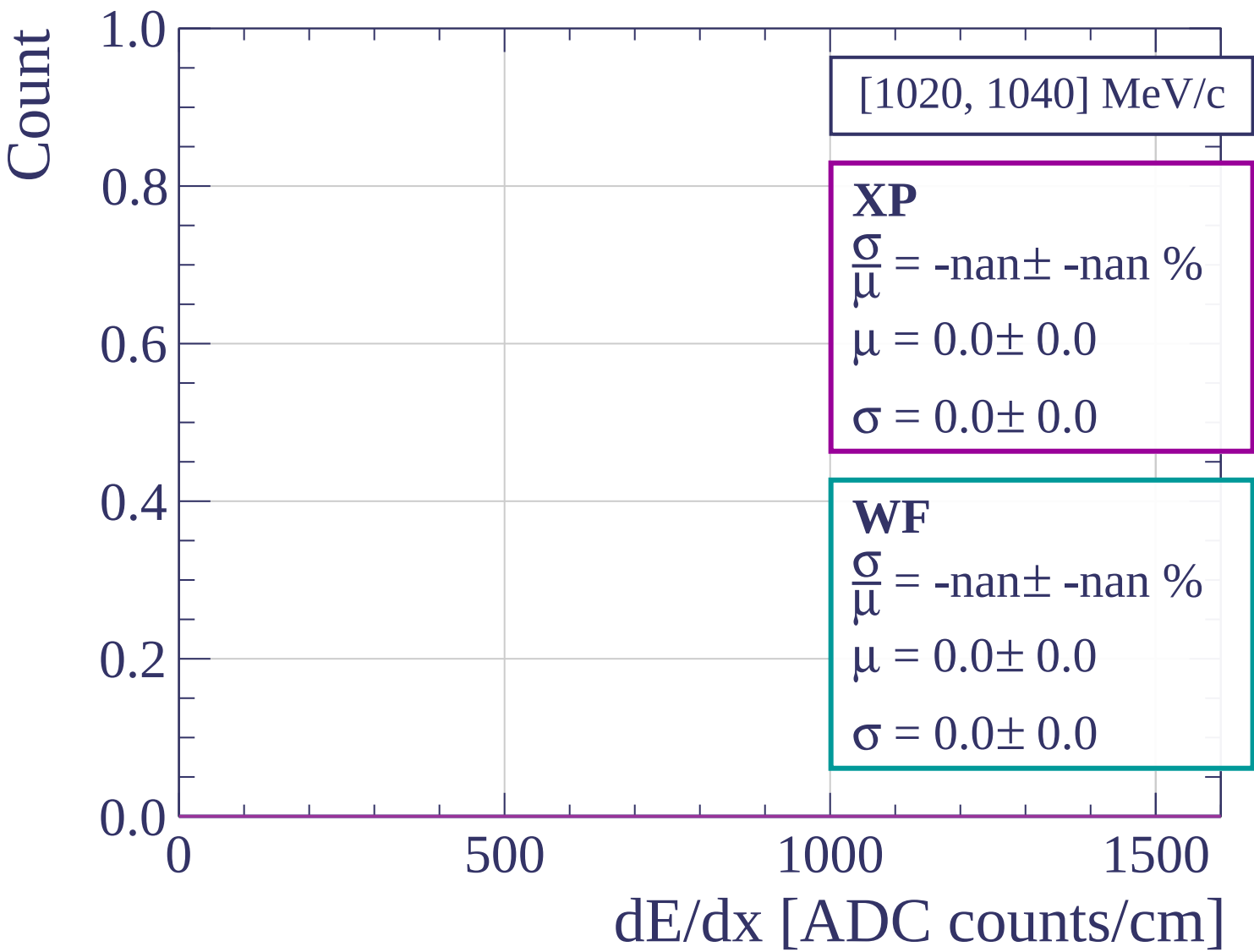


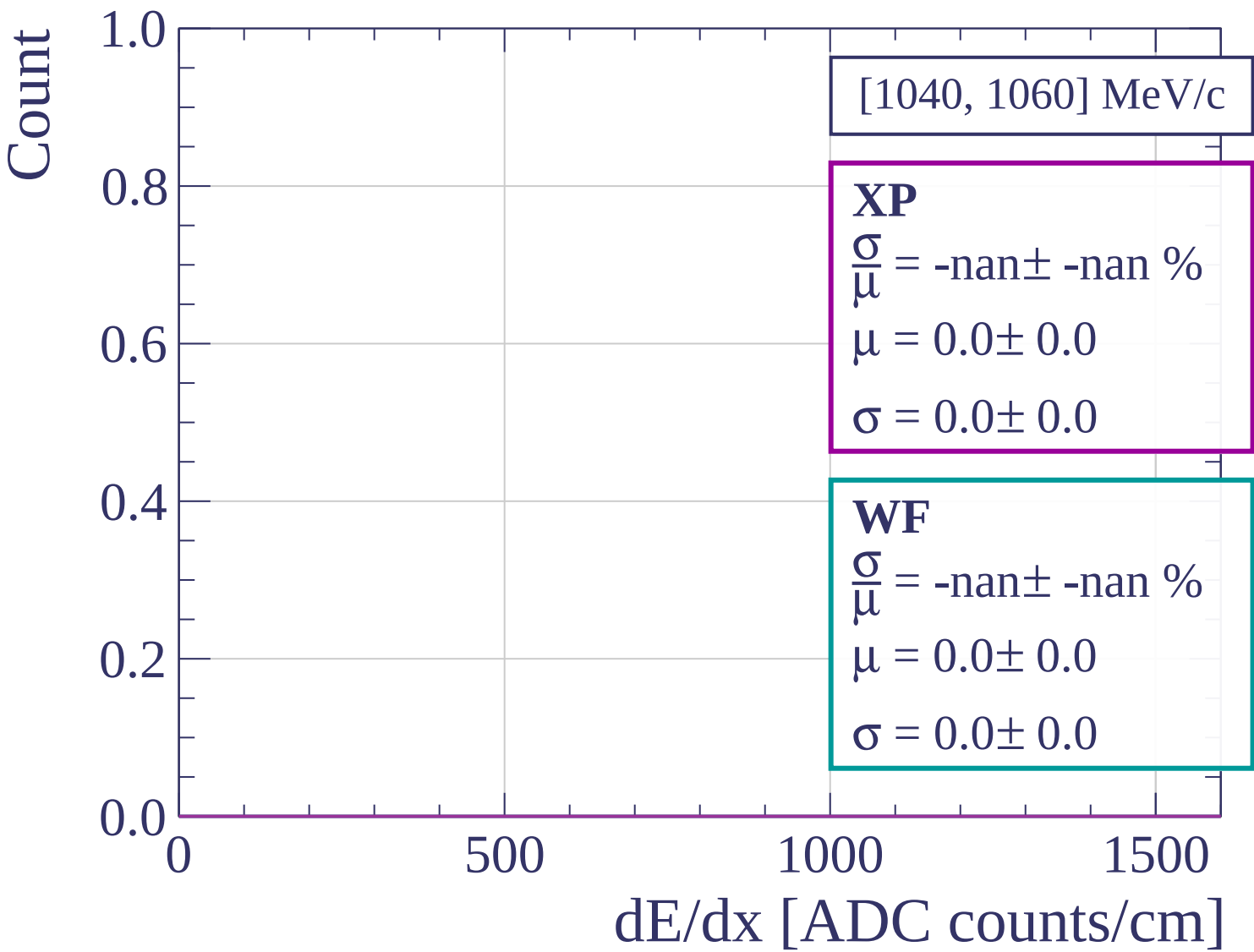


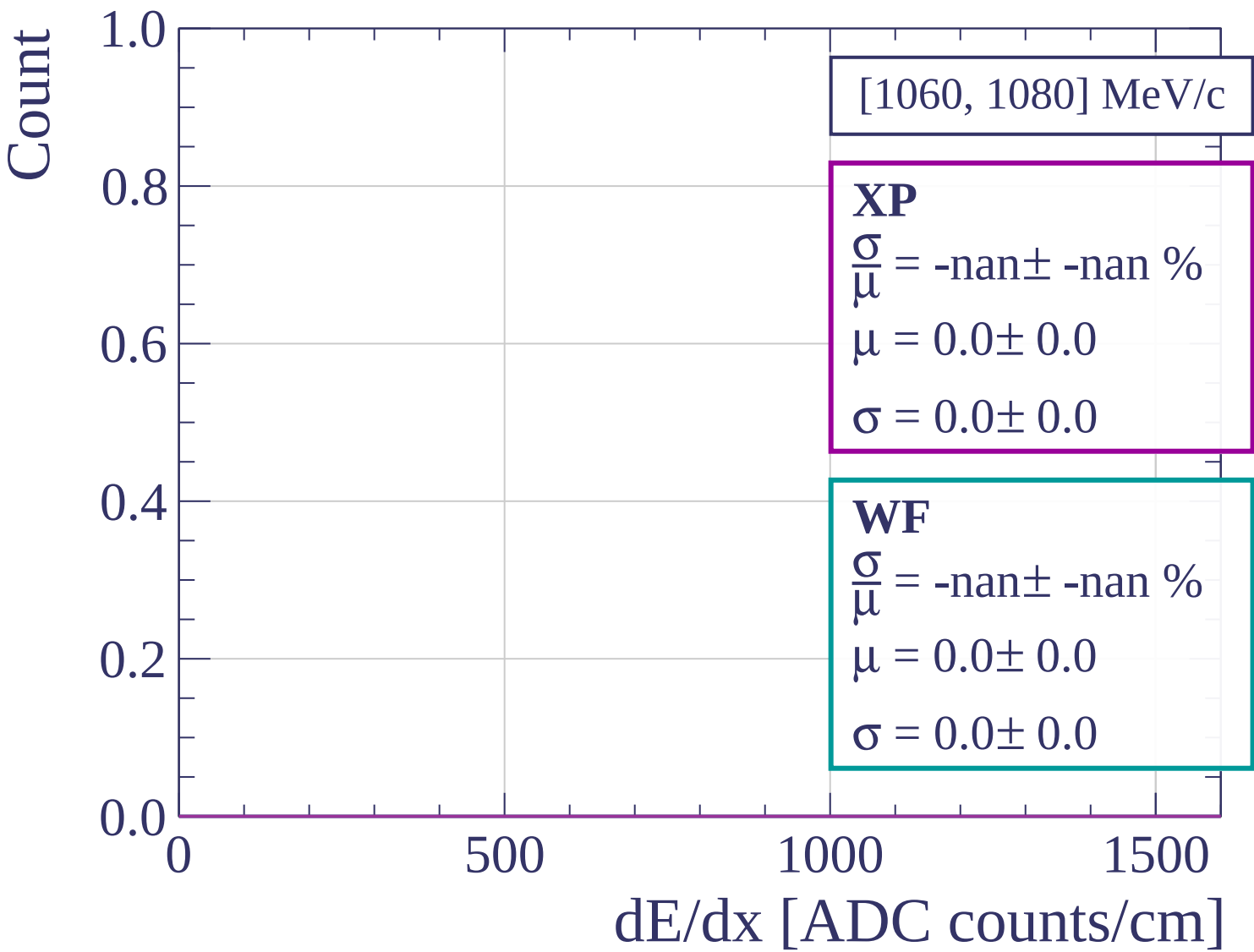


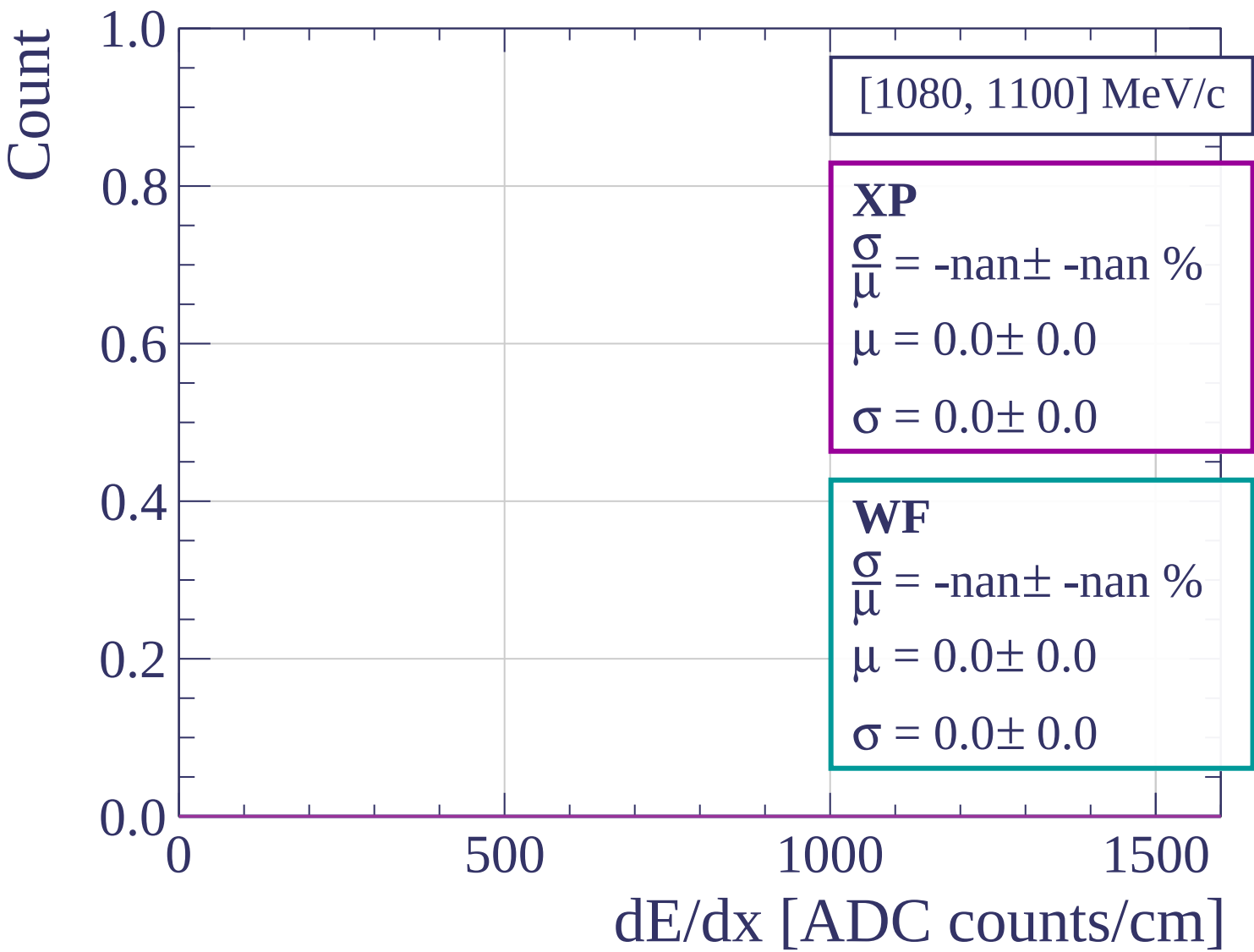


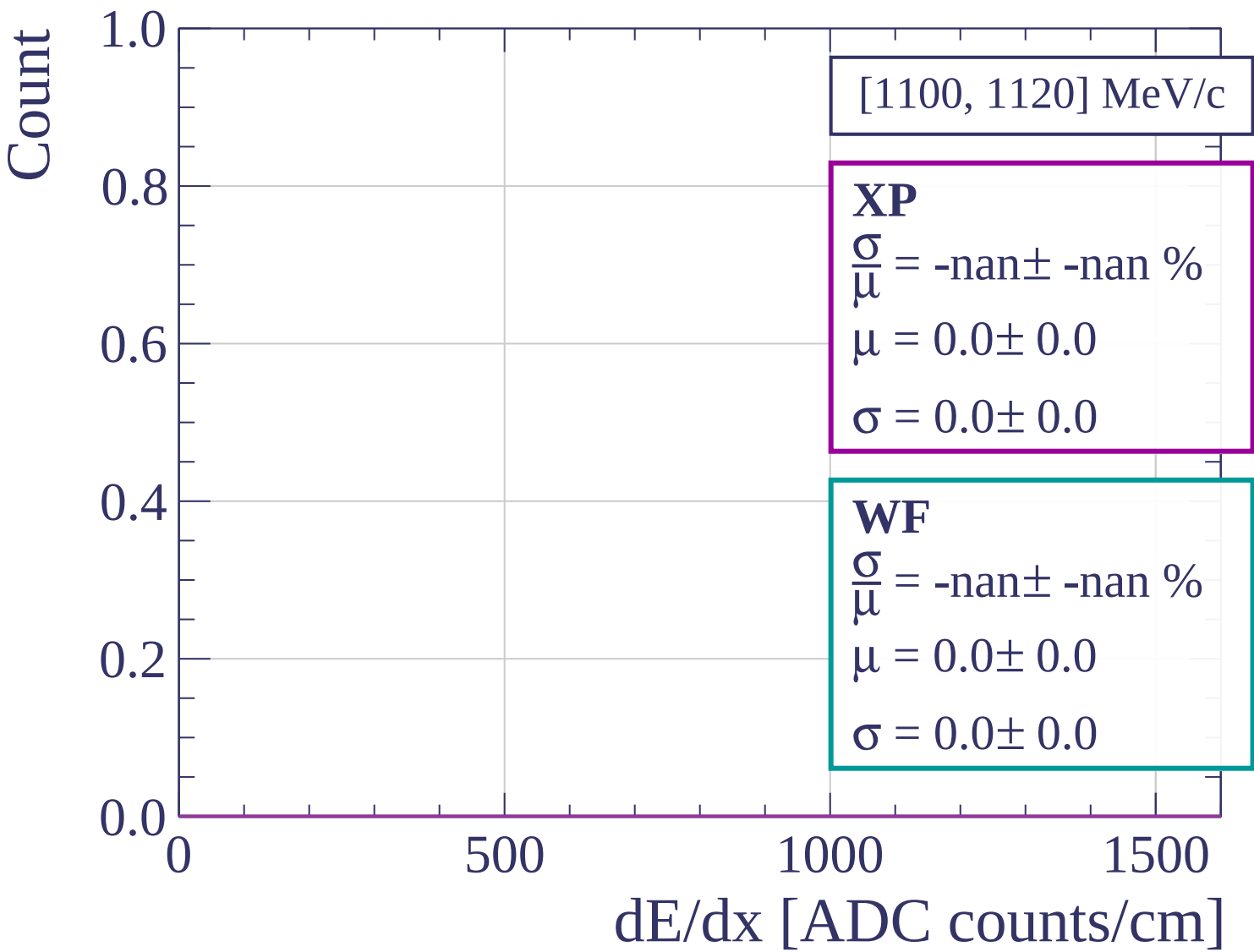


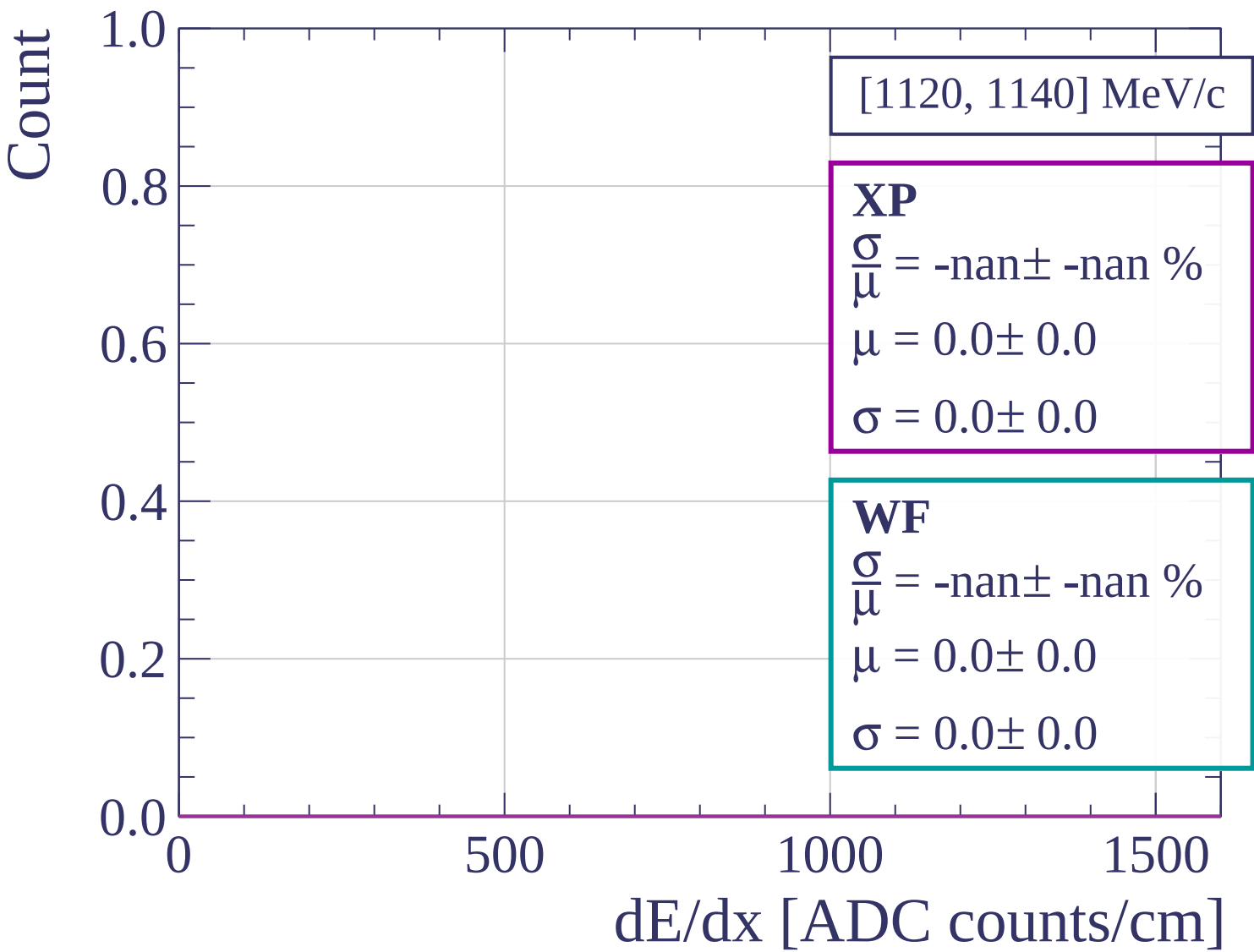


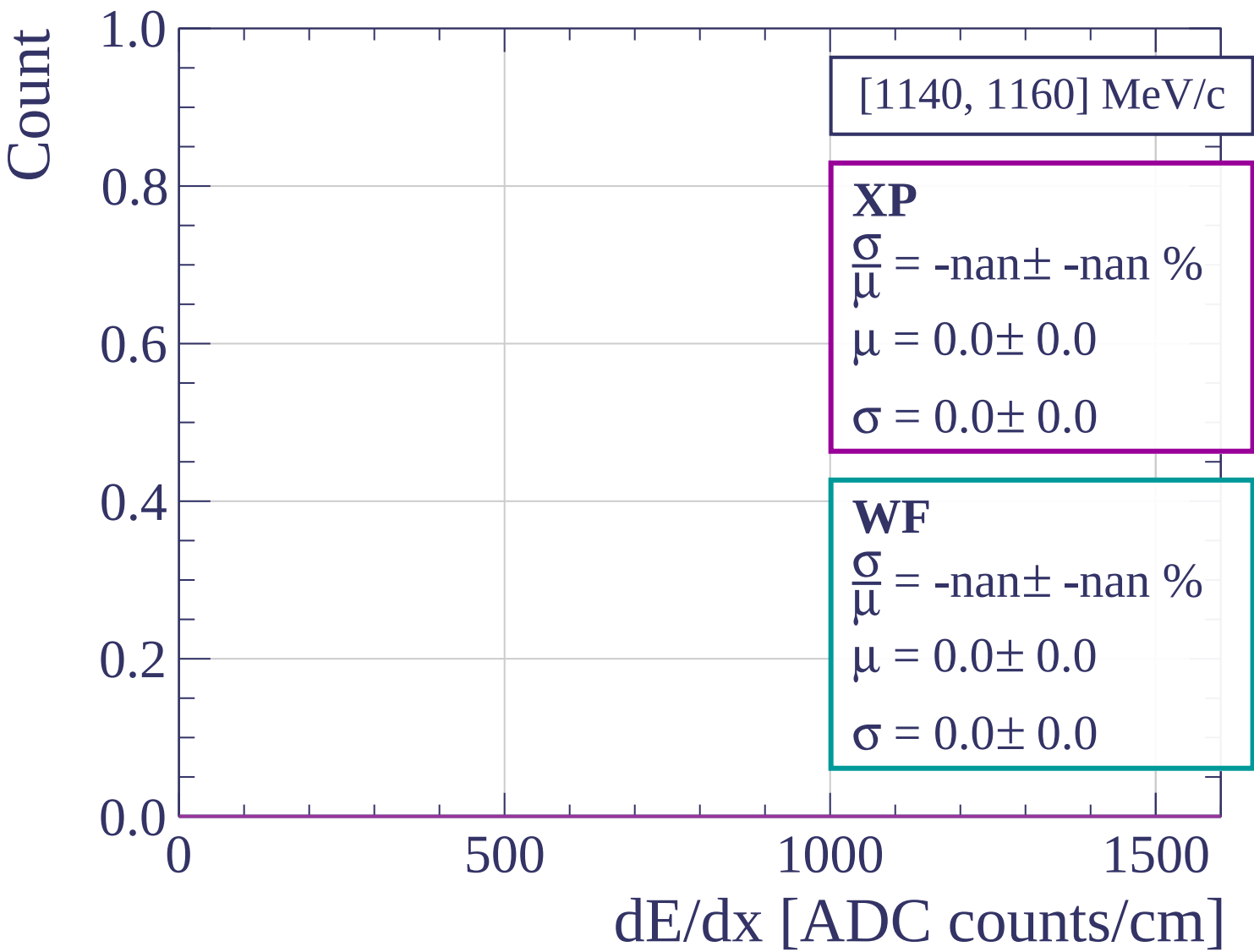


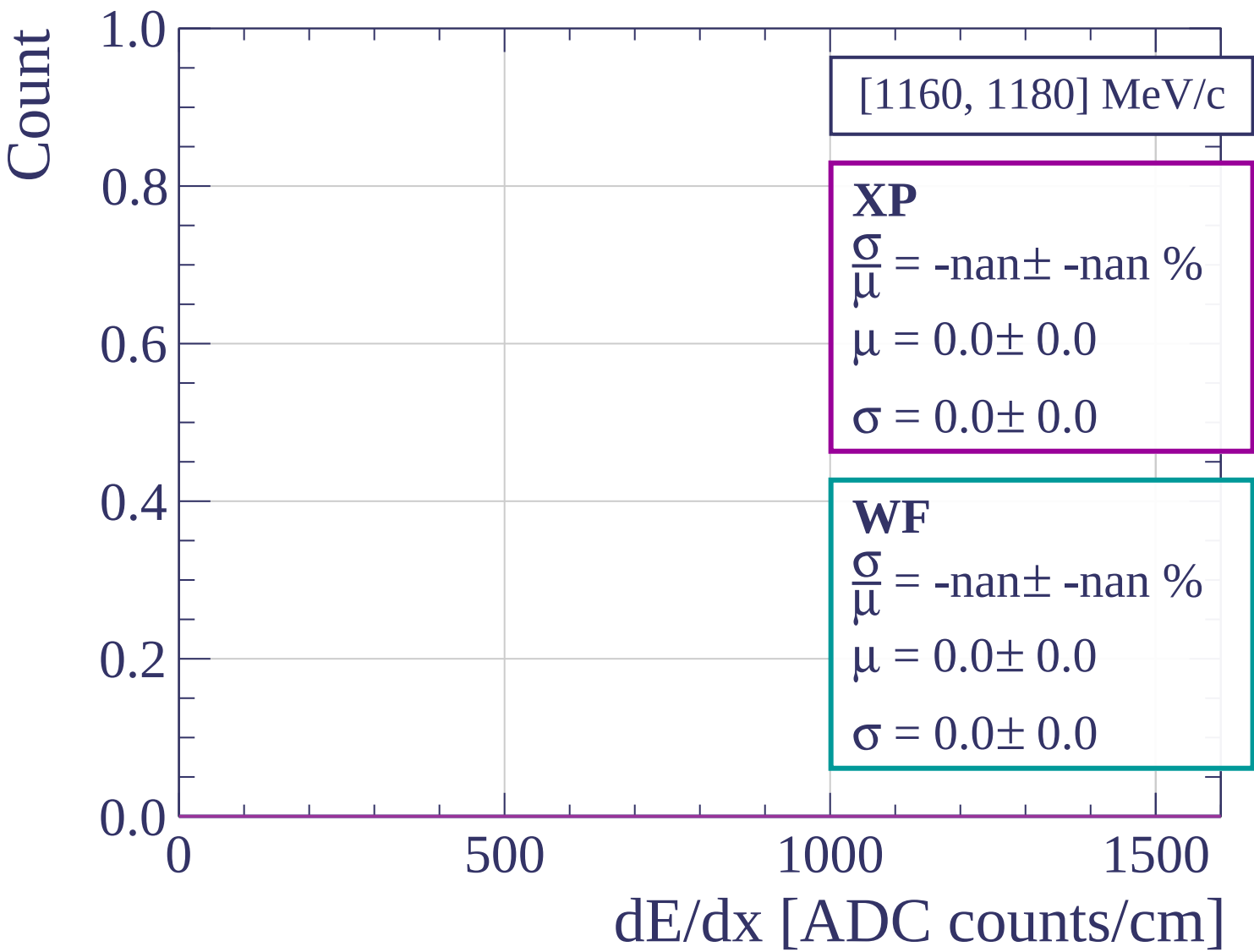


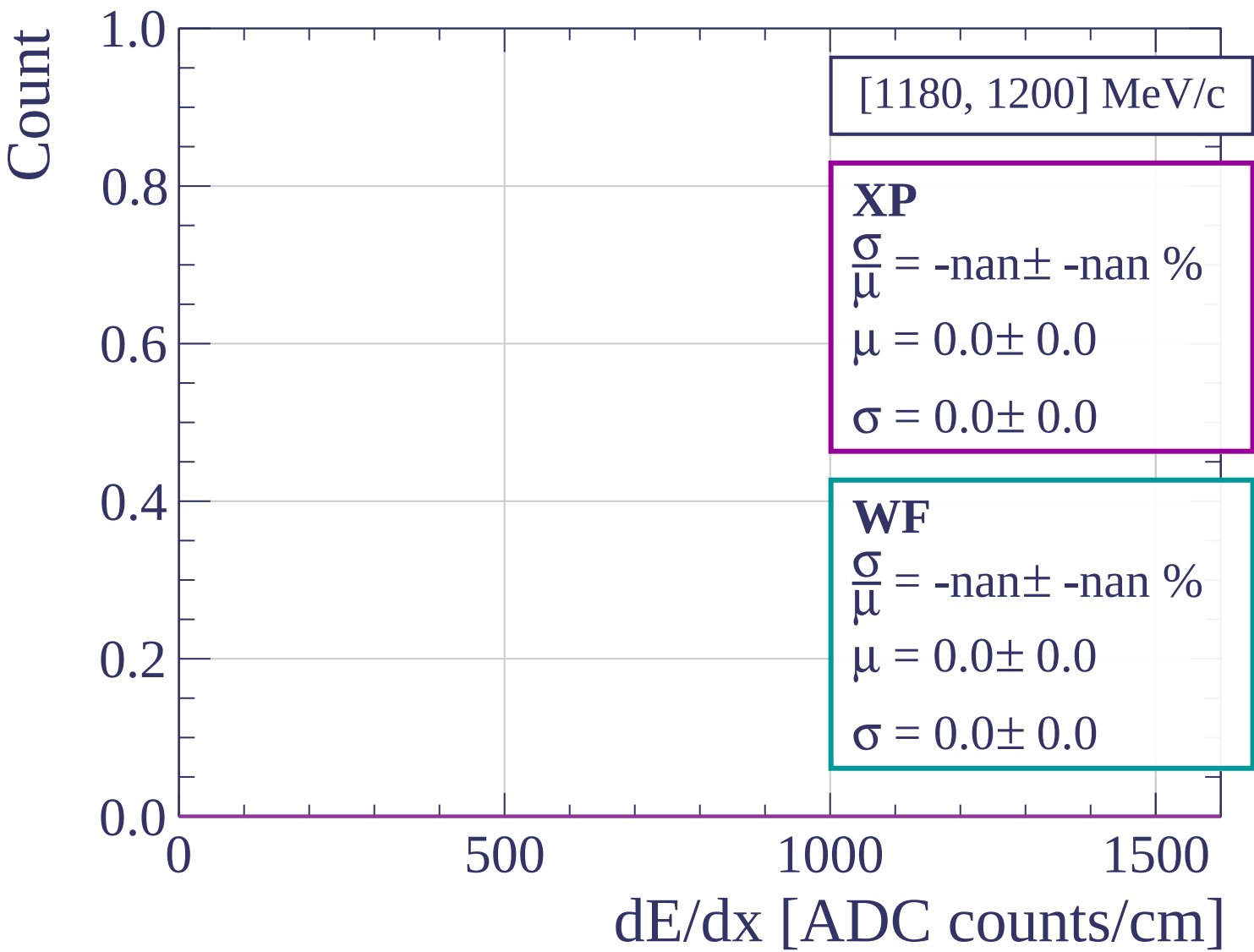


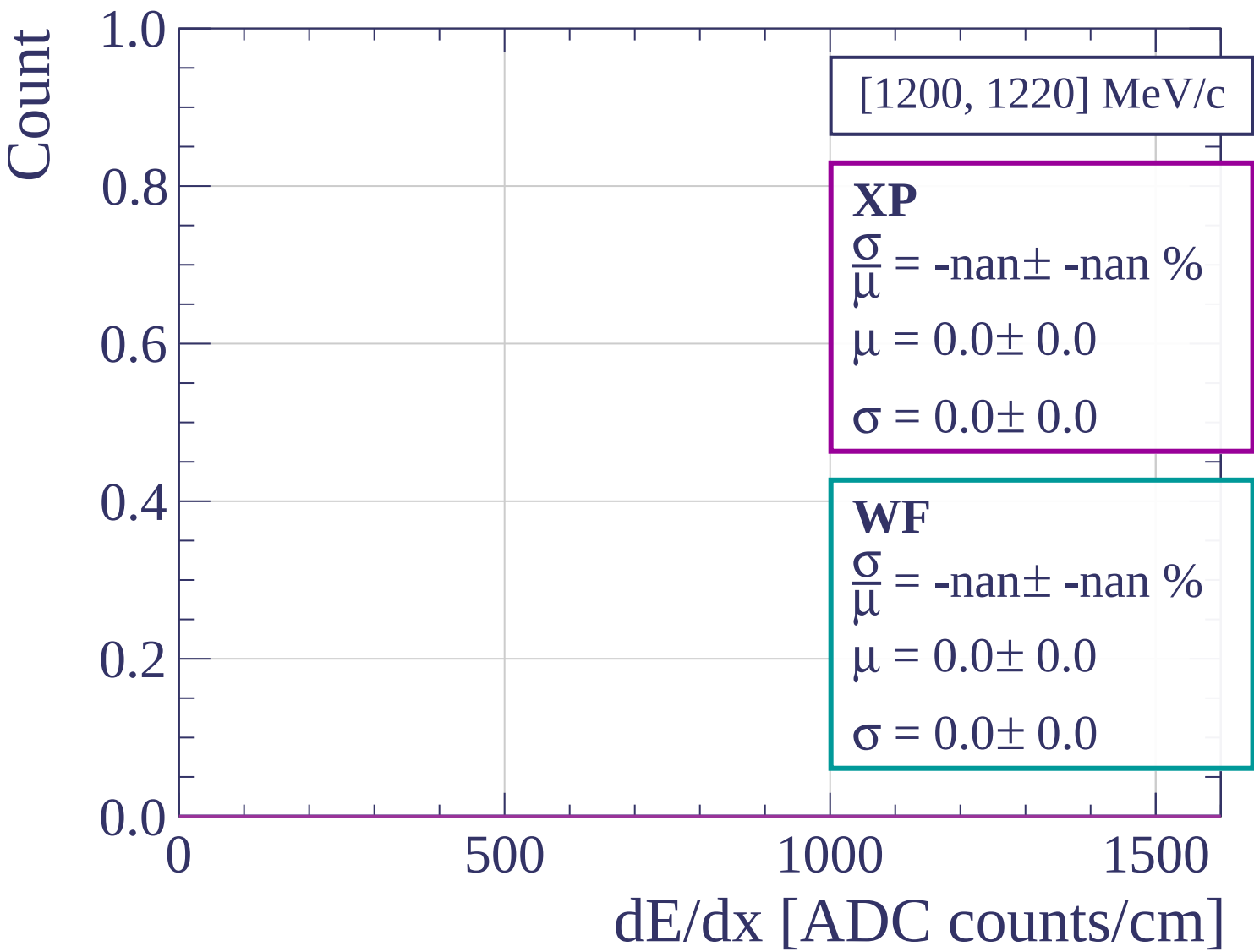


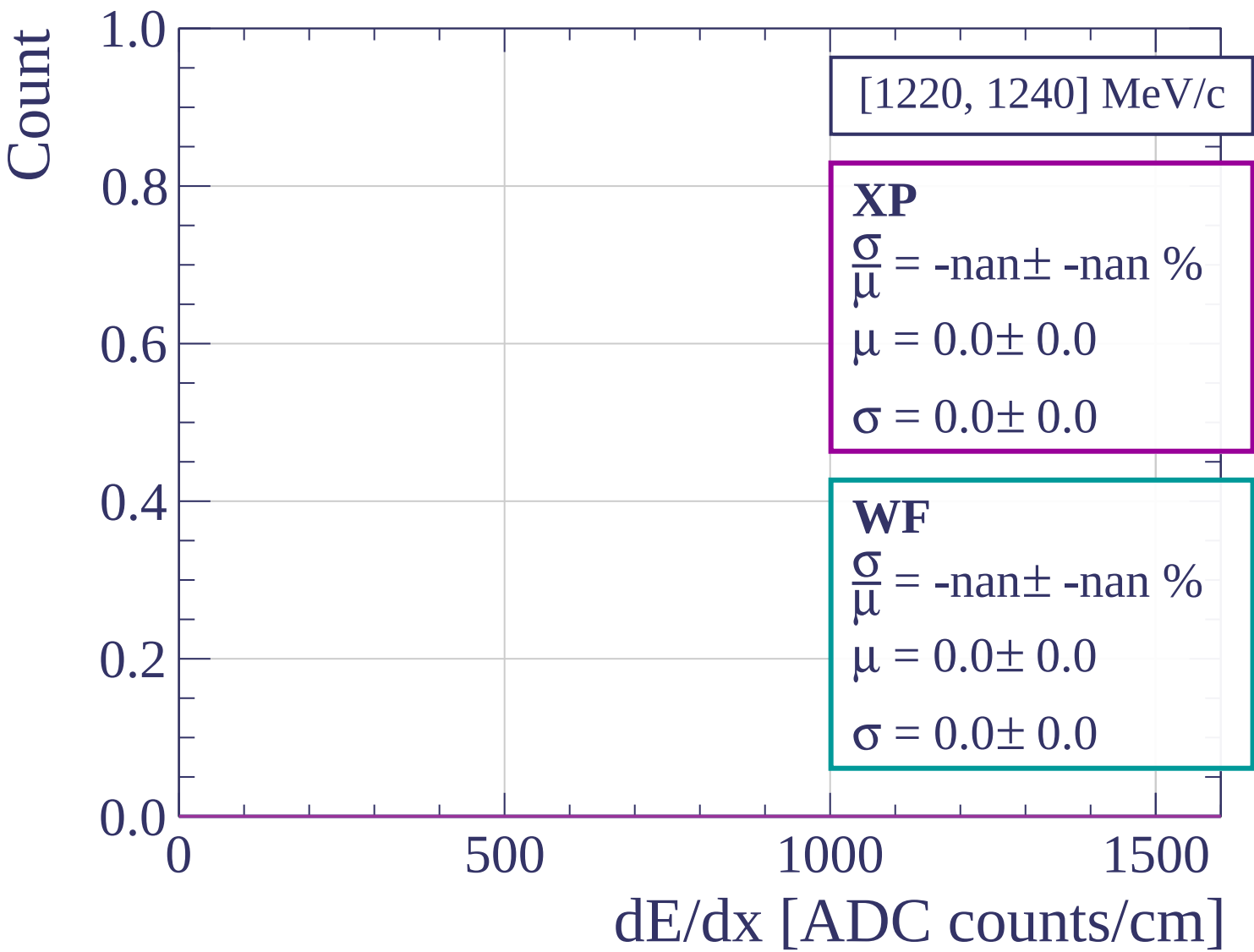


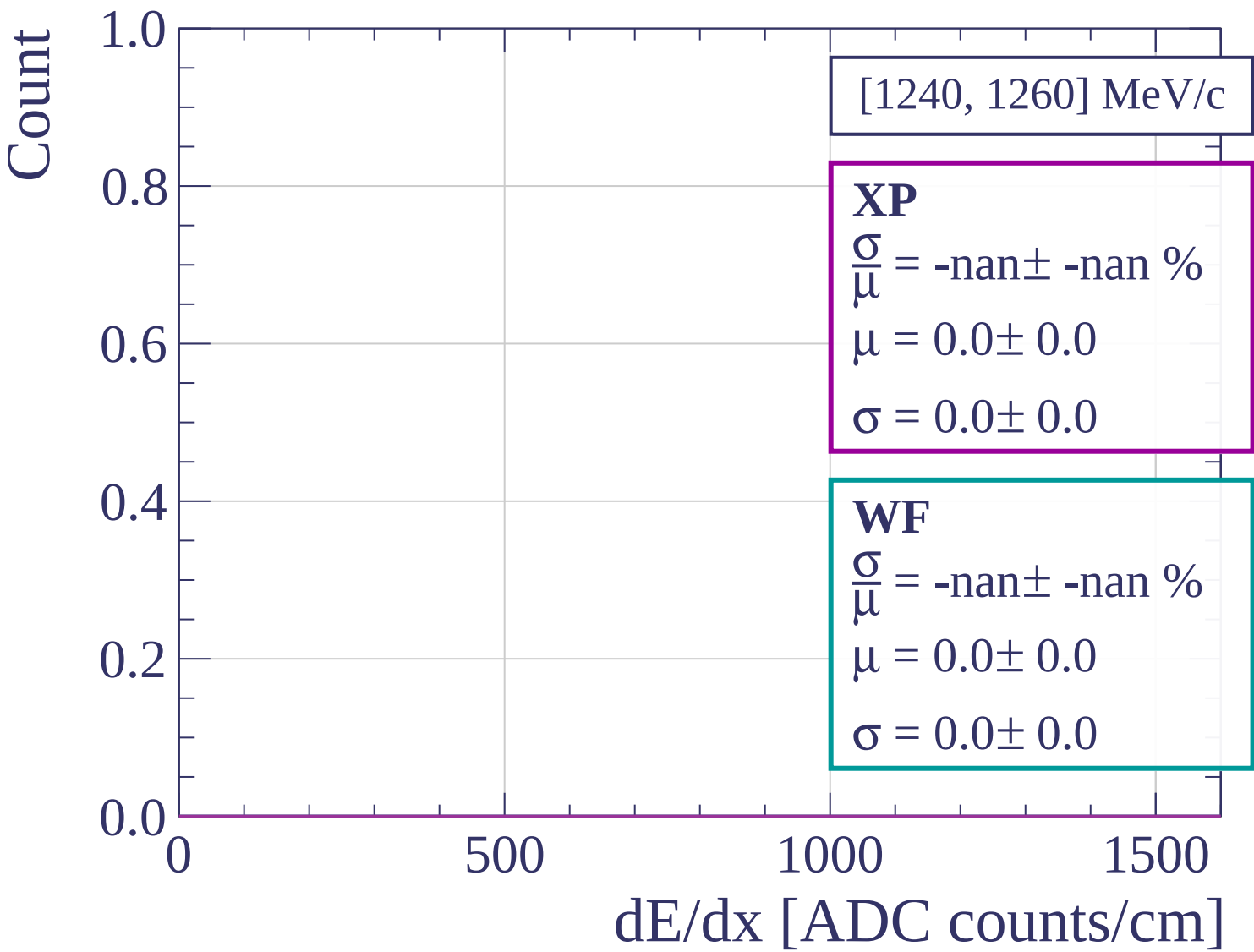


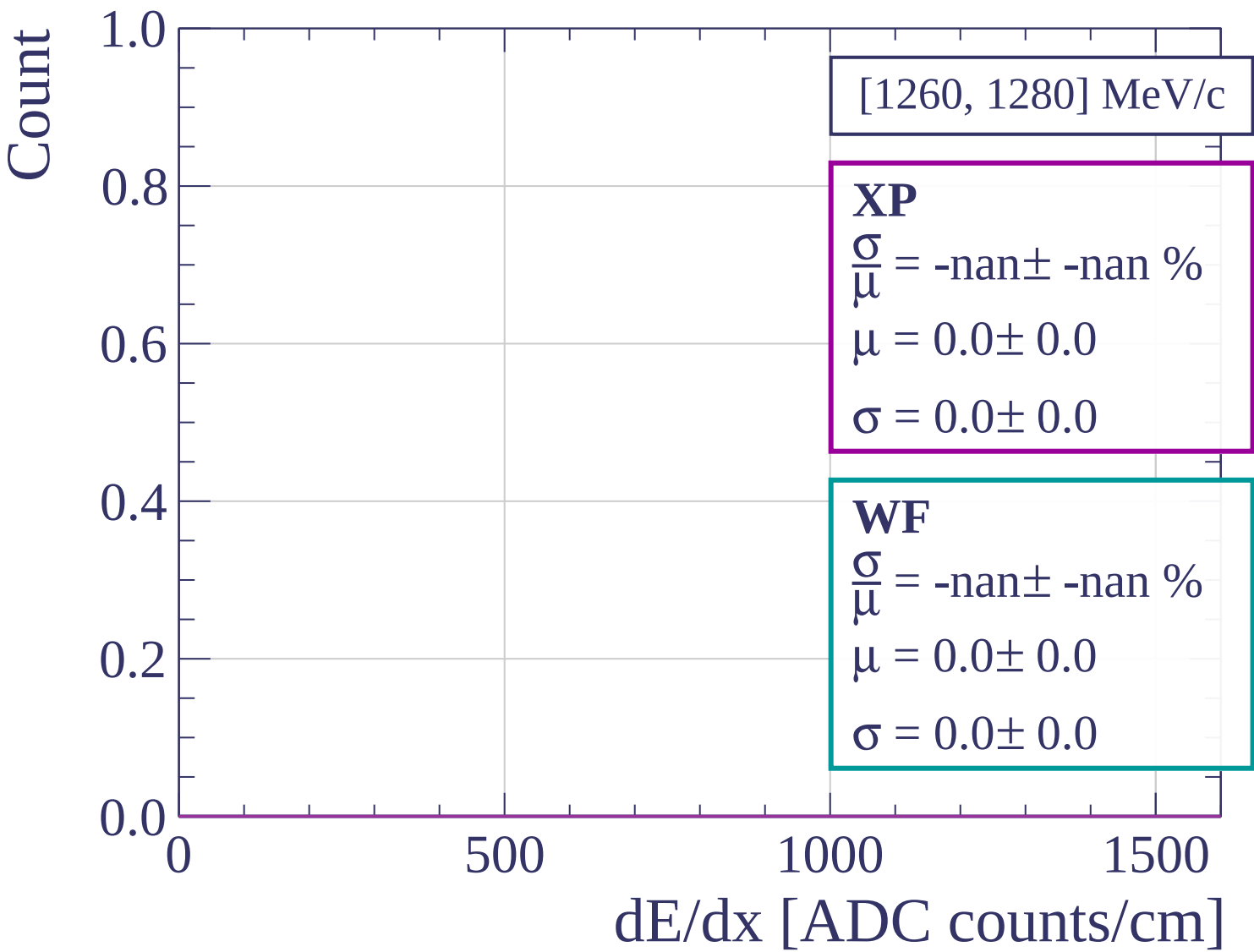


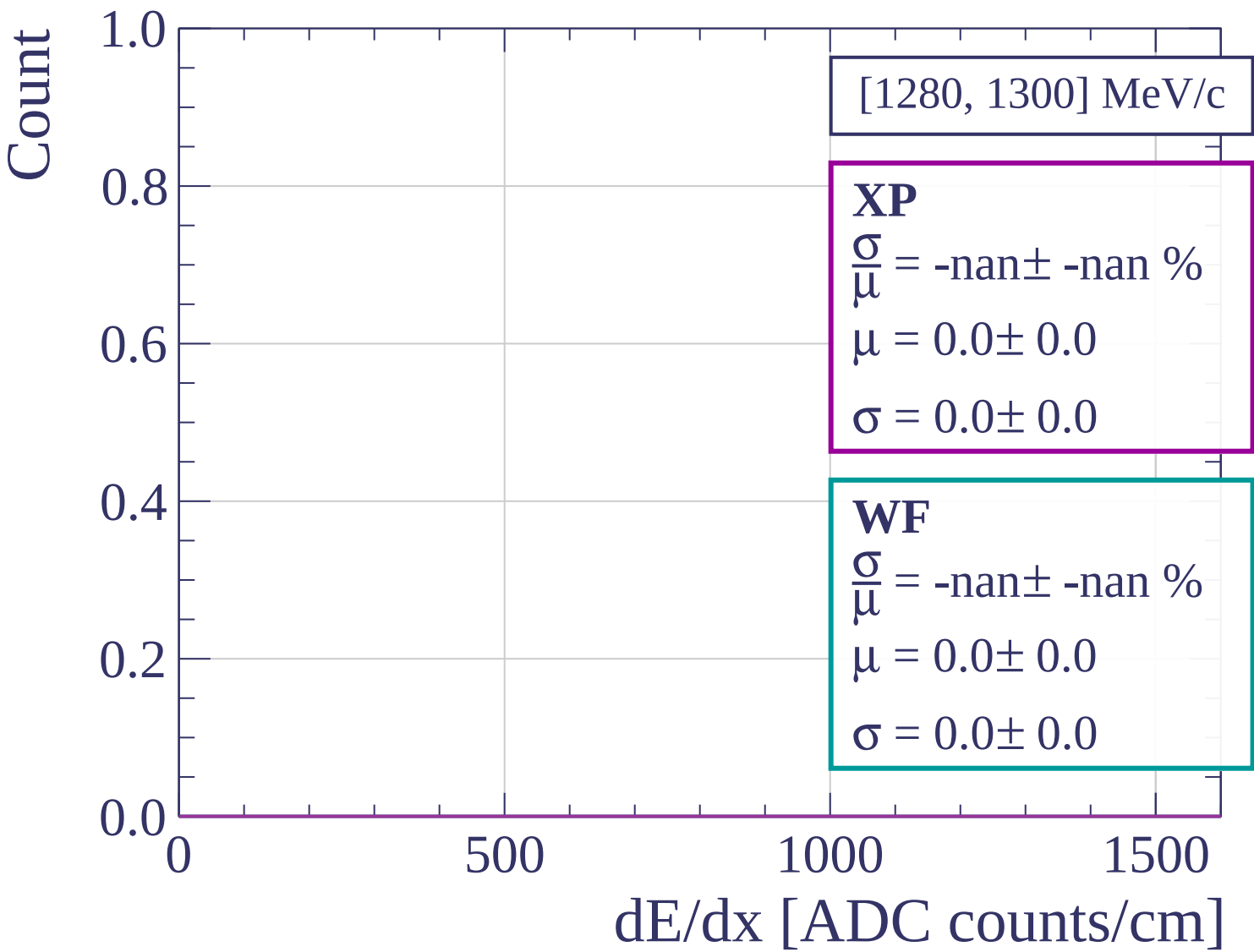


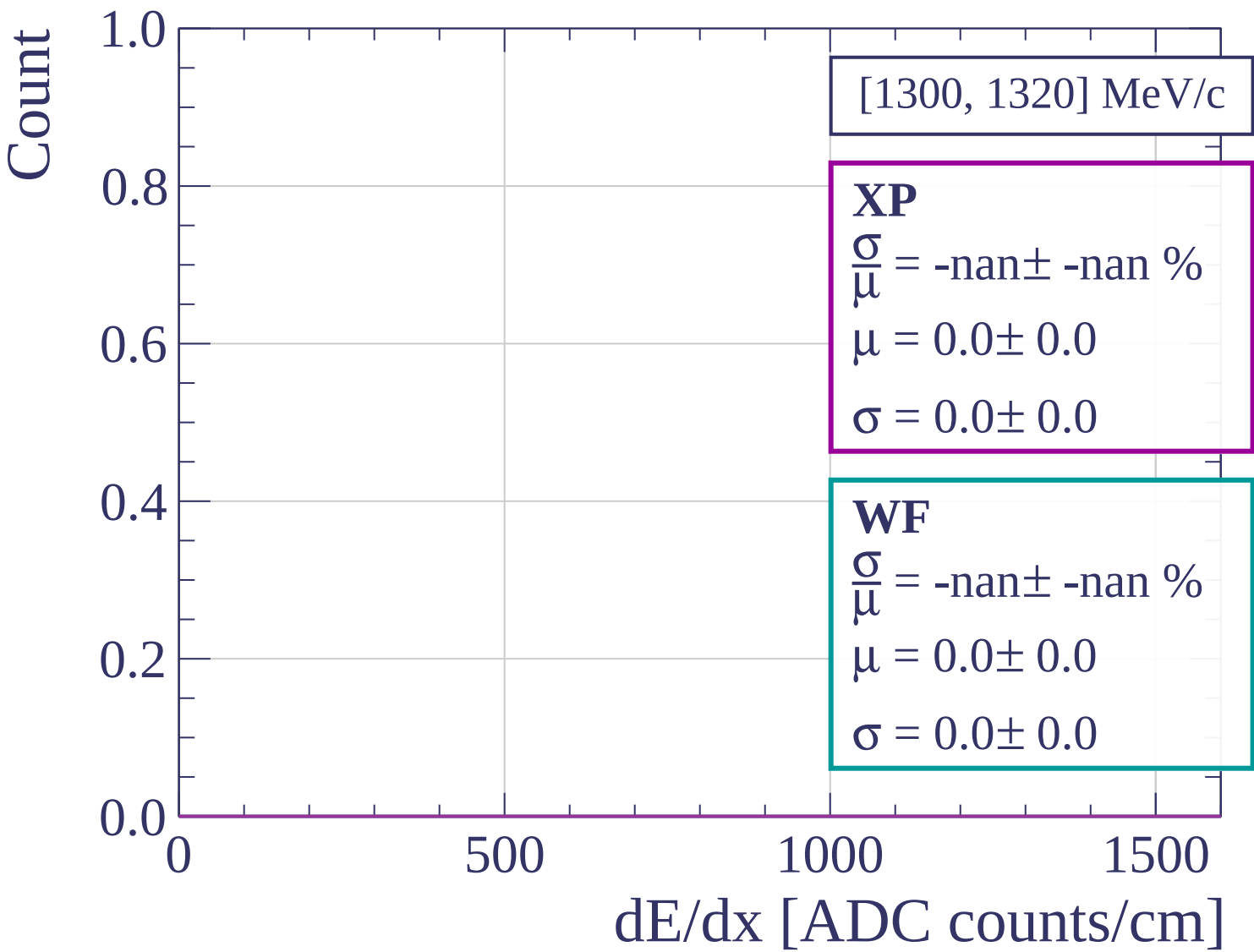


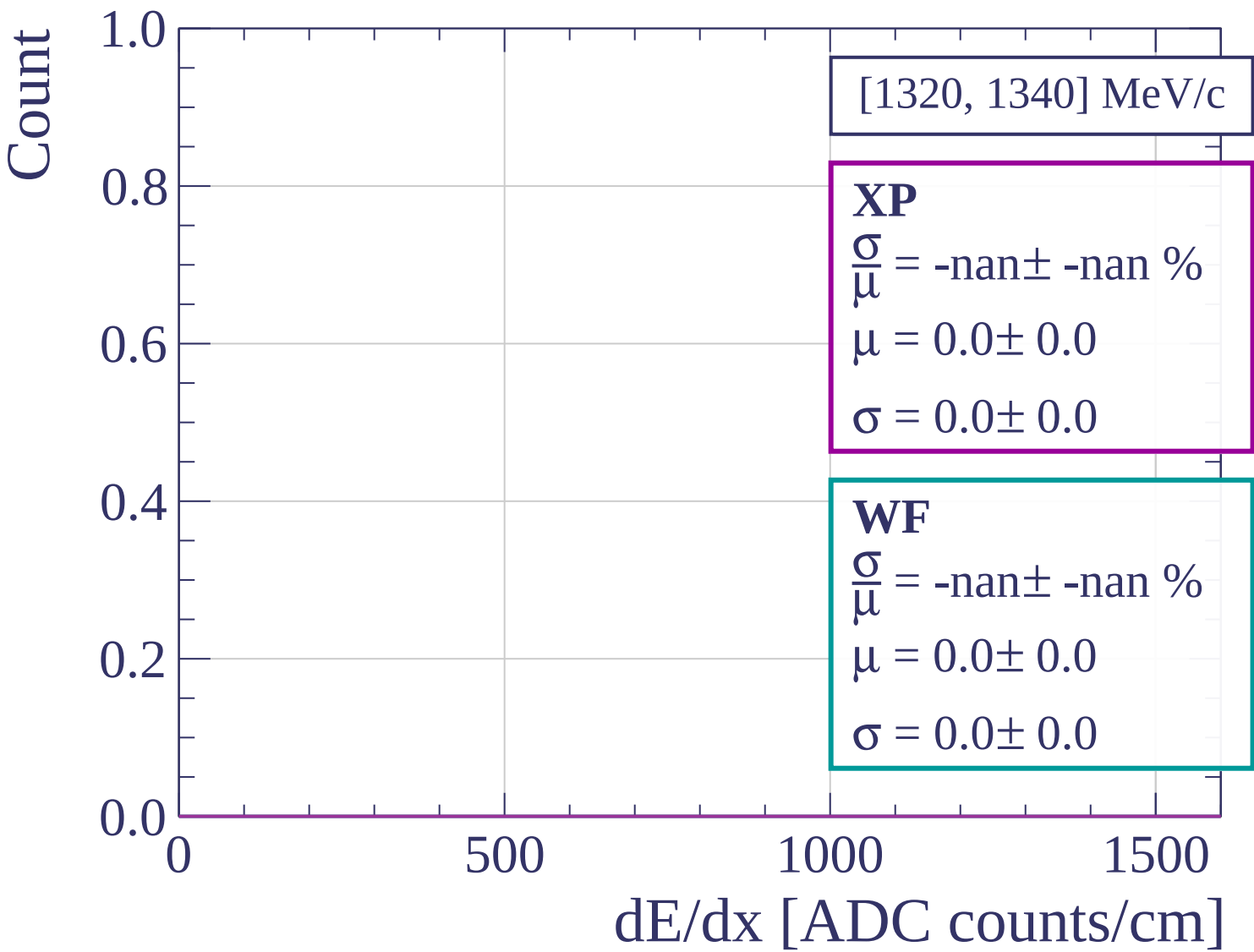


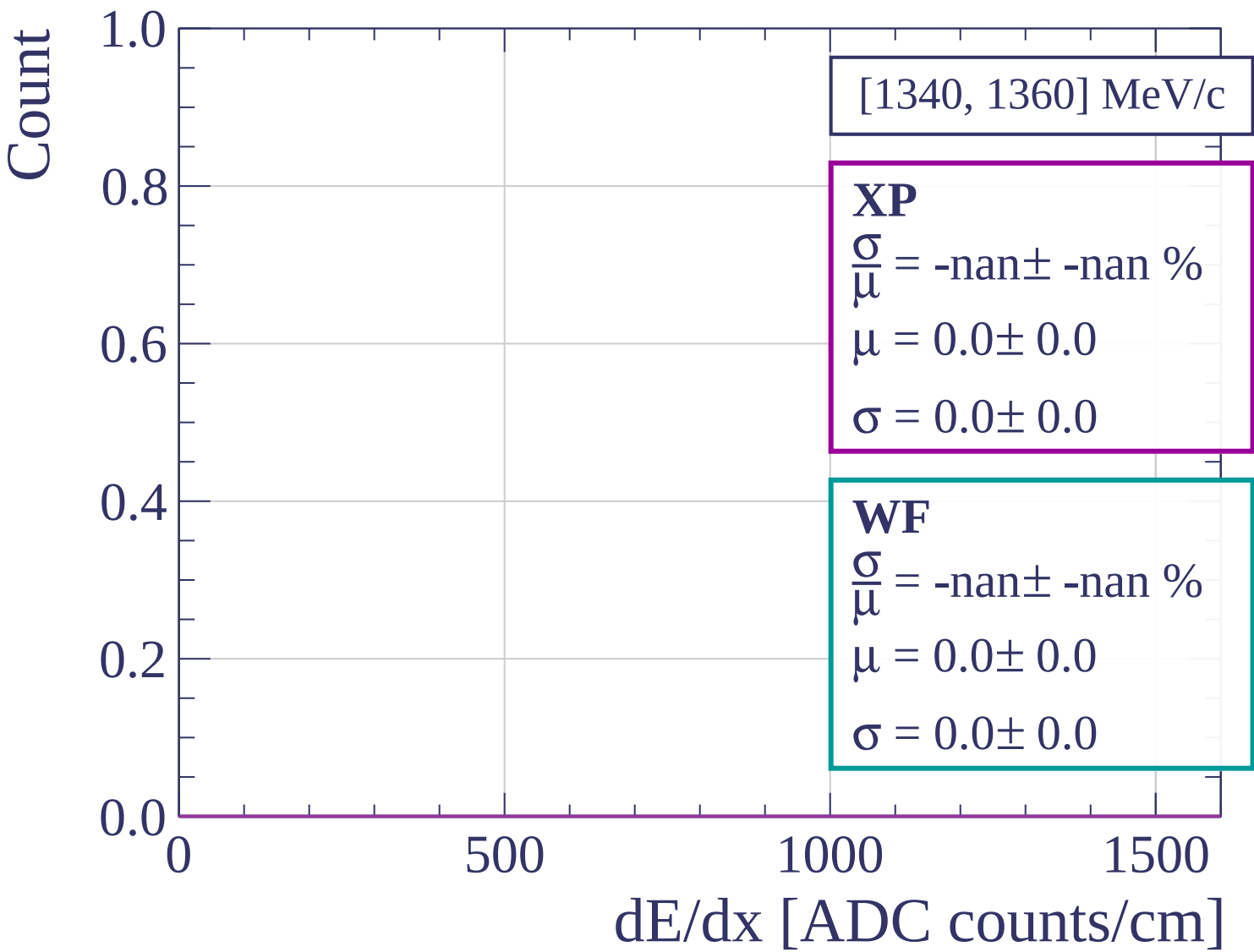


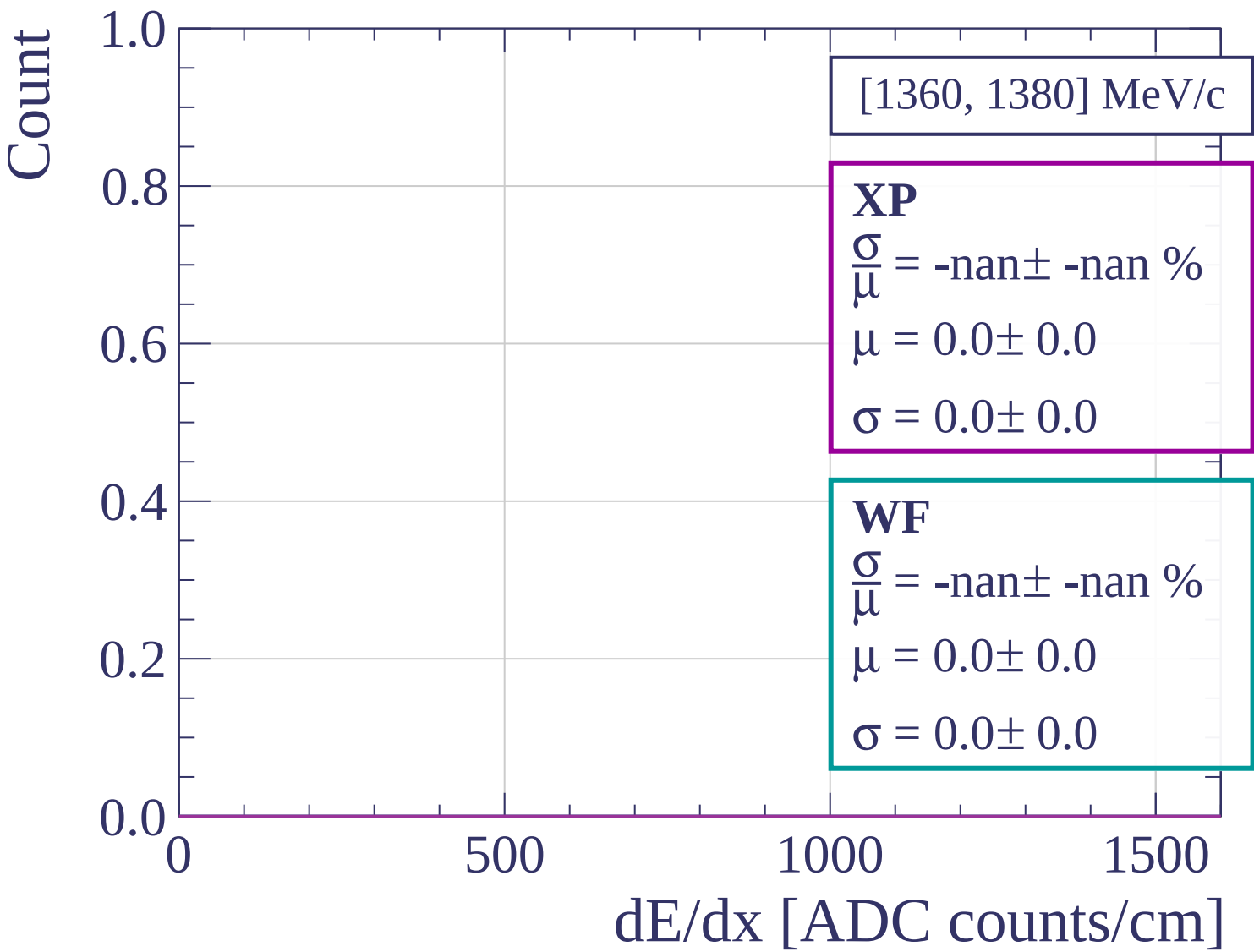


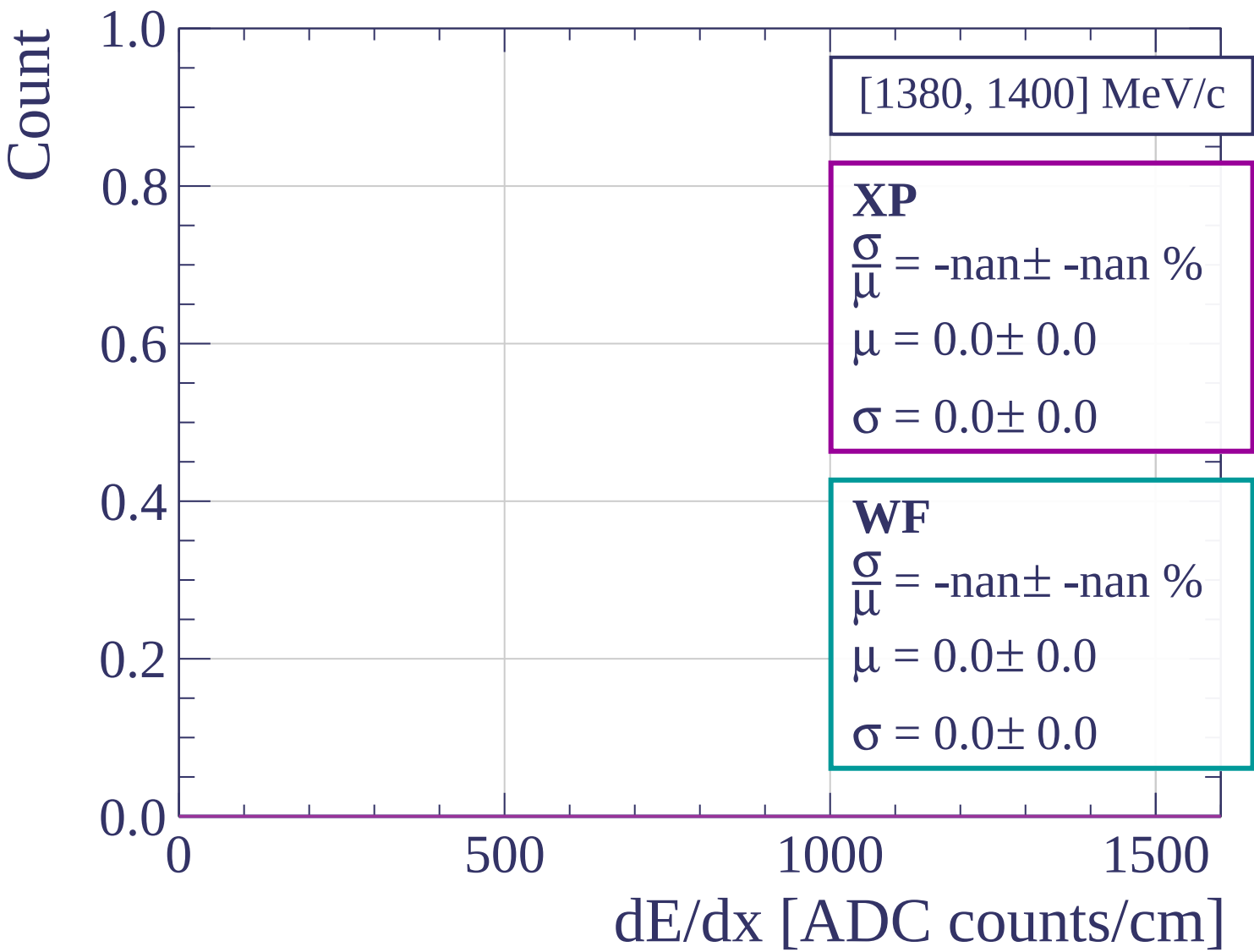


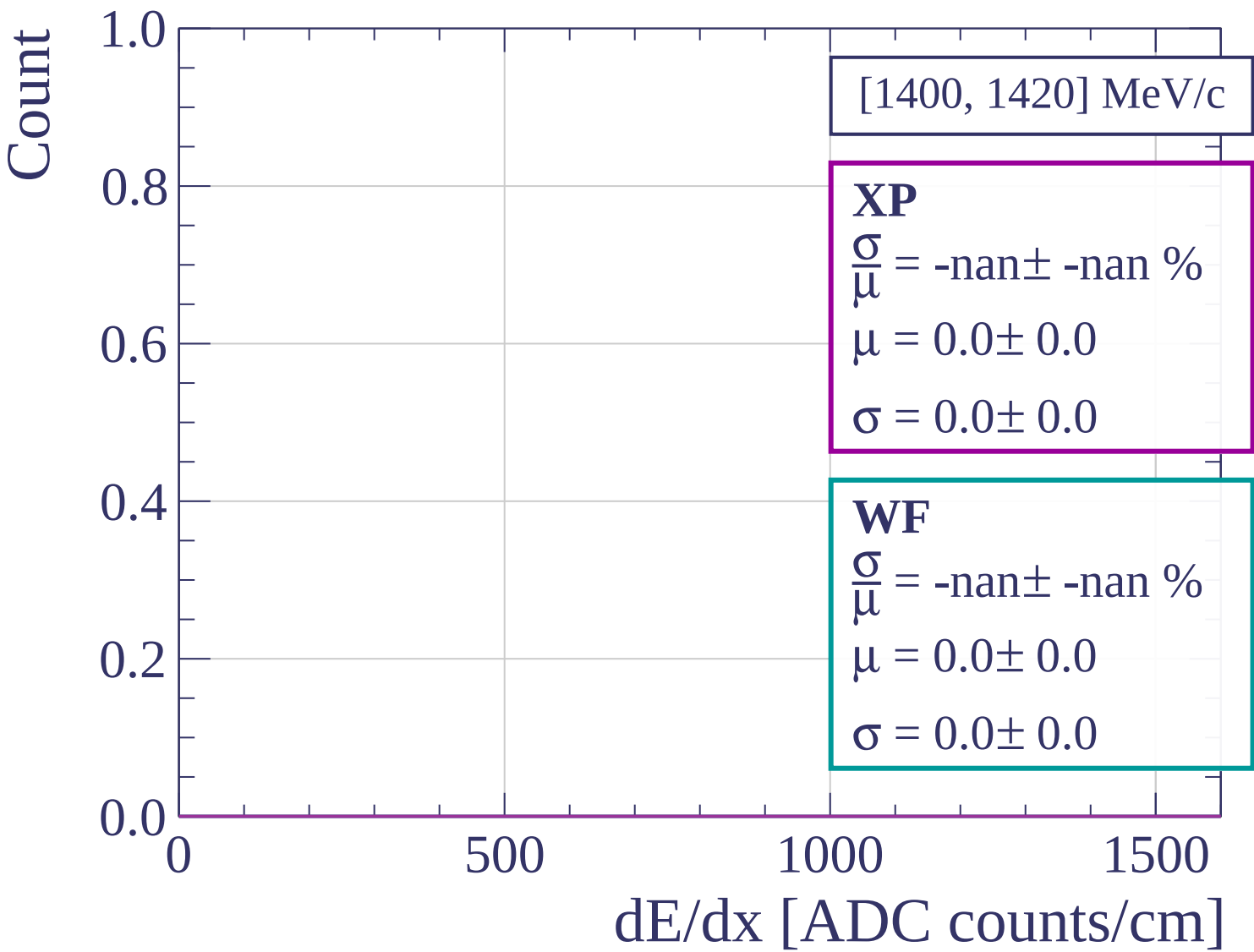


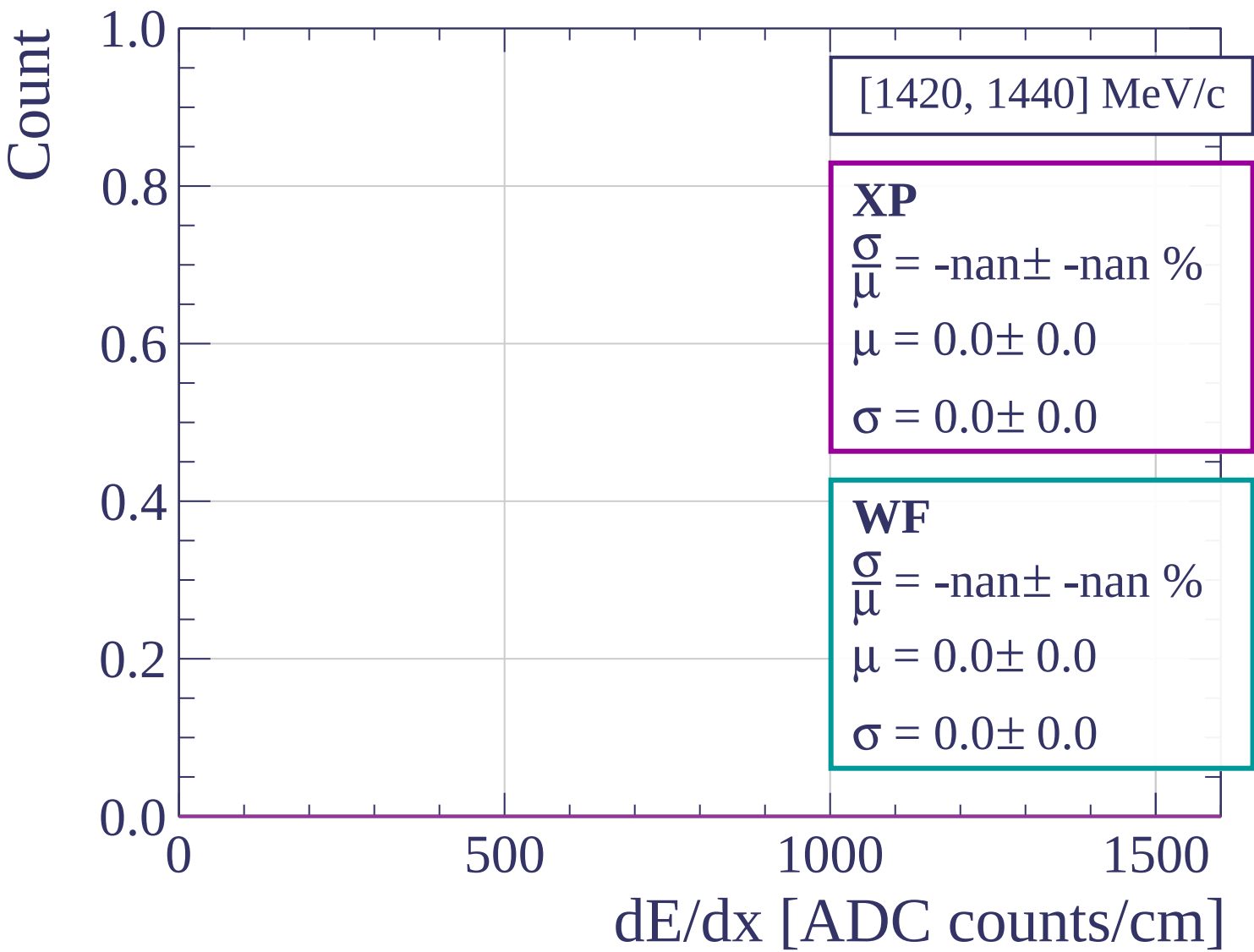


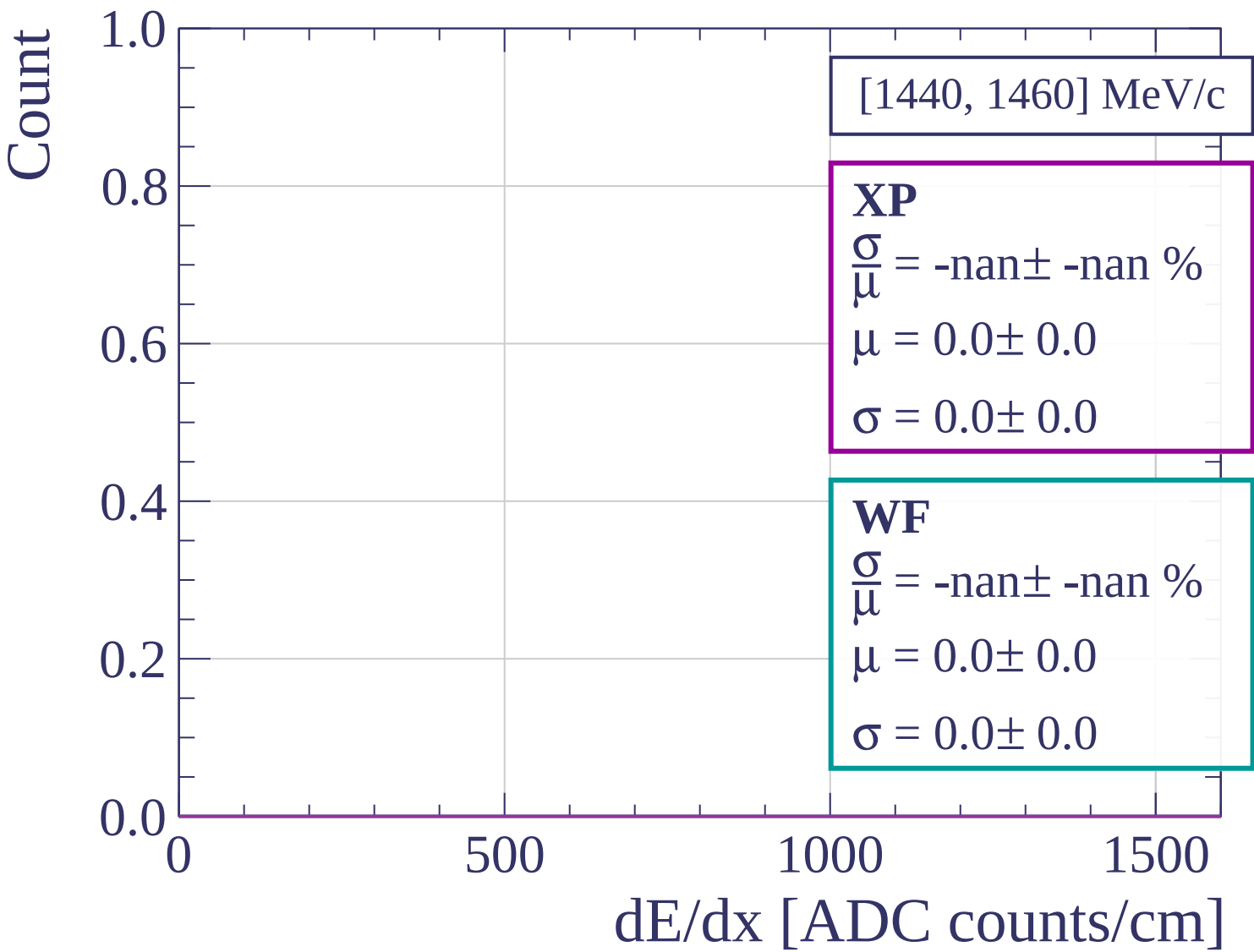


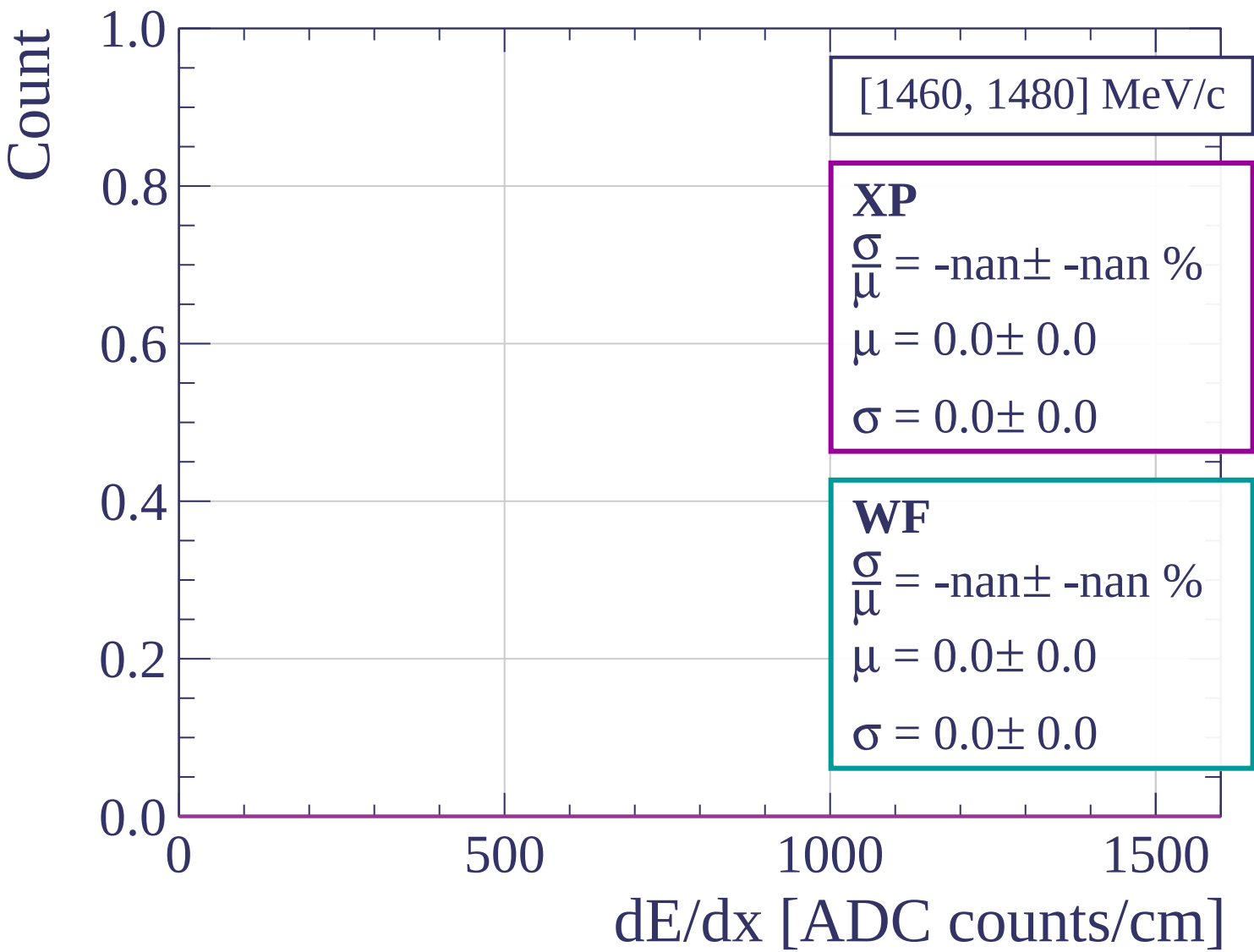


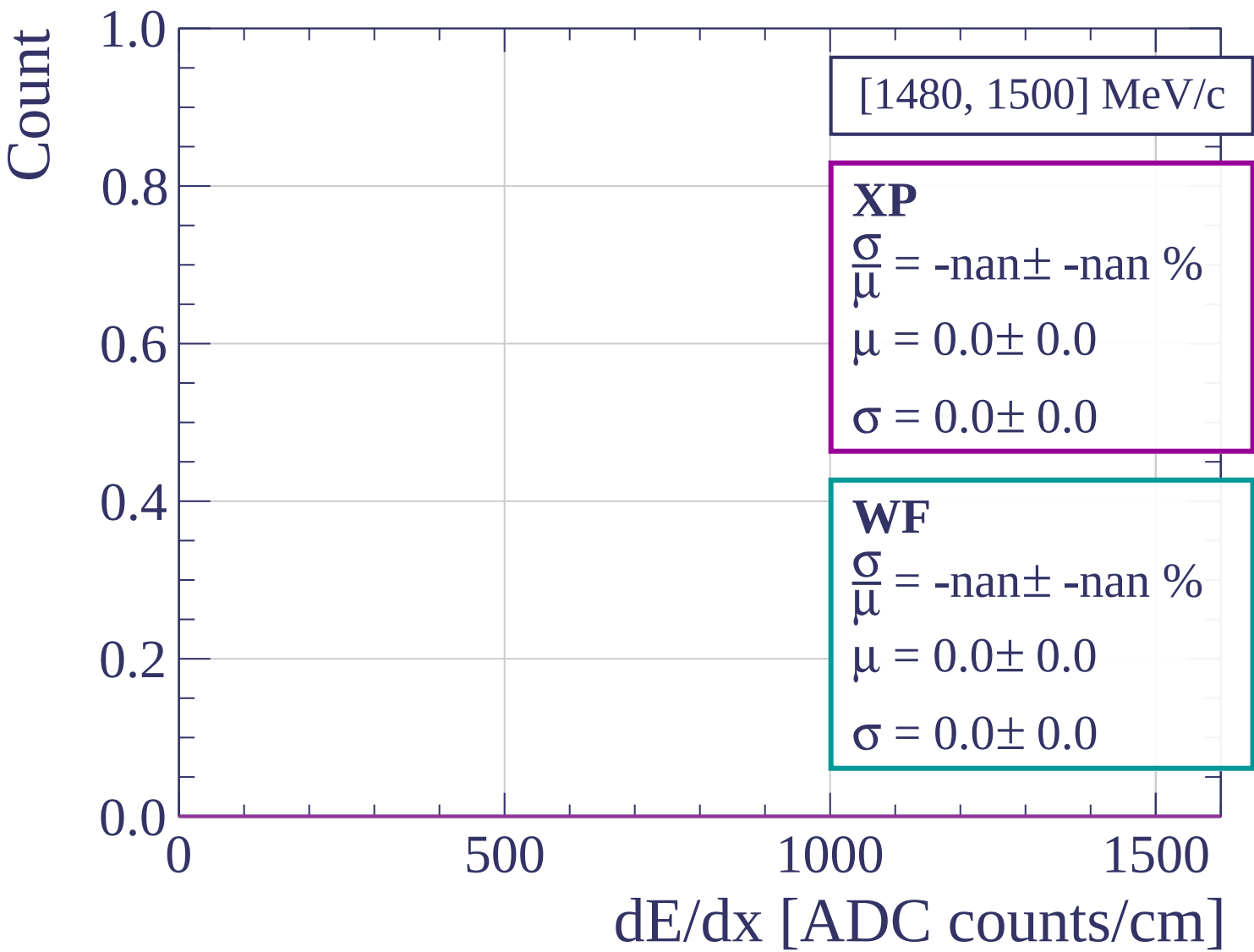


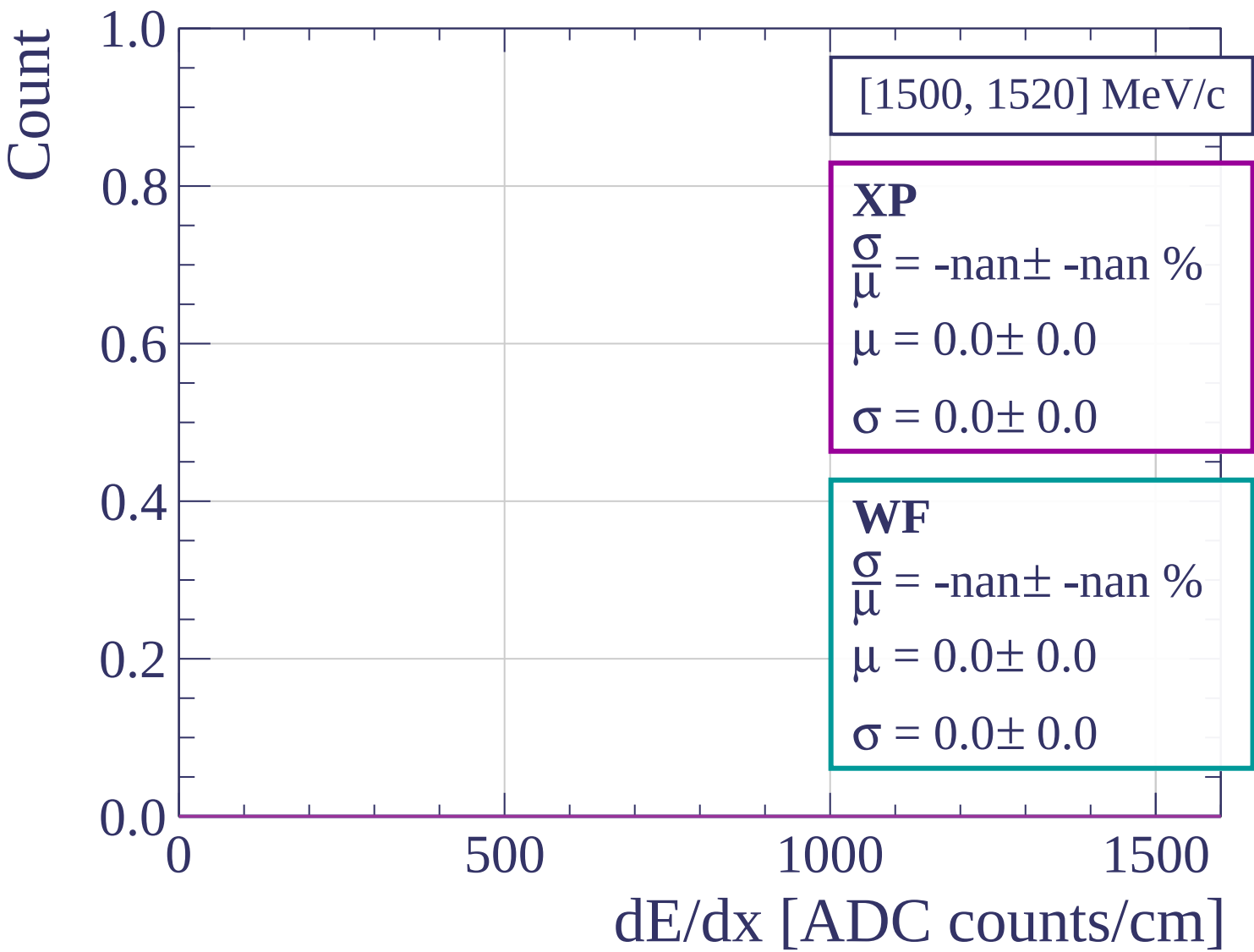


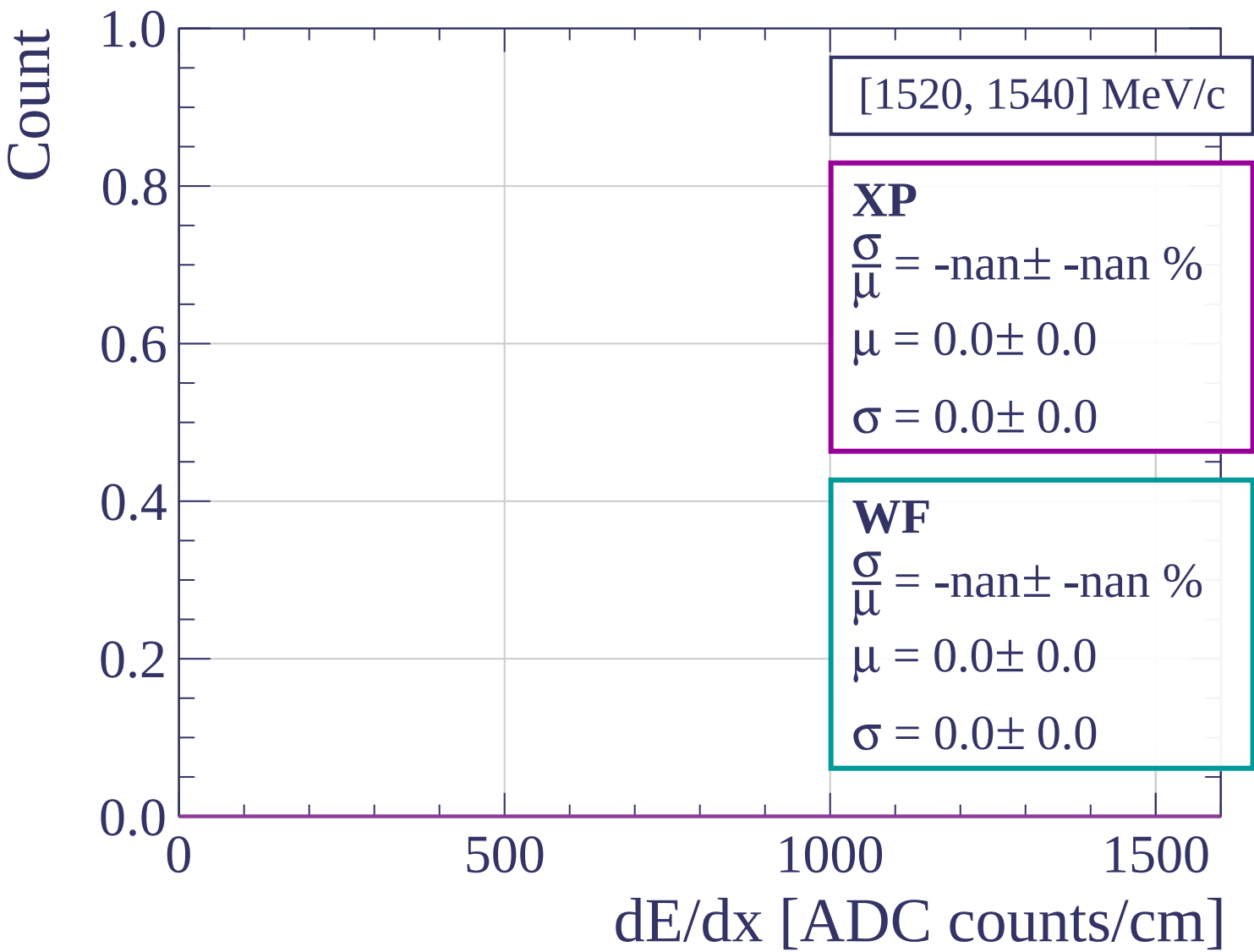


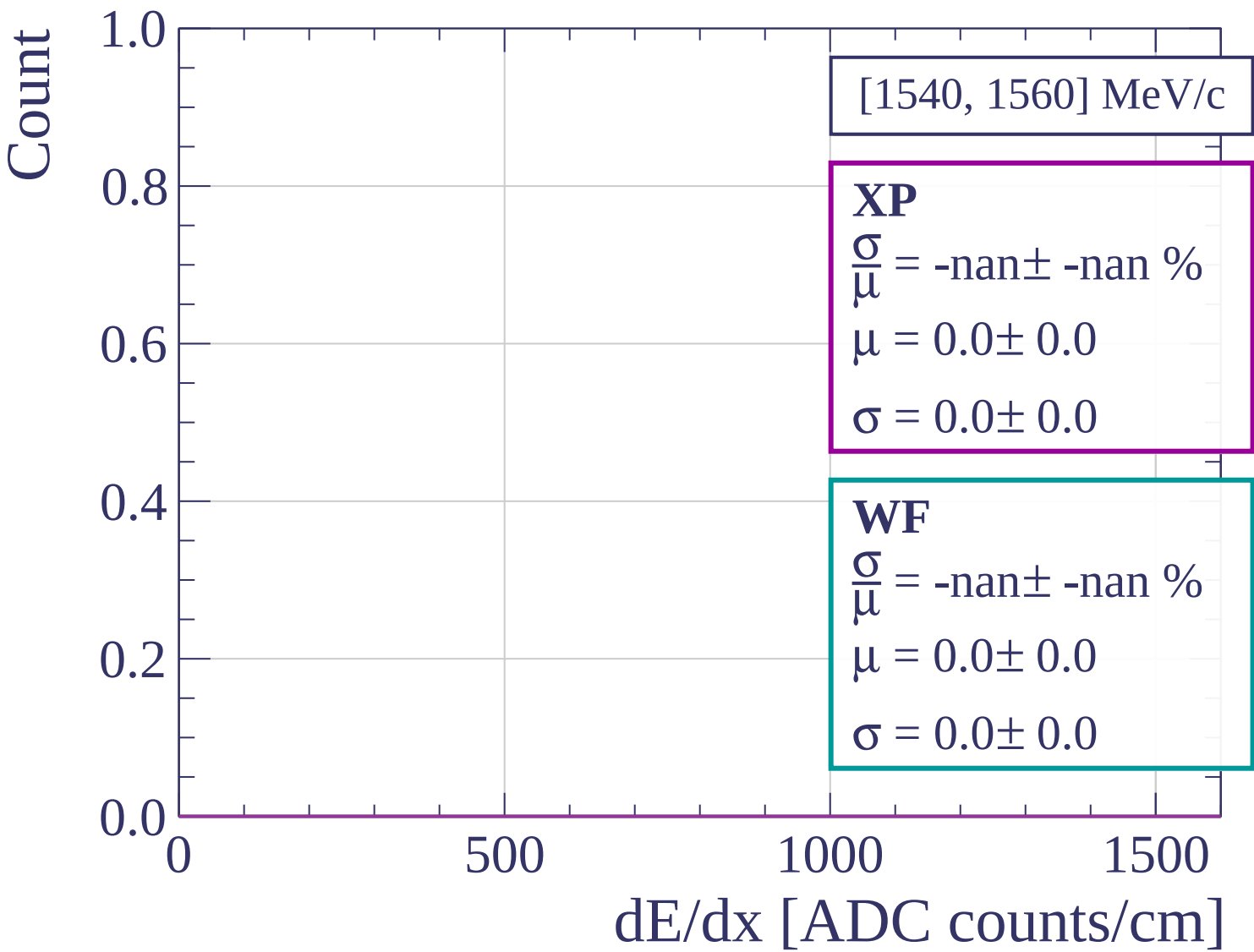


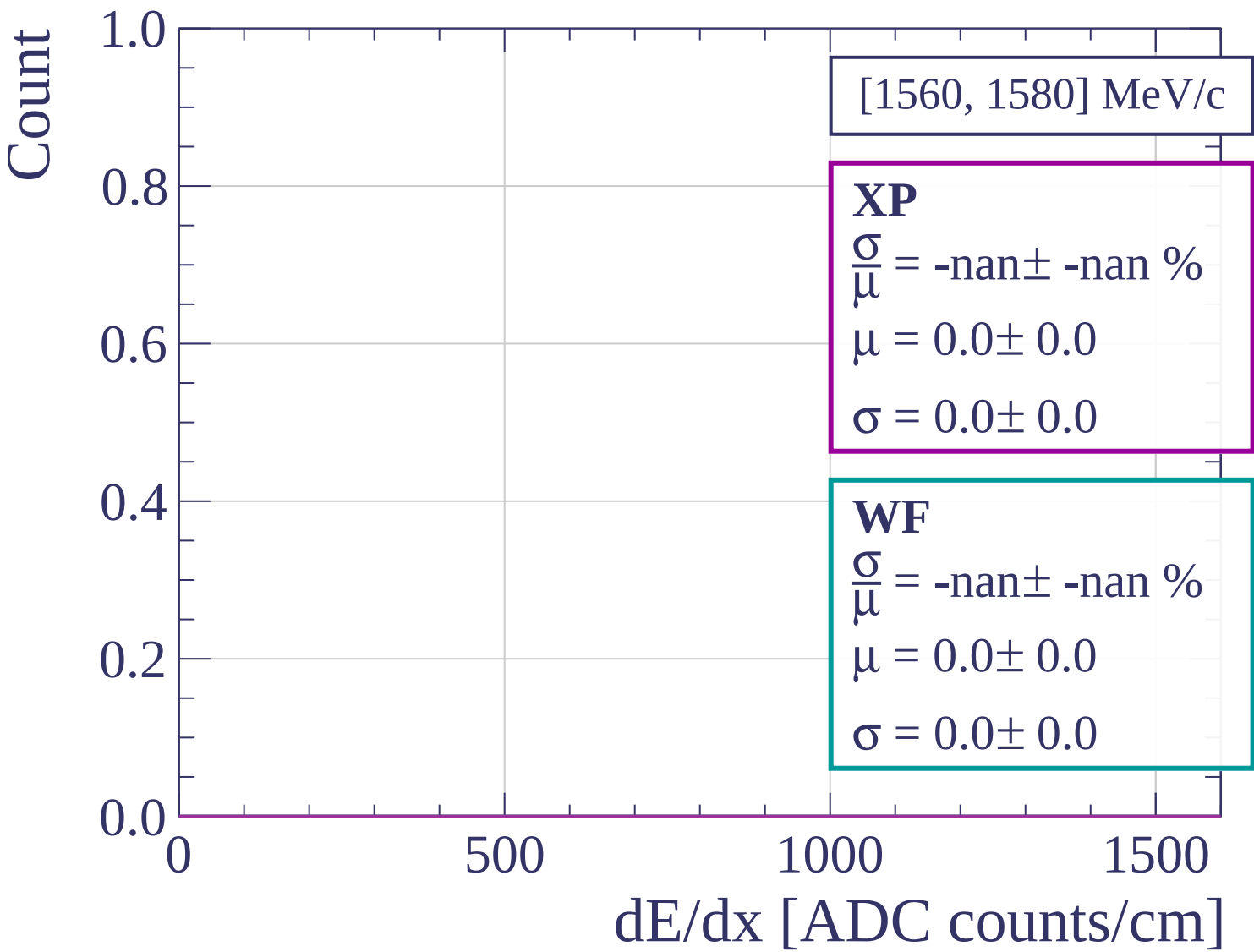


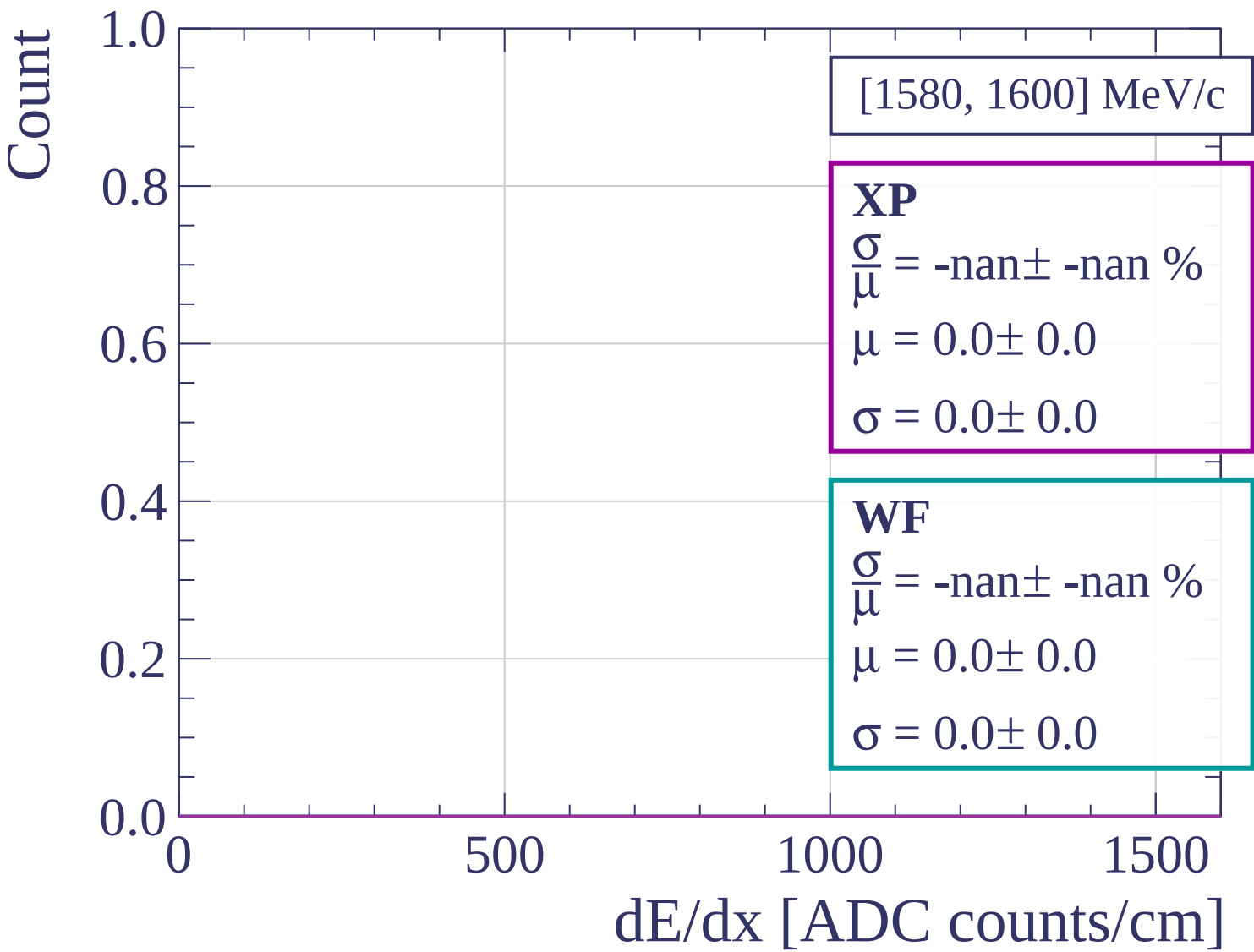


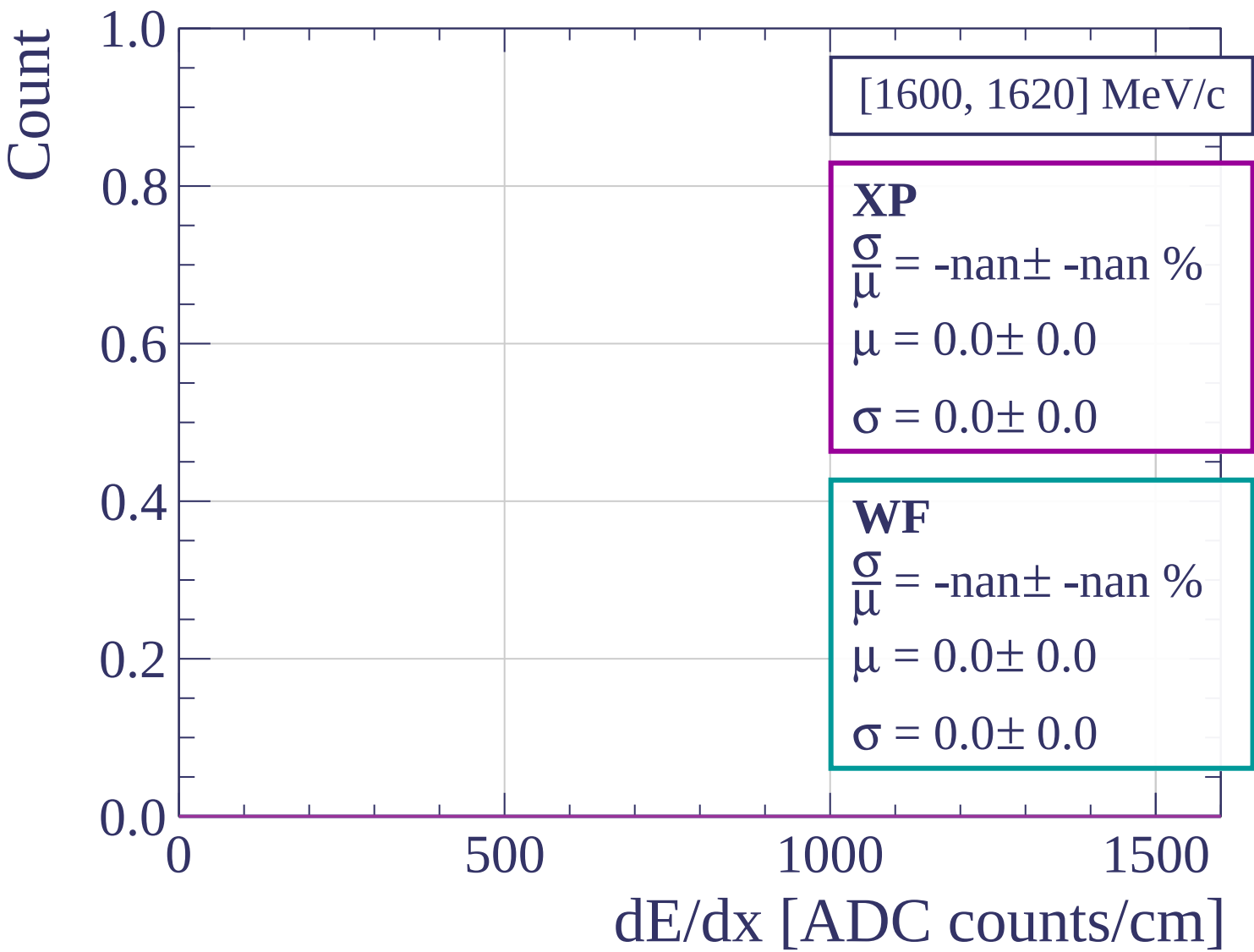


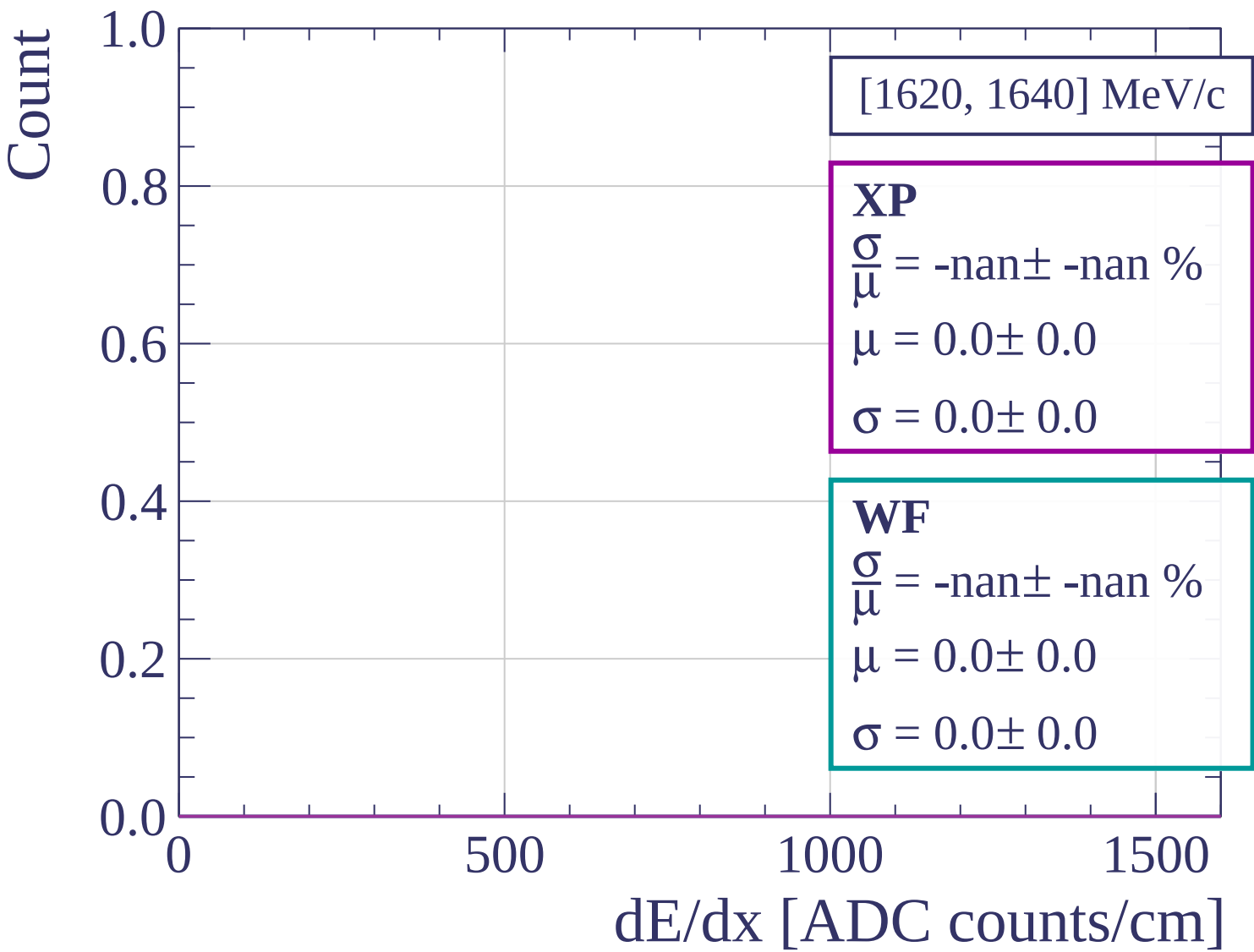


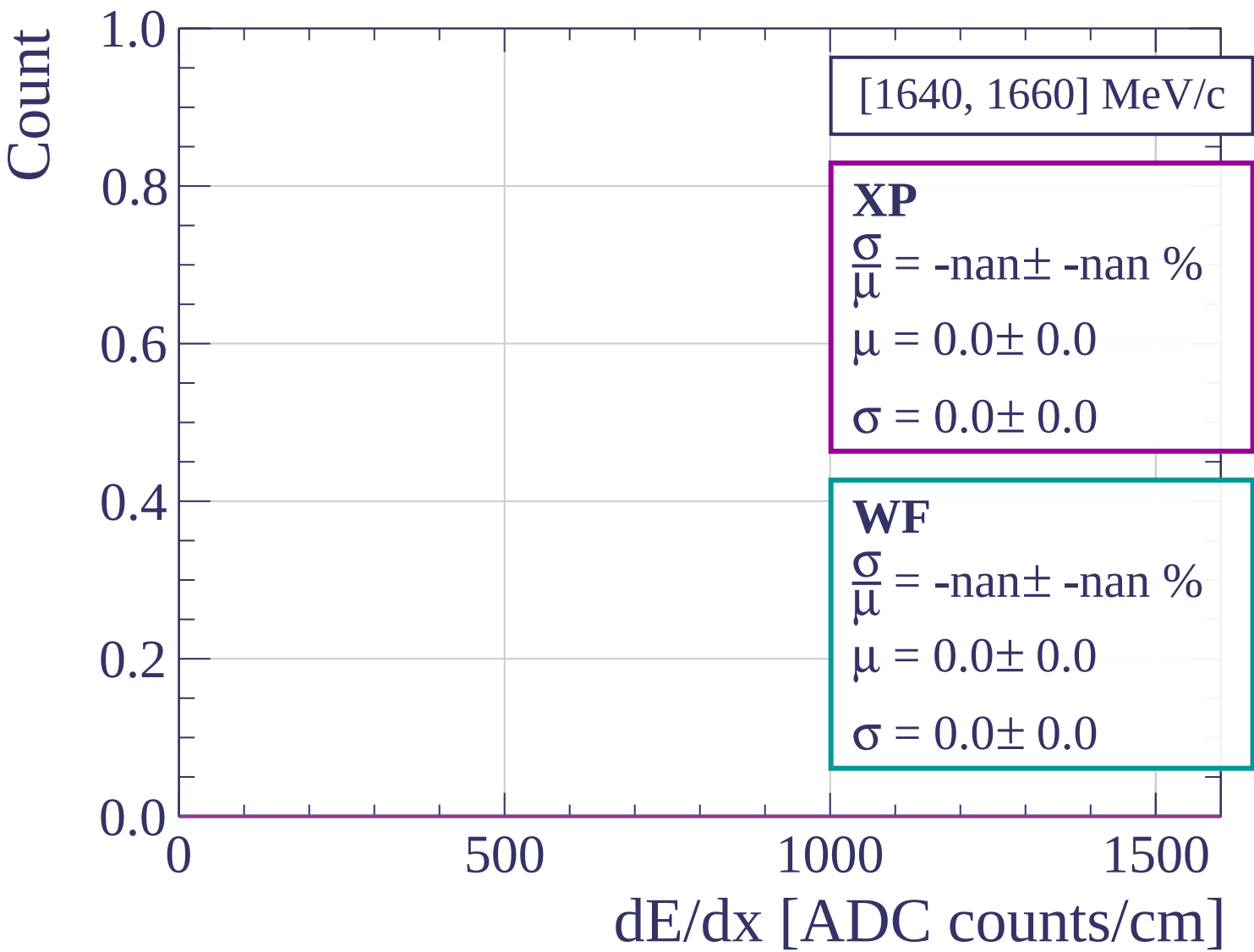


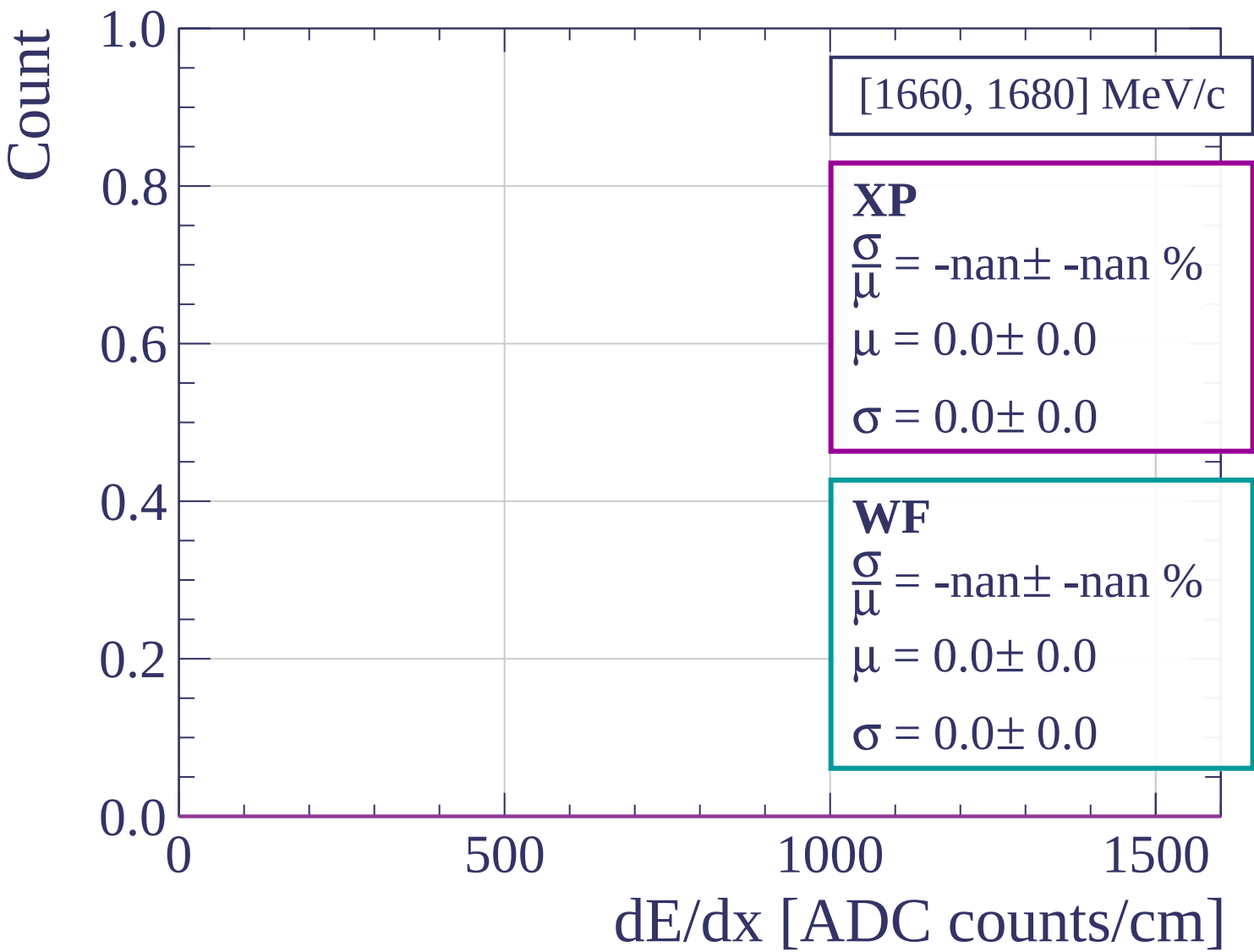


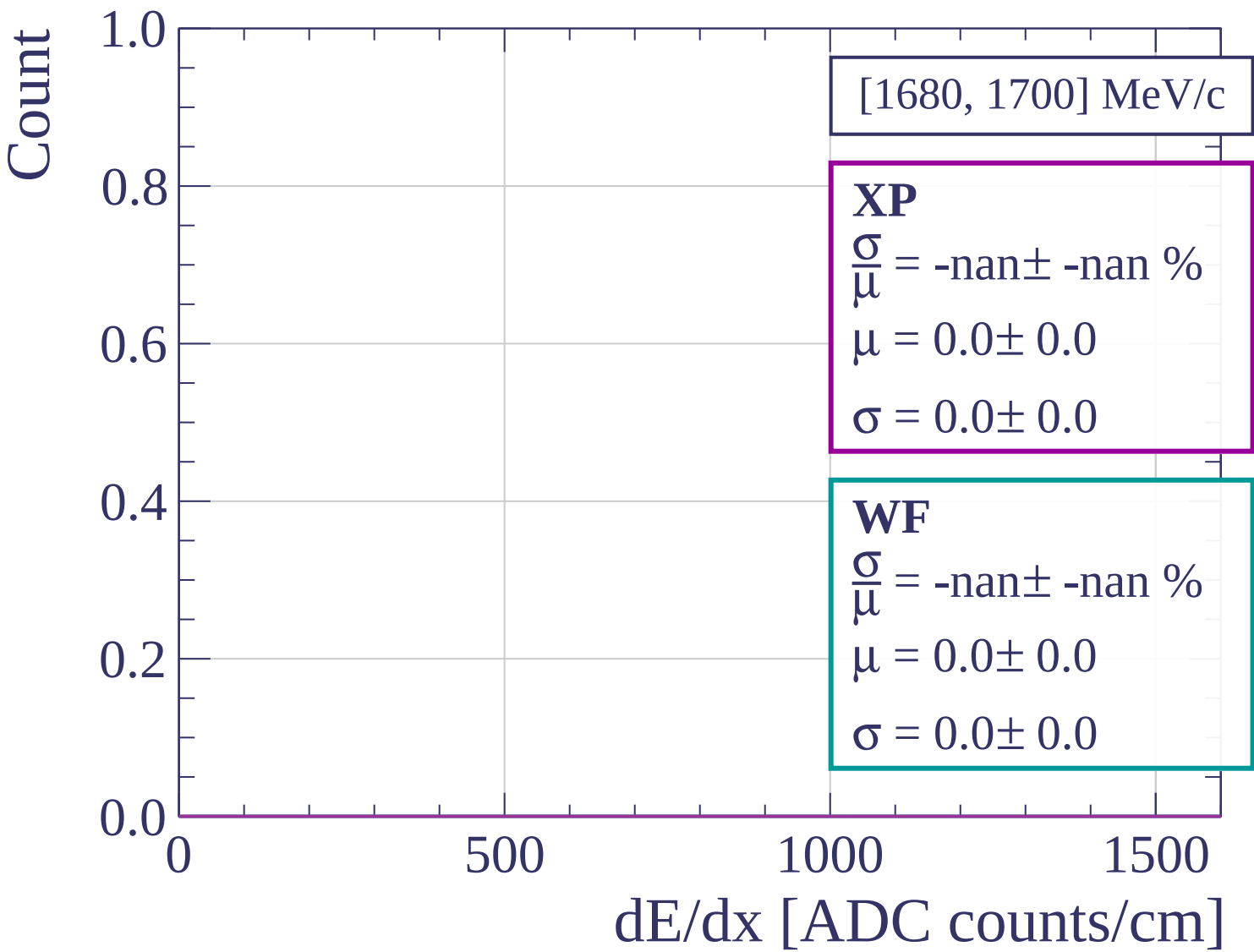


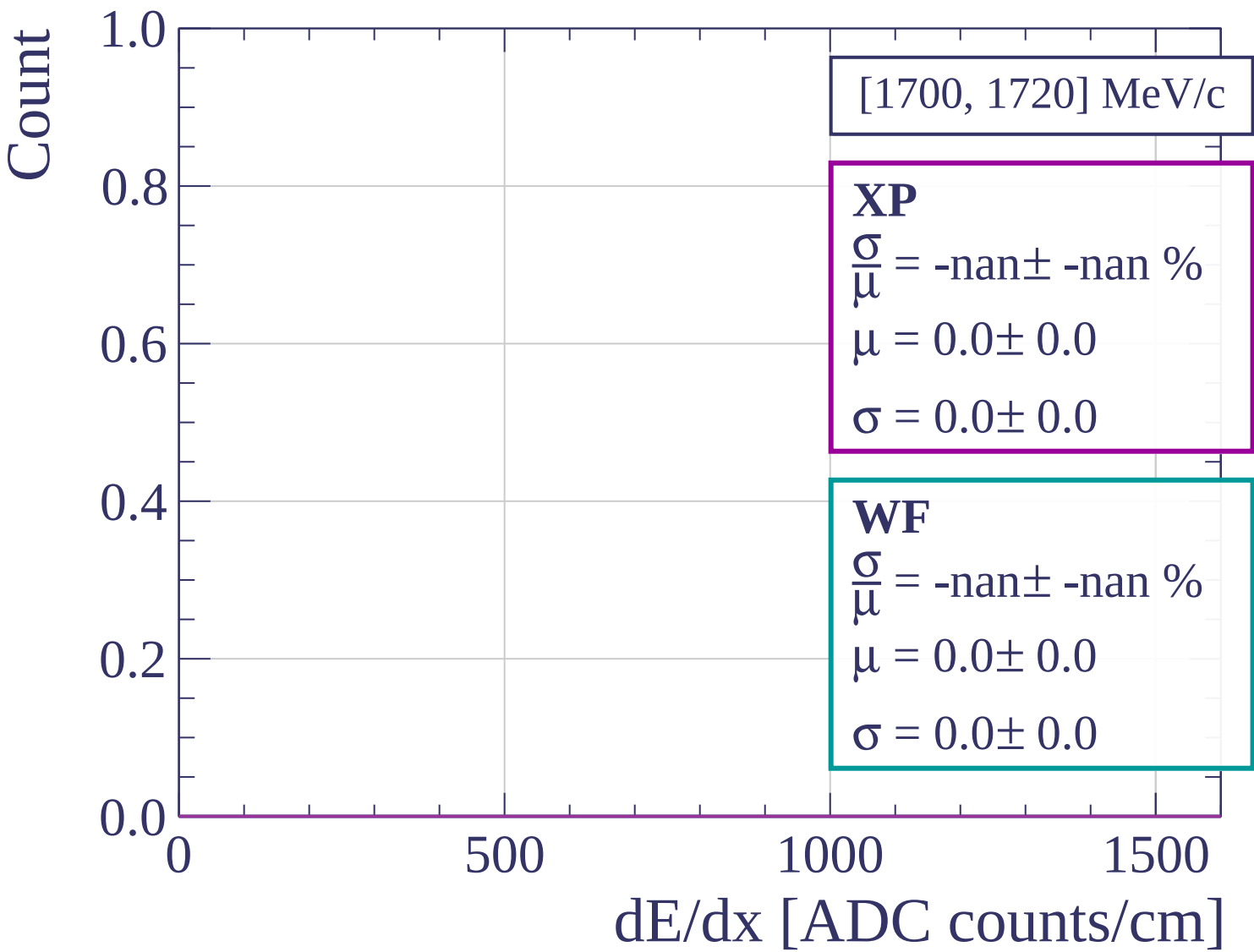


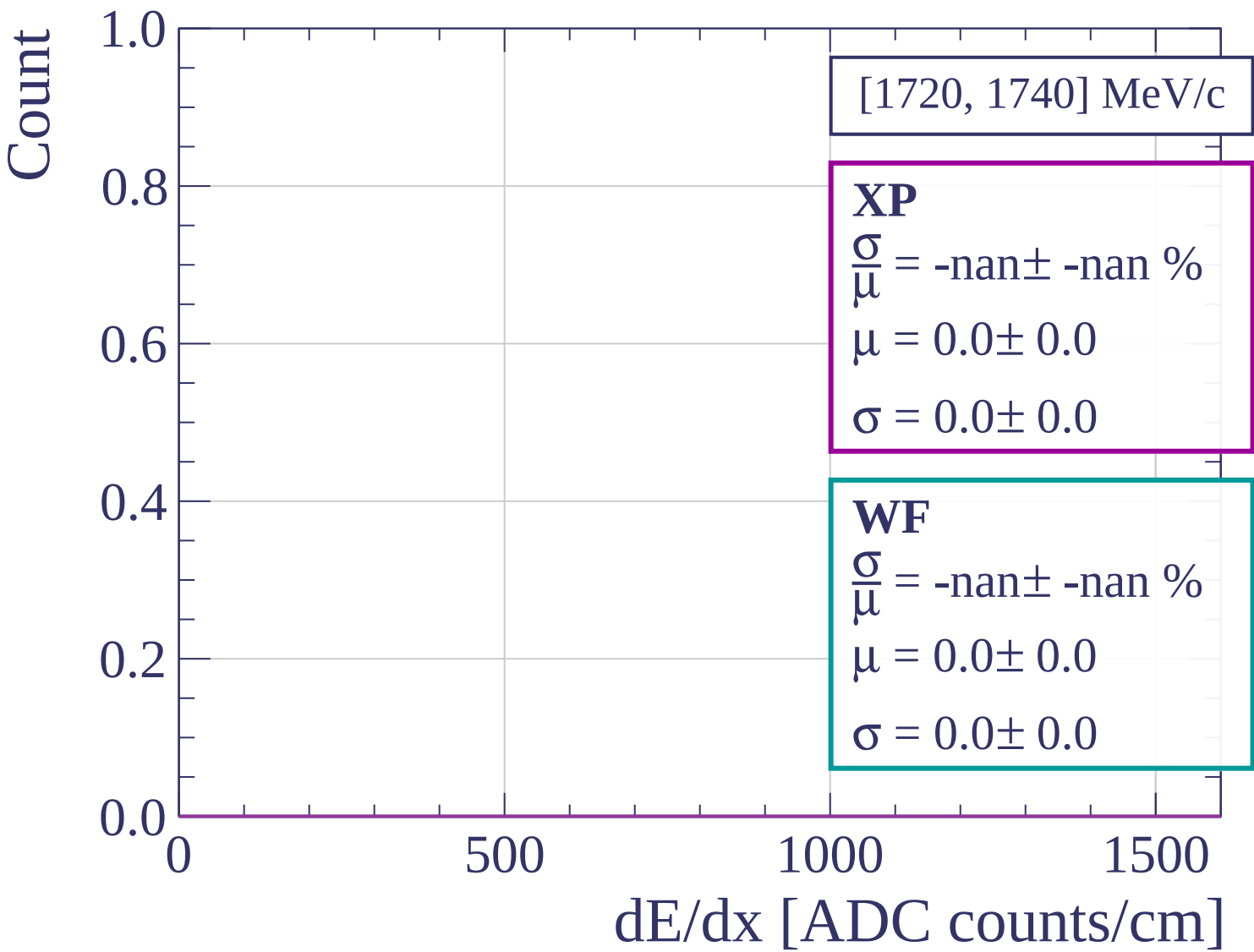


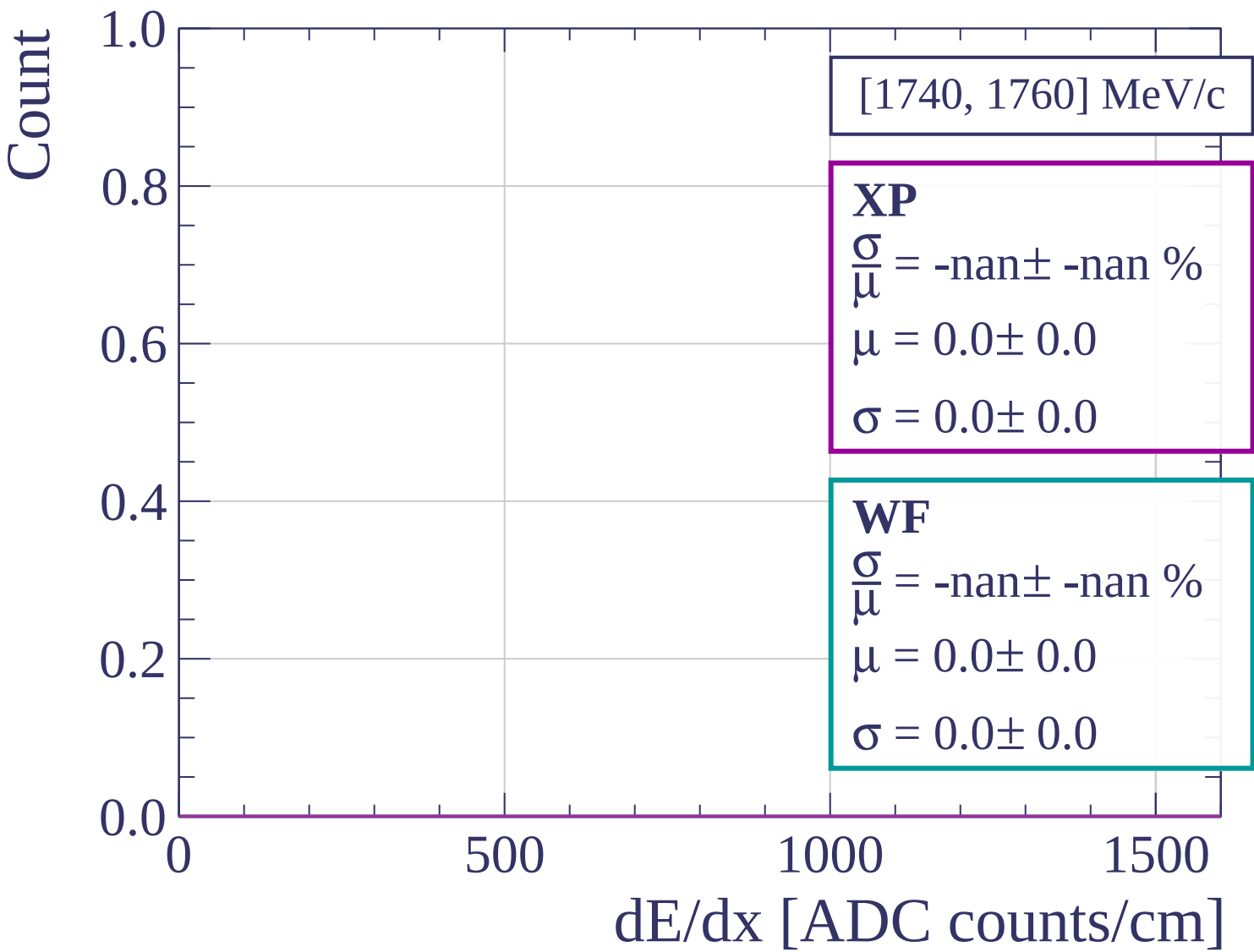


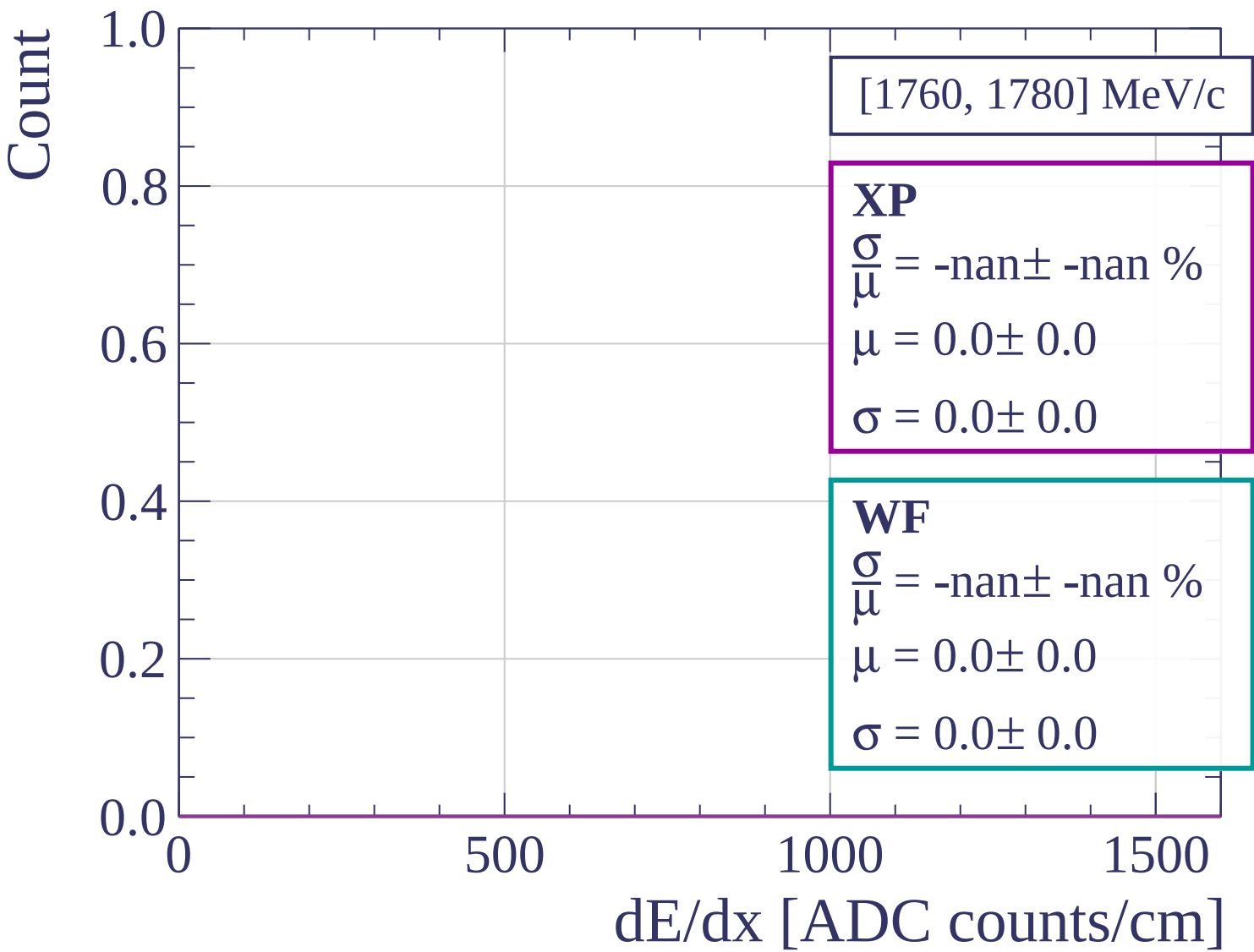


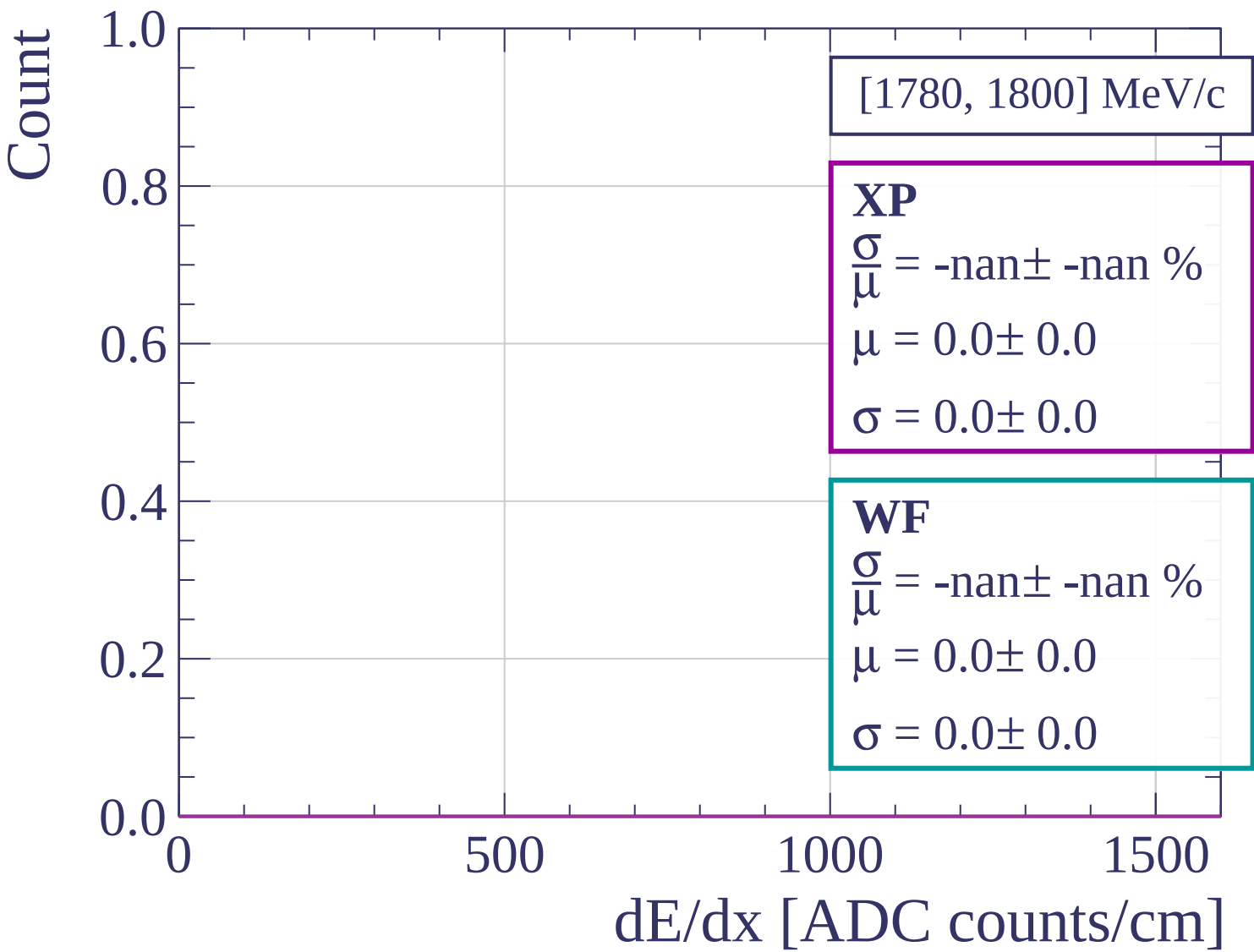


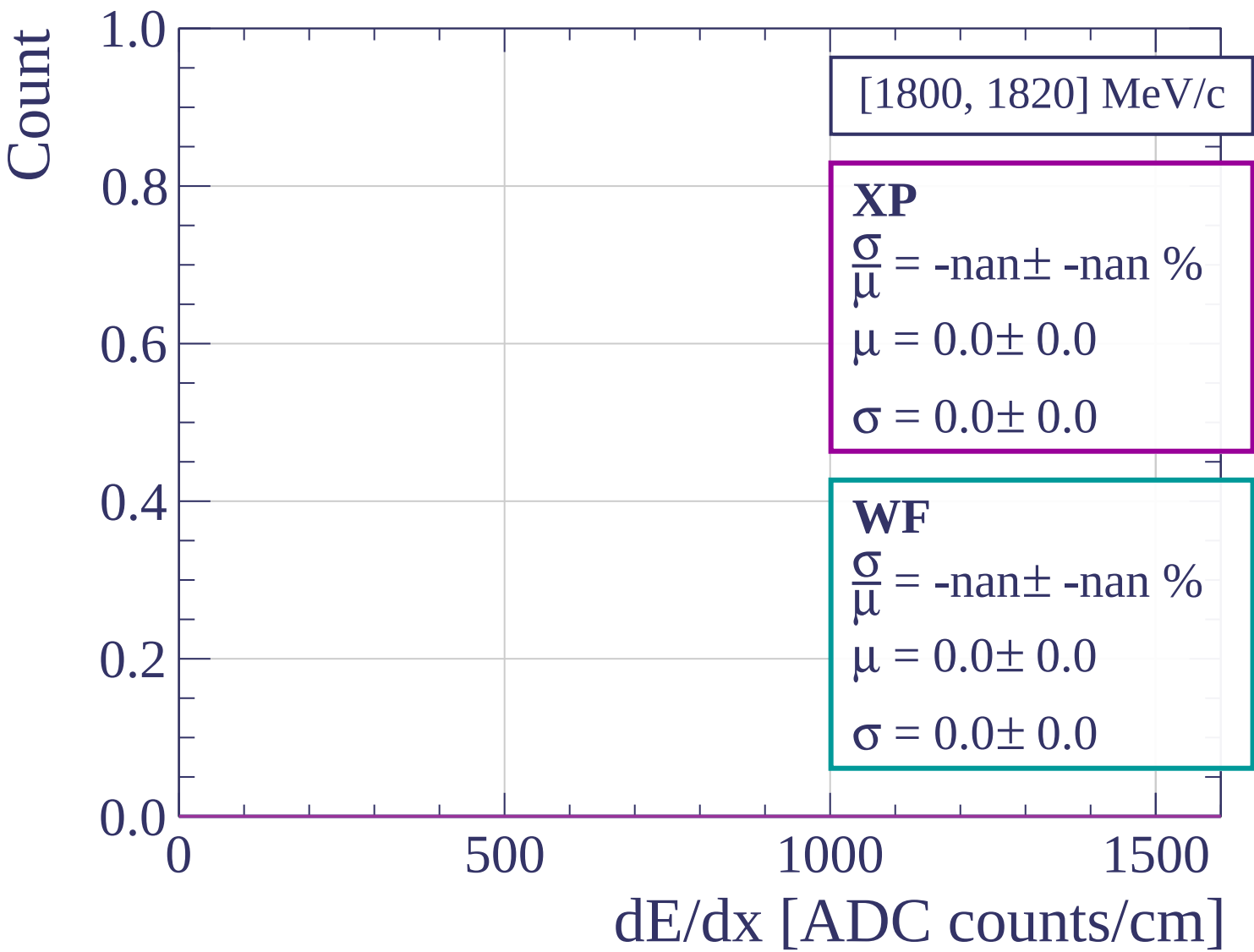


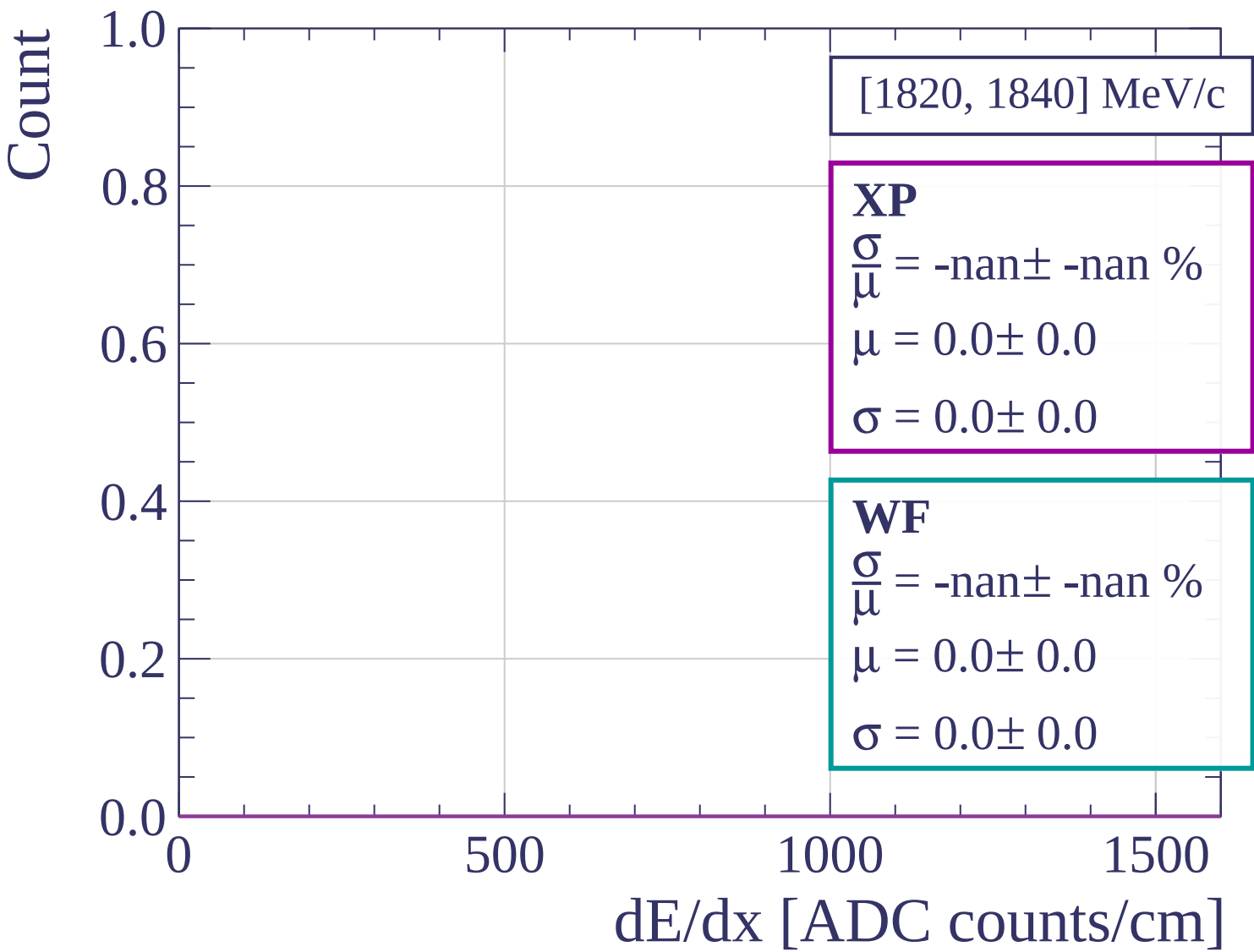


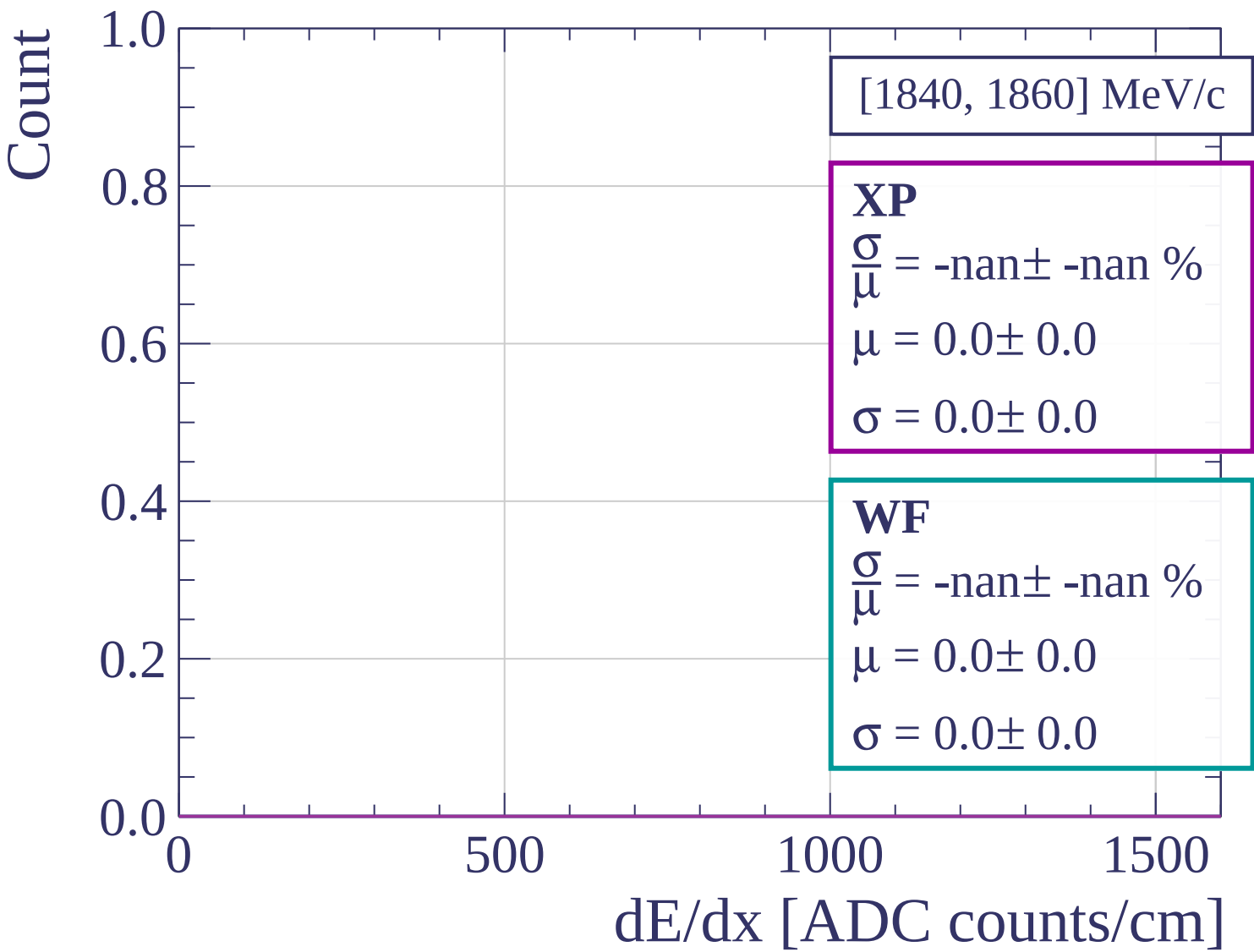


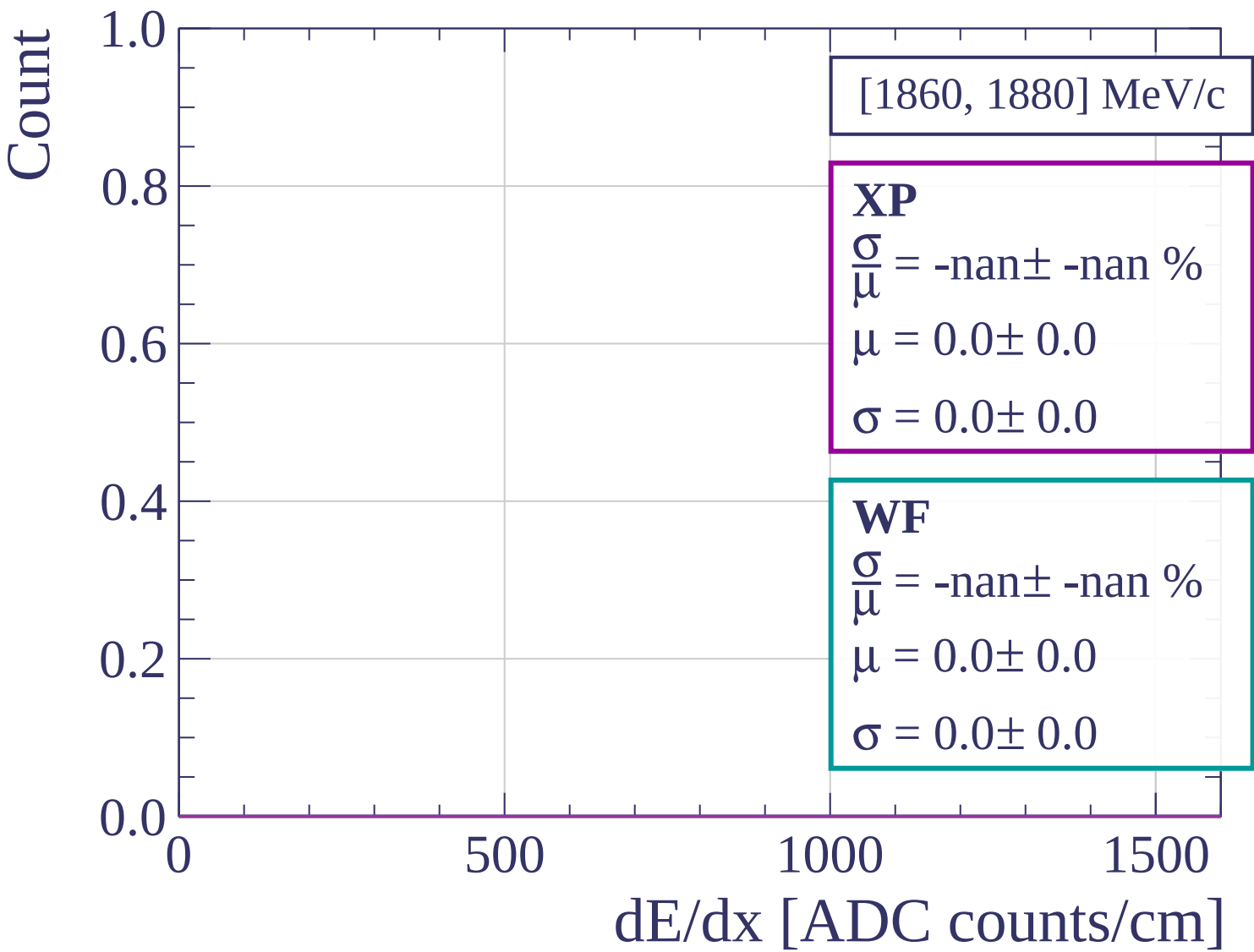


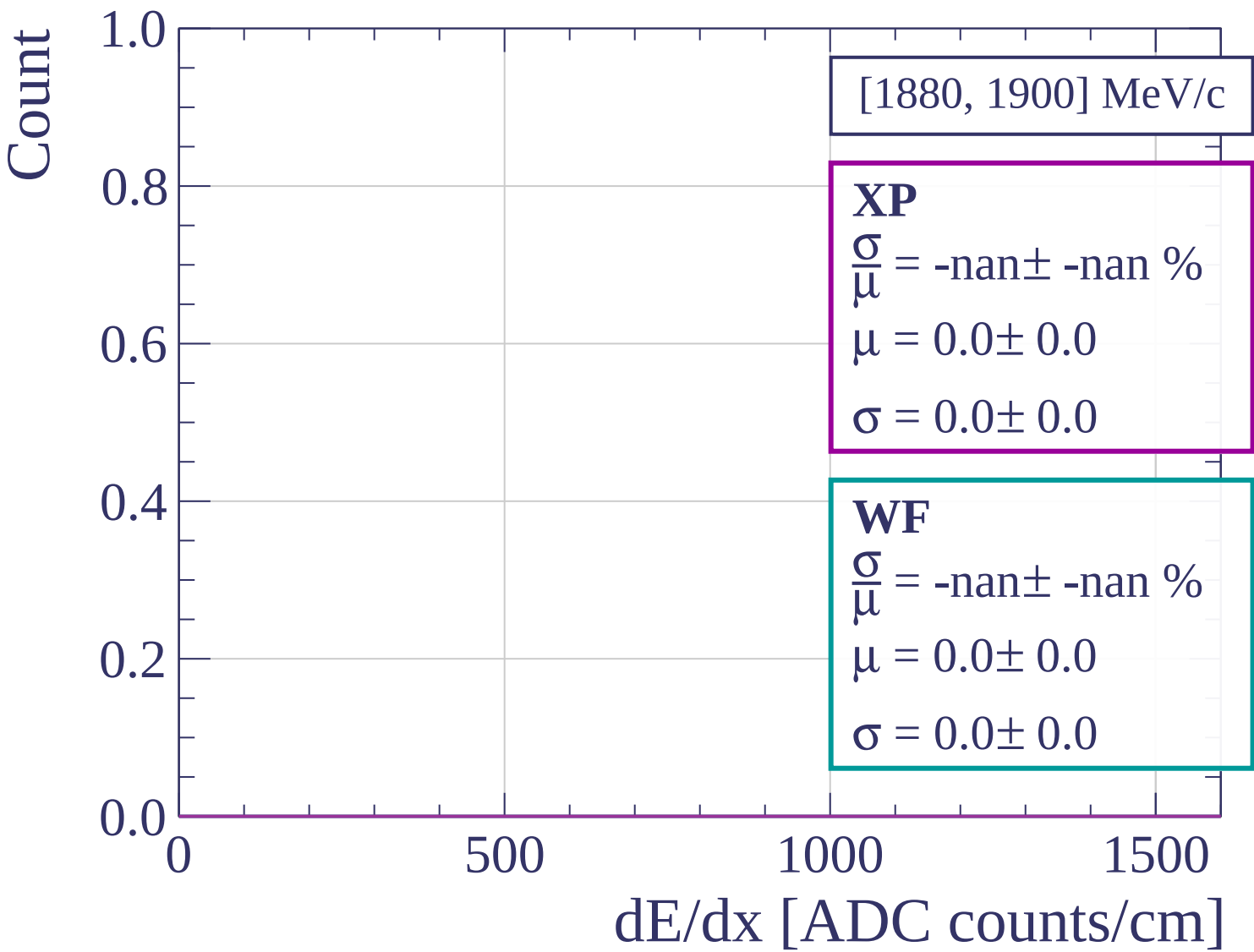


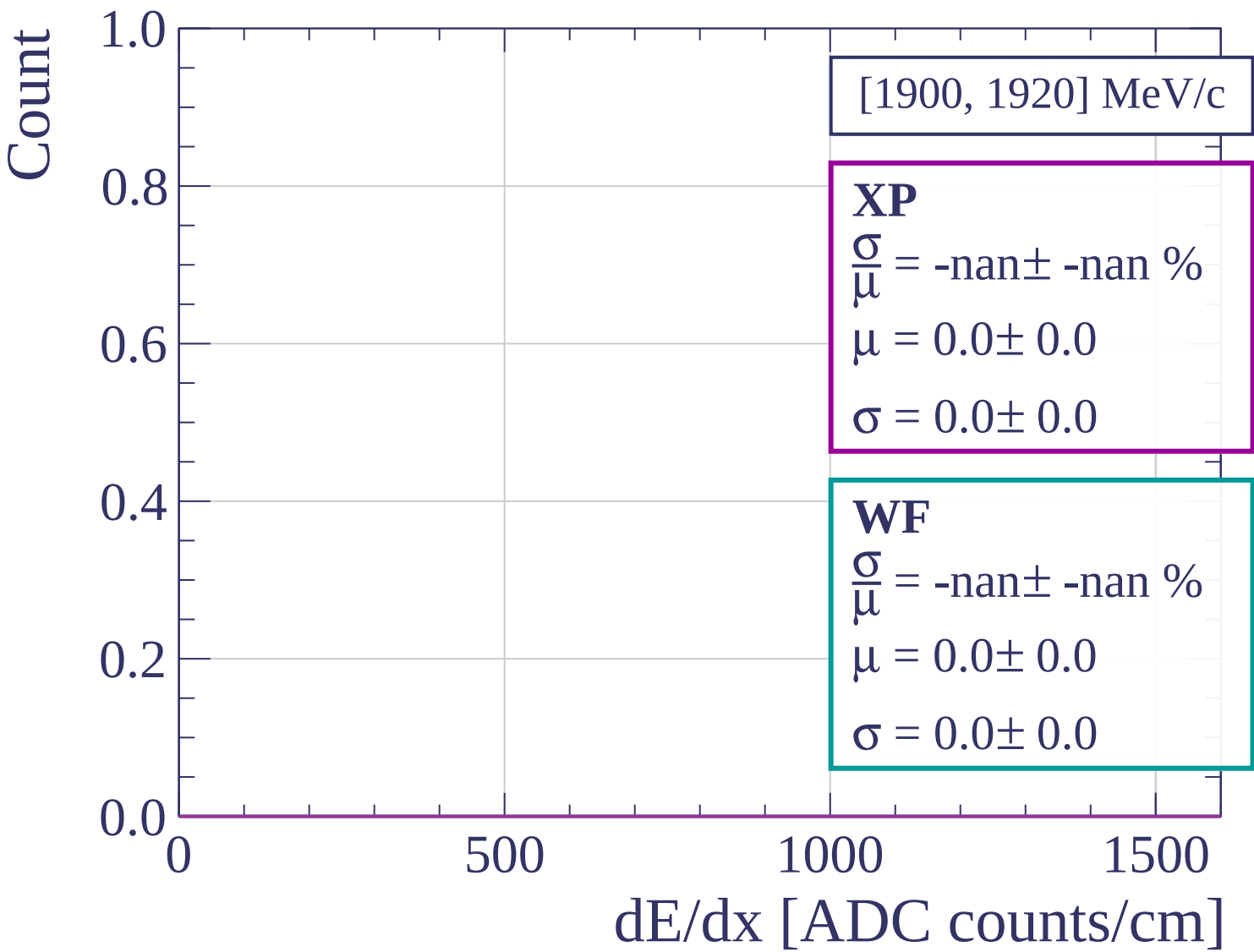


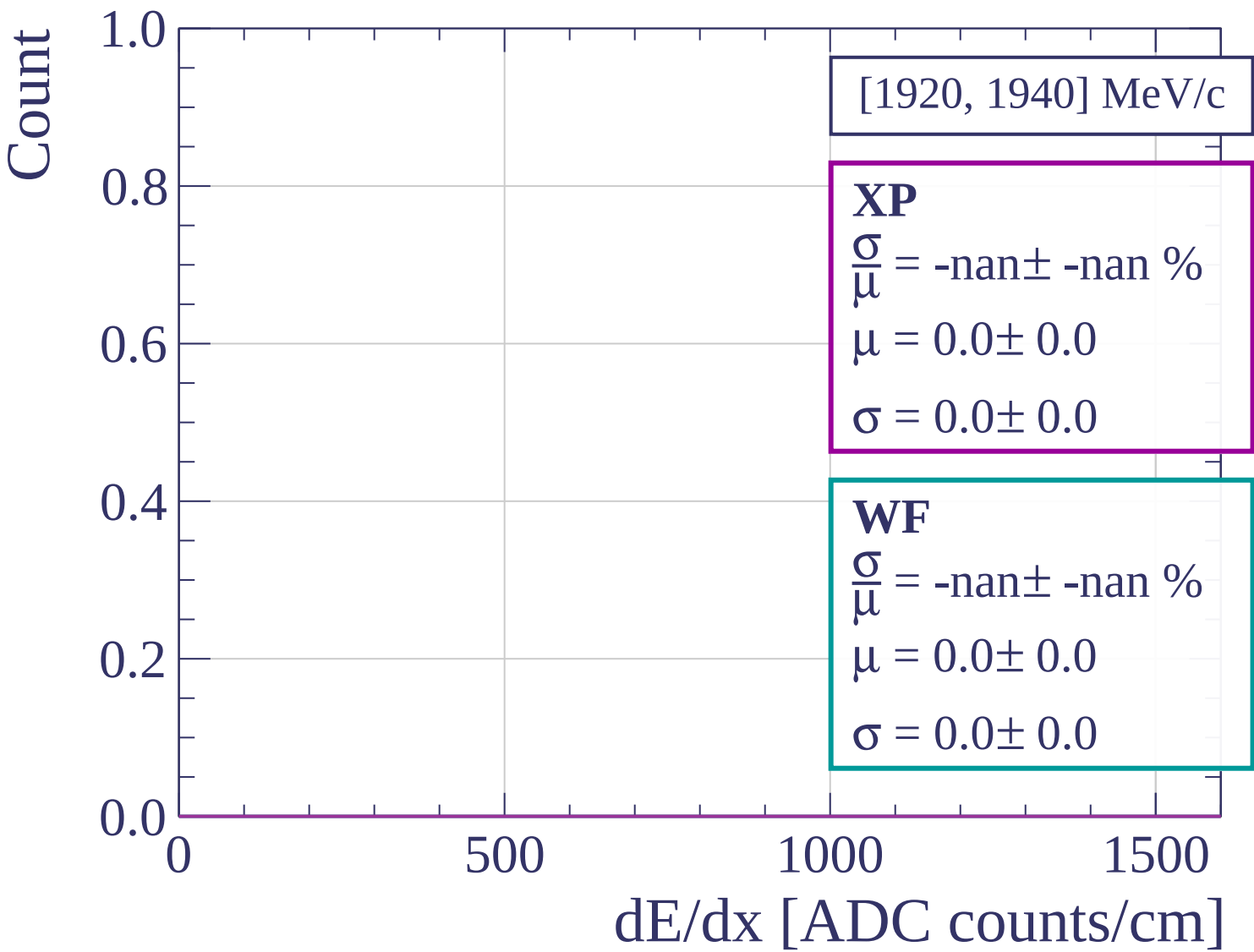


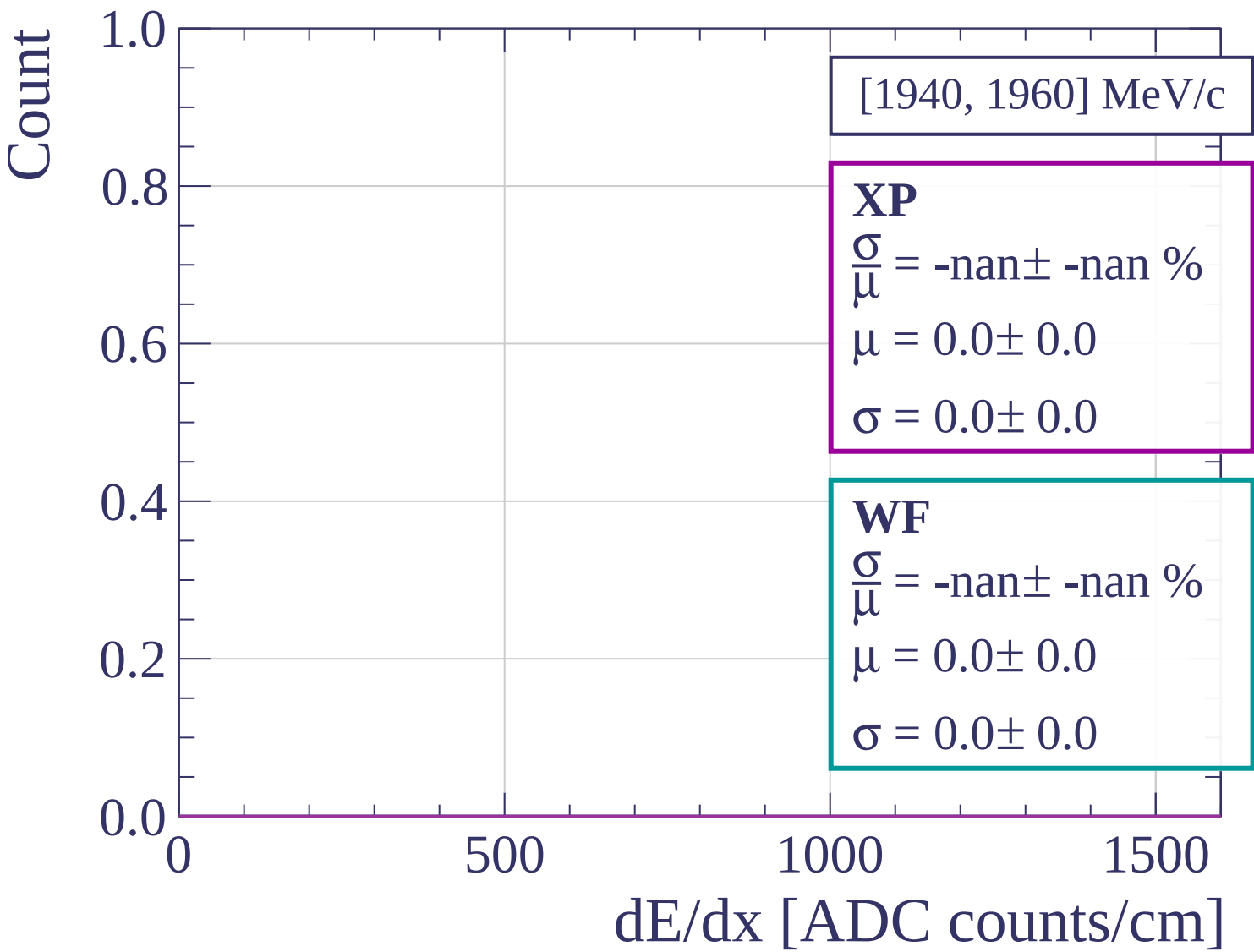


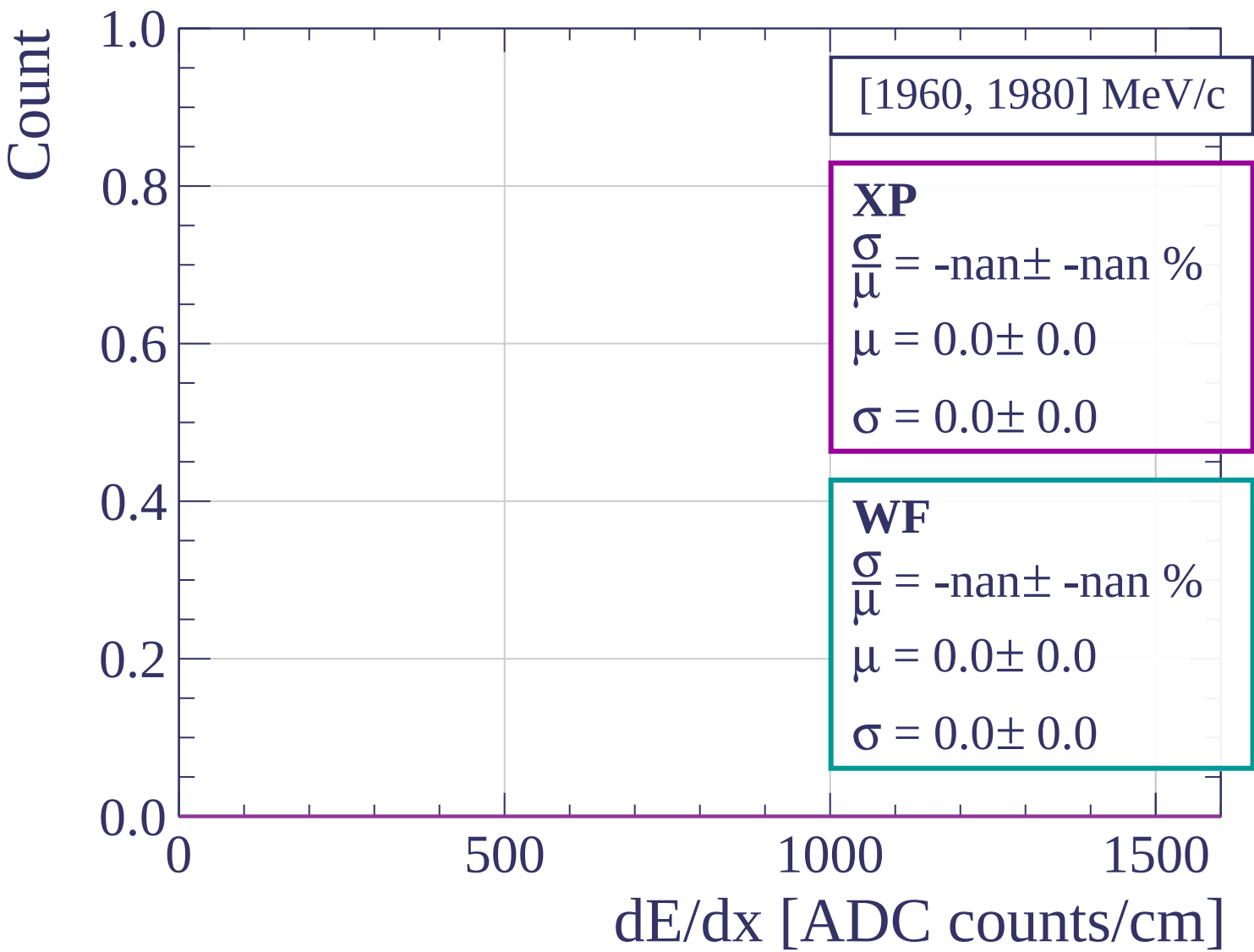


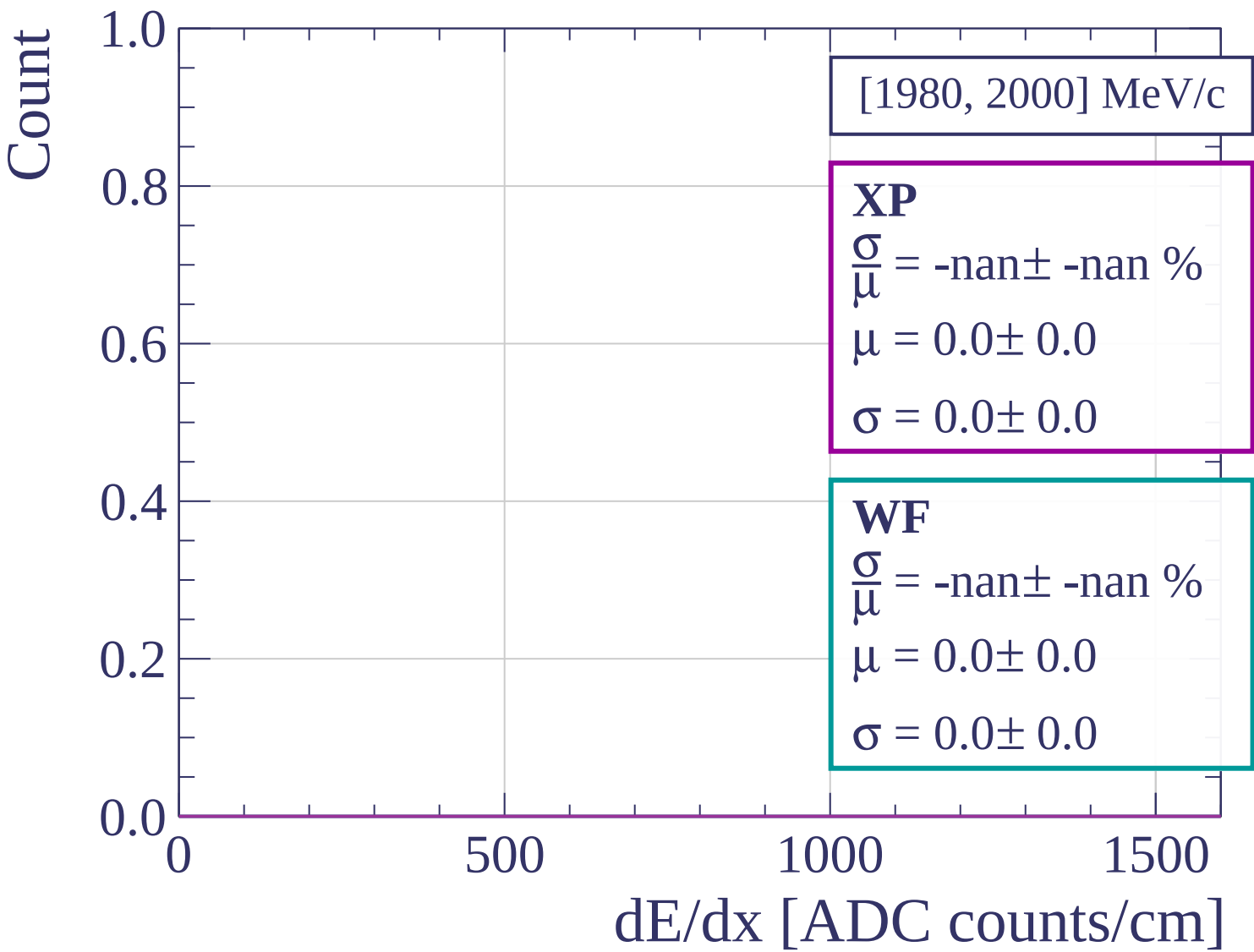


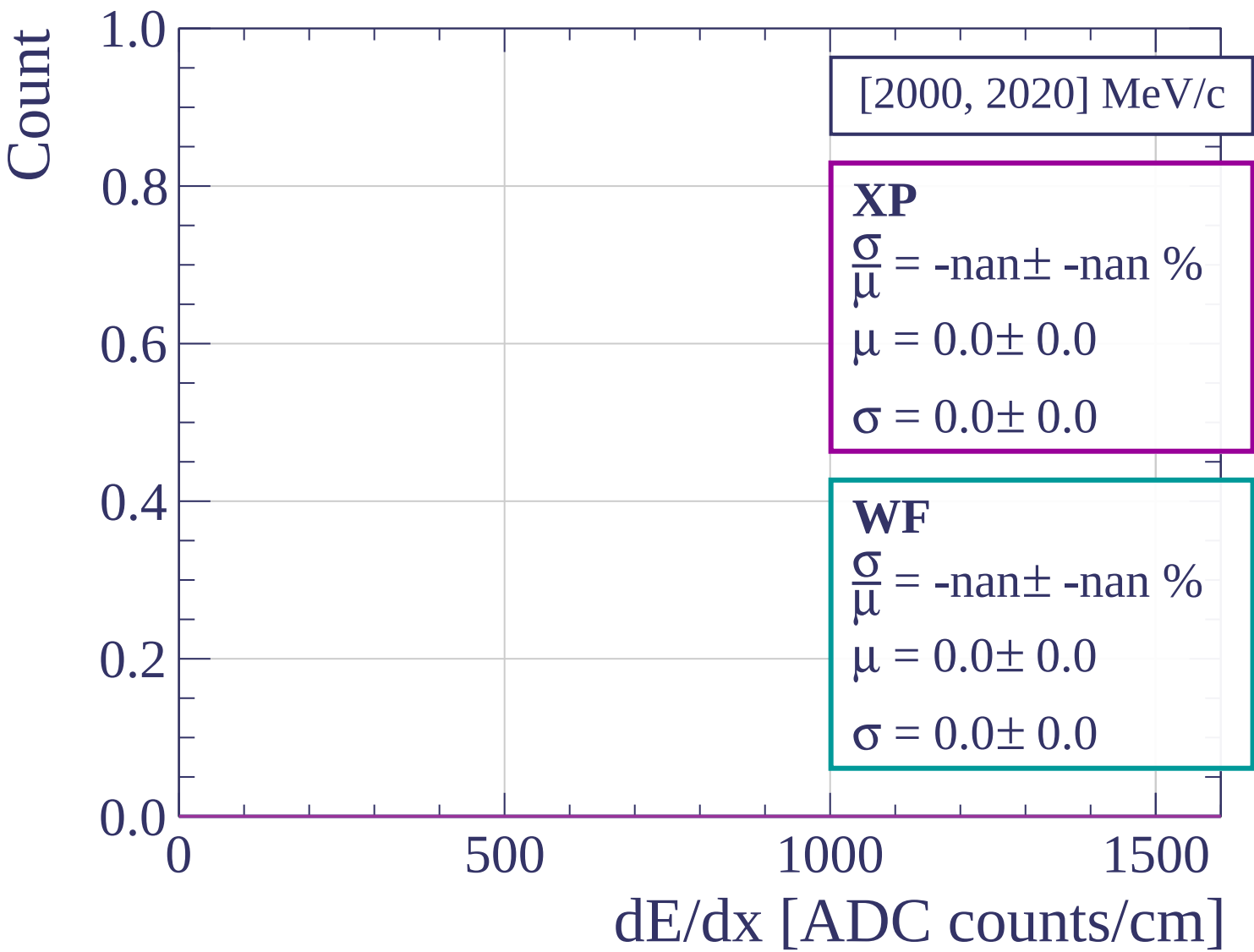


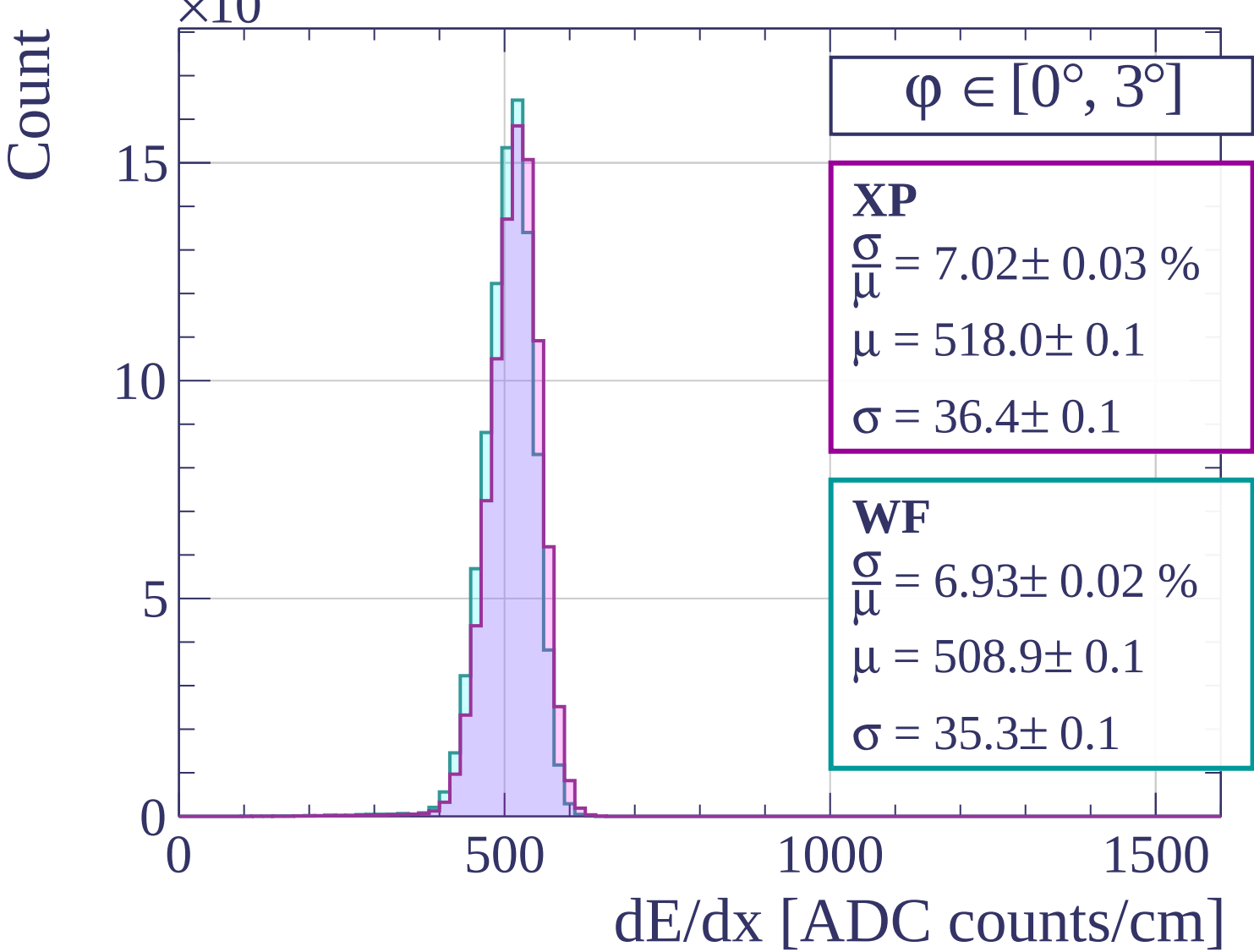


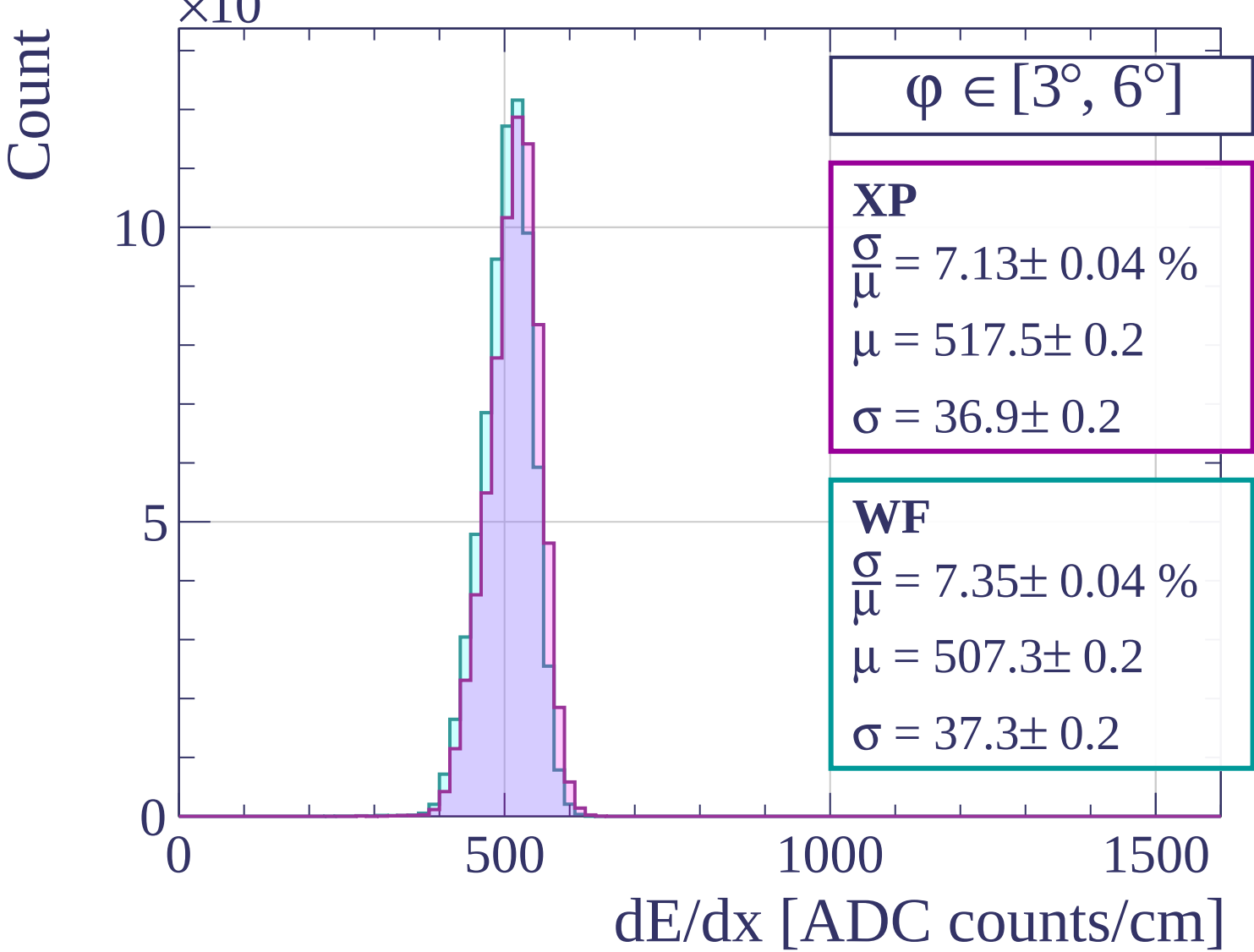


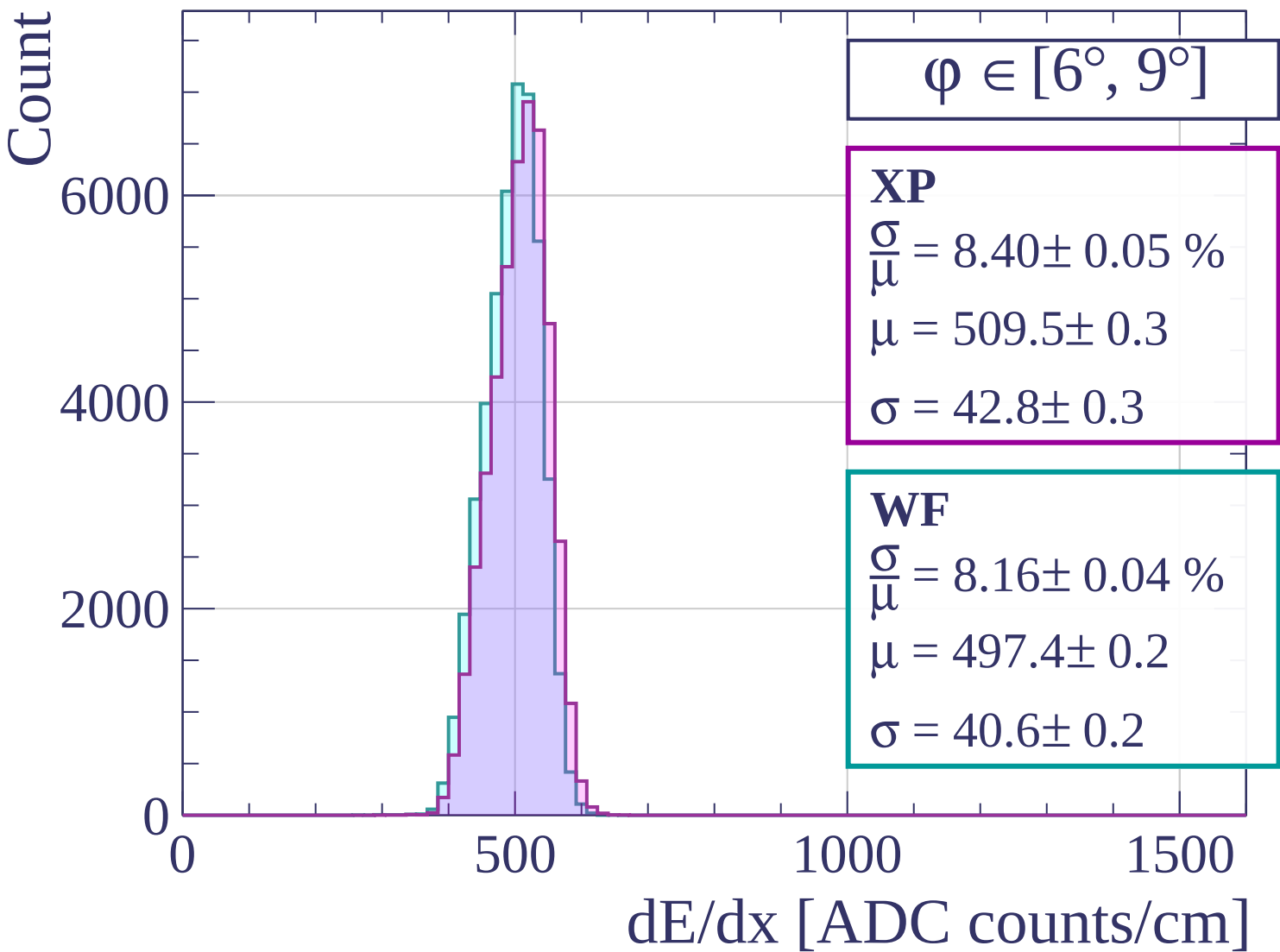


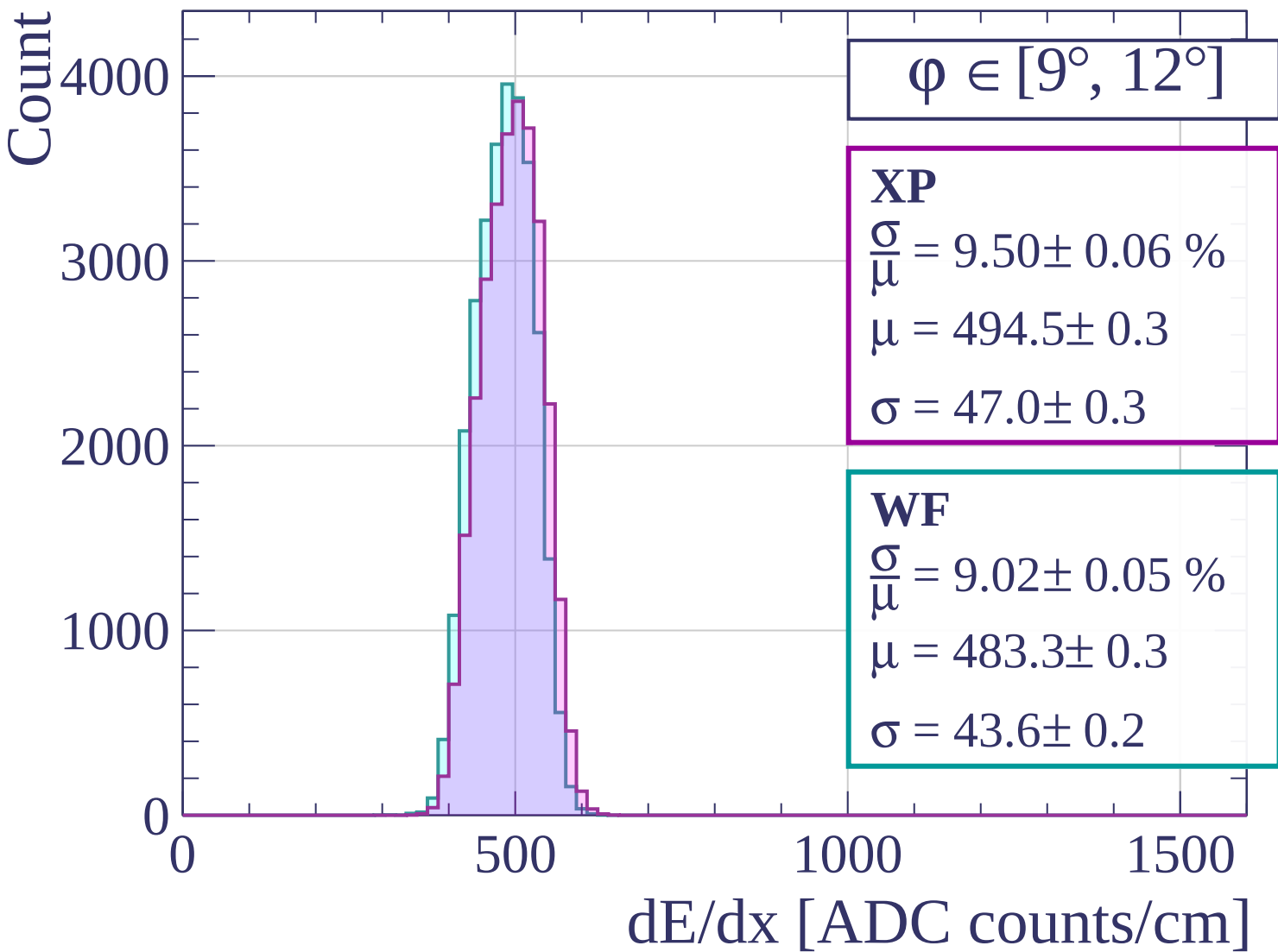


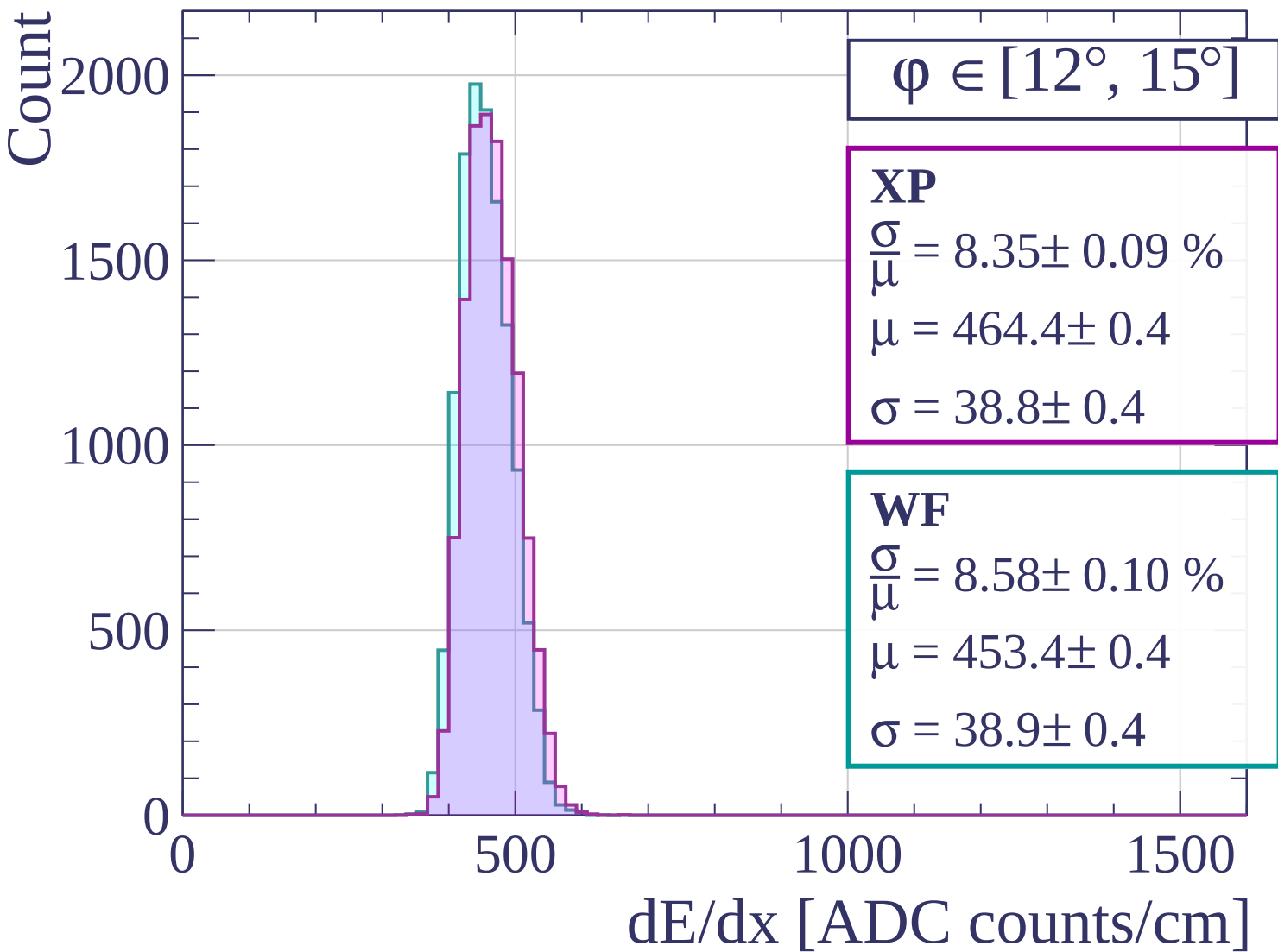


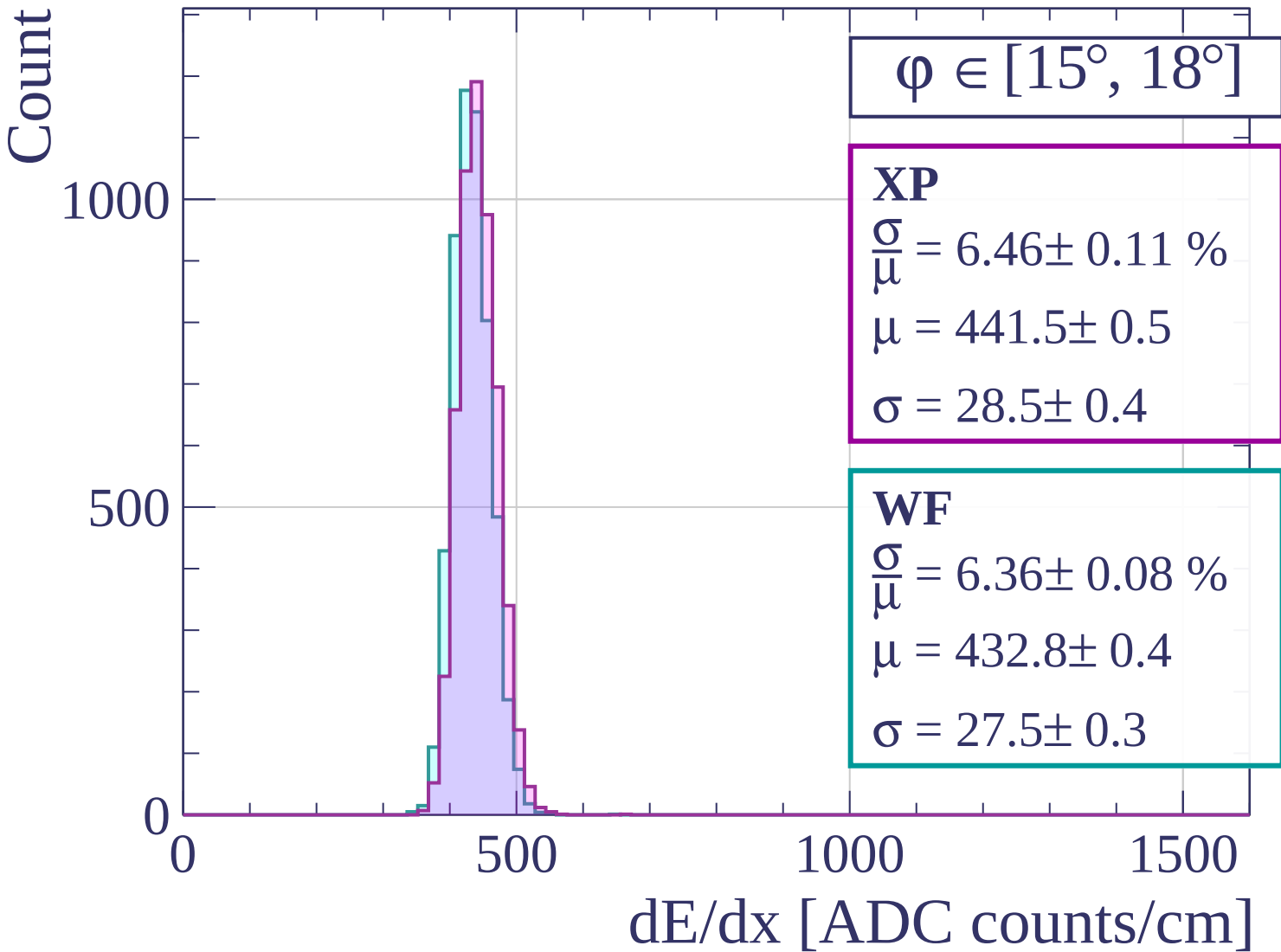


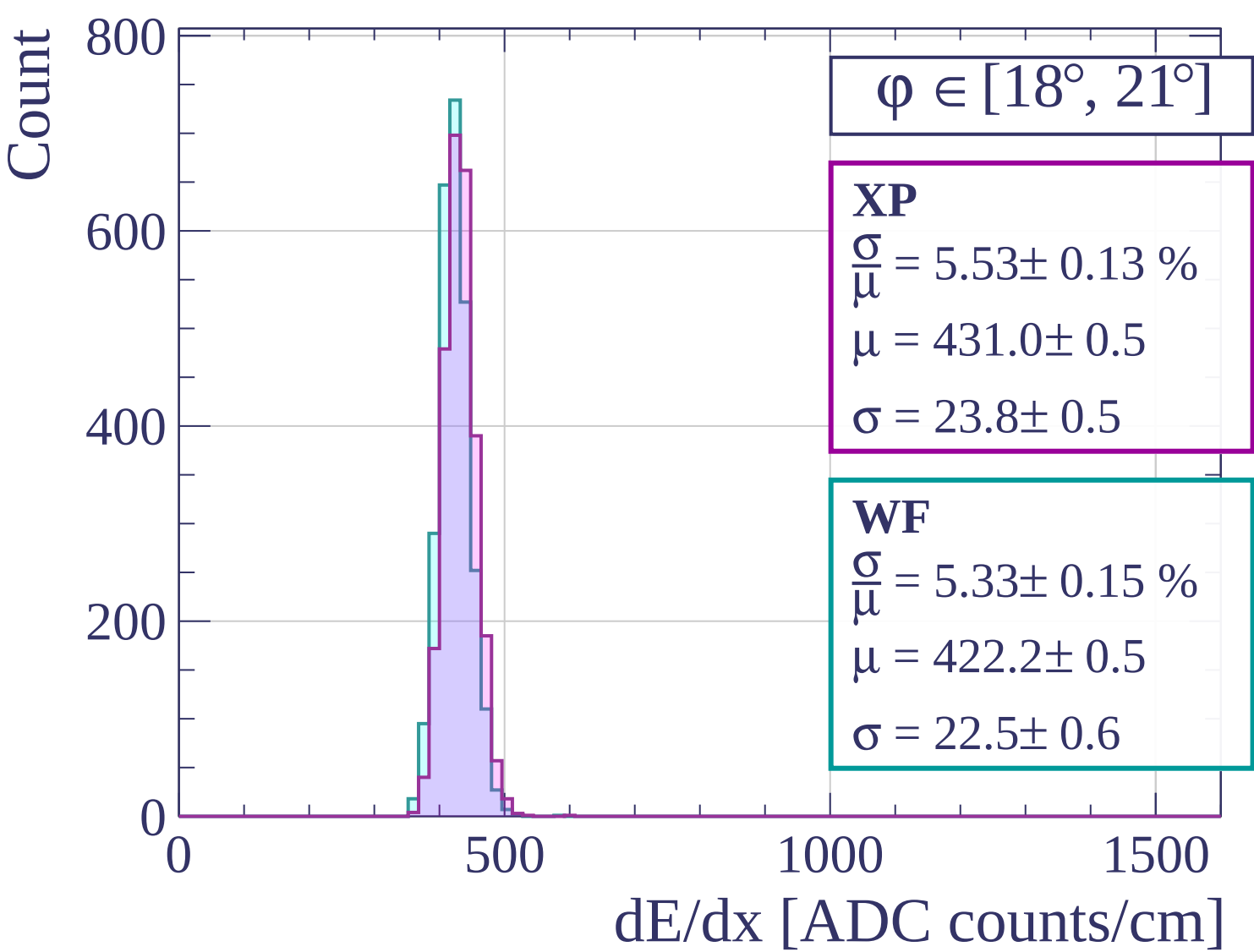












Count

$\varphi \in [21^\circ, 24^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.32 \pm 0.22 \%$$

$$\mu = 425.6 \pm 0.8$$

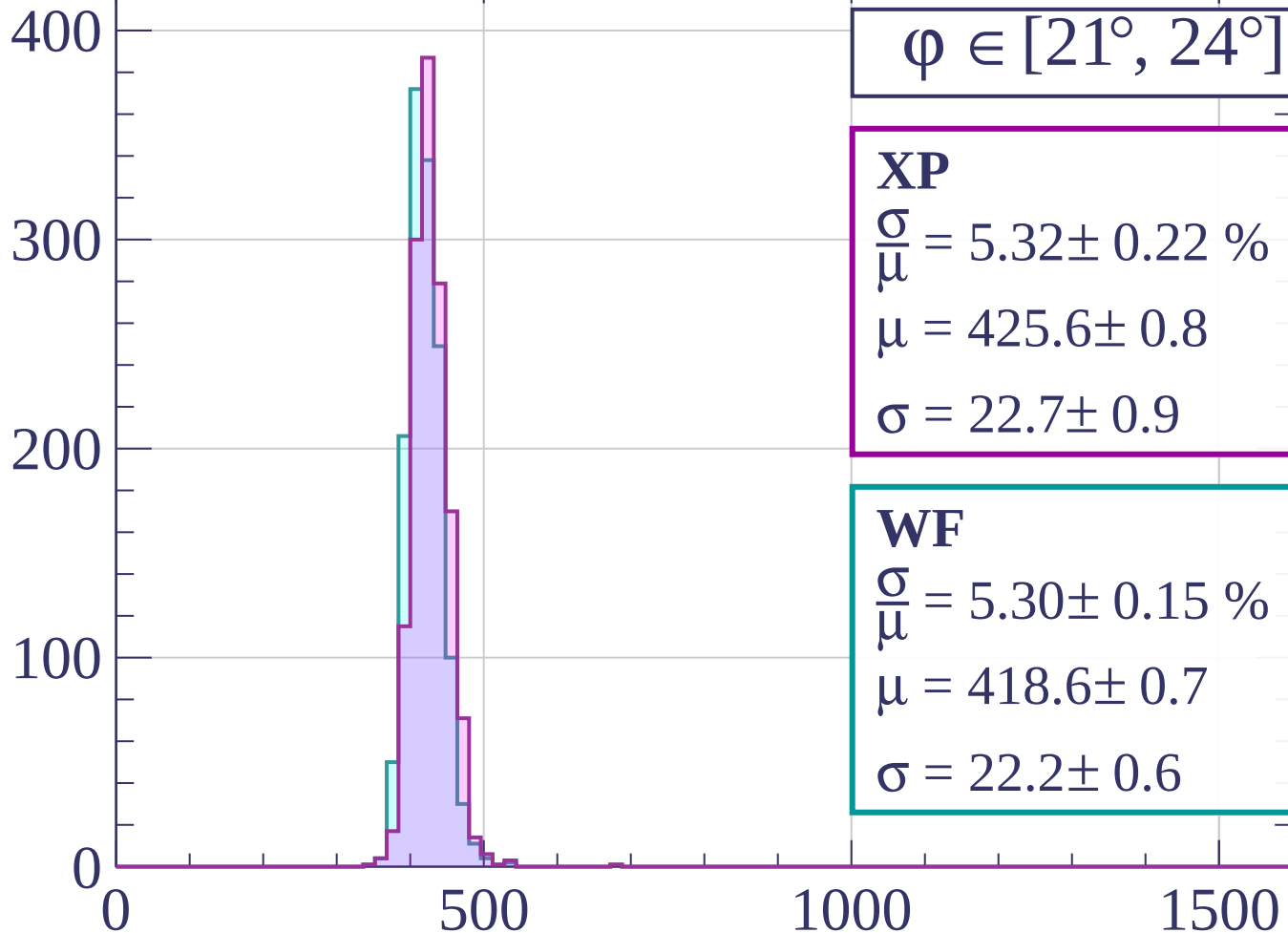
$$\sigma = 22.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.30 \pm 0.15 \%$$

$$\mu = 418.6 \pm 0.7$$

$$\sigma = 22.2 \pm 0.6$$



dE/dx [ADC counts/cm]

Count

$\phi \in [24^\circ, 27^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.14 \pm 0.31 \%$$

$$\mu = 423.4 \pm 1.1$$

$$\sigma = 21.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 4.69 \pm 0.34 \%$$

$$\mu = 416.9 \pm 1.1$$

$$\sigma = 19.6 \pm 1.4$$

150

100

50

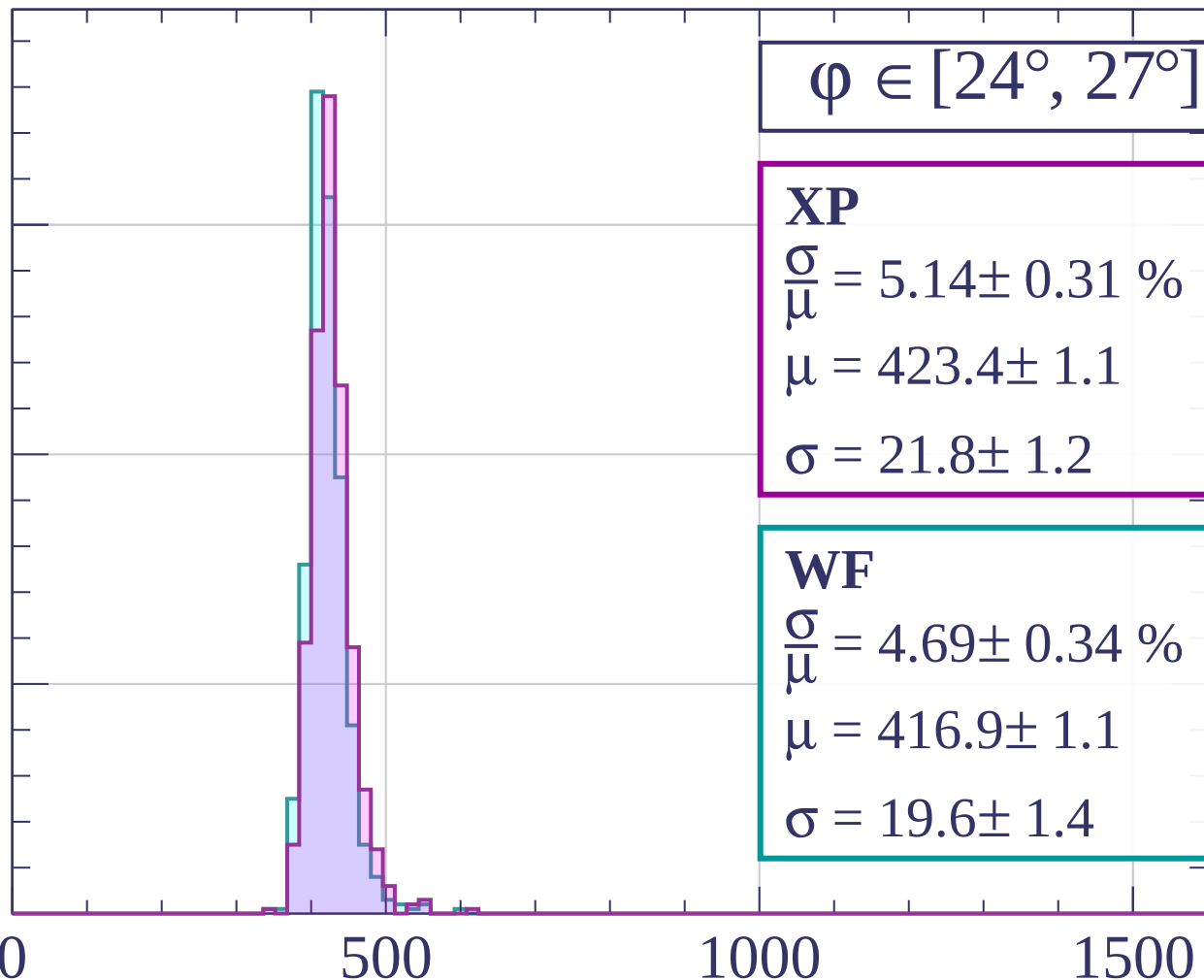
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

$\phi \in [27^\circ, 30^\circ]$

XP

$$\frac{\sigma}{\mu} = 5.71 \pm 0.38 \%$$

$$\mu = 427.6 \pm 1.6$$

$$\sigma = 24.4 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 5.96 \pm 0.35 \%$$

$$\mu = 422.3 \pm 1.6$$

$$\sigma = 25.2 \pm 1.4$$

80

60

40

20

0

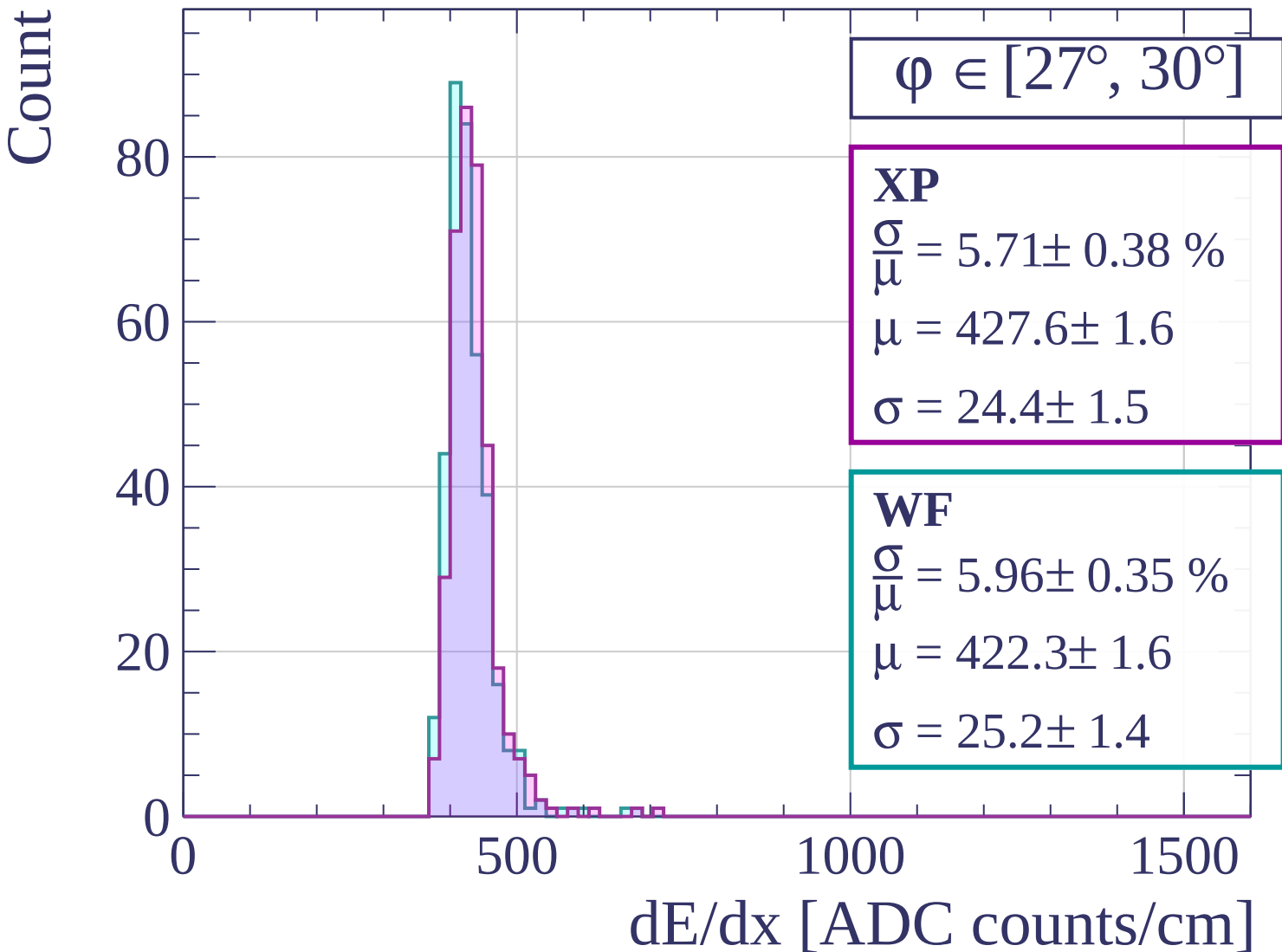
0

500

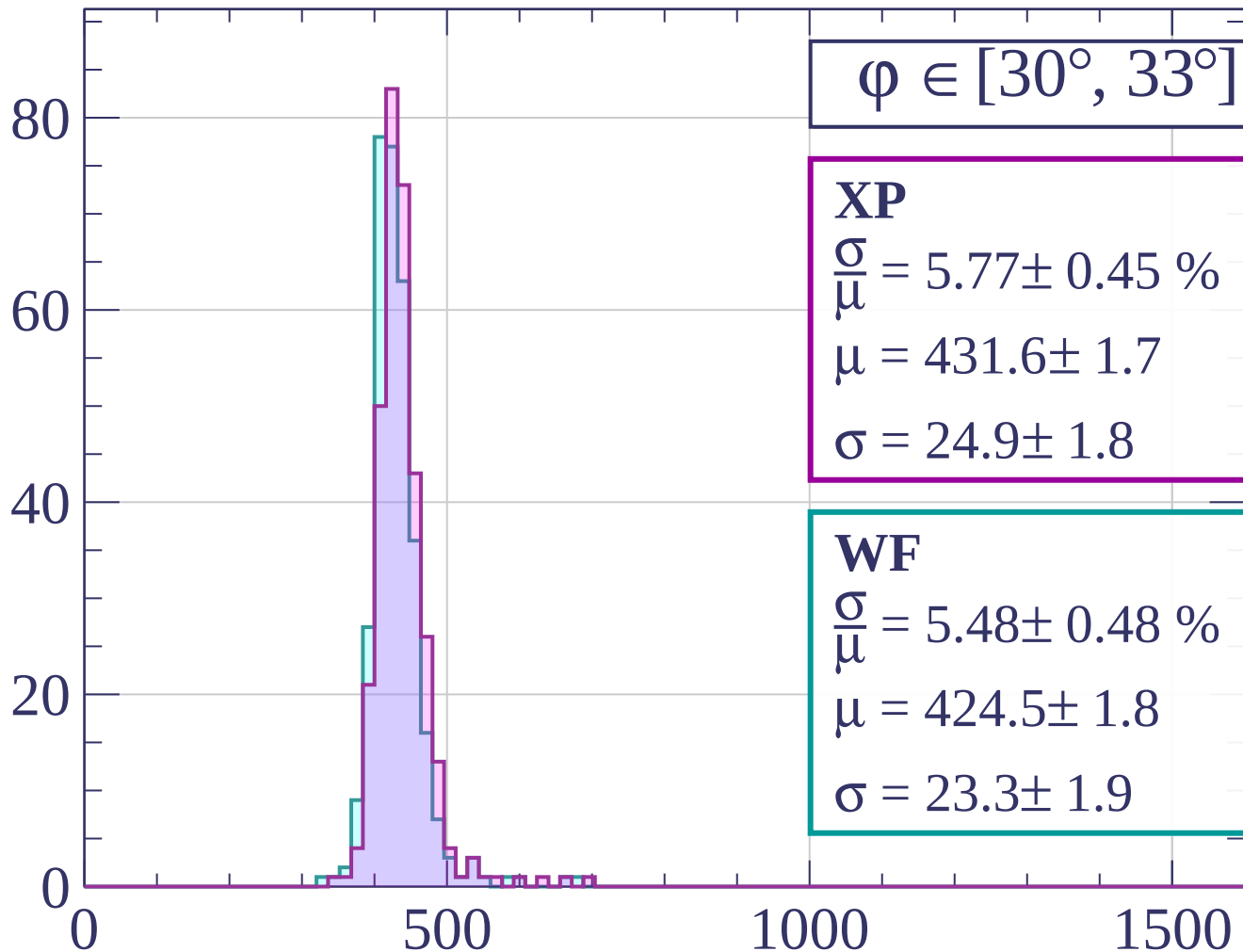
1000

1500

dE/dx [ADC counts/cm]

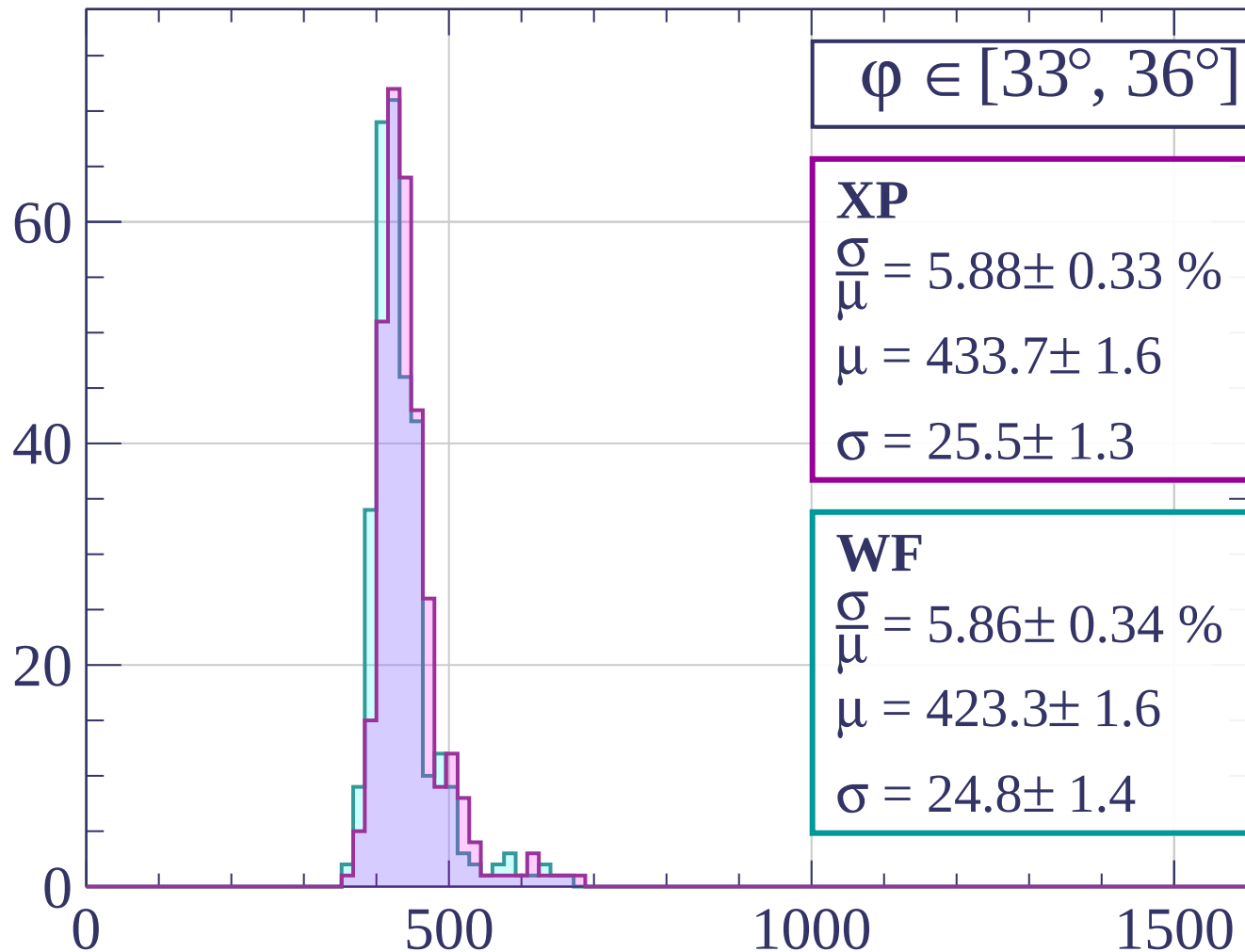


Count



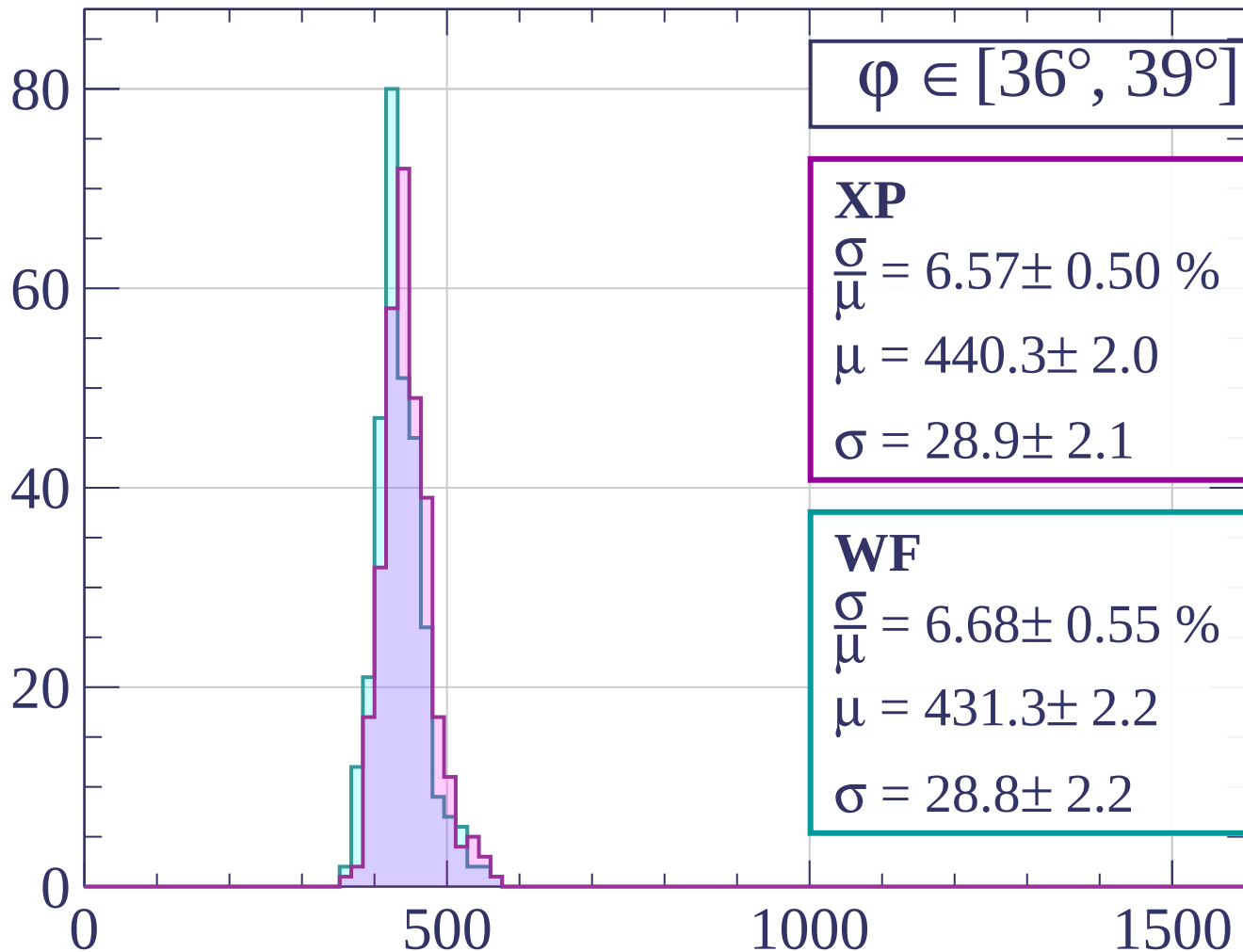
dE/dx [ADC counts/cm]

Count



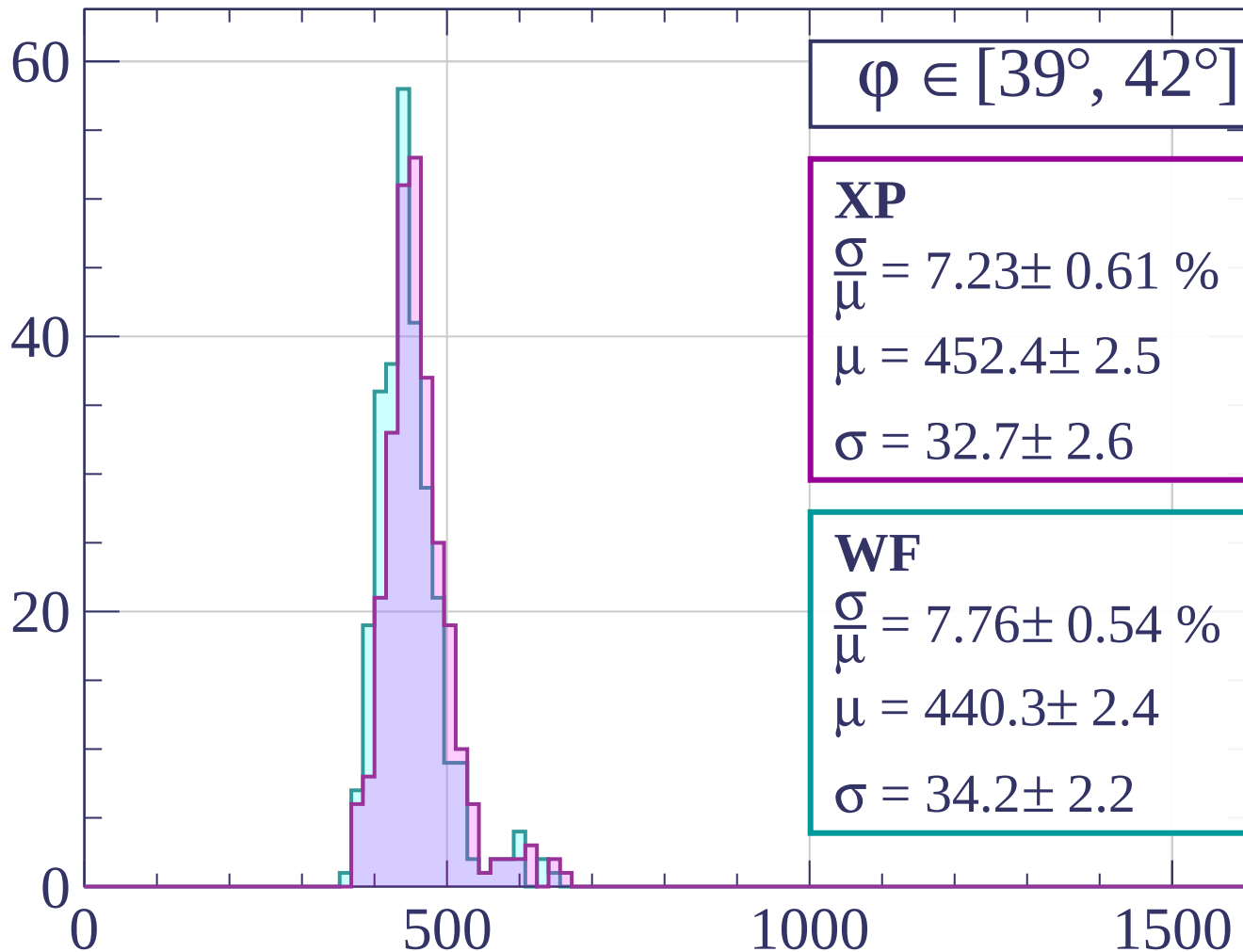
dE/dx [ADC counts/cm]

Count



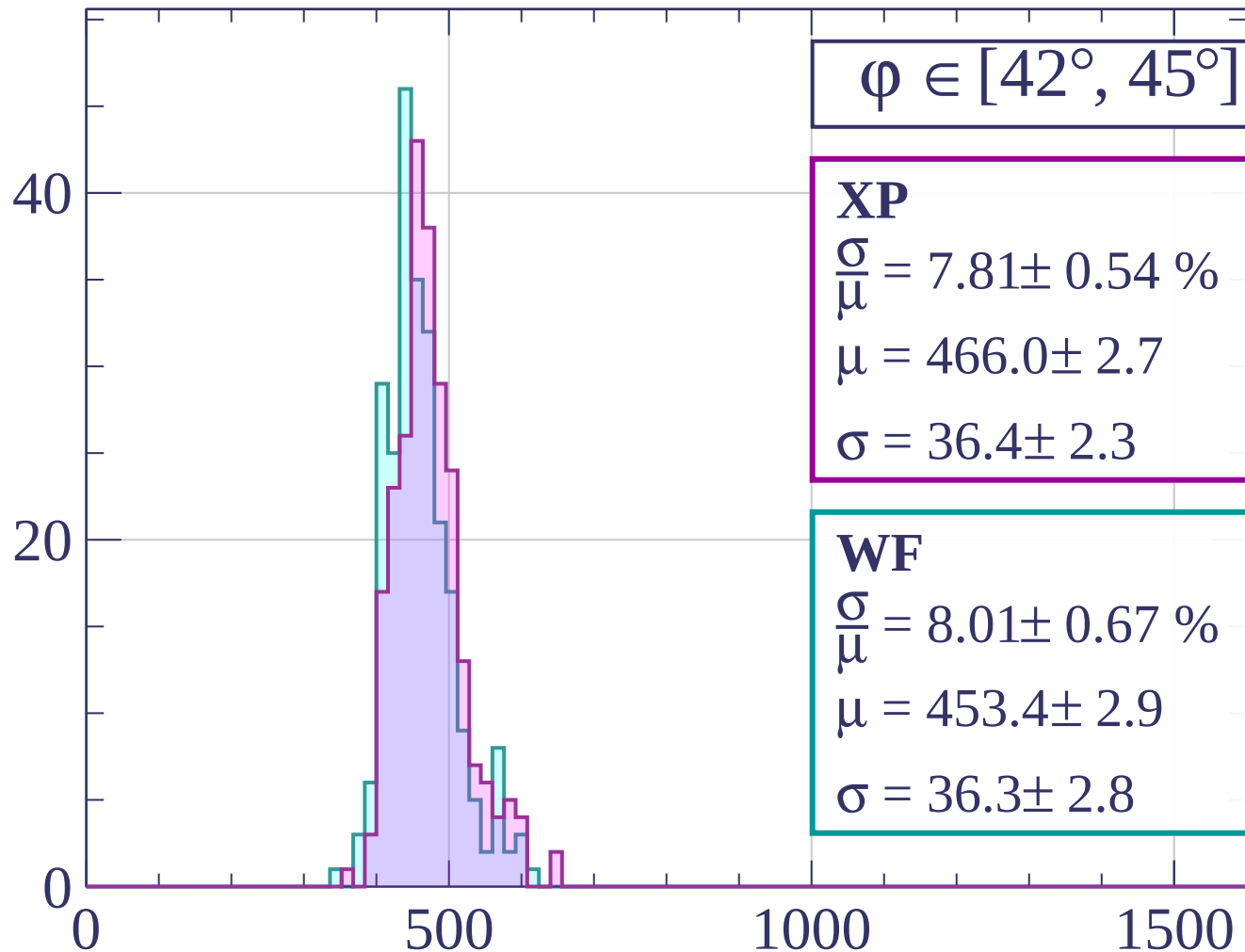
dE/dx [ADC counts/cm]

Count



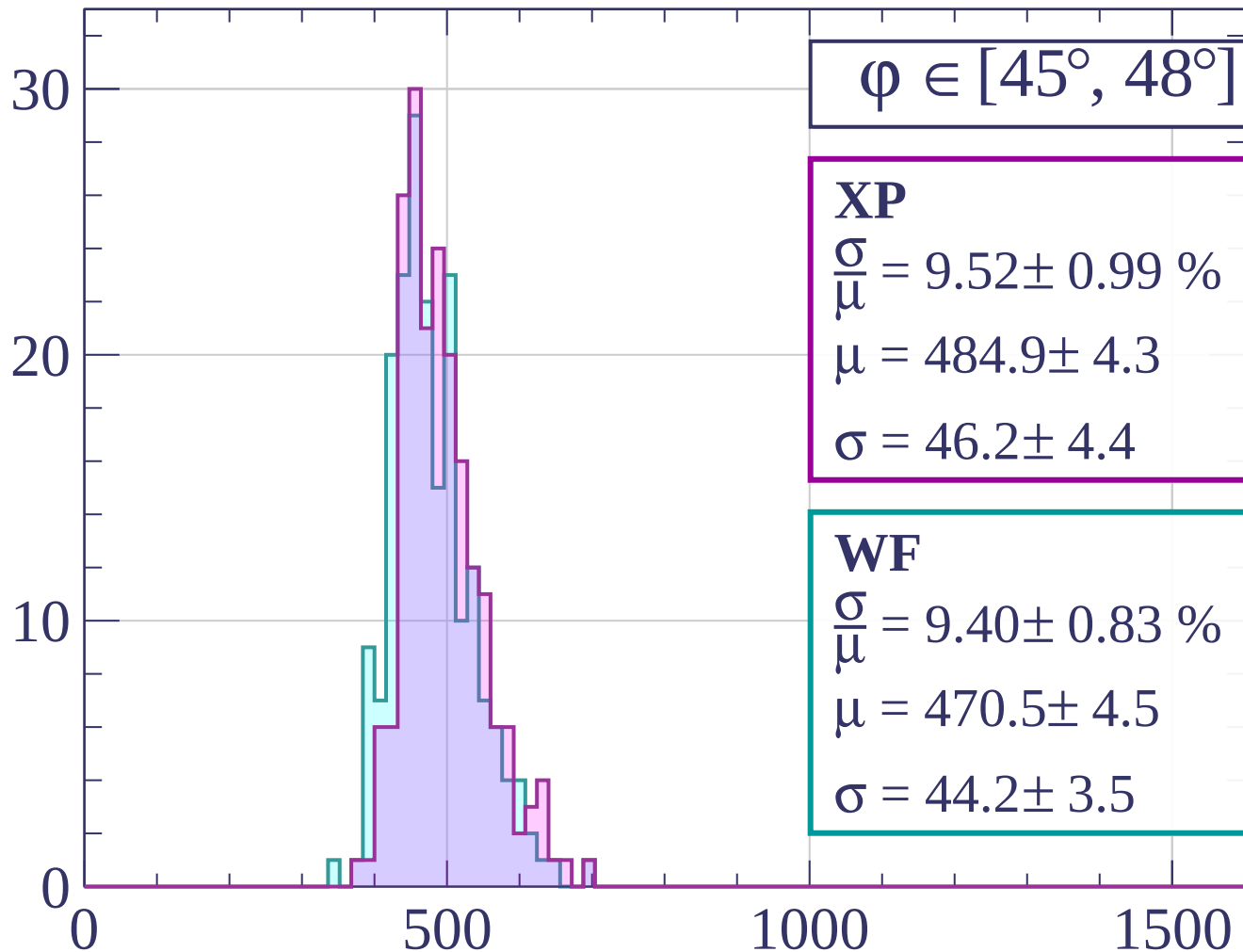
dE/dx [ADC counts/cm]

Count



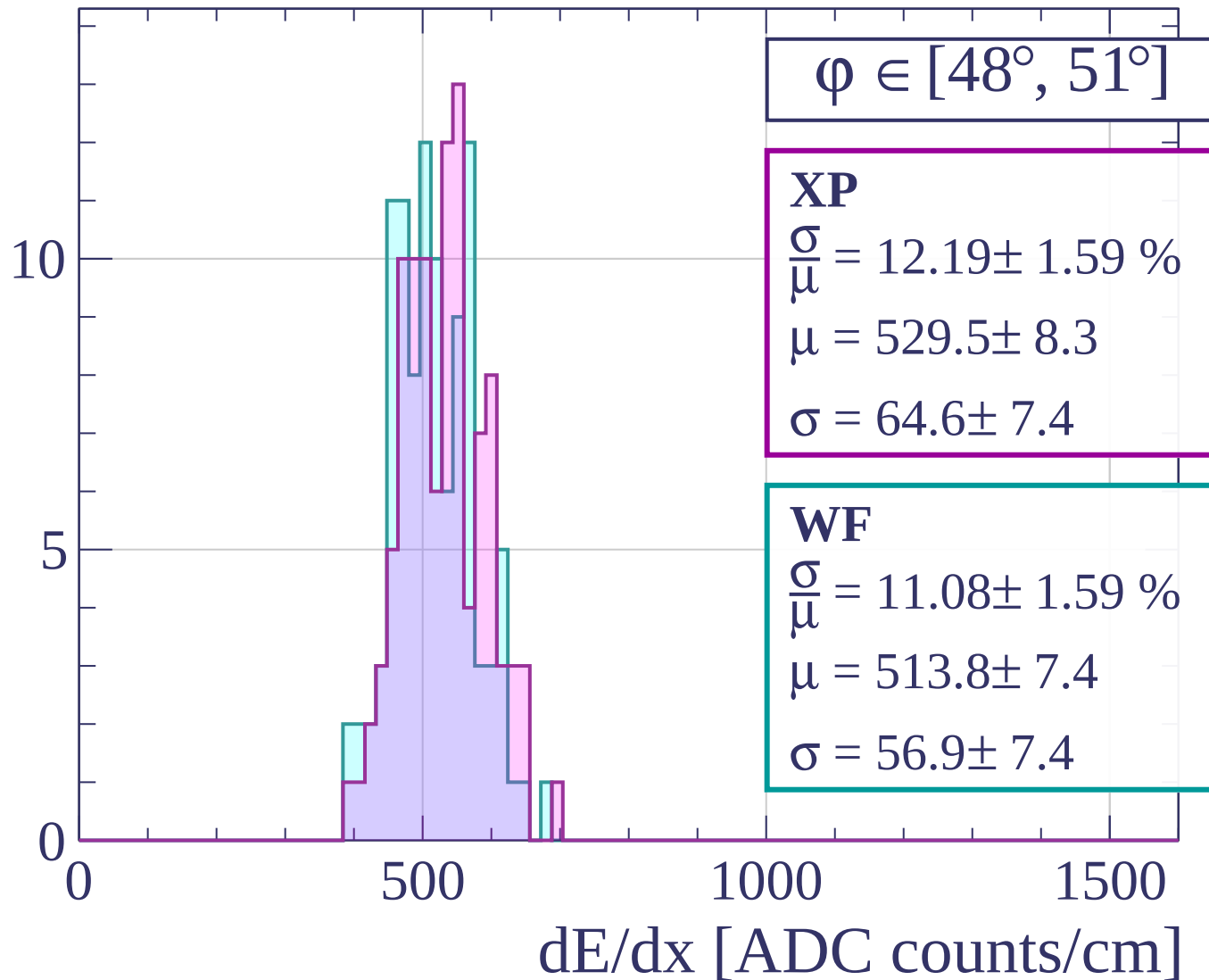
dE/dx [ADC counts/cm]

Count

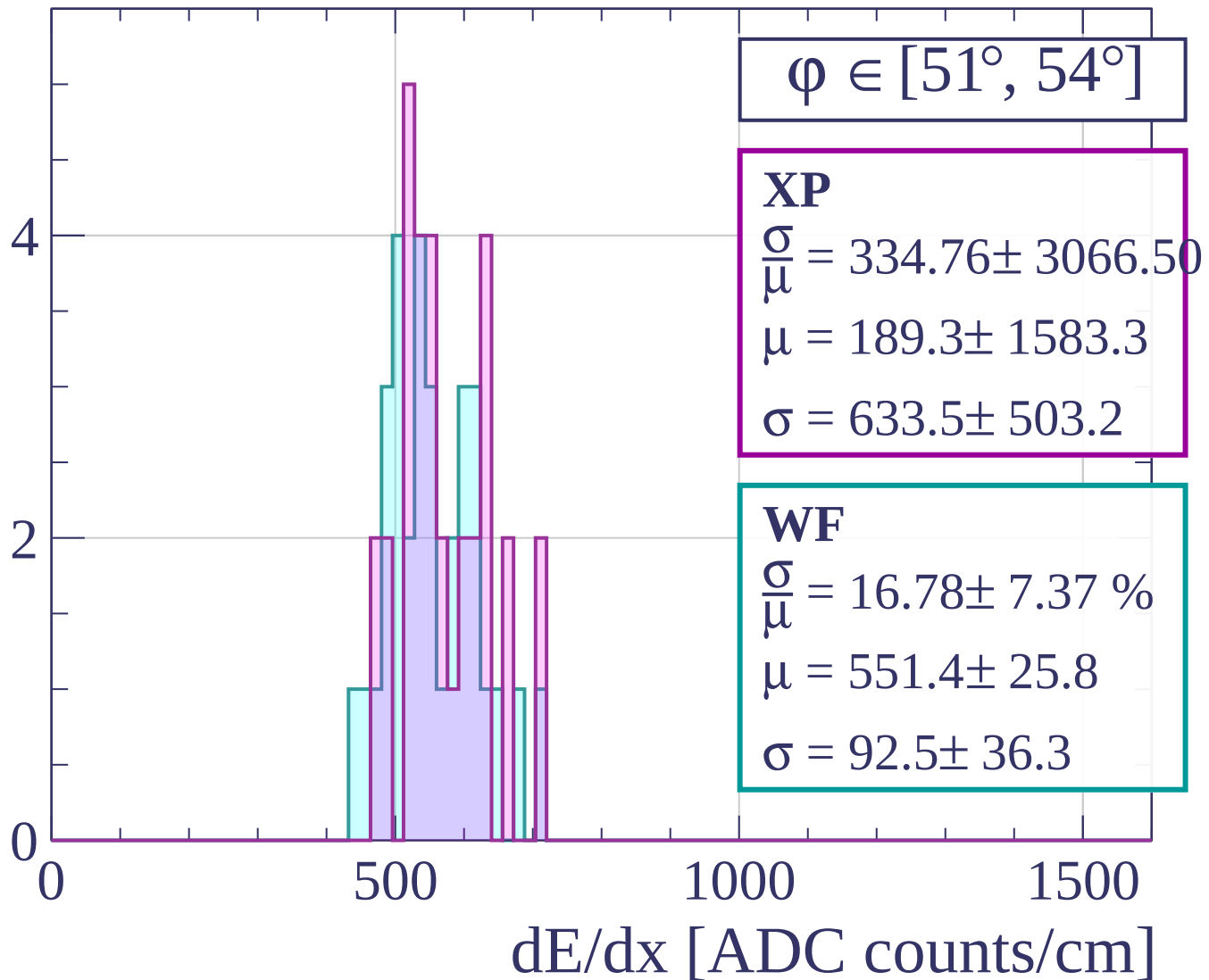


dE/dx [ADC counts/cm]

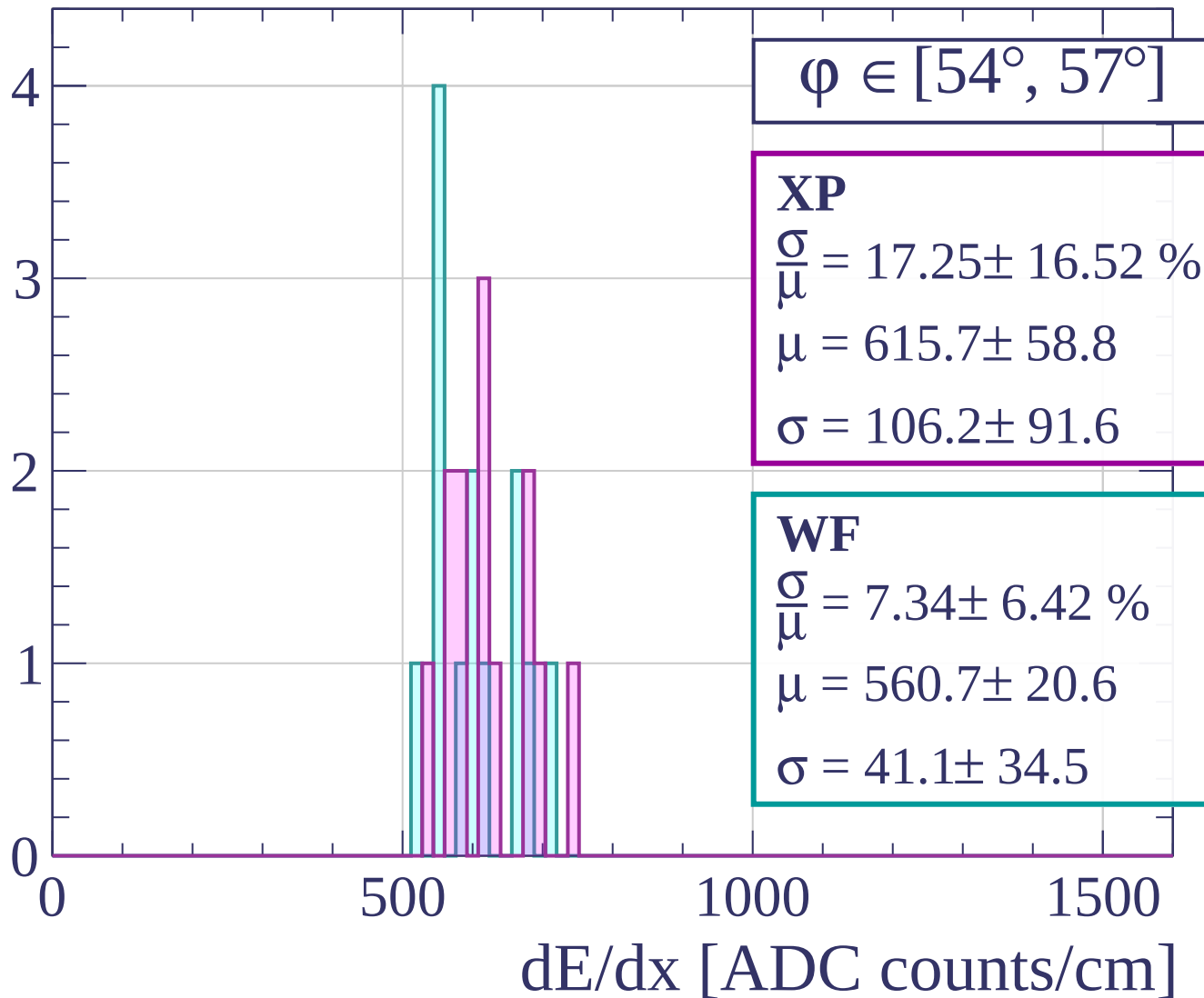
Count



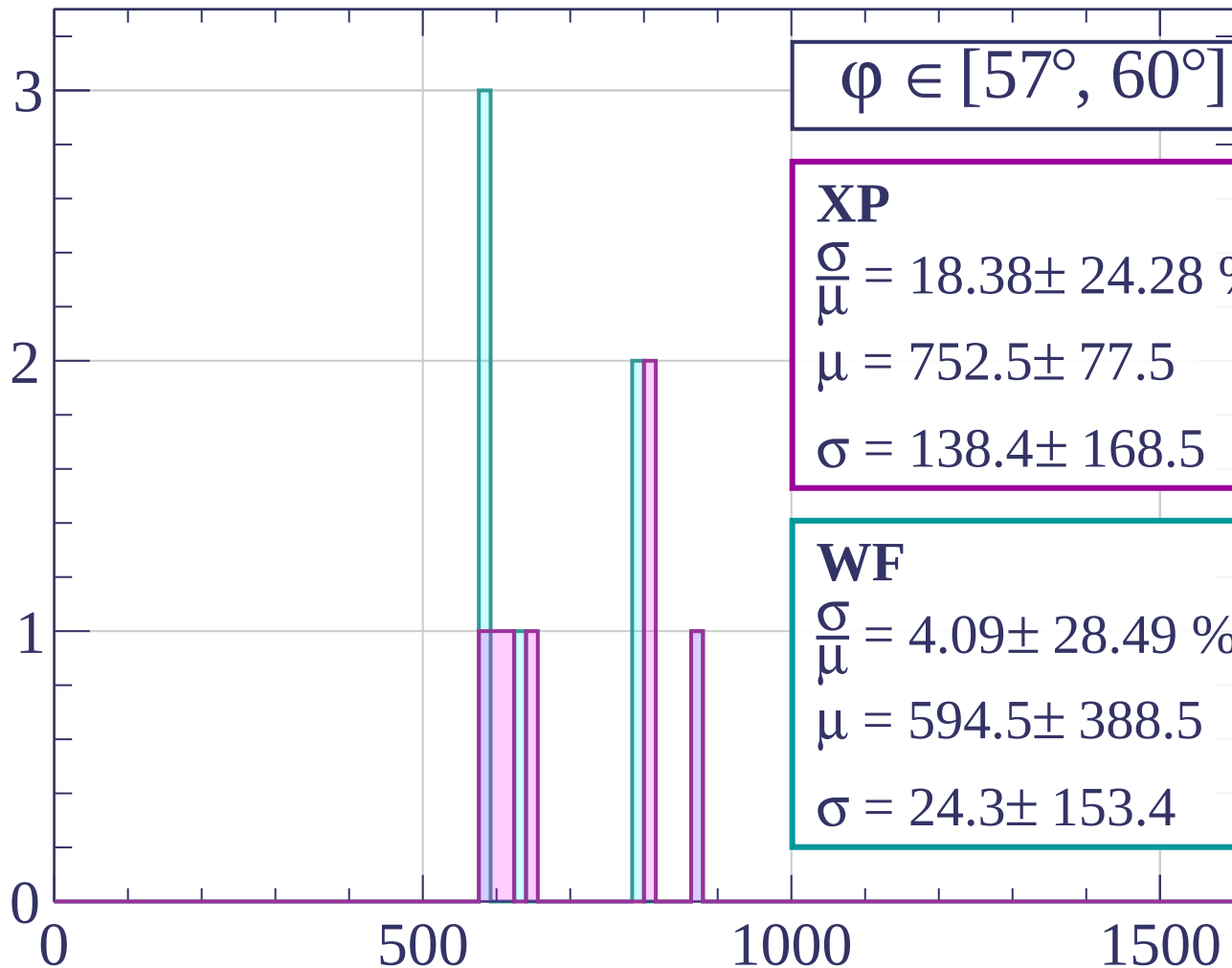
Count



Count



Count



dE/dx [ADC counts/cm]

Count

$\phi \in [60^\circ, 63^\circ]$

XP

$$\frac{\sigma}{\mu} = 2.60 \pm 24.57 \%$$

$$\mu = 616.0 \pm 55.4$$

$$\sigma = 16.0 \pm 149.9$$

WF

$$\frac{\sigma}{\mu} = 0.05 \pm 0.31 \%$$

$$\mu = 567.7 \pm 2.4$$

$$\sigma = 0.3 \pm 1.8$$

1.0

0.5

0.0

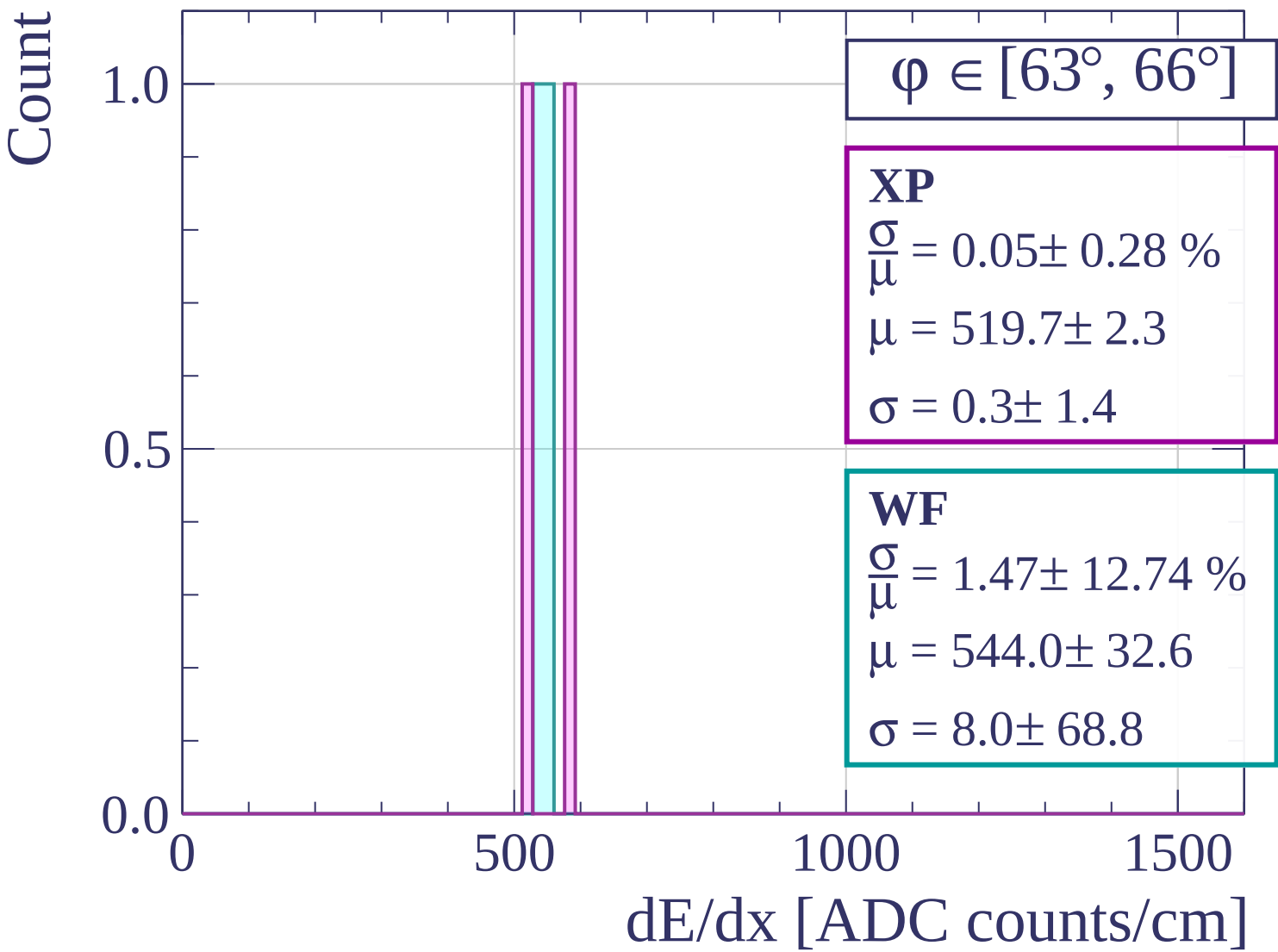
0

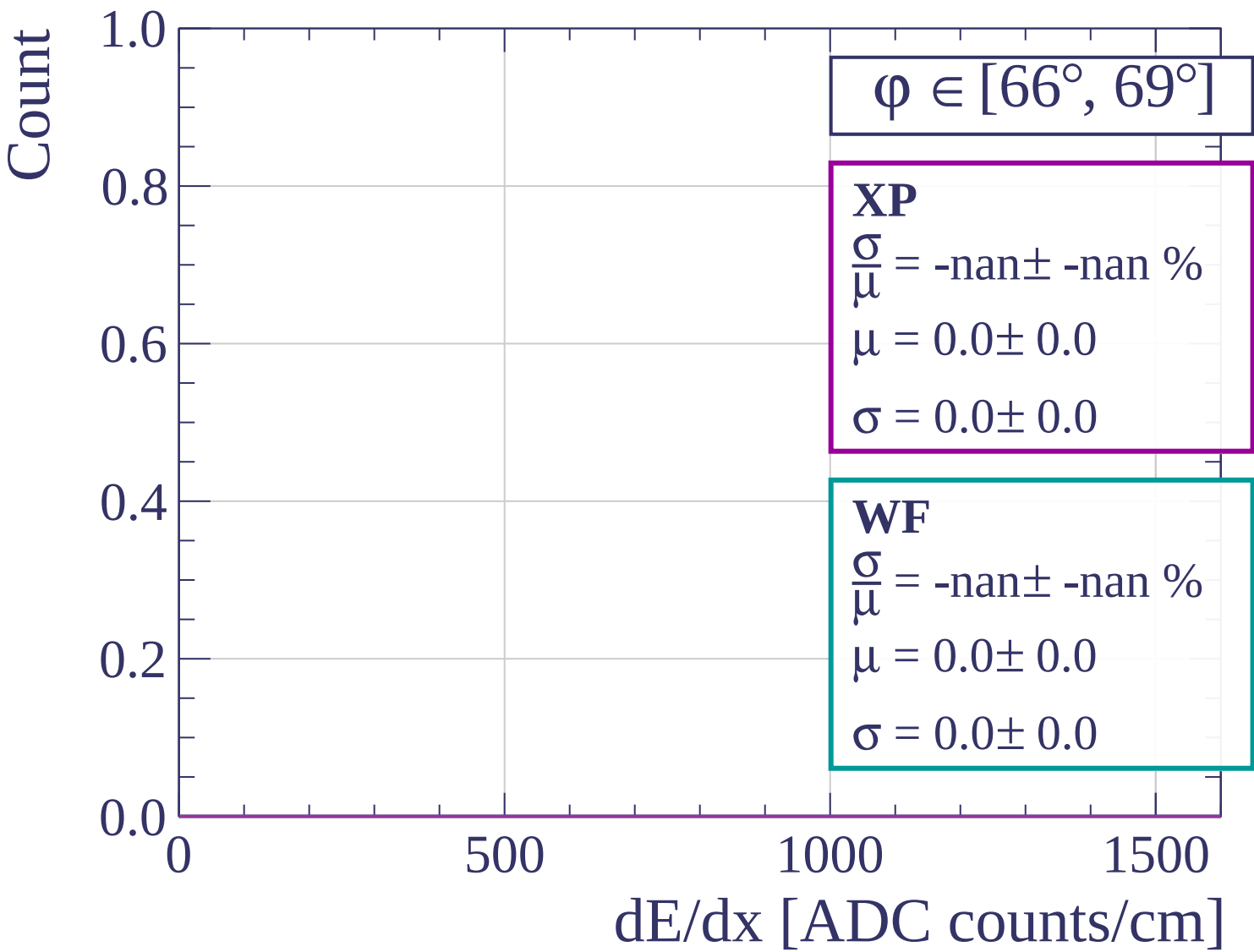
500

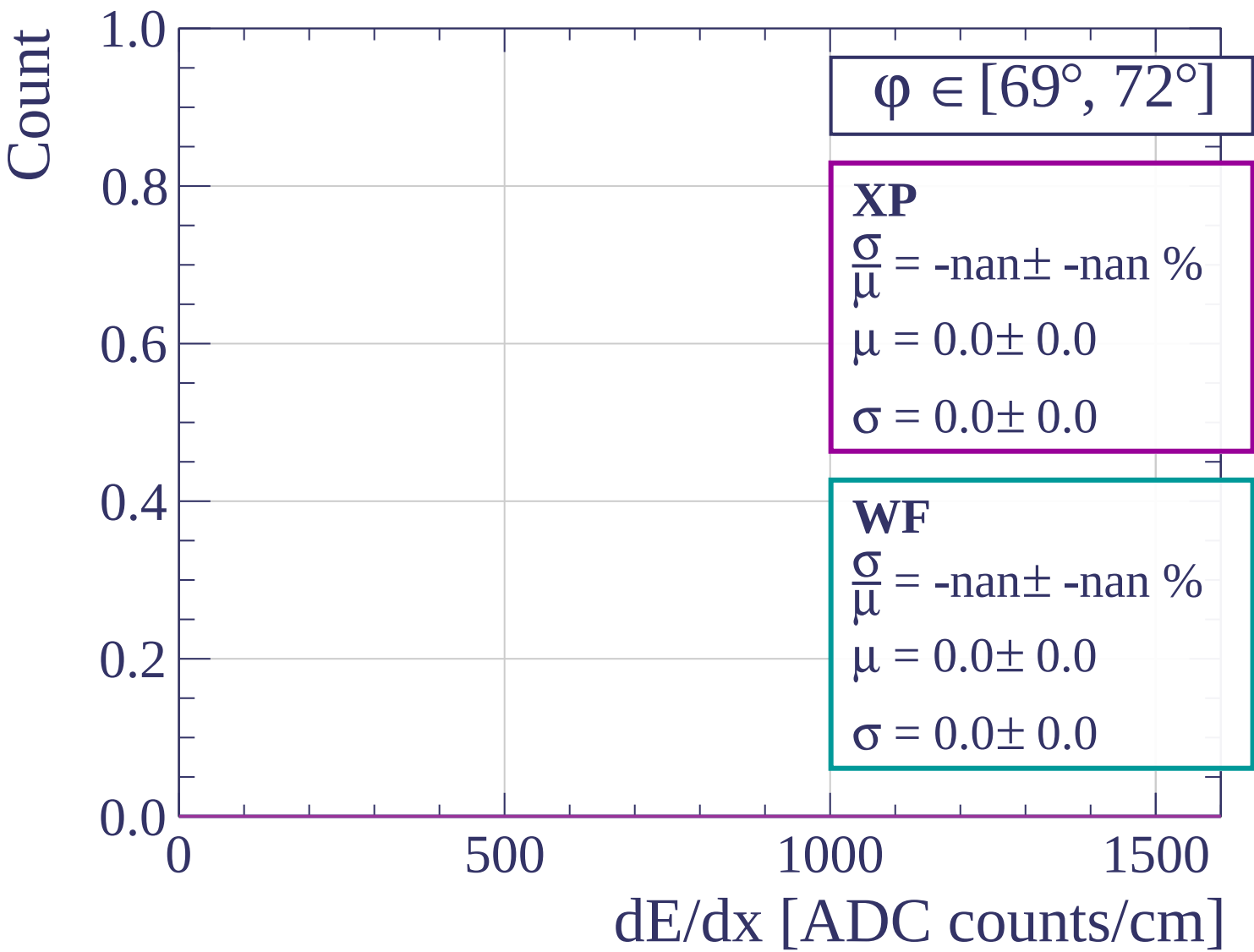
1000

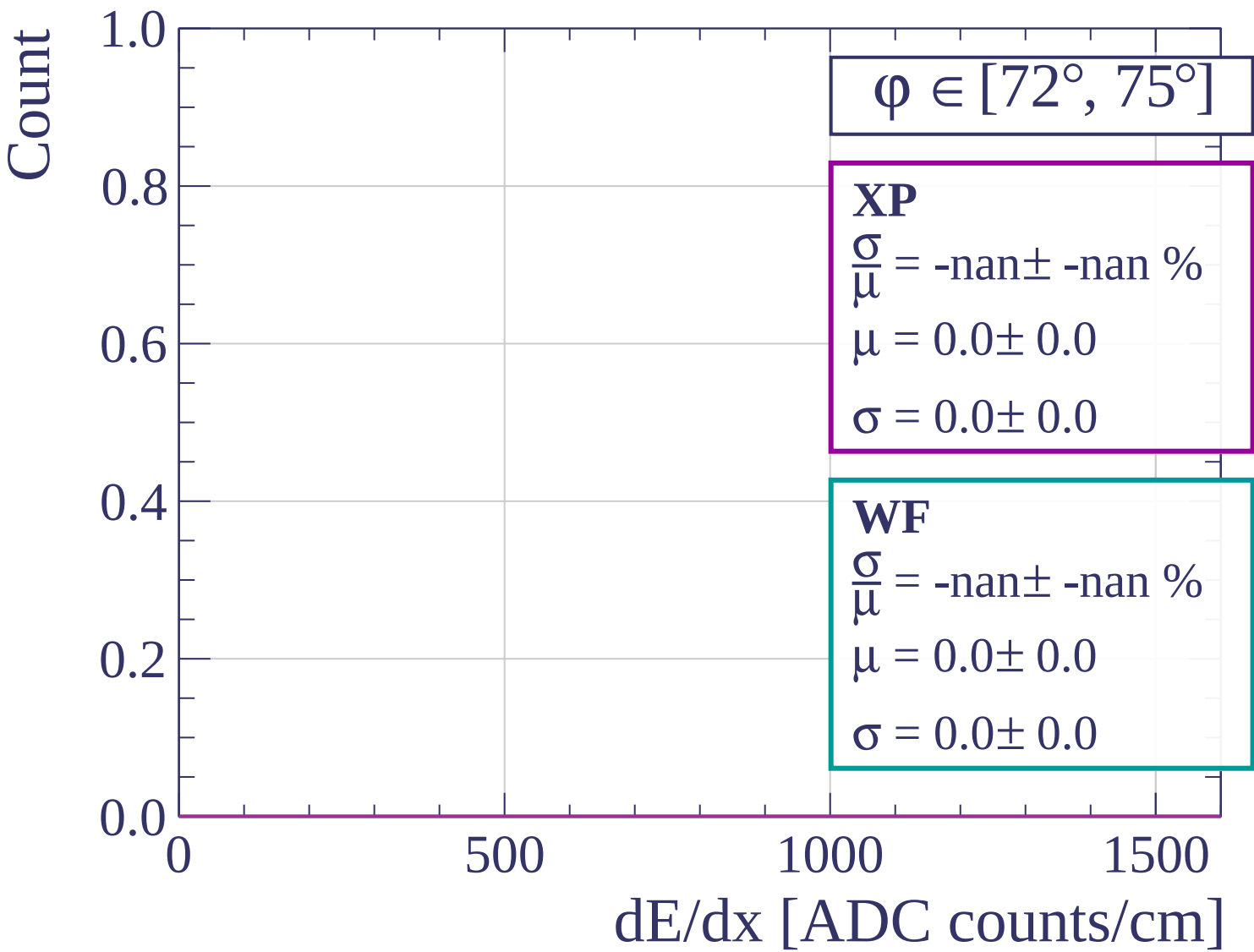
1500

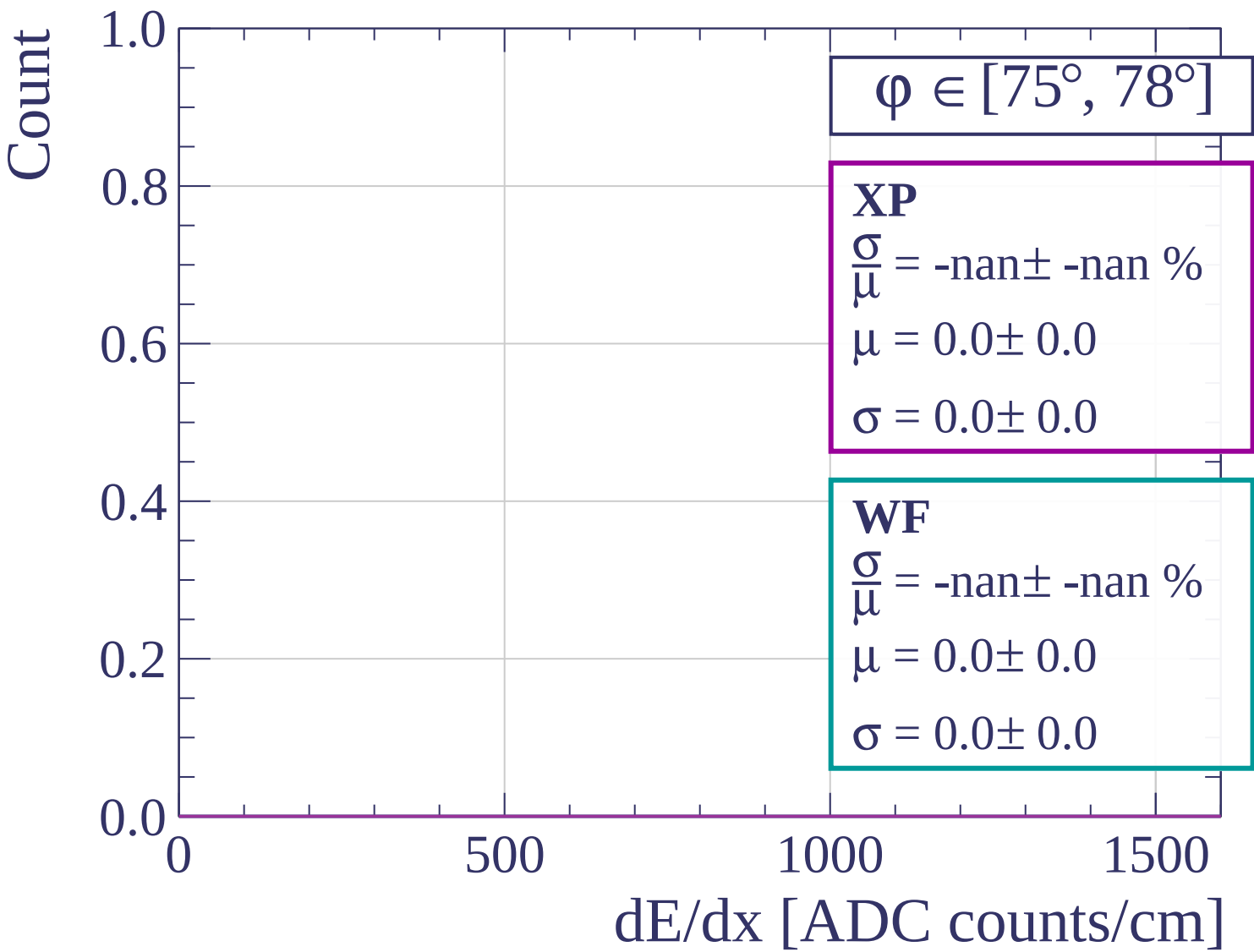
dE/dx [ADC counts/cm]

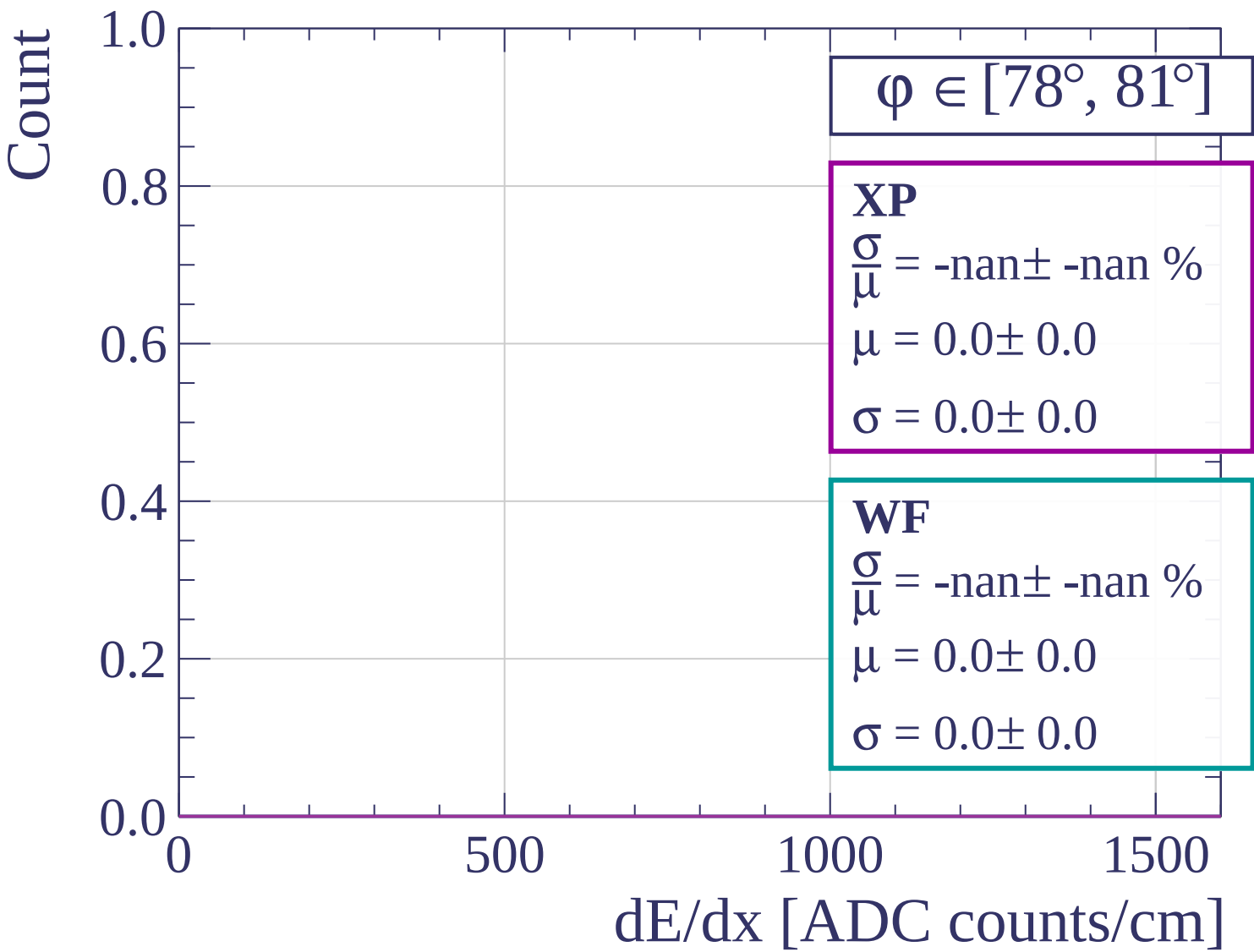


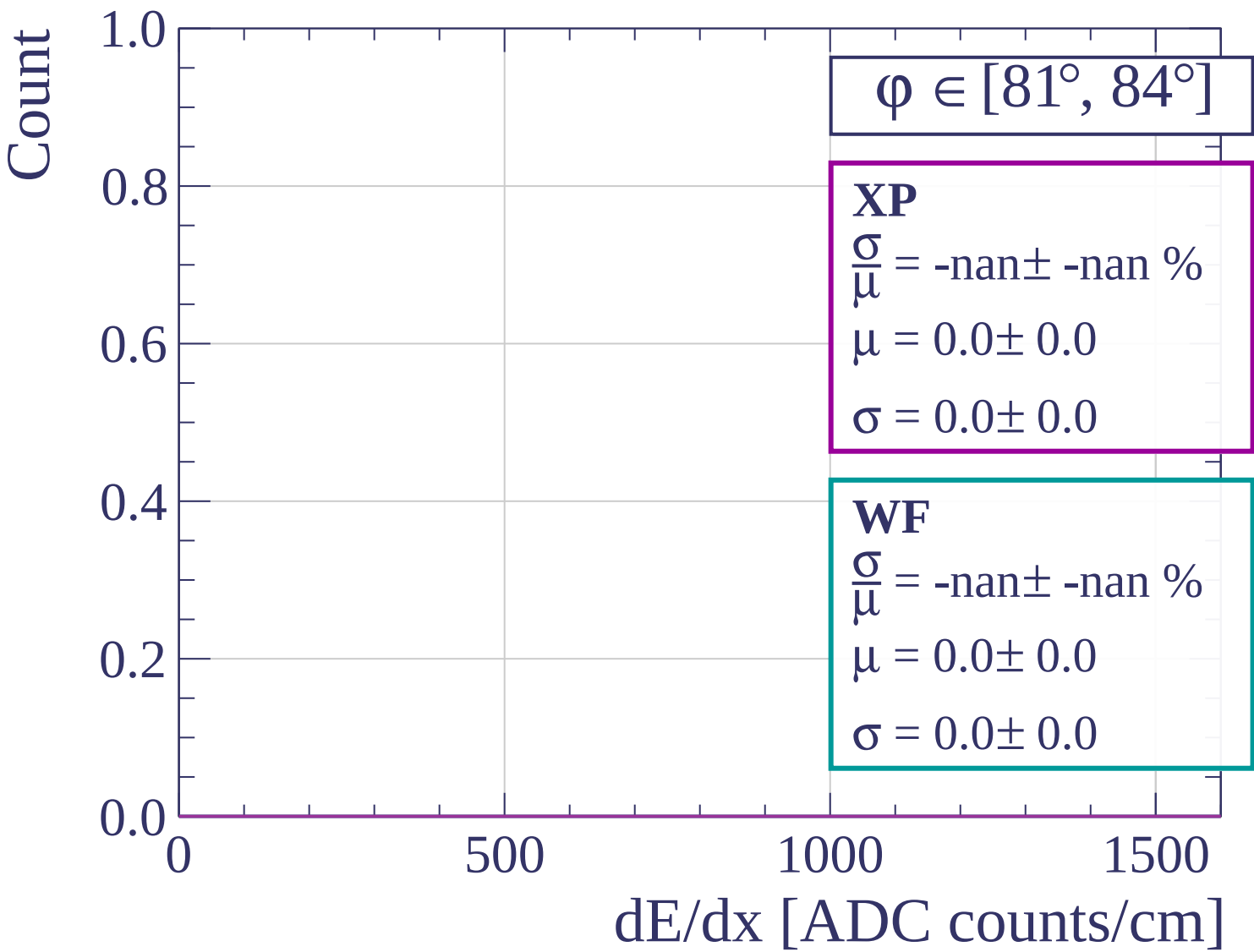


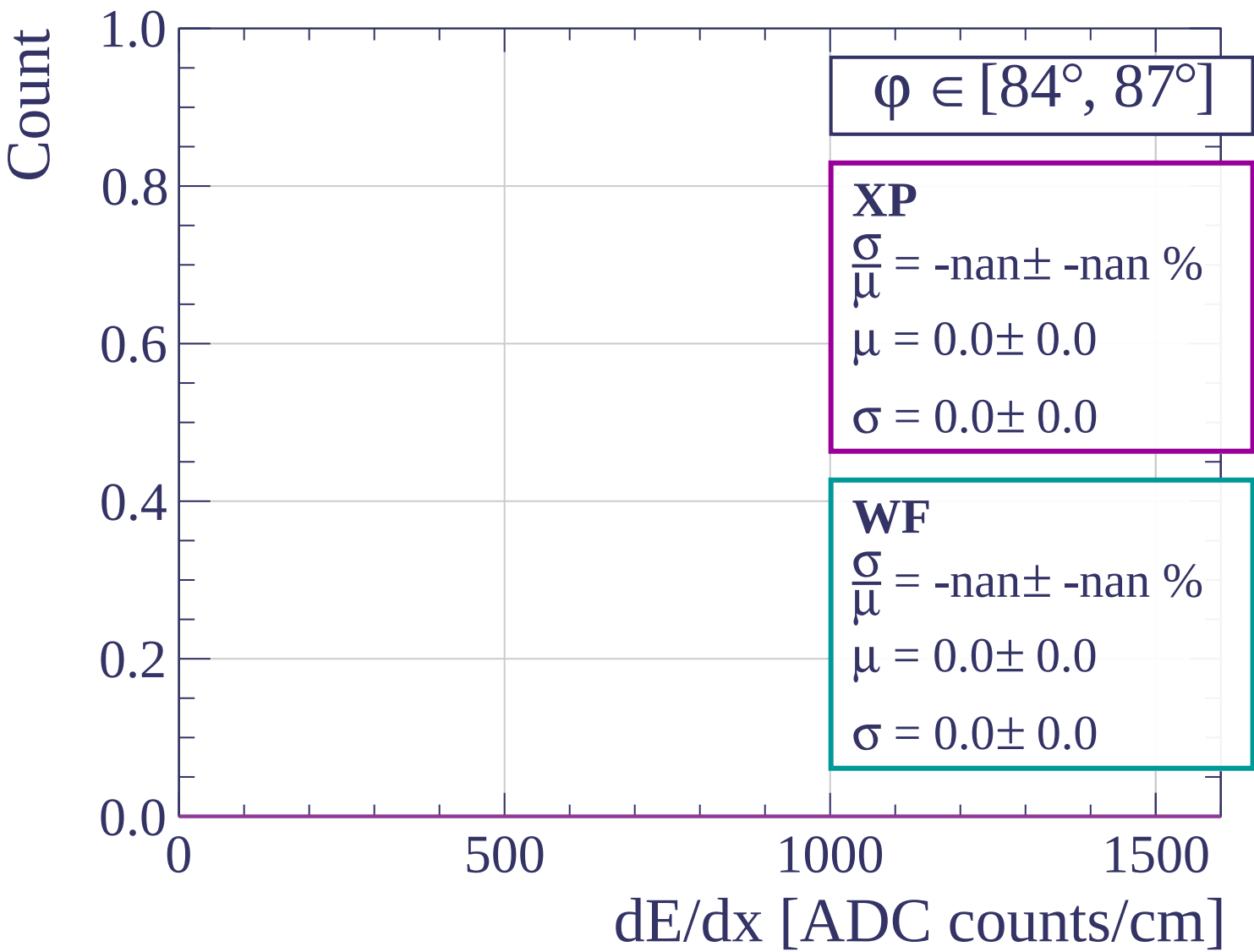


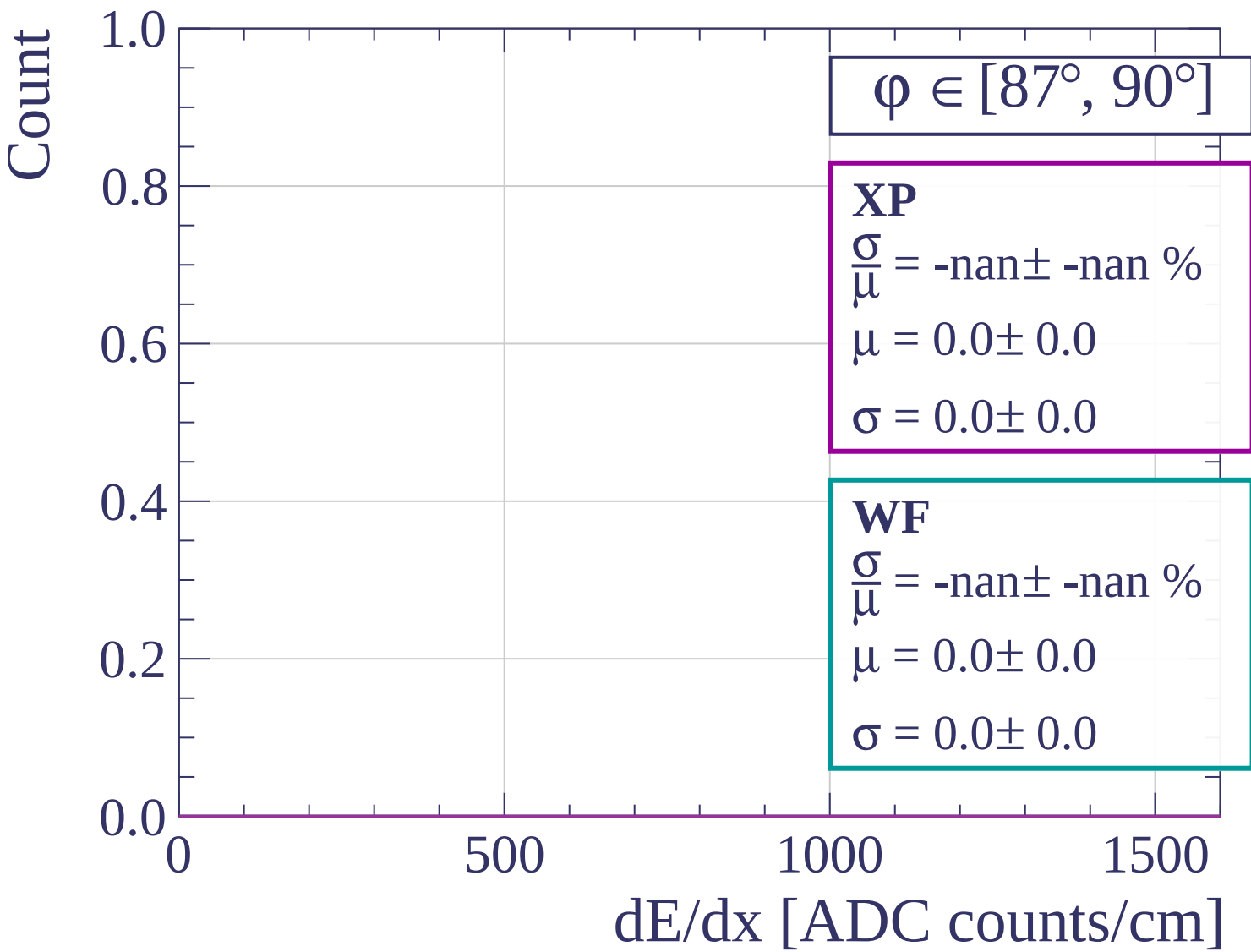


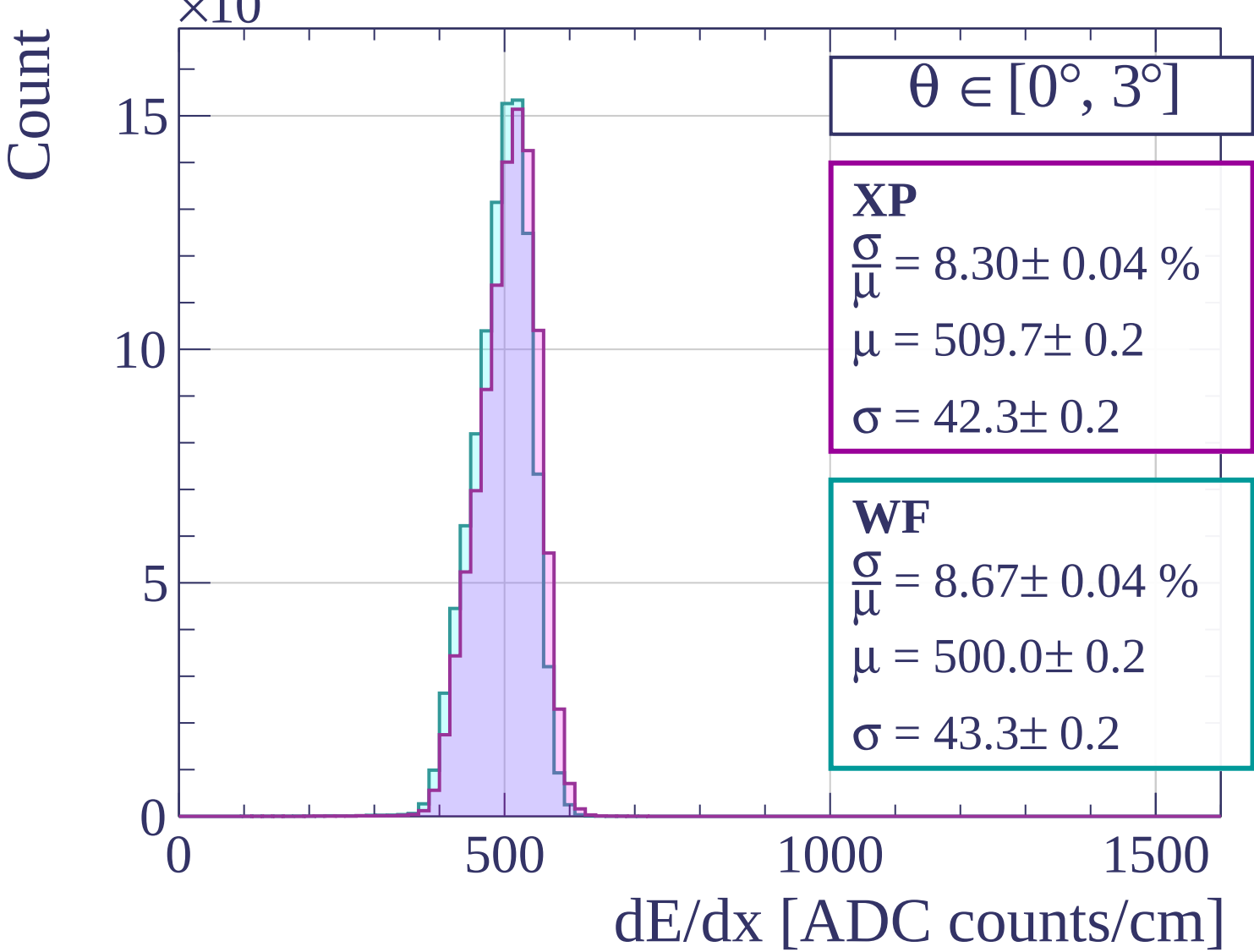


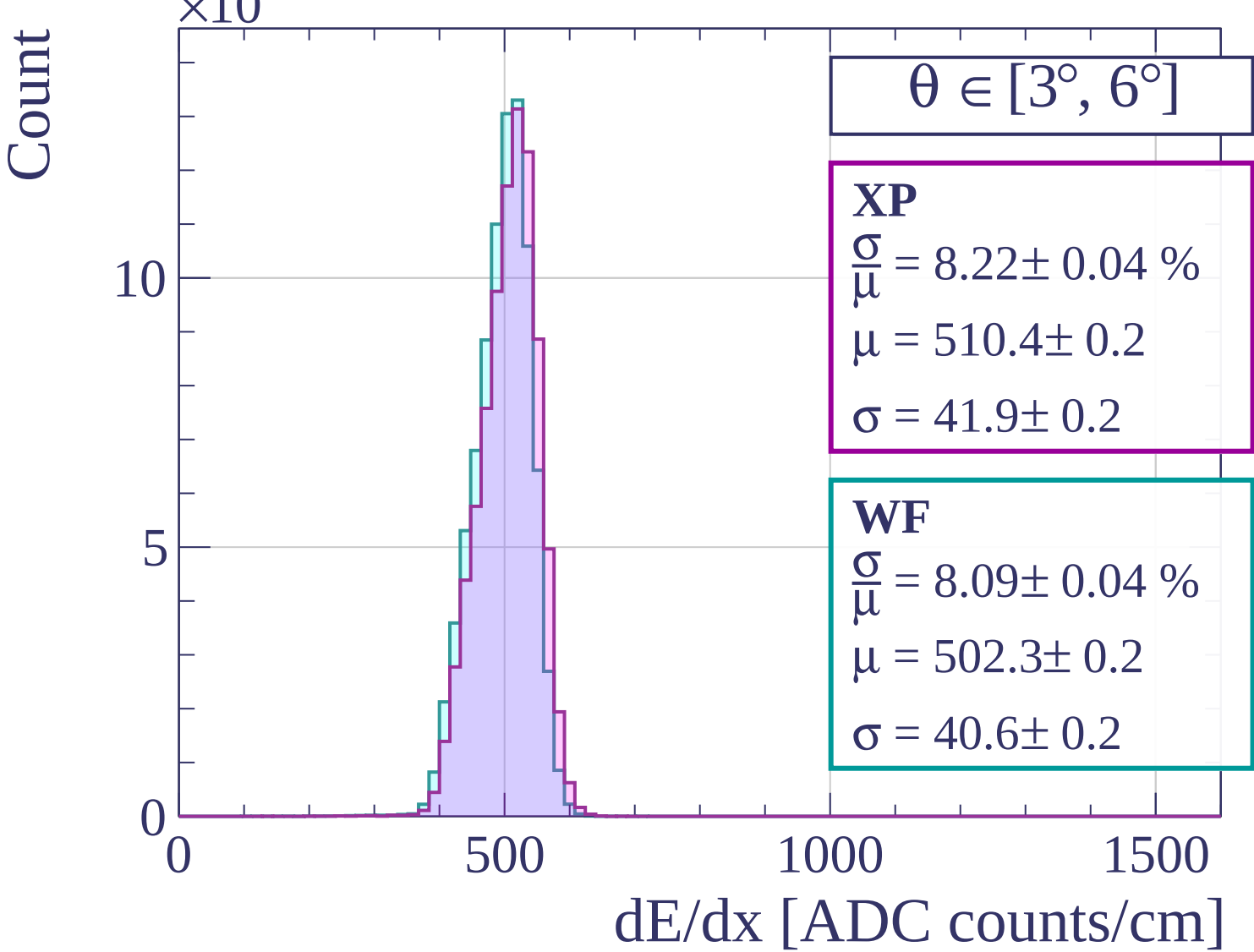


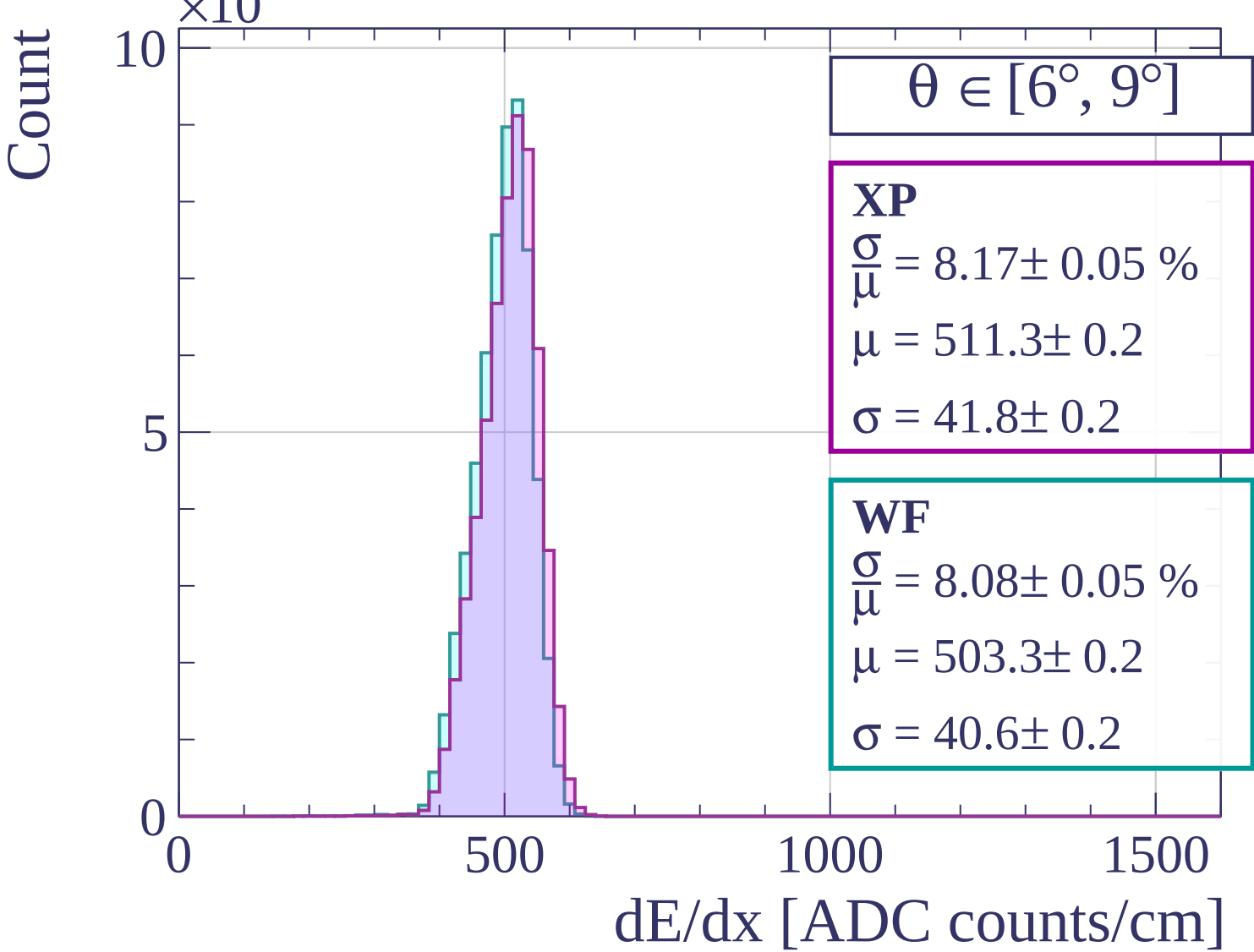


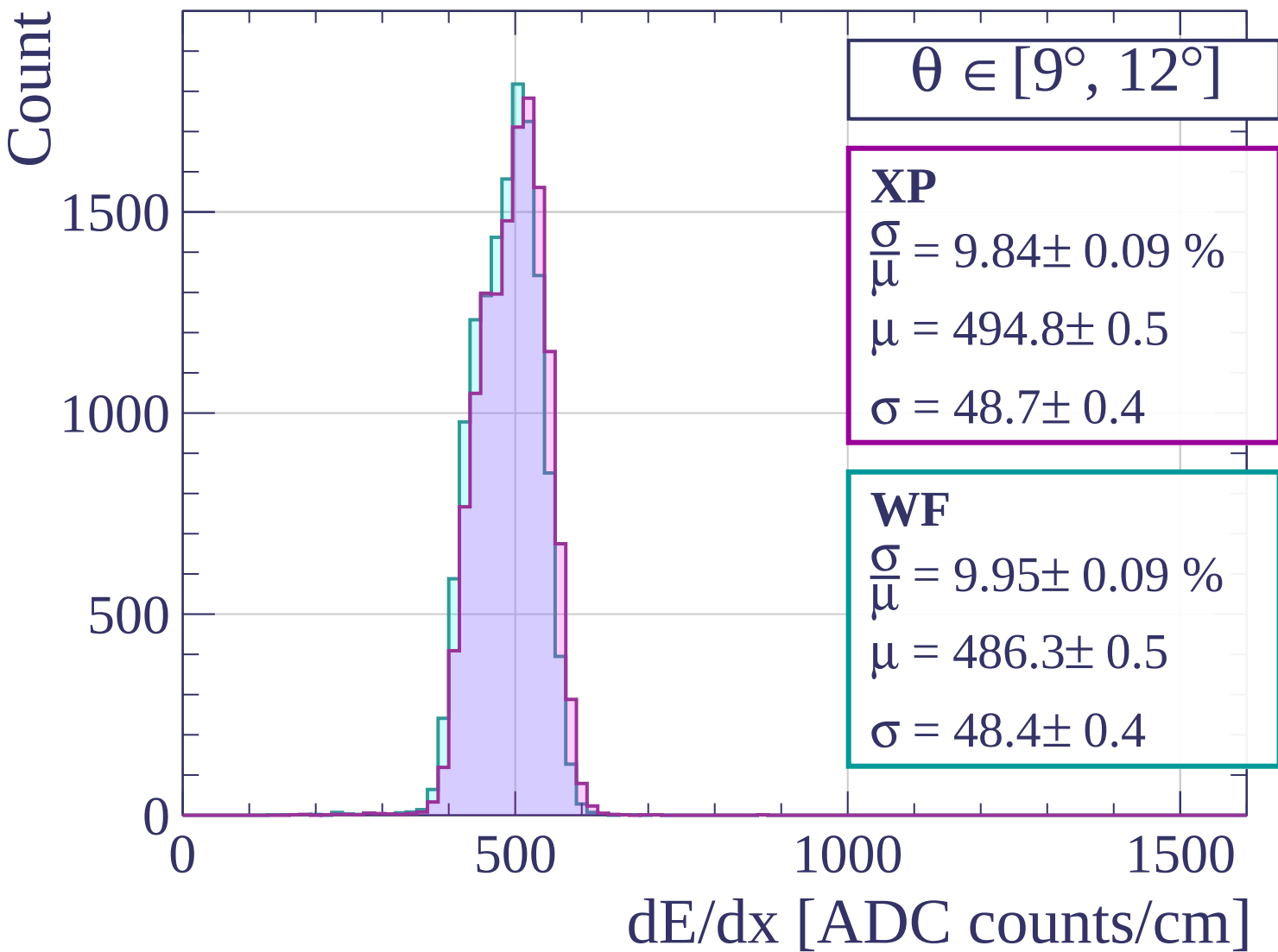












Count

$\theta \in [12^\circ, 15^\circ]$

XP

$$\frac{\sigma}{\mu} = 6.41 \pm 0.27 \%$$

$$\mu = 439.0 \pm 1.1$$

$$\sigma = 28.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.28 \pm 0.29 \%$$

$$\mu = 430.3 \pm 1.1$$

$$\sigma = 27.0 \pm 1.2$$

200

150

100

50

0

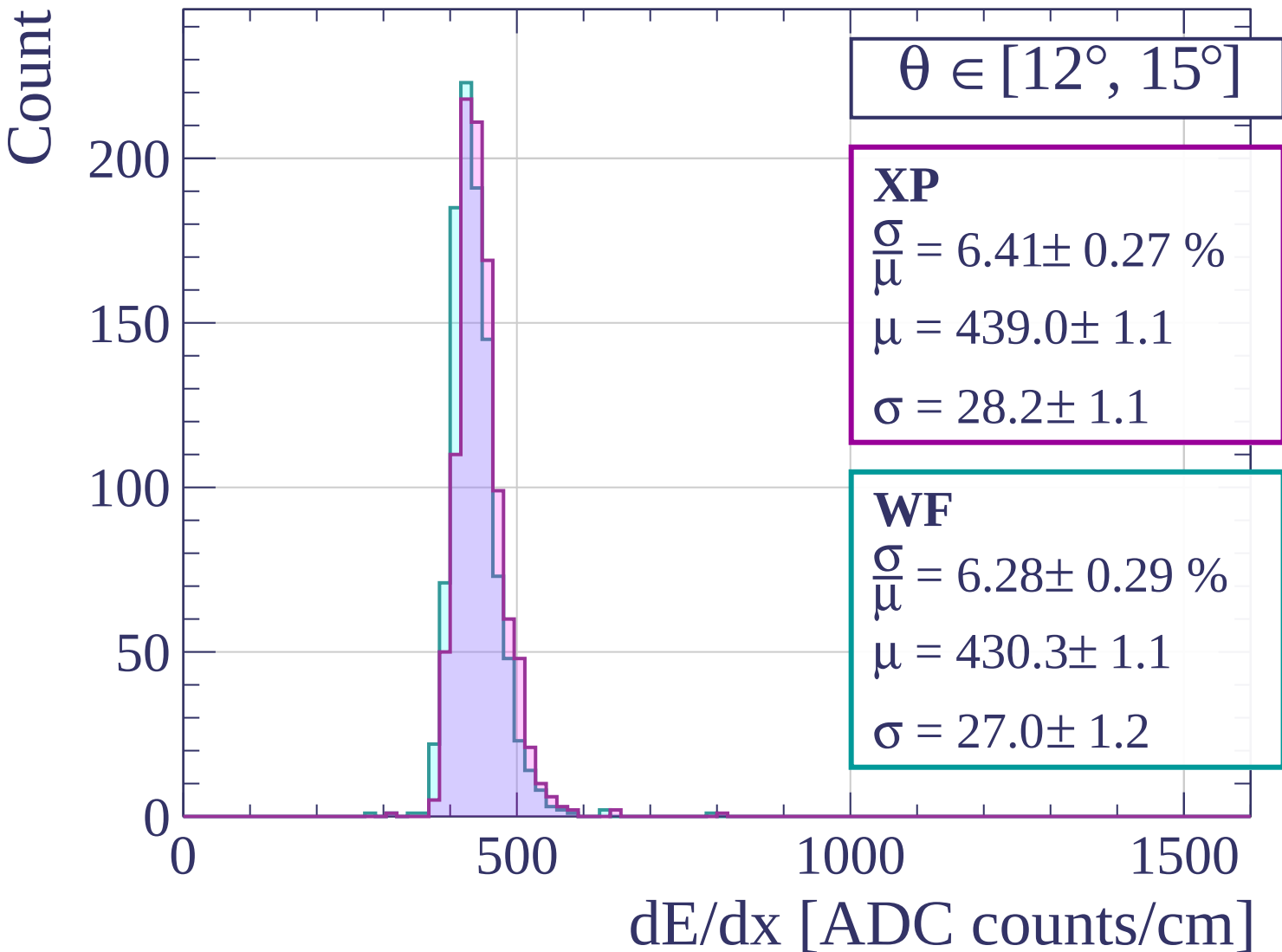
0

500

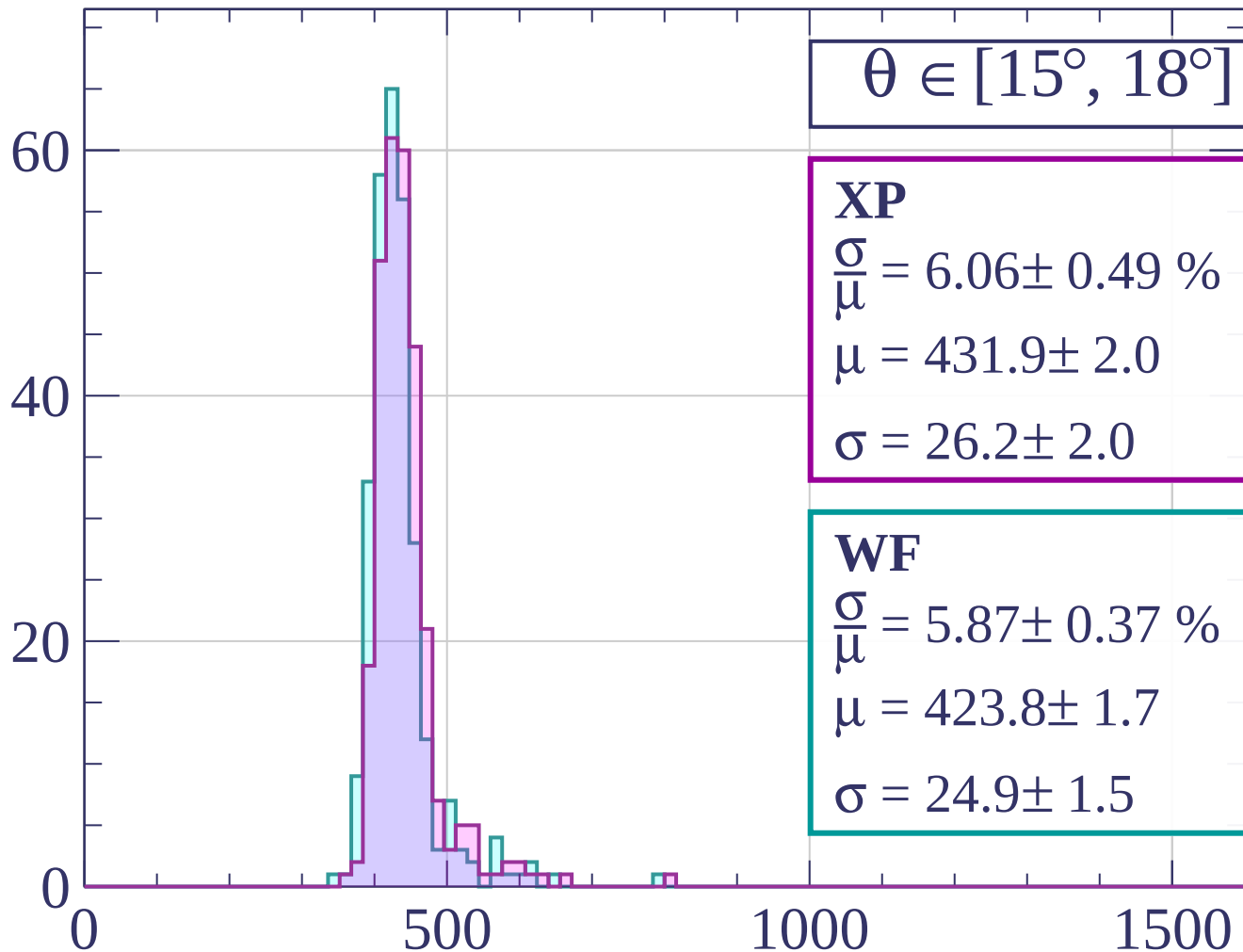
1000

1500

dE/dx [ADC counts/cm]

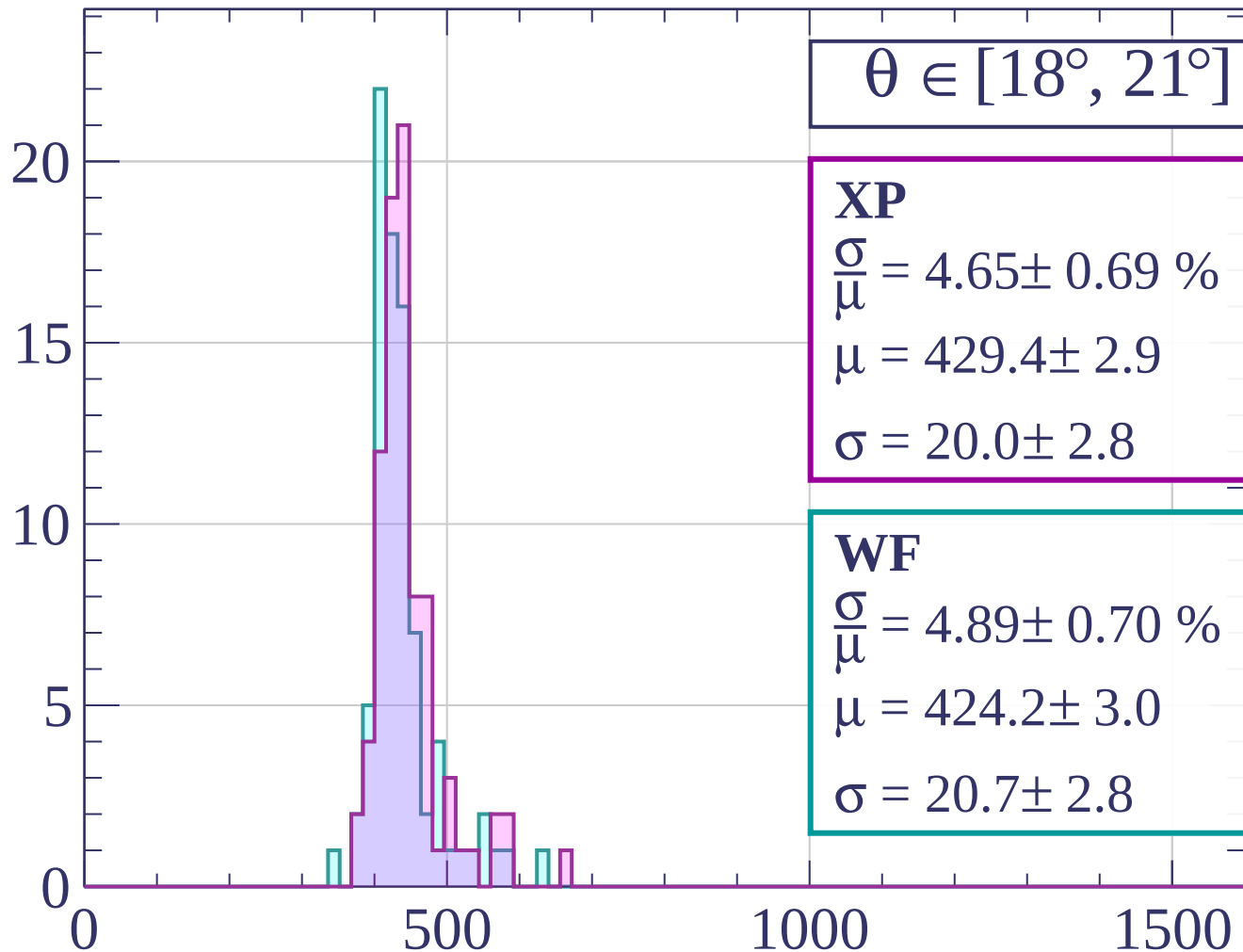


Count



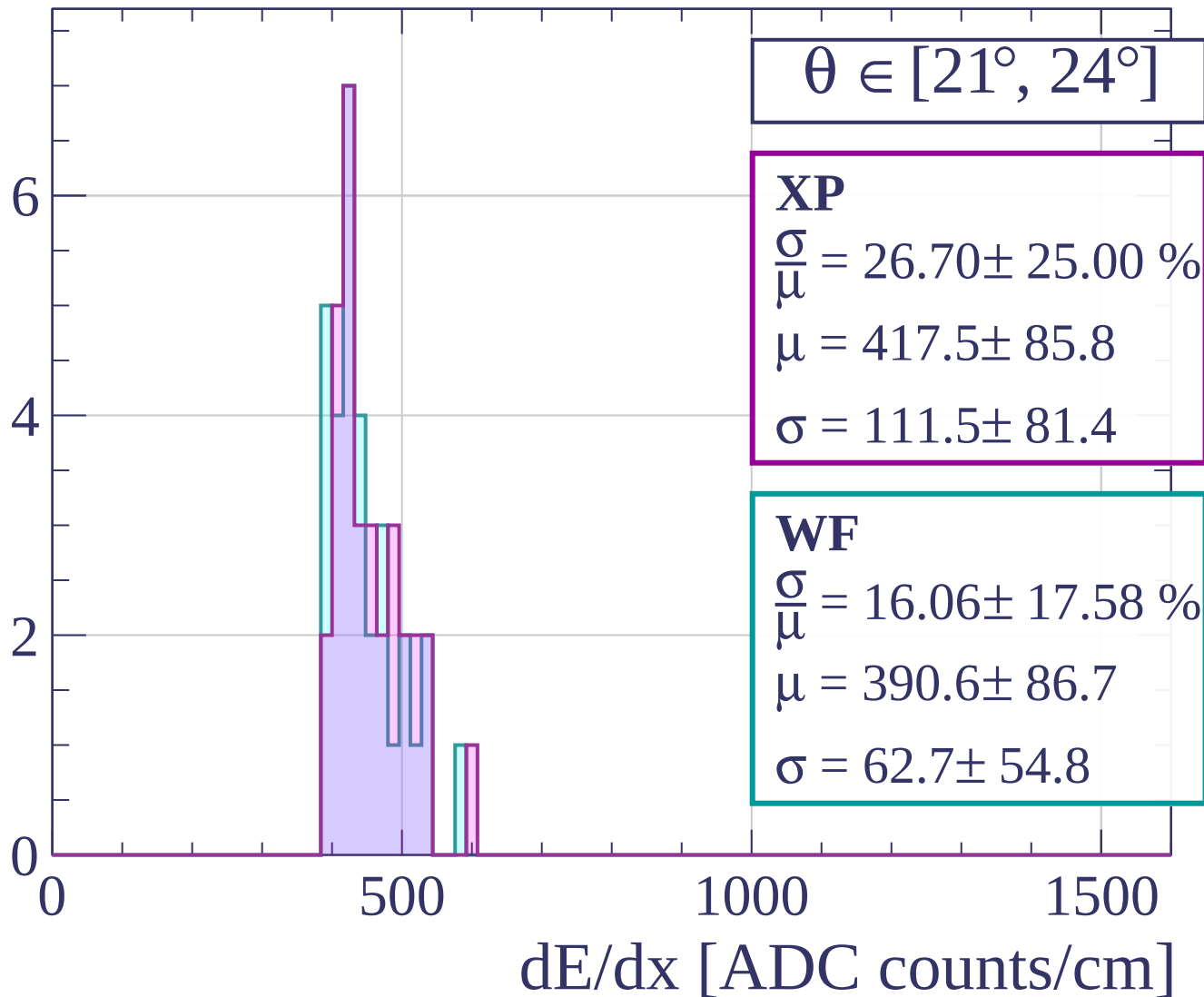
dE/dx [ADC counts/cm]

Count

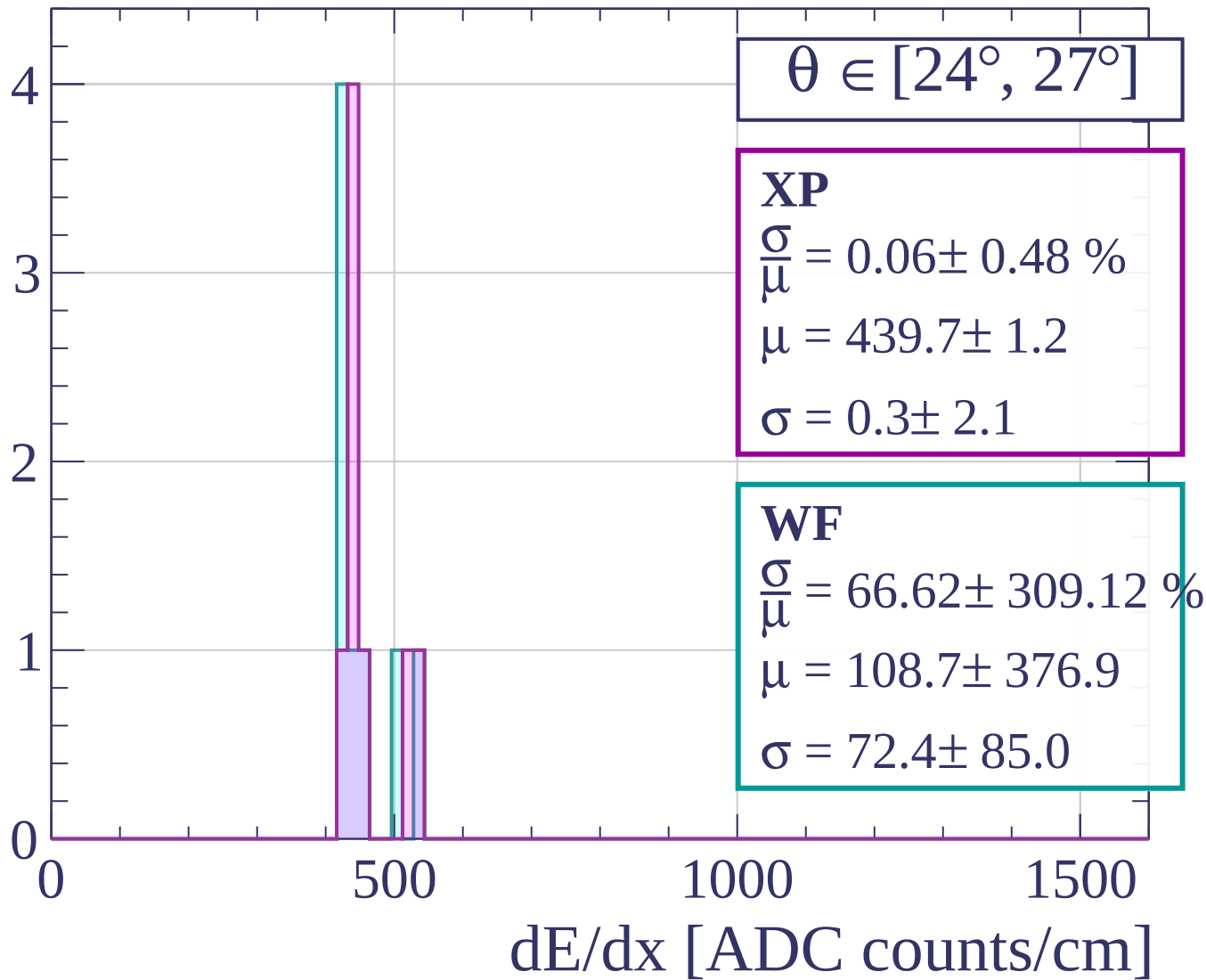


dE/dx [ADC counts/cm]

Count



Count



Count

$\theta \in [27^\circ, 30^\circ]$

XP

$$\frac{\sigma}{\mu} = 97.47 \pm 430.38 \%$$

$$\mu = 479.9 \pm 1796.5$$

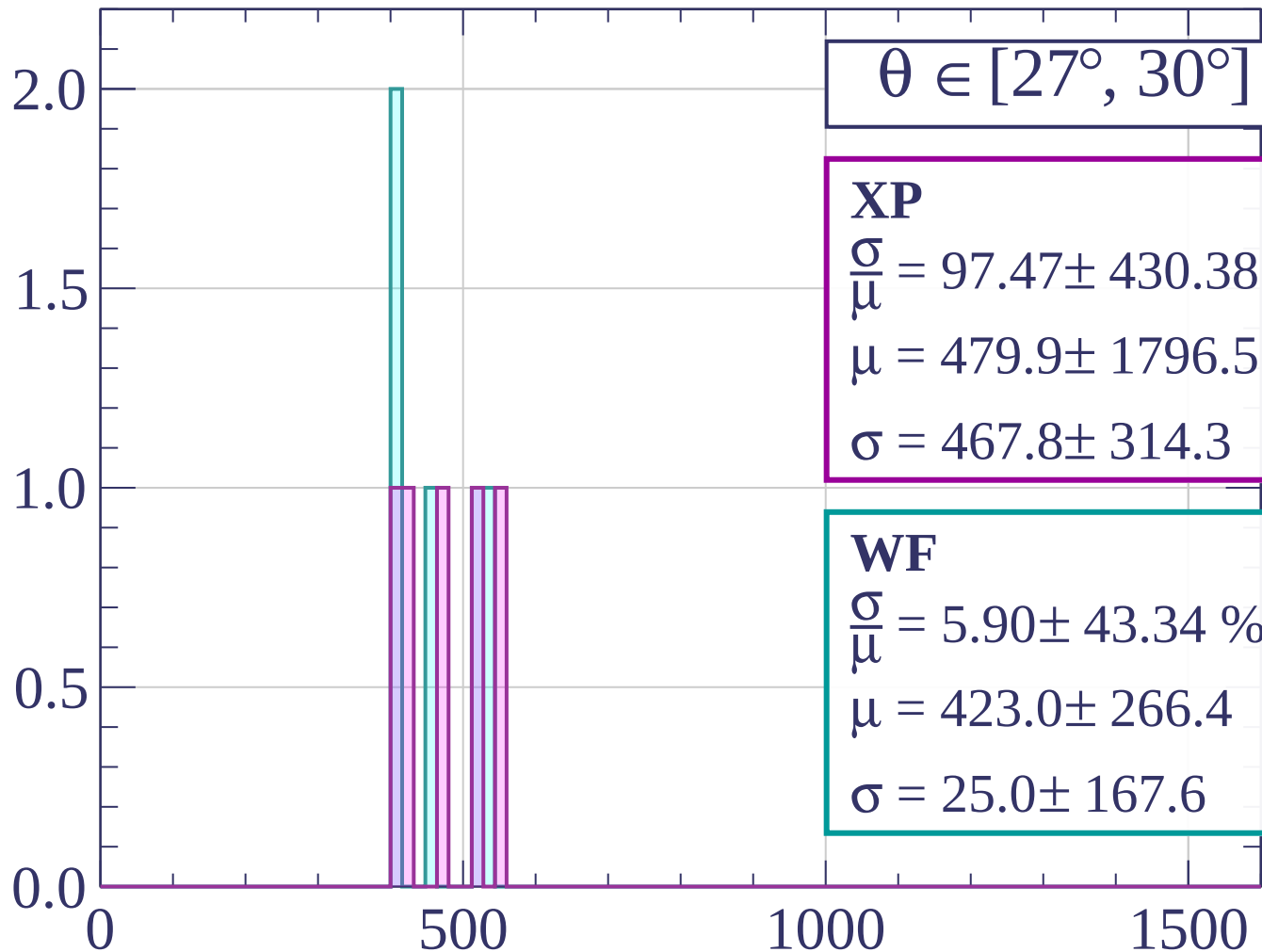
$$\sigma = 467.8 \pm 314.3$$

WF

$$\frac{\sigma}{\mu} = 5.90 \pm 43.34 \%$$

$$\mu = 423.0 \pm 266.4$$

$$\sigma = 25.0 \pm 167.6$$



dE/dx [ADC counts/cm]

